



ACSM's Guidelines for Exercise Testing and Prescription

Twelfth Edition



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ACSM's Guidelines for Exercise Testing and Prescription

Twelfth Edition

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ACSM's Guidelines for Exercise Testing and Prescription

Twelfth Edition



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The authors, editors, and publisher have exerted every effort to ensure that drug selection and dosage set forth in this text are in accordance

with the current recommendations and practice at the time of publication. However, in view of ongoing research, changes in government regulations, and the constant flow of information relating to drug therapy and drug reactions, the reader is urged to check the package insert for each drug for any change in indications and dosage and for added warnings and precautions. This is particularly important when the recommended agent is a new or infrequently employed drug.

Some drugs and medical devices presented in this publication have U.S. Food and Drug Administration (FDA) clearance for limited use in restricted research settings. It is the responsibility of the health care provider to ascertain the FDA status of each drug or device planned for use in their clinical practice.

This book is dedicated to the hundreds of volunteer professionals who have, since 1975, contributed thousands of hours developing these internationally adopted Guidelines. Now in its twelfth edition, it is the most widely circulated set of guidelines established for exercise professionals. This edition is dedicated to the editors, the writing teams, and the reviewers of this and previous editions who have not only provided their collective expertise but also sacrificed precious time with their colleagues, friends, and families to make sure that these Guidelines meet the highest standards in both science and practice.

INTRODUCTION

The American College of Sports Medicine (ACSM) *Guidelines* origins are within the ACSM Committee on Certification and Registry Boards (CCRB, formerly known as the Certification and Education Committee and the Preventive and Rehabilitative Exercise Committee). Today, the *Guidelines* are reviewed by a combination of certified professionals and content experts to provide the most relevant information to individuals who conduct exercise testing or develop exercise programs. The *Guidelines* provide the foundation of content for its supporting companion texts produced by ACSM, which include *ACSM's Certification Review*, *ACSM's Resources for the Personal Trainer*, *ACSM's Resources for the Exercise Physiologist*, *ACSM's Fitness Assessment Manual*, and several other key ACSM titles.

The first edition of the *Guidelines* was published in 1975, with updated editions published approximately every 4–6 years. The outstanding scientists and clinicians who have served in leadership positions as chairs and editors of the *Guidelines* since 1975 are as follows:

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Foreword

HOW TO USE THE GUIDELINES: CEMAL OZEMEK, PHD, FACSM, FAACVPR, FCEPA, ACSM-CEP

The twelfth edition of the *ACSM's Guidelines for Exercise Testing and Prescription* celebrates 50 years since its inception in 1975. Since then, numerous scientific contributions have shaped our understanding of the benefits associated with exercise across populations and the strategies that can be employed to elicit physiologic adaptations. The growth of the field can be appreciated when considering that the first edition of the *Guidelines* was only 116 pages and 4 chapters, while this latest edition is over 600 pages with 12 chapters. The *Guidelines* have maintained their intended purpose of being used as a quick, evidence-based resource for applying exercise testing and prescription across populations. However, this text has expanded to provide information on foundational concepts within topic areas, making it an invaluable resource in the classroom and in the field. Acknowledging that the use of the *Guidelines* has evolved over the past half-century, this Foreword aims to propose how this text can be optimally utilized.

Use Recommendations

Students — Information presented in these *Guidelines* are done so in a manner that provides surface-level support and explanation of various practices in the absence of an exhaustive review. To gain mastery within the topics of your interest and career trajectory, reading the referenced

resources at the end of respective chapters and seeking out additional text is highly recommended.

Course Instructors — For the reasons stated *above*, the GETP is an excellent resource that can be used to introduce and reinforce foundational skills in exercise testing and exercise prescription across populations. This text is particularly useful when implemented in *practical* learning settings where students have the opportunity to apply their knowledge and skills. Moreover, it may be safe to presume that *an objective* of a course or an academic program is to prepare students to pass an ACSM Certification or Certificate examination while simultaneously striving for specialization and mastery of the material. To successfully accomplish both, we advise that this text be used as a supplement in courses that specialize in areas of exercise science while making use of focused text options listed in the *Introduction* paragraph.

Professionals in Applied Settings — These *Guidelines* are meant to guide the delivery of an assessment, exercise prescription, and/or supervision of exercise within a population of interest. After working in the field, it should become evident that “one size fits all” methods of testing and prescribing exercise can limit an individual’s potential outcomes. Drawing information from each chapter and section can enhance exercise testing and prescription efficacy for the client or patient that may have a myriad of physical, personal, psychological, and health considerations. Furthermore, use the *Guidelines* as a good place to start an intervention and mindfully deviate relative to the considerations and goals identified by you and the individual. It is highly recommended to continuously expand your skillset by reading

additional textbooks, journal articles, and regularly enrolling in continuing education opportunities through professional organizations.

Preface

The 12th edition of *ACSM's Guidelines for Exercise Testing and Prescription* will continue the efforts of the editors and contributing authors of recent editions to make it a true *Guidelines* book rather than a sole and inclusive *resource*. It was the original intent of the *Guidelines* to be user-friendly, easily accessible, and a current primary resource for exercise and other health professionals who conduct exercise testing and exercise programs. Accordingly, edits were made in an effort to support the application of said practices in applied settings, while expanding sections on special populations. Additional online resources conclude individual sections or chapters with the purpose of facilitating the mastery of the material.

The reader of this edition of *ACSM's Guidelines for Exercise Testing and Prescription* will notice several innovations. The 12th edition of the *Guidelines* presents new sections on sex differences, transgender and gender diverse individuals, spontaneous coronary artery dissection, postural orthostatic tachycardia syndrome, pediatric patients with cardiac conditions, metabolic dysfunction–associated steatotic liver disease, and myalgic encephalomyelitis/chronic fatigue syndrome. Also, a section on respiratory muscle testing and training was added to the section on pulmonary disease as much evidence has shown clinically meaningful outcomes. Moreover, [Chapter 12](#) on Behavioral Theories and the resistance training section in [Chapter 5](#) were significantly

expanded upon to reflect contemporary, evidence-based practices. We have integrated the most recent guidelines and recommendations available from ACSM position stands and other relevant professional organizations' scientific statements, so that the *Guidelines* are the most current, primary resource for exercise testing and prescription. It is important for the readership to know that new themes and innovations included in the 12th edition were developed with input from ACSM student, instructor, and professional members prior to the initiation of this project via an electronic survey and virtual focus groups that asked respondents and individuals, respectively, for their suggestions regarding the content.

Any updates made in this edition of the *Guidelines* after their publication and prior to the publication of the next edition of the *Guidelines* can be accessed from the ACSM website (<https://www.acsm.org>). Furthermore, the reader is referred to the ACSM Get Certified link for a listing of ACSM Certifications at <https://www.acsm.org/certification/get-certified> for detailed examination content outlines.

ACKNOWLEDGMENTS

This editorial team must first express our deepest gratitude for the patience, expertise, guidance, and friendship of Angie Chastain, ACSM Director of Publishing, and Katie Feltman, ACSM Chief Operating Officer and Chief Executive Officer. Their consistent support allowed us to comfortably navigate through the editorial process, giving us the time and energy to focus our attention on the content. We would also like to acknowledge the extraordinary work of the ACSM Publications Committee and its Chair, Jeffrey Potteiger, and for entrusting in us with the *Guidelines*.

The *Guidelines* is among the premier texts in our field and meets the highest standards because of the internationally recognized contributing authors. We are indebted to the contributing authors of the 12th edition for volunteering their expertise and valuable time to ensure the *Guidelines* meet the highest standards in exercise science and practice. It was an honor for us to work with each and every contributor. Moreover, the *Guidelines* review process undergoes many layers of expert scrutiny to ensure the highest quality of content, and we thank the many reviewers for their careful reviews of the 12th edition.

We thank our publisher, Wolters Kluwer, and in particular Lindsey Porambo, Acquisitions Editor; Amy Millholen, Senior Development Editor; and Kirsten Watrud, Marketing Manager.

Upon accepting this role, I solicited advice from past Senior Editors. Perhaps the most impactful advice came from Walt Thompson, PhD (Senior Editor of the ninth edition). “Choose your Associate Editors

wisely.” I was humbled by Drs. Amanda Bonikowske, Jeff Christle, and Paul Gallo’s enthusiastic acceptance of the position, inspired by their work ethic and mastery of topics within the field, and simply appreciative of their selflessness, professionalism, and authenticity. They have embodied the essence of teamwork in our collective effort to maintain the highest standards for the *Guidelines*. Above all, I am privileged to have had the opportunity to work with these dear friends on a text that holds significant personal and professional meaning. Lastly, but of greatest importance, we are grateful for the love and support we receive from our respective partners and children on a daily basis. They have always kept us centered and have given us the motivation and courage in our pursuit of excellence.

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Nota Bene

The views and information contained in this edition of *ACSM’s Guidelines for Exercise Testing and Prescription* are provided as guidelines — as opposed to *standards of practice*. This distinction is an important one because specific legal connotations may be attached to standards of practice that are not attached to guidelines. This distinction is critical inasmuch as it gives the professional in exercise testing and programmatic settings the freedom to deviate from these guidelines when necessary and appropriate in the course of using independent and prudent judgment. *ACSM’s Guidelines for Exercise Testing and Prescription* presents a framework whereby the professional may certainly — and in some cases has the obligation to —

tailor to individual needs while balancing institutional or legal requirements.

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AACE	American Association of Clinical Endocrinologists and American College of Endocrinology
AACVPR	American Association of Cardiovascular and Pulmonary Rehabilitation
AAP	American Academy of Pediatrics
ABI	ankle/brachial pressure index
ACC	American College of Cardiology
ACE-I	angiotensin-converting enzyme inhibitors
ACS	Acute coronary syndrome
ACSM	American College of Sports Medicine
ACSM-CEP	ACSM Certified Clinical Exercise Physiologist
ACSM-CPT	ACSM Certified Personal Trainer
ACSM-EP	ACSM Certified Exercise Physiologist
ACSM-GEI	ACSM Certified Group Exercise Instructor
ACOG	American Congress of Obstetricians and Gynecologists
ADHD	attention-deficit/hyperactivity disorder
ADL	activities of daily living
ADP	air displacement plethysmography
ADT	androgen deprivation therapy

AED	automated external defibrillators
AHA	American Heart Association
AHFS	American Hospital Formulary Service
AI	artificial intelligence
AIDS	acquired immunodeficiency syndrome
AIS	American Spinal Cord Injury Impairment Scale
ALP	alkaline phosphatase
ALT	alanine transaminase
AMI	acute myocardial infarction
AMRAP	as-many-reps-as-possible
AMS	acute mountain sickness
apoB	apolipoprotein B
ARBs	angiotensin II receptor blockers
ART	antiretroviral therapy
AS	ankylosing spondylitis
ASA	American Stroke Association
ASCVD	atherosclerotic cardiovascular disease
ASD	autism spectrum disorder
AST	aspartate transaminase
ATP III	Adult Treatment Panel III
AV	atrioventricular
BCI	brain-computer interface
BESS	balance error scoring system
BIA	bioelectrical impedance analysis

BMD	bone mineral density
BMI	body mass index
BMT	bone marrow transplantation
BP	blood pressure
BWSTT	body weight supported treadmill training
CABG(S)	coronary artery bypass graft (surgery)
CAD	coronary artery disease
CCB	calcium channel blockers
CDC	Centers for Disease Control and Prevention
CHF	congestive heart failure
CKD	chronic kidney disease
CM	cardiomyopathy
CNS	central nervous system
COPD	chronic obstructive pulmonary disease
CO ₂	carbon dioxide
CP	cerebral palsy
CPET	cardiopulmonary exercise test
CPR	cardiopulmonary resuscitation
CR	cardiac rehabilitation
CR10	Borg category-ratio 10 scale
CRF	cardiorespiratory fitness
CT	computed tomography
CVA	cerebrovascular accident

CVD	cardiovascular disease
CWR	constant work rate
Db	body density
DBP	diastolic blood pressure
DBS	deep brain stimulation
DM	diabetes mellitus
DMARDS	disease-modifying antirheumatic drugs
DOMS	delayed onset muscle soreness
DRS	diastasis recti abdominis
DS	Down syndrome
DVR	dynamic variable resistance
DXA	dual-energy x-ray absorptiometry
EAMC	exercise-associated muscle cramps
EAS	European Atherosclerosis Society
ECG	electrocardiogram (electrocardiographic)
EDSS	Kurtzke Expanded Disability Status Scale
EE	energy expenditure
EI	energy intake
EIB	exercise-induced bronchoconstriction
ESRD	end-stage renal disease
ETT	exercise tolerance test
Ex R _x	exercise prescription
FBG	fasting blood glucose

FES-LCE	functional electrical stimulation-leg cycle ergometry
FEV _{1.0}	forced expiratory volume in one second
FFBd	fat-free body density
FFM	fat-free mass
FIO ₂	fraction of inspired oxygen
FIQ	Fibromyalgia Impact Questionnaire
FITT	Frequency, Intensity, Time, Type
FM	fat mass
FMD	fibromuscular dysplasia
FMS	fundamental movement skill
FN	false negative
FP	false positive
FPG	fasting plasma glucose
FRAX	Fracture Risk Algorithm
FRIEND	Fitness Registry and the Importance of Exercise National Database
FVC	forced vital capacity
GAQ- Pregnancy	get active questionnaire for pregnancy
GFR	glomerular filtration rate
GLP	glucagon like peptide
GMFCS	gross motor function classification system
GnRHa	gonadotropin-releasing hormone agonists
GOLD	Global Initiative for Chronic Obstructive Lung Disease

GREX	Global Renal Exercise Network
GXT	graded exercise test
HACE	high-altitude cerebral edema
HAPE	high-altitude pulmonary edema
HbA1C	glycated hemoglobin
HBM	health belief model
HDL-C	high-density lipoprotein cholesterol
HeFH	heterozygous familial hypercholesterolemia
HFpEF	heart failure with preserved ejection fraction
HFrEF	heart failure with reduced ejection fraction
HHQ	Health History Questionnaire
HIIT	high-intensity interval training
HIPAA	Health Insurance Portability and Accountability Act
HIV	human immunodeficiency virus
HMG-CoA	hydroxymethylglutaryl coenzyme A
HR	heart rate
HR _{max}	maximal heart rate
HR _{peak}	peak heart rate
HRR	heart rate reserve
HR _{rest}	resting heart rate
hs-CRP	high-sensitivity C-reactive protein
HSCT	hematopoietic stem cell transplantation
ICD	implantable cardioverter defibrillator

ID	intellectual disability
ID	iron deficiency
IDF	International Diabetes Federation
IFG	impaired fasting glucose
IGT	impaired glucose tolerance
IHD	ischemic heart disease
K	interstitial lung disease
IMT	inspiratory muscle training
IRD	inter-rectus distance
ISH	International Society of Hypertension
ISPD	International Society for Peritoneal Dialysis
IVCD	intraventricular conduction delay
JTA	job task analysis
KDIGO	Kidney Diseases: Improving Global Outcomes
KSs	knowledge and skills
LABS	Longitudinal Assessment of Bariatric Surgery
LBBS	left bundle branch block
LBP	low back pain
LDL-C	low-density lipoprotein cholesterol
LGBTQ	lesbian, gay, bisexual, transgender, queer or questioning
L-G-L	Lown-Ganong-Levine
LVAD	left ventricular assist device
LVEF	left ventricular ejection fraction
LVH	left ventricular hypertrophy

MAP	mean arterial pressure
MASLD	metabolic dysfunction-associated steatotic liver disease
MCR	metabolic chronotropic reserve
MDS-UPDRS	Movement Disorder Society Unified Parkinson's Disease Rating Scale
ME/CFS	myalgic encephalomyelitis/chronic fatigue syndrome
MEP	maximal expiratory pressure
MET	metabolic equivalent
MetSyn	metabolic syndrome
MI	myocardial infarction
MIP	maximal inspiratory pressure
MRI	magnetic resonance imaging
MS	multiple sclerosis
MSI	musculoskeletal injury
MVC	maximal voluntary contraction
MVPA	moderate-to-vigorous physical activity
6-MWT	6-min walk test
NAFLD	nonalcoholic fatty liver disease
NATA	National Athletic Trainer's Association
NCEP	National Cholesterol Education Program
NDRI	Norepinephrine-dopamine reuptake inhibitor
NFCI	nonfreezing cold injuries
NIH	National Institutes of Health
NHANES	National Health and Nutrition Examination Survey

NHLBI	National Heart, Lung, and Blood Institute
NMES	neuromuscular electrical stimulation
NOTF	National Obesity Task Force
NR	not recommended
NSAIDs	nonsteroidal anti-inflammatory drugs
NYHA	New York Heart Association
OA	osteoarthritis
OGTT	oral glucose tolerance test
OUES	oxygen uptake efficiency slope
O ₂	oxygen
PA	physical activity
PACER	progressive aerobic cardiovascular endurance run
PAD	peripheral artery disease
P _a CO ₂	partial pressure of carbon dioxide
PAH	pulmonary arterial hypertension
PEM	post exertional malaise
P _a O ₂	partial pressure of arterial oxygen
PAR-Q+	Physical Activity Readiness Questionnaire for Everyone
PCI	percutaneous coronary intervention
PD	Parkinson disease
PG	plasma glucose
PH	pulmonary hypertension
PNF	proprioceptive neuromuscular facilitation

POTS	postural orthostatic tachycardia syndrome
PR	pulmonary rehabilitation
PVC	premature ventricular contraction
PWH	people with HIV
$\dot{Q}$	cardiac output
QTc	QT corrected for heart rate
RA	rheumatoid arthritis
RED-S	relative energy deficiency in sport
RER	respiratory exchange ratio
RHR	resting heart rate
RIR	repetitions in reserve
1-RM	one repetition maximum
RMD	rheumatic and musculoskeletal disease
RMT	respiratory muscle training
ROM	range of motion
RPE	rating of perceived exertion
RPP	rate pressure product
RVH	right ventricular hypertrophy
SaO ₂	percent saturation of arterial oxygen
SBP	systolic blood pressure
SCAD	spontaneous coronary artery dissection
SCD	sudden cardiac death
SCI	spinal cord injury

SCIM	spinal cord injury independence measures
SCT	social cognitive theory
SD	standard deviation
SDT	self-determination theory
SEE	standard error of the estimate
SEM	social ecological model
SFT	senior fitness test
SIPE	swimming-induced pulmonary edema
SIT	sprint interval training
SMARTS	specific, measurable, action-oriented, realistic, timely, self-determined
SNRI	Serotonin and norepinephrine reuptake inhibitor
SpO ₂	percent saturation of arterial oxygen
SPPB	Short Physical Performance Battery
T1DM	Type 1 diabetes mellitus
T2DM	Type 2 diabetes mellitus
TG	triglycerides
THR	target heart rate
TN	true negative
TP	true positive
TPB	theory of planned behavior
TTM	transtheoretical model
TUG	timed up and go
UKKA	United Kingdom Kidney Association

USPSTF	United States Preventive Services Task Force
VAT	ventilatory-derived anaerobic threshold
$\dot{V}CO_2$	volume of carbon dioxide per minute
$\dot{V}E$	expired ventilation per minute
VF	ventricular fibrillation
$\dot{V}O_2$	volume of oxygen consumed per minute
$\dot{V}O_{2max}$	maximal volume of oxygen consumed per minute (maximal oxygen uptake, maximal oxygen consumption)
$\dot{V}2_{peak}$	peak oxygen uptake
$\dot{V}O_{2R}$	oxygen uptake reserve
% $\dot{V}O_{2R}$	percentage of oxygen uptake reserve
VT	ventilatory threshold
WADA	World Anti-Doping Agency
WBGT	wet-bulb globe temperature
WCT	Wind Chill Temperature Index
WHR	waist-to-hip ratio
W-P-W	Wolff-Parkinson-White
YLD	years lived with disease

CHAPTER 1

Benefits and Risks Associated With Physical Activity

INTRODUCTION

This chapter summarizes information regarding the general benefits and risks of physical activity (PA) and/or exercise. Information related to the benefits of PA and exercise specific to a disease, disability, or health condition are explained within the respective chapters of this edition of the guidelines. PA continues to take on an increasingly important role in the prevention and treatment of multiple chronic diseases, health conditions, and their associated risk factors. Thus, this chapter focuses on the public health perspective that forms the basis for the current PA recommendations ([1–6](#)). This chapter concludes with recommendations for reducing the incidence and severity of exercise-related complications for primary and secondary prevention programs.

PHYSICAL ACTIVITY AND FITNESS

TERMINOLOGY

PA and exercise are often used interchangeably; however, these terms are not synonymous. *PA* is defined as any bodily movement produced by the contraction of skeletal muscles that results in an increase in caloric requirements over resting energy expenditure ([7](#)). *Exercise*, on the other hand, is a type of PA consisting of planned, structured, and repetitive bodily movement done to improve and/or maintain one or more components of physical fitness ([7](#)). *Physical fitness*, although defined in several ways, has generally been described as a set of attributes or characteristics individuals have or achieve that relate to their ability to perform PA and activities of daily living ([7](#)). These attributes or characteristics are commonly separated into health- and skill-related components of physical fitness. Nonetheless, recent evidence suggests that these components of physical fitness may not be mutually exclusive, as several skill-related components can be important for achieving health goals and, therefore, should be incorporated when designing exercise prescription programs with different populations (*e.g.*, power and balance activities with older adults) ([Box 1.1](#)).

Box 1.1 Health- and Skill-Related Components of Physical Fitness

Health-Related Physical Fitness Components

- Cardiorespiratory endurance: the ability of the circulatory and respiratory system to supply oxygen during sustained physical activity
- Body composition: the relative amounts of muscle, fat, bone, and other vital parts of the body
- Muscular strength: the ability of muscle to exert force
- Muscular endurance: the ability of muscle to continue to perform without fatigue
- Flexibility: the range of motion available at a joint

Skill-Related Physical Fitness Components

- Agility: the ability to change the position of the body in space with speed and accuracy
- Coordination: the ability to use the senses, such as sight and hearing, together with body parts in performing tasks smoothly and accurately
- Balance: the maintenance of equilibrium while stationary or moving
- Power: the ability or rate at which one can perform work
- Reaction time: the time elapsed between stimulation and the beginning of the reaction to it
- Speed: the ability to perform a movement within a short period of time

Adapted from [\(7\)](#). Originally printed in Corbin, Charles B., et al., Concepts in physical education with laboratories and experiments, 4th ed. © 1970, 1974, 1978, 1980 Wm. C. Brown Publishers, Dubuque, IA.

In addition to defining type of PA, it is important to clearly define the wide range of intensities associated with PA (see [Table 5.2](#)) and with different methods for estimating intensities, which include percentage of oxygen uptake reserve ($\dot{V}O_{2R}$), heart rate reserve (HRR), volume of oxygen consumed per minute ($\dot{V}O_2$), heart rate (HR), or metabolic equivalents (METs; see [Box 5.2](#) and [Appendix C](#)). Several chapters throughout the guidelines provide the methodology and guidance for selecting a suitable estimation method based on individual circumstances.

METs are a practical, convenient, and standardized method for quantifying the intensity of various behaviors and activities. One MET is equivalent to resting metabolic rate ($\sim 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) or the amount of oxygen consumed while sitting at rest. Among adults, light-intensity PA is defined as 1.6–2.9 METs, moderate as 3.0–5.9 METs, and vigorous as ≥ 6.0 METs ([6](#)). [Table 1.1](#) gives specific examples of MET values for activities in each of the described intensity ranges. A comprehensive catalog of absolute intensity values for various behaviors and activities can be found in the Compendium of Physical Activities ([8–10](#)).

TABLE 1.1 Metabolic Equivalents (METs) Values of Common Physical Activities Classified as Light, Moderate, or Vigorous Intensity

Light (1.6–2.9 METs)	Moderate (3.0–5.9 METs)	Vigorous (≥6.0 METs)
Walking Walking slowly around home, store, or office = 2.0^a Household and occupation Standing performing light work, such as making bed, washing dishes, ironing, preparing food, or store clerk = 2.0–2.5 Leisure time and sports Billiards = 2.5 Boating — power = 2.5 Croquet = 2.5 Darts = 2.5 Fishing — sitting = 2.5 Playing most musical instruments = 2.0–2.5	Walking Walking 3.0 mi · h⁻¹ = 3.0^a Walking at very brisk pace (4 mi · h⁻¹) = 5.0^a Household and occupation Cleaning, heavy — washing windows, car, clean garage = 3.0 Sweeping floors or carpet, vacuuming, mopping = 3.0–3.5 Carpentry — general = 3.6 Carrying and stacking wood = 5.5 Mowing lawn — walk power mower = 5.5 Leisure time and sports Badminton — recreational = 4.5 Basketball — shooting around = 4.5 Dancing — ballroom slow = 3.0; ballroom fast = 4.5 Fishing from riverbank and walking = 4.0	Walking, jogging, and running Walking at very, very brisk pace (4.5 mi · h⁻¹) = 6.3^a Walking/hiking at a moderate pace and grade with no or light pack (<10 lb) = 7.0 Hiking at steep grades and pack 10–42 lb = 7.5–9.0 Jogging at 5 mi · h⁻¹ = 8.0^a Jogging at 6 mi · h⁻¹ = 10.0^a Running at 7 mi · h⁻¹ = 11.5^a Household and occupation Shoveling sand, coal, etc. = 7.0 Carrying heavy loads, such as bricks = 7.5 Heavy farming, such as bailing hay = 8.0 Shoveling, digging ditches = 8.5 Leisure time and sports Bicycling on flat — light effort (10–12 mi · h⁻¹) = 6.0 Basketball game = 8.0 Bicycling on flat — moderate effort (12–14 mi · h⁻¹) = 8.0; fast (14–16 mi · h⁻¹) = 10.0 Skiing cross-country — slow (2.5 mi · h⁻¹) = 7.0; fast (5.0–7.9 mi · h⁻¹) = 9.0

Light (1.6–2.9 METs)	Moderate (3.0–5.9 METs)	Vigorous (≥6.0 METs)
	Golf — walking, pulling clubs = 4.3 Sailing boat, wind surfing = 3.0 Table tennis = 4.0 Tennis doubles = 5.0 Volleyball — noncompetitive = 3.0–4.0	Soccer — casual = 7.0; competitive = 10.0 Swimming leisurely = 6.0 ^b ; swimming — moderate/hard = 8.0–11.0 ^b Tennis singles = 8.0 Volleyball — competitive at gym or beach = 8.0

^a On flat, hard surface.

^b MET values can vary substantially from individual to individual during swimming as a result of different strokes and skill levels. Adapted from (8–10).

Because of age-related declines in cardiorespiratory fitness (CRF, or the integrated capacity to transport oxygen from the atmosphere to the mitochondria to perform physical work) (2, 4, 11), when older and younger individuals work at the same MET level, the relative exercise intensity (*e.g.*, $\dot{V}O_{2\max}$) will usually be different (see [Chapter 5](#)), as an older individual will typically work at a greater relative percentage of maximal oxygen consumption ($\dot{V}O_{2\max}$) than a younger individual. Nonetheless, physically active older adults may have CRF comparable to or greater than those of physically inactive younger adults. This relationship will be similar when comparing individuals with different fitness levels, where those with lower $\dot{V}O_{2\max}$ will work at a higher percentage of their maximal ability compared to their more fit counterparts at the same absolute MET value.

PUBLIC HEALTH PERSPECTIVE FOR CURRENT RECOMMENDATIONS

In the mid-1990s, the American College of Sports Medicine (ACSM), in conjunction with the U.S. Centers for Disease Control and Prevention (CDC) ([12](#)), the U.S. Surgeon General ([13](#)), and the National Institutes of Health (NIH) ([14](#)), issued landmark publications on PA and health. An important goal of these reports was to clarify for exercise professionals and the public the amount and intensity of PA needed to improve health, lower susceptibility to disease (morbidity), and decrease premature mortality ([12–14](#)). In addition, these reports documented the positive dose-response relationship between PA and health.

In 1995, the CDC and ACSM recommended that “every U.S. adult should accumulate 30 minutes or more of moderate physical activity on most, preferably all, days of the week” ([12](#)). This recommendation was quickly followed in 1996 by *Physical Activity and Health: A Report of the Surgeon General*, a landmark report detailing the myriad health benefits associated with regular PA ([13](#)). Collectively, the intent of these statements was to increase public awareness of the health-related benefits of moderate-intensity PA. As a result of increasing awareness of the adverse health effects of physical inactivity and confusion around the original PA recommendations, the ACSM and American Heart Association (AHA) issued updated recommendations for PA and health in 2007 ([Box 1.2](#)) ([3](#)).

Box 1.2 The ACSM-AHA Primary Physical Activity Recommendations (3)

- All healthy adults aged 18–65 years should participate in moderate-intensity aerobic PA for a minimum of 30 minutes on 5 d · wk⁻¹ or vigorous-intensity aerobic activity for a minimum of 20 minutes on 3 d · wk⁻¹.
- Combinations of moderate- and vigorous-intensity exercise can be performed to meet this recommendation.
- Moderate-intensity aerobic activity can be accumulated to total 30 minutes by performing bouts, each lasting ≥10 minutes.
- Every adult should perform activities that maintain or increase muscular strength and endurance for a minimum of 2 d · wk⁻¹.
- Because of the dose-response relationship between PA and health, individuals who wish to further improve their fitness, reduce their risk for chronic diseases and disabilities, and/or prevent unhealthy weight gain may benefit by exceeding the minimum recommended amounts of PA.

ACSM, American College of Sports Medicine; AHA, American Heart Association; PA, physical activity.

Shortly after the publication of the joint ACSM and AHA recommendations, the federal government convened an expert panel, the 2008 Physical Activity Guidelines Advisory Committee (15), to review the scientific evidence on PA and health published since the 1996 U.S. Surgeon General's Report (13). This committee found compelling evidence regarding the benefits of PA for health as well as the presence of a dose-response relationship for many diseases and health conditions. The 2008 Physical Activity Guidelines Advisory Committee provided the *Physical Activity Guidelines Advisory Committee Report* (15), which resulted in two important conclusions that influenced the development of subsequent PA recommendations:

1. Important health benefits can be obtained by performing a moderate amount of PA on most, if not all, days of the week.
2. Individuals who maintain a regular program of PA that is longer in duration, of greater intensity, or both are likely to derive greater benefits than those who engage in lesser amounts.

These recommendations from the *Physical Activity Guidelines Advisory Committee Report* ultimately resulted in the *2008 Physical Activity Guidelines for Americans*, which were the first comprehensive guidelines related to PA published by the Federal government (see <https://health.gov/paguidelines/2008/>) (5).

After a decade of research and implementation of the *2008 Physical Activity Guidelines for Americans*, the federal government convened another expert panel, the 2018 Physical Activity Guidelines Advisory Committee (16), to further review the scientific evidence on PA and health published since the *2008 Physical Activity Guidelines for Americans* (5) were released. The committee identified further evidence supporting the beneficial effects of PA on health. Beyond

attenuating chronic disease risks, additional conclusions regarding the benefits of PA for health from the *2018 Physical Activity Guidelines Advisory Committee Scientific Report* include, but are not limited to, the following:

1. For those who are inactive, PA of any intensity (light, moderate, or vigorous) can provide health benefits.
2. Health benefits can be experienced following just a single bout of moderate-to-vigorous physical activity (MVPA).
3. Any bout of MVPA, regardless of duration, can be included when quantifying daily or weekly volumes of PA intended to be counted toward meeting recommendations.

The collective findings from the *2018 Physical Activity Guidelines Advisory Committee Scientific Report* ([16](#)) have been further summarized and incorporated into the most recent federal PA guidelines — the *Physical Activity Guidelines for Americans, Second Edition* (<https://health.gov/paguidelines/second-edition/>) ([Box 1.3](#)) ([6](#)).

Box 1.3 The Primary Physical Activity Recommendations for Adults From the *Physical Activity Guidelines for Americans, Second Edition* (6)

- Adults should move more and sit less throughout the day. Some physical activity is better than none. Adults who sit less and do any amount of moderate-to-vigorous physical activity gain some health benefits.
- For substantial health benefits, adults should do at least $150\text{--}300\text{ min} \cdot \text{wk}^{-1}$ of moderate-intensity or $75\text{--}150\text{ min} \cdot \text{wk}^{-1}$ of vigorous-intensity aerobic physical activity, or an equivalent combination of moderate- and vigorous-intensity aerobic activity. Preferably, aerobic activity should be spread throughout the week.
- Additional health benefits are gained by engaging in physical activity beyond the equivalent of 300 minutes of moderate-intensity physical activity a week.
- Adults should also do muscle strengthening activities of moderate or greater intensity and that involve all major muscle groups on 2 or more $\text{d} \cdot \text{wk}^{-1}$, as these activities provide additional health benefits.

Notable among changes reflected in the 2018 guidelines is the elimination of the previous requirement that aerobic activities must occur in bouts of at least 10 minutes in duration to elicit significant health benefits (6). There is now overwhelming evidence indicating that bouts of PA less than 10 minutes in duration are also associated with favorable outcomes for a variety of health-related indicators (16). Furthermore, it is seen as counterproductive to discourage being physically active for any duration.

Since the release of the 1996 Surgeon General's Report (13), several position statements have advocated for PA levels above the minimum CDC-ACSM recommendations (17, 18). There are data supporting that PA that is sufficient to reduce chronic disease and mortality risks may be insufficient to prevent or reverse weight gain and/or obesity given the typical American lifestyle. Therefore, PA beyond the minimum recommendations combined with proper nutrition is likely needed for many individuals to manage and/or prevent weight gain and obesity (17, 19). These recommendations for the volume of PA likely to prevent weight gain and/or obesity should not be viewed as contradictory but rather supplementary to the general guidelines.

Numerous large-scale epidemiological studies have been performed that document the inverse relationship between PA and cardiovascular disease (CVD) incidence and all-cause or CVD-related mortality (20–22). A meta-analysis of 23 sex-specific cohorts representing 1,325,004 person-years of follow-up showed a strong inverse dose-response relationship between PA and risk for CVD independent of baseline levels of PA or CRF (Figure 1.1) (22). In general, findings from this study indicated that even low or moderate levels of PA can provide some benefits; however, higher levels of PA or exercise provide additional health benefits (22).

[Table 1.2](#) provides the strength of evidence for the dose-response relationships among PA and numerous health outcomes.

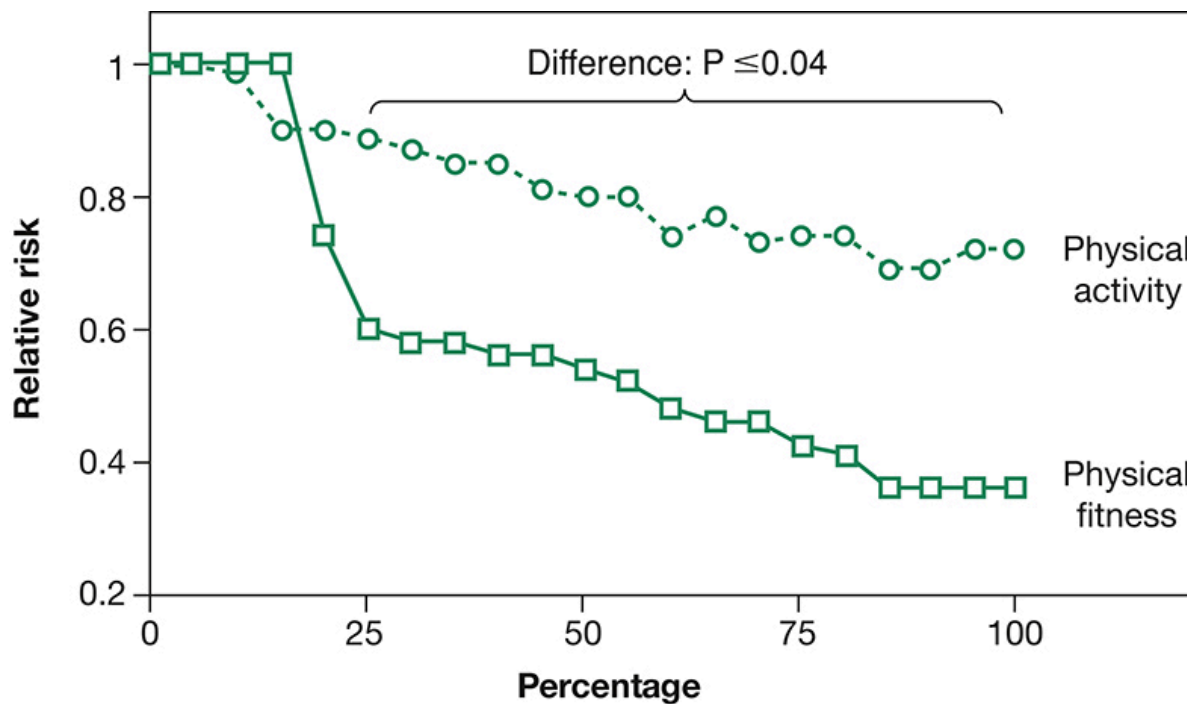


Figure 1.1 Estimated dose-response curve for the relative risk of atherosclerotic cardiovascular disease by sample percentages of fitness and physical activity. Studies weighted by individual-years of experience. Used with permission from ([22](#)).

TABLE 1.2 Evidence for Dose-Response Relationship Between Physical Activity and Health Outcomes in Adults

Variable	Evidence for a Dose-Response Relationship ^a	Strength of Evidence ^b
All-cause mortality ^c	Yes	Strong
Cardiorespiratory health ^c	Yes	Strong
Metabolic health ^c	Yes	Strong
Energy balance:		
Weight gain prevention ^c	Yes	Limited
Weight loss ^d	Yes	Strong
Weight maintenance following weight loss ^d	Yes	Moderate
Abdominal obesity ^d	Yes	Moderate
Musculoskeletal health:		
Bone ^d	Yes	Moderate
Muscular ^d	Yes	Strong
Functional health ^d	Yes	Moderate
Specific cancer risk:		
Bladder ^c	Yes	Moderate
Breast ^c	Yes	Strong
Colon ^c	Yes	Strong
Endometrial ^c	Yes	Moderate
Lung ^c	Yes	Limited
Ovarian ^c	No	Limited
Prostate ^c	Not assignable	Insufficient evidence
Mental health:		
Anxiety ^c	Yes	Limited
Cognition ^c	Not assignable	Insufficient evidence
Depression ^c	Yes	Limited
Well-being:		

Variable	Evidence for a Dose-Response Relationship ^a	Strength of Evidence ^b
Quality of life ^c	Not assignable	Insufficient evidence
Sleep ^c	Yes	Moderate

^a Evidence for a dose-response relationship was classified as follows:

“Yes” — Evidence supports a dose-response relationship

“No” — Evidence does not support a dose-response relationship

“Not assignable” — Evidence inadequate or too disparate to generate a meaningful conclusion

^b Strength of the evidence was classified as follows:

“Strong” — Consistent findings across many studies

“Moderate” — Some inconsistent findings across a moderate number of studies

“Limited” — Inconsistent findings across few studies

“Insufficient evidence” — Inadequate, too few, or no studies to assess strength of evidence

^c Adapted from (16).

^d Adapted from (15).

Older and younger individuals have important differences, which are reflected in exercise testing and prescription guidelines. The ACSM and AHA have released two publications examining the relationship between PA and public health in older adults (1, 4). In general, these publications offer recommendations that are similar to the updated guidelines for adults (2, 3). In addition, age-specific recommendations are made concerning the importance of flexibility, neuromotor, and resistance training exercises (1, 4). The *2008 Physical Activity Guidelines for Americans* (5) initially made three age-specific recommendations for PA targeted at children and adolescents (6–17 years), adults (18–64 years), and older adults (≥65 years) that are similar to recommendations by the ACSM and AHA. However, the *Physical Activity Guidelines for Americans, Second Edition* (6) has provided further age specificity by delineating PA recommendations for preschool-aged children (3–5 years), school-aged children and adolescents (6–17 years), adults (18–64 years), and older adults (≥65 years) to account for physiological, maturational, and health differences across these age groups.

SEDENTARY BEHAVIOR AND HEALTH

Despite the well-known health benefits of regular PA, physical inactivity remains a global health concern and has been identified as a leading contributor to premature mortality ([23–25](#)). Globally, greater than 30% of adults are physically inactive ([23](#)). In the United States, self-report data indicate that about half of adults meet endurance training guidelines, 30% meet resistance training guidelines, and only 20% meet both ([26](#)).

Over the past 10–15 years, the role of sedentary behavior in disease risk and progression has become an increasingly prominent public health concern ([27, 28](#)). To clarify, *sedentary behavior* is defined as any waking behavior characterized by an energy expenditure of ≤ 1.5 METs while in a sitting, reclining, or lying posture ([29](#)). Despite this standardized definition, scientific efforts to measure free-living sedentary behavior have varied considerably in terms of the assessment methodologies employed.

Initial efforts to better understand contemporary sedentary behavior levels and their associated health risks largely relied on self-reported sitting and/or screen viewing time data collected with the use of questionnaires ([30–32](#)), whereas subsequent studies have used motion sensing technologies (*e.g.*, pedometers, accelerometers) to estimate sedentary behaviors ([33–35](#)). Due to these differing methodologies, it remains unknown how much time Americans spend daily in sedentary behaviors. Self-reported data from the National Health and Nutrition Examination Survey (NHANES) indicate that American adults accumulate an average total sitting duration of $5.5–7.1 \text{ h} \cdot \text{d}^{-1}$ ([36](#)). In contrast, average estimates for sedentary time measured via waist-worn accelerometry have been higher, at $7.7–8.0 \text{ h} \cdot \text{d}^{-1}$ resulting in a range of approximately $5.5–8.0 \text{ h} \cdot \text{d}^{-1}$ ([37, 38](#)). Regardless of these discrepant measurement methodologies, a large proportion of people are spending too much time being sedentary, and the collective body of research evidence clearly indicates that high levels of sedentary behavior can be detrimental to health ([16](#)).

The deleterious relationships between sedentary behavior and various health indicators and outcomes have been demonstrated using a variety of research designs. Short-duration clinical experiments have shown that prolonged sitting elicits unfavorable effects on glucose and insulin control ([39–42](#)), lipid metabolism ([40, 41](#)), and vascular function ([43, 44](#)). Numerous cross-sectional studies have indicated that sedentary behavior is positively associated with various cardiometabolic risk factors ([45–49](#)). Moreover, a series of prospective longitudinal studies have demonstrated that high levels of sedentary behavior are associated with increased risks for incidence of diabetes ([31, 50–52](#)), heart disease ([53](#)), and cancer ([54, 55](#)) as well as increased risks for mortality from all causes ([32, 56–59](#)), CVD ([32, 56, 60–62](#)), and cancer ([63–65](#)). Follow-up systematic reviews and meta-analyses have largely confirmed findings from

previous longitudinal studies and further indicate that high levels of sedentary behavior pose significant health risks ([66–70](#)). Initially, it was believed that the relationship between sedentary behavior and various health outcomes, including mortality, was independent of time spent in higher-intensity activities (*i.e.*, MVPA) ([66](#)). However, a recent meta-analysis demonstrated that high levels of MVPA (≥ 4.3 times the minimum recommended amount — about 60–75 min $\cdot$ d⁻¹ of moderate-intensity PA) appear to eliminate the mortality risk associated with high levels of sedentary behavior ([67](#)). Collectively, there exists compelling evidence that time spent in sedentary behavior and MVPA influence health, and it is important to understand each of these behaviors to more fully describe health and mortality risks.

HEALTH BENEFITS OF REGULAR PHYSICAL ACTIVITY AND EXERCISE

Evidence supporting the inverse relationship between regular PA and/or exercise and premature mortality, CVD/coronary artery disease (CAD), hypertension, stroke, osteoporosis, Type 2 diabetes mellitus, metabolic syndrome, obesity, certain cancers, depression, functional health, falls, and cognitive function continues to accumulate ([6](#)). For many of these diseases and health conditions, there is also strong evidence of a dose-response relationship with PA (see [Table 1.2](#)).

Several large-scale epidemiological studies have documented a dose-response relationship between PA and risk of CVD and premature mortality in men and women and in ethnically diverse samples ([71–76](#)). It is also important to note that CRF has an inverse relationship with many adverse health outcomes, including risk of premature death from all causes and specifically from CVD ([77–81](#)), and higher levels of habitual PA are associated with higher levels of CRF ([82–86](#)), which in turn are related to many health benefits ([6](#)). [Box 1.4](#) summarizes the benefits of regular PA and/or exercise.

Box 1.4 Benefits of Regular Physical Activity and/or Exercise

Improvement in Cardiovascular and Respiratory Function

- Increased maximal oxygen uptake resulting from both central and peripheral adaptations
- Decreased minute ventilation at a given absolute submaximal intensity
- Decreased myocardial oxygen cost for a given absolute submaximal intensity
- Decreased heart rate and blood pressure at a given submaximal intensity
- Increased capillary density in skeletal muscle
- Increased exercise threshold for the accumulation of lactate in the blood
- Increased exercise threshold for the onset of disease signs or symptoms (*e.g.*, angina pectoris, ischemic ST-segment depression, claudication)

Reduction in Cardiovascular Disease Risk Factors

- Reduced resting systolic/diastolic pressure
- Increased serum high-density lipoprotein cholesterol and decreased serum triglycerides
- Reduced total body fat and intra-abdominal fat
- Reduced insulin needs; improved glucose tolerance
- Reduced blood platelet adhesiveness and aggregation
- Reduced inflammation

Decreased Morbidity and Mortality

- Primary prevention (*i.e.*, interventions to prevent the initial occurrence).
- Higher activity and/or fitness levels are associated with lower death rates from CAD.
- Higher activity and/or fitness levels are associated with lower incidence rates for CVD; CAD; stroke; Type 2 diabetes mellitus; metabolic syndrome; osteoporotic fractures; cancer of the bladder, breast, colon, endometrium, and lung; and gallbladder disease.
- Secondary prevention (*i.e.*, interventions after a cardiac event to prevent another).
- Based on meta-analyses (*i.e.*, pooled data across studies), cardiovascular and all-cause mortality are reduced in patients with post-MI who participate in cardiac rehabilitation exercise training, especially as a component of multifactorial risk factor reduction.

Other Benefits

- Decreased anxiety and depression
- Improved cognitive function
- Enhanced physical function and independent living in older individuals
- Enhanced feelings of well-being
- Enhanced quality of life
- Improved sleep quality and efficiency
- Enhanced performance of work, recreational, and sport activities
- Reduced risk of falls and injuries from falls in older individuals
- Prevention or mitigation of functional limitations in older adults
- Effective therapy for many chronic diseases in older adults

CAD, coronary artery disease; CVD, cardiovascular disease, MI, myocardial infarction.

Adapted from ([4](#), [13](#), [15](#), [16](#), [87](#)).

HEALTH BENEFITS OF IMPROVING MUSCULAR FITNESS

The health benefits of enhancing muscular fitness (*i.e.*, the functional parameters of muscle strength, endurance, and power) are well established ([16](#)) and are associated with significantly better cardiometabolic risk profiles, lower risk of all-cause mortality, fewer CVD events, lower risk of developing physical function limitations, and lower risk for nonfatal disease ([2](#)). Significant beneficial changes in health-related biomarkers can be realized with regular participation in resistance training, including improvements in body composition, blood glucose levels, insulin sensitivity, and blood pressure in individuals with mild or moderate hypertension ([2](#), [88](#), [89](#)). Recent evidence suggests that resistance training is as effective as endurance training in the management and treatment of Type 2 diabetes mellitus ([90](#), [91](#)) and in improving the blood lipid profiles of individuals who are overweight or obese ([92](#)). Resistance training positively affects walking distance and velocity in those with peripheral artery disease ([91](#), [93](#)). Further health benefits attributed to resistance training are supported by several meta-analyses indicating that regimens featuring mild- to moderate-intensity isometric muscle actions are more effective in reducing blood pressure in both normal people and people with hypertension than aerobic training or dynamic resistance training alone or in combination ([94](#), [95](#)). Accordingly, resistance training may be effective for preventing and treating conditions related to metabolic syndrome ([2](#)) (see [Chapter 9](#)).

Exercise that enhances muscle strength and mass also increases bone mass (*i.e.*, bone mineral density and content) and bone strength and may serve as a valuable measure to prevent, slow, or reverse the loss of bone mass experienced by individuals with aging or diseases such as osteoporosis ([15](#), [16](#)) (see [Chapter 10](#)). Resistance training can reduce pain and disability in individuals with osteoarthritis ([2](#), [96](#)) and has been shown to be effective in the treatment of chronic back pain ([97](#), [98](#)). Additionally, preliminary work suggests that resistance exercise may prevent and improve depression and anxiety, increase vigor, and reduce fatigue ([2](#), [99](#)). Beyond maintenance or improvements in muscle size and strength, previous and emerging research has demonstrated that resistance training can positively influence other indicators of musculoskeletal fitness and mood while also serving as a primary treatment strategy for several chronic conditions (*e.g.*, osteoporosis, back pain, etc.).

RISKS ASSOCIATED WITH PHYSICAL ACTIVITY AND EXERCISE

In general, the benefits of engaging in regular PA far outweigh the risks ([15](#), [16](#), [100](#)). However, participation in PA or exercise is associated with an increased risk for musculoskeletal injury (MSI) ([101](#)) and potential cardiovascular complications ([2](#)). MSI is the most common exercise-related complication and is often associated with exercise intensity, the nature of the activity, preexisting conditions, and musculoskeletal anomalies. Adverse cardiovascular events such as sudden cardiac death (SCD) and acute myocardial infarction (AMI) are usually associated with vigorous-intensity exercise ([102](#), [103](#)). While SCD and AMI are more likely to lead to long-term morbidity and mortality, they are rare and much less common than MSI ([100](#)).

EXERCISE-RELATED MUSCULOSKELETAL INJURY

Walking and moderate-intensity PA are associated with a very low risk of MSI compared to more vigorous activities including jogging, running, and competitive sports ([104](#), [105](#)). The risk of MSI is higher in activities where there is direct contact between participants or where the body may be thrown to the ground (*e.g.*, football, wrestling) compared to activities where the contact between participants or with the ground is less severe or nonexistent (*i.e.*, baseball, running, walking) ([106](#)). In 2014, over 6.3 million Americans received medical attention for sport-related MSI, with the highest rates found in individuals 12–17 years (83 injury episodes per 1,000 population) and 18–44 years (22 injury episodes per 1,000 population) ([107](#)). The most common anatomical sites for MSI are the lower extremities, with higher rates in the knees, followed by the foot and ankle ([104](#), [105](#)).

The literature on MSI consequences of PA participation often focuses on men from nonrepresentative populations (*e.g.*, military personnel, athletes) ([108](#)). A prospective study of community-dwelling women found that meeting the national PA guidelines of $\geq 150 \text{ min} \cdot \text{wk}^{-1}$ of MVPA resulted in a modest increase in PA-related MSI compared to women not meeting the guidelines ([109](#)). However, the risk of incurring an MSI is inversely related to physical fitness and PA level ([15](#)). In other words, for any given dose of PA, unfit or physically inactive individuals are more likely to experience MSI when compared to their more fit or more active counterparts ([15](#)).

Commonly used methods to reduce MSI (*e.g.*, stretching, warm-up, cool-down, and gradual progression of exercise intensity and volume) may be helpful in some situations; however, there are few studies confirming the effectiveness of these methods ([2](#)). A comprehensive list of strategies that may assist in preventing MSI can be found elsewhere ([110](#), [111](#)).

SUDDEN CARDIAC DEATH AMONG YOUNG INDIVIDUALS

The cardiovascular causes of exercise-related sudden death in young athletes are shown in [Table 1.3](#) ([100](#)). These data clearly indicate that the most common causes of SCD in young individuals are congenital and hereditary cardiovascular abnormalities, including hypertrophic cardiomyopathy, coronary artery abnormalities, and aortic stenosis. An early report evaluating sudden death among younger individuals indicated the absolute annual risk of exercise-related death among high school and college athletes at 1:133,000 men and 1:769,000 women ([112](#)). It should be noted that these rates, although low, included all sports-related nontraumatic deaths. Of the 136 total identifiable causes of death, 100 were caused by CVD. More recent data among young competitive U.S. athletes (age: 19 ± 6 years) from 1980 to 2011 have indicated somewhat higher annual incidence rates for all-cause sudden death at 1:62,439 men and 1:523,093 women ([116](#)). Additionally, Maron and colleagues ([116](#)) reported that 40% of recorded sudden deaths were attributable to SCD, with annualized incidence rates of 1:121,691 men and 1:787,392 women. Some evidence, however, suggests that the incidence of SCD in young sports participants may be higher, ranging from 1:40,000 to 1:80,000 athletes annually ([117](#)). Furthermore, death rates seem to be higher in African American male athletes and basketball players specifically ([115–117](#)). There remains some debate regarding the between-study variability of incidence rates for exercise-related sudden death. These variances are likely due to differences in the (a) populations studied, (b) estimation of the number of sport participants, and (c) subject and/or incident case assignment. To reduce the risk of SCD incidence in young individuals, well-recognized organizations such as the International Olympic Committee and AHA have endorsed the practice of preparticipation cardiovascular screening ([118–120](#)). The position stand by the American Medical Society for Sports Medicine presents the latest evidence-based research on cardiovascular preparticipation screening in athletes ([121](#)).

TABLE 1.3 Cardiovascular Causes of Exercise-Related Sudden Death in Young Athletes^a

	Van Camp et al. (n = 100) ^b (112)	Maron et al. (n = 134) (113)	Corrado et al. (n = 55) ^c (114)	Maron et al. (n = 842) ^b (115)
Hypertrophic CM	51	36	1	302
Probable hypertrophic CM	5	10	0	77
Coronary anomalies	18	23	9	158
Valvular and subvalvular aortic stenosis	8	4	0	20
Possible myocarditis	7	3	5	57
Dilated and nonspecific CM	7	3	1	18
Atherosclerotic CVD	3	2	10	38
Aortic dissection/rupture	2	5	1	23
Arrhythmogenic right ventricular CM	1	3	11	43
Myocardial scarring	0	3	0	0
Mitral valve prolapse	1	2	6	31
Other congenital abnormalities	0	1.5	0	8
Long QT syndrome	0	0.5	0	18
Wolff-Parkinson-White syndrome	1	0	1	8
Cardiac conduction disease	0	0	3	2
Cardiac sarcoidosis	0	0.5	0	4
Coronary artery aneurysm	1	0	0	0
Normal heart at necropsy	7	2	1	18
Pulmonary thromboembolism	0	0	1	15

^a Ages ranged from 13 to 24 years (112), 12 to 40 years (113), 12 to 35 years (114), and 15 to 24 years (115). References (112, 113) used the same database and include many of the same athletes. All (112), 90% (113), and 89% (114) had symptom onset during or within an hour of training or competition.

^b Total exceeds 100% because several athletes had multiple abnormalities.

^c Includes some athletes whose deaths were not associated with recent exertion. Includes aberrant artery origin and course, tunneled arteries, and other abnormalities.

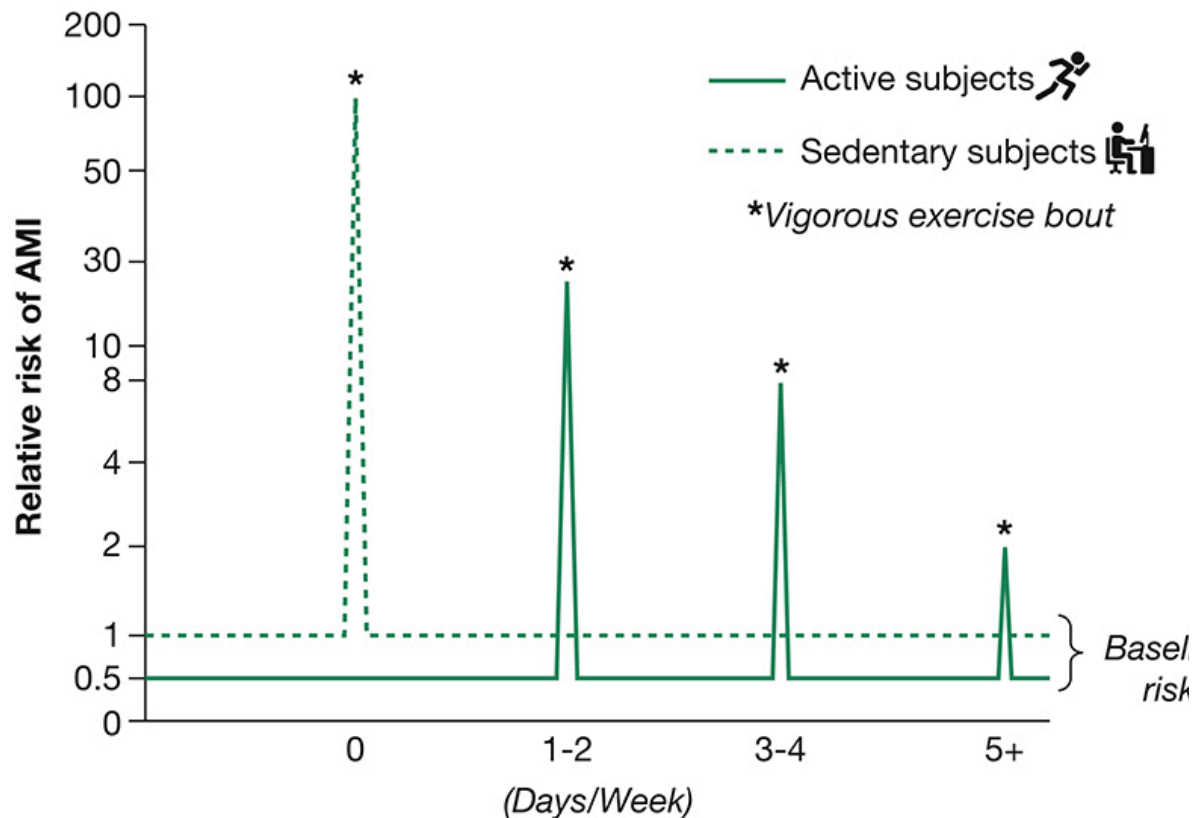
CM, cardiomyopathy; CVD, cardiovascular disease.

Used with permission from [\(100\)](#).

EXERCISE-RELATED CARDIAC EVENTS IN ADULTS

In general, exercise does not elicit cardiovascular events in healthy individuals without congenital or inherited cardiovascular predispositions ([122](#)). The risk of SCD and AMI is very low in apparently healthy (*i.e.*, no signs or symptoms of CVD) individuals performing moderate-intensity PA ([123](#), [124](#)). There is an acute and transient increase in the risk of SCD and AMI in individuals performing vigorous-intensity PA, particularly in sedentary men and women with diagnosed or occult CVD ([100](#), [125](#)). However, this risk decreases with long-term compliance to an exercise regimen and increasing volumes of regular PA ([100](#), [122](#)). [Chapter 2](#) includes an exercise preparticipation health screening algorithm to help identify individuals who may be at risk for exercise-related cardiovascular events.

It is well established that the transient risks of SCD and AMI are substantially higher during and immediately (~ 1 hour) after acute vigorous PA than rest ([100](#), [103](#), [126](#), [127](#)). A meta-analysis of published studies reported a fivefold increased risk of SCD and a 3.5-fold increased risk of AMI during or shortly after vigorous-intensity PA ([125](#)). The risk of SCD or AMI is higher in middle-aged and older adults than in younger individuals due to the higher prevalence of CVD in the older population. The rates of SCD and AMI are disproportionately more elevated in the most inactive individuals when they perform unaccustomed or infrequent exercise ([100](#), [103](#)). As an example, the Myocardial Infarction Onset Study ([102](#)) showed that the risk of AMI during or immediately following vigorous-intensity exercise was 50 times higher for the habitually inactive compared to individuals who exercised vigorously for 1-hour sessions ≥ 5 d $\cdot$ wk⁻¹ ([Figure 1.2](#)).



Habitual frequency of vigorous physical activity

Figure 1.2 The relationship between habitual frequency of vigorous physical activity and the relative risk of acute myocardial infarction (AMI). Used with permission from (128).

Although the *relative* risks of SCD and AMI are higher during sudden vigorous PA compared to rest, the *absolute* risk of these events is very low (100, 122). Prospective evidence from the Physicians' Health Study and Nurses' Health Study suggests that SCD occurs every 1.5 million episodes of vigorous PA in men (103) and every 36.5 million episodes of MVPA in women (124). Retrospective analyses also support the rarity of these events. Thompson and colleagues (129) reported 1 death per 396,000 hours of jogging in 30- to 60-year-old men from Rhode Island who reported jogging at least twice weekly. An analysis of exercise-related cardiovascular events among participants at YMCA sports centers found 1 death per 2,897,057 person-hours, although exercise intensity was not documented (130). Kim et al. (131) studied over 10 million marathon and half-marathon runners and identified an overall cardiac arrest incidence rate of 1 per 184,000 runners and an SCD incidence rate of 1 per 256,000 runners, which translates to 0.20 cardiac arrests and 0.14 SCDs per 100,000 estimated runner-hours.

Although the risk is extremely low, vigorous-intensity exercise has a small but measurable acute risk of CVD complications; therefore, mitigating this risk in susceptible individuals is important (see Chapter 2). The exact mechanisms of SCD and AMI during vigorous-intensity

exercise are not completely understood. Among those older than 30-35 years of age, exercise-related SCD can most often be attributed to acute complications of atherosclerosis ([100](#), [122](#)). Specifically, atherosclerosis-related complications are associated with exercise-related SCD in more than 80% of cases among those older than 35 years of age and greater than 95% of cases among those older than 40 years of age ([132–136](#)). Generally, exercise-related stress is believed to be a risk factor for vulnerable atherosclerotic plaque ([122](#)). Secondary to mechanisms that remain unclear, this exercise-related stress may accelerate the fissuring of fragile and nonocclusive plaque. Subsequent geometric and hemodynamic changes in the epicardial arteries may then lead to plaque disruption ([122](#)). Moreover, plaque rupture may spontaneously be induced via increased fibrinolytic activity after heightened thrombogenicity in response to vigorous exercise ([122](#), [137](#)).

EXERCISE TESTING AND THE RISK OF CARDIAC EVENTS

As with vigorous-intensity exercise, the risk of cardiac events during maximal or near-maximal exercise testing varies directly with the prevalence of diagnosed or occult CVD in the study population. Numerous studies spanning more than four decades have documented these risks during exercise testing ([138–149](#)). [Table 1.4](#) summarizes the risks of various cardiac events, including AMI, ventricular fibrillation, hospitalization, and death. These data indicate that in a mixed population, the risk of exercise testing is low, with approximately six cardiac events per 10,000 tests. One of these studies includes data for which nonphysicians supervised exercise testing, indicating low risk even in less controlled exercise testing settings ([144](#)).

TABLE 1.4 Cardiac Complications During Exercise Testing^a

Reference	Year	Site	No. of Tests	MI	VF	Death	Hospitalization	Comment
Rochmis and Blackburn (147)	1971	73 U.S. centers	170,000	NA	NA	1	3	34% of tests were symptom limited; 50% of deaths in 8 h; 50% over the next 4 d.
Irving et al. (141)	1977	15 Seattle facilities	10,700	NA	4.67	0	NR	
McHenry (145)	1977	Hospital	12,000	0	0	0	0	
Atterhög et al. (138)	1979	20 Swedish centers	50,000	0.8	0.8	0.4	5.2	
Stuart and Ellestad (149)	1980	1,375 U.S. centers	518,448	3.58	4.78	0.5	NR	VF includes other dysrhythmias requiring treatment.
Gibbons et al. (139)	1989	Cooper Clinic	71,914	0.56	0.29	0	NR	Only 4% of men and 2% of women had CVD.
Knight et al. (144)	1995	Geisinger Cardiology Service	28,133	1.42	1.77	0	NR	25% were inpatient tests supervised by non-MDs.
Myers et al. (146)	2000	72 Veterans Affairs medical centers in the United States	75,828	0.40	0.13	0	NR	VF includes other dysrhythmias requiring treatment.
Kane et al. (142)	2008	Mayo Clinic	8,592	0	4.66	0	5.82	Tests were supervised by registered nurses. VF includes other dysrhythmias

Reference	Year	Site	No. of Tests	MI	VF	Death	Hospitalization	Comment
								requiring treatment.
Keteyian et al. (143)	2009	82 clinical sites in the United States, Canada, and France	4,411	0	0	0	0	All patients were diagnosed with heart failure.
Skalski et al. (148)	2012	Mayo Clinic	5,060	0	7.91	0	11.9	All patients were diagnosed with CVD prior to testing. VF includes other dysrhythmias resulting in hospitalization.

^a Events are per 10,000 tests.

CVD, cardiovascular disease; MD, medical doctor; MI, myocardial infarction; NA, not applicable; NR, not reported; VF, ventricular fibrillation.

RISKS OF CARDIAC EVENTS DURING CARDIAC REHABILITATION

The highest risk of cardiovascular events occurs in those individuals diagnosed with CAD. Older but still relevant research found 1 nonfatal complication per 34,673 patient-hours and 1 fatal cardiovascular complication per 116,402 patient-hours of cardiac rehabilitation ([150](#)). Collectively, average rates from other studies have been lower: 1 cardiac arrest per 116,906 patient-hours, 1 AMI per 219,970 patient-hours, 1 fatality per 752,365 patient-hours, and 1 major complication per 81,670 patient-hours ([151–154](#)). Summary statistics from these studies are presented in [Table 1.5](#) ([100](#)). More recent studies have demonstrated an even lower rate of cardiovascular complications during cardiac rehabilitation with ≤ 1 cardiac arrest per 169,344–743,471 patient-hours, ≤ 1 AMIs per 338,638–743,471 patient-hours, and ≤ 1 fatality per 338,638–743,471 patient-hours ([155–157](#)). Of note, the mortality rate is estimated to be six times higher when patients exercise in facilities without the ability to successfully manage cardiac arrest ([100](#)). Interestingly, however, a review of home-based cardiac rehabilitation programs found no increase in cardiovascular complications versus formal center-based exercise programs ([158](#)). Although complication rates are typically low in cardiac rehabilitation settings, the lowest risks have been demonstrated among patients screened and exercised in medically supervised settings equipped to handle cardiac emergencies.

TABLE 1.5 Summary of Exercise-Based Cardiac Rehabilitation Program Complication Rates

Investigator	Year	Patient Exercise Hours	Cardiac Arrest	Myocardial Infarction	Fatal Events	Major Complications ^a
Van Camp and Peterson (153)	1980–1984	2,351,916	1/111,996 ^b	1/293,990	1/783,972	1/81,101
Digenio et al. (151)	1982–1988	480,000	1/120,000 ^c		1/160,000	1/120,000
Vongvanich et al. (154)	1986–1995	268,503	1/89,501 ^d	1/268,503 ^d	0/268,503	1/67,126
Franklin et al. (152)	1982–1998	292,254	1/146,127 ^d	1/97,418 ^d	0/292,254	1/58,451
Average			1/116,906	1/219,970	1/752,365	1/81,670

^a Myocardial infarction and cardiac arrest.

^b Fatal 14%.

^c Fatal 75%.

^d Fatal 0%.

Used with permission from [\(100\)](#).

PREVENTION OF EXERCISE-RELATED CARDIAC EVENTS

Because of the low incidence of cardiac events related to vigorous-intensity exercise, it is very difficult to test the effectiveness of strategies to reduce the occurrence of these events.

Therefore, in a 2007 joint report by the ACSM and AHA, the authors suggested that “physicians should not overestimate the risks of exercise because the benefits of habitual physical activity substantially outweigh the risks” ([100](#)). This report also recommends several strategies to reduce risk of these cardiac events during vigorous-intensity exercise ([100](#)):

- Health care professionals should know the pathologic conditions associated with exercise-related events to appropriately evaluate physically active children and adults.
- Physically active individuals should know the nature of cardiac prodromal symptoms (*e.g.*, excessive, unusual fatigue and pain in the chest and/or upper back) and seek prompt medical care if such symptoms develop (see [Table 2.1](#)).
- High school and college athletes should undergo preparticipation screening by health care professionals.
- Athletes with known cardiac conditions or a family history should be evaluated by members of the health care team prior to competition using established guidelines.
- Health care facilities should ensure that their staff are trained in managing cardiac emergencies and have a specified plan and appropriate resuscitation equipment.
- Physically active individuals should modify their exercise program in response to variations in their exercise capacity, habitual activity level, and the environment (see [Chapters 5](#) and [7](#)).

Although strategies for reducing the number of cardiovascular events during vigorous-intensity exercise have not been systematically studied, it is incumbent on the exercise professional to take reasonable precautions when working with individuals who wish to become more physically active/fit and/or increase their PA/fitness levels. These precautions are particularly true when the exercise program is of vigorous intensity. Although many sedentary individuals can safely begin a light- to moderate-intensity exercise program, all individuals should participate in the exercise preparticipation screening process to determine the need for medical clearance (see [Chapter 2](#)).

Exercise professionals supervising exercise and fitness programs should have current certification in at least basic and ideally advanced cardiac life support, emergency procedures, and for ACSM Certified Professionals, first aid. These emergency procedures should be reviewed and practiced at regular intervals. Finally, individuals should be educated on the signs

and symptoms of CVD and should be referred to a health care provider for further evaluation should these symptoms occur.

ONLINE RESOURCES

American College of Sports Medicine position stand on the quantity and quality of exercise:

[https://journals.lww.com/acsm-
msse/Fulltext/2011/07000/Quantity_and_Quality_of_Exercise_for_Developing.26.aspx](https://journals.lww.com/acsm-msse/Fulltext/2011/07000/Quantity_and_Quality_of_Exercise_for_Developing.26.aspx)

Physical Activity Guidelines for Americans, Second Edition:

<https://health.gov/paguidelines/second-edition/>

2008 Physical Activity Guidelines for Americans: <https://health.gov/paguidelines/2008/>

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CHAPTER 2

Preparticipation Evaluation

INTRODUCTION

This chapter contains the recommended steps that should be taken by an exercise professional prior to someone engaging in physical activity (PA). Specific components of the preparticipation evaluation, recommended by the American College of Sports Medicine (ACSM), include (a) informed consent, (b) exercise preparticipation health screening, (c) health history, (d) and cardiovascular (CV) risk factor analysis. This chapter provides details of these components, and where available, based on the most current evidence.

The ACSM exercise preparticipation recommendations described in this chapter are designed to identify individuals who are at risk for adverse exercise-related CV events and to provide guidance about which individuals should be referred for medical clearance (*i.e.*, approval from a health care provider to engage in exercise). These recommendations emphasize the public health message of PA for all, while at the same time acknowledge that vigorous exercise is associated with a small but measurable acute risk of CV disease (CVD) complications, as described in [Chapter 1](#). These recommendations also decrease the need to see a care provider prior to initiating exercise, thereby reducing or removing unnecessary barriers to adopting and maintaining habitual PA ([1](#)).

INFORMED CONSENT

In many settings, the informed consent process is the first step when working with a new individual. Obtaining adequate informed consent from participants is an important ethical and legal consideration and should be completed prior to (a) the collection of any personal and confidential information, (b) any form of fitness testing, or (c) exercise participation. Although the content and extent of consent forms may vary, enough information must be present in the informed consent process to ensure that the participant knows and understands the purpose(s) and risks associated with screening, assessment, and exercise participation. The consent form must be verbally explained to the participant and include a statement indicating the individual has been given an opportunity to ask questions about the exercise preparticipation health screening, exercise testing/fitness assessment, or the exercise program and has been given sufficient information to provide their informed consent. It is important to document specific questions from the participant on the form along with the responses provided. This provides additional information for consideration when developing an individualized exercise program. The consent form must indicate the participant is free to withdraw at any time. Also, all reasonable efforts must be made to protect the privacy of the individual's health information (*e.g.*, medical history, test results) as described in the Health Insurance Portability and Accountability Act ([2](#), [3](#)). A sample consent form for exercise testing is provided in [Figure 2.1](#). However, it is advisable to check with authoritative bodies (*e.g.*, hospital risk management, institutional review boards, facility legal counsel) to determine what is appropriate for an acceptable informed consent process. No sample form should be adopted for a specific test or program unless approved by legal counsel and/or the appropriate institutional review board.

Informed Consent for an Exercise Test

1. Purpose and Explanation of the Test

You will perform an exercise test on a cycle ergometer or a motor-driven treadmill. The exercise intensity will begin at a low level and will be advanced in stages depending on your fitness level. We may stop the test at any time because of signs of fatigue or changes in your heart rate, electrocardiogram, or blood pressure, or symptoms you may experience. It is important for you to realize that you may stop when you wish because of feelings of fatigue or any other discomfort.

2. Attendant Risks and Discomforts

There exists the possibility of certain changes occurring during the test that increase risk. These include abnormal blood pressure; fainting; irregular, fast, or slow heart rhythm; and, in rare instances, heart attack, stroke, or death. Every effort will be made to minimize these risks by evaluation of preliminary information relating to your health and fitness and by careful observations during testing. Emergency equipment and trained personnel are available to deal with unusual situations that may arise.

3. Responsibilities of the Participant

Information you possess about your health status or previous experiences of heart-related symptoms (e.g., shortness of breath with low-level activity; pain; pressure; tightness; heaviness in the chest, neck, jaw, back, and/or arms) with physical effort may affect the safety of your exercise test. Your prompt reporting of these and any other unusual feelings with effort during the exercise test itself is very important. You are responsible for fully disclosing your medical history as well as symptoms that may occur during the test. You are also expected to report all medications (including nonprescription) taken recently and, in particular, those taken today to the testing staff.

4. Benefits to be Expected

The results obtained from the exercise test may assist in the diagnosis of your illness, in evaluating the effect of your medications, or in evaluating what type of physical activities you might do with low risk.

5. Inquiries

Any questions about the procedures used in the exercise test or the results of your test are encouraged. If you have any concerns or questions, please ask us for further explanations.

6. Use of Medical Records

The information that is obtained during exercise testing will be treated as privileged and confidential as described in the Health Insurance Portability and Accountability Act of 1996. It is not to be released or revealed to any individual except your referring physician without your written consent. However, the information obtained may be used for statistical analysis or scientific purposes with your right to privacy retained.

7. Freedom of Consent

I hereby consent to voluntarily engage in an exercise test to determine my exercise capacity and state of cardiovascular health. My permission to perform this exercise test is given voluntarily. I understand that I am free to stop the test at any point if I so desire.

I have read this form, and I understand the test procedures that I will perform and the attendant risks and discomforts. Knowing these risks and discomforts, and having had an opportunity to ask questions that have been answered to my satisfaction, I consent to participate in this test.

Date

Signature of Patient

Date

Signature of Witness

Figure 2.1 Sample of informed consent form for a symptom-limited exercise test.

When an informed consent is being used in a research setting, applicable policies for the testing of human subjects must be implemented. Thus, health care professionals and

research scientists should obtain approval from their institutional review board when conducting an exercise test for research purposes.

The program must ensure available personnel are properly trained and authorized to carry out emergency procedures that use such equipment. Written emergency policies and procedures that address possible medical emergencies such as common orthopedic injuries, CV events, stroke, or hypoglycemia should be in place and reviewed quarterly (4). These policies need to clearly outline the locations of all emergency equipment (automated external defibrillators, phones, first aid kits) as well as the entrances and exits throughout the building (4). Emergency drill practice is recommended to take place at least twice a year, or more frequently when there is a change in staff (4, 5). Should an emergency event occur, it is important to document it appropriately. Suggested components of an incident report include (a) day, date, time, and location of the event, (b) person(s) involved (including staff) and their contact information, (c) witnesses and their contact information, (d) a detailed description of the event and actions taken, and (e) outcomes of the event and/or actions taken as shown in [Figure 2.2](#) (4).

Sample Incident Report Form

Date of accident _____ Time of accident _____

Member's name _____ Member number _____

Address _____

Home phone _____ Business phone _____

Location of accident _____

Staff attending _____

Witness (nonstaff) _____

Details of accident _____

Action taken by staff _____

Staff reporting _____ Date _____

Department head's signature _____ Date _____

Figure 2.2 Sample of an incident report form ([33](#)).

EXERCISE PREPARTICIPATION HEALTH SCREENING

The overarching purpose of the ACSM exercise preparticipation health screening process is to identify individuals who are at risk for adverse exercise-related CV events. More specifically, the goals are to identify individuals (a) who require medical clearance before initiating a moderate- to vigorous-intensity exercise program or increasing the intensity of their current program, (b) with clinically significant disease who may benefit from participating in a medically supervised exercise program, and (c) with medical conditions that may require exclusion from exercise programs until those conditions are abated or better controlled. It is important to highlight that exercise preparticipation health screening recommendations are not a replacement for sound clinical judgment and decisions about referral to a health care provider for medical clearance prior to the initiation of an exercise program, or adjustments to an exercise program to include vigorous exercise. Exercise preparticipation health screening recommendations should continue to be made on an individual basis.

The ACSM exercise preparticipation health screening process is a screening algorithm with recommendations for medical clearance based on an individual's current level of exercise participation, presence of signs or symptoms and/or known CV, metabolic, or renal disease, and the anticipated or desired exercise intensity ([1](#)). These factors are included because the risk for activity-associated sudden cardiac death (SCD) and acute myocardial infarction (AMI) is known to be highest among those with underlying CVD who perform unaccustomed vigorous PA ([6–8](#)). The relative risk of SCD and AMI during vigorous- to near maximal-intensity exercise is directly related to the presence of CVD and/or exertional symptoms ([9](#)) and is inversely related to the habitual level of PA ([7, 10–13](#)). The relative risk of a CV event is transiently increased during vigorous-intensity exercise as compared with rest, but the absolute risk of an exercise-related acute cardiac event is low in healthy asymptomatic individuals (see [Figure 1.2](#)) ([14](#)).

Exercise testing is often a useful tool for the development of an individualized exercise program; however, ACSM does not recommend the inclusion of exercise testing or any other type of medical examination as part of medical clearance; rather, those decisions are left to the clinical judgment of qualified health care providers. This is intended to better align with relevant evidence that exercise testing is not a uniformly recommended screening procedure, as it is a poor predictor of acute cardiac events in asymptomatic individuals ([15](#)). Even though exercise testing may detect flow-limiting coronary lesions via the provocation of

ischemic ST-segment depression, angina pectoris, or both, SCD and AMI are usually triggered by the rapid progression of a previously nonobstructive lesion (14). Furthermore, there is a lack of consensus regarding the extent of the medical evaluation (*i.e.*, physical examination; peak or symptom-limited exercise testing) needed as part of the preparticipation health screening process prior to initiating an exercise program, even when the program will be of vigorous intensity. The American College of Cardiology (ACC)/American Heart Association (AHA) recommends that exercise testing may be considered in intermediate-risk asymptomatic adults. This includes sedentary adults starting a vigorous-intensity exercise program if nonelectrocardiogram markers such as exercise capacity, chronotropic incompetence, and heart rate recovery are considered; however, it is acknowledged that the efficacy of this recommendation is not well established (16). The U.S. Preventive Services Task Force (USPSTF) recommends against the use of routine diagnostic testing or exercise electrocardiography as a screening tool in asymptomatic individuals who are at low risk for CVD events. The USPSTF concluded that there is insufficient evidence to evaluate the benefits and harm of exercise testing before initiating a PA program. Furthermore, the USPSTF does not make specific recommendations regarding the need for exercise testing for individuals at intermediate and high risk for CVD events (17). Similarly, randomized trial data on the clinical value of exercise testing for screening purposes are absent; it is not known if exercise testing in asymptomatic adults reduces the risk of premature mortality or major cardiac morbidity (18).

Insufficient evidence is available to suggest that the presence of CVD risk factors *without* underlying disease confers substantial risk of adverse exercise-related CV events. The high prevalence of CVD risk factors among adults (19), combined with the rarity of exercise-related SCD and AMI (14, 12), suggest that the ability to predict these rare events by assessing risk factors is low, especially among otherwise healthy adults (14). Furthermore, conventional CVD risk factor–based exercise preparticipation health screening may be overly conservative due to the high prevalence of risk factors among the population. This may generate excessive referrals, particularly in adults older than 40 years (19). Therefore, elimination of CVD risk factors from the prescreening procedure substantially lowers the proportion of individuals referred for medical clearance (20). Nonetheless, it is recommended that CVD risk factor assessment be conducted and shared with the participant and health care provider. Therefore, clearance to engage in a moderate-to-vigorous exercise program among those with signs/symptoms of CVD should not be solely based on CVD risk factor assessment, but as part of a complete medical evaluation as determined by the medical professional.

Overall, there is a low risk of SCD and AMI associated with participation in a light- to moderate-intensity exercise program (21). Much of the risk of participating in vigorous

exercise is mitigated by adopting a “progressive transitional phase” (~2–3 months) during which the duration and intensity of exercise are gradually increased ([1](#), [22](#)). When previously sedentary individuals initiate an exercise program, such individuals are strongly recommended to begin with light intensity (*e.g.*, 2–3 metabolic equivalents [METs]) and gradually increase to moderate intensity (*e.g.*, 3–5 METs) over time, provided that the individual remains symptom free. Such a gradual progression appears prudent because these intensities are below the vigorous-intensity threshold (≥ 6 METs) that is commonly associated with the triggering of acute CV events in susceptible individuals ([14](#)). This progressive transitional phase will help to minimize the risk of musculoskeletal injury and allow sedentary individuals to improve their cardiorespiratory fitness without going through a period during which each session of vigorous exercise is associated with spikes in relative CV risk ([23](#)).

American College of Sports Medicine Preparticipation Screening Process

The following section provides guidance for preparticipation screening for exercise professionals working with the general, nonclinical population. Recommendations for those individuals who are working in a clinical or medical fitness setting are presented separately in [Chapter 4](#).

Preparticipation health screening before initiating a moderate-to-vigorous exercise program is a two-stage process that should be distinguished from a periodic medical examination or formal collection of a detailed health history and should be encouraged as an additional part of routine health maintenance:

- The need for medical clearance before initiating or progressing exercise programming is determined using the ACSM screening algorithm and the help of a qualified exercise or health care professional. In the absence of professional assistance, interested individuals may use the Physical Activity Readiness Questionnaire Plus (PAR-Q+) ([24](#)) to determine whether or not it is advised to seek medical advice prior to engaging in PA or structured exercise.
- If indicated during screening or if there is any suspicion of underlying risk, medical clearance from a health care provider should be recommended. The manner of clearance, however, should be determined by the clinical judgment and discretion of said health care provider.

The following section provides guidance for using the preparticipation health screening algorithm with respect to:

- Determining current PA levels
- Identifying signs and symptoms of underlying CV, metabolic, and renal disease ([*Table 2.1*](#))

TABLE 2.1 Major Signs or Symptoms Suggestive of Cardiovascular, Metabolic, and Renal Disease^a

Signs or Symptoms	Clarification/Significance
Pain; discomfort (or other anginal equivalent) in the chest, neck, jaw, arms, or other areas that may result from myocardial ischemia; or other recent onset pain of unknown origin	One of the cardinal manifestations of cardiac disease; in particular, coronary artery disease Key features <i>favoring an ischemic origin</i> include the following: • <i>Character</i>: constricting, squeezing, burning, “heaviness,” or “heavy feeling” • <i>Location</i>: substernal, across midthorax, anteriorly; in one or both arms, shoulders; in neck, cheeks, teeth; in forearms, fingers, in interscapular region • <i>Provoking factors</i>: exercise or exertion, excitement, other forms of stress, cold weather, occurrence after meals Key features against an ischemic origin include the following: • <i>Character</i>: dull ache; “knifelike,” sharp, stabbing; “jabs” aggravated by respiration • <i>Location</i>: in left submammary area; in left hemithorax • <i>Provoking factors</i>: after completion of exercise, provoked by a specific body motion
Shortness of breath at rest or with mild exertion	Dyspnea (defined as an abnormally uncomfortable awareness of breathing) is one of the principal symptoms of cardiac and pulmonary disease. It commonly occurs during strenuous exertion in healthy, well-trained individuals and during moderate exertion in healthy, untrained individuals. However, it should be regarded as abnormal when it occurs at a level of exertion that is not expected to evoke this symptom in a given individual. Abnormal exertional dyspnea suggests the presence of cardiopulmonary disorders; in particular, left ventricular dysfunction or chronic obstructive pulmonary disease.
Dizziness or syncope	Syncope (defined as a loss of consciousness) is most commonly caused by a reduced perfusion of the brain. Dizziness and, in particular, syncope <i>during</i> exercise may result from cardiac disorders that prevent the normal rise (or an actual fall) in cardiac output. Such cardiac disorders are potentially life-

Signs or Symptoms	Clarification/Significance
	threatening and include severe coronary artery disease, hypertrophic cardiomyopathy, aortic stenosis, and malignant ventricular dysrhythmias. Although dizziness or syncope shortly <i>after</i> cessation of exercise should not be ignored, these symptoms may occur even in healthy individuals as a result of a reduction in venous return to the heart.
Orthopnea or paroxysmal nocturnal dyspnea	Orthopnea refers to dyspnea occurring at rest in the recumbent position that is relieved promptly by sitting upright or standing. Paroxysmal nocturnal dyspnea refers to dyspnea, beginning usually 2–5 h after the onset of sleep, which may be relieved by sitting on the side of the bed or getting out of bed. Both are symptoms of left ventricular dysfunction. Although nocturnal dyspnea may occur in individuals with chronic obstructive pulmonary disease, it differs in that it is usually relieved following a bowel movement rather than specifically by sitting up.
Ankle edema	Bilateral ankle edema that is most evident at night is a characteristic sign of heart failure or bilateral chronic venous insufficiency. Unilateral edema of a limb often results from venous thrombosis or lymphatic blockage in the limb. Generalized edema (known as anasarca) occurs in individuals with the nephrotic syndrome, severe heart failure, or hepatic cirrhosis.
Palpitations or tachycardia	Palpitations (defined as an unpleasant awareness of the forceful or rapid beating of the heart) may be induced by various disorders of cardiac rhythm. These include tachycardia, bradycardia of sudden onset, ectopic beats, compensatory pauses, and accentuated stroke volume resulting from valvular regurgitation. Palpitations also often result from anxiety states and high cardiac output (or hyperkinetic) states, such as anemia, fever, thyrotoxicosis, arteriovenous fistula, and the so-called idiopathic hyperkinetic heart syndrome.

Signs or Symptoms	Clarification/Significance
Intermittent claudication	Intermittent claudication refers to the pain that occurs in the lower extremities with an inadequate blood supply (usually as a result of atherosclerosis) that is brought on by exercise. The pain does not occur with standing or sitting, is reproducible from day to day, is more severe when walking upstairs or up a hill, and is often described as a cramp, which disappears within 1–2 min after stopping exercise. Coronary artery disease is more prevalent in individuals with intermittent claudication. Patients with diabetes are at increased risk for this condition.
Known heart murmur	Although some may be innocent, heart murmurs may indicate valvular or other cardiovascular disease. From an exercise safety standpoint, it is especially important to exclude hypertrophic cardiomyopathy and aortic stenosis as underlying causes because these are among the more common causes of exertion-related sudden cardiac death.
Unusual fatigue or shortness of breath with usual activities	Although there may be benign origins for these symptoms, they also may signal the onset of or change in the status of cardiovascular disease or metabolic disease.

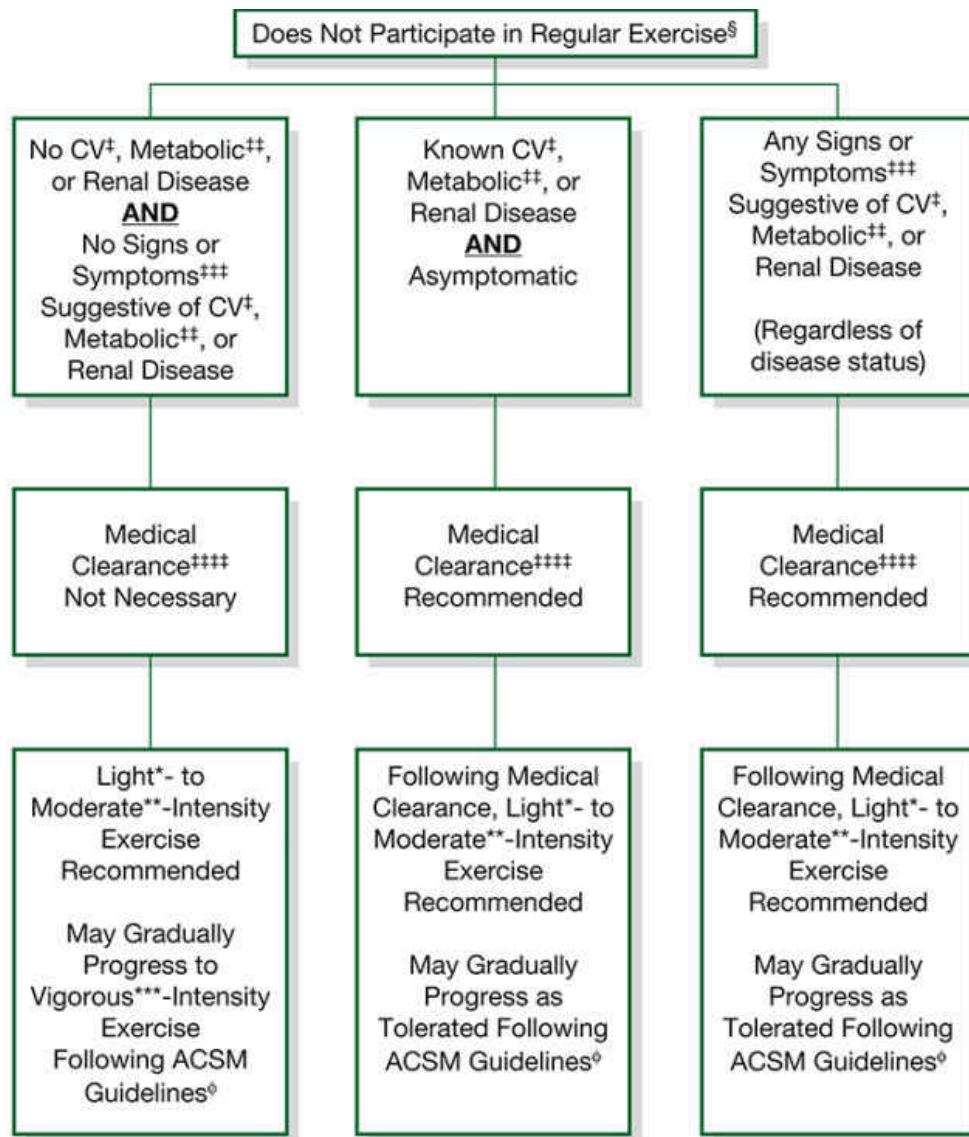
^a These signs or symptoms must be interpreted within the clinical context in which they appear because they are not all specific for cardiovascular, metabolic, or renal diseases.

Modified from (34).

- Identifying individuals with diagnosed CV, metabolic, and/or renal disease
- Using signs and symptoms, disease history, current exercise participation, and desired exercise intensity to guide recommendations for preparticipation medical clearance

American College of Sports Medicine Preparticipation Screening Algorithm

The ACSM preparticipation screening algorithm ([Figure 2.3](#)) is designed to identify individuals at risk for CV complications as a direct result of a bout of endurance exercise. Although preparticipation screening is recommended before participation with any type of exercise program, current evidence regarding prescreening for CV complications during resistance training is limited; thus, the inherent risk cannot currently be determined but appears to be low ([21](#)).



[§]Exercise Participation

Performing planned, structured physical activity at least 30 min at moderate intensity on at least 3 d · wk⁻¹ for at least the last 3 mo

*Light-Intensity Exercise

30%–39% HRR or $\dot{V}O_2R$, 2–2.9 METs, RPE 9–11, an intensity that causes slight increases in HR and breathing

**Moderate-Intensity Exercise

40%–59% HRR or $\dot{V}O_2R$, 3–5.9 METs, RPE 12–13, an intensity that causes noticeable increases in HR and breathing

***Vigorous-Intensity Exercise

≥60% HRR or $\dot{V}O_2R$, ≥6 METs, RPE ≥14, an intensity that causes substantial increases in HR and breathing

[‡]Cardiovascular (CV) Disease

Cardiac, peripheral vascular, or cerebrovascular disease

^{‡‡}Metabolic Disease

Type 1 and 2 diabetes mellitus

^{‡‡‡}Signs and Symptoms

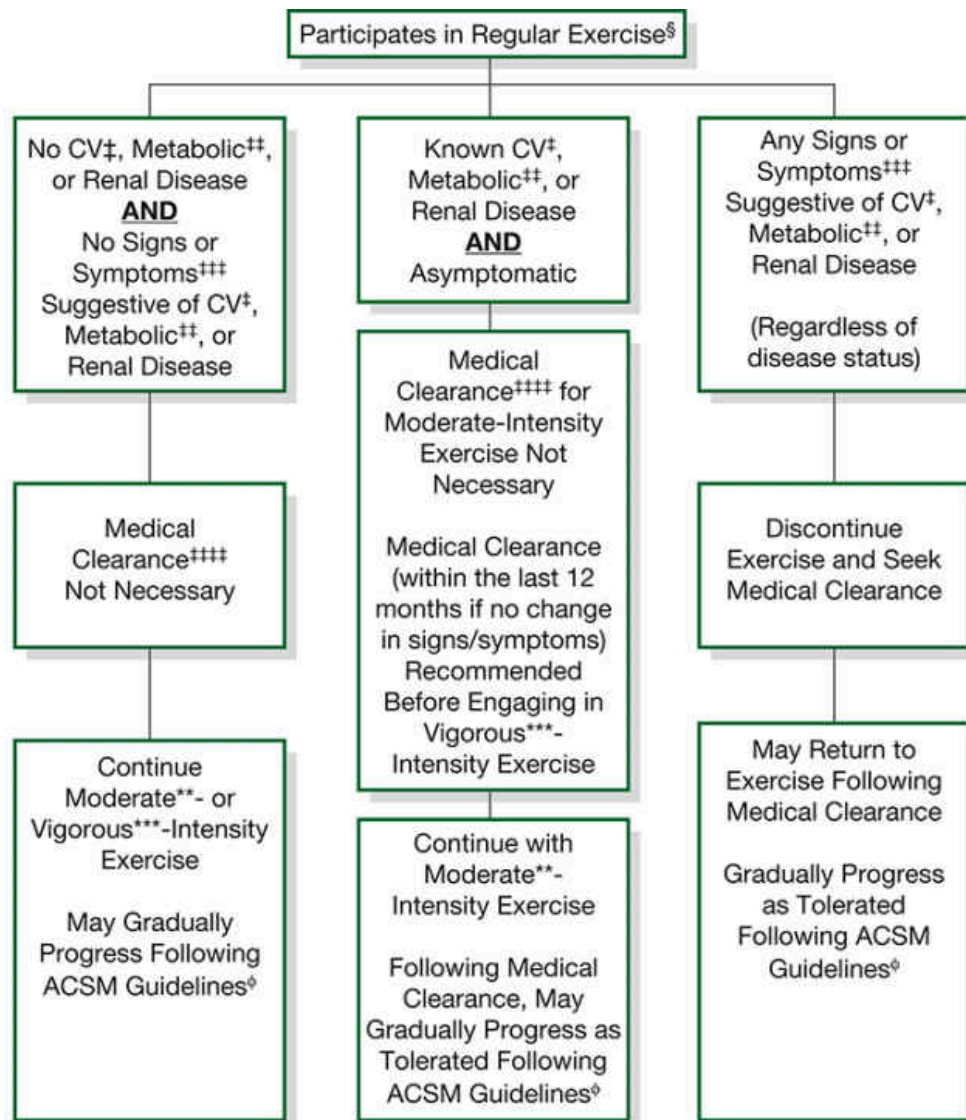
At rest or during activity. Includes pain, discomfort in the chest, neck, jaw, arms, or other areas that may result from ischemia; shortness of breath at rest or with mild exertion; dizziness or syncope; orthopnea or paroxysmal nocturnal dyspnea; ankle edema; palpitations or tachycardia; intermittent claudication; known heart murmur; unusual fatigue or shortness of breath with usual activities.

^{‡‡‡‡}Medical Clearance

Approval from a health care professional to engage in exercise

[°]ACSM Guidelines

See the most current edition of *ACSM's Guidelines for Exercise Testing and Prescription*



[§] Exercise Participation	Performing planned, structured physical activity at least 30 min at moderate intensity on at least 3 d · wk ⁻¹ for at least the last 3 mo
*Light-Intensity Exercise	30%–39% HRR or $\dot{V}O_2R$, 2–2.9 METs, RPE 9–11, an intensity that causes slight increases in HR and breathing
**Moderate-Intensity Exercise	40%–59% HRR or $\dot{V}O_2R$, 3–5.9 METs, RPE 12–13, an intensity that causes noticeable increases in HR and breathing
***Vigorous-Intensity Exercise	≥60% HRR or $\dot{V}O_2R$, ≥6 METs, RPE ≥14, an intensity that causes substantial increases in HR and breathing
[‡] Cardiovascular (CV) Disease	Cardiac, peripheral vascular, or cerebrovascular disease
^{‡‡} Metabolic Disease	Type 1 and 2 diabetes mellitus
^{‡‡‡} Signs and Symptoms	At rest or during activity. Includes pain, discomfort in the chest, neck, jaw, arms, or other areas that may result from ischemia; shortness of breath at rest or with mild exertion; dizziness or syncope; orthopnea or paroxysmal nocturnal dyspnea; ankle edema; palpitations or tachycardia; intermittent claudication; known heart murmur; unusual fatigue; or shortness of breath with usual activities.
^{‡‡‡‡} Medical Clearance	Approval from a health care professional to engage in exercise
[¶] ACSM Guidelines	See the most current edition of <i>ACSM's Guidelines for Exercise Testing and Prescription</i>

Figure 2.3 The American College of Sports Medicine (ACSM) preparticipation screening algorithm. HR, heart rate; HRR, heart rate reserve; METs, metabolic equivalents; RPE, rating of perceived exertion; O₂R, oxygen uptake reserve. Used with permission from (1).

Algorithm Components

The screening algorithm (see [Figure 2.3](#)) begins by classifying individuals who do or do not currently participate in regular PA. The intent is to better identify those individuals unaccustomed to regular physical exertion for whom exercise may place disproportionate demands on the CV system and increase the risk of complications. As designated, participants classified as current exercisers should have a history of performing planned, structured PA of at least moderate intensity for at least 30 minutes on 3 or more d · wk⁻¹ during the past 3 months. This volume of activity is not the minimum threshold representing the PA Guidelines but rather the volume of activity that significantly reduces one's risk of CV events compared to sedentary individuals.

The next level of classification involves identifying individuals with known CV, metabolic, or renal diseases or those with signs or symptoms suggestive of cardiac, peripheral vascular, renal, or cerebrovascular diseases, and diabetes mellitus (DM). During the preparticipation screening process, participants should be asked if a physician or other qualified health care provider has ever diagnosed them with any of these conditions. During preparticipation health screening, hypertension is considered a CVD risk factor and not a cardiac disease ([25](#)). Once an individual's disease status has been ascertained, attention should shift toward signs and symptoms suggestive of CV, metabolic, or renal disease. To better identify those who may have undiagnosed disease, individuals need to be screened for the presence or absence of signs and symptoms suggestive of these diseases, as described in [Table 2.1](#). Care must be taken to interpret the signs and symptoms within the context of the individual's recent history, and additional information should be sought to clarify vague or ambiguous responses. For example, an individual may describe recent periods of noticeable breathlessness; however, this occurrence is a nonspecific symptom of CVD, as many factors can cause shortness of breath. Pertinent follow-up questions may include the following:

- What were you doing during these periods?
- Were you more breathless than you would have expected for this activity?
- Compared to recent similar activities, were you more fatigued following the activity?

These questions may provide clarification to better distinguish expected signs and symptoms from potentially pathological signs and symptoms. An exercise preparticipation

health screening checklist is included to guide the exercise professional through the prescreening process ([Figure 2.4](#)).

Exercise Preparticipation Health Screening Questionnaire for Exercise Professionals

Assess your client's health needs by marking all *true* statements.

Step 1

SIGNS AND SYMPTOMS

Does your client experience:

- ☐ chest discomfort with exertion
- ☐ unreasonable breathlessness
- ☐ dizziness, fainting, blackouts
- ☐ ankle swelling
- ☐ unpleasant awareness of a forceful, rapid or irregular heart rate
- ☐ burning or cramping sensations in your lower legs when walking short distance
- ☐ known heart murmur

If you **did** mark any of these statements under the symptoms, **STOP**, your client should seek medical clearance before engaging in or resuming exercise. Your client may need to use a facility with a **medically qualified staff**.

If you **did not** mark any symptoms, continue to steps 2 and 3.

Step 2

CURRENT ACTIVITY

Has your client performed planned, structured physical activity for at least 30 min at moderate intensity on at least 3 days per week for at least the last 3 months?

Yes ☐ No ☐

Continue to Step 3.

Step 3

MEDICAL CONDITIONS

Has your client had or do they currently have:

- ☐ a heart attack
- ☐ heart surgery, cardiac catheterization, or coronary angioplasty
- ☐ pacemaker/implantable cardiac defibrillator/rhythm disturbance
- ☐ heart valve disease
- ☐ heart failure
- ☐ heart transplantation
- ☐ congenital heart disease
- ☐ diabetes
- ☐ renal disease

Evaluating Steps 2 and 3:

- If you **did not** mark any of the statements in **Step 3**, medical clearance is not necessary.
- If you marked Step 2 "**yes**" and **marked any of the statements in Step 3**, your client may continue to exercise at light to moderate intensity without medical clearance. However, medical clearance is recommended before engaging in vigorous exercise.
- If you marked Step 2 "**no**" and **marked any of the statements in Step 3**, medical clearance is recommended. Your client may need to use a facility with a **medically qualified staff**.

Figure 2.4 Exercise preparticipation health screening questionnaire for exercise professionals. Used with permission ([33](#)).

In the prescreening process, individuals with pulmonary disease are not automatically referred for medical clearance because pulmonary disease does not increase the risks of CV complications during or immediately after exercise; on the contrary, it is the associated inactive and sedentary lifestyle of many individuals with pulmonary disease that may increase the risk of these events ([26](#)). However, chronic obstructive pulmonary disease (COPD) and CVD are often comorbid due to the common risk factor of smoking, and the presence of COPD in current or former smokers is an independent predictor of overall CV events ([27](#)). Thus, careful attention to the presence of signs and symptoms of CV and metabolic disease is warranted in individuals with COPD during the exercise preparticipation health screening process. The presence of pulmonary or other diseases increases the complexity of screening and remains an important consideration for determining the safest and most effective exercise program ([1](#)).

Desired exercise intensity is the final component in the preparticipation screening algorithm. Because vigorous-intensity exercise is more likely to trigger acute CV events, versus light- to moderate-intensity exercise ([14](#)), identifying the intensity at which a participant intends to exercise is important. Guidance is offered in the footnotes of the algorithm on the aforementioned designations as well as what constitutes light-, moderate-, and vigorous-intensity exercise (for additional information regarding exercise intensity please see [Chapter 5](#)).

Using the Algorithm

According to the preparticipation screening algorithm, participants are grouped into one of six categories (see [Figure 2.3](#)). Importantly, exercise professionals using this algorithm should monitor individuals for changes that may alter their categorization and recommendations. For example, an individual who initially declares no signs or symptoms of disease may develop signs or symptoms after beginning an exercise program; this would necessitate more aggressive screening recommendations.

When medical clearance is warranted for individuals, they should be referred to a physician or health care provider. Importantly, the type of medical evaluation is left to the discretion and clinical judgment of the provider to whom the participant is referred because there is no single, universally recommended screening test. The type of procedures conducted during clearance may vary widely from provider to provider and may include verbal consultations, resting or stress electrocardiogram (ECG), computed tomography for the assessment of coronary artery calcium, or even nuclear medicine imaging studies or coronary angiography. Exercise professionals may request written clearance along with special instructions or restrictions (*e.g.*, exercise intensity) for the individual in question, and

continued communication between health care providers and exercise professionals is strongly encouraged. To better understand the preparticipation screening algorithm, case studies are presented in [*Box 2.1*](#).

Box 2.1 Preparticipation Screening Case Studies

The following case studies are presented as an example how to utilize the American College of Sports Medicine preparticipation screening algorithm.

CASE STUDY I

A 50-year-old nonsmoking male was recently invited by colleagues to participate in a 10-km trail run. Currently, he walks at a moderate intensity for 40 minutes every Monday, Wednesday, and Friday — something he has done “for years.” His goal is to run the entire race without stopping, and he is seeking training services. He reports having what he describes as a “mild heart attack” at 45 years old, completed cardiac rehabilitation, and has had no problems since. He takes a statin, an angiotensin-converting enzyme inhibitor, and aspirin daily. During the last visit with his cardiologist, which took place 2 years ago, the cardiologist noted no changes in his medical condition.

CASE STUDY II

A 22-year-old recent college graduate is joining a gym. Since becoming an accountant 6 months ago, she no longer walks across campus or plays intramural soccer and has concerns about her sedentary lifestyle. Although her body mass index is slightly above normal, she reports no significant medical history and no symptoms of any diseases, even when walking up three flights of stairs to her apartment. She would like to begin playing golf.

CASE STUDY III

A 45-year-old former collegiate swimmer turned avid lifelong triathlete who trains at least $60 \text{ min} \cdot \text{d}^{-1}$, $6 \text{ d} \cdot \text{wk}^{-1}$ requests assistance with run training. His only significant medical history is a series of overuse injuries to his shoulders and Achilles tendon. In recent weeks, he notes his vigorous-intensity workouts are unusually difficult and reports feeling constriction in his chest with exertion — something he attributes to deficiencies in core strength. Upon further questioning, he explains that the chest constriction is improved with rest and that he often feels dizzy during recovery.

CASE STUDY IV

A 60-year-old female is beginning a professionally led walking program. Two years ago, she had a drug-eluting stent placed in her left anterior descending coronary artery after a routine exercise stress test revealed significant ST-segment depression. She completed a brief cardiac rehabilitation program in the 2 months following the procedure but has been inactive since. She reports no signs or symptoms and takes a cholesterol-lowering statin and antiplatelet medications as directed by her cardiologist.

CASE STUDY V

A 35-year-old business consultant is in town for 2 weeks and seeking a temporary membership at a fitness club. She and her friends have been training at a moderate- to vigorous-intensity for a long-distance charity bike ride for the past 16 weeks; she is unable to travel with her bike and she does not want to lose her fitness. She reports no current symptoms of cardiovascular (CV) or metabolic disease and has no medical history except hyperlipidemia, for which she takes a cholesterol-lowering statin daily.

	Case Study I	Case Study II	Case Study III	Case Study IV	Case Study V
Currently participates in regular exercise?	Yes	No	Yes	No	Yes
Known CV, metabolic, or renal disease?	Yes	No	No	Yes	No
Signs or symptoms suggestive of disease?	No	No	Yes	No	No
Desired intensity?	Vigorous	Moderate	Vigorous	Moderate	Vigorous
Medical clearance needed?	Yes	No	Yes	Yes	No

Alternative Self-Guided Method

In the absence of a qualified exercise or health care professional, preparticipation health screening using a self-screening tool should be completed by any individual wishing to initiate an exercise program. The PAR-Q+ (see *Online Resources*) includes seven questions followed by several additional follow-up questions to guide preparticipation recommendations. The PAR-Q+ is an evidence-based tool and was developed in part to reduce barriers for exercise and false-positive screenings. The tool uses follow-up questions to better tailor preexercise recommendations based on relevant medical history and symptomatology. The PAR-Q+ may be used as a self-guided exercise preparticipation health screening tool or as a supplemental tool for professionals that may want additional screening resources beyond the ACSM algorithm. Considering the cognitive ability required to fully answer the PAR-Q+, some individuals may require assistance completing the assessment. Fitness professionals should follow up with the individual to make sure all questions are answered accurately or ask follow-up questions if needed.

Exercise Testing

The ACSM preparticipation algorithm provides guidance regarding the need for a medical clearance and for determining appropriate exercise intensity for individuals starting or progressing an exercise program. However, this procedure is not intended as a sole screening tool prior to conducting exercise testing and should accompany other screening tools (Health History Questionnaire, PAR-Q+, etc.), particularly when conducting maximal exercise tests, which are commonly done in research, clinical, fitness, and performance-based settings. A review of 51 studies found the risks of fatal (0.2–0.8:10,000 tests) and nonfatal (1.4:10,000 tests) adverse events during maximal exercise testing in apparently healthy individuals are rare (6). Submaximal exercise testing (see [Chapter 3](#)), which is commonly used in health fitness settings with nonclinical populations, likely confers an even lower risk of untoward events; however, there are no data that specifically describe this risk (6). As such, individuals conducting maximal exercise testing should follow suggested guidelines as outlined in [Chapter 4](#), which outlines the indications and contraindications for testing (see *Box 4.1*),

pretest likelihood of ischemic heart disease, utility of testing, how to best conduct the test, staffing and monitoring the test, testing protocols, terminating the test, and interpreting the test. In addition, clinical and research facilities, in coordination with their respective ethical review boards, typically have their own set of testing guidelines, or protocol-specific guidelines, to conduct maximal exercise testing, which need to be followed at all times.

Risk Stratification for Individuals With Clinical Conditions and Medical Fitness Facilities

Previous sections in this chapter presented the ACSM preparticipation screening algorithm for the general public. The next section will discuss considerations for reviewing additional information when assessing one's need for medical clearance. Such information may include an extensive medical history as well as a client's CVD risk factors. These details can provide the exercise professional with valuable information about each individual and may discover conditions which may lead to referring the individual for medical evaluation and/or medical clearance. Individuals with chronic disease and other health challenges are often encountered in health fitness settings; the exercise professional is urged to be prudent in identifying those who may need a clinical exercise prescription (Ex R_x) or have other specific considerations. In clinical settings, or medical fitness centers, a more comprehensive medical history may more often be warranted and exercise professionals working with individuals with known CVD or any other clinical condition are advised to use more in-depth risk stratification procedures, such as those defined by the American Association of Cardiovascular and Pulmonary Rehabilitation and presented in [Box 2.2 \(28\)](#). Regardless of the extent of the preparticipation evaluation, the findings should guide the decision about the need for medical clearance, exercise testing, and Ex R_x.

Box 2.2 American Association of Cardiovascular and Pulmonary Rehabilitation Risk Stratification Algorithm for Risk of Event

Patient is at **HIGH RISK** if ANY ONE OR MORE of the following factors are present:

- Left ventricular ejection fraction less than 40%
- Survivor of cardiac arrest or sudden death
- Complex ventricular dysrhythmias (ventricular tachycardia, frequent [$>6 \text{ beats} \cdot \text{min}^{-1}$] multiform PVCs) at rest or with exercise
- MI or cardiac surgery complicated by cardiogenic shock, CHF, and/or signs/symptoms of postprocedure ischemia
- Abnormal hemodynamics with exercise, especially flat or decreasing systolic blood pressure or chronotropic incompetence with increasing workload
- Significant silent ischemia (ST depression 2 mm or greater without symptoms) with exercise or in recovery
- Signs/symptoms including angina pectoris, dizziness, light-headedness, or dyspnea at low levels of exercise (<5.0 METs) or in recovery
- Maximal functional capacity of less than 5.0 METs
 - If measured functional capacity is not available, this factor can be excluded.
- Clinically significant depression or depressive symptoms

Patient is at **MODERATE RISK** if they meet neither high-risk nor low-risk standards:

- Left ventricular ejection fraction = 40%–50%
- Signs/symptoms including angina at “moderate” levels of exercise (60%–75% of maximal functional capacity) or in recovery
- Mild-to-moderate silent ischemia (ST depression <2 mm) with exercise or in recovery

The patient is at **LOW RISK** if ALL of the following factors are present:

- Left ventricular ejection fraction greater than 50%
- No resting or exercise-induced complex dysrhythmias
- Uncomplicated MI, CABG, angioplasty, atherectomy, or stent:
 - Absence of CHF or signs/symptoms indicating postevent ischemia
- Normal hemodynamic and ECG responses with exercise and in recovery
- Asymptomatic with exercise or in recovery, including absence of angina
- Maximal functional capacity at least 7.0 METs
 - If measured functional capacity is not available, this factor can be excluded.
- Absence of clinical depression or depressive symptoms

CABG, coronary artery bypass graft; CHF, congestive heart failure; ECG, electrocardiogram; METs, metabolic equivalents; MI, myocardial infarction; PVCs, premature ventricular contractions.

Reprinted with permission from ([28](#)).

MEDICAL HISTORY AND CARDIOVASCULAR DISEASE RISK FACTOR ASSESSMENT

The medical history and CVD risk factor assessment ([Table 2.2](#)) are used to provide exercise professionals with more in-depth information regarding the health and well-being of each individual. This additional information can help identify individuals who may need referral for medical clearance due to comorbidities not included in the preparticipation screening procedure.

TABLE 2.2 Cardiovascular Disease (CVD) Risk Factors and Defining Criteria

Positive Risk Factors^a	Defining Criteria
Age	Men ≥ 45 yr; women ≥ 55 yr (35)
Family history	Myocardial infarction, coronary revascularization, or sudden death before 55 yr in father or other male first-degree relative or before 65 yr in mother or other female first-degree relative (36)
Cigarette smoking	Current cigarette smoker or those who quit within the previous 6 mo or exposure to environmental tobacco smoke (36, 37)
Physical inactivity	Not meeting the minimum threshold of 500–1,000 MET-min of moderate-to-vigorous physical activity or 75–150 min $\cdot$ wk ⁻¹ of moderate- to vigorous-intensity physical activity (21)
Body mass index/waist circumference	Body mass index ≥ 30 kg $\cdot$ m ⁻² or waist girth > 102 cm (40 in) for men and > 88 cm (38 in) for women (38)
Blood pressure	Systolic blood pressure ≥ 130 mm Hg and/or diastolic ≥ 80 mm Hg, based on an average of ≥ 2 readings obtained on ≥ 2 occasions, or on antihypertensive medication (29)
Lipids	Low-density lipoprotein cholesterol (LDL-C) ≥ 130 mg $\cdot$ dL ⁻¹ (3.37 mmol $\cdot$ L ⁻¹) or high-density lipoprotein cholesterol (HDL-C) < 40 mg $\cdot$ dL ⁻¹ (1.04 mmol $\cdot$ L ⁻¹) in men and < 50 mg $\cdot$ dL ⁻¹ (1.30 mmol $\cdot$ L ⁻¹) in women or non-HDL-C < 130 (3.37 mmol $\cdot$ L ⁻¹) or on lipid-lowering medication. If total serum cholesterol is all that is available, use ≥ 200 mg $\cdot$ dL ⁻¹ (5.18 mmol $\cdot$ L ⁻¹) (30)
Blood glucose	Fasting plasma glucose ≥ 100 mg $\cdot$ dL ⁻¹ (5.5 mmol $\cdot$ L ⁻¹); or 2 h plasma glucose values in oral glucose tolerance test (OGTT) ≥ 140 mg $\cdot$ dL ⁻¹ (7.77 mmol $\cdot$ L ⁻¹); or HbA1C $\geq 5.7\%$ (39)
Negative Risk Factors	Defining Criteria
HDL-C ^b	≥ 60 mg $\cdot$ dL ⁻¹ (1.55 mmol $\cdot$ L ⁻¹) (30)

^a If the presence or absence of a CVD risk factor is not disclosed or is not available, that CVD risk factor should be counted as a risk factor.

^b High HDL-C is considered a negative risk factor. For individuals having high HDL ≥ 60 mg $\cdot$ dL⁻¹ (1.55 mmol $\cdot$ L⁻¹), one positive risk factor is subtracted from the sum of positive risk factors.

HbA1C, glycated hemoglobin; MET, metabolic equivalent; non-HDL-C, total cholesterol minus HDL-C.

The preparticipation medical history should be thorough and include past and current information. Appropriate components of medical history are presented in [Box 2.3](#). Identifying and controlling CVD risk factors continue to be an important objective of overall CV and

metabolic disease prevention and management ([16, 29](#)). CVD risk factor assessment provides important information for the development of an individual's Ex R_x, as well as for lifestyle modification, and is an important opportunity for one-to-one education about CVD risk reduction. Criteria for hypertension ([Table 2.3](#)) and dyslipidemia ([Tables 2.4](#) and [2.5](#)) are essential parts of developing an appropriate Ex R_x. Therefore, exercise professionals are encouraged to complete a CVD risk factor assessment with each individual to determine if the individual meets any of the criteria for CVD risk factors (see [Table 2.2](#)). If the presence of a CVD risk factor is uncertain or is not available, it should be considered as a risk factor until it can be verified. Because of the cardioprotective effect of high-density lipoprotein cholesterol (HDL-C), high levels [$\geq 60 \text{ mg} \cdot \text{dL}^{-1}$ ($1.55 \text{ mmol} \cdot \text{L}^{-1}$)] of this lipoprotein are considered oppositional to CVD risk factors ([31](#)). Therefore, for individuals with HDL-C levels greater than $60 \text{ mg} \cdot \text{dL}^{-1}$ ($1.55 \text{ mmol} \cdot \text{L}^{-1}$), one positive CVD risk factor is subtracted from the sum of risk factors. [Box 2.4](#) provides a framework for conducting CVD risk factor assessment. While counting the number of positive CVD risk factors in an algorithm to identify the need for medical clearance prior to engaging in exercise, identifying risk factors is important for the overall health evaluation process and can provide essential information for exercise program development. Therefore, it is up to the exercise professional to use their best clinical judgement when evaluating CVD risk factors and be aware of any possible contraindications or special considerations needed prior to exercise participation.

TABLE 2.3 Classification and Management of Blood Pressure (BP) for Adults^{a,b}

ACC/AHA Criteria^c				
BP Classification	Normal	Elevated	Stage 1 Hypertension	Stage 2 Hypertension
SBP (mm Hg)	<120	120–129	130–139	≥140
DBP (mm Hg)	<80	<80	80–89	≥90
Treatment recommendations	Promote optimal lifestyle habits; reassess yearly.	Nonpharmacological therapy; reassess in 3–6 mo.	Estimate 10-yr CVD risk. If <10% risk, start with healthy lifestyle recommendation; reassess in 3–6 mo. If ≥10% risk or ASCVD, DM, kidney disease, recommend lifestyle change, and pharmacological treatment; reassess within 1 mo.	Nonpharmacological treatment and BP-lowering medication; follow-up monthly until BP is controlled.
JNC Criteria^d				
BP Classification	Normal	Prehypertension	Hypertension, Stage 1	Hypertension, Stage 2
SBP (mm Hg)	<120	120–139	140–159	≥160
DBP (mm Hg)	<80	80–89	90–99	≥100
Treatment recommendations	Promote optimal lifestyle habits; reassess yearly.	Lifestyle modifications.	Lifestyle modifications and BP-lowering medication.	Lifestyle modifications and BP-lowering medication.

^a Individuals are classified in any given category by meeting either of SBP or DBP thresholds.

^b Individuals with SBP and DBP in two classifications should be designated to the higher BP classification.

^c Whelton PK, Carey RM, Aronow WS, et al. 2017 ACC/AHA/AAPA/ABC/ACPM/AGS/APhA/ASH/ASPC/NMA/PCNA guideline for the prevention, detection, evaluation, and management of high blood pressure in adults: a report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines. Hypertension. 2018;71:1269–1324. doi: 10.1161/HYP.0000000000000065

^d Chobanian AV, Bakris GL, Black HR, et al. Seventh report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure. Hypertension. 2003;42(6):1206–1252.

ACC, American College of Cardiology; AHA, American Heart Association; ASCVD, atherosclerotic cardiovascular disease; DBP, diastolic BP; SBP, systolic BP.

TABLE 2.4 Classifications of Cholesterol and Triglycerides Levels (mg · dL⁻¹)
(41)

Non-HDL-C	
<130	Desirable
130–159	Above desirable
160–189	Borderline high
190–219	High
≥220	Very high
LDL-C	
<100	Desirable
100–129	Above desirable
130–159	Borderline high
160–189	High
≥190	Very high
HDL-C	
<40 (males)	Low
<50 (females)	Low
Triglycerides	
<150	Normal
150–199	Borderline high
200–499	High
≥500	Very high ^a

^a Severe hypertriglyceridemia is another term used for very high triglycerides in pharmaceutical product labeling.

HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; non-HDL-C, total cholesterol minus HDL-C.

TABLE 2.5 Atherosclerotic Cardiovascular Disease Risk Categories and LDL-C Treatment Goals (41)

			Treatment Goals		
	Risk Category	Risk Factors ^a /10-yr Risk ^b	LDL-C (mg · dL ⁻¹)	Non-HDL-C (mg · dL ⁻¹)	apoB (mg · dL ⁻¹)
Extreme risk	AACE	- Progressive ASCVD after achieving an LDL-C < 70 mg · dL⁻¹ - Established clinical cardiovascular disease in patients with DM, CKD stage 3 or 4, or HeFH - History of premature ASCVD (<55 male, <65 female)	<55	<80	<70
	EAS	No recommendations made	—	—	—
Very high risk	AACE	- Established or recent hospitalization for ACS; coronary, carotid, or peripheral vascular disease; 10-yr risk >20% - Diabetes or CKD stage 3 or 4 with one or more risk factor(s) - HeFH	<70	<100	<80
	EAS	- Established ASCVD - Severe CKD (GFR < 30) - DM with target organ damage or major risk factor	<70	<100	<80
High risk	AACE	>2 risk factors and 10-yr risk 10%–20% - Diabetes or CKD stage 3 or 4 with no other risk factors	<100	<130	<90
	EAS	Diabetes, moderate CKD (GFR 30–50), 10-yr risk 5%–10%, familial hypercholesterolemia	<100	<130	<100

			Treatment Goals		
	Risk Category	Risk Factors ^a /10-yr Risk ^b	LDL-C (mg · dL ⁻¹)	Non-HDL-C (mg · dL ⁻¹)	apoB (mg · dL ⁻¹)
Moderate risk	AACE	<2 risk factors and 10-yr risk <10%	<100	<130	<90
	EAS	10-yr risk 1%–5%	<115	—	—
	AACE	No risk factors	<130	<160	NR
	EAS	10-yr risk <1%	<115	—	—

^a Major independent risk factors are high LDL-C, polycystic ovary syndrome, cigarette smoking, hypertension (blood pressure $\geq 140/90$ mm Hg or on hypertensive medication), low HDL-C (<40 mg · dL⁻¹), family history of coronary artery disease (in male, first-degree relative younger than 55 yr; in female, first-degree relative younger than 65 yr), chronic renal disease (CKD) stage 3 or 4, evidence of coronary artery calcification, and age (men ≥ 45 yr; women ≥ 55 yr). Subtract 1 risk factor if the person has high HDL-C.

^b Framingham risk scoring is applied to determine 10-yr risk.

AACE, American Association of Clinical Endocrinologists and American College of Endocrinology; ACS, acute coronary syndrome; apoB, apolipoprotein B; ASCVD, atherosclerotic cardiovascular disease; CKD, chronic kidney disease; DM, diabetes mellitus; EAS, European Atherosclerotic Society; GFR, glomerular filtration rate; HDL-C, high-density lipoprotein cholesterol; HeFH, heterozygous familial hypercholesterolemia; LDL-C, low-density lipoprotein cholesterol; NR, not recommended.

Box 2.3 Components of the Medical History

Appropriate components of the medical history may include the following:

- Medical diagnoses and history of medical procedures: cardiovascular disease risk factors including hypertension, obesity, dyslipidemia, and diabetes; cardiovascular disease including heart failure, valvular dysfunction (*e.g.*, aortic stenosis/mitral valve disease), myocardial infarction, and other acute coronary syndromes; percutaneous coronary interventions including angioplasty and coronary stent(s), coronary artery bypass surgery, and other cardiac surgeries such as valvular surgeries; cardiac transplantation; pacemaker and/or implantable cardioverter defibrillator; ablation procedures for dysrhythmias; peripheral vascular disease; pulmonary disease including asthma, emphysema, and bronchitis; cerebrovascular disease including stroke and transient ischemic attacks; anemia and other blood dyscrasias (*e.g.*, lupus erythematosus); phlebitis, deep vein thrombosis, or emboli; cancer; pregnancy; osteoporosis; musculoskeletal disorders; emotional disorders; and eating disorders
- Previous physical examination findings: murmurs, clicks, gallop rhythms, other abnormal heart sounds, and other unusual cardiac and vascular findings; abnormal pulmonary findings (*e.g.*, wheezes, rales, crackles), high blood pressure, and edema
- Laboratory findings (*i.e.*, plasma glucose, HbA1C, hs-CRP, serum lipids, and lipoproteins)
- History of symptoms: discomfort (*e.g.*, pressure, tingling sensation, pain, heaviness, burning, tightness, squeezing, and numbness) in the chest, jaw, neck, back, or arms; light-headedness, dizziness, or fainting; temporary loss of visual acuity or speech; transient unilateral numbness or weakness; shortness of breath; rapid heartbeat or palpitations, especially if associated with physical activity, eating a large meal, emotional upset, or exposure to cold (or any combination of these activities)
- Recent illness, hospitalization, new medical diagnoses, or surgical procedures
- Orthopedic problems including arthritis, joint swelling, and any condition that would make ambulation or use of certain test modalities difficult
- Medication use (including dietary/nutritional supplements) and drug allergies
- Other habits including caffeine, alcohol, tobacco, or recreational (illicit) drug use
- Exercise history: information on readiness for change and habitual level of activity: frequency, duration or time, type, and intensity or FITT of exercise
- Work history with emphasis on current or expected physical demands, noting upper and lower extremity requirements
- Family history of cardiac, pulmonary, or metabolic disease, stroke, or sudden death

FITT, frequency, intensity, time, and type; HbA1C, glycosylated hemoglobin; hs-CRP, high-sensitivity C-reactive protein.

Box 2.4 Case Studies to Conduct Cardiovascular Disease Risk Factor Assessment

CASE STUDY I

Female, age 21 years, smokes socially on weekends (~10–20 cigarettes). Drinks alcohol one or two nights a week, usually on weekends. Height = 63 in (160 cm), weight = 124 lb (56.4 kg), BMI = $22.0 \text{ kg} \cdot \text{m}^{-2}$. RHR = $76 \text{ beats} \cdot \text{min}^{-1}$, resting BP = 118/72 mm Hg. Total cholesterol = $178 \text{ mg} \cdot \text{dL}^{-1}$ ($4.61 \text{ mmol} \cdot \text{L}^{-1}$), LDL-C = $98 \text{ mg} \cdot \text{dL}^{-1}$ ($2.54 \text{ mmol} \cdot \text{L}^{-1}$), HDL-C = $62 \text{ mg} \cdot \text{dL}^{-1}$ ($1.60 \text{ mmol} \cdot \text{L}^{-1}$), FBG = $96 \text{ mg} \cdot \text{dL}^{-1}$ ($5.33 \text{ mmol} \cdot \text{L}^{-1}$). Currently taking oral contraceptives. Attends 45-minute moderate-intensity group exercise class two to three times a week; both parents living and in good health.

CASE STUDY II

Male, age 45 years, nonsmoker. Height = 72 in (182.9 cm), weight = 168 lb (76.4 kg), BMI = $22.8 \text{ kg} \cdot \text{m}^{-2}$. RHR = $64 \text{ beats} \cdot \text{min}^{-1}$, resting BP = 124/78 mm Hg. Total cholesterol = $187 \text{ mg} \cdot \text{dL}^{-1}$ ($4.84 \text{ mmol} \cdot \text{L}^{-1}$), LDL-C = $103 \text{ mg} \cdot \text{L}^{-1}$ ($2.67 \text{ mmol} \cdot \text{L}^{-1}$), HDL-C = $39 \text{ mg} \cdot \text{dL}^{-1}$ ($1.01 \text{ mmol} \cdot \text{L}^{-1}$), FBG = $88 \text{ mg} \cdot \text{dL}^{-1}$ ($4.84 \text{ mmol} \cdot \text{L}^{-1}$). Recreationally competitive runner, runs $4\text{--}7 \text{ d} \cdot \text{wk}^{-1}$, completes one to two marathons and numerous other road races every year. No medications other than over-the-counter ibuprofen as needed. Father died at age 51 years of a heart attack; mother died at age 81 years of cancer.

CASE STUDY III

Male, age 44 years, nonsmoker. Height = 70 in (177.8 cm), weight = 216 lb (98.2 kg), BMI = $31.0 \text{ kg} \cdot \text{m}^{-2}$. RHR = $62 \text{ beats} \cdot \text{min}^{-1}$, resting BP = 128/84 mm Hg. Total serum cholesterol = $184 \text{ mg} \cdot \text{dL}^{-1}$ ($4.77 \text{ mmol} \cdot \text{L}^{-1}$), LDL-C = $106 \text{ mg} \cdot \text{dL}^{-1}$ ($2.75 \text{ mmol} \cdot \text{L}^{-1}$), HDL-C = $44 \text{ mg} \cdot \text{dL}^{-1}$ ($1.14 \text{ mmol} \cdot \text{L}^{-1}$), FBG = $130 \text{ mg} \cdot \text{dL}^{-1}$ ($7.22 \text{ mmol} \cdot \text{L}^{-1}$). Reports that he does not have time to exercise. Father had Type 2 diabetes and died at age 67 years of a heart attack; mother living, no CVD; no medications.

CASE STUDY IV

Female, age 36 years, nonsmoker. Height = 64 in (162.6 cm), weight = 108 lb (49.1 kg), BMI = $18.5 \text{ kg} \cdot \text{m}^{-2}$. RHR = $61 \text{ beats} \cdot \text{min}^{-1}$, resting BP = 142/86 mm Hg. Total cholesterol = $174 \text{ mg} \cdot \text{dL}^{-1}$ ($4.51 \text{ mmol} \cdot \text{L}^{-1}$), blood glucose normal with insulin injections. Type 1 diabetes mellitus diagnosed at age 7 years. She teaches high-intensity cardio kickboxing classes three times per week and walks at a moderate intensity for approximately 45 minutes four times a week; both parents in good health with no history of CVD.

	Case Study I	Case Study II	Case Study III	Case Study IV
CVD risk factors:				
Age?	No	Yes	No	No
Family history?	No	Yes	No	No
Cigarette smoking?	Yes	No	No	No
Physical inactivity?	No	No	Yes	No
Obesity?	No	No	Yes	No
Hypertension?	No	No	Yes	Yes
Dyslipidemia?	No	Yes	No	No
Diabetes?	No	No	Yes	Yes
Negative risk factor:				
HDL-C $\geq 60 \text{ mg} \cdot \text{dL}^{-1}$	Yes	No	No	No
Number of CVD risk factors	Zero	Three	Four	Two

BMI, body mass index; BP, blood pressure; CVD, cardiovascular disease; FBG, fasting blood glucose; HDL-C, high-density lipoprotein cholesterol; LDL-C, low-density lipoprotein cholesterol; RHR, resting heart rate.

Additional Recommendations

If warranted, a physical examination should be performed by a physician or other qualified health care provider. Appropriate components of the physical examination specific to subsequent exercise testing are presented in [Box 2.5](#). Recommended laboratory tests, depending on individual risk factors, signs, and symptoms, could include fasting serum total cholesterol, fasting plasma glucose, 12-lead ECG, Holter monitoring, cardiac echocardiography, chest radiography, pulmonary function, and oximetry. Although detailed descriptions of all the physical examination procedures and the recommended laboratory tests are beyond the scope of the *Guidelines*, additional basic information related to assessment of CVD risk factors are provided in subsequent chapters. The reader is encouraged to read the sections pertinent to the risk factor of interest and information related to exercise testing and prescription related to that condition. Additionally, for more detailed descriptions of these assessments, the reader is referred to the work of Bickley and colleagues ([31](#)).

Box 2.5 Components of the Preparticipation Physical Examination*

Appropriate components of the physical examination may include the following:

- Body weight; in many instances, determination of body mass index, waist girth, and/or body composition (body fat percentage) is desirable.
- Apical pulse rate and rhythm
- Resting blood pressure: seated, supine, and standing
- Auscultation of the lungs with specific attention to uniformity of breath sounds in all areas (absence of rales, wheezes, and other breathing sounds)
- Palpation of the cardiac apical impulse and point of maximal impulse
- Auscultation of the heart with specific attention to murmurs, gallops, clicks, and rubs
- Palpation and auscultation of carotid, abdominal, and femoral arteries
- Evaluation of the abdomen for bowel sounds, masses, visceromegaly, and tenderness
- Palpation and inspection of lower extremities for edema and presence of arterial pulses
- Absence or presence of tendon xanthoma and skin xanthelasma
- Follow-up examination related to orthopedic or other medical conditions that would limit exercise testing
- Tests of neurologic function including reflexes and cognition (as indicated)
- Inspection of the skin, especially of the lower extremities in known patients with diabetes mellitus

* For more detailed information see reference ([31](#)).

ONLINE RESOURCES

American College of Cardiology: <http://tools.acc.org/ASCVD-Risk-Estimator-Plus/#!/calculate/estimate/>

American College of Sports Medicine Exercise is Medicine:
<http://www.exerciseismedicine.org>

American Diabetes Association: <http://www.diabetes.org>

American Heart Association: <http://www.americanheart.org>

European Society of Cardiology: <http://www.escardio.org>

National Heart, Lung, and Blood Institute Health Information for Professionals:
<http://www.nhlbi.nih.gov/health/indexpro.htm>

Physical Activity Readiness Questionnaire for Everyone (PAR-Q+):
https://www.nasm.org/docs/pdf/parqplus-2020.pdf?sfvrsn=401bf1af_24

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CHAPTER 3

Health-Related Physical Fitness Testing and Interpretation

INTRODUCTION

Numerous health benefits result from regular participation in physical activity (PA) and structured exercise programs ([1](#)). Health-related components of physical fitness have a strong relationship with overall health, are characterized by an ability to perform activities of daily living (ADL) with vigor, and are associated with a lower prevalence of chronic disease and health conditions and their associated risk factors ([2](#)). Measures of health-related physical fitness are closely aligned with both disease prevention and health promotion. Skill-related components typically associated with sport performance (*i.e.*, power and agility) are also important, particularly in supporting an independent living status as one ages ([3–5](#)). A fundamental goal of primary and secondary prevention and rehabilitation programs should be the promotion of health; hence, exercise programs should focus on enhancement of the health-related components of physical fitness. This chapter therefore focuses on the health-related components of physical fitness testing and interpretation.

PURPOSES OF HEALTH-RELATED PHYSICAL FITNESS TESTING

Physical fitness measurement is a common and appropriate practice in preventive and rehabilitative exercise programs. Minimally, a health-related physical fitness test must be both reliable and valid, and ideally, it should be relatively inexpensive. The information obtained from health-related physical fitness testing, in combination with the individual's medical and exercise history, is used for the following:

- Collecting baseline data and educating individuals about their present health/fitness status relative to health-related standards and age- and sex-matched norms.
- Providing data that are helpful in the development of individualized exercise programs to address all health/fitness components.
- Collecting follow-up data that allow evaluation of short- and long-term progress following an exercise prescription (Ex R_x).
- Motivating individuals by establishing reasonable and attainable health/fitness goals (see [Chapter 12](#)).

BASIC PRINCIPLES AND GUIDELINES

Pretest Instructions for Fitness Testing

All pretest instructions should be provided and adhered to prior to arrival at the testing facility. For information on prescreening, informed consent, and pretest instructions, please refer to [Chapter 2](#).

Organizing the Fitness Test

The following should be accomplished before the individual begins a fitness test:

- Ensure that all consent and screening forms, data recording sheets, and any related testing documents are available in the individual's file prior to test administration.
- Ensure that selected testing equipment (*e.g.*, cycle ergometer, treadmill, sphygmomanometer, skinfold caliper) has been calibrated according to manufacturer's recommendation, or more frequently, based on use (*e.g.*, ventilatory expired gas analysis systems, blood pressure [BP] sphygmomanometer), and document all equipment calibration. Skinfold calipers should be regularly checked for accuracy and sent to the manufacturer for calibration when needed.
- Ensure that room temperature is between 68 °F and 72 °F (20 °C and 22 °C) and humidity is less than 60% with adequate airflow ([6](#)).

When multiple tests are to be administered during the same appointment, the sequencing of the testing session can be important, depending on which

physical fitness components are to be evaluated. Resting measurements such as heart rate (HR), BP, height, weight, and body composition should be obtained first. An optimal testing order for exertional fitness components (*i.e.*, cardiorespiratory fitness [CRF], muscular fitness, and flexibility) has not been established, but sufficient time should be allowed for HR and BP to return to baseline between tests conducted serially. Additionally, tests should be arranged in an order that does not result in stressing the same muscle group repeatedly. To ensure reliability, the chosen order should be followed on subsequent testing sessions. Because certain medications (*e.g.*, β -blockers) will affect some physical fitness test results, use of these medications should be noted (see [Appendix A](#)).

Test Environment

The environment is important for test validity and reliability. Test anxiety, emotions, room temperature, and ventilation should be controlled as much as possible. To minimize individual anxiety, the test procedures should be explained adequately and should not be rushed, and the test environment should be quiet and private. The room should be equipped with a comfortable seat and/or examination table to be used for resting HR and BP. The demeanor of personnel should be one of relaxed confidence to put the individual at ease. Finally, the exercise professional should be familiar with the facility's emergency response plan ([Z](#)).

A COMPREHENSIVE HEALTH FITNESS EVALUATION

A comprehensive health/fitness assessment can usually be completed in a single session and includes the following: (a) informed consent, exercise preparticipation health screening, and preexercise evaluation (see [Chapter 2](#)); (b) resting measurements; (c) circumference measurements and body composition analysis; (d) measurement of CRF; (e) measurement of muscular fitness; and (f) measurement of flexibility. Additional evaluations, such as static and dynamic balance and postural measurements, may be administered. The data accrued from the evaluation should be interpreted by a competent exercise professional and conveyed to the individual in terms that they can understand. This information is central to educate the individual about their current health-related physical fitness status and to the development of the individual's short- and long-term goals. Testing results form the basis for the individualized Ex Rx and subsequent evaluations to monitor progress.

The number of transgender individuals is increasing, and they are therefore more likely to be encountered by exercise professionals in their daily work. The current estimated number of adults reporting as transgender in the United States is 1.6 million ([8](#)). In regard to fitness testing and health risk status, many norms and criteria are based on sex recorded at birth, along with age and other discriminating factors. A challenge exercise professionals face is what sex-based criteria and norms to use for transgender individuals undergoing medical treatment. Although it may seem intuitive to use the current gender identification to align with health and fitness norms, there is little current evidence to support this. It has been suggested that individuals wait at least 2 years as females and 3 years as males until they are compared to their current gender identification ([9](#)); however, this has not been

reproduced in the nonmilitary population. Therefore, because there are few data and few norms for individuals in the midst of a medical treatment for gender incongruence, it may be best to use collected data only on an intrapersonal basis while norms are being developed.

For certain individuals, the risks of health-related physical fitness testing may outweigh the potential benefits. Some assessments pose little risk (*e.g.*, body composition), whereas others may have higher risks (*e.g.*, CRF and one-repetition maximum [1-RM]) for some individuals. It is important to carefully assess risk versus benefit when deciding on whether a fitness test should be performed. Preexercise evaluations performed with a careful review of current and prior medical history will help to identify the possible need for referral to a medical provider, potential contraindications, and also increase the safety of the health-related physical fitness assessment. *Box 4.1* provides a list of conditions that preclude exercise testing (absolute contraindications) or permit exercise testing if the benefits outweigh the risks (relative contraindications).

MEASUREMENT OF RESTING HEART RATE AND BLOOD PRESSURE

A comprehensive physical fitness assessment includes the measurement of resting HR and BP. HR can be determined using several techniques including pulse palpation, auscultation with a stethoscope, or the use of a HR monitor. Although more commonly used in clinical assessments, electrocardiogram (ECG) monitoring is also an option for monitoring HR. Upon arrival to the testing facility, each individual should relax and remain seated or assume a supine position on a nonconductive surface for at least 5 minutes — preferably up to 15 minutes — in a quiet area, before assessing HR and BP, allowing these measures to stabilize ([10](#)).

The pulse palpation technique involves “feeling” the pulse by placing the index and middle fingers over the radial artery, located near the thumb side of the wrist. The pulse is counted for 30 or 60 seconds. The 30-second count is multiplied by 2 to determine the 1-minute resting HR (beats per minute). For the auscultation method, the bell of the stethoscope should be placed to the left of the sternum just above the level of the nipple. The auscultation method is most accurate when the heart sounds are clearly audible, and the individual’s torso is stable. If a HR monitor (*i.e.*, chest strap or wristwatch) is used, it should fit snugly, be in direct contact with the skin, and be positioned on the body as recommended by the manufacturer.

The measurement of resting BP is described in [Box 3.1](#). Potential sources of error in BP measurement are listed in [Box 3.2](#).

Box 3.1 Procedures for Assessment of Resting Blood Pressure

1. Those being tested should be seated quietly for at least 5 minutes in a chair with back support (rather than on an examination table) with their feet on the floor and their arms supported at heart level. Individuals should refrain from smoking cigarettes or ingesting caffeine for at least 30 minutes preceding the measurement.
2. Measuring supine and standing values may be indicated under special circumstances.
3. Wrap cuff firmly around upper arm at heart level; align cuff with brachial artery.
4. The appropriate cuff size must be used to ensure accurate measurement. The bladder within the cuff should encircle at least 80% of the upper arm. Many adults require a large adult cuff.
5. Place stethoscope chest piece below the antecubital space over the brachial artery. Bell and diaphragm side of chest piece appear equally effective in assessing BP ([11](#)).
6. Quickly inflate cuff pressure to 20 mm Hg above the first Korotkoff sound.
7. Slowly release pressure at rate equal to $2\text{--}3 \text{ mm Hg} \cdot \text{s}^{-1}$.
8. SBP is the point at which the first of two or more Korotkoff sounds is heard (phase 1), and DBP is the point before the disappearance of Korotkoff sounds (phase 5).
9. At least two measurements should be made (minimum of 1 minute apart) and the average should be taken.
10. BP should be measured in both arms during the first examination. Higher pressure should be used when there is consistent interarm difference.
11. Provide to individuals, verbally and in writing, their specific BP numbers and BP goals.

BP, blood pressure; DBP, diastolic blood pressure; SBP, systolic blood pressure. Data from ([10](#)). For additional, more detailed recommendations, see ([12](#)).

Box 3.2 Potential Sources of Error in Blood Pressure Assessment

- Inaccurate sphygmomanometer
- Improper cuff size
- Auditory acuity of technician
- Rate of inflation or deflation of cuff pressure
- Experience of technician
- Faulty equipment
- Improper stethoscope placement or pressure
- Not having the cuff at heart level
- Certain physiologic abnormalities (*e.g.*, damaged brachial artery, subclavian steal syndrome, arteriovenous fistula)
- Reaction time of technician^{*}
- Background noise
- Allowing individuals to hold treadmill handrails or flex elbow^{*}

^{*} Applies specifically during exercise testing.

BODY COMPOSITION

It is well established that excess body fat, particularly when located centrally around the abdomen, is associated with many chronic conditions including hypertension, metabolic syndrome, Type 2 diabetes mellitus (T2DM), stroke, cardiovascular disease (CVD), dyslipidemia, and many types of cancer ([13–17](#)). More than two-thirds (70.2%) of American adults are classified as either overweight or obese (body mass index [BMI] $\geq 25 \text{ kg} \cdot \text{m}^{-2}$), and more than a third (37.7%) are classified as obese (BMI $\geq 30 \text{ kg} \cdot \text{m}^{-2}$) ([18](#)). Nearly one-third (31.8%) of American children and adolescents are overweight or obese ([17](#)) (see [Chapter 9](#)). The data on overweight/obesity prevalence among the adult and pediatric populations and its health implications have precipitated an increased awareness in the value of identifying and treating individuals with excess body weight ([19](#)). Indeed, in 2013, the American Medical Association labeled obesity as a disease ([20](#)).

It is important to recognize the health-related changes in body composition that accompany aging. Such changes may include increasing body fat, decreasing bone mineral density, and loss of muscle mass. Sarcopenia, the degenerative loss of muscle mass and strength as a result of aging and reduced PA, is associated with a reduced ability to perform ADL and increases the risk and fear of falling and risk of musculoskeletal injury ([21](#)). Thus, body composition measurement can be used to monitor changes in lean body mass, particularly among older adults (see [Chapter 6](#)).

Basic body composition can be expressed as the relative percentage of body mass that is fat and fat-free tissue using a two-component model. Body composition can be estimated with methods that vary in terms of complexity, cost, and accuracy ([22](#)). Different assessment techniques are briefly reviewed in this section, but details associated with obtaining measurements and calculating estimates of body fat for all of these techniques are beyond the

scope of the *Guidelines*. Additional detailed information is available ([23–25](#)). Before collecting data for body composition assessment, the technician must be trained, experienced in the techniques, and already have demonstrated reliability in obtaining measurements, independent of the technique being used.

Anthropometric Methods

Height, Weight, and Body Mass Index

Body weight should be measured using a calibrated balance beam or electronic scale with the individual wearing minimal clothing and having empty pockets. Shoes should be removed prior to these assessments ([24](#)).

BMI, or the Quetelet index, is used to assess weight relative to height and is calculated by dividing body weight in kilograms by height in meters squared ($\text{kg} \cdot \text{m}^{-2}$). Common practice is to define a BMI of less than $18.5 \text{ kg} \cdot \text{m}^{-2}$ as underweight, $18.5\text{--}24.9 \text{ kg} \cdot \text{m}^{-2}$ as normal, $25.0\text{--}29.9 \text{ kg} \cdot \text{m}^{-2}$ as overweight, and $\geq 30.0 \text{ kg} \cdot \text{m}^{-2}$ as obese ([26](#)). Because Asian populations develop health problems at lower BMI values in comparison to other population subgroups, using lower cut points for defining overweight and obesity for Asian populations (≥ 23.0 and $\geq 25.0 \text{ kg} \cdot \text{m}^{-2}$, respectively) is recommended ([27](#)). Moreover, the distribution of body weight and composition proportions are known to vary across different population subgroups ([28](#)), thereby raising questions about the suitability of BMI as a proxy for adiposity. Although BMI fails to distinguish between body fat, muscle mass, or bone, it is well accepted that with the exception of individuals with large amounts of muscle mass, those with a BMI $\geq 30 \text{ kg} \cdot \text{m}^{-2}$ have excess body fat. An increased risk of obesity-related diseases, health conditions, and mortality are associated with a BMI $\geq 30.0 \text{ kg} \cdot \text{m}^{-2}$ ([Table 3.1](#)) ([30](#), [31](#)). This association is not perfect, as there is compelling evidence to indicate individuals diagnosed with

congestive heart failure (CHF) actually have improved survival when BMI is $\geq 30.0 \text{ kg} \cdot \text{m}^{-2}$, a phenomenon known as the “obesity paradox” (32).

TABLE 3.1 Classification of Disease Risk Based on Body Mass Index (BMI) and Waist Circumference

		Disease Risk^a Relative to Normal Weight and Waist Circumference	
	BMI ($\text{kg} \cdot \text{m}^{-2}$)	Males, $\leq 102 \text{ cm}$ (40 in) Females, $\leq 88 \text{ cm}$ (35 in)	Males, $> 102 \text{ cm}$ (40 in) Females, $> 88 \text{ cm}$ (35 in)
Underweight	< 18.5	—	—
Normal	18.5–24.9	—	—
Overweight	25.0–29.9	Increased	High
Obesity, class			
I	30.0–34.9	High	Very high
II	35.0–39.9	Very high	Very high
III	≥ 40.0	Extremely high	Extremely high

^a Disease risk for Type 2 diabetes, hypertension, and cardiovascular disease.

Dashes (—) indicate that no additional risk at these levels of BMI was assigned. Increased waist circumference can also be a marker for increased risk even in individuals of normal weight.

Modified from (29).

Compared to individuals classified as obese, the link between BMI in the overweight range ($25.0\text{--}29.9 \text{ kg} \cdot \text{m}^{-2}$) and higher mortality risk is less clear. However, a BMI of $25.0\text{--}29.9 \text{ kg} \cdot \text{m}^{-2}$ is convincingly linked to an increased risk for other health issues such as T2DM, dyslipidemia, hypertension, and certain cancers (33). A BMI of less than $18.5 \text{ kg} \cdot \text{m}^{-2}$ also increases mortality risk from causes other than cancer and CVD (34) and may be indicative of malnutrition, disordered eating, osteoporosis, severe pulmonary disease, and

metabolic abnormalities ([35](#), [36](#)). Despite the relationship between BMI and health risks, other methods of body composition assessment should be used to estimate percent body fat during a physical fitness assessment ([37](#)).

Circumferences

Regional body circumference measurement can be important to quantify body fat distribution, especially of the waist and hip. The pattern of body fat distribution is an important indicator of health and prognosis ([38](#)). Android obesity that is characterized by more fat on the trunk (*i.e.*, abdominal fat) increases the risk of hypertension, metabolic syndrome, T2DM, dyslipidemia, CVD, cancer, and premature death compared with individuals who demonstrate gynoid or gynecoid obesity (*i.e.*, fat distributed in the hip/thigh region) ([16](#), [39](#)). Moreover, increased visceral fat (*i.e.*, fat within and surrounding thoracic and abdominal cavities) confers a higher risk for development of metabolic syndrome compared to distribution of fat within the subcutaneous compartment ([40](#)). Because of this, circumference (or girth) measurements can provide a general representation of body fat distribution and subsequent risk. Equations are available for both biological sexes and a range of age groups to predict body fat percentage from circumference measurements (standard error of estimate [SEE] = 2.5%–4.0%) ([24](#), [41](#), [42](#)). A cloth tape measure with a spring-loaded handle (*e.g.*, Gulick tape measure) is recommended, allowing for the standard tension of the tape on the skin and improved measurement consistency. However, other types of tape measures can be used. Duplicate measurements are recommended at each site and should be obtained in a rotational instead of a consecutive order (*i.e.*, take one measurement at all sites being assessed, then repeat the sequence). An average of the two measures is used provided they do not differ by more than 5 mm ([43](#)). [Box 3.3](#) contains a description of the common measurement sites.

Box 3.3 Standardized Description of Circumference Sites and Procedures

Abdomen:	With the individual standing, a horizontal measure is taken at the height of the iliac crest, usually at the level of the umbilicus.
Arm:	With the individual standing and arms hanging freely at the sides with hands facing the thigh, a horizontal measure is taken midway between the acromion and olecranon processes.
Buttocks/hips:	With the individual standing and feet together, a horizontal measure is taken at the maximal circumference of the buttocks. This measure is used for the hip measure in the waist-to-hip ratio.
Calf:	With the individual standing (feet apart ~20 cm), a horizontal measure is taken at the level of the maximum circumference between the knee and the ankle, perpendicular to the long axis.
Forearm:	With the individual standing, arms hanging downward but slightly away from the trunk and palms facing anteriorly, a measure is taken perpendicular to the long axis at the maximal circumference.
Hips/thigh:	With the individual standing, legs slightly apart (~10 cm), a horizontal measure is taken at the maximal circumference of the hip/proximal thigh, just below the gluteal fold.
Midthigh:	With the individual standing and one foot on a bench so the knee is flexed at 90°, a measure is taken midway between the inguinal crease and the proximal border of the patella, perpendicular to the long axis.
Waist:	With the individual standing, arms at the sides, feet together, and abdomen relaxed, a horizontal measure is taken at the narrowest part of the torso (above the umbilicus and below the xiphoid process). The National Obesity Task Force (NOTF) suggests obtaining a horizontal measure directly above the iliac crest as a method to enhance standardization (29). Unfortunately, current formulas are not predicated on the NOTF suggested site.

Procedures

- All measurements should be made with a flexible yet inelastic tape measure.
- The tape should be placed on the skin surface without compressing the subcutaneous adipose tissue.
- If a Gulick-type spring-loaded tape measure is used, the handle should be extended to the same marking with each trial.
- Take duplicate measures at each site, and retest if duplicate measurements are not within 5 mm.
- Rotate through measurement sites or allow time for skin to regain normal texture.

Modified from ([43](#)).

The waist-to-hip ratio (WHR) is the circumference of the waist divided by the circumference of the hips (see [Box 3.3](#) for waist and buttocks/hips measures) and has traditionally been used as a simple method for assessing body fat distribution patterns. WHR identifies individuals with higher amounts of abdominal fat or central adiposity ([44](#)). Health risk increases as WHR increases, and the standards for risk vary with age and sex. For example, for those younger than 60 years of age, health risk is *very high* for males when WHR is greater than 0.95 and for females when WHR is greater than 0.86. For individuals aged 60–69 years, the WHR cutoff values are greater than 1.03 for males and greater than 0.90 for females for the same high-risk classification as young adults ([24](#)).

The waist or abdominal circumference alone measured at the largest waist circumference above the iliac crests and below the last rib may be used as an indicator of obesity-related health risk as central obesity is a primary health issue ([45](#), [46](#)). Waist circumference and WHR are reported to be superior to BMI for this purpose ([47](#)). Although BMI and waist circumference are correlated, the correlation between BMI and WHR is weak and implies the information provided by the two measures is different ([44](#)). Waist circumference is a proxy for the combination of visceral fat and abdominal fat. Because visceral adiposity increases the risk for obesity-related diseases, waist circumference is an important measure for health risk assessments ([48](#)). The Expert Panel on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults provides a classification of disease risk based on both BMI and waist circumference as shown in [Table 3.1](#) ([29](#)). Previous research demonstrated that the waist circumference thresholds shown in [Table 3.1](#) effectively identify individuals at increased health risk across the different BMI categories ([49](#)). Furthermore, risk criteria for adults based on more specific waist circumferences have been developed ([Table 3.2](#)) ([50](#)). It is important to note that these risk criteria are based on data derived from Caucasian males and females and may be different for other

racial/ethnic groups. For example, South Asian ([51](#)) and African American males and females may have different cut points for specific BMI and circumferences ([52, 53](#)). The Pennington Center Longitudinal Study found that BMI, along with other measures of obesity (*i.e.*, body adiposity index, waist-to-height ratio, and WHR), all correlated with mortality in Caucasians but not in African Americans. However, the risk of mortality associated with waist circumference was almost identical between races ([54](#)). Furthermore, the optimal BMI and waist circumference thresholds to identify cardiometabolic health risk differ between Caucasian and African American males and females ([51](#)).

TABLE 3.2 Risk Criteria for Waist Circumference in Adults

	Waist Circumference cm (in)	
Risk category	Females	Males
Very low	<70 (<27.5)	<80 (31.5)
Low	70–89 (27.5–35.0)	80–99 (31.5–39.0)
High	90–110 (35.5–43.0)	100–120 (39.5–47.0)
Very high	>110 (>43.5)	>120 (>47.0)

Reprinted with permission from ([50](#)).

Several methods for waist circumference measurement involving different anatomical sites are available. Practitioners should be aware of which anatomical location the waist circumference is measured in order to be consistent with certain disease risk stratification and for follow-up assessments. For example, [Table 3.2](#) is based on data where the waist circumference was taken at the level of the iliac crest ([50, 55](#)), whereas the Pennington Longitudinal Studies take waist circumference at the midpoint between the inferior border of the ribcage and the superior aspect of the iliac crest ([52–54](#)). Evidence indicates that waist circumference taken just inferior to the lowest rib is preferred over waist circumference at the midpoint as a

predictor of T2DM and cardiometabolic risk in adults between 46 and 73 years of age ([56](#)).

Skinfold Measurements

Although BMI and circumferences are anthropometric measures that may be used to assess health risk, they are not true measures of body composition. The skinfold technique is a body composition method that estimates body fat percentage by determining the thickness of several folds of skin across the body. Body fat percentage determined from skinfold thickness measurements correlates well ($r = 0.70\text{--}0.93$) with hydrodensitometry ([57](#)), air displacement plethysmography (ADP) ([58](#)), and dual-energy x-ray absorptiometry (DXA) ([59](#)). The principle behind the skinfold technique is that the amount of subcutaneous fat is proportional to the total amount of body fat. It is assumed that approximately one-third of the total fat in the body is located subcutaneously ([60](#)), but there is considerable variation in intramuscular, intermuscular, and internal organ fat deposits among individuals ([61](#), [62](#)). The exact proportion of subcutaneous to total fat also varies with sex, age, and race ([14](#), [62](#)). Therefore, regression equations used to convert sum of skinfolds to body density and to convert body density to percent body fat should consider these variables for reducing prediction error. [Box 3.4](#) presents a standardized description of skinfold sites and procedures. Additional detail of skinfold technique is described elsewhere ([23](#), [24](#)). Skinfold assessment of body composition is dependent on the expertise of the technician, so proper training (*i.e.*, knowledge of anatomical landmarks) and ample practice of the technique is necessary to obtain accurate measurements. The accuracy of predicting percent body fat from skinfolds is approximately $\pm 3.5\%$, assuming appropriate techniques and equations have been used ([24](#)).

Box 3.4 Standardized Description of Skinfold Sites and Procedures

Skinfold Site

Abdominal	Vertical fold; 2 cm to the right side of the umbilicus
Triceps	Vertical fold; on the posterior midline of the upper arm, halfway between the acromion and olecranon processes, with the arm held freely to the side of the body
Biceps	Vertical fold; on the anterior aspect of the arm over the belly of the biceps muscle, 1 cm above the level used to mark the triceps site
Chest/pectoral	Diagonal fold; one-half the distance between the anterior axillary line and the nipple (males) or one-third of the distance between the anterior axillary line and the nipple (females)
Medial calf	Vertical fold; at the maximum circumference of the calf on the midline of its medial border
Midaxillary	Vertical fold; on the midaxillary line at the level of the xiphoid process of the sternum. An alternate method is a horizontal fold taken at the level of the xiphoid/sternal border on the midaxillary line.
Subscapular	Diagonal fold (45° angle); 1–2 cm below the inferior angle of the scapula
Suprailiac	Diagonal fold; in line with the natural angle of the iliac crest taken in the anterior axillary line immediately superior to the iliac crest
Thigh	Vertical fold; on the anterior midline of the thigh, midway between the proximal border of the patella and the inguinal crease (hip)

Procedures

- All measurements should be made on the right side of the body with the individual standing upright.
- Caliper should be placed directly on the skin surface, 1 cm away from the thumb and finger, perpendicular to the skinfold, and halfway between the crest and the base of the fold.
- Pinch should be maintained while reading the caliper.
- Wait 1–2 seconds before reading caliper.
- Take duplicate measures at each site, and retest if duplicate measurements are not within 1–2 mm.
- Rotate through measurement sites or allow time for skin to regain normal texture and thickness.

Factors that may contribute to measurement error within skinfold assessment include poor anatomical landmark identification, poor

measurement technique, an inexperienced evaluator, an extremely obese or extremely lean individual, perspiration, and an improperly calibrated caliper ([63](#), [64](#)). Various regression equations have been developed to predict body density or percent body fat from skinfold measurements. [Box 3.5](#) lists generalized equations that allow calculation of body density for a wide range of individuals ([64](#), [65](#)). Other equations have been published that are sex, age, race, fat, and sport specific ([24](#), [66](#)). A useful alternative to using skinfolds to predict body fat is to track change in measurements at individual skinfold sites or use the sum of skinfolds. It is also important to note the type of skinfold calipers used in each investigation, as different manufacturers of skinfold calipers may provide different estimations of skinfold thickness (*e.g.*, Lange vs. Harpenden calipers) ([67](#)).

Box 3.5 Generalized Skinfold Equations

Males

- **Seven-site formula** (chest, midaxillary, triceps, subscapular, abdomen, suprailiac, thigh)

Body density = $1.112 - 0.00043499 \sum \text{seven skinfolds} + 0.00000055 \sum \text{seven skinfolds}^2 - 0.00028826 \text{age}$ SEE 0.008 or ~3.5% fat
StartLayout 1st Row Body density equals 1.112 minus 0.00043499 left parenthesis s u m of seven skinfolds right parenthesis 2nd Row plus 0.00000055 left parenthesis s u m of seven skinfolds right parenthesis squared 3rd Row minus 0.00028826 left parenthesis a g e right parenthesis left bracket italic upper S upper E upper E 0.008 italic or tilde 3.5 percent sign italic f a t right bracket EndLayout

- **Three-site formula** (chest, abdomen, thigh)

Body density = $1.10938 - 0.0008267 \sum \text{three skinfolds} + 0.0000016 \sum \text{three skinfolds}^2 - 0.0002574 \text{age}$ SEE 0.008 or ~3.4% fat
StartLayout 1st Row Body density equals 1.10938 minus 0.0008267 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000016 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.0002574 left parenthesis a g e right parenthesis left bracket italic upper S upper E upper E italic 0.008 italic or tilde italic 3.4 percent sign italic f a t right bracket EndLayout

- **Three-site formula** (chest, triceps, subscapular)

Body density = $1.1125025 - 0.0013125 \sum \text{three skinfolds} + 0.0000055 \sum \text{three skinfolds}^2 - 0.000244 \text{age}$ SEE 0.008 or ~3.6% fat
StartLayout 1st Row Body density equals 1.1125025 minus 0.0013125 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000055 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.000244 left parenthesis a g e right parenthesis left bracket italic upper S upper E upper E italic 0.008 italic or tilde italic 3.6 percent sign italic f a t right bracket EndLayout

Females

- **Seven-site formula** (chest, midaxillary, triceps, subscapular, abdomen, suprailiac, thigh)

Body density = $1.097 - 0.00046971 \sum \text{seven skinfolds} + 0.00000056 \sum \text{seven skinfolds}^2 - 0.00012828 \text{age}$ SEE 0.008 or ~3.8% fat
StartLayout 1st Row Body density equals 1.097 minus 0.00046971 left parenthesis s u m of seven skinfolds right parenthesis 2nd Row plus 0.00000056 left parenthesis s u m of seven skinfolds right parenthesis squared 3rd Row minus 0.00012828 left parenthesis a g e right parenthesis left bracket italic upper S upper E upper E italic 0.008 italic or tilde italic 3.8 percent sign italic f a t right bracket EndLayout

- **Three-site formula** (triceps, suprailiac, thigh)

Body density = $1.0994921 - 0.0009929 \sum \text{three skinfolds} + 0.0000023 \sum \text{three skinfolds}^2 - 0.0001392 \text{age} - \text{SEE} \cdot 0.009$ or $\sim 3.9\% \text{fat}$

1st Row Body density equals 1.0994921 minus 0.0009929 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000023 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.0001392 left parenthesis italic a g e right parenthesis left bracket italic upper S upper E upper E italic 0.009 italic or tilde italic 3.9 percent sign italic f a t right bracket EndLayout

- **Three-site formula** (triceps, suprailiac, abdominal)

Body density = $1.089733 - 0.0009245 \sum \text{three skinfolds} + 0.0000025 \sum \text{three skinfolds}^2 - 0.0000979 \text{age} - \text{SEE} \cdot 0.009$ or $\sim 3.9\% \text{fat}$

1st Row Body density equals 1.089733 minus 0.0009245 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000025 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.0000979 left parenthesis a g e right parenthesis left bracket italic upper S upper E upper E italic 0.009 italic or tilde italic 3.9 percent sign italic f a t right bracket EndLayout

SEE, standard error of estimate.

Adapted from ([65](#), [68](#)).

Densitometry

The estimate of total body fat percentage can be derived from a measurement of whole-body density using the ratio of body mass to body volume. Densitometry has been used as a reference or criterion standard for assessing body composition for many years; although, DXA and multicomponent modeling have recently gained popularity as a criterion measure. The limiting factor in the measurement of body density is the accuracy of the body volume measurement because the measurement of body mass (weight) is considered to be highly accurate. Body volume can be measured by hydrodensitometry (underwater weighing), plethysmography, or calculated using DXA algorithms ([69](#)). However, the DXA method of body volume determination requires additional development and validation ([57](#), [70](#)).

Hydrodensitometry (underwater) weighing is based on Archimedes principle that states when a body is immersed in water, it is buoyed by a

counterforce equal to the weight of the water displaced. This loss of weight in water allows for calculation of body volume. Bone and muscle tissues are denser than water, whereas fat tissue is less dense. Therefore, when two individuals have the same total body mass, the person with more fat-free mass (FFM) weighs more in water and has a higher body density and lower percentage of body fat compared to the person with less FFM (FFM = body mass – fat mass [FM]). Although hydrostatic weighing is a standard method for measuring body volume and, hence, body composition, it requires special equipment, the accurate measurement of residual volume, body density conversion formulas, and significant cooperation by the individual (71).

Body volume can also be measured by ADP. Albeit expensive, ADP is well established and is thought to reduce the challenges associated with submersion in water during hydrodensitometry in some individuals (66). ADP uses the same principles as the gold standard method of underwater weighing, but representing a densitometric method that is based on air displacement rather than on water immersion. ADP offers several advantages over established body composition reference methods, including a quick, comfortable, automated, noninvasive, and safe measurement process, and accommodation of various subject types (*e.g.*, children, obese, older adults, and people with disabilities) (72). However, its accuracy declines at the extremes of body fat percentages, tending to slightly understate the percent body fat in overweight and obese persons (by 1.68%–2.94% depending on the method of calculation) and to overstate to a much larger degree the percent body fat in very lean subjects (by an average of 6.8%, with up to a 13% overstatement of the reported body percentage of one individual — *i.e.*, 2% body fat by DXA but 15% by ADP; https://en.wikipedia.org/wiki/Body_fat_percentage#cite_note-LowryTomiyama-11) (73). For a more detailed explanation of these densitometric techniques, see Gibson et al. (66).

Conversion of Body Density to Body Composition

Percent body fat can be estimated once body density has been determined. The most commonly used prediction equation to estimate percent body fat from body density was derived from the two-component model of body composition ([69](#)):

$$4.95/D_b - 4.50 \times 100$$

The prediction of body fat from body density assumes the density of FM and FFM are consistent for the studied population. However, age, sex, race, training status, and certain disease states may affect the density of FFM, with much of this variance related to the bone mineral density component of FFM. Because of this variance, population-specific, two-component model conversion formulas are also available for specific age, sex, race, training status, and disease condition ([Table 3.3](#)). Because of the significant effect of these factors on the validity of the conversion of body density to body fat, exercise professionals are encouraged to select the most specific formula possible for each individual ([66](#)).

TABLE 3.3 Population-Specific Formulas for Conversion of Body Density to Percent Body Fat

	Population	Age	Sex	%BF ^a	FFBd ^b (g · cm ⁻³)
Ethnicity	African American	9–17	Females	(5.24/Db) – 4.82	1.088
		19–45	Males	(4.85/Db) – 4.39	1.106
		24–79	Females	(4.86/Db) – 4.39	1.106
	American Indian	18–62	Males	(4.97/Db) – 4.52	1.099
		18–60	Females	(4.81/Db) – 4.34	1.108
	Asian Japanese Native	18–48	Males	(4.97/Db) – 4.52	1.099
			Females	(4.76/Db) – 4.28	1.111
		61–78	Males	(4.87/Db) – 4.41	1.105
			Females	(4.95/Db) – 4.50	1.1
	Singaporean (Chinese, Indian, Malay)		Males	(4.94/Db) – 4.48	1.102
			Females	(4.84/Db) – 4.37	1.107
	Caucasian	8–12	Males	(5.27/Db) – 4.85	1.086
			Females	(5.27/Db) – 4.85	1.086

	Population	Age	Sex	%BF ^a	FFBd ^b (g · cm ⁻³)
		13–17	Males	(5.12/Db) – 4.69	1.092
			Females	(5.19/Db) – 4.76	1.09
		18–59	Males	(4.95/Db) – 4.50	1.1
			Females	(4.96/Db) – 4.51	1.101
		60–90	Males	(4.97/Db) – 4.52	1.099
			Females	(5.02/Db) – 4.57	1.098
	Hispanic		Males	NA	NA
		20–40	Females	(4.87/Db) – 4.41	1.105
Athletes	Resistance trained	24 ± 4	Males	(5.21/Db) – 4.78	1.089
		35 ± 6	Females	(4.97/Db) – 4.52	1.099
	Endurance trained	21 ± 2	Males	(5.03/Db) – 4.59	1.097
		21 ± 4	Females	(4.95/Db) – 4.50	1.1
	All sports	18–22	Males	(5.12/Db) – 4.68	1.093
		18–22	Females	(4.97/Db) – 4.52	1.099

	Population	Age	Sex	%BF ^a	FFBd ^b (g · cm ⁻³)
Clinical populations ^c	Anorexia nervosa	15–44	Females	(4.96/Db) – 4.51	1.101
	Cirrhosis				
	Childs A			(5.33/Db) – 4.91	1.084
	Childs B			(5.48/Db) – 5.08	1.078
	Childs C			(5.69/Db) – 5.32	1.07
	Obesity	17–62	Females	(4.95/Db) – 4.50	1.1
	Spinal cord injury (paraplegic/quadruplegic)	18–73	Males	(4.67/Db) – 4.18	1.116
		18–73	Females	(4.70/Db) – 4.22	1.114

^a Multiply by 100 for percentage of body fat.

^b FFBd, fat-free body density based on average values reported in selected research articles.

^c There are insufficient multicomponent model data to estimate the average FFBd of the following clinical populations: coronary artery disease, heart/lung transplants, chronic obstructive pulmonary disease, cystic fibrosis, diabetes mellitus, thyroid disease, human immunodeficiency virus/acquired immunodeficiency syndrome, cancer, kidney failure (dialysis), multiple sclerosis, and muscular dystrophy. %BF, percentage of body fat; Db, body density; NA, no data available for this population subgroup.

Adapted with permission from (24).

Other Techniques

DXA is now a common body composition assessment method used in clinical research settings but has limited applicability in routine health/fitness testing because of cost, specialized equipment, and the need for highly

trained personnel ([23](#)). To estimate % body fat from DXA, a total-body x-ray is performed with two extremely low-level energy beams. As the x-rays pass through the subject, the density of each different body area is determined. DXA can measure fat and muscle distribution in localized areas of the body (*e.g.*, the trunk and extremities). Although radiation exposure is a concern, the exposure from a typical scan is minimal, being less than a cross continental airplane flight ([60](#), [74](#)). Prediction of the deleterious visceral adipose tissue volume, as correlated with a computed tomography (CT) criterion, is reported to be similar for anthropometric (*e.g.*, waist circumference and BMI) and standard DXA abdominal fat measurements ([75](#)). Sources of error for DXA include incorrect body positioning, hydration status, disagreement between the subject's measured body weight and the DXA weight, the variance of DXA estimation formulas between manufacturers, food intake, and the limitation of being unable to assess the % fat of individuals with a very large body mass ([60](#)). The reported prediction error with this technique has been reported to be 1.2%–4.8% ([60](#)).

Bioelectrical impedance analysis (BIA) has now gained wide acceptance in the fitness industry as an assessment technique for body fat analysis in routine health/fitness testing settings. This technique is a simple, quick, noninvasive method that is based on the principal that as adipose tissue contains little water, fat impedes the flow of electrical current. Conversely, muscle conducts electrical current very well. As the electrical current passes throughout the body, the voltage decreases. This voltage drop (impedance) is used to calculate % body fat. Typically, other information such as sex, height, age, and ethnicity are programmed into the BIA device to more accurately predict % body fat ([74](#)). Generally, the accuracy of BIA is similar to skinfolds, as long as stringent protocol adherence (*e.g.*, assurance of normal hydration status) is followed, and the equations programmed into the analyzer are valid for the populations being tested. For example, differences in BIA

accuracy between obese and normal weight individuals may be due to how body water is distributed in these groups. Reliance on the results from empirical linear regression models is a limitation of both single frequency and multifrequency BIA ([31](#), [76](#)).

Ultrasound uses sound waves to provide a noninvasive, direct measure of subcutaneous fat thickness, as opposed to skinfold calipers that yield an indirect estimate of fat thickness from a compressed fold of fat and skin. Ultrasound devices vary in cost and complexity, but proprietary prediction equations from inexpensive devices have been validated against skinfolds, DXA, and ADP ([77](#), [78](#)). Intertester reliability is higher for ultrasound compared to skinfolds ([79](#)). Smartphone applications that provide estimates of body fat percentage from photos are also available, but they lack rigorous independent validation at this point.

CT and magnetic resonance imaging (MRI) are also imaging techniques that more accurately identify the amount of fat (particularly deep fat) found in the trunk. As deep visceral fat has a strong association with CVD, these two techniques can more accurately quantify fat distribution patterns. CT scans use X-rays to generate images of fat and fat-free tissues, while MRI scans use a strong magnetic field to produce images of regional body fat. These two methods require expensive equipment used only in clinical settings.

Body Composition Norms

There are no universally accepted norms for body composition; however, [Tables 3.4](#) and [3.5](#), which were developed using skinfold measurements, provide percentile values for percent body fat in males and females, respectively. A consensus opinion for an exact percent body fat value associated with optimal health risk has yet to be defined. However, based on these skinfold reference values, the range of 12%–23% and 17%–26% for males and females, respectively, spans the “good” category of body fat values across a wide age spectrum (see [Tables 3.4](#) and [3.5](#)). Other research supports

this range, although age, sex, race, and athletic level impact what may be construed as a healthy percent body fat. Method of assessment also influences results and further complicates the definition of healthy body fat percentages. DXA percent body fat ranges for a Caucasian adult sample from the United States ([80](#)) are higher (19%–26% and 28%–36%, respectively, for males and females) compared to the same “good” category percentiles in [Tables 3.4](#) and [3.5](#). Sex- and age-specific lean mass data derived from DXA scans ([81](#)) are also available.

TABLE 3.4 Fitness Categories for Body Composition (% Body Fat) for Males by Age

		Age (yr)					
%		20-29	30-39	40-49	50-59	60-69	70-79
99	Very lean ^a	4.2	7.3	9.5	11.1	12.0	13.6
95		6.4	10.3	13.0	14.9	16.1	15.5
90	Excellent	7.9	12.5	15.0	17.0	18.1	17.5
85		9.1	13.8	16.4	18.3	19.2	19.0
80		10.5	14.9	17.5	19.4	20.2	20.2
75	Good	11.5	15.9	18.5	20.2	21.0	21.1
70		12.6	16.8	19.3	21.0	21.7	21.6
65		13.8	17.7	20.1	21.7	22.4	22.3
60		14.8	18.4	20.8	22.3	23.0	22.9
55	Fair	15.8	19.2	21.4	23.0	23.6	23.6
50		16.7	20.0	22.1	23.6	24.2	24.1
45		17.5	20.7	22.8	24.2	24.9	24.5
40		18.6	21.6	23.5	24.9	25.6	25.2
35	Poor	19.8	22.4	24.2	25.6	26.4	25.7
30		20.7	23.2	24.9	26.3	27.0	26.3
25		22.1	24.1	25.7	27.1	27.9	27.1
20		23.3	25.1	26.6	28.1	28.8	28.0
15	Very poor	25.1	26.4	27.7	29.2	29.8	29.3
10		26.6	27.8	29.1	30.6	31.2	30.6
5		29.3	30.2	31.2	32.7	33.5	32.9
1		33.7	34.4	35.2	36.4	37.2	37.3
<i>n</i>		1,938	10,457	16,032	9,976	3,097	571

^a Very lean, no less than 3% body fat is recommended for males.

Total $n = 42,071$.

Adapted with permission from Physical Fitness Assessments and Norms for Adults and Law Enforcement.

Dallas (TX): The Cooper Institute; 2013. For more information: <http://www.cooperinstitute.org>.

TABLE 3.5 Fitness Categories for Body Composition (% Body Fat) for Females by Age

		Age (yr)					
%		20-29	30-39	40-49	50-59	60-69	70-79
99	Very lean ^a	11.4	11.0	11.7	13.8	13.8	13.7
95		14.1	13.8	15.2	16.9	17.7	16.4
90	Excellent	15.2	15.5	16.8	19.1	20.1	18.8
85		16.1	16.5	18.2	20.8	22.0	21.2
80	Good	16.8	17.5	19.5	22.3	23.2	22.6
75		17.7	18.3	20.5	23.5	24.5	23.7
70		18.6	19.2	21.6	24.7	25.5	24.5
65		19.2	20.1	22.6	25.7	26.6	25.4
60		20.0	21.0	23.6	26.6	27.5	26.3
55	Fair	20.7	22.0	24.6	27.4	28.3	27.1
50		21.8	22.9	25.5	28.3	29.2	27.8
45		22.6	23.7	26.4	29.2	30.1	28.6
40		23.5	24.8	27.4	30.0	30.8	30.0
35	Poor	24.4	25.8	28.3	30.7	31.5	30.9
30		25.7	26.9	29.5	31.7	32.5	31.6
25		26.9	28.1	30.7	32.8	33.3	32.6
20		28.6	29.6	31.9	33.8	34.4	33.6
15	Very poor	30.9	31.4	33.4	34.9	35.4	35.0
10		33.8	33.6	35.0	36.0	36.6	36.1
5		36.6	36.2	37.0	37.4	38.1	37.5
1		38.4	39.0	39.0	39.8	40.3	40.0
<i>n</i>		1,342	4,376	6,392	4,496	1,576	325

^a Very lean, no less than 10%–13% body fat is recommended for females.

Total n = 18,507.

Adapted with permission from Physical Fitness Assessments and Norms for Adults and Law Enforcement.

Dallas (TX): The Cooper Institute; 2013. For more information: <http://www.cooperinstitute.org>.

CARDIORESPIRATORY FITNESS

CRF, estimated by peak oxygen consumption ($\dot{V}O_{2\max}$), is related to the ability to perform large muscle, dynamic, submaximal intensity exercise for prolonged periods of time. Performance of exercise at this level of physical exertion depends on the integrated physiologic and functional state of the respiratory, cardiovascular, and musculoskeletal systems. CRF is considered a health-related component of physical fitness because (a) low levels of CRF have been associated with a markedly increased risk of premature death from all causes and specifically from CVD; (b) increases in CRF are associated with a reduction in death from all causes; and (c) high levels of CRF are associated with higher levels of habitual PA, which in turn are associated with many health benefits ([82](#)). The American Heart Association (AHA) recognizes CRF as a clinical vital sign and potentially a stronger predictor of mortality than other established risk factors in adults across the health and disease spectrum ([82](#)). As such, the assessment of CRF is an important part of any primary or secondary prevention and rehabilitation program, and the knowledge and skills to complete the assessment and interpret the subsequent results are an important responsibility of the exercise professional.

The Concept of Maximal Oxygen Uptake

Maximal volume of oxygen consumed per unit time $\dot{V}O_{2\max}$ is accepted as the criterion measure of CRF. This variable is typically expressed clinically in relative ($\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) as opposed to absolute ($\text{mL} \cdot \text{min}^{-1}$) terms, allowing for meaningful comparisons between/among individuals with differing body habitus. $\dot{V}O_{2\max}$ is the product of the maximal cardiac output ($\text{SV} \cdot \text{HR} = \dot{Q}$; $\text{L blood} \cdot \text{min}^{-1}$) and arterial-venous oxygen difference ($\text{mL O}_2 \cdot \text{L blood}^{-1}$). In apparently healthy individuals, $\dot{V}O_{2\max}$ is primarily limited by

the capacity of the cardiorespiratory system to deliver oxygenated blood to the exercising tissues thus highlighting the important role of cardiac function for determining CRF (83). The designation of $\dot{V}O_{2\max}$ implies an individual's true physiologic limit has been reached, and a plateau in volume of oxygen consumed per minute ($\dot{V}O_2$) may be observed between the final two work rates of a progressive exercise test. This plateau is not consistently observed during maximal exercise testing, and the endpoint criteria chosen to designate maximal exertion impacts the ultimate $\dot{V}O_{2\max}$ value (84). A verification bout of exercise at a workload at 105%–110% of the highest workload attained in the maximal exertion exercise test is an alternative to the reliance on secondary criteria historically used to verify achievement of $\dot{V}O_{2\max}$ (85, 86). The absence of a verification protocol or in specific populations such as individuals with CVD or pulmonary disease where a plateau infrequently occurs (87), peak $\dot{V}O_2$ ($\dot{V}O_{2\text{peak}}$) may be the preferred term although acceptance of the $\dot{V}O_{2\text{peak}}$ term as a descriptor of CRF is equivocal, particularly in clinical populations (86, 88). A plateau in HR (≤ 2 beats $\cdot$ min⁻¹) in the final minute of data collection is reported to hold promise as a singular method for confirming $\dot{V}O_{2\max}$ without need for a verification trial (89).

Cardiopulmonary exercise testing (CPET) incorporating open circuit spirometry during a graded incremental or ramp exercise test to exhaustion is the gold standard measure of $\dot{V}O_{2\max}$. In this procedure, the individual breathes either through a low-resistance valve with their nose occluded or breaths normally without restriction through a nonlatex mask, while ventilation and expired fractions of oxygen (O_2) and carbon dioxide (CO_2) are measured. In addition, the use of open circuit spirometry during maximal exercise testing may allow for the accurate assessment of ventilatory thresholds and direct measurement of $\dot{V}O_{2\max}/\dot{V}O_{2\text{peak}}$. CPET also provides objective measures (*i.e.*, respiratory exchange ratio) that can help determine if $\dot{V}O_{2\max}$ was achieved. Automated systems are typically used to

collect these data; however, system calibration and trained personnel are essential to obtain accurate results ([90, 91](#)). The mode selected (*e.g.*, leg ergometer, treadmill) for the exercise test can impact the result, as performance is related to familiarity with the exercise modality as well as the amount of muscle mass involved (see [Chapter 4](#)). Administration of the test and interpretation of results should be reserved for trained professional personnel with a thorough understanding of exercise physiology. Because of costs associated with the equipment, space, and personnel needed to carry out these tests, direct measurement of $\dot{V}O_{2\max}$ may not always be possible. Several equations predict $\dot{V}O_{2\max}$ from submaximal $\dot{V}O_2$ values acquired via direct measurements and produce results that do not differ significantly from the measured $\dot{V}O_{2\max}$ ([92](#)). Although these prediction protocols may reduce the time and risk of a maximal exertion exercise test, they still require direct measurement of O_2 consumption.

When direct measurement of $\dot{V}O_{2\max}$ is not feasible, $\dot{V}O_{2\max}$ can be estimated from prediction equations or a variety of maximal and submaximal exercise tests that do not use spirometry. These tests have been validated by examining (a) the correlation between directly measured $\dot{V}O_{2\max}$ and the $\dot{V}O_{2\max}$ estimated from physiologic responses to submaximal exercise (*e.g.*, HR at a specified power output) or (b) the correlation between directly measured $\dot{V}O_{2\max}$ and field test performance (*e.g.*, time to run 1 or 1.5 mi [1.6 or 2.4 km]) or time to volitional fatigue using a standard graded exercise test (GXT) protocol. It should be noted that there is the potential for a significant underestimation or overestimation of $\dot{V}O_{2\max}$ by these types of indirect measurement techniques. Overestimation is more likely to occur when the exercise protocol (see [Chapter 4](#)) chosen for testing is too aggressive for a given individual (*i.e.*, Bruce treadmill protocol in individuals with CHF) or when an individual heavily relies on treadmill handrail support ([87](#)). Every effort must be taken to select the appropriate exercise protocol

given an individual's characteristics, and handrail use should be minimized during testing on a treadmill as much as possible and at all times ([93](#)).

Maximal Versus Submaximal Exercise Testing

The decision to use a maximal or submaximal exercise test depends largely on the reasons for the test, risk level of the individual, and availability of appropriate equipment and personnel (see [Chapter 4](#)). Maximal tests require an individual to exercise to the point of volitional fatigue, which may be inappropriate for some individuals and may require the need for emergency equipment to be available ([94](#)). The AHA identified competencies, roles, and responsibilities of GXT team members that limit the risk of adverse events from happening during exercise test administration ([95](#)).

Exercise professionals often rely on submaximal exercise tests to assess CRF because maximal exercise testing is not always feasible in the health/fitness setting. The primary purpose of submaximal exercise testing is to determine the HR response to one or more submaximal work rates and use the results to predict $\dot{V}O_{2\max}$; however, it is important to obtain additional indices of the individual's response to exercise. The exercise professional should use the various submaximal measures of HR, BP, workload, rating of perceived exertion (RPE), and other individual indices as valuable information regarding one's functional response to exercise. This information can be used to evaluate submaximal exercise responses over time, in a controlled environment, and make modifications to the Ex R_x as needed.

The most accurate estimate of $\dot{V}O_{2\max}$ is achieved from the HR response to submaximal exercise tests based on the following assumptions ([66](#)):

- A steady state HR (two consecutive stages with HRs within 5 beats of each other) is obtained for each exercise work rate.

- A linear relationship exists between HR and work rate.
- The difference between actual and predicted maximal HR is minimal.
- Mechanical efficiency (*i.e.*, $\dot{V}O_2$ at a given work rate) is the same for everyone.
- The individual is not on any HR-altering medications (see [Appendix A](#)).
- The individual is not using high quantities of caffeine, ill, or in a high-temperature environment, all of which may alter the HR response.

Cardiorespiratory Fitness Test Sequence and Measures

After the initial screening process, baseline measurements of HR and BP should be obtained prior to the start of the exercise test. A minimum of HR, BP, and RPE should be measured during exercise tests. Monitoring exercise HR via ECG or HR monitor is most common. In situations where an HR monitor or ECG is unavailable, it may be necessary to palpate or auscultate HR using 10-s, 15-s, or 30-s intervals with conversion to beats per minute. Most protocols that use postexercise HR to assess CRF also use these shorter time intervals due to the rapid and immediate decline in HR following the cessation of exercise. HR telemetry monitors with chest electrodes or radio telemetry have proven to be accurate and reliable, provided there is no outside electrical interference ([96](#)). Light-based technology based on photoplethysmography is now integrated into wearable devices and monitors pulsatile blood flow at the nail bed, wrist, or forearm. Proprietary algorithms convert the blood flow to HR with varying degrees of accuracy, adequately measuring HR in laboratory-based activities ([97–100](#)).

BP should be measured at heart level with the individual in the exercise position (*i.e.*, standing or seated) and individual's arm relaxed and supported and not grasping a handrail (treadmill) or handlebar (cycle ergometer). Automated brachial cuffs typically allow for auditory confirmation of the BP measurement, which may improve confidence in the resting BP value

obtained. Use of automated brachial or finger cuffs during exercise is discouraged, as they are susceptible to erroneous results attributable to artifacts ([101](#)). To obtain accurate BP measures during exercise, follow the stethoscope placement and cuff inflation and deflation guidelines in [Box 3.1](#). If an automated BP system is used during exercise testing, calibration checks with manual BP measurements must be routinely performed to confirm accuracy of the automated readings ([101](#)). Systolic BP (SBP) and diastolic BP (DBP) measurements should be used as indicators for stopping an exercise test ([Box 3.6](#)).

Box 3.6 General Indications for Stopping an Exercise Test^{*}

- Onset of angina or anginalike symptoms
- Drop in SBP ≥ 10 mm Hg with an increase in work rate or if SBP decreases below the value obtained in the same position prior to testing
- Excessive rise in BP: systolic pressure greater than 250 mm Hg and/or diastolic pressure greater than 115 mm Hg
- Shortness of breath, wheezing, leg cramps, or claudication
- Signs of poor perfusion: light-headedness, confusion, ataxia, pallor, cyanosis, nausea, or cold and clammy skin
- Failure of HR to increase with increased exercise intensity
- Noticeable change in heart rhythm by palpation or auscultation
- Individual requests to stop
- Physical or verbal manifestations of severe fatigue
- Failure of the testing equipment

^{*} Assumes that testing is nondiagnostic and is being performed without electrocardiogram monitoring. For clinical testing, *Box 4.3* provides more definitive and specific termination criteria. BP, blood pressure; HR, heart rate; SBP, systolic blood pressure.

RPE can be a valuable indicator for monitoring an individual's exercise tolerance, as supplemental to other tools or alone in cases where other tools are not available. The RPE scale was developed to allow the exerciser to individually rate their physical strain during exercise ([102](#)). An individual's perception of exertion can be influenced by a multitude of factors such as personal health and exercise history, demographics, and testing environment. Therefore, exercise professionals should control as many exercise-related variables as possible and refrain from comparing RPE responses across modalities and individuals. The correlation between RPE and indicators of exercise intensity is strong, but interindividual variability of HR and blood lactate at specific RPEs is high ([103](#)). The relationship between RPE and objective measures of exercise intensity (HR and blood lactate) is

reported as being independent of PA level, exercise modality, age, sex, and medical history of coronary artery disease ([103](#)). The exercise professional's explanation of how to use the RPE scale is of utmost importance in helping an individual convey their own assessment of the level of exertion ([102](#), [104](#)). Two RPE scales are widely used: (a) the original Borg or category scale, which rates exercise intensity from 6 to 20 ([Table 3.6](#)), and (b) the category-ratio scale of 0–10 (see [Figure 4.2](#)). OMNI 0–10 scales with pictorial representations of exertion are also available and appropriate for various age groups, exercise modalities, and exercise intensities between 50% and 70% $\dot{V}O_2$ reserve [$\% \times (\dot{V}O_{2\text{max}} - \dot{V}O_2 \text{ at rest}) + \dot{V}O_2 \text{ at rest}$] ([106–109](#)).

TABLE 3.6 The Borg Rating of Perceived Exertion Scale

6	No exertion at all
7	Extremely light
8	
9	Very light
10	
11	Light
12	
13	Somewhat hard
14	
15	Hard (heavy)
16	
17	Very hard
18	
19	Extremely hard
20	Maximal exertion

© Gunnar Borg. Reproduced with permission ([105](#)). The scale with correct instructions can be obtained from Borg Perception, Radisvagen 124, 16573 Hasselby, Sweden. See also the home page: <http://www.borgperception.se>.

During exercise testing, the RPE can be used as an indication of impending fatigue. Most apparently healthy individuals reach their limit of fatigue at an RPE of 18–19 out of 20 (very, very hard) on the category Borg scale, or 9–10 out of 10 (very, very strong) on the category-ratio scale; therefore, RPE can be used to monitor progress toward maximal exertion during exercise testing. Reference standards for peak RPE across age, sex, and test modality have recently been published ([110](#)).

Test Termination Criteria

GXT, whether maximal or submaximal, is a safe procedure when individual prescreening and testing guidelines are adhered to and when the GXT is administered by trained exercise professionals ([95](#)). Occasionally, for safety reasons, the test may have to be terminated prior to the individual reaching a measured or estimated $\dot{V}O_{2\max}$, volitional fatigue, or a predetermined endpoint (*i.e.*, 50%–70% heart rate reserve [HRR] or 70%–85% age-predicted maximal heart rate [$HR_{\max}$]). Because of the individual variation in $HR_{\max}$, the upper limit of 85% of an estimated $HR_{\max}$ may result in a maximal effort for some individuals and submaximal effort in others. Thus, maximal effort should be determined by a compilation of objective and subjective markers. General indications for stopping an exercise test are outlined in [Box 3.6](#).

Modes of Testing

Commonly used modes for exercise testing include treadmills, cycle ergometers, and field tests. The mode of exercise testing used is dependent

on the setting, equipment available, and training of personnel. There are advantages and disadvantages of each exercise testing mode:

- **Treadmills** can be used for submaximal and maximal testing and are often employed for diagnostic testing. Treadmills provide a familiar form of exercise to many and, if the correct protocol is chosen (*i.e.*, aggressive vs. conservative adjustments in workload), can accommodate individuals across the fitness continuum of walking to running speeds. Nevertheless, a familiarization session might be necessary in some cases to permit habituation and reduce anxiety. On the other hand, treadmills are fairly expensive, and some measurements (*e.g.*, BP, ECG) become more difficult at higher speeds, particularly while running. Treadmills must also be calibrated periodically to ensure the accuracy of the test when estimating $\dot{V}O_{2\max}$. In addition, holding on to the support rail(s) should be discouraged to ensure accuracy of metabolic work output. Lack of calibration and extensive handrail use are among factors leading to errors in predictions of $\dot{V}O_{2\max}$ ([111](#)).
- **Mechanically braked cycle ergometers** are also a viable test modality for submaximal and maximal testing and are frequently used for diagnostic testing ([93](#)). Advantages of this testing mode include lower expense, transportability, and greater ease in obtaining stable BP and ECG (if appropriate) measurements. Cycle ergometers also provide a non-weight-bearing test modality in which work rates are easily adjusted in small increments. The main disadvantage is cycling may be a less familiar mode of exercise to some individuals, often resulting in localized muscle fatigue and an underestimation of $\dot{V}O_{2\max}$. The cycle ergometer must be calibrated, and the individual must maintain the proper pedal rate because most tests require HR to be measured at specific work rates ([93](#)). Electronic cycle ergometers can deliver the same work rate across a range of pedal rates (*i.e.*, revolutions per minute [rpm]), but calibration might require special equipment not available in some laboratories. If a cycle

ergometer cannot be calibrated for any reason or if it does not provide a reasonable estimate of workload, it should not be used for fitness testing to predict CRF.

- **Field tests** are those that are conducted outside of a laboratory to predict CRF by measuring HR response, duration of test, or distance travelled, depending on the protocol. These tests may consist of walking, running, and/or stepping, following predetermined protocols. Ease of administration to large numbers of individuals at one time, the relatively low skill needed to complete the tests, low cost, and little equipment needed (*e.g.*, a stopwatch) are the main advantages. Additionally, these tests can be implemented in rehabilitation settings as well as with general populations. However, there are several disadvantages, including the inability to control individual effort, identify test termination criteria (see [Box 3.6](#)), and monitor BP and HR responses during the test. Additionally, an individual may not be able to complete the test, as some tests can be near-maximal or maximal intensity for some, particularly in individuals with low CRF. Therefore, these tests may be inappropriate for sedentary individuals or those at increased risk for cardiovascular and/or musculoskeletal complications. Many factors can have a profound impact on test results, including an individual's sex, training status, age, test procedures, and required distance ([112](#)).

Treadmill Tests

The primary exercise modality for submaximal exercise testing traditionally has been the cycle ergometer, largely due to the ability to tightly control workload, although treadmills are used in many settings. Similar to submaximal cycling protocols, submaximal treadmill tests use the same submaximal definition (70% HRR or 85% of age-predicted HR_{max}) as the HR-based test termination criterion, and the stages of the test should be 3 minutes or longer to ensure a steady state HR response at each stage. The HR

values are extrapolated to age-predicted HR_{max} , and $\dot{V}O_{2max}$ is estimated using the formula in [Appendix D](#) from the highest speed and/or grade that would have been achieved if the individual had worked to maximum. The most common treadmill protocols are presented in [Figure 4.1](#).

Cycle Ergometer Tests

The Åstrand-Rhyming cycle ergometer test is a single-stage test lasting 6 minutes ([113](#)). The pedal rate is set at 50 rpm. The goal is to obtain HR values between 125 and 170 beats $\cdot$ min⁻¹, with HR measured during the 5th and 6th minute of work. Due to the HR requirements, this test may not be suitable in HR-altering medications such as β -blockers. The average of the two HRs is then used to estimate $\dot{V}O_{2max}$ from a nomogram ([Figure 3.1](#)). The suggested work rate is based on sex and an individual's fitness status as follows:

Pulse rate
Male Female

170
166
162
158
154
150
146
142
138
134
130
126
122
128
124
120

$\dot{V}O_{2max}$, L

1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0
3.2
3.4
3.6
3.8
4.0
4.2
4.4
4.6
4.8
5.0
5.2
5.4
5.6
6.0

Step test
33 cm 40 cm
Female Male
(weight, kg)

40
50
60
70
80
90

$\dot{V}O_2$, L

Work load
(kg/min)
Female Male

0.8
0.9
1.0
1.1
1.2
1.3
1.4
1.5
1.6
1.7
1.8
1.9
2.0
2.1
2.2
2.3
2.4
2.5
2.6
2.7
2.8
2.9
3.0
3.1
3.2
3.3
3.4
3.5
300
450
600
750
900
1.050
1.200
1.500

Figure 3.1 Modified Åstrand-Rhyming nomogram. Used with permission from ([114](#)).

- Males, unconditioned: 300 or 600 kg · m · min⁻¹ (50 or 100 W)
- Males, conditioned: 600 or 900 kg · m · min⁻¹ (100 or 150 W)
- Females, unconditioned: 300 or 450 kg · m · min⁻¹ (50 or 75 W)
- Females, conditioned: 450 or 600 kg · m · min⁻¹ (75 or 100 W)

Because HR_{max} decreases with age, the value from the nomogram must be adjusted for age by multiplying the $\dot{V}O_{2\max}$ value by the following correction factors ([113](#)):

Age	Correction Factor
15	1.10
25	1.00
35	0.87
40	0.83
45	0.78
50	0.75
55	0.71
60	0.68
65	0.65

In contrast to the Åstrand-Rhyming cycle ergometer single-stage test, Maritz et al. ([115](#)) devised a test where HR was measured at a series of submaximal work rates and extrapolated the HR response to the individual's age-predicted HR_{max}. This multistage method is a well-known assessment technique to estimate $\dot{V}O_{2\max}$. The HR measured during the last minute of two steady state stages is plotted against work rate. The line generated from the plotted points is then extrapolated to the age-predicted HR_{max}, and a perpendicular line is dropped to the x-axis to estimate the work rate that

would have been achieved if the individual had worked to maximum. As with all tests that are based on HR response, special consideration should be made for individuals on HR-altering medications or conditions that may influence the true maximal HR. HR measurements below $110 \text{ beats} \cdot \text{min}^{-1}$ should not be used to estimate $\dot{V}O_{2\text{max}}$ because there is more day-to-day and individual variability at lower HR levels, which reduces the accuracy of prediction, and submaximal exercise tests are terminated if an individual reaches 70% HRR (85% HR_{max}). Therefore, two consecutive HR measurements between $110 \text{ beats} \cdot \text{min}^{-1}$ and 70% HRR (85% HR_{max}) should be obtained to predict $\dot{V}O_{2\text{max}}$ using this method.

[Figure 3.2](#) presents an example of graphing the HR response to two submaximal workloads to estimate $\dot{V}O_{2\text{max}}$. The two lines noted as ± 1 standard deviation (SD) show what the estimated $\dot{V}O_{2\text{max}}$ would be if the individual's true HR_{max} were 168 or 192 $\text{beats} \cdot \text{min}^{-1}$, rather than 180 $\text{beats} \cdot \text{min}^{-1}$. $\dot{V}O_{2\text{max}}$ is estimated from the work rate using the formula in [Appendix D](#). This equation is valid to estimate $\dot{V}O_2$ at submaximal steady state workloads (from 300 to 1,200 $\text{kg} \cdot \text{m} \cdot \text{min}^{-1}$) (50–200 W); therefore, caution must be used if extrapolating to workloads outside this range. However, a larger part of the error involved in estimating $\dot{V}O_{2\text{max}}$ from submaximal HR responses occurs as the result of estimating HR_{max} (see [Table 5.3](#)) (116, 117). Accurate submaximal HR recording is also critical, as extrapolation magnifies even the smallest of errors. In addition, errors can be attributed to inaccurate pedaling cadence (workload), imprecise achievement of steady state HR, and the extrapolation of work rate to $\dot{V}O_2$ at maximal intensities. Finally, the test administrator should recognize the error associated with age-predicted HR_{max} (see [Table 5.3](#)) and should monitor the individual throughout the test to ensure that the test remains submaximal.

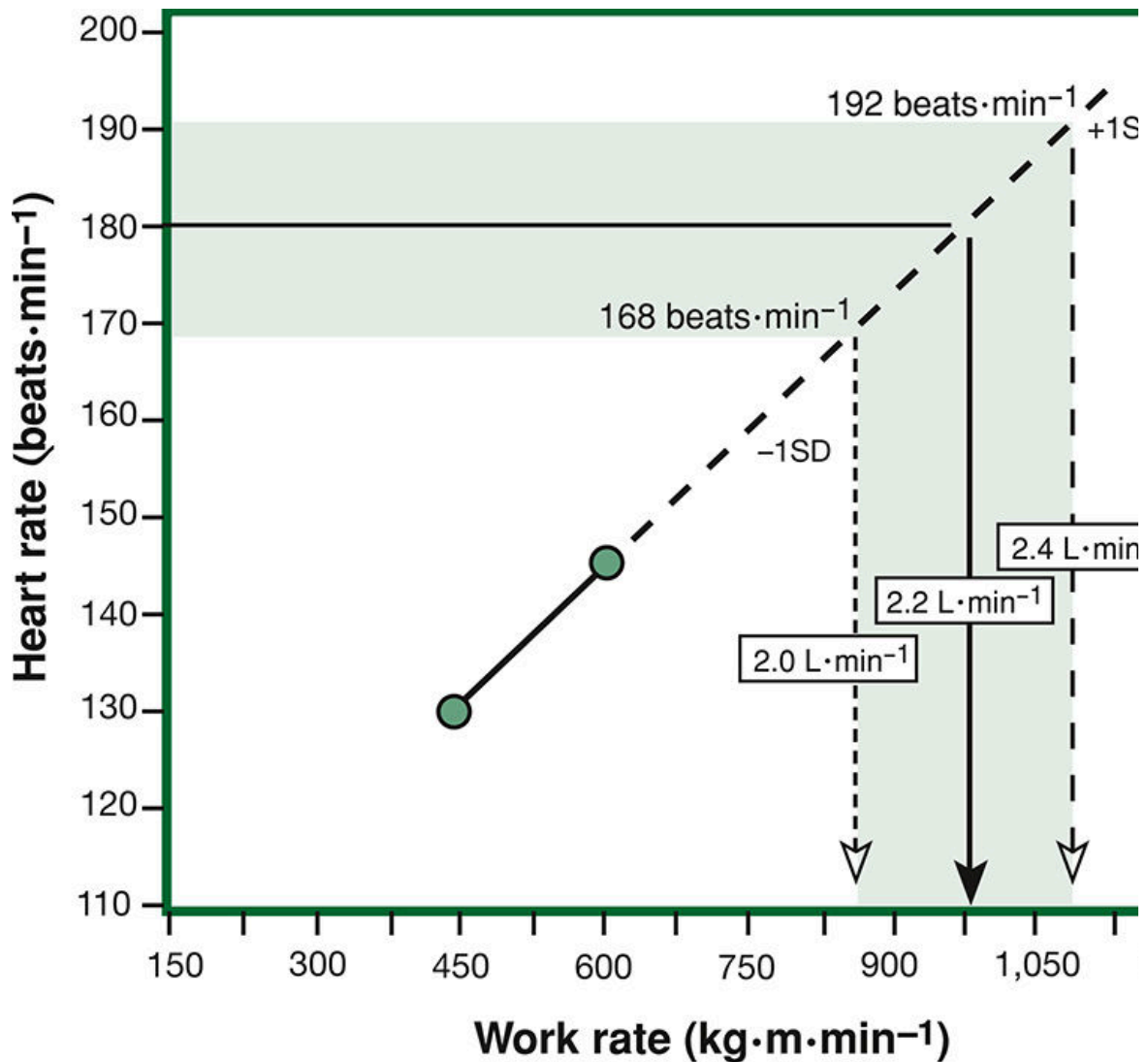


Figure 3.2 Heart rate (HR) responses to two submaximal work rates for a sedentary female 40 years of age weighing 64 kg. Maximal workload was estimated by extrapolating the HR response to the age-predicted maximal heart rate (HR_{max}) of $180 \text{ beats} \cdot \text{min}^{-1}$ (based on $220 - \text{age}$). The work rate that would have been achieved at that HR was determined by dropping a line from that HR value to the x-axis. The other two lines estimate what the maximal workload would have been if the individual's true HR_{max} was ± 1 standard deviation (SD) from the $180 \text{ beats} \cdot \text{min}^{-1}$ value. $\dot{V}O_{2max}$ estimated using the formula in [Table 5.2](#) and expressed in $\text{L} \cdot \text{min}^{-1}$ was $2.2 \text{ L} \cdot \text{min}^{-1}$.

The modified YMCA protocol is a good example of a multistage submaximal cycle ergometer test that uses two to four 3-minute stages of continuous exercise with a constant pedal rate of 50 rpm on a Monark cycle ergometer ([118](#)). Stage 1 requires individuals to pedal against 0.5 kg of resistance (25 W; 150 kgm · min⁻¹). The workload for stage 2 is based on the steady state HR measured during the last minute of the initial stage:

- HR less than 80 beats · min⁻¹ — change the resistance to 2.5 kg (125 W; 750 kgm · min⁻¹)
- HR 80–89 beats · min⁻¹ — change the resistance to 2.0 kg (100 W; 600 kgm · min⁻¹)
- HR 90–100 beats · min⁻¹ — change the resistance to 1.5 kg (75 W; 450 kgm · min⁻¹)
- HR greater than 100 beats · min⁻¹ — change the resistance to 1.0 kg (50 W; 300 kgm · min⁻¹)

Use stages 3 and 4 as needed to elicit two consecutive steady state HRs between 110 beats · min⁻¹ and 70% HRR (85% HR_{max}). For stages 3 and 4, the resistance used in stage 2 is increased by 0.5 kg (25 W; 150 kgm · min⁻¹) per stage. Normative tables for the YMCA protocol are published elsewhere ([118](#)).

Field Tests

Two of the most widely used run/walk tests (individuals may run, walk, or use a combination of both to complete the test) for assessing CRF are the 1.5-mi (2.4 km) test for time and the Cooper 12-minute test. The objective of the 1.5-mi (2.4 km) test is to cover the distance in the shortest period of time, whereas the Cooper 12-minute test requires the individual to cover the greatest distance in the allotted time period. $\dot{V}O_2\text{max}$ can be estimated from using the following equations:

1.5-mi run/walk test

$\dot{V}O_{2\max} \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 3.5 + 483 / 1.5 \text{ min}$
 Modifying Above normal upper V With' normal upper O Subscript 2 max Baseline left parenthesis m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts right parenthesis equals 3.5 plus 483 divided by 1.5 m i time left parenthesis min right parenthesis

12-min walk/run test

$\dot{V}O_{2\max} \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = \text{distance in meters} - 504.9 / 44.73$
 Modifying Above normal upper V With' normal upper O Subscript 2 max Baseline left parenthesis m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts right parenthesis left parenthesis distance in meters negative 504.9 right parenthesis divided by 44.73

The Rockport One-Mile Fitness Walking Test is another well-recognized field test for estimating CRF. In this test, an individual walks 1 mi (1.6 km) as fast as possible, preferably on a track or a level surface, and HR is obtained in the final minute. An alternative is to measure a 10-second HR (multiply by 6 for BPM) immediately on completion of the 1-mi (1.6 km) walk, but this may overestimate the $\dot{V}O_{2\max}$ compared to when HR is measured during the walk. $\dot{V}O_{2\max}$ is estimated using the following regression equation ([119](#)):

$\dot{V}O_{2\max} \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 132.853 - 0.1692 \times \text{body mass in kg} - 0.3877 \times \text{age in years} + 6.315 \times \text{gender} - 3.2649 \times \text{time in minutes} - 0.1565 \times \text{HR}$
 SEE=5.0 mL · kg⁻¹ · min⁻¹ sex=0 for female 1 for male
 StartLayout 1st Row Modifying Above normal upper V With' normal upper O Subscript 2 max Baseline left parenthesis m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts right parenthesis equals 132.853 minus left parenthesis 0.1692 times body mass in k g right parenthesis minus 2nd Row left parenthesis 0.3877 times a g e in years right parenthesis plus left parenthesis 6.315 times gender right parenthesis minus left parenthesis 3.2649 times time in minutes right parenthesis minus 3rd Row left parenthesis 0.1565 times upper H upper R

right parenthesis 4th Row left parenthesis upper S upper E upper E equals 5.0 m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts semicolon s e x equals 0 for female comma 1 for male right parenthesis EndLayout

In addition to independently predicting morbidity and mortality ([120](#), [121](#)), the 6-minute walk test has been used to evaluate CRF in populations considered to have reduced CRF, such as older adults and some clinical populations (*e.g.*, individuals with CHF or pulmonary disease). The American Thoracic Society and European Respiratory Society have published technical standards for field walking tests, including the 6-minute walk test ([122](#)). Even though the test is considered submaximal, it may result in near-maximal performance for those with low physical fitness levels or disease ([123](#)). Individuals completing less than 300 m (~984 ft) during the 6-minute walk demonstrate a poorer short-term survival compared to those surpassing this threshold ([124](#)). Several multivariate equations are available to predict $\dot{V}O_{2peak}$ from the 6-minute walk; however, the following equation requires minimal clinical information ([125](#)):

$\dot{V}O_{2peak} = \dot{V}O_{2mL \cdot kg^{-1} \cdot min^{-1}} = 0.02 \times distance_{cm} - 0.191 \times age_{yr} - 0.07 \times weight_{kg} + 0.09 \times height_{cm} + 0.26 \times RPP \times 10^{-3} + 2.45$

ModifyingAbove normal upper V With' normal upper O Subscript 2 peak Baseline equals ModifyingAbove normal upper V With' normal upper O 2 m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts equals left parenthesis 0.02 times distance left bracket normal m right bracket right parenthesis minus 2nd Row left parenthesis 0.191 times a g e left bracket y r right bracket right parenthesis minus left parenthesis 0.07 times weight left bracket k g right bracket right parenthesis plus left parenthesis 0.09 times height left bracket c m right bracket right parenthesis plus 3rd Row left parenthesis 0.26 times upper R upper P upper P left bracket times 10 Superscript negative 3 Baseline right bracket right parenthesis plus 2.45 EndLayout

where m = distance in meters; yr = year; kg = kilogram; cm = centimeter;
RPP = rate pressure product (HR $\times$ SBP in mm Hg); SEE = 2.68 mL $\cdot$ kg⁻¹ $\cdot$ min⁻¹

Step tests are also used to estimate $\dot{V}O_{2\max}$. Protocols with fixed stepping rates and step heights tend to produce less accurate CRF values compared to protocols individualized for stepping rate and stature, as the required cadence and step height may be inappropriate for the individual ([126](#)). In addition to their ease to complete, the results are also easy to explain to individuals ([127](#)). Special precautions may be needed for those who have balance problems or are extremely deconditioned. Some single-stage step tests require an energy cost of 7–9 metabolic equivalents (METs), which may exceed the maximal capacity of some individuals. Therefore, the protocol chosen must be appropriate for the physical fitness level of the individual. In addition, inadequate compliance to the step cadence and excessive fatigue in the lead limb may diminish the value of a step test. Most tests do not monitor HR and BP while stepping because of the difficulty of these measures during the test. Some tests include RPE along with postexercise HR; others record the time required to complete a specific number of steps at a fixed step height in combination with select anthropometric variables. A summary of common step test protocols is available in [Table 3.7](#).

TABLE 3.7 Summary of Common Step Tests

Test	Population	Step Height (cm)	Step Rate (steps · min ⁻¹)	Duration
Åstrand-Rhyming (114)	Healthy adults	Females: 33 Males: 40	22.5	5 min
Webb (128)	Young adults	Individualized (0.19 × stature in cm)	Variable; dependent on perceived functional ability; increased by 5 steps · min ⁻¹ every 2 min	75% of age-predicted HR _{max}
YMCA (118)	Healthy adults	30.5	24	3 min
Queens College (129)	Young adults	41.3	Females: 22 Males: 24	3 min
STEP Tool (130)	All adults	20	Variable	20 stepping cycles

HR_{max}, maximal heart rate.

Åstrand and Rhyming ([131](#)) used a single-step height of 33 cm (13 in) for females and 40 cm (15.7 in) for males at a rate of 22.5 steps · min⁻¹ (when counting just the leading leg) for 5 minutes. These tests require $\dot{V}O_2$ of about 25.8 and 29.5 mL · kg⁻¹ · min⁻¹, respectively. Because of this, step tests may not be a good choice of modality for individuals who are less fit, have a contraindication for testing, or are taking a medication that affects HR. HR is measured in the last minute as described for the Åstrand-Rhyming cycle ergometer test, and $\dot{V}O_{2\max}$ is estimated from a nomogram (see [Figure 3.1](#)). Multistage step tests are also possible. Specifically designed for college-aged

adults, the Webb protocol stepping cadence is increased by 5 steps $\cdot$ min⁻¹ every 2 minutes until 75% of the age-predicted HR_{max} (HR_{max} = 207 – [0.7 $\times$ age]) is attained ([128](#)). Step height is individualized based on a person's height; final stepping cadence, recovery HR at 45 seconds, and the individualized perceived functional ability are needed to estimate $\dot{V}O_{2\max}$. Such step tests should be modified to suit the population being tested. The Canadian Home Fitness Test has demonstrated that such testing can be performed on a large scale and at low cost ([132](#)).

Instead of estimating $\dot{V}O_{2\max}$ from HR responses to submaximal work rates, a wide variety of step tests have been developed to categorize CRF based on an individual's recovery HR following a standardized step test, as postexercise (recovery) HR decreases with improved CRF. This eliminates the potential problem of taking HR during stepping. The 3-Minute YMCA Step Test is a good example of such a test. This test uses a 12-in (30.5-cm) bench, with a stepping rate of 24 steps $\cdot$ min⁻¹ (estimated $\dot{V}O_2$ of 25.8 mL $\cdot$ kg⁻¹ $\cdot$ min⁻¹). After stepping is completed, the individual immediately sits down, and HR is counted for 1 minute. Counting must start within 5 seconds at the end of exercise. HR values are used to obtain a qualitative rating of fitness from published normative tables ([118](#)). Additionally, the Queens College Step Test (also called the McArdle Step Test) requires individuals to step at a rate of 24 steps $\cdot$ min⁻¹ for males and 22 steps $\cdot$ min⁻¹ for females for 3 minutes. The bench height is 16.25 in (41.25 cm). After stepping is completed, the individual remains standing. Wait 5 seconds, take a 15-second HR count, and multiply the HR by 4 to convert to beats $\cdot$ min⁻¹. $\dot{V}O_{2\max}$ is calculated using the formulas below ([129](#)):

For males

$\dot{V}O_{2\max} \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 111.33 - 0.42 \times \text{HR}$
 Above normal upper V With normal upper O Subscript 2 max Baseline left parenthesis m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1

Endscripts right parenthesis equals 111.33 minus left parenthesis 0.42 times upper H upper R right parenthesis

For females

$\dot{V}O_{2\max} \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 65.81 - 0.1847 \times \text{HR}$ Modifying Above normal upper V With normal upper O Subscript 2 max Baseline left parenthesis m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts right parenthesis equals 65.81 minus left parenthesis 0.1847 times upper H upper R right parenthesis

where HR = heart rate ($\text{beats} \cdot \text{min}^{-1}$)

There are self-paced step test alternatives that do not rely on a metronome for controlling stepping rate. These single-stage step tests provide health care practitioners with a feasible opportunity to quickly estimate CRF during an office visit. The STEP Tool protocol requires a standard set of two contiguous steps, each 20 cm (~8 in) in height, and $\dot{V}O_{2\max}$ is based on the individual's time to complete 20 stepping cycles, body mass, age, sex, and the 6-second recovery HR ([130](#)). For older adults completing this protocol, sex-specific estimations of $\dot{V}O_{2\max}$ are based on time to completion, O_2 pulse, age, BMI, and HR immediately following the last stepping cycle ([133](#)). For this protocol, O_2 pulse is calculated as the O_2 cost of stepping/HR immediately postexercise, with O_2 cost of stepping = $[(\text{stepping frequency} \times \text{step height (cm)} \times 1.78 \times 1.3) + 13 \text{ stepping frequency}]$. Exercise professionals and clinicians should screen each person carefully and familiarize them with the stepping procedure prior to having them undertake these self-paced step tests.

Submaximal Exercise Tests

Single-stage and multistage submaximal exercise tests are available to estimate $\dot{V}O_{2\max}$ from simple HR measurements. Accurate measurement of

HR is critical for valid testing, and devices to accurately assess HR are almost ubiquitous. Occasionally it will be difficult to use devices to assess HR, and although HR obtained by palpation can be used, the accuracy of this method depends on the experience and technique of the evaluator and is likely less accurate than most other methods. It is recommended that an ECG, a stethoscope, or a HR monitor be used to determine HR. The submaximal HR response is easily altered by a number of environmental (*e.g.*, heat, humidity; see [Chapter 7](#)), dietary (*e.g.*, caffeine, time since last meal), and behavioral (*e.g.*, anxiety, smoking, previous PA) factors. These variables must be controlled to have a valid estimate that can be used as a reference point in an individual's fitness program. In addition, the test mode (*e.g.*, cycle, treadmill, step) should be consistent with the primary exercise modality used by the individual to address specificity of training issues. Standardized procedures for submaximal testing are presented in [Box 3.7](#). See [Figure 4.1](#) for a list of incremental treadmill protocols that may be used to assess submaximal exercise responses.

Box 3.7 General Procedures for Submaximal Testing of Cardiorespiratory Fitness

1. Obtain resting HR and BP immediately prior to exercise in the exercise posture.
2. The individual should be familiarized with the ergometer or treadmill. If using a cycle ergometer, properly position the individual on the ergometer (*i.e.*, upright posture, $\sim 25^\circ$ bend in the knee at maximal leg extension, and hands in proper position on handlebars) ([134](#), [135](#)).
3. The exercise test should begin with a 2- to 3-minute warm-up to acquaint the individual with the cycle ergometer or treadmill and prepare him or her for the exercise intensity in the first stage of the test.
4. A specific protocol should consist of 2- or 3-minute stages with appropriate increments in work rate.
5. HR should be monitored at least two times during each stage, near the end of the 2nd and 3rd minutes of each stage. If HR is greater than $110 \text{ beats} \cdot \text{min}^{-1}$, steady state HR (*i.e.*, two HRs within $5 \text{ beats} \cdot \text{min}^{-1}$) should be reached before the workload is increased.
6. BP should be monitored in the last minute of each stage and repeated (verified) in the event of a hypotensive or hypertensive response.
7. RPE (using either the Borg category or category-ratio scale [see [Table 3.6](#) and [Figure 4.2](#)]) and additional rating scales should be monitored near the end of the last minute of each stage.
8. Individual's appearance and symptoms should be monitored and recorded regularly.
9. The test should be terminated when the individual reaches 70% heart rate reserve (85% of age-predicted HR_{max}), fails to conform to the exercise test protocol, experiences adverse signs or symptoms, requests to stop, or experiences an emergency situation.
10. An appropriate cool-down/recovery period should be initiated consisting of either
 - a. Continued exercise at a work rate equivalent to that of the first stage of the exercise test protocol or lower or
 - b. A passive cool-down if the individual experiences signs of discomfort or an emergency situation occurs
11. All physiologic observations (*e.g.*, HR, BP, signs, and symptoms) should be continued for at least 5 minutes of recovery unless abnormal responses occur, which would warrant a longer posttest surveillance period. Continue low-level exercise until HR and BP stabilize but not necessarily until they reach preexercise levels.

BP, blood pressure; HR, heart rate; HR_{max} , maximal heart rate; RPE, rating of perceived exertion.

Interpretation of Results

[Tables 3.8](#) and [3.9](#) provide normative fitness categories and percentiles by age group for CRF from CPET on a treadmill and cycle ergometer, respectively, with directly measured $\dot{V}O_{2\max}$. These data were obtained from the Fitness Registry and the Importance of Exercise National Database (FRIEND) Registry, which provides age- and sex-specific reference values for males and females who were considered free from known CVD ([136](#), [137](#)) and in individuals with CVD diagnosis ([138](#)). Additionally, FRIEND has provided age- and sex-specific reference values for ventilatory threshold ([139](#)), ventilatory efficiency ([140](#)), RPE ([110](#)), peak BP ([141](#)), peak circulatory power ([142](#)), peak ventilation ([143](#)), and peak oxygen pulse ([144](#)) that may aid in interpreting exercise test results. Finally, FRIEND estimated peak $\dot{V}O_2$ was validated against external cohorts for clinical endpoints of predicting prognosis in heart failure ([145](#)). Research suggests that low CRF, usually defined as the lowest quartile or quintile on an exercise test, is associated with marked increases in CVD or all-cause mortality, independent of other CVD risk factors ([82](#), [146](#)).

TABLE 3.8 Treadmill-Based Cardiorespiratory Fitness Classifications ($\dot{V}O_2\text{max}$) by Age and Sex

$\dot{V}O_2\text{max}$ (mL $O_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)								
MALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
95	Superior	63.5	58.8	54.6	47.6	40.6	34.0	24.1
90	Excellent	58.6	55.5	50.8	43.4	37.1	29.4	22.8
85		56.3	53.1	47.8	40.6	34.3	27.4	22.2
80		54.5	50.0	45.2	38.3	32.0	25.9	21.4
75	Good	53.1	48.2	43.1	35.8	30.1	24.8	20.8
70		51.9	46.4	40.9	34.3	28.7	23.8	20.0
65		50.1	44.9	39.3	32.9	27.6	22.8	18.9
60		49.0	43.4	37.9	31.8	26.5	22.2	18.4
55	Fair	47.9	41.6	36.5	30.6	25.5	21.4	18.0
50		46.5	39.7	35.3	29.2	24.6	20.6	17.6
45		45.0	38.2	33.8	27.9	23.6	19.7	17.0
40		43.6	37.0	32.4	26.9	22.8	19.1	16.6
35	Poor	41.7	35.3	31.2	25.8	21.8	18.2	16.3
30		40.0	33.5	29.7	24.5	20.7	17.3	16.1
25		37.6	31.8	28.2	23.4	19.6	16.7	15.6
20		35.2	29.8	26.7	22.2	18.5	15.9	14.8
15	Very poor	32.2	27.3	24.7	20.7	17.3	14.9	13.9
10		28.6	24.9	22.1	18.6	15.8	13.6	12.9
5		24.2	20.6	18.8	16.1	13.0	11.6	11.8

V O₂ max (mL O₂ · kg⁻¹ · min⁻¹)								
MALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
		(n = 1,278)	(n = 1,473)	(n = 2,119)	(n = 2,082)	(n = 1,663)	(n = 776)	(n = 173)
FEMALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
95	Superior	52.1	45.9	41.1	36.0	30.0	24.3	22.7
90	Excellent	49.0	42.1	37.8	32.4	27.3	22.8	20.8
85		46.8	39.4	35.0	30.0	25.6	21.5	19.3
80		44.8	37.0	33.0	28.4	24.3	20.8	18.4
75	Good	43.3	35.0	31.2	27.3	23.3	20.1	17.7
70		41.8	33.6	30.0	26.3	22.4	19.6	17.3
65		40.7	32.2	28.8	25.5	21.6	19.0	16.6
60		39.0	31.0	27.7	24.6	20.9	18.3	16.0
55	Fair	37.9	29.7	26.6	23.7	20.2	17.8	15.7
50		36.6	28.3	25.7	22.9	19.6	17.2	15.4
45		35.3	27.3	24.8	22.1	18.8	16.8	15.1
40		34.0	26.4	23.9	21.5	18.3	16.2	14.7
35	Poor	32.4	25.4	22.8	20.9	17.7	15.6	14.2
30		30.8	24.2	21.8	20.1	17.0	15.2	13.7
25		29.2	23.0	20.7	19.3	16.3	14.7	13.2

$\dot{V} O_2 \text{ max (mL O}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1})$								
MALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
20		27.2	21.9	19.7	18.5	15.4	14.0	12.6
15	Very poor	25.1	20.4	18.6	17.5	14.5	13.5	11.9
10		22.5	18.6	17.2	16.5	13.4	12.3	11.4
5		19.1	16.1	15.1	14.5	11.9	11.2	10.7
		(n = 1,142)	(n = 1,043)	(n = 1,372)	(n = 1,457)	(n = 1,045)	(n = 549)	(n = 106)

Percentiles from cardiopulmonary exercise testing on a treadmill with measured maximal volume of oxygen consumed per unit time $\dot{V}O_{2\text{max}}$ ($\text{mL O}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$).

Data obtained from the Fitness Registry and the Importance of Exercise National Database (FRIEND) Registry for males and females who were considered free from known cardiovascular disease.

Adapted with permission from ([136](#)).

TABLE 3.9 Cycle Ergometer–Based Cardiorespiratory Fitness Classifications ($\dot{V}O_2\text{max}$) by Age and Sex

$\dot{V}O_2\text{max}$ ($\text{mL O}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$)								
MALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
95	Superior	66.1	60.3	47.3	44.0	37.7	29.0	19.0
90	Excellent	62.2	50.5	41.9	37.1	31.4	26.2	18.7
85		59.3	44.4	37.2	33.7	28.6	23.7	17.8
80		57.0	39.0	35.1	31.6	27.0	22.6	17.3
75	Good	54.8	37.0	32.9	30.2	25.8	21.6	17.2
70		52.8	35.5	31.4	28.4	24.5	20.6	16.2
65		50.6	34.0	30.0	27.2	23.8	20.1	15.3
60		48.3	31.6	29.0	26.3	23.3	19.4	14.6
55	Fair	46.2	30.8	28.1	25.5	22.6	18.7	13.9
50		44.0	30.2	27.4	24.5	21.7	18.3	13.2
45		42.5	29.2	26.3	23.9	21.3	17.6	12.4
40		40.8	27.9	25.4	23.1	20.7	17.1	12.2
35	Poor	38.9	27.0	24.9	22.6	19.9	16.7	11.6
30		37.4	25.7	23.8	22.0	19.1	16.0	11.1
25		36.0	24.1	22.6	21.3	18.4	15.6	10.4
20		34.5	22.6	21.9	20.2	17.5	14.7	9.7
15	Very poor	32.6	21.2	21.1	19.0	16.3	12.9	9.3
10		28.8	19.1	19.8	17.2	14.7	11.0	8.4
5		23.7	15.7	18.0	13.8	12.4	9.5	8.0

V O ₂ max (mL O ₂ · kg ⁻¹ · min ⁻¹)								
MALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
		(n = 367)	(n = 251)	(n = 446)	(n = 601)	(n = 465)	(n = 257)	(n = 52)
FEMALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
95	Superior	50.7	38.3	30.6	25.1	22.1	19.1	18.9
90	Excellent	46.0	32.0	27.3	22.4	20.3	18.0	18.1
85		42.7	29.0	24.9	21.2	19.1	17.3	15.9
80		40.9	27.0	23.5	20.4	18.5	16.8	14.3
75	Good	39.1	25.8	22.6	19.6	17.9	16.3	13.2
70		37.5	24.5	21.8	18.9	17.4	15.9	12.9
65		35.6	23.5	21.1	18.3	16.8	15.4	11.9
60		34.3	22.9	20.3	17.8	16.4	15.0	11.3
55	Fair	33.2	22.2	19.5	17.3	16.0	14.7	11.1
50		31.6	21.6	18.8	16.9	15.7	14.5	10.9
45		30.4	20.8	18.4	16.5	15.3	13.9	10.4
40		28.9	19.9	17.9	16.1	15.0	13.6	10.1
35	Poor	27.0	19.1	17.1	15.6	14.6	13.3	9.6
30		25.6	18.6	16.6	15.2	14.2	12.9	9.4
25		23.6	17.9	15.9	14.8	13.8	12.7	9.1

V O 2 max (mL O ₂ · kg ⁻¹ · min ⁻¹)								
MALES								
		Age Group (yr)						
Percentile		20–29	30–39	40–49	50–59	60–69	70–79	80–89
20		21.9	17.0	15.4	14.3	13.4	12.0	8.7
15	Very poor	20.6	15.8	14.7	13.7	12.8	11.5	8.6
10		18.8	15.0	13.7	13.0	12.2	10.7	7.8
5		16.3	13.5	12.4	11.6	10.9	9.8	6.5
		(n = 411)	(n = 377)	(n = 674)	(n = 1,115)	(n = 750)	(n = 308)	(n = 27)

Percentiles from cardiopulmonary exercise testing on a cycle ergometer with measured maximal volume of oxygen consumed per unit time ($\dot{V}O_{2\text{max}}$) ($\text{mL O}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$).

Data obtained from the Fitness Registry and the Importance of Exercise National Database (FRIEND) Registry for males and females who were considered free from known cardiovascular disease.

Adapted with permission from ([137](#)).

Although submaximal exercise testing is not as precise as maximal exercise testing, it provides a general reflection of an individual's physical fitness at a lower cost, potentially reduced risk for adverse events, and requires less time and effort on the part of the individual. Some of the assumptions inherent in a submaximal test are more easily met (*e.g.*, steady state HR can be verified), whereas others (*e.g.*, estimated HR_{max}) introduce errors into the prediction of $\dot{V}O_{2\text{max}}$. Despite this, when an individual is given repeated submaximal exercise tests over the course of an Ex R_x , the HR response to a fixed work rate decreases, which indicates the individual's CRF has improved, independent of the accuracy of the $\dot{V}O_{2\text{max}}$ prediction. Despite differences in test accuracy and methodology, virtually all evaluations can establish a baseline and be used to track relative progress during exercise training.

Alongside the FRIEND equations, several other regression equations exist for estimating CRF without the need for an exercise test. These age- and sex-based equations produce a single expected aerobic capacity value for comparison to a measured response as opposed to percentiles. Of the available regression equations, research indicates prediction formulas derived from a Veterans Affairs cohort (predicted METs = $18 - 0.15 \times \text{age}$), and the St. James Women Take Heart Project (predicted METs = $14.7 - 0.13 \times \text{age}$) may provide somewhat better prognostic information in males and females, respectively ([147](#)). These prediction equations may be useful when CRF testing is not possible.

MUSCULAR FITNESS

Muscular fitness includes muscular strength, endurance, and power and is an integral part of total health-related fitness ([148](#)). *Muscular strength* refers to the muscle's ability to exert maximal force, *muscular endurance* is the muscle's capacity to continue to perform successive exertions (or repetitions) against a submaximal load, and *muscular power* is the ability to exert force per unit of time (*i.e.*, rate) ([149](#)). Traditionally, tests allowing few repetitions (≤ 3) of a task prior to reaching muscular fatigue have been considered strength measures, whereas those in which numerous repetitions (> 12) are performed prior to fatigue were considered measures of muscular endurance. However, the performance of maximal repetitions (*i.e.*, 4, 6, or 8 repetitions) across a wider range can also be used to predict muscle strength.

Muscular fitness is a health-related fitness component and also a fundamental performance-related fitness component. It should be noted, however, this section will focus on improving or maintaining the following health-related fitness characteristics in apparently healthy individuals ([148](#), [150](#), [151](#)):

- FFM and resting metabolic rate, which are related to weight management
- Bone mass, which is related to osteoporosis
- Muscle mass, which is related to sarcopenia
- Glucose tolerance, which is pertinent in both the prediabetic and diabetic state
- Musculotendinous integrity, which is related to a lower risk of injury (including low back pain)
- Ability to carry out ADL, which is related to perceived quality of life and self-efficacy among other indicators of mental health

Furthermore, participating in muscular fitness-promoting exercise reduces all-cause mortality risk beyond what is observed with aerobic exercise alone ([152](#)).

Rationale

Tests of muscular strength, endurance, and power before commencing exercise training can provide valuable information on an individual's baseline physical fitness level. For example, muscular fitness test results can be compared to established standards and can be helpful in identifying weaknesses in certain muscle groups or muscle imbalances that could be targeted in exercise training programs. The information obtained during baseline muscular fitness assessments can also serve as a basis for designing individualized exercise training programs. An equally useful application of physical fitness testing is to show an individual's progressive improvement over time as a result of the training program and thus provide feedback that is often beneficial in promoting long-term exercise adherence.

Principles

Muscle function tests are very specific to the muscle group(s) and joint(s) tested, the type of muscle action (*i.e.*, dynamic or static), velocity of muscle movement, type of equipment, and joint range of motion (ROM). Results of any one test are specific to the procedures used, and no single test exists for evaluating total body muscular strength, endurance, or power. Individuals should participate in familiarization/practice sessions with test equipment and adhere to a specific protocol including a predetermined repetition duration and ROM in order to obtain a reliable score that can be used to track true physiologic adaptations over time. Moreover, a warm-up consisting of 5–10 minutes of light-intensity aerobic exercise (*e.g.*, treadmill or cycle ergometer), dynamic stretching, and several low-intensity repetitions of the

specific testing exercise should precede muscular fitness testing (see [Chapter 5](#) for more detailed information). If muscular fitness testing is occurring after CRF testing, a warm-up may not be necessary. These warm-up activities increase muscle temperature and localized blood flow and promote appropriate cardiovascular responses for exercise. Standardized conditions for muscular fitness assessment include the following:

- Aerobic warm-up
- Equipment familiarization
- Ensuring correct form and technique with all movement patterns
- Consistent repetition duration (movement speed)
- Full ROM
- Use of spotters (when necessary)

Change in muscular fitness over time can be based on the absolute value of the external load or resistance (*e.g.*, newtons, kilograms, or pounds), but when comparisons are made between individuals, the values should be expressed as relative values (per kilogram of body mass [$\text{kg} \cdot \text{kg}^{-1}$]). In both cases, caution must be used in the interpretation of the scores because the normative data may not include a representative sample of the individual being measured, a standardized protocol may be absent, or the exact test being used (*e.g.*, free weight vs. machine weight) may differ. In addition, the biomechanics for a given resistance exercise may differ significantly when using equipment from different manufacturers, further impacting generalizability.

Muscular Strength

Muscular strength refers to the external force generated by a specific muscle or muscle group (properly expressed in newtons, but kilograms and pounds are commonly used as well). It is commonly expressed in terms of resistance

met or overcome. Strength can be assessed either dynamically (*i.e.*, movement of an external load or body part in which the muscle changes length) or statically (*i.e.*, no overt muscular movement at a given joint[s]). Dynamic strength tests include concentric (muscle shortening) and/or eccentric (muscle lengthening) movements. Static (or isometric) strength can be measured conveniently using a variety of devices including cable tensiometers and handgrip dynamometers. Measures of static strength are specific to the muscle group and joint angle involved in testing and thus may be limited in describing overall muscular strength. Despite this limitation, simple measurements, such as handgrip strength, have predicted mortality and functional status in older individuals ([153](#), [154](#)). Peak force development in such tests is commonly referred to as the maximal voluntary contraction. Procedures for the grip strength test are described in [Box 3.8](#), and grip strength normative data are provided in [Table 3.10](#).

TABLE 3.10 Fitness Categories for Grip Strength^a by Sex and Age

	5th	10th	25th	50th	75th	90th	95th
Age (yr)	MALES						
20-24	32	34	38	43	48	52	55
25-29	34	37	41	45	50	54	57
30-34	36	38	42	47	52	56	59
35-39	37	39	43	48	53	57	60
40-44	37	40	44	48	53	57	60
45-49	37	39	43	48	53	57	60
50-54	36	39	43	47	52	56	59
55-59	34	37	41	46	50	54	57
60-64	32	35	39	44	48	52	55
65-69	29	32	36	41	45	49	52
70-74	25	29	33	38	42	46	49
75-79	21	25	29	34	38	42	44
	FEMALES						
20-24	20	22	24	27	29	32	34
25-29	21	22	25	28	30	33	35
30-34	22	23	25	28	31	34	35
35-39	22	23	26	28	31	34	36
40-44	22	23	26	29	31	34	36
45-49	22	23	25	28	31	34	36
50-54	21	23	25	28	31	33	35
55-59	20	22	24	27	30	32	34
60-64	19	21	23	26	29	31	33

	5th	10th	25th	50th	75th	90th	95th
Age (yr)	MALES						
65–69	17	19	22	25	27	30	31
70–74	15	18	21	23	25	28	29
75–79	13	16	19	21	23	26	27

^a Norms use the best score measured in kilograms for left or right hands.

Adapted with permission from ([155](#)).

Box 3.8 Static Handgrip Strength Test Procedures

1. Adjust the grip bar so the second joint of the fingers fits snugly over the handle. Set the dynamometer to zero.
2. The individual stands with feet slightly apart and holds the handgrip dynamometer in line with the forearm at the level of the thigh, away from the body.
3. The individual squeezes the handgrip dynamometer as hard as possible without holding the breath (to avoid the Valsalva maneuver). Neither the hand nor the handgrip dynamometer should touch the body or any other object.
4. Repeat the test twice with each hand. Maximum grip strength is the highest value attained from either hand (to the nearest kilogram).

Adapted from ([155](#)).

Traditionally, the 1-RM, the greatest resistance that can be moved through the full ROM in a controlled manner with good posture, has been the standard for dynamic strength assessment. The exercise professional should be aware that 1-RM measurements may vary between different types of equipment ([156](#)), although with appropriate testing familiarization, 1-RM is a reliable indicator of muscle strength ([157–159](#)). A multiple RM, such as 5- or 10-RM, can also be used as a measure of muscular strength. It is important when performing 5- to 10-RM that the exercise be performed to failure. When using a multiple RM (*i.e.*, 5- to 10-RM) to estimate the 1-RM, the prediction accuracy improves as the resistance increases and the RM gets closer to 1-RM ([156](#), [160](#)). Tables and prediction equations are available to estimate 1-RM from multiple RM ([156](#), [160](#)). It is also possible to track strength gains over time without the need to estimate 1-RM. For example, if one were training with 6- to 8-RM, the performance of a 6-RM to muscular fatigue would provide an index of strength changes over time, independent of the true 1-RM.

A conservative approach to assessing maximal muscle strength should be considered in individuals at high risk for or with known CVD, pulmonary, and

metabolic diseases and health conditions that present relative contraindications to muscular strength testing. For these groups, assessment of 10- to 15-RM that approximates training recommendations may be prudent ([151](#)).

Valid measures of general upper body strength include the 1-RM values for bench press or shoulder press. Corresponding indices of lower body strength include 1-RM values for the leg press or squat. Compared to previous editions of the Guidelines, the present version of [Chapter 3](#) does not include some normative data, as the organization's policy for permission has changed. Normative tables for the 1-RM bench press and 1-RM leg press are published elsewhere ([161](#), [162](#)). Proposed fitness standards for the upper body strength and lower body strength are provided in [Figures 3.3](#) and [3.4](#), respectively. The normative data must be interpreted with caution, and strict adherence to proper lifting technique must be followed for the comparisons to be valid. There are inherent differences between 1-RM testing done using machines versus free weights; thus, it is important to use the same type of resistance modality when tracking an individual's performance over time. The basic steps in 1-RM (or any multiple RM) testing following familiarization/practice sessions are presented in [Box 3.9](#).

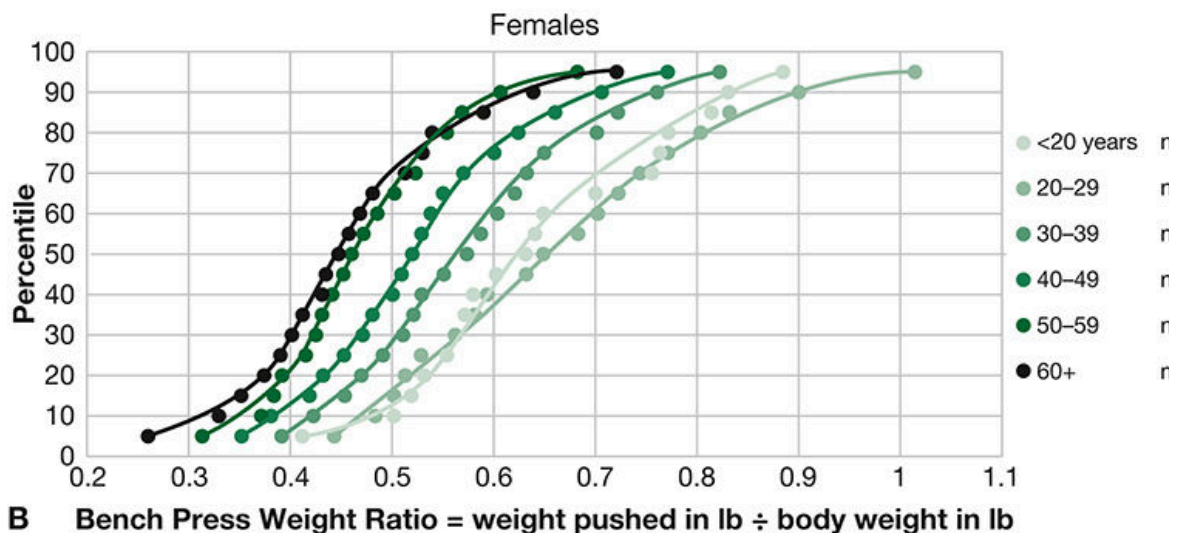


Figure 3.3 Fitness categories for upper body strength for males (A) and females (B) by age. One repetition maximum (1-RM) bench press, with bench press weight ratio = weight pushed in pounds per body weight in pounds. 1-RM was measured using a Universal Dynamic Variable Resistance (DVR) machine. Bench Press Weight Ratio = weight pushed in lb ÷ body weight in lb. Percentiles can be categorized as 90 = excellent; 75 = good; 55 = fair; 35 = poor; 15 = very poor. Data from ([161](#)).

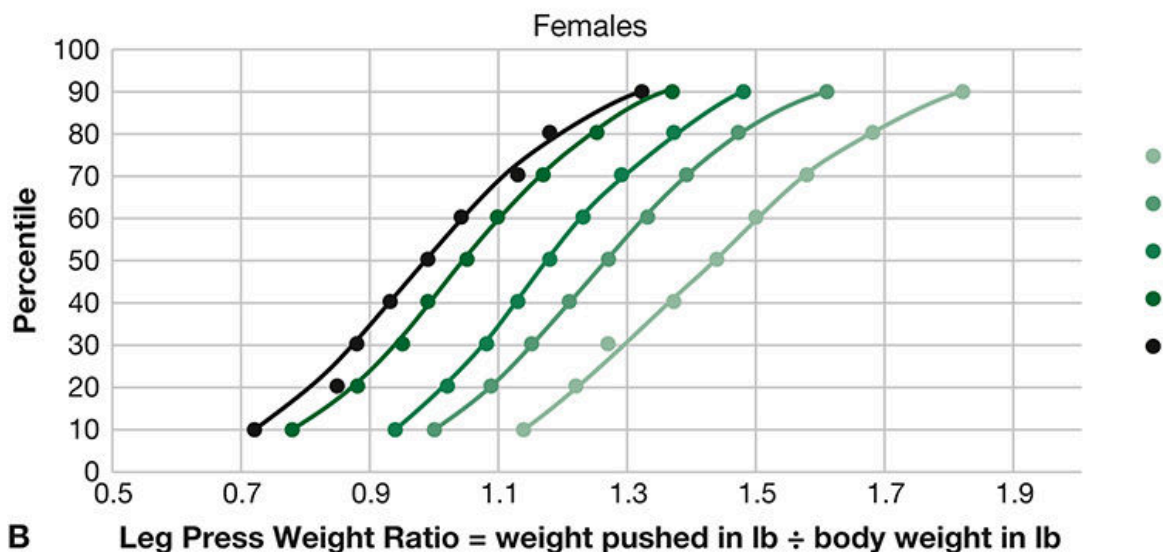
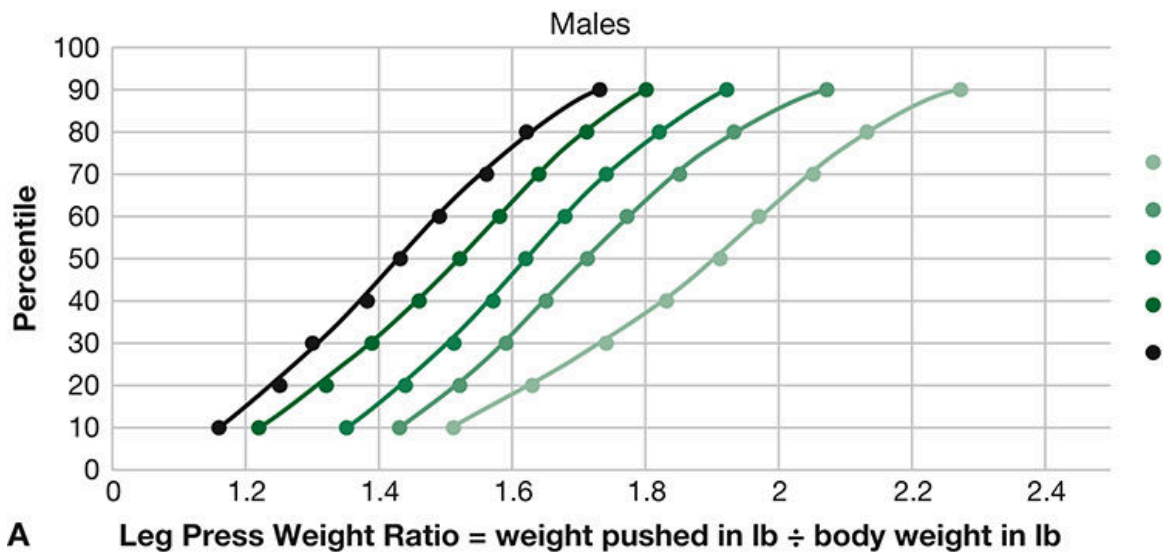


Figure 3.4 Fitness categories for leg strength for males (A) and females (B) by age. One repetition maximum (1-RM) leg press with leg press weight ratio = weight pushed per body weight. 1-RM was measured using a Universal Dynamic Variable Resistance (DVR) machine. Study population for the data set was predominantly white and college educated. Leg press weight ratio = weight pushed in lb ÷ body weight in lb. Percentiles can be categorized as 90 = well above average; 80 = above average; 60 = average; 40 = below average; 20 = well below average. Data from ([162](#)).

Box 3.9 One-Repetition Maximum (1-RM) and Multiple Repetition Maximum (RM) Test Procedures for Measurement of Muscular Strength

1. Testing should be completed only after the individual has participated in familiarization/practice sessions.
2. The individual should warm up by completing a number of submaximal repetitions of the specific exercise that will be used to determine the 1-RM.
3. Determine the 1-RM (or any multiple of 1-RM) within four trials with rest periods of 3 to 5 minutes between trials.
4. Select an initial weight that is within the individual's perceived capacity (~50%–70% of capacity).
5. Resistance is progressively increased by 5%–10% for upper body or 10%–20% for lower body exercise from the previous successful attempt until the individual cannot complete the selected repetition(s); all repetitions should be performed at the same speed of movement and ROM to instill consistency between trials.
6. The final weight lifted successfully is recorded as the absolute 1-RM or multiple RM.

ROM, range of motion.

Adapted from ([163](#), [164](#)).

Isokinetic testing involves the assessment of maximal muscle tension throughout an ROM set at a constant angular velocity (*e.g.*, $60 \text{ degrees} \cdot \text{s}^{-1}$). Equipment that allows control of the speed of joint rotation ($\text{degrees} \cdot \text{s}^{-1}$), as well as the ability to test movement around various joints (*e.g.*, knee, hip, shoulder, elbow), is available from commercial sources (*e.g.*, isokinetic dynamometers). Such devices accurately measure peak rotational force or torque, but an important drawback is that they are substantially more expensive compared to other strength testing modalities ([165](#)).

Muscular Endurance

Muscular endurance is the ability of a muscle group to execute repeated muscle contractions for a duration of time sufficient enough to cause muscular fatigue or to maintain a specific percentage of the 1-RM. If the total number of repetitions at a given amount of resistance is measured, the result is termed *absolute muscular endurance*. If the number of repetitions performed at a percentage of the 1-RM (*e.g.*, 70%) is used pre- and posttesting, the result is termed *relative muscular endurance*.

A simple field test such as the maximum number of push-ups that can be performed without rest may be used to evaluate the endurance of upper body muscles ([166](#)). Procedures for conducting this push-up endurance test are presented in [Box 3.10](#), and [Table 3.11](#) shows physical fitness categories based on sex and age. These categories are based on standard push-ups for males and modified push-ups for females. Recent research on college age females indicates the suitability of standard push-up assessments for both males and females ([168](#)), yet further investigations spanning various age brackets are necessary to establish normative data charts for females performing standard push-ups.

Box 3.10 Push-up Test Procedures for Measurement of Muscular Endurance

1. The push-up test is administered with males starting in the standard “down” position (fingers pointing forward and under the shoulder; back straight, head up, using the toes as the pivotal point) and females in the modified “knee push-up” position (lower legs in contact with mat with ankles plantarflexed and using the knees as the pivotal point).
2. The individual must raise the body by straightening the elbows and return to the “down” position, until the chin touches the mat. The stomach should not touch the mat.
3. For both males and females, the individual’s back must be straight at all times, and the individual must push-up to a straight arm position.
4. The maximal number of push-ups performed consecutively without rest is counted as the score.
5. The test is stopped when the individual strains forcibly or unable to maintain the appropriate technique within two repetitions.

TABLE 3.11 Fitness Categories for the Push-up by Age and Sex

	Age (yr)											
Category	15-19		20-29		30-39		40-49		50-59		60-69	
Sex	M	F	M	F	M	F	M	F	M	F	M	F
Excellent	≥39	≥33	≥36	≥30	≥30	≥27	≥25	≥24	≥21	≥21	≥18	≥17
Very good	29-38	25-32	29-35	21-29	22-29	20-26	17-24	15-23	13-20	11-20	11-17	12-16
Good	23-28	18-24	22-28	15-20	17-21	13-19	13-16	11-14	10-12	7-10	8-10	5-11
Fair	18-22	12-17	17-21	10-14	12-16	8-12	10-12	5-10	7-9	2-6	5-7	2-4
Poor	≤17	≤11	≤16	≤9	≤11	≤7	≤9	≤4	≤6	≤1	≤4	≤1

M, males; F, females.

Reprinted with permission from the Canadian Society for Exercise Physiology ([167](#)).

Muscular Power

Muscular power is the rate of performing work (*i.e.*, the combination of strength and speed). Traditionally, muscular power has been considered a performance-related component of fitness rather than a health-related variable, and consequently, it has not been included in health-related fitness batteries. However, muscular power declines at a faster rate than muscular strength or muscular endurance with aging (see [Chapter 5](#)) ([169](#)), and it may be the most valuable of the muscular fitness variables for predicting maintenance of functional independence and improving quality of life ([170](#)). Therefore, fitness professionals should consider including a measurement of muscular power in their assessments of muscular fitness.

Commercially available linear transducers and accelerometers can be used to assess movement velocity of a person or barbell in order to determine muscular power. For example, a linear transducer attached to the waist was used to accurately measure relative power in older adults (>65 years) rising as quickly as possible from a seated to standing position ([171](#)).

Unfortunately, standardized field tests that do not require any specialized equipment and corresponding norms have yet to be established for measuring power of older adults. Alternatively, jump height from a countermovement vertical jump is a commonly used surrogate for estimating muscular power in a younger population in whom impact loading is less of a concern. Procedures for measuring the vertical jump are presented in [Box 3.11](#), and age-sex norms for the countermovement vertical jump are provided in [Table 3.12](#).

Box 3.11 Countermovement Vertical Jump Test Procedures for Measurement of Muscular Power

1. The individual stands flat-footed and reaches as high as possible with the dominant hand. If a commercial vertical jump testing apparatus is being used, measure the individual's reach height with the highest vane that can be touched. If no jump testing apparatus is being used, the individual can put chalk on the fingertips and touch the wall to mark reach height.
2. Without a running start or preparatory steps, the individual makes a ballistic countermovement from a standing position by rapidly flexing the hips and knees and swinging the arms backward, followed immediately by an explosive arm swing upward and extension of the hips and knees to jump as high as possible.
3. During the jump, the dominant hand should reach as high as possible at the height of the jump. If using a jump testing apparatus, swat the vane with the outstretched dominant hand at the highest point of the jump. Alternatively, chalked fingers can place a mark on the wall.
4. The examiner records the difference between the jump height (highest vane moved on a jump apparatus or highest chalk mark on the wall) and the reach height. This is the individual's vertical jump.
5. The best of three trials is used for comparison to normative values.

TABLE 3.12 Fitness Categories for the Countermovement Vertical Jump by Age and Sex

	Age (yr)					
	15-19	20-29	30-39	40-49	50-59	60-69
Males						
Excellent	≥56	≥58	≥52	≥43	≥41	≥33
Very good	51-55	54-57	46-51	36-42	34-40	29-32
Good	46-50	48-53	40-45	32-35	28-33	25-28
Fair	42-45	42-47	31-39	26-31	18-27	18-24
Poor	≤41	≤41	≤30	≤25	≤17	≤17
Females						
Excellent	≥40	≥38	≥36	≥31	≥25	≥19
Very good	36-39	34-37	32-35	27-30	21-24	15-18
Good	32-35	29-33	28-31	23-26	16-20	11-14
Fair	28-31	25-28	24-27	18-22	10-15	7-10
Poor	≤27	≤24	≤23	≤17	≤9	≤6

All values are in centimeters.

Adapted with permission from ([172](#)).

FLEXIBILITY

Flexibility is the ability to move a joint through its complete, pain-free ROM and is important in both athletic performance and the ability to carry out ADLs. Maintaining flexibility of all joints facilitates movement and may prevent injury; however, it is impossible to state with confidence that preactivity stretching unequivocally reduces injuries associated with PA ([173](#)). Nevertheless, when an activity moves the structures of a joint beyond its full ROM, tissue damage can occur.

Flexibility depends on a number of specific variables including distensibility of the joint capsule, adequate warm-up, and muscle viscosity. In addition, compliance (*i.e.*, tightness) of various other tissues such as ligaments and tendons affect ROM. Just as muscular strength and endurance are specific to the muscles involved, flexibility is joint specific; therefore, no single flexibility test can be used to evaluate total body flexibility. Laboratory tests usually quantify flexibility in terms of ROM expressed in degrees; therefore, measuring ROM in degrees is considered a direct measurement. Common devices for direct ROM measurement include goniometers, electrogoniometers, the Leighton flexometer, and inclinometers. Goniometers and inclinometers are now available with digital displays that might reduce reading errors. Comprehensive instructions are available for the evaluation of flexibility of most anatomic joints ([174](#), [175](#)). Basic instructions for using a goniometer and inclinometer are provided in [Box 3.12](#). Accurate joint ROM measurements require in-depth knowledge of bone, muscle, and joint anatomy as well as experience in administering the evaluation. [Table 3.13](#) provides normative ROM values for select anatomic joints based on age and sex. Additional information can be found elsewhere ([66](#), [177](#)).

Box 3.12 General Guidelines for Range of Motion Testing

1. Each individual should participate in a general warm-up followed by static stretching prior to range of motion testing.
2. If using a goniometer, the axis of the goniometer should be placed at the center of the joint being evaluated. The fixed arm of the goniometer should be aligned with a bony landmark of the stationary body part, and the movable arm of the goniometer should be aligned with a bony landmark of the body segment that is going to be moving. For anatomical landmarks, refer to Gibson et al. ([66](#)). If using an inclinometer, it should be held on the distal end of the moveable body segment.
3. Record the range of motion in degrees.
4. Administer multiple trials; three are recommended.
5. Use the best score to compare to reference values.

TABLE 3.13 Range of Motion in Degrees at Select Joints by Age and Sex

	Age (yr)					
	9-19		20-44		45-69	
	M	F	M	F	M	F
Hip extension	18 (17-20)	21 (19-22)	17 (16-19)	18 (17-19)	14 (13-15)	17 (16-18)
Hip flexion	135 (133-137)	135 (133-137)	130 (129-132)	134 (133-135)	127 (126-129)	131 (129-132)
Knee flexion	142 (140-144)	142 (141-144)	138 (137-139)	142 (141-143)	133 (132-134)	138 (137-139)
Knee extension	2 (1-3)	2 (2-3)	1 (1-1)	2 (1-2)	1 (0-1)	1 (1-2)
Ankle dorsiflexion	16 (15-18)	17 (16-19)	13 (12-14)	14 (13-15)	12 (11-13)	12 (11-13)
Ankle plantarflexion	53 (51-55)	57 (55-60)	55 (53-56)	62 (61-64)	49 (48-51)	57 (55-58)
Shoulder flexion	171 (169-173)	172 (170-174)	169 (167-170)	172 (171-173)	164 (162-166)	168 (167-170)
Elbow flexion	148 (147-150)	150 (149-151)	145 (144-146)	150 (149-151)	144 (142-145)	148 (147-149)
Elbow extension	5 (4-7)	6 (5-8)	1 (0-2)	5 (4-6)	-1 (-2-0)	4 (3-5)
Elbow pronation	80 (79-82)	81 (80-83)	77 (76-78)	82 (81-83)	78 (77-79)	81 (80-82)
Elbow supination	88 (86-90)	90 (88-92)	85 (84-86)	91 (89-92)	82 (81-84)	87 (86-88)

M, males; F, females. Data are means (95% confidence interval).

Adapted with permission from ([176](#)).

Previous editions of the *Guidelines* included the standard sit-and-reach test (an indirect linear measurement of ROM) as a simple field test for assessment of low back and hamstring flexibility. However, this test has been shown to be unrelated to low back pain ([178](#), [179](#)) and is questionable as a measure of hamstring flexibility ([180](#)). Given these findings, the standard sit-and-reach test is no longer recommended for assessment of hamstring or low back flexibility. A better alternative would be a test such as the chair sit-and-reach test. This lower body ROM test has established normative values for older adults. This test may also be useful for providing ROM assessment and ROM exercise prescriptions for older adults ([74](#), [181](#)). The modified back-saver sit-and-reach test is another version of the traditional sit-and-reach and is included in a number of standard elementary school fitness test evaluations. This test has been shown to be a reliable test for low back and hamstring flexibility ([74](#)). Nevertheless, given the low cost, portability, and simplicity of the laboratory tests (*i.e.*, goniometer and inclinometer), combined with the questionable validity of the sit-and-reach test, direct ROM measures are recommended rather than indirect methods.

BALANCE

Individuals with enhanced balance: (a) maximize their athletic performance ([182–185](#)), (b) prevent sport-related injuries, particularly ankle injuries ([186, 187](#)), (c) decrease their risk of falling, especially for older adults or individuals with neurological conditions ([188](#)), or (d) improve their recovery from injury ([189, 190](#)). Due to these benefits, balance should be incorporated within a fitness assessment test battery.

Balance assessments are typically divided into static and dynamic balance. Static balance is the ability to maintain an upright posture while standing still. Dynamic balance is the ability to maintain stability while moving the body through space ([191](#)). Analytic systems that consist of computerized force plates provide center of pressure data and are considered direct measures of balance. Technological advances in inertial sensors, such as those found on smartwatches and phones, provide portable alternatives. However, due to limited research, practical alternatives, such as field and clinical testing, are still recommended ([192, 193](#)).

Common field balance assessments are the Balance Error Scoring System (BESS) and the Y-balance test or timed up and go (TUG) test for static and dynamic balance, respectively. The BESS involves counting the number of balance errors that a person makes while standing in three different positions: (a) feet side-by-side, (b) single limb stance on the nondominant foot, and (c) heel-to-toe tandem stance. Each stance position is performed on a firm surface and then repeated on a foam pad, and the eyes are closed for each trial. Procedures for the BESS are presented in [Box 3.13](#) ([194](#)).

Box 3.13 Instructions for Administering the Balance Error Scoring System (BESS) Test

A stopwatch and a medium-density ($45 \text{ cm}^2 \times 13 \text{ cm}$ thick) foam pad or an Airex blue pad are needed.

1. The first three tests are performed in bare feet while standing on the floor. The three tests are then repeated while standing on a foam pad. All tests are done with the eyes closed and hands placed on the hips. All tests positions are to be held for 20 seconds.
2. Test 1 (stance 1): Individual stands with feet side by side (Romberg stance).
3. Test 2 (stance 2): Individual stands on nondominant foot (single leg stance).
4. Test 3 (stance 3): Individual stands heel-to-toe with the nondominant foot placed behind the dominant foot (tandem stance).
5. Repeat the three stance positions while standing on a foam pad.
6. The examiner counts the number of errors the individual makes on each of the six tests. Errors include the following: (a) lifting hands off of the hips, (b) opening eyes, (c) stepping/stumbling, (d) excessive hip flexion or abduction ($>30^\circ$) to correct balance, and (e) lifting foot or heel. If an error is made, the individual should correct it as soon as possible. The number of errors made are summed. The maximal number of errors counted for each of the 20-second trials is 10. Failure to maintain the stance position for at least 5 seconds of the 20-second trial results in a maximal error score of 10.

The Y-balance test is recommended as a progression for individuals who can complete the BESS. This test uses a commercially available apparatus to measure the distance a person can reach with one foot while maintaining balance on the other foot. The individual slides moveable blocks with their toes as far as possible in the anterior, posteromedial, and posterolateral directions, forming a “Y.” Procedures for the Y-balance test are explained in [Box 3.14 \(195\)](#).

Box 3.14 Instructions for Administering the Y-Balance Test

1. A Y-balance testing apparatus is needed. These are commercially available.
2. Individual stands bare foot on the center block with one foot. The leg supporting the body weight is the one being tested.
3. The toes of the opposite foot are used to push the anterior block as far forward as possible while maintaining balance on the center block. The distance the block is moved is recorded by the examiner.
4. The individual moves the posteromedial and posterolateral blocks in the same manner.
5. Each direction is tested three times. Switch feet on the center block and repeat the test. Thus, the anterior, posteromedial, and posterolateral directions are each challenged three times with both the dominant and nondominant leg.
6. The best of the three trials for each direction and each foot is used for evaluation.
7. Leg length is measured (anterior superior iliac spine to the medial malleolus).
8. The best anterior, posteromedial, and posterolateral score for the right leg are summed and divided by 3 times the leg length to get a composite score (percentage of leg length) for the right leg. This is repeated for the left leg.

The TUG test is recommended for individuals who are unable to complete the BESS ([196](#)). Within this timed test, individuals are asked to stand up from a chair, walk 3 m (9.84 ft) toward a line marked on the floor, turn 180°, walk back, and sit down. Procedures for the TUG test are explained in [Box 3.15](#) ([197](#)). A modified version of the TUG is the 8-Foot Up-and-Go test, which is part of the Senior Fitness Test ([198](#)). For the 8-Foot Up-and-Go test, the individual walks 2.44 m (8 ft) and turns around a cone instead of a marked line on the floor. These modifications resulted in the test requiring less space and increasing clarity about when and how to turn ([199](#)). Both the TUG and 8-Foot Up-and-Go tests provide insight into the individual's balance, strength, mobility, and risk of falling.

Box 3.15 Instructions for Administering the Timed Up and Go (TUG) Test

1. A standard chair (46 cm), 3-m walk distance, line on floor, and stopwatch are needed.
2. Individual wears regular footwear and can use walking aid (*e.g.*, cane or walker).
3. Individual sits with back against the chair and arms resting on the chair's arm. If using cane, cane starts in hand. If using walker, walker starts in front of them.
4. Individual is instructed to get up on the word "go" and walk at a comfortable and safe pace to a line on the floor 3 m away, turn around, walk back to chair, and sit back down again.
5. The individual completes a test trial before being timed to familiarize oneself with the test.
6. The individual is timed, in seconds, on the time it takes for them to stand up from a standard armchair, walk a distance of 3 m, turns, walk back to the chair, and sit back down.

ONLINE RESOURCES

ACSM certifications: <https://www.acsm.org/get-stay-certified>

ACSM Exercise is Medicine — Exercise Professionals:

https://www.exerciseismedicine.org/support_page.php?p=91

American Heart Association: <https://www.heart.org/en/>

Clinical Guidelines on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults: The Evidence Report (Clinical Guidelines on the Identification, Evaluation, and Treatment of Overweight and Obesity in Adults: The Evidence Report [Internet] 2008):

<https://wayback.archive-it.org/5774/20220414053414/https://www.healthypeople.gov/2020/tools-resources/evidence-based-resource/clinical-guidelines-on-the-identification-evaluation-and-treatment-of-overweight-and-obesity-in>

National Heart, Lung, and Blood Institute Health Information for

Professionals: <https://www.nhlbi.nih.gov/health/indexpro.htm>

Physical Activity Guidelines for Americans, 2nd edition:

<https://health.gov/our-work/physical-activity/current-guidelines>

The Cooper Institute Fitness Adult Education:

<https://www.cooperinstitute.org/blog>

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CHAPTER 4

Clinical Exercise Testing and Interpretation

INTRODUCTION

Clinical exercise testing has been part of the differential diagnosis of individuals with ischemic heart disease (IHD) for more than 50 years. Although there are several indications for clinical exercise testing, most tests are performed as part of the diagnosis and evaluation of IHD. There are several evidence-based statements from professional organizations related to the conduct and application of clinical exercise testing. This chapter briefly summarizes these statements with a focus on noninvasive, symptom-limited, maximal exercise tests in adults with or suspected of having heart disease. Professionals who regularly perform or supervise clinical exercise tests should be familiar with the professional statements referenced in this chapter, especially those related to the conditions that are regularly presented in their clinic.

During a clinical exercise test, individuals are monitored continually while performing incremental (most common) or constant work rate exercise using standardized protocols and procedures and typically using a treadmill or a stationary cycle ergometer ([1–4](#)). The purpose is to observe physiological responses to increasing or sustained metabolic demand. The clinical exercise test typically continues until the individual reaches a sign (*e.g.*, ST-segment depression) or symptom-limited (*e.g.*, angina) maximal level of exertion. A clinical exercise test is often referred to as a graded exercise test (GXT), an exercise stress test, or an exercise tolerance test (ETT). When an exercise test includes the analysis of expired gases during exercise, it is termed a *cardiopulmonary exercise test* (most often abbreviated CPX or CPET) or a metabolic exercise test.

INDICATIONS FOR A CLINICAL EXERCISE TEST

Indications for clinical exercise testing encompass three general categories: (a) diagnosis (*e.g.*, presence of disease or abnormal physiologic response), (b) prognosis (*e.g.*, risk for an adverse event), and (c) evaluation of the physiologic response to exercise (*e.g.*, blood pressure [BP] and peak exercise capacity). These indications are useful for evaluating therapy for individuals with cardiovascular or pulmonary disease. The most common diagnostic indication is the assessment of symptoms suggestive of IHD. The American College of Cardiology (ACC) and the American Heart Association (AHA) recommend a logistic approach to determining the type of test to be used in the evaluation of someone presenting with chest pain ([3](#)). In this approach, a symptom-limited maximal exercise test with electrocardiographic monitoring only (*i.e.*, without adjunctive cardiac imaging) should initially be considered when the diagnosis of IHD is not certain, the individual has an interpretable resting electrocardiogram (ECG) (see “Electrocardiogram” section), and the individual can tolerate exercise ([5](#), [6](#)).

Current evidence does not support the routine use of exercise testing (with or without imaging) to screen for IHD or the risk of IHD-related events in asymptomatic individuals who have a very low or low pretest probability of IHD ([5](#), [7](#), [8](#)). This is because the exercise test is less accurate in low-risk individuals ([3](#), [5](#), [8](#)). Evidence also does not support the routine use of the test among individuals with a high pretest probability of IHD based on age, symptoms, and sex, given that the diagnosis is generally already known ([5](#)). Pretest likelihood of IHD is described in [Table 4.1](#). In addition, the exercise ECG is less accurate in diagnosing IHD among individuals on digitalis therapy (see [Appendix A](#)) with ST-segment depression on their resting ECG and among those who meet the ECG criteria for left ventricular hypertrophy with ST-segment depression on their resting ECG ([5](#)). Additionally, the exercise test with ECG alone is not useful for the diagnosis of IHD in individuals with Wolff-Parkinson-White (W-P-W), ventricular pacing, greater than 1.0 mm of ST-segment depression on their resting ECG, or left bundle-branch block (LBBB) ([5](#)). Although these ECG abnormalities limit the utility of the exercise test with ECG alone in the diagnosis of

IHD, there may be other indications in which the exercise test is appropriate, such as the assessment of symptoms or the measurement of exercise capacity.

TABLE 4.1 Pretest Likelihood of Ischemic Heart Disease^a

Age	Sex	Typical/Definite Angina Pectoris	Atypical/Probable Angina Pectoris	Nonanginal Chest Pain	Asymptomatic
30–39 yr	Male	Intermediate	Intermediate	Low	Very low
	Female	Intermediate	Very low	Very low	Very low
40–49 yr	Male	High	Intermediate	Intermediate	Low
	Female	Intermediate	Low	Very low	Very low
50–59 yr	Male	High	Intermediate	Intermediate	Low
	Female	Intermediate	Intermediate	Low	Very low
60–69 yr	Male	High	Intermediate	Intermediate	Low
	Female	High	Intermediate	Intermediate	Low

^a No data exist for patients who are less than 30 or greater than 69 years, but it can be assumed that prevalence of ischemic heart disease increases with age. In a few cases, patients with ages at the extremes of the decades listed may have probabilities slightly outside the high or low range. High indicates greater than 90%; intermediate, 10%–90%; low, less than 10%; and very low, less than 5%.

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The clinical utility of exercise testing is described in several evidence-based guideline statements aimed at specific cardiac diagnoses (9–15). In addition, an exercise test can be useful in the evaluation of individuals who present to emergency departments with chest pain. This practice (a) appears to be safe in individuals who are at low-to-intermediate risk for IHD and have been appropriately screened by a physician, (b) may improve the accuracy of diagnosing acute coronary syndrome, and (c) may reduce the cost of care by reducing the need for additional tests and length of

stay (16). Generally, exercise testing may be appropriate for individuals whose symptoms have resolved, have a normal ECG, and had no change in enzymes reflecting cardiac muscle damage. Exercise testing in this setting should be performed only as part of a well-defined clinical care pathway (16).

Additional indications that might warrant the use of a clinical exercise test include the assessment of various pulmonary diseases (*e.g.*, chronic obstructive pulmonary disease) (1, 17), exercise intolerance and unexplained dyspnea (1, 17, 18), exercise-induced bronchoconstriction (1, 17, 19), exercise-induced arrhythmias (5), a pacemaker or an impaired heart rate (HR) response to exercise (5), preoperative risk evaluation (1, 12, 17), claudication in peripheral arterial disease (20), disability evaluation (1, 18), and physical activity (PA) counselling (1, 5, 17, 21).

In addition to diagnostic utility, data from a clinical exercise test can be useful to predict prognosis. There is an inverse relationship between cardiorespiratory fitness (CRF) measured from an exercise test and the risk of mortality among apparently healthy individuals (2, 21); individuals at risk for IHD (21, 22); and those with diagnosed heart disease (1, 18, 22, 23), heart failure (1, 18, 24), and pulmonary disease (2, 25). In addition to CRF, other measures from an exercise test have been associated with prognosis, such as the chronotropic response to exercise or HR recovery after an exercise test (17, 26–30).

Clinical exercise testing is useful in guiding recommendations for return-to-work after a cardiac event (see [Chapter 8](#)) as well as developing an exercise prescription (Ex Rx) in those with known heart disease (5). In addition, the maximal exercise test is the gold standard to objectively measure exercise capacity. Although exercise time and/or peak workload achieved during an exercise test can be used to estimate peak metabolic equivalents (METs), the best measurement of exercise capacity is via respiratory gas analysis using open circuit indirect calorimetry for the determination of maximal volume of oxygen consumed per unit of time ($\dot{V}O_{2\max}$) (2, 5, 18, 24).

CONDUCTING THE CLINICAL EXERCISE TEST

When administering clinical exercise tests, it is important to consider and screen for contraindications, the exercise test protocol and mode, test endpoint indicators, safety, medications, and staff and facility emergency preparedness ([3](#), [4](#)). The AHA ([3](#)) has outlined both absolute and relative contraindications to clinical exercise testing ([Box 4.1](#)). These contraindications are intended to avoid unstable ischemic, rhythm, or hemodynamic conditions or situations in which the risk associated with undergoing the exercise test is likely to exceed the information to be gained from it and includes determining and preventing the risk of postexercise exacerbation and symptom aggravation.

Box 4.1 Contraindications to Symptom-Limited Maximal Exercise Testing

Absolute Contraindications

- Acute myocardial infarction within 2 d
- Ongoing unstable angina
- Uncontrolled cardiac arrhythmia with hemodynamic compromise
- Active endocarditis
- Symptomatic severe aortic stenosis
- Decompensated heart failure
- Acute pulmonary embolism, pulmonary infarction, or deep venous thrombosis
- Acute myocarditis or pericarditis
- Acute aortic dissection
- Physical disability that precludes safe and adequate testing

Relative Contraindications

- Known obstructive left main coronary artery stenosis
- Moderate to severe aortic stenosis with uncertain relationship to symptoms
- Tachyarrhythmias with uncontrolled ventricular rates
- Acquired advanced or complete heart block
- Recent stroke or transient ischemia attack
- Mental impairment with limited ability to cooperate
- Resting hypertension with systolic greater than 200 mm Hg or diastolic greater than 110 mm Hg
- Uncorrected medical conditions, such as significant anemia, important electrolyte imbalance, and hyperthyroidism

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Before the exercise test, informed consent needs to be obtained to ensure that the individual understands the purpose, expectations, and risks associated with the test (see [Chapter 2](#)) [\(4\)](#). The extent and quality of data obtained from a symptom-limited maximal exercise test depends on the individual's ability and willingness to provide maximal exertion; therefore, it is important to educate the individual about what they may experience during the test (*e.g.*, fatigue, dyspnea, chest pain) [\(4\)](#). Prior to performing an exercise test, the medical history (including current and recent symptoms), current medications (see [Appendix A](#)), and indications for the test should be screened for and recorded [\(4\)](#). Lastly, the resting ECG should be examined for

abnormalities that may preclude testing, such as new-onset atrial fibrillation or new repolarization changes ([4](#), [5](#)). In addition, if the purpose of the exercise test is the assessment of exercise-induced myocardial ischemia, the resting ECG must allow for interpretation of exercise-induced repolarization changes ([4](#), [5](#)); otherwise, consideration should be given to adjunctive imaging procedures such as nuclear or echocardiographic imaging ([5](#)). These additional imaging procedures are not necessary if the exercise test is being conducted for reasons other than the assessment of myocardial ischemia.

Testing Staff

Over several decades, there has been a transition in many exercise testing laboratories from tests being administered by physicians to nonphysician allied health professionals, such as clinical exercise physiologists, nurses, physical therapists, and physician assistants. This shift from physician to nonphysician staff has occurred to reduce staffing costs and improve the utilization of physician time ([31](#)). These allied health care professionals are not intended to replace the knowledge and skills of a physician ([31](#)) but a move toward the use and implementation of other disciplines such as clinical exercise physiologists acknowledges a suitable set of skills and expertise that can be used to support diagnostic procedures. The overall supervision of clinical exercise testing laboratories, as well as the interpretation of test results, remains the legal responsibility of the supervising physician ([4](#), [31](#), [32](#)).

According to the ACC and AHA, the nonphysician allied health care professional who administers clinical exercise tests should have cognitive skills and suitable experience similar to, although not as extensive as, the physician who provides the final interpretation ([31](#), [32](#)). These skills are presented in [Box 4.2](#). In addition, this individual should perform at least 50 exercise tests with preceptor supervision ([32](#)). However, 200 supervised exercise tests before independence are recommended ([31](#)). Recommendations for maintenance of competency vary between 25 ([32](#)) and 50 ([31](#)) exercise tests per year. Appropriately trained nonphysician staff can safely administer maximal clinical exercise tests when a qualified physician is “in the immediate vicinity ... and available for emergencies” ([31](#)) and who later reviews and provides final interpretation of the test results ([4](#)). There are no differences in morbidity and mortality rates related to maximal exercise testing when the testing is performed by

an appropriately trained allied health professional compared to a physician (31). The AHA has defined high-risk groups in which they recommend that a physician provides “personal supervision” (*i.e.*, the physician is directly present in the exercise testing room) (Table 4.2) (31). Empirical evidence suggests, however, that “direct supervision” (*i.e.*, the physician is available within the vicinity of the exercise testing room) and “general supervision” (*i.e.*, the physician is available by phone) (31) are the models employed in the majority of noninvasive clinical exercise testing laboratories in the United States, regardless of the disease severity of individuals being tested.

TABLE 4.2 Recommendations for Patients Requiring Personal Physician Supervision Based on Clinical Safety Criteria^a

- Moderate-to-severe aortic stenosis in an asymptomatic or questionably symptomatic patient
- Moderate-to-severe mitral stenosis in an asymptomatic or questionably symptomatic patient
- Hypertrophic cardiomyopathy: risk stratification and exercise gradient assessment
- History of malignant or exertional arrhythmias; sudden cardiac death
- History of exertional syncope or presyncope
- Intracardiac shunts
- Genetic channelopathies
- Within 7 days of myocardial infarction or other acute coronary syndrome
- New York Heart Association class III heart failure
- Severe left ventricular dysfunction (particularly patients whose clinical status has recently deteriorated and those who have never undergone prior exercise testing)
- Severe pulmonary arterial hypertension
- Broader context of potential instability resulting from noncardiovascular comorbidities (*e.g.*, frailty, dehydration, orthopedic limitations, chronic obstructive lung disease)

^a Personal supervision defined as physical presence in the room.

Reprinted from (31).

Box 4.2 Cognitive Skills Required to Competently Supervise Clinical Exercise Tests

- Knowledge of appropriate indications for exercise testing
- Knowledge of alternative physiologic cardiovascular tests
- Knowledge of appropriate contraindications, risks, and risk assessment of testing
- Knowledge to promptly recognize and treat complications of exercise testing
- Competence in cardiopulmonary resuscitation and successful completion of an American Heart Association–sponsored course in advanced cardiovascular life support and renewal on a regular basis
- Knowledge of various exercise protocols and indications for each
- Knowledge of basic cardiovascular and exercise physiology including hemodynamic response to exercise
- Knowledge of cardiac arrhythmias and the ability to recognize and treat serious arrhythmias (see [Appendix B](#))
- Knowledge of cardiovascular drugs and how they can affect exercise performance, hemodynamics, and the electrocardiogram (see [Appendix A](#))
- Knowledge of the effects of age and disease on hemodynamic and the electrocardiographic response to exercise
- Knowledge of principles and details of exercise testing including proper lead placement and skin preparation
- Knowledge of endpoints of exercise testing and indications to terminate exercise testing

Adapted from (32).

In addition to the test administrator (physician or nonphysician), at least one support technician should assist with testing (4). This individual should have knowledge and skills in obtaining informed consent and medical history, skin preparation and ECG electrode placement, equipment operation, the measurement of BP at rest and during exercise, and effective interaction skills (4).

Testing Mode and Protocol

The mode selected for the exercise test can impact the results and should be selected based on the purpose of the test and the individual being tested (3). In the United States, the treadmill is the most frequently used mode, whereas a cycle ergometer is more common in Europe. With the potential exception of highly trained cyclists, peak exercise capacity (*e.g.*, peak oxygen consumption [$\dot{V}O_{2peak}$]) is typically 5%–20% lower during a maximal exercise test performed on a cycle ergometer compared to a treadmill due to regional muscle fatigue (1–4). This range of 5%–20% suggests interstudy and interindividual variability. Based on anecdotal evidence, a 10%

difference is typically used by clinicians when comparing peak exercise responses between cycle ergometry and treadmill exercise. Optimally, the same exercise mode would be used at each time point when tracking an individual's response over time. Other exercise testing modes may be considered as needed, such as arm ergometry, combined (upper and lower body) ergometry, or seated stepping ergometry. These can be useful options for individuals with balance issues, amputation, extreme obesity, and other mobility deficiencies. Notably, when using modalities that require less muscle mass, there will be a concomitant lower peak oxygen consumption ($\dot{V}O_2$).

The use of a standardized exercise protocol, such as those shown in [Figure 4.1](#), represents a convenient and repeatable way to conduct the exercise test for both the individual and the clinician supervising the test. Many exercise laboratories use the same exercise protocol regardless of the individual tested because of convenience. However, guidelines have consistently recommended that the protocol should be individualized based on a given individual's age, exercise tolerance, medical history, and disease state including relevant symptoms ([2-5](#)). Protocols must be adapted as necessary for clinical exercise testing and should include a low-intensity warm-up phase followed by progressive, continuous exercise in which the demand is elevated to an individual's maximal level. The Bruce treadmill protocol is the most widely used exercise protocol in the United States ([33](#)). This will likely continue due to physician familiarity and the breadth of research based on the Bruce protocol ([27](#), [28](#), [34](#), [35](#)).

		TREADMILL PROTOCOLS												
METS	CYCLE ERGOMETER	RAMP		MODIFIED BRUCE 3 min Stages		BRUCE 3 min Stages		NAUGHTON 2 min Stages		MODIFIED NAUGHTON (CH 2 min Stages)				
	FOR 70 KG BODY WEIGHT 1 WATT = 6.1 Kpm/min			MPH	%GR	MPH	%GR							
21				6.0	22	6.0	22							
20														
19														
18				5.5	20	5.5	20							
17														
16														
15				Kpm/min	5.0	18	5.0					18		
14				1500	PER 30 SEC									MPH
					MPH	%GR								
13	1350	3.0	25.0	4.2	16	4.2	16			3.0				
		3.0	24.0							3.0				
12		3.0	23.0							3.0				
		3.0	22.0							3.0				
11	1200	3.0	21.0							3.0				
		3.0	20.0							3.0				
		3.0	19.0							3.0				
10	1050	3.0	18.0	3.4	14	3.4	14			3.0				
		3.0	17.0							3.0				
9	900	3.0	16.0					MPH	%GR	3.0				
		3.0	15.0					2	17.5	3.0				
		3.0	14.0							3.0				
8	750	3.0	13.0					2	14.0	3.0				
		3.0	12.0							3.0				
		3.0	11.0	2.5	12	2.5	12			3.0				
7	600	3.0	10.0							3.0				
		3.0	9.0							3.0				
		3.0	8.0							3.0				
6	450	3.0	7.0					2	10.5	2.0				
		3.0	6.0	1.7	10	1.7	10			2.0				
		3.0	5.0					2	7.0	2.0				
4	300	3.0	4.0							2.0				
		3.0	3.0					2	3.5	2.0				
		3.0	2.0							2.0				
3	150	3.0	1.0	1.7	5					1.5				
		3.0	0					2	0	1.5				
		2.5	0							1.5				
2		2.0	0	1.7	0					1.0				
		1.5	0					1	0	1.0				
		1.0	0							1.0				
1		0.5	0							1.0				

Figure 4.1 Common treadmill and stationary cycle ergometry protocols used in symptom-limited maximal exercise testing with exercise workload and metabolic demand. METs reflect the estimated value for each stage. CHF, chronic heart failure; kg, kilogram; KPM, kilopond meter; METs, metabolic

equivalents; min, minute; MPH, miles per hour; %GR, percent grade. Modified with permission from (3).

When performing a sign- and symptom-limited maximal exercise test, it is often recommended that the selected exercise testing protocol results in a total exercise duration between 8 and 12 minutes (3–5). To assist in the protocol selection, the individual's medical and PA history and symptomology should be considered. The aerobic requirements associated with the first stage of the Bruce protocol (~5 METs) and the large increases between stages (~3 METs) make it less than optimal for individuals with cardiovascular or pulmonary disease who may have a low functional capacity. As such, the Bruce protocol can result in extensive handrail support and an overestimation of the individual's peak exercise capacity based on the exercise duration or peak workload achieved (36–38). In response to these limitations, modifications of the Bruce protocol and other treadmill and cycle ergometry protocols have been developed, including individual-specific ramping protocols (37–40). [Figure 4.1](#) shows some common protocols and the estimated metabolic requirement for each.

Monitoring and Test Termination

Variables that are typically monitored during clinical exercise testing include HR, ECG, cardiac rhythm, BP, perceived exertion (*i.e.*, rating of perceived exertion [RPE]; Borg 6–20 scale), clinical signs/symptoms or clinician-observed symptoms, and self-reported symptoms suggestive of myocardial ischemia, inadequate blood perfusion, inadequate gas diffusion, and limitations in pulmonary ventilation (3–5).

Measurement of expired gases through open circuit spirometry during a CPET and oxygen saturation of blood through pulse oximetry and/or arterial blood gases may also be obtained when indicated (1, 2, 4, 18).

[Table 4.3](#) outlines best practices for monitoring during a symptom-limited maximal exercise test. A high-quality ECG tracing should be obtained during an exercise test. However, this requires more attention to preparation of the individual and lead placement than is typically required for a resting ECG. A thorough discussion of ECG preparation is provided in the AHA Exercise Standards for Exercise Testing and Training (3). HR and BP should be measured, and an ECG should be recorded regularly during the test (*e.g.*, each minute or each stage), at peak exercise, and

regularly through at least 6 minutes of recovery (3–5). Monitoring should continue into the recovery period until HR, BP, symptoms, and ECG changes stabilize. It can also be helpful to assess the individual’s RPE regularly during the exercise test and at peak exercise. Throughout the test, the ECG should be continuously monitored for repolarization changes suggestive of myocardial ischemia and dysrhythmias (3–5).

TABLE 4.3 Best Practices for Monitoring During a Symptom-Limited Maximal Exercise Test

Variable	Before Exercise Test	During Exercise Test	After Exercise Test
Electrocardiogram	Monitor continuously; record in supine position and position of exercise (<i>e.g.</i> , standing).	Monitor continuously; record during the last 5–10 s of each stage or every 2 min (ramp protocol).	Monitor continuously; record immediately postexercise, after 60 s of recovery and then every 2 min.
Heart rate ^a	Monitor continuously; record in supine position and position of exercise (<i>e.g.</i> , standing).	Monitor continuously; record during the last 5–10 s of each minute.	Monitor continuously; record during the last 5–10 s of each minute.
Blood pressure ^{a,b}	Monitor continuously; record in supine position and position of exercise (<i>e.g.</i> , standing).	Measure and record during the last 30–60 s of each stage or every 2 min (ramp protocol).	Measure and record immediately postexercise, after 60 s of recovery and then every 2 min.
Signs and symptoms	Monitor continuously; record as observed.	Monitor continuously; record as observed.	Monitor continuously; record as observed or as symptoms resolve.
Rating of perceived exertion	Explain scale.	Record during the last 5–10 s each stage or every 2 min (ramp protocol).	Obtain peak exercise shortly after exercise is terminated.

^a In addition, heart rate and blood pressure should be assessed and recorded whenever adverse symptoms or abnormal electrocardiogram changes occur.

^b An unchanged or decreasing systolic blood pressure with increasing workloads should be retaken (*i.e.*, verified immediately).

Adapted and used with permission from Brubaker PH, Kaminsky LA, Whaley MH. Coronary artery disease: essentials of prevention and rehabilitation programs. Champaign (IL): Human Kinetics; 2002. p. 364.

During the test and postexercise recovery, the clinician must monitor the individual for untoward symptoms, such as light-headedness, angina, dyspnea, claudication (if suspected by history), and fatigue (see [Table 2.1](#)) ([3–5](#)). In the case of chest pain that is suspected to be angina pectoris, the timing, character, magnitude, and resolution should be described ([4](#)). The appearance of symptoms should be correlated with HR, BP, and ECG abnormalities (when present). Standardized scales to assess RPE (see [Table 3.6](#) and [Figure 4.2](#)), angina, dyspnea, and claudication ([Figure 4.3](#)) are available. For each of these symptoms, a rating of three out of four is an indication to stop the test. Although scales to assess these symptoms have been recommended by the AHA ([4](#)), some clinical exercise testing laboratories use a 10-point visual analog pain scale (see [Figure 10.1](#)).

Borg CR10 Scale®

0	Nothing at all	
0.3		
0.5	Extremely weak	Just noticeable
0.7		
1	Very weak	
1.5		
2	Weak	Light
2.5		
3	Moderate	
4		
5	Strong	Heavy
6		
7	Very strong	
8		
9		
10	Extremely strong	“Maximal”
11		
}		
•	Absolute maximum	Highest possible

Figure 4.2 The Borg category-ratio scale. Printed with permission from Borg G, Borg E. The Borg CR scales folder. Hässelby (Sweden): Borg Perception; 2010.

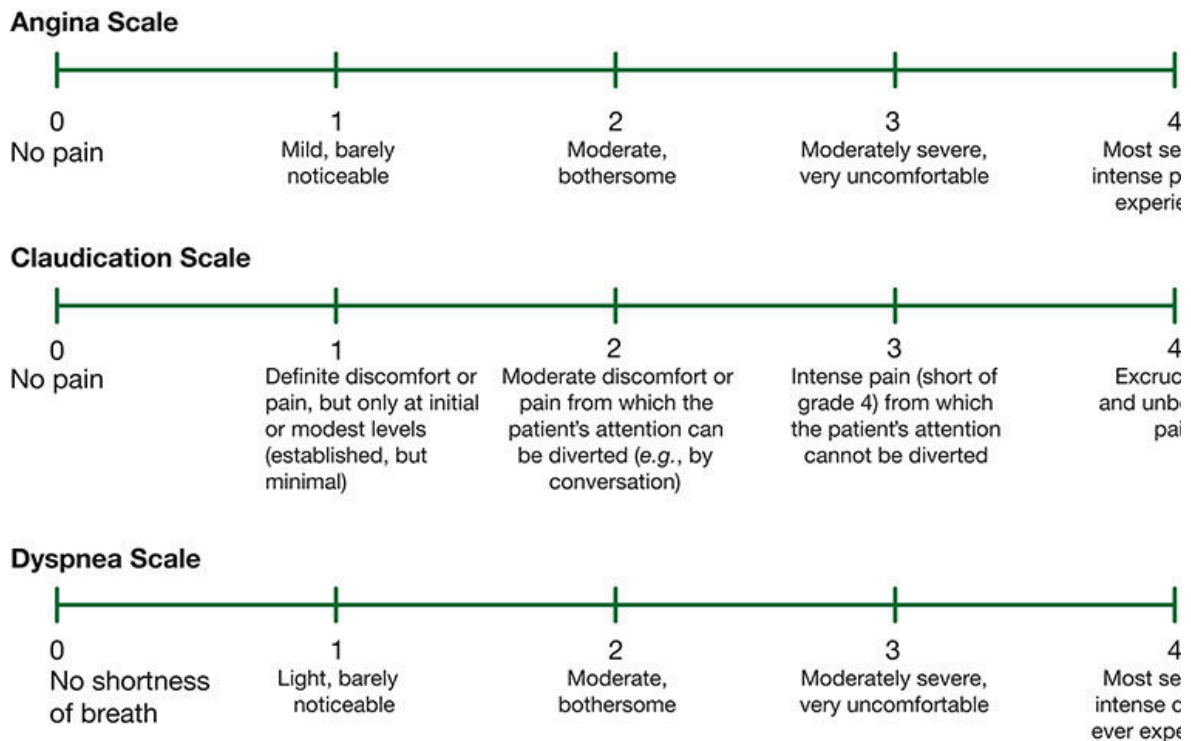


Figure 4.3 Frequently used scales for assessing the individual's level of angina (*top*), claudication (*middle*), and dyspnea (*bottom*).

The analysis of expired gas during a CPET overcomes the potential inaccuracies associated with estimating exercise capacity from peak workload (*e.g.*, treadmill speed and grade). The direct measurement of $\dot{V}O_2$ is the most accurate measure of exercise capacity and is a useful index of overall cardiopulmonary health ([1](#), [18](#), [24](#)). The CPET provides additional data that are not available without expired gas analysis, such as the respiratory exchange ratio (RER), ventilatory-derived anaerobic threshold (VAT), and the rate of change of minute ventilation (volume of expired air per unit time [$\dot{V}E$]) to change in volume of carbon dioxide exhaled ($\dot{V}CO_2$) during exercise (*i.e.*, $\dot{V}E/\dot{V}CO_2$ slope; an indicator of ventilatory efficiency). Responses from the ETT, and the CPET in particular, are useful in the differentiation between a normal response and cardiovascular and pulmonary limitations to exercise ([Table 4.4](#)). The CPET is also useful in assessing the cause of exertional dyspnea and in risk stratification of many individual groups, particularly those with heart failure ([1](#), [2](#), [18](#), [24](#)). There are several extensive resources available on CPET ([1](#), [18](#), [24](#), [41](#), [42](#)).

TABLE 4.4 Typical Exercise Test Responses in Differentiating a Normal Response and Patients Limited by Cardiovascular and Pulmonary Disease

Exercise Response	Condition		
	Deconditioning	Cardiovascular Disease	Pulmonary Disease
Typical limiting symptom	Fatigue/leg fatigue	May have angina	Dyspnea
Peak $\dot{V}O_2$	Decreased	Decreased	Decreased
Ventilatory threshold	Normal	Decreased	Decreased or normal
Oxygen pulse	Normal	Decreased or plateau	Normal
Exercise ECG	Normal	May be abnormal	Normal
$\dot{V}O_2$ /work rate slope	Normal	Decreased	Normal
Breathing reserve	Normal	Normal	<30%
O_2 saturation	Normal	Normal	May decrease
$\dot{V}E / \dot{V}CO_2$ slope	Normal	May be abnormal	Abnormal

Oxygen desaturation may be a cause of exertional dyspnea in some individuals. Although measurement of the partial pressure of arterial oxygen (P_aO_2) and partial pressure of carbon dioxide in arterial blood (P_aCO_2) via the measurement of arterial blood gases is the gold standard, pulse oximetry provides a noninvasive, indirect measure of arterial oxygen saturation (SpO_2). In individuals with pulmonary disease, direct measurements of percent saturation of arterial oxygen (SaO_2) correlate reasonably well with SpO_2 ($\pm 2\% - 3\%$), provided SpO_2 remains greater than 85% (1, 18). An absolute decrease in $SpO_2 \geq 5\%$ during exercise is considered an abnormal response suggestive of exercise-induced hypoxemia, and follow-up testing with arterial blood gases may be indicated (1, 18). A $SpO_2 \leq 80\%$ with signs or symptoms of hypoxemia is an indication to stop a test (1). The measurement of SpO_2 with pulse oximetry through a fingertip probe can be affected by low perfusion or low pulse wave, hemoglobin abnormalities, low oxygen saturation, very dark skin tone, nail

polish, acrylic nails ([43](#)), and movement during exercise. Alternate probe locations such as the earlobe or forehead can be helpful.

Termination criteria for clinical exercise testing have been established by the AHA and ACC ([Box 4.3](#)) ([3](#)). When the goal is a symptom-limited maximal exercise test, a predetermined intensity, such as 85% of the age-predicted maximal heart rate (HR_{max}), should not be used as a reason to end the test ([3](#), [5](#)). Failure to continue a test until the individual attains maximal exertion or a clinical limitation will result in an underestimation of the person's peak exercise capacity. Some clinicians view the achievement of 85% of the age-predicted HR_{max} as an adequate level of stress for revealing exertional ischemia. However, age-predicted HRs are highly inaccurate, and the sensitivity of exercise test results is increased when the HR achieved is greater than 85% of the predicted ([3](#)).

Box 4.3 Indications for Terminating a Symptom-Limited Maximal Exercise Test

Absolute Indications

- ST elevation (>1.0 mm) in leads without preexisting Q waves because of prior MI (other than aVR, aVL, or V₁)
- Drop in systolic blood pressure of greater than 10 mm Hg, despite an increase in workload, when accompanied by other evidence of ischemia
- Moderate-to-severe angina
- Central nervous system symptoms (*e.g.*, ataxia, dizziness, or near syncope)
- Signs of poor perfusion (cyanosis or pallor)
- Sustained ventricular tachycardia or other arrhythmia, including second- or third-degree atrioventricular block, that interferes with normal maintenance of cardiac output during exercise
- Technical difficulties monitoring the ECG or systolic blood pressure
- The individual's request to stop

Relative Indications

- Marked ST displacement (horizontal or downsloping of >2 mm, measured 60–80 ms after the J point in an individual with suspected ischemia)
- Drop in systolic blood pressure greater than 10 mm Hg (persistently below baseline) despite an increase in workload, *in the absence* of other evidence of ischemia
- Increasing chest pain
- Fatigue, shortness of breath, wheezing, leg cramps, or claudication
- Arrhythmias other than sustained ventricular tachycardia, including multifocal ectopy, ventricular triplets, supraventricular tachycardia, and bradyarrhythmias that have the potential to become more complex or to interfere with hemodynamic stability
- Exaggerated hypertensive response (systolic blood pressure >250 mm Hg or diastolic blood pressure >115 mm Hg)
- Development of bundle branch block that cannot be distinguished from ventricular tachycardia
- $\text{SpO}_2 \leq 80\%$ ([3](#))

ECG, electrocardiogram; MI, myocardial infarction; SpO_2 , percent saturation of arterial oxygen.

Postexercise

The sensitivity of the exercise test for the diagnosis of IHD can be maximized when the individual is placed in a supine position immediately following exercise ([3](#), [4](#)). Therefore, if the primary indication of the test is suspected IHD and nonsignificant repolarization changes are observed at peak exercise, then placing the individual in

the supine position without active recovery should be considered. However, exercise cessation can cause an excessive drop in venous return resulting in profound hypotension during recovery and ischemia secondary to decreased perfusion pressure into the myocardium. Therefore, the continuation of low-intensity active recovery during the postexercise period is often practiced to support venous return and hemodynamic stability. Each laboratory should develop standardized procedures for the postexercise recovery period (active vs. inactive and monitoring duration) with the laboratory's medical director that considers the indication for the exercise test and the individual's status during the test.

Safety

Although untoward events do occur, clinical exercise testing is very safe when performed by appropriately trained clinicians. The classic data of Rochmis and Blackburn ([44](#)) in 1971 reported a rate of serious complications (morbidity or mortality) of 34 events per 10,000 tests. Numerous studies have shown that the event rate has improved considerably in the years since. Excluding studies of individuals tested with a history of life-threatening ventricular arrhythmias, among 17 studies, serious complications (while the definition has varied, this has been typically defined as an event serious enough to require hospitalization) during clinical exercise tests ranged from 0 to 35 events per 10,000 tests, with rates typically higher among individuals known to have higher risk, such as individuals with heart failure ([31](#)). However, prior studies likely overestimate the risk of today's individuals given advances in medicine, such as the implantable cardioverter defibrillator ([31](#)). Current consensus suggests that an event rate of approximately 1–2 per 10,000 tests occurs with modern exercise testing.

In tests that are performed to assess the likelihood of IHD, some physicians might request that select individuals withhold medications that are known to limit the hemodynamic response to exercise (*e.g.*, β -adrenergic blocking agents) because they may limit test sensitivity ([3](#), [5](#)). However, for most test indications, individuals are encouraged to continue to take their medications on the day of testing ([5](#)). If the indication for the exercise test is to evaluate the effectiveness (*e.g.*, change in exercise capacity) of medical therapy, then individuals should be instructed to continue their normal medical regimen ([5](#)).

INTERPRETING THE CLINICAL EXERCISE TEST

Multiple factors should be considered during the interpretation of exercise test data including individual symptoms, ECG responses, exercise capacity, hemodynamic responses, and the combination of multiple responses, as reflected by exercise test scores such as the Duke Treadmill Score (discussed later).

Heart Rate Response

The normal HR response to incremental exercise is an increase with progressive workloads at a rate of $\approx 10 \text{ beats} \cdot \text{min}^{-1}$ per 1 MET (3). HR_{max} decreases with age and is attenuated in individuals on β -adrenergic blocking agents. Several equations have been published to predict HR_{max} in individuals who are not taking a β -adrenergic blocking agent (see [Table 5.3](#)) (3). All of these estimates have large interindividual variability with standard deviations of 10 beats or more (5, 45), thus limiting the utility of age-predicted HR during exercise testing.

Among individuals referred for testing secondary to IHD and in the absence of β -adrenergic blocking agents, failure to achieve an age-predicted $\text{HR}_{\text{max}} \geq 85\%$ in the presence of maximal effort is an indicator of chronotropic incompetence and is independently associated with increased risk of morbidity and mortality (3). The metabolic chronotropic reserve (MCR), also known as the chronotropic index, can be calculated from the ratio of the HR reserve to the metabolic reserve during submaximal exercise. The advantage of using the MCR is that it adjusts for age and fitness and appears to be unaffected by the exercise testing mode or protocol (26). An abnormal chronotropic response ($\text{MCR} < 0.8$) provides prognostic information that is independent of myocardial perfusion. The combination of a myocardial perfusion abnormality and an abnormal chronotropic response suggests a worse prognosis than either abnormality alone (46).

The rate of decline in HR following exercise (HR recovery) provides independent information related to prognosis (3, 30). A failure of the HR to decrease by at least 12 beats during the first minute or 22 beats by the end of the second minute of

postexercise recovery is strongly associated with an increased risk of mortality in individuals diagnosed with or at increased risk for IHD ([3](#), [30](#), [46](#), [47](#)). The failure of HR to recover adequately may be related to the inability of the parasympathetic nervous system to reassert vagal control of HR, which is known to predispose individuals to ventricular dysrhythmias ([46](#)).

Blood Pressure Response

The normal systolic blood pressure (SBP) response to exercise is an increase with increasing workloads at a rate of approximately 10 mm Hg per 1 MET ([3](#)). On average, this response is greater among men; increases with age; and is attenuated in individuals on vasodilators, calcium channel blockers, angiotensin-converting enzyme inhibitors, and α - or β -adrenergic blockers. Specific SBP responses are outlined in the following:

- Hypertensive response: An SBP greater than 250 mm Hg is a relative indication to stop a test (see [Box 4.3](#)) ([3](#)). An SBP $\geq$ 210 mm Hg in males and $\geq$ 190 mm Hg in females during exercise is considered an exaggerated response ([3](#)). A peak SBP greater than 250 mm Hg or an increase in SBP greater than 140 mm Hg during exercise above the pretest resting value is predictive of future resting hypertension ([48](#)).
- Hypotensive response: A decrease of SBP below the pretest resting value or by greater than 10 mm Hg after a preliminary increase, particularly in the presence of other indices of ischemia, is abnormal and often associated with myocardial ischemia, left ventricular dysfunction, and an increased risk of subsequent cardiac events ([3](#)). Either of these responses are indications to stop the test (see [Box 4.3](#)).
- Blunted response: In individuals with a limited ability to augment cardiac output, the response of SBP during exercise will be slower compared to normal.
- Postexercise response: SBP typically returns to preexercise levels or lower by 6 minutes of recovery ([3](#)). Studies have demonstrated that a delay in the recovery of SBP is highly related both to ischemic abnormalities and to a poor prognosis ([49](#), [50](#)).

There is normally no change or a slight decrease in diastolic blood pressure (DBP) during an exercise test. A peak DBP greater than 90 mm Hg or an increase in DBP

greater than 10 mm Hg during exercise above the pretest resting value is considered an abnormal response (3) and may occur with exertional ischemia. A DBP greater than 115 mm Hg is an exaggerated response and a relative indication to stop a test (see [Box 4.3](#)) (3).

Rate-Pressure Product

Rate-pressure product (RPP) (also known as *double product*) is calculated by multiplying the values for HR and SBP that occur at the same time during rest or exercise. RPP is a surrogate for myocardial oxygen uptake. There is a linear relationship between myocardial oxygen uptake and both coronary blood flow and exercise intensity (3). Coronary blood flow increases due to increased myocardial oxygen demand as a result of increases in HR and myocardial contractility. If coronary blood supply is impaired, which can occur in obstructive IHD, then signs or symptoms of myocardial ischemia may be present. The point during exercise when this occurs is the ischemic threshold. RPP is a repeatable estimate of the ischemic threshold and is more reliable than external workload. The normal range for peak RPP is 25,000–40,000 mm Hg · beats⁻¹ · min⁻¹ (3). RPP at peak exercise and at the ischemic threshold (when applicable) should be reported.

Electrocardiogram

The normal response of the ECG (see [Appendix B](#)) during exercise includes the following (3):

- P wave: increased magnitude among inferior leads
- PR segment: shortens and slopes downward among inferior leads
- QRS: duration decreases, septal Q waves increase among lateral leads, R waves decrease, and S waves increase among inferior leads
- J point (J junction): depresses below the isoelectric line with upsloping ST segments that reach the isoelectric line within 80 ms
- T wave: decreases amplitude in early exercise, returns to preexercise amplitude at higher exercise intensities, and may exceed preexercise amplitude in recovery
- QT interval: absolute QT interval decreases. The QT interval corrected for HR increases with early exercise and then decreases at higher HRs

ST-segment changes (*i.e.*, depression and elevation) are the cornerstone of the exercise test clinically and are associated with myocardial ischemia and injury. The interpretation of ST segments may be affected by the resting ECG configuration and the presence of digitalis therapy and should be interpreted in the context of the pretest probability of IHD ([3](#), [5](#)). Considerations that may indicate that an exercise test with only ECG would be inadequate for the diagnosis of IHD are shown in [Box 4.4](#).

Box 4.4 Considerations That May Necessitate Adjunctive Imaging When the Indication Is the Assessment of Ischemic Heart Disease (5)

- Resting ST-segment depression greater than 1.0 mm
- Ventricular paced rhythm
- Left ventricular hypertrophy with repolarization abnormalities
- Left bundle-branch block
- Leads V1–V3 will not be interpretable with right bundle-branch block.
- Wolff-Parkinson-White
- Digitalis therapy

Abnormal responses of the ST segment during exercise include the following (3):

- To be clinically meaningful, ST-segment depression or elevation should be present in at least three consecutive cardiac cycles within the same lead. The level of the ST segment should be compared relative to the end of the PR segment. Automated computer-averaged complexes should not be relied on for interpretation and should be visually confirmed by the raw ECG.
- Horizontal or downsloping ST-segment depression ≥ 1.0 mm (0.1 mV) at 80 ms after the J point is a strong indicator of myocardial ischemia.
- Clinically significant ST-segment depression that occurs during postexercise recovery is an indicator of myocardial ischemia.
- ST-segment depression at a low workload or low RPP is associated with a worse prognosis and increased likelihood for multivessel disease.
- When ST-segment depression is present on the upright resting ECG, only additional ST-segment depression during exercise is considered for ischemia.
- When ST-segment elevation is present in the upright resting ECG, only ST-segment depression below the isoelectric line during exercise is considered for ischemia.
- Upsloping ST-segment depression ≥ 2.0 mm (0.2 mV) at 80 ms after the J point may represent myocardial ischemia, especially in the presence of angina. However, this response has a low positive predictive value; it is often categorized as equivocal.
- Among individuals after myocardial infarction (MI), exercise-induced ST-segment elevation (>1.0 mm or >0.1 mV for 60 ms) in leads with Q waves is an abnormal

response and may represent reversible ischemia or wall motion abnormalities.

- Among individuals without prior MI, exercise-induced ST-segment elevation most often represents transient combined endocardial and subepicardial ischemia but may also be due to acute coronary spasm.
- Repolarization changes (ST-segment depression or T-wave inversion) that normalize with exercise may represent exercise-induced myocardial ischemia but is considered a normal response in young individuals with early repolarization on the resting ECG.

Dysrhythmias

In general, dysrhythmias that increase in frequency or complexity with progressive exercise intensity and are associated with ischemia or hemodynamic instability are more likely to cause a poor outcome than isolated dysrhythmias (3). The clinical significance of ventricular ectopy during exercise has varied. Although ventricular ectopy is more common with some pathologies, such as cardiomyopathy, in general, frequent and complex ventricular ectopy during exercise and especially in recovery are associated with an increased risk for cardiac arrest (3). Sustained ventricular tachycardia is an absolute criterion to terminate a test. There are several relative termination criteria related to atrial and ventricular dysrhythmias and blocks that should be considered based on the presence of signs or symptoms of myocardial ischemia or inadequate perfusion (see [Box 4.3](#)) (3).

Symptoms

Symptoms that are consistent with myocardial ischemia (*e.g.*, angina, dyspnea) or hemodynamic instability (*e.g.*, light-headedness) should be noted and considered in context with ECG, HR, and BP abnormalities (when present). Exercise-induced angina is associated with an increased risk for IHD (3). The occurrence of angina during exercise is generally considered to supersede other exercise test responses as an indicator of IHD, but the risk for IHD is greater when angina and ST-segment depression occur together (3). It is important to recognize that dyspnea can be an anginal equivalent. Compared to leg fatigue or general exhaustion, an exercise test that is limited by dyspnea has been associated with a worse prognosis (3).

Exercise Capacity

Evaluating exercise capacity is an important component of exercise testing. A high exercise capacity is generally indicative of good overall cardiopulmonary health and the absence of serious limitations in left ventricular function. Over the past three decades, many studies have been published demonstrating the importance of exercise capacity relative to prognosis among both individuals with and without cardiovascular disease ([1](#), [2](#), [18](#), [21](#), [24](#), [34](#), [35](#), [41](#)). Both absolute and age- and sex-normalized exercise capacities are strongly related to survival ([2](#), [21](#), [41](#)). A significant issue relative to exercise capacity is the imprecision of estimating exercise capacity from exercise time or peak workload ([2](#)). The error in estimating exercise capacity from various published prediction equations is at least ± 1 MET ([37–40](#), [51](#)). This measurement error is less meaningful in young, healthy individuals with a peak exercise capacity greater than 13 METs (7%–8% error), but more significant in individuals with reduced exercise capacities typical of those observed among individuals with cardiovascular or pulmonary disease (4–8 METs; 13%–25% error). Estimating exercise capacity on a treadmill is confounded by several factors, including treadmill experience, walking efficiency, presence of disease, the exercise protocol used, and the use of handrails for support. In addition, there is an inherent error associated with regression equations developed to estimate $\dot{V}O_{2\max}$ by extrapolation of $\dot{V}O_{2\max}$ achieved at submaximal, steady state workloads ([2](#), [5](#), [36](#), [41](#), [51](#)).

Together, these factors tend to result in an overestimation of exercise capacity ([2](#), [36](#)). Although equations exist to predict exercise capacity from an exercise test using handrail support, the standard error of the estimate remains large ([36](#)). The safety of treadmill walking is always an important consideration, and allowing an individual to use the handrails should be determined on a case-by-case basis. Because of these limitations associated with estimating exercise capacity from work rate, it is important to state when exercise capacity is measured directly (*i.e.*, $\dot{V}O_{2\text{peak}}$) or estimated from the work rate. More accurate equations for treadmill and cycle ergometry have recently been developed based on the Fitness Registry and the Importance of Exercise National Database (FRIEND). The overall error of the FRIEND equations was significantly lower for both treadmill and cycle ergometer testing compared to the existing equations ([52](#), [53](#)).

In addition to describing an individual's exercise capacity as estimated peak METs or measured $\dot{V}O_{2\text{peak}}$, exercise capacity is frequently expressed relative to age- and sex-based normal standards ([1](#), [18](#), [41](#), [54](#)). This is especially true for $\dot{V}O_{2\text{peak}}$. Expressing $\dot{V}O_{2\text{peak}}$ as a percentage of age- and sex-predicted normal value is advantageous because exercise capacity declines with age and is higher among males compared to females. Accurate knowledge of an individual's $\dot{V}O_{2\text{peak}}$ relative to their peers provides a context with which to optimize the prognostic applications of the test and the assessment of therapy and provides support for activity counseling. Several equations exist to estimate $\dot{V}O_{2\text{peak}}$ based on select demographics (*e.g.*, sex, age, height, weight) ([1](#), [18](#), [54](#)). These equations vary due to population specificity, and thus, it can be problematic to determine precisely which equation is best for a particular individual. Reference tables are also available that provide percentile rankings for an individual's measured exercise capacity by sex and age categories (for treadmill, see [Table 3.8](#); for cycle ergometer, see [Table 3.9](#)) ([55](#)), and nomograms have been developed to enable estimation of an individual's age-predicted exercise capacity at a glance ([56–58](#)).

The most widely used prediction equations for directly measured $\dot{V}O_{2\text{peak}}$ were developed by Hansen, Sue, and Wasserman (commonly termed the *Wasserman equations*) ([59](#)). Although these equations have been commonly used for more than 30 years, they were derived from a relatively limited sample and often do not function well in particular populations ([18](#), [54](#), [60](#), [61](#)). When expressing exercise capacity as a percentage of age-predicted normal value, it is important that the equation used is derived from a population representative of the individual tested, that it reflects whether exercise capacity was measured directly or estimated from the work rate, that it reflects the appropriate exercise mode used (cycle ergometer or treadmill), and that exercise capacity is expressed appropriately ($\dot{V}O_{2\text{peak}}$ in $\text{mL} \cdot \text{min}^{-1}$, $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, or estimated METs). Examples of commonly used regression equations for age-predicted exercise capacity are presented in [Box 4.5](#).

Box 4.5 Examples of Regression Equations for Age-Predicted Normal Standards for Exercise Capacity

Directly measured versus estimated values are indicated for each equation.

Hansen, Sue, and Wasserman equations for measured $\dot{V}O_{2\text{peak}}$ (59):

	Mode	Overweight	Predicted $\dot{V}O_{2\text{max}}$ ($\text{mL} \cdot \text{min}^{-1}$)
Males	Cycle ^a	No	$\text{Wt} \times [50.72 - (0.372 \times \text{A})]$
		Yes	$[0.79 \times (\text{Ht} - 60.7)] \times [50.72 - (0.372 \times \text{A})]$
	Treadmill ^b	No	$\text{Wt} \times [56.36 - (0.413 \times \text{A})]$
		Yes	$[0.79 \times (\text{Ht} - 60.7)] \times [56.36 - (0.413 \times \text{A})]$
Females	Cycle ^a	No	$(42.8 + \text{Wt}) \times [22.78 - (0.17 \times \text{A})]$
		Yes	$\text{Ht} \times [14.81 - (0.11 \times \text{A})]$
	Treadmill ^c	No	$\text{Wt} \times [44.37 - (0.413 \times \text{A})]$
		Yes	$[0.79 \times (\text{Ht} - 68.2)] \times [44.37 - (0.413 \times \text{A})]$

^a Overweight is $\text{Wt} > [0.79 \times (\text{Ht} - 60.7)]$.

^b Overweight is $\text{Wt} > [0.65 \times (\text{Ht} - 42.8)]$.

^c Overweight is $\text{Wt} > [0.79 \times (\text{Ht} - 68.2)]$.

Fitness Registry and the Importance of Exercise National Data Base (FRIEND) equation for measured $\dot{V}O_{2\text{peak}}$ (54):

$\dot{V}O_{2\text{max, mL O}_2 \cdot \text{kg}^{-1} \cdot \text{min}^{-1}}(\text{treadmill or bike; single equation}) = 45.2 - 0.35 \times \text{age} - 10.9 \times \text{sex} (\text{male} = 1; \text{female} = 2) - 0.15 \times \text{weight} (\text{lb}) + 0.68 \times \text{height} (\text{in}) - 0.46 \times \text{exercise mode} (\text{treadmill} = 1; \text{bike} = 2)$

StartLayout 1st Row Body density equals 1.112 minus 0.00043499 left parenthesis s u m of seven skinfolds right parenthesis 2nd Row plus 0.00000055 left parenthesis s u m of seven skinfolds right parenthesis squared 3rd Row minus 0.00028826 left parenthesis a g e right parenthesis left bracket upper S upper E upper E 0.008 or tilde 3.5 percent sign f a t right bracket EndLayout

Percentage exercise capacity achieved in estimated METs among male veterans (58):

All individuals:	METs = $14.7 - 0.11 (\text{age})$
Active individuals:	METs = $16.4 - 0.13 (\text{age})$
Sedentary individuals:	METs = $11.9 - 0.07 (\text{age})$

Percentage predicted $\dot{V}O_{2\text{peak}}$ in males and females with a medical or surgical diagnosis ($\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) (52):

Male: $33.97 - 0.242 \times \text{age}$

Female: $21.69 - 0.116 \times \text{age}$

Percentage predicted exercise capacity achieved in healthy females (estimated METs) (57):

METs = $14.7 - (0.13 \times \text{age})$.

A, age in years; Ht, height in cm; Wt, weight in kg.

Cardiopulmonary Exercise Testing

A major advantage of measuring ventilatory gas exchange during exercise is a more accurate measurement of exercise capacity. Several thorough reviews on CPET are available ([1](#), [18](#), [24](#), [41](#), [42](#), [62](#)). In addition to a more accurate measurement of exercise capacity, the CPET is an integral tool for clinicians. It can provide important insight when evaluating the risk of perioperative death and postoperative complications while also supporting preoperative planning algorithms, developing management strategies and in disease prognostication. Peak $\dot{V}O_2$ was recognized more than 30 years ago as an important selection criterion to help determine which patients would benefit most from cardiac transplantation. Patients with heart failure who achieved a peak $\dot{V}O_2$ value $\geq 14 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ had survival rates that were similar to those who received transplantation, while patients who achieved less than $14 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ had very low survival rates ([63](#)). This suggested that the scarce donor hearts available for transplantation are best reserved for patients achieving less than $14 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. The threshold of $14 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ has been confirmed in many studies since that time and has been applied in heart failure guidelines as part of determining which patients should be listed for transplantation ([18](#), [64](#), [65](#)). Current guidelines suggest that this threshold be modified to $12 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ for patients who are receiving β -blockers ([64](#), [65](#)). In recent years, other CPET responses, such as indices of ventilatory inefficiency (e.g., the $\dot{V}E/\dot{V}CO_2$ slope, oxygen uptake efficiency slope, oscillatory breathing) are strong predictors of adverse outcomes ([18](#), [24](#), [62](#), [65](#)). These responses have also been shown to complement peak $\dot{V}O_2$ in estimating risk, and multivariable scores using CPET have been applied and shown to optimize risk stratification ([18](#), [24](#), [62](#), [66](#), [67](#)). Other important variables that can be determined through the measurement of respiratory gas exchange include the VAT, oxygen pulse, the slope of the change in work rate to changes in oxygen uptake, partial pressure of end-tidal CO_2 , breathing reserve, and the RER.

Guidelines for various surgical interventions have increasingly applied peak $\dot{V}O_2$ and other CPET responses in the context of assessing the suitability of patients for

surgery, although specific cut points for different surgeries or transplantation for organs other than the heart are lacking. Across the spectrum of surgical interventions, patients with higher peak $\dot{V}O_2$ values are better equipped to handle the physiological demands of transplantation or surgery ([68](#), [69](#)). This has engendered the concept of “prehabilitation,” which involves exercise intervention to optimize peak $\dot{V}O_2$ and other CPET responses before surgery to improve short- and long-term outcomes ([69](#), [70](#)).

The CPET is particularly useful in identifying whether the cause of dyspnea has cardiac or pulmonary etiology (see [Table 4.4](#)) ([1](#), [18](#)). For example, patients with cardiovascular disease generally have a breathing reserve of greater than 30% (the ratio of maximal exercise ventilation to maximal voluntary ventilation at rest), whereas patients with a pulmonary limitation to exercise tend to have little if any breathing reserve. Algorithms from CPET-derived data are available to assist the interpretation of the CPET for both diagnostic ([71](#)) and prognostic ([72](#)) purposes. Common CPET responses and their applications are shown in [Table 4.5](#).

TABLE 4.5 Common CPET-Derived Responses and Their Applications

CPET Response	Application
Peak $\dot{V}O_2$	Defines the limits of the cardiopulmonary system; gold standard for fitness; prognostic marker
Breathing reserve Absolute MVV – $\dot{V}E_{\max}$ Relative ($\dot{V}E_{\max}/MVV$) $\times$ 100%	Marker of ventilatory limitation; distinguish between cardiac and pulmonary limitation to exercise
$\dot{V}E / \dot{V}CO_2$ slope	Ventilatory efficiency; prognostic marker
Oxygen uptake efficiency slope	Ventilatory efficiency; prognostic marker
Oscillatory ventilation	Impaired ventilatory control and ventilatory efficiency
O_2 pulse	Surrogate for stroke volume and ventricular function
End-tidal CO_2	Ventilation-perfusion mismatch, impaired lung function, and control of ventilation
Ventilatory threshold	Prognostic marker, ability to perform ADLs, distinguish cardiac and pulmonary limitation to exercise
Respiratory exchange ratio	Indicator of effort proportion of fuel used
Ventilatory equivalent for O_2	Ventilatory efficiency, detection of VAT

MVV, maximal voluntary ventilation; VAT, ventilatory anaerobic threshold; $\dot{V}E$, minute ventilation.

Gas Exchange Data Sampling

Since the advent of computerized metabolic systems in the 1970s and 1980s, there has been a great deal of discussion related to how to optimally express data derived from the CPET. Once limited to mixing chambers or Douglas bags to collect data each minute, computerized systems and rapidly responding gas analyzers enhanced precision and permitted the acquisition of breath-by-breath data. These technological advances have facilitated precision and convenience, but they have led to confusion regarding data sampling. Collecting data on each breath is precise, and it has particular applications such as research on $\dot{V}O_2$ kinetics or when end-tidal pressures are needed, but it is inappropriate for clinical use because of the variability inherent

in each breath. There are numerous breath averaging techniques in use by the various metabolic systems, and many research groups have compared these sampling techniques to find the proper balance between precision and variability. Guidelines have suggested that 20- or 30-second “rolling” samples should be used that are listed frequently (such as every 10 seconds, with each sample representing, for example, an overlapping 30 seconds of data) ([18](#)). This smooths the data, yet permits adequate resolution for choosing test endpoints, the VAT, and other relevant analysis points. Regardless of the sample chosen, clinicians should report the sampling interval used, and the intervals should be consistent when serially testing patients or throughout a given trial when studying an intervention.

Maximal Versus Peak Cardiorespiratory Stress

When an exercise test is performed as part of the evaluation of IHD, individuals should be encouraged to exercise to their maximal level of exertion or until a clinical indication to stop the test is observed. However, the determination of what constitutes “maximal” effort, although important for interpreting test results, can be difficult. Various criteria have been used to confirm that a maximal effort has been elicited during a GXT:

- A plateau in $\dot{V}O_2$ (or failure to increase $\dot{V}O_2$ by $150 \text{ mL} \cdot \text{min}^{-1}$) with increased workload ([73](#), [74](#)). This criterion has fallen out of favor because a plateau is not consistently observed during maximal exercise testing, particularly in individuals with cardiovascular or pulmonary disease ([75](#), [76](#)).
- Failure of HR to increase with increases in workload.
- A postexercise venous lactate concentration greater than $8.0 \text{ mmol} \cdot \text{L}^{-1}$.
- A RPE at peak exercise greater than 17 on the 6–20 scale or greater than 7 on the 0–10 scale.
- A peak RER ≥ 1.10 . Peak RER is perhaps the most accurate and objective noninvasive indicator of individual effort during a GXT ([18](#)).

There is no consensus on the number of criteria that should be met in order to call a test maximal ([76](#)). In addition, interindividual and interprotocol variability may limit the validity of these criteria ([76](#)). In the absence of data supporting that an individual reached their physiologic maximum, data at maximal exercise are

commonly described as “peak” (*e.g.*, HR_{peak} , $\dot{\text{V}}\text{O2peak}$) instead of “maximal” (*e.g.*, HR_{max} , $\dot{\text{V}}\text{O2max}$) ([1–3](#)).

DIAGNOSTIC VALUE OF EXERCISE TESTING FOR THE DETECTION OF ISCHEMIC HEART DISEASE

The diagnostic value of the clinical exercise test for the detection of IHD is influenced by the principles of conditional probability (*i.e.*, the probability of identifying an individual with IHD given the probability of IHD in the underlying population). Key metrics that describe the diagnostic value of exercise testing (and other diagnostic tests) are the sensitivity, specificity, and predictive value of the test procedure. Each of these is affected by the prevalence of IHD in the population tested ([5](#)).

Sensitivity, Specificity, and Predictive Value

Sensitivity refers to the ability to positively identify individuals who truly have IHD. Exercise ECG sensitivity for the detection of IHD has traditionally been based on angiographic evidence of a coronary artery stenosis $\geq 70\%$ in at least one vessel. In a true-positive (TP) test, the test is positive for myocardial ischemia (*e.g.*, ≥ 1.0 mm of horizontal or downsloping ST-segment depression), and the individual truly has IHD. Conversely, in a false-negative (FN) test, the test is negative for myocardial ischemia, but the individual truly has IHD ([5](#)).

Common factors that contribute to FN exercise tests are summarized in [Box 4.6](#). The sensitivity of an exercise test is decreased by inadequate myocardial stress, medications that attenuate the cardiac demand to exercise or reduce myocardial ischemia (*e.g.*, β -adrenergic blockers, nitrates, calcium channel blocking agents), and insufficient ECG lead monitoring ([3](#), [5](#)). Preexisting ECG changes such as left ventricular hypertrophy, LBBB, or the preexcitation syndrome (W-P-W) limit the ability to interpret exercise-induced ST-segment changes ([5](#)).

Box 4.6 Causes of False-Negative Symptom-Limited Maximal Exercise Test Results for the Diagnosis of Ischemic Heart Disease

- Failure to reach an ischemic threshold
- Monitoring an insufficient number of leads to detect ECG changes
- Failure to recognize non-ECG signs and symptoms that may be associated with underlying CVD (*e.g.*, exertional hypotension)
- Angiographically significant CVD compensated by collateral circulation
- Musculoskeletal limitations to exercise preceding cardiac abnormalities
- Technical or observer error

CVD, cardiovascular disease; ECG, electrocardiogram.

Specificity refers to the ability to correctly identify individuals who do not have IHD. In a true-negative (TN) test, the test is negative for myocardial ischemia and the individual is free of IHD ([5](#)). Conversely, in a false-positive (FP) test result, the test is positive for myocardial ischemia, but the individual does not have IHD. Conditions that may cause an abnormal exercise ECG response in the absence of significant IHD are shown in [Box 4.7](#) ([5](#)).

Box 4.7 Causes of False-Positive Symptom-Limited Maximal Exercise Test Results for the Diagnosis of Ischemic Heart Disease

- ST-segment depression greater than 1.0 mm at rest
- Left ventricular hypertrophy
- Accelerated conduction defects (*e.g.*, Wolff-Parkinson-White syndrome)
- Digitalis therapy
- Nonischemic cardiomyopathy
- Hypokalemia
- Vasoregulatory abnormalities
- Mitral valve prolapse
- Pericardial disorders
- Technical or observer error
- Coronary spasm
- Anemia

Reported values for the specificity and sensitivity of exercise testing with ECG vary because of the differences in disease prevalence of the cohort studied, test protocols, ECG criteria for a positive test, and the angiographic definition of IHD. In studies that accounted for these variables, the pooled results show a sensitivity of 68% and specificity of 77% ([5](#)). Sensitivity, however, is somewhat lower, and specificity is higher when workup bias (*i.e.*, only assessing individuals with a higher likelihood for IHD) is removed ([77](#)).

The *predictive value* of clinical exercise testing is a measure of how accurately a test result (positive or negative) correctly identifies the presence or absence of IHD in individuals and is calculated from sensitivity and specificity. The positive predictive value is the percentage of individuals with an abnormal test who truly have IHD. The negative predictive value is the percentage of individuals with a negative test who are free of IHD ([5](#)). Definitions of diagnostic test characteristics, including sensitivity, specificity, and predictive value, are shown in [Box 4.8](#).

Box 4.8 Sensitivity, Specificity, and Predictive Value of Symptom-Limited Maximal Exercise Testing for the Diagnosis of Ischemic Heart Disease

$$\text{Sensitivity} = [\text{TP}/(\text{TP} + \text{FN})] \times 100$$

- The percentage of individuals with IHD who have a positive test

$$\text{Specificity} = [\text{TN}/(\text{TN} + \text{FP})] \times 100$$

- The percentage of individuals without IHD who have a negative test

$$\text{Positive predictive value} = [\text{TP}/(\text{TP} + \text{FP})] \times 100$$

- The percentage of positive tests that correctly identify individuals with IHD

$$\text{Negative predictive value} = [\text{TN}/(\text{TN} + \text{FN})] \times 100$$

- The percentage of negative tests that correctly identify individuals without IHD

FN, false negative; FP, false positive; IHD, ischemic heart disease; TN, true negative; TP, true positive.

Clinical Exercise Test Data and Prognosis

First introduced in 1987 when the Duke Treadmill Score was published ([78](#), [79](#)), the implementation of various exercise test scores that combine information derived during the exercise test into a single prognostic estimate has gained popularity. The most widely applied prognostic score is the Duke Treadmill Score or the related Duke Treadmill Nomogram ([3](#), [5](#)). Both are appropriate for individuals with or without a history of IHD being considered for coronary angiography without a history of an MI or revascularization procedure. The Duke Score/Nomogram ([Figure 4.4](#)) considers exercise capacity, the magnitude of ST-segment depression, and the presence and severity of angina pectoris. The calculated score is related to annual and 5-year survival rates and allows the categorization of individuals into low-, moderate-, and high-risk subgroups. This categorization may help the physician choose between more conservative or more aggressive therapies. Physicians may also use prognostic estimates based on other hemodynamic findings, such as chronotropic incompetence

or an abnormal HR recovery, to guide their clinical decisions (3, 5). Each of these abnormalities from exercise testing contributes independent prognostic information. Although there is a general belief that physicians informally integrate much of this information without the specific calculation of an exercise test score, estimates of the presence of IHD provided by scores are superior to physician estimates and analysis of ST-segment changes alone (80).

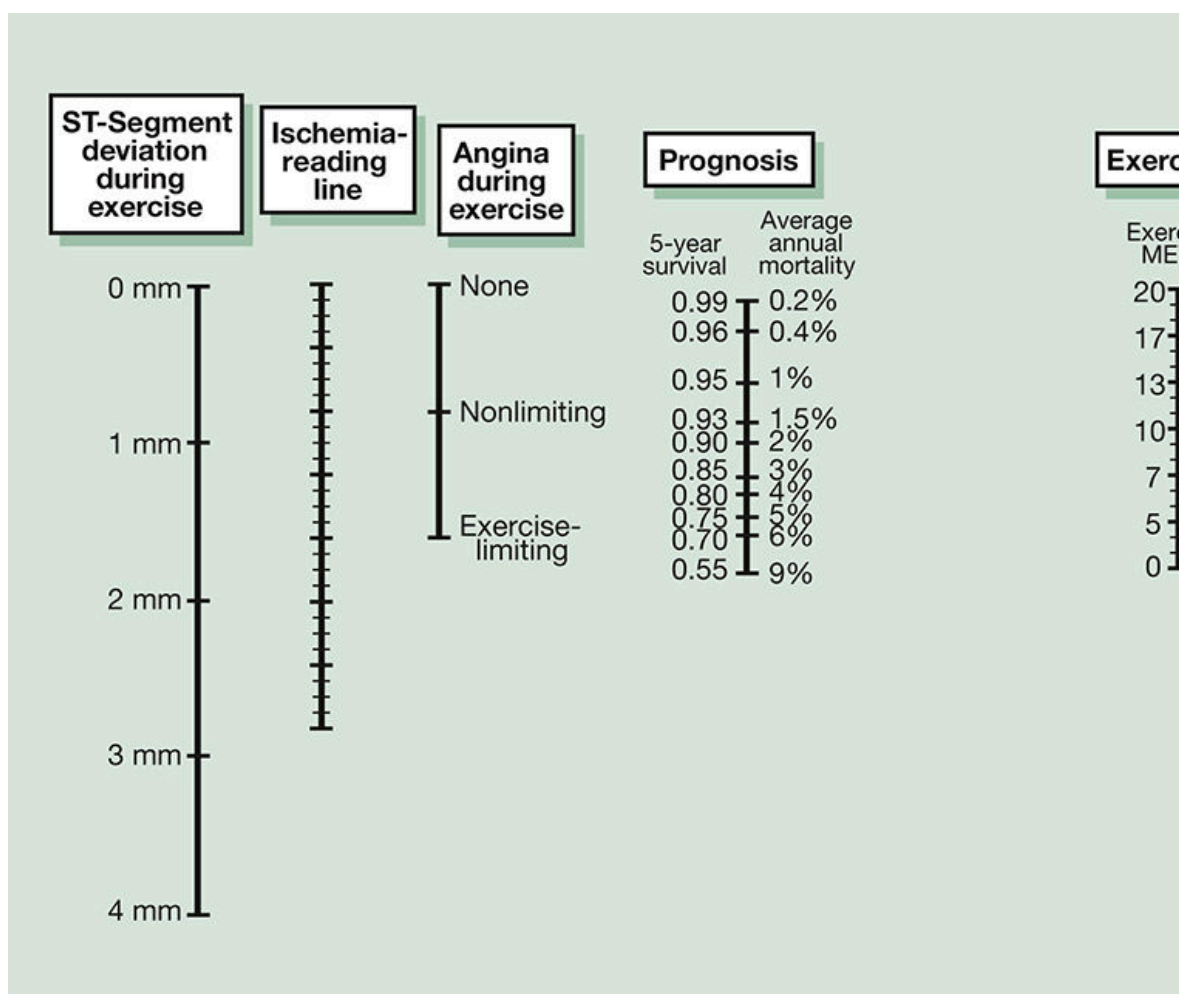


Figure 4.4 The Duke Nomogram. This nomogram uses five variables to estimate prognosis for a given individual. First, the observed amount of ST-segment depression is marked on the ST-segment deviation line. Second, the observed degree of angina is marked on the line for angina, and these two points are connected. Third, the point where this line intersects the ischemia reading line is noted. Fourth, the observed exercise tolerance is marked on the line for exercise capacity. Finally, the mark on the ischemia reading line is connected to the mark on the exercise capacity line, and the estimated 5-year survival or average annual mortality rate is read from the point at

which this line intersects the prognosis scale. Reprinted with permission from ([79](#)).

CLINICAL EXERCISE TESTS WITH IMAGING

When the resting ECG is abnormal, exercise testing may be coupled with other techniques designed to either augment the information provided by the ECG or to replace the ECG when resting abnormalities (see [Box 4.4](#)) make evaluation of changes during exercise impossible. Various radioisotopes can be used to evaluate the presence of a myocardial perfusion abnormality, which is the initiating event in exertional ischemia and the beginning of the “ischemic cascade,” or abnormalities of ventricular function that often occur with MI or myocardial ischemia ([3, 5](#)). When exercise testing is coupled with myocardial perfusion imaging (*e.g.*, nuclear stress test) or echocardiography, all other aspects of the exercise test should remain the same, including HR and BP monitoring during and after exercise, symptom evaluation, rhythm monitoring, and symptom-limited maximal exertion.

Myocardial perfusion imaging can be performed with a variety of agents and imaging approaches, although the two most common isotopes are ^{201}Tl and $^{99\text{m}}\text{Tc}$ sestamibi (Cardiolite). Delivery of the isotope is proportional to coronary blood flow. These agents cross cell membranes of metabolically active tissue either actively (thallium) or passively (sestamibi). In the case of an MI, the isotope does not cross the cell membrane of the necrotic tissue, and thus, a permanent reduction of isotope activity is observed on the image, referred to as a nonreversible, or fixed, perfusion defect. In the case of exertional myocardial ischemia, the tissue uptake in the ischemic region is reduced during exercise by virtue of the relative reduction of blood flow (and thus isotope) to the ischemic tissue. This abnormality is reversed when myocardial perfusion is evaluated at rest. This is called a reversible, or transient, perfusion defect and is diagnostic of exertional myocardial ischemia.

Echocardiography can also be used as an adjunct during an exercise test and is often called *stress echocardiography*. An echocardiographic examination allows the evaluation of wall motion, wall thickness, and valve function. Although it is theoretically possible to perform echocardiography during the course of upright cycle ergometer exercise, it is technically challenging. Typical practice is to have the individual lie down on their left side immediately following completion of the exercise

test (treadmill or upright cycle ergometer) or for exercise to involve recumbent cycle ergometry. This allows optimization of the echocardiographic window to the heart. Regional wall motion is assessed for various segments of the left ventricle. Deterioration in regional wall motion with exercise (compared to rest) is a sign of myocardial ischemia. Left ventricular ejection fraction is also measured before and after exercise.

Imaging techniques, such as radionuclide myocardial perfusion imaging and echocardiography, allow the physician to identify the location and magnitude of myocardial ischemia. For individuals who are incapable of exercising, it is also possible to perform either myocardial perfusion imaging or stress echocardiography with pharmacologic stress. These techniques are beyond the scope of this chapter.

FIELD WALKING TESTS

This chapter focuses on the traditional sign- and symptom-limited, maximal exercise test with ECG monitoring that is performed in a clinical laboratory, often with a treadmill or cycle ergometer. However, non-laboratory-based clinical exercise tests are also frequently used in individuals with chronic disease (*see [Chapter 3](#)*). These are generally classified as field or hallway walking tests and are typically considered submaximal. Similar to maximal exercise tests, field walking tests are used to evaluate exercise capacity, estimate prognosis, and evaluate response to treatment ([1](#), [2](#), [81](#), [82](#)). The most common among the field walking tests is the 6-minute walk test (6-MWT), but evidence has been building for other field walking tests, such as the incremental and endurance shuttle walk tests ([81](#), [82](#)). The 6-MWT was originally developed to assess individuals with pulmonary disease ([81](#)); however, it has been applied in various individual groups and is a popular tool to assess individuals with heart failure.

The advantages of field walking tests are their simplicity and minimal cost, often requiring just a hallway. In addition, because the individual walks at a self-selected pace, a field walking test might be more representative of an individual's ability to perform activities of daily living ([2](#), [67](#), [68](#)).

Given the increasing recognition of the importance of exercise capacity in prognosis, and the call for the recognition of exercise capacity as a “vital sign” ([21](#)), there have been numerous “nonexercise” approaches to estimate CRF ([21](#), [83](#), [84](#)). Because it is inappropriate as well as impractical to perform a maximal exercise test on most individuals without a specific indication, these approaches permit a reasonable estimate of CRF without performing a maximal test. Nonexercise estimates of CRF can be useful for activity counseling as well as for providing information on risk stratification ([21](#), [83](#), [84](#)).

ONLINE RESOURCES

American College of Cardiology, guidelines: <http://www.acc.org/guidelines>

American Heart Association, guidelines and statements:

https://professional.heart.org/professional/GuidelinesStatements/UCM_316885_Guidelines-Statements.jsp

American Thoracic Society, statements, guidelines, and reports:

<https://www.thoracic.org/statements/>

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CHAPTER 5

General Principles of Exercise Prescription

The scientific evidence demonstrating the beneficial effects of exercise is indisputable ([1](#)). Moreover, the benefits of exercise outweigh the risks for most adults (see [Chapters 1](#) and [2](#)) ([1-3](#)). In addition, sedentary behaviors increase adverse health outcomes, even among those who exercise regularly ([4-8](#)). Although activity breaks, or frequent daily activity bouts of any duration, are recommended by the *2018 Physical Activity Guidelines* ([1](#)) and believed to mitigate some of the negative consequences of sedentary behavior, gaps still exist in the literature related to sedentary behavior and activity breaks ([9, 10](#)).

Therefore, in addition to minimizing sedentary activities, the optimal exercise prescription (Ex Rx) should address components of cardiorespiratory (aerobic) and muscular fitness, mobility/flexibility, and body composition. Thus, a well-crafted Ex Rx should aim to improve at least one health-related component of fitness and also include a plan to decrease periods of physical inactivity ([2, 3, 8, 11](#)). The exercise professional is also required to determine appropriate modifications of exercise programming that may include progression and regression to help maximize movement competence and safety of individuals in each of the components of fitness.

AN INTRODUCTION TO THE PRINCIPLES OF EXERCISE PRESCRIPTION

This chapter employs the FITT principles of Ex R_x:

- F: Frequency (how often)
- I: Intensity (level of exertion)
- T: Time (duration of exercise sessions)
- T: Type (mode of activity)

In addition, components such as volume (V) (total amount of exercise) and progression (P) (exercise advancement) should also be considered when designing an individualized Ex R_x consistent with American College of Sports Medicine (ACSM) recommendations.

The FITT principles of Ex R_x presented in this chapter are based on the application of the existing evidence on the physiologic, psychological (see [Chapter 12](#)), and health benefits of exercise (see [Chapter 1](#)). Nonetheless, some individuals may not respond as expected, considering appreciable individual variability in the magnitude of responses to a particular exercise regimen ([3, 12–15](#)). Furthermore, the FITT principles of Ex R_x may not apply in the same way in certain cases because of individual characteristics (*e.g.*, health status, physical ability, age) or athletic and performance goals. Appropriate modifications to the Ex R_x should be made for individuals with clinical conditions and healthy individuals with special considerations and as indicated in other related chapters of the *Guidelines* (see [Chapters 4, 6–11](#)). Although considered a separate population, the Exercise Professional must consider certain training principles for individuals who are competitive or recreational athletes throughout their life.

For most adults, an exercise program including aerobic and resistance training (RT) is necessary to improve and maintain physical fitness and health ([1](#)). The FITT Ex R_x guidelines present recommended targets for exercise derived from the available scientific evidence showing most individuals will realize benefit when following the stated quantity and quality of exercise ([3](#)). However, some individuals will want or need to include only some of the health-related components of physical fitness in their training regimen or exercise less than suggested by the guidelines presented in this chapter. Furthermore, certain exercises may be regressed as necessary for the different levels of movement competence displayed by the individual (see CRF and RT sections below). Even if an individual cannot meet the recommended targets in this chapter, performing some exercise is beneficial, especially in inactive or deconditioned individuals, and, for that reason, should be encouraged except where there are safety concerns.

GENERAL CONSIDERATIONS FOR EXERCISE PRESCRIPTION

Even though exercise is primarily safe ([1](#)), the risk of cardiovascular disease (CVD) and musculoskeletal complications can be minimized by (a) following the preparticipation health screening and evaluation procedures outlined in [Chapter 2](#); (b) beginning a new program of exercise at light-to-moderate intensity, with a gradual progression of volume and intensity ([3](#)); and (c) employing an individualized exercise program that adheres to the Ex Rx guidelines set forth in this chapter. Moreover, behavioral interventions that may reduce barriers and enhance the adoption and adherence to exercise participation are also important to the implementation of the Ex Rx (see [Chapter 12](#)).

Aside from the components of the Ex Rx, variables such as an individual's goals, physical ability, physical fitness, health status, schedule, physical and social environment, as well as the available facilities and equipment should be considered when designing an Ex Rx. This chapter presents evidence-informed recommendations for aerobic, RT, or flexibility exercise, as based on a combination of the FITT principles. The following sections present specific recommendations for the Ex Rx to improve health and fitness.

COMPONENTS OF THE EXERCISE TRAINING SESSION

A single exercise training session is typically composed of the following phases:

- Warm-up
- Conditioning
- Cool-down

Regardless of the individual goal, a training program should be divided into one of three phases — warm-up, conditioning, and cool-down ([16](#)). Each exercise session should be designed with a training goal in mind and include some type of conditioning exercise, preceded by a movement-specific warm-up ([17](#)). Flexibility exercises may also be incorporated into the training session, either before or after the conditioning phase, and/or separately, in order to improve range of motion (ROM) and enhance muscular coordination ([18](#)).

The warm-up phase is a transitional phase that allows the body to adjust to the changing physiologic, biomechanical, and bioenergetic demands of the specific exercise session and should include light- to moderate-intensity activities specific to the muscle groups that will be employed during exercise ([3](#), [17](#)). Warming up also improves ROM and may reduce the risk of injury during the exercise session ([3](#), [17](#)). A dynamic warm-up, primarily involving large muscle groups (*e.g.*, walking, jogging, cycling), is superior to static flexibility exercises for the purpose of enhancing the performance of cardiorespiratory/aerobic exercise, sports, or resistance exercise, especially activities that are of long duration or with many repetitions ([3](#), [17](#), [19](#)). The specific time dedicated to a warm-up may vary depending on the metabolic demands of the activity as well as the individual; however, evidence suggests that this period should be between 5 and 15 minutes ([17](#)).

During the conditioning phase, training exercises could include aerobic, resistance, flexibility, and/or sports activities, depending on the specific goals of the exercise session. Specifics about these modes of exercise are discussed in subsequent sections of this chapter; however, the conditioning portion of any exercise program may last anywhere between 10 and 60 minutes, depending on the intensity and type of the activity.

Although cool-downs have been suggested as an integral part of any exercise session, recent evidence suggests that cool-downs have limited impact on improving psychobiological markers of recovery ([20](#)). Nonetheless, cool-downs may be useful to allow the body to return to near-resting levels (*e.g.*, volume of oxygen consumed per unit time $\dot{V}O_2$, heart rate [HR]) after the exercise session. Low- to moderate-intensity flexibility exercises, such as static stretching, may also be performed during the cool-down phase to help facilitate a more relaxed physiologic state ([21](#)).

SEX DIFFERENCE CONSIDERATIONS FOR EXERCISE PRESCRIPTION AND TRAINING

The guidelines for Ex Rx outlined after the sex difference considerations are appropriate for and directly applicable to males and females. Because males and females do not differ in their maximal HRs achieved during exercise (22–24) for example, intensity of exercise based on HR is appropriate to each sex but may have implications for energy balance as outlined below. Males and females, however, differ in their baseline strength, power, and aerobic endurance, so Ex Rx should be based on progression relative to their baseline (starting) values (*e.g.*, load of weight lifted) as is the case for people of varying levels of fitness. There are some sex differences, however, that need to be considered and are addressed below. The following section highlights some of the fundamental sex differences that influence exercise performance, the adaptations of males and females in response to training, and any recommended differences in Ex Rx.

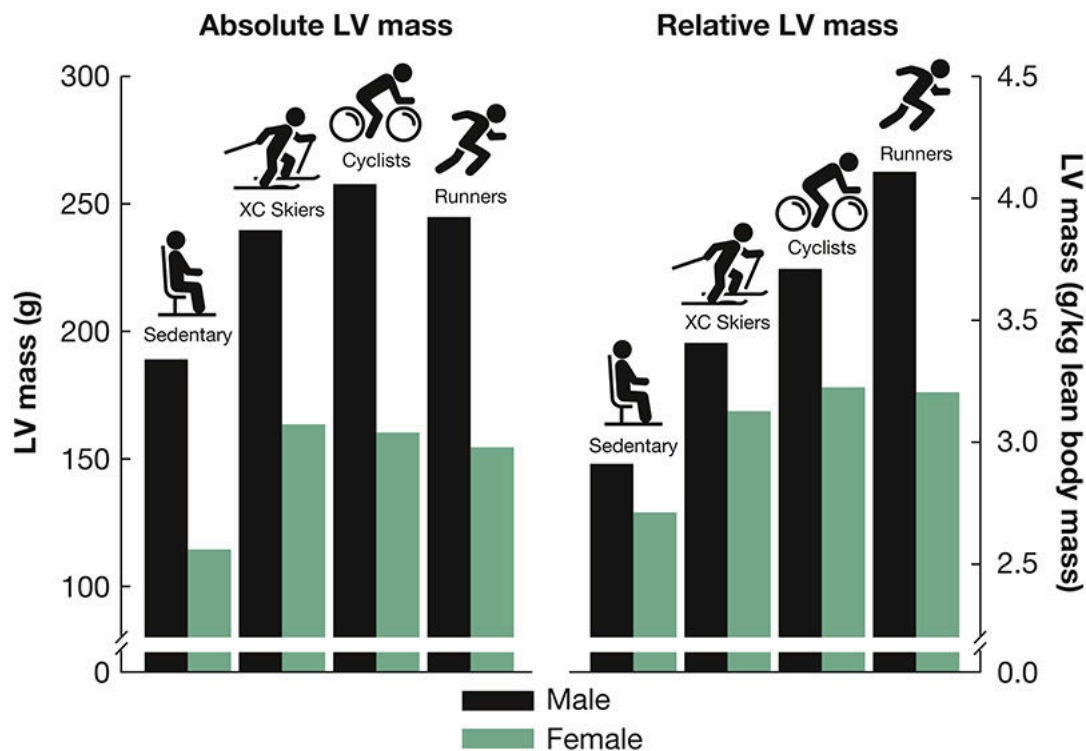
The following guidelines are based on sex differences, which is a primary determinant of athletic performance (25) and more predictive than gender (26). Sex is a biological construct and based on sex chromosomes (male, XY, and female, XX), hormonal, and genital differences (27). Gender is fluid and a cultural-social construct that can change over time and between cultures (27) (<https://orwh.od.nih.gov/sex-gender>). *Chapter 6* examines considerations for the transgender person in greater detail.

SEX DIFFERENCES IN EXERCISE PERFORMANCE

Sex differences in athletic performance involving muscle strength, muscle power, and aerobic power are due to fundamental differences in physiology and anatomy between males and females that diverge markedly during puberty. While males and females increase strength, power, and endurance during puberty, males improve more due to the presence of the Y chromosome and the sex steroid testosterone (a powerful anabolic hormone) that grows to approximately 15-fold higher in males than females during and after puberty ([28, 29](#)). Consequently, adult males are stronger, more powerful, and faster than females ([25, 30](#)). Males, on average, are taller, with longer limbs and have larger and faster skeletal muscle. The skeletal muscles of males produce more force and power (product of force and velocity of the muscle) than females of similar age and training status ([25](#)). The strength and power differences can be large in the upper limb (males can be ~50% stronger) and lower limb (males ~30% stronger) with the largest differences in the upper limb muscles ([25, 31](#)). Thus, when lifting weights or performing exercise with body weight, the strength and absolute loads typically able to be lifted are greater for males than females when starting an exercise program. A more comprehensive review highlighting the magnitude of the sex differences and the mechanisms can be found in the 2023 ACSM consensus statement on the Biological Sex Differences in Athletic Performance ([25](#)).

Males also have greater aerobic power (*i.e.*, maximal oxygen utilization or $\dot{V}O_{2\max}$, both as liters per minute and when normalized to body weight as $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) than females ([25](#)). $\dot{V}O_{2\max}$ in males is approximately 10%–30% higher than females, which allows for a greater absolute critical power allowing them to run and cycle, for example, at higher speeds over short and long distances compared with females of similar age and training status ([25, 32](#)). The larger $\dot{V}O_{2\max}$ of males occurs because males possess a bigger heart (ventricular mass) allowing a larger stroke volume and cardiac output than females ([33](#)). This is the case for sedentary males and females and athletes (see [Figure 5.1](#)). Males also have a higher hemoglobin concentration and mass in their blood and so a larger blood oxygen-carrying capacity than females and bigger lungs for oxygen exchange ([36, 37](#)). Consequently, males have greater oxygen utilization and can perform aerobic exercise such as running and cycling at higher speeds than females ([32](#)).

Athletes Have Big Muscular Hearts



data adapted from Milikan, et al. *Am J Cardiol.* 1988;62:301-05. Riley-Hagan, et al., *Am J Cardiol.* 1992;69:1067-1074.

Figure 5.1 Cardiac mass (absolute and relative to lean body mass) of male and female athletes and sedentary controls. LV, left ventricular. Data adapted from (34, 35) and published in Hunter et al. (25) with permission.

Sex differences in fuel metabolism during exercise are also apparent, although effects on absolute performance are likely small. Females oxidize greater amounts of fat and lower amounts of carbohydrate and protein at similar intensities compared with males (38–41). Males crossover sooner from fat to carbohydrate metabolism compared with females across the exercise intensity continuum. This is likely due to sex differences in the proportional area of type I and II muscle fibers (males have larger proportional area of the fast type II fiber than females (25)) with males demonstrating greater glycolytic capacity compared with females (39).

SEX DIFFERENCES IN RESPONSE TO EXERCISE TRAINING

Given the sex difference in muscle strength, muscle power, and aerobic power at baseline, it is important to consider whether males and females respond similarly to training regimes and whether Ex Rx needs to differ to obtain similar adaptations ([25](#)). Generally, males and females adapt similarly (similar percentage increase) to short term (~12 weeks) high-resistance strength training ([42](#)) and aerobic endurance training ([43–45](#)) with profound performance and health benefits in males and females: the physiological adaptations and thus changes in performance are similar relative to starting values. In the long term (>12 weeks), there may be larger cardiac adaptations in males than females with a more comprehensive review found in the 2023 ACSM consensus statement ([25](#)) and summarized below.

Resistance Strength Training and High-Intensity Interval Training

Males and females demonstrate similar relative (percentage) muscle hypertrophy and improvements in muscle strength and power in response to high-load RT (~6–12 weeks) in young adults ([42, 46–49](#)) and older adults >50 years ([50](#)). The moderating effects of frequency, volume, and intensity are similar in both sexes ([47, 49](#)). Similarly, after 12 weeks of cycling, high-intensity interval training in males and females exhibited similar relative muscle hypertrophy (5%) and increased fatigue resistance (4.8%) ([51](#)). As noted before, because males are stronger and more powerful than females before training, the absolute increase in muscle mass and strength or power are typically larger than that of females, but the increases are similar when expressed as a percentage of pretraining values.

Recovery from acute bouts of training, however, may differ between males and females. Progressive overload and fatigability, in combination with adequate recovery of muscles, are a foundation of RT to increase strength and power ([52, 53](#)). Studies on sex differences in fatigability of limb muscles, however, show that females are less fatigable than males especially for isometric contractions and slow dynamic contractions ([54](#)). Additionally, some females recover more quickly than males between sets of fatiguing contractions ([55](#)). Thus, when prescribing RT with slow dynamic or static contractions, females may experience less fatigability of the muscles (*i.e.*, less percentage reduction in force or power at the end of a fatiguing bout of exercise) or recover more quickly than males.

Cardiovascular and Aerobic Endurance Training

As indicated previously, $\dot{V}O_{2\max}$ (both absolute and relative) is higher in males compared with females, and one of the principal drivers for this is the greater cardiac compliance/mass/chamber dimensions ([34, 35](#)). See [Figure 5.1](#). Males have greater absolute and relative left ventricular (LV) mass than females that coupled with higher blood volumes, results in greater utilization of the Frank-Starling mechanism leading to a larger stroke volume compared to females. Given that maximal HRs are largely similar between the sexes, this allows for greater maximal cardiac output, which is a major determinant of $\dot{V}O_{2\max}$ ([23](#)).

Further, significant sex differences exist in response to aerobic training. Specifically, females exhibit a blunted cardiovascular response to endurance training ([56](#)). This is characterized by greater improvements in absolute $\dot{V}O_{2\max}$ in males than females over a year of aerobic training with earlier plateaus in females (3 months after initiation of training) than males (9 months after initiation of training). This larger increase in $\dot{V}O_{2\max}$ in males appeared to be driven by a greater LV and right ventricular (RV) hypertrophic remodeling response in males compared to females. LV stroke volume in males continues to increase over 1-year of training; however, increases in LV stroke volume appears to plateau in females after 9-months of training ([56](#)). The mechanisms underpinning sex differences in cardiovascular adaptation to aerobic training are not fully understood however, gonadal sex steroid hormones, and the greater concentration of testosterone in males than females appear to be involved ([44](#), [57](#), [58](#)).

Sex Differences in Body Weight Responses to Aerobic Exercise Training

Males generally exhibit greater weight loss in response to aerobic exercise training than females when exercise is prescribed using a traditional frequency, intensity, and duration prescription where intensity is based on heart rate reserve (HRR) ([59](#)). This is likely due to greater compensatory responses to energy balance in females versus males ([60](#), [61](#)). These differences in weight change largely disappear when exercise prescriptions are based on energy expenditure of exercise ([62](#)). However, this necessitates resource-intensive verification of exercise energy expenditure monthly, which may not be feasible.

OTHER SEX-BASED SPECIAL CONSIDERATIONS

The menstrual cycle in eumenorrheic females (regular menstrual cycle) has the potential to affect performance and responses to training because of the fluctuations in the dominant hormones, estrogen and progesterone. While hormonal fluctuations across the menstrual cycle in eumenorrheic females can alter physiological responses such as metabolism and thermoregulation, the effect on performance on average is quite small ([63](#), [64](#)). Regardless, for some females, menstrual pain (dysmenorrhea) or premenstrual symptoms can be substantial with potential effects on their ability to exercise and perform ([65–69](#)). Individual responses to menstrual cycle–related symptoms can affect mood, pain, motivation, injury risk, and anemia among females, although these effects are highly variable with exercise performance ([64](#), [70](#), [71](#)). It is recommended that exercise training regimes be conducted within the limits of the females' symptoms. Furthermore, some females, particularly athletes, experience secondary amenorrhea (absence of a menstrual period for >3 months) or oligomenorrhea (irregular menstrual cycles) ([69](#)). These disordered menstrual cycles in female athletes are often associated with relative energy deficiency in sport (RED-S) due to high amounts of exercise and low energy intake ([72](#)) and are considered a central part of the Female triad ([73](#)). Exercise and athletic performance can be deleteriously affected in females who experience RED-S and the associated impairments in the reproductive system, and it is recommended that medical advice and prevention be sought ([72](#)).

Another question that has generated interest and several studies is whether different types of training are more advantageous at different phases of the menstrual cycle (*e.g.*, follicular vs. luteal phases) ([74](#)). The evidence is sparse with minimal data. Importantly, any emerging recommendations are likely difficult to implement because most training regimes extend more than 2 weeks (the approximate length of the follicular or luteal phases) and thus will span many menstrual cycles. For continued benefits, it is recommended that females regularly exercise across the entire menstrual cycle. It is important to ensure females who experience menstrual pain, and other premenstrual symptoms can manage their symptoms so as to stay comfortable and regularly exercise. Furthermore, oral contraceptive use does not appear to significantly influence physiological or anatomical adaptations and performance in response to exercise training ([75](#)).

CARDIORESPIRATORY FITNESS

The term cardiorespiratory fitness (CRF) reflects the functional capabilities of the heart, blood vessels, lungs, and skeletal muscles to transport and utilize oxygen to perform physical work (76). In population-based studies, the least fit cohort (bottom 20% of CRF) has the highest relative risk of death. A significant survivor benefit occurs by progressing to a level of “good” fitness (upper 60th percentile) (77–81). An Ex R_x featuring moderate- to vigorous-intensity continuous training consisting of large rhythmic movement for a sustained duration is the traditional method to improve CRF (3, 82). However, emerging evidence points to the efficacy of interval training, in which moderate- to high-intensity efforts interspersed with brief periods of rest or low-intensity activity performed repetitively demonstrate similar improvements in CRF as observed in traditional endurance training, with a reduced total exercise volume and time commitment (83–90). Whereas improvements in CRF are primarily driven by intensity, frequency, and duration, all types and subsequent combinations of aerobic activities (*e.g.*, walking, jogging, running, sprinting) can contribute toward meeting physical activity (PA) recommendations (1) and improving health outcomes, independent of improvements in CRF (91, 92); this is especially true for individuals with lower CRF. However, for the purpose of enhancing CRF, an Ex R_x consisting solely of 150 min · wk⁻¹ of moderate-intensity exercise may not be sufficient to improve CRF in a large portion of the population, thus requiring a marked increase in the dose-response exercise stimulus, specifically increases in intensity and/or duration (93–95).

To optimize CRF, the FITT principle of Ex R_x is presented as *Frequency* (d · wk⁻¹), *Intensity* (% of HRR, or maximal volume of oxygen consumed per unit time [$\dot{V}O_{2max}$]), *Time* (duration per session), and *Type* (mode of PA), with additional recommendations for quantifying volume (product of training frequency, exercise intensity, and exercise duration) and the implementation of progressive overload. The FITT principle of Ex R_x for CRF is summarized in [Table 5.1](#).

TABLE 5.1 Aerobic (Cardiovascular Endurance) Exercise Recommendations

FITT	Recommendation
<i>Frequency</i>	- At least 3 d · wk⁻¹ - For most adults, spreading the exercise sessions across 3–5 d · wk⁻¹ may be the most conducive strategy to reach the recommended amounts of PA.
<i>Intensity</i>	Moderate (40%–59% HRR) and/or vigorous (60%–89% HRR) intensity is recommended for most adults.
<i>Time</i>	Most adults should accumulate 30–60 min · d⁻¹ (≥150 min · wk⁻¹ and up to 300 min · wk⁻¹) of moderate-intensity exercise, 20–60 min · d⁻¹ (≥75 min · wk⁻¹ and up to 150 min · wk⁻¹) of vigorous-intensity exercise, or a combination of moderate- and vigorous-intensity exercise daily to attain the recommended targeted volumes and health-related benefits of exercise.
<i>Type</i>	Aerobic exercise performed in a continuous or intermittent manner that involves major muscle groups is recommended for most adults.

HRR, heart rate reserve; PA, physical activity.

Frequency of Aerobic Exercise

The frequency of PA (*i.e.*, the number of days per week dedicated to an exercise program) is an important contributor to health and fitness benefits. According to the *2018 Physical Activity Guidelines*, aerobic exercise is recommended to be performed on at least 3 d · wk⁻¹ ([1](#)). For most adults, spreading the exercise sessions across 3–5 d · wk⁻¹ may be best to reach the recommended amount of PA ([82](#)). The *World Health Organization (WHO) Guidelines on Physical Activity and Sedentary Behavior* support the aforementioned guidelines but further state that adults who can should undertake 150–300 minutes of moderate-intensity or 75–150 minutes of vigorous-intensity PA or some equivalent combination of moderate-intensity and vigorous-intensity aerobic PA throughout the week ([80, 82](#)). The frequency of exercise can vary, as can the intensity and duration, as these three variables are interdependent on each other ([3](#)). The current evidence on PA frequency/intensity is not conclusive to state that 5 d · wk⁻¹ of 30 minutes of aerobic exercise is superior to 3 d · wk⁻¹ of 50 minutes; therefore, multiple combinations of frequency and duration can be combined across the week to meet recommendations ([80, 82, 96](#)). Furthermore, a weekly combination of 3–5 d · wk⁻¹ of moderate- and vigorous-intensity exercise options can be performed, which may be more suitable for most individuals ([3, 96](#)).

Aerobic exercise performed at a frequency of only once or twice per week at moderate-to-vigorous intensity can still result in substantial health/fitness benefits ([3, 97–99](#)). This exercise pattern may be sufficient to reduce risks for all-cause, CVD, and cancer mortality in some individuals ([80, 97](#)).

Intensity of Aerobic Exercise

There is a positive dose response of health/fitness benefits that results from increasing exercise intensity ([3, 80, 82, 100](#)). The overload principle of training states that exercising below a minimum intensity, or threshold, will not challenge the body sufficiently to result in changes in physiologic parameters ([3](#)). However, the minimum threshold of intensity for benefit seems to vary depending on an individual's current CRF level and other factors such as age, health status, physiological differences, genetics, habitual PA, physical ability, and social and psychological factors; therefore, precisely defining an exact threshold to improve CRF may be difficult ([3, 80, 82, 101](#)). For example, highly trained individuals may need to exercise at an intensity of ≥90% $\dot{V}O_{2R}$ or HRR to improve $\dot{V}O_{2max}$, whereas 60%–89% $\dot{V}O_{2R}$ or HRR may provide a sufficient stimulus in moderately trained individuals ([3, 101](#)). Whereas both moderate and vigorous-intensity exercise can be used to meet PA recommendations, to improve CRF, vigorous-intensity exercise (>6 metabolic equivalents [METs]; 60%–89% HRR) is more effective at increasing $\dot{V}O_{2max}$ than moderate-intensity exercise (3.0–5.9 METs; 40%–59% HRR) ([80, 82, 101–103](#)). Interval training, broadly defined as intermittent periods of intense exercise separated by periods of recovery, is an emerging area of interest for Ex R_x ([104](#)). Interval training typically consists of alternating bouts of vigorous- to supramaximal-intensity exercise equivalent to 90%–120% of $\dot{V}O_{2max}$ for 20 seconds to four minutes followed by bouts of light- to moderate-intensity exercise (60 seconds to six minutes) and can entail several types of aerobic and bodyweight exercise ([99, 105, 106](#)). Athletes may use 5-minute bouts or 10-second bouts when performing exercises at near-maximal and supramaximal efforts, respectively. However, for detrained individuals, a much less demanding implementation of interval training could simply incorporate walking or other enjoyable activities in an interval manner, in which periods of brisk walking or other activities are alternated with periods of walking or other activities at a reduced pace ([107–110](#)).

Previous research has found that interval training elicits physiologic adaptations similar to traditional endurance training despite a lower total workload and superior physiological adaptations when total exercise dose is matched ([83](#), [84](#), [104](#), [111](#), [112](#)).

Interval training can be classified as either high-intensity interval training (HIIT) or sprint interval training (SIT) ([87–89](#), [113](#)). HIIT, characterized by vigorous efforts, is often performed at an intensity close to that which elicits $\geq 60\%$ – 89% of HRR; SIT is characterized by an all-out effort at a workload or pace which elicits responses ranging from $\geq 90\%$ of HRR ($\dot{V}O_{2R}$) to $\geq 100\%$ peak oxygen uptake ([86](#), [87](#), [89](#), [104](#), [113](#)). However, interval training is not limited to only aerobic-based exercises (*e.g.*, cycling, running). Although not traditionally considered as rigorous, a CRF stimuli as aerobic-type interval training modes, resistance-based circuit interval training combining body weight exercises, plyometrics, and RT can provide an enjoyable, comprehensive, time efficient exercise stimulus that can increase muscular fitness and CRF ([99](#), [114–116](#)). The magnitude of benefits are dependent on the length and relative intensity of the interval bouts performed and subsequent recovery bouts ([109](#), [117–120](#)). See [Box 5.1](#) for examples of interval protocols.

Box 5.1 Examples of Interval Training Protocols

High-intensity interval training (HIIT): 4 × 4 protocol (four intervals at 4 min each completed at 90%–95% peak heart rate, with 3 min rest between intervals) ([113](#)).

Sprint interval training (SIT): 3 × 20 protocol (three intervals at 20 s, each completed with an all-out effort, with 2 min rest between intervals) ([121](#)).

Resistance-based interval training: barbell complex — five repetitions of the following five exercises performed back-to-back: Romanian deadlift, bent-over row, hang clean, front squat, overhead press; rest 2 min, repeat for one to two additional rounds.

High-intensity functional training (HIFT): three rounds for time of: 400-m run, 10 clean and press, 10 burpees.

During interval training, several aspects of the Ex R_x can be varied based on the incorporation of aerobic, resistance, or both aerobic- and resistance-based exercises, all of which are ultimately dependent on the goals of the training session and physical fitness level of the individual ([99](#), [105](#), [106](#), [114](#)). However, the primary factors of interval-based Ex R_x concern the intensity and duration of the exercise and recovery interval and the total number of intervals performed ([122](#), [123](#)).

Methods of Estimating Intensity of Aerobic Exercise

Several effective methods for prescribing exercise intensity result in improvements in CRF that can be recommended for individualized Ex R_x ([3](#)). [Table 5.2](#) shows the approximate classification of exercise intensity commonly used in practice.

TABLE 5.2 Methods of Estimating Intensity of Cardiorespiratory Exercise

					Cardiorespiratory Endurance Exercise				
Relative Intensity					Intensity (% $\dot{V}O_2$ max) Relative to Maximal Exercise Capacity in MET			Absolute Intensity	
Intensity	%HRR or % $\dot{V}O_2R$	%HR _{max}	% $\dot{V}O_2$ max	Perceived Exertion (Rating on 6–20 RPE Scale)	20 METs % $\dot{V}O_2$ max	10 METs % $\dot{V}O_2$ max	5 METs % $\dot{V}O_2$ max	METs	Y _{max} (23)
Very light	<30	<57	<37	Very light (RPE <9)	<34	<37	<44	<2.0	<2.0
Light	30–39	57–63	37–45	Very light to fairly light (RPE 9–11)	34–42	37–45	44–51	2.0–2.9	2.0–4.0
Moderate	40–59	64–76	46–63	Fairly light to somewhat hard (RPE 12–13)	43–61	46–63	52–67	3.0–5.9	4.0–7.0
Vigorous	60–89	77–95	64–90	Somewhat hard to very hard (RPE 14–17)	62–90	64–90	68–91	6.0–8.7	7.0–10.0
Near-maximal to maximal	≥90	≥96	≥91	≥Very hard (RPE ≥18)	≥91	≥91	≥92	≥8.8	≥10.0

HR_{max}, maximal heart rate; HRR, heart rate reserve; MET, metabolic equivalent; RPE, rating of perceived exertion; $\dot{V}O_2$ max, maximal volume of oxygen consumed per unit time; $\dot{V}O_2R$, oxygen uptake reserve.

Adapted from (3).

A summary of methods for calculating exercise intensity is presented in [Box 5.2](#). Intensity of exercise training is usually determined as a range, so the calculation using the formulas presented in [Box 5.2](#) need to be repeated twice (*i.e.*, once for the lower limit of the desired intensity range and once for the upper limit of the desired intensity range). The prescribed exercise intensity range for an individual should be determined by taking various factors into consideration, including age, habitual PA level, physical fitness level, and health status. The accuracy of any of these methods may be influenced by the method of measurement or estimation used; therefore, direct measurement of the physiological responses to exercise through an incremental (graded) cardiopulmonary exercise test is the preferred method or “Gold standard” for Ex R_x whenever possible (3). Examples illustrating the use of several methods for prescribing exercise intensity are found in [Appendix D](#).

Box 5.2 Summary of Methods for Prescribing Exercise Intensity Using Heart Rate (HR), Oxygen Uptake $\dot{V}O_2$, and Metabolic Equivalents (METs)

- HRR method: target HR (THR) = $[(HR_{\max}/peak^a - HR_{\text{rest}}) \times \% \text{ intensity desired}] + HR_{\text{rest}}$
- $\dot{V}O_2$ method: target $\dot{V}O_{2R}^b = [(\dot{V}O_{2\max}/rest^c - \dot{V}O_{2\text{rest}}) \times \% \text{ intensity desired}] + \dot{V}O_{2\text{rest}}$
- HR method: target HR = $HR_{\max}/peak^a \times \% \text{ intensity desired}$
- $\dot{V}O_2$ method: target $\dot{V}O_2^b = \dot{V}O_{2\max}/peak^c \times \% \text{ intensity desired}$
- MET method: target MET^b = $[(\dot{V}O_{2\max}/peak^c)/3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}] \times \% \text{ intensity desired}$

^a $HR_{\max}/peak$ is the highest value obtained during maximal/peak exercise or it can be estimated via a prediction equation (see [Table 5.3](#)).

^bActivities at the target $\dot{V}O_2$ and MET can be determined using a compendium of physical activity ([128, 129](#)) or metabolic calculations ([130](#)) (see [Table D.1](#)).

^c $\dot{V}O_{2\max}/peak$ is the highest value obtained during maximal/peak exercise, or it can be estimated from a submaximal exercise test. See “The Concept of Maximal Oxygen Uptake” section in [Chapter 3](#) for the distinction between $\dot{V}O_{2\max}$ and $\dot{V}O_{2\text{peak}}$. $HR_{\max}/peak$, maximal or peak heart rate; HRR, heart rate reserve; HR_{rest} , resting heart rate; $\dot{V}O_{2\max}/peak$, maximal or peak volume of oxygen consumed per unit of time; $\dot{V}O_{2R}$, oxygen uptake reserve; $\dot{V}O_{2\text{rest}}$, resting volume of oxygen consumed per unit of time.

Results from recent studies indicate that discrepancies between predicted and measured maximal HR values exist for most prediction equations and reinforce the importance of utilizing population-specific equations when values from actual maximal exercise tests are not available ([100, 131–133](#)). The formula $(220 - \text{age})$ has been commonly used to predict $HR_{\max}$ ([100, 134](#)). This formula is simple to use but can underestimate or overestimate measured $HR_{\max}$ and therefore is not recommended ([125, 133, 135](#)). Cundric et al. found that the application of machine learning significantly enhanced the accuracy of maximal HR prediction in a large cohort of apparently healthy adults with the Fox equation $(220 - \text{age})$ and recommend its similar application with clinical populations ([131](#)). Machine learning is a branch of artificial intelligence and computer science, which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy ([136](#)). It involves the addition of measurable variables like age, weight, height, resting HR, and systolic and diastolic blood pressure (BP) to the algorithm ([131](#)). Specialized regression equations for estimating $HR_{\max}$ may be superior to the equation of $220 - \text{age}$ in some individuals, but exercise professionals should carefully select equations best fitting the specific population they serve ([125, 126, 131–133, 135](#)). [Table 5.3](#) shows the more commonly used equations to estimate $HR_{\max}$. It is important to recognize that these population-specific equations still have prediction errors of 10–15 bpm, which can significantly impact the quality of the $\text{Ex } R_x$. For greater accuracy in determining exercise intensity for the $\text{Ex } R_x$, using the directly measured $HR_{\max}$ is preferred to estimated methods.

TABLE 5.3 Commonly Used Equations for Estimating Maximal Heart Rate (HR_{max})

Author	Equation	Population
Astrand (124)	$HR_{max} = 216.6 - (0.84 \times \text{age})$	Males and females age 4–34 yr
Tanaka et al. (125)	$HR_{max} = 208 - (0.7 \times \text{age})$	Healthy males and females
Gellish et al. (126)	$HR_{max} = 207 - (0.7 \times \text{age})$	Males and females in an adult fitness program with broad range of age and fitness levels
Gulati et al. (127)	$HR_{max} = 206 - (0.88 \times \text{age})$	Asymptomatic middle-aged females referred for stress testing

Measured or estimated measures of absolute exercise intensity include caloric expenditure ($\text{kcal} \cdot \text{min}^{-1}$), absolute oxygen uptake ($\text{mL} \cdot \text{min}^{-1}$ or $\text{L} \cdot \text{min}^{-1}$), and METs. These absolute measures can result in misclassification of exercise intensity (*e.g.*, moderate and vigorous intensity) because they do not take into consideration individual factors such as body weight, sex, age, and fitness level ([128](#), [129](#), [137](#)). Measurement error, and consequently misclassification, is greater when using estimated rather than directly measured absolute energy expenditure (EE) and under free living conditions compared to laboratory conditions ([128](#), [129](#), [137](#)). For example, an older individual working at 6 METs may be exercising at a vigorous-to-maximal intensity, whereas a younger individual working at the same absolute intensity may be exercising at a moderate intensity ([137](#)). Therefore, for individual Ex R_x , a *relative* measure of intensity (*i.e.*, the energy cost of the activity relative to the individual's peak or maximal capacity such as % $\dot{V}O_2$ [*i.e.*, $\dot{V}O_2 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$], HRR, and oxygen uptake reserve [$\dot{V}O_{2R}$]) is more appropriate, especially for deconditioned individuals ([137](#), [138](#)). Although calculating individualized exercise training intensities from peak or maximal physiological responses from peak or maximal exercise is considered the “Gold standard,” the need for recalibration with repeated maximal exercise testing may not be practical, feasible, or desirable. Other practical methods for safely recalibrating exercise intensity are appropriate for healthy participants in a field setting ([81](#), [100](#)). They can eliminate the need for frequent exercise retesting. According to Franklin et al., level walking at 2 and 3 mph approximates 2 and 3 METs, respectively. At a 2-mph speed, each 3.5% grade increment adds an additional MET to the gross energy expenditure. For the 3-mph pace, each 2.5% increase in treadmill grade adds an additional MET ([100](#)). Additionally, MET levels can be quickly estimated during exercise or physical activities, employing the resting and exercise HRs using the HR index equation: $\text{METs} = (6 \times \text{heart rate index}) - 5$, where the HR index equals the HR measured during a specific point during the exercise or activity divided by the resting HR ([100](#), [139](#), [140](#)). Franklin et al. developed age- and sex-based regression equations corresponding to 70% $\dot{V}O_{2R}$ (between the range of 60% to 80% $\dot{V}O_{2R}$) for healthy males and females aged 30–79 years, that signify the training MET level requirements needed to achieve age- and sex-specific “good” fitness levels representing the 60th percentile and can be attained over a 3- to 12-month period. They are shown below ([81](#)). Franklin et al. used the mean peak or maximum CRF of 6,860 healthy males and females during maximal treadmill testing from the FRIEND Registry Database when developing the equations ([141](#)).

Quick method for determining exercise MET levels

- $\text{METs} = 6 \times \text{heart rate index} - 5$
Heart rate index = exercise heart rate ÷ resting heart rate
- $\text{METs} = 6 \times \text{heart rate index} - 5$
Heart rate index = exercise heart rate ÷ resting heart rate

- Training intensity levels in METs recommended to achieve a “good” fitness level (81).
- Males training METs = $12.480 - 0.092 \times \text{age in years}$ Females training METs = $8.975 - 0.065 \times \text{age in years}$
- As an example, a 35-year-old female would strive to attain a training intensity of approximately 6.7 METs over a 3- to 12-month period to achieve a “good” age and sex-specific CRF level of approximately 9 METs (81).
- Female training METs = $8.975 - 0.065 \times 35 \text{ yr} = 6.7 \text{ METs}$

Safe, enjoyable, and realistic activity choices within the desired intensity range can be identified by using a compendium of PAs (128, 129) or metabolic calculations (130) when using $\dot{V}O_2$ or METs to prescribe exercise. The 2023 edition of the compendium of PAs includes a new interactive Web site (www.pacompendium.com) and will include a separate compendium for older adults and another for wheelchair users. Metabolic equations for estimation of EE during common physical activities are found in [Appendix C](#).

Measures of perceived effort and affective valence (*i.e.*, the pleasantness of exercise) can be used to modulate or refine the prescribed exercise intensity (100). The talk test is a valid and reliable measure of exercise intensity that is a reasonable surrogate of the lactate threshold, ventilatory threshold, and respiratory compensation point across a broad range of individuals and can now be recommended as an effective primary method for prescribing and monitoring exercise intensity (142–145). This test can be administered by asking the exercising individual to recite something familiar or read from a text that is roughly 30 words long. Upon completion of the task, ask whether they can speak comfortably, if their answer is “yes,” their exercise intensity is likely below the ventilatory threshold. If they answer equivocally, their exercise intensity is likely near the ventilatory threshold, and if they answer no, their exercise intensity is likely above the ventilatory threshold. Other methods (*i.e.*, rating of perceived exertion, OMNI, Feeling Scale) are recommended as adjunct methods for prescribing and monitoring exercise, which require proper explanation of and practice using them with specific populations (3, 144).

Time (Duration) of Aerobic Exercise

Exercise time/duration is prescribed as a measure of the amount of time PA is performed. The recommended time/duration of PA may be performed continuously (*i.e.*, one session) or intermittently and can be accumulated over the course of a day in one or more sessions (1, 82). Most adults are recommended to accumulate $30\text{--}60 \text{ min} \cdot \text{d}^{-1}$ of moderate-intensity exercise, $20\text{--}60 \text{ min} \cdot \text{d}^{-1}$ of vigorous-intensity exercise, or a combination of moderate- and vigorous-intensity exercise per day (1, 3). A general guideline for aerobic Ex R_x is that 1 minute of vigorous-intensity aerobic activity can provide equivalent health benefits as 2 minutes of moderate-intensity aerobic exercise (1).

Any amount of PA can lead to health benefits, which is particularly encouraging for individuals who are currently sedentary or minimally active, because moving from a state of physical inactivity to any level of activity can result in significant mortality risk reductions (146). Evidence suggests that PA accumulated in bouts of less than 10 minutes, also referred to as micro-exercise, are associated with favorable health-related outcomes, thus engaging in PA, regardless of length of the bout, has health-enhancing effects, reinforcing the benefits of PA regardless of the duration (1, 82).

Type (Mode)

The principle of specificity of training (the physiologic adaptations to exercise are specific to the type of exercise performed) should be kept in mind when selecting the exercise modalities to be included in the Ex Rx (3). Exercise mode or type can be classified as types A–D. Mode or type is based on the nature of the activity, including major body parts used, skill level needed, etc. Exercise results can be negatively affected by lower fitness levels, pain, and by selecting exercises that exceed the individual's movement competency (147, 148). Movement competency pertains to the individual's ability to perform fundamental movement patterns like squatting, lunging, and ambulating. Poor movement has been shown to alter the loading of singular joints and potentially limits the individual's ability to complete movement tasks effectively (148). Therefore, careful consideration should be given to selecting exercise mode(s) that individuals can perform comfortably and safely (147, 148). Observations of an individual's movement competency during exercise can help the exercise professional determine whether exercise modifications, regressions, or substitutions are warranted and the individual's readiness to perform other modes of exercise or physical activity (149). Although a detailed discussion of signs of reduced movement competency during exercise is beyond the scope of this chapter, exercise professionals should recognize general signs of difficulty with performing tasks like walking, stepping, climbing, cycling, single or multi joint exercises, and modify/regress as needed (150, 151). Some examples of poor movement competency can include excessive trunk flexion, forward head protrusion, rounded shoulders, and/or limb favoring during walking, running, stepping, cycling, or rowing (150, 152). When using a treadmill, excessive gripping of the handrails or frequency gazing at the feet during walking can increase the risk of falling (152). Additionally, using the compendium of PAs for Ex Rx can facilitate a variety of exercise modes that are enjoyable, place varying impact stresses on the body (*e.g.*, running, cycling), and using different muscle groups (www.pacompendium.com). The modes of exercise that result in improvement and maintenance of CRF are found in [Table 5.4](#).

TABLE 5.4 Modes of Aerobic (Cardiorespiratory Endurance) Exercises to Improve Physical Fitness

Exercise Group	Exercise Description	Recommended for	Examples
A	Endurance activities requiring minimal skill or physical fitness to perform	All adults	Walking, leisurely cycling, aqua-aerobics, slow dancing
B	Vigorous-intensity endurance activities requiring minimal skill	Adults (as per the preparticipation screening guidelines in Chapter 2) who are habitually physically active and/or at least average physical fitness	Jogging, running, rowing, aerobics, spinning, elliptical exercise, stepping exercise, fast dancing
C	Endurance activities requiring skill to perform	Adults with acquired skill and/or at least average physical fitness levels	Swimming, cross-country skiing, skating
D	Recreational sports	Adults with a regular exercise program and at least average physical fitness	Racquet sports, basketball, soccer, downhill skiing, hiking

Volume (Quantity) of Aerobic Exercise

Exercise volume is the product of training frequency, exercise intensity, and the duration of the exercise session. Evidence supports the important role of exercise volume in realizing health/fitness outcomes, particularly with respect to weight management. For additional and more extensive health benefits, exercise may need to exceed the PA recommendations ([1](#)), with the goal of adults increasing aerobic PA to 300 min · wk⁻¹ of moderate-intensity aerobic exercise, or 150 min · wk⁻¹ of vigorous-intensity aerobic exercise, or an equivalent combination of moderate- and vigorous-intensity aerobic exercise ([1](#), [82](#), [100](#)).

Exercise volume may also be used to estimate the gross EE of an individual's Ex R_x. MET-min · wk⁻¹ and kcal · wk⁻¹ can be used to estimate exercise volume in a standardized manner. [Box 5.3](#) shows the definition and calculations for METs, MET-min, and kcal · min⁻¹ for a wide array of PAs. These variables can also be estimated using previously published tables ([128](#), [129](#), [153](#)). MET-min and kcal · min⁻¹ can then be used to calculate MET-min · wk⁻¹ and kcal · wk⁻¹ that are accumulated as part of an exercise program to evaluate whether the exercise volume is within the ranges described later in this chapter that will likely result in health/fitness benefits.

Box 5.3 Calculation of Metabolic Equivalents (METs), MET·min⁻¹, and kcal · min⁻¹

METs: an index of energy expenditure (EE). “A MET is the ratio of the rate of energy expended during an activity to the rate of energy expended at rest. ... [One] MET is the rate of EE while sitting at rest ... by convention ... [1 MET is equal to] an oxygen uptake of 3.5 [mL · kg⁻¹ · min⁻¹]” (1).

MET·min: an index of EE that quantifies the total amount of physical activity performed in a standardized manner across individuals and types of activities (1). Calculated as the product of the number of METs associated with one or more physical activities and the number of minutes the activities were performed (*i.e.*, METs × min), usually standardized per week or per day as a measure of exercise volume.

Kilocalorie (kcal): the energy needed to increase the temperature of 1 kg of water by 1 °C. To convert METs to kcal · min⁻¹, it is necessary to know an individual's body weight, $\text{kcal} \cdot \text{min}^{-1} = [(\text{METs} \times 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \times \text{body weight in kg}) / 1,000] \times 5$. Usually standardized as kilocalorie per week or per day as a measure of exercise volume.

Jogging (at ~7 METs) for 30 minutes on 3 d · wk⁻¹ for a 70-kg male:

$7 \text{ METs} \times 30 \text{ minutes} \times 3 \text{ times per week} = 630 \text{ MET} \cdot \text{min} \cdot \text{wk}^{-1}$

$[(7 \text{ METs} \times 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \times 70 \text{ kg}) / 1,000] \times 5 = 8.575 \text{ kcal} \cdot \text{min}^{-1}$

$8.575 \text{ kcal} \cdot \text{min}^{-1} \times 30 \text{ minutes} \times 3 \text{ times per week} = 771.75 \text{ kcal} \cdot \text{wk}^{-1}$

Data from (3).

The results of epidemiologic studies and randomized clinical trials have demonstrated a dose-response association between the volume of exercise and health/fitness outcomes (*i.e.*, with greater amounts of PA, the health/fitness benefits also increase) (1, 3). Whether or not there is a minimum or maximum amount of exercise that is needed to attain health/fitness benefits is not clear. However, a total EE of ≥500–1,000 MET·min · wk⁻¹ is consistently associated with lower rates of CVD and premature mortality. Thus, ≥500–1,000 MET·min · wk⁻¹ is a reasonable target volume for an exercise program for most adults (1, 3, 82, 100). This volume is approximately equal to (a) 1,000 kcal · wk⁻¹ of moderate-intensity PA (or about 150 min · wk⁻¹), (b) an exercise intensity of 3–5.9 METs (for individuals weighing ~68–91 kg [~150–200 lb]), and (c) 10 MET·h · wk⁻¹ (1, 3). Lower volumes of exercise (*i.e.*, 4 kcal · kg⁻¹ · wk⁻¹ or 330 kcal · wk⁻¹) can result in health/fitness benefits in some individuals, especially in those who are deconditioned (1, 3, 82, 100).

Pedometers and PA monitors are effective tools for promoting PA and can be used to approximate exercise volume in steps per day (146). With the advances in wearable technologies allowing for improved tracking of PA, a walking pattern or cadence of at least 100 steps · min⁻¹ in adults appears to meet the minimum threshold for moderate-intensity PA (154). The goal of 10,000 steps · d⁻¹ is often cited as a target; however, a daily step count of 7,000–8,000 steps · d⁻¹, with at least 3,000 steps · d⁻¹ at a brisk pace (3 METs/>100 steps · min⁻¹), is a reasonable minimum daily threshold associated with health benefits (155).

Progression of Aerobic Exercise

The recommended rate of progression in an exercise program depends on the individual's health status, physical fitness, training responses, and exercise program goals. Progression may consist of increasing any of the components of the FITT principle of Ex R_x as tolerated by the individual. During the initial phase of the exercise program, applying the adage “start low and go slow” is prudent to reduce risks of adverse cardiovascular events and injury, as well as to enhance adoption and adherence to exercise (see [Chapters 1 and 2](#) (3, 100)). Initiating exercise at a light-to-moderate intensity in currently inactive individuals and then

increasing exercise time/duration (*i.e.*, minutes per session) as tolerated is recommended. Total time can also be accumulated with multiple bouts of intermittent exercise. An increase in exercise time/duration per session of 5–10 minutes every 1–2 weeks over the first 4–6 weeks of an exercise training program is reasonable for the average adult (3). Improvements in movement quality and reductions in reported ratings of perceived exertion during exercise sessions can serve as proxy signs of improvements in fitness levels and exercise tolerance. After the individual has been exercising regularly for ≥ 1 month, the Ex R_x is gradually adjusted upward over the next 4–8 months or longer for very deconditioned individuals. Any progression in Ex R_x should be made gradually, avoiding large increases in any of the FITT components to minimize risks of muscular soreness, injury, undue fatigue, and the long-term risk of overtraining (3). Following any adjustments in the Ex R_x, the individual should be monitored for any adverse effects of the increased volume, such as excessive shortness of breath, fatigue, and muscle soreness, and downward adjustments should be made if the exercise is not well tolerated (3, 100).

RESISTANCE TRAINING

Muscular fitness which includes muscular strength, hypertrophy, power, and local endurance (see [Chapter 3](#)) is optimized through the implementation of RT, which can encompass free weights, machines, body weight, bands/tubing, or any other object that requires one to exert force against a resistance to improve strength, endurance, and power ([156](#)).

As stated in the 2018 Physical Activity Guidelines for Americans, for additional benefits beyond those associated with aerobic activity, a minimum recommendation for RT is at least $2 \text{ d} \cdot \text{wk}^{-1}$ at moderate to heavy loads and work the major muscle groups of the body. It is important to note that the individual's RT goals should guide the components of the RT program and should incorporate the following aspects of Ex R_x : frequency, intensity, time (i.e., rest intervals), type volume, and progression.

Regular exposure to an appropriate RT program can have numerous positive health benefits across the lifespan (see [Chapter 6](#)). These include improved balance, physical function, coordination, body composition, bone health, reduced risk of developing chronic diseases, and aerobic benefits (*e.g.*, cardiac output, capillary density, etc.). For additional information, readers are encouraged to read work by Maestroni et al. ([157](#)) and Mang et al. ([158](#)). Furthermore, loss of muscle mass with age (*i.e.*, sarcopenia) is generally not counteracted by endurance training, whereas RT is highly effective for this. In addition, while low muscle mass may be related to increased morbidity and mortality ([159](#)), researchers have concluded that decreased muscular strength may have a greater impact on physical function as well as morbidity and mortality as age progresses ([159–161](#)). Finally, RT can positively influence muscular power (force $\times$ velocity) capabilities, which is a main predictor of muscle function and independence in old age ([162](#)).

For those seeking to enhance athletic performance, engaging in RT can provide additional benefits that may increase their potential to execute various skills, mitigate fatigue, and decrease the likelihood of injury ([163](#)) or in the event of injury may reduce the severity and total time to return to full function and competition. For additional reading on the importance of muscular strength and its role in athletic performance, readers are encouraged to read the review by Suchomel et al. ([164](#)).

Frequency of Resistance Training Exercise

Regardless of one's RT experience, frequency of training is often represented by the number of days per week an individual engages in the activity. It is important to consider that there is indeed an ideal range of training frequency; that is, training too frequently — before adequate recovery has occurred between bouts of exercise — over an extended period of time may increase risk of injury and illness and may accrue into a state of elevated fatigue ([165](#)). Training infrequently may prevent an individual from receiving a sufficient training stimulus to achieve the desired adaptations. In response to a bout of heavy RT, a cascade of events occurs within the body that stimulates recovery and repair of the tissues worked in that session. It can be advantageous to allow 48–72 hours to pass between bouts of heavy resistance exercise that repeatedly and directly target the same muscle group ([166, 167](#)). Therefore, for most individuals, it is recommended to engage in RT $2\text{--}4 \text{ d} \cdot \text{wk}^{-1}$.

A distinction between the general versus athletic population lies in the frequency of RT workouts and the individual's availability. For example, the considerations with the general population are grounded largely in the individual's availability around their work and/or family schedule; however, for the competitive athlete,

further consideration must be given to include the timing within the athletic year, often referred to as an annual plan.

The annual plan is multilevel and serves as a guide for outlining the training for the sporting year and includes important dates, such as competition and testing sessions ([Table 5.5](#)). At the broadest level, the macrocycle serves to outline the four seasons of the sport year: off-season, preseason, in-season, and postseason. Once these seasons are identified, each may then be broken down into mesocycles, or blocks of training. These blocks are commonly prescribed in 4-week bouts but may range from 2 to 6 weeks depending on the competition schedule, proximity to peaking for a competition, and the individual's tolerance to accumulated fatigue. Within each mesocycle exists microcycles, where individual weeks are broken down into days. Microcycles are generally where the programming exists within the workout plan. Based on the specific needs of the individual, many RT principles for athletes may apply for those individuals who wish to compete or engage in recreational sports throughout the lifespan. For more information on RT principles specific to the athletic population, the exercise professional is recommended to read additional references for a full scope and broader perspective with regard to developing an annual plan ([168–170](#)).

TABLE 5.5 Developing the Annual Plan

Macrocycle	• Identify the sport phases: Off-season, preseason, in-season, postseason • Outline the mesocycles within each sport phase of the macrocycle
Mesocycle	• Based upon the sport season, assign the training objective to the mesocycle • Early off-season: Strength-endurance/hypertrophy • Late off-season: Basic strength progressing to maximal strength • Preseason: Maximal strength and power • In-season: Maintain strength and power • Postseason: Active rest
Microcycle	• Identify the number of microcycles within each mesocycle • Within each microcycle, identify the number of resistance training days • Within each day, determine the exercise selection, intensity, volume (sets × repetitions), and rest periods

While the general population may resistance train between 2 and 4 d · wk⁻¹ for a prolonged period, those who are competitive athletes or performance-based should alter their RT frequency with regard to the sport season and sport-training demands. For example, during the off-season, sport training may be deemphasized while RT is emphasized and may occur 4 d · wk⁻¹ to promote the physiological and anatomical adaptations for subsequent RT phases. While transitioning toward preseason, RT sessions may decrease to 2 or 3 d · wk⁻¹ as the emphasis shifts toward sport training sessions. During the in-season, it would be appropriate to engage in RT 1 or 2 d · wk⁻¹ to maintain the RT adaptations that occurred in the preceding phases. Finally, during the postseason phase, the emphasis should be on recovering from the competitive season and minimizing additional stress to the body, and any RT should not be strenuous, whether in intensity or the total volume. Therefore, RT in the postseason should be 0–3 d · wk⁻¹ where the emphasis should strictly lie in low-intensity, low-stress exercises and may not include RT exercises.

Intensity of Resistance Training Exercise

Intensity has commonly been used synonymously with the weight, or load, lifted when referring to RT. However, in addition to load, intensity should also consider the level of technical difficulty, number of muscle groups and joints involved, speed of movement, and ROM. Thus, intensity should not be used interchangeably with load as the stress imposed by an exercise exists on a spectrum with consideration to the above facets of intensity (*e.g.*, loaded lunges vs. 30-m sprinting vs. overhead press).

Multiple methods exist for prescribing load and intensity, including as many reps as possible (AMRAP), 1-repetition maximum (1-RM), repetition maximum (RM), rating of perceived exertion (RPE), and repetitions in reserve (RIR) ([171](#)). It is important to select a method that makes sense for the exercise, the RT goal, and the individual.

A benefit of AMRAP is ease of implementation, whereby one load (may also be bodyweight) is moved in an exercise until failure or until a set time has lapsed, for example, AMRAP in 45 seconds. A drawback is when instructed to complete a task as quickly as possible, individuals may focus more on the number of repetitions more than the quality or technique of the movement. While this is an efficient method for RT when working in group settings, it is not necessarily appropriate when working with competitive athletes or individuals with specific RT goals.

A common method of loading RT exercises is by a percentage of 1-RM (%1-RM), or the greatest load an individual can lift with sound technique, according to the 1-RM testing standards for that exercise (see [Chapter 3](#)). 1-RM use is best suited for individuals with more RT experience; in novices and those with intermediate experience, there are more rapid increases in the maximal amount one can lift due to the resulting neural ([172](#)) and subsequent muscle architectural adaptations ([173](#)).

Similar to the 1-RM, RM testing can be performed and then calculated to estimate an individual's 1-RM. Examples include 3-RM, 5-RM, 10-RM, etc. ([Table 5.6](#)). However, the heavier the loads, for example, 3-RM, the more accurate the estimated 1-RM becomes ([174](#), [175](#)). An advantage is being able to implement this testing in individuals who may not yet have the strength to maintain technical composure under heavy loads. This method may also be easily incorporated into an RT session, though if testing multiple exercises, it is advised to follow the recommended order of exercise implementation that is presented within Types of Resistance Training Exercises; if testing one exercise, it is advised to perform that exercise first in the session. When selecting exercises for 1-RM testing, exercises that recruit one or more large muscle groups and involve multiple joints should be chosen as they are better able to handle the heavy loads and may be a more practical indication of one's abilities.

TABLE 5.6 RM Percentage of 1-RM

Number of completed repetitions	Percentage of 1-RM
1	100
2	95
3	93
4	90
5	87
6	85
7	85
8	80
9	77
10	75
11	70
12	67
15	65

Sheppard JM, Triplett NT. Program design for resistance training. In: Haff GG, Triplett NT, editors. Essentials of strength training and conditioning. Champaign (IL): Human Kinetics; 2016. p. 439–468; Chapman PP, Whitehead JR, Binkert RH. The 225-lb reps-to-fatigue test as a submaximal estimate of 1RM bench press performance in college football players. *J Strength Cond Res.* 1998;12(4):258–261; Lander J. Maximum based on reps. *NSCA J.* 1984;6(6):60–61; Mayhew JL, Ball TE, Arnold ME, et al. Relative muscular endurance performance as a predictor of bench press strength in college men and women. *J Appl Sport Sci Res.* 1992;6(4):200–206; Morales J, Sobonya S. Use of submaximal repetition tests for predicting 1-RM strength in class athletes. *J Strength Cond Res.* 1996;10(3):186–189.

Beyond RM testing, other intensity prescription methods have been developed that enable an individual to subjectively rate their perceived exertion during an exercise or activity. Two methods of this are RPE and RIR. RPE is a scaled method, where the individual rates the degree of difficulty on a 1–10 scale where 1 is at rest and 10 is maximal effort. RIR is based upon RPE and is adapted to RT, where a maximal effort is 0 RIR and no further repetitions can be completed with sound technique. Unlike RPE, RIR does not exist on a scale, and accuracy of how many repetitions are “in reserve” after completing a set decreases as the number of repetitions and the load are farther from set-failure ([176](#)). While the primary limitation to these methods involves the individual’s familiarity with a maximal effort, these are safe methods to begin incorporating RT load in a manner that enables the individual to adjust load based upon their abilities for that session. For more on RPE and RIR, readers are encouraged to read Zourdos et al. ([177](#)) and Helms et al. ([178](#)). To learn more about using RM and RIR to predict RT 1-RM, readers are encouraged to read DeWeese et al. ([179](#)) and McKeever and Howard ([180](#)).

When training for specific adaptations, the load is important for adequately stimulating the adaptation. To train muscular endurance, the load should remain relatively light to permit the high number of repetitions per set, estimated to be at or below 67% of an individual’s 1-RM ([168](#)). Due to the high repetitions and low loads — below the threshold of a strength stimulus — muscular endurance RT is not recommended for competitive athletes. For strength endurance and hypertrophy, loads range between 67% and 85% 1-RM to target increases in work capacity — or repeatability of a task — while also meeting the minimum recommended threshold (60% 1-RM) for a strength stimulus ([168](#), [181](#)). To increase strength, loads should

be at or above 85% of 1-RM (168). To train for power, loads should be exercise specific and may range from 0% to 95% 1-RM (182). For people who are not accustomed to RT, lighter loads with higher repetitions may afford greater practice for improved form and technique and, initially, enough of a stimulus for strength gains (183).

Time (Rest Periods of Resistance Training Exercises)

Rest periods are an underpinning component of an appropriate RT program and play a crucial role in allowing the body to recover from a set of exercise. It is important to highlight that resistance-based interval training and RT are not synonymous, and a primary concern of those implementing resistance-based interval training should be maintaining safe and proper technique and movement competency throughout the full workout (see CRF section above). Appropriate inter-set rest periods encourage sustained technique and ability to tolerate increased loads throughout the session, furthering the stimulus for strength adaptations. Schoenfeld et al. (184) compared strength and hypertrophy adaptations between 1- and 3-minute rest periods, finding that the longer rest periods resulted in greater strength and hypertrophy adaptations. Generally, the more intense or taxing the exercise and load, the more rest should take place between sets. Muscular endurance rest periods may range from 30 seconds to 1 minute; strength endurance and hypertrophic rest periods range from 90 seconds to 3 minutes (184–186) and maximal strength and power should range from 2 to 5 minutes (168, 187, 188) (Table 5.7).

TABLE 5.7 Rest Periods by Training Goal

Muscular Endurance	Strength-Endurance and Hypertrophy	Strength	Power
30–60 s	90 s to 3 min	2–5 min	2–5 min

Adapted from McKeever and Howard (180).

Types of Resistance Training Exercises

Within RT, there exists a wide range of equipment modalities, including bodyweight (exercises without additional resistance), suspension devices, resistance bands and tubes, machines, and free weights (Table 5.8). It is paramount to consider the individual’s abilities and one’s own ability to correctly demonstrate and/or cue when choosing the equipment modality and exercise.

TABLE 5.8 Resistance Training Modalities and Their Benefits and Drawbacks

	Benefits	Drawbacks
Bodyweight	• No additional equipment • Relatively safe	• No load versatility • May plateau easily
Suspension devices	• Small equipment footprint • Relatively safe • Easily adaptable exercise difficulty • High exercise versatility	• Limited load versatility • May plateau easily
Resistance bands/tubes	• Small equipment footprint • Relatively safe • High exercise versatility	• Limited load versatility • Level of difficulty limited to each band's resistance • As resistance increases, overall elasticity decreases
Machine weights	• Relatively safe • Restricts plane(s) of movement (strength sticking points or point of weakness, working around an injury) • High load versatility (up to a point)	• Large equipment footprint • Most are not versatile for multiple exercises • Restricts plane(s) of movement (decreased transfer to daily activities and movements) • Load versatility often limited to the load built into the machine
Free weights	• Most have small equipment footprint • Relatively safe <i>when implemented and supervised correctly</i> • Can challenge all planes of motion (exercise dependent) • High exercise versatility • High load versatility	• Must be supervised more closely • Use of barbells requires larger footprint

Bodyweight exercises are beneficial for individuals new to an RT program, are learning a new exercise or technique modification, and in scenarios where there may be limited equipment, space, or supervision (*e.g.*, moderate to large group sessions).

Suspension devices are a great method for enabling the individual to adjust their position for an increased or decreased challenge, have a reasonably small footprint during use, and accommodate many different exercises.

Resistance bands and tubes offer many exercise options and range in resistance from very light to very heavy, depending on the band width and thickness. It is important to check the bands or tubes for wear on a regular basis to prevent instances of the band or tubing breaking during use.

RT machines are those where the degree of motion is limited to a particular path or ROM and at least one plane of motion has been restricted (189). The resistance is often set with specific load increments, and the inherent stability provided by the device is both a benefit and drawback as it offers support of one or more areas of the body but simultaneously decreases the need for the individual to contribute to stability (e.g., leg extension machine vs. squats). Due to the restrictive nature of machine-based exercises, there is limited carryover into activities of daily living and athletic skills (189).

Free weights exist in a multitude of forms, including barbells, dumbbells, kettlebells, and medicine balls. An advantage of free weights lies in the versatility of most of the equipment to be used for many exercises and the relatively small footprint of most pieces. However, extra supervision and cueing may be necessary with use of free weights to ensure safe implementation. As the load is not stabilized by an outside component, there is increased demand for stability and control by the individual while using free weights, particularly with heavy loads.

RT programs may consist of various classifications of exercises, such as multijoint, single-joint, and trunk stabilizing. An important consideration while developing an RT program is the efficiency and effectiveness of the exercises prescribed to achieve the individual’s goals.

Multijoint exercises are those that involve a coordinated, synergistic effort of multiple muscle groups (both agonists and antagonists) that act upon multiple joints simultaneously. These include, but are not limited to, squats, deadlifts, lunges, bench press, push-ups, pull-ups, and bent over row. This exercise classification is effective for inducing a greater stress stimulus compared to single-joint and trunk stability due to the number of muscle groups simultaneously engaged and involvement of the nervous system to coordinate the motion.

Single-joint exercises involve movement of a single joint to perform the exercise. Commonly prescribed examples include leg extensions, leg curls, calf raises, biceps curls, and triceps extensions. Due to the nature of single-joint exercises, there is a reduced coordination effort to perform the exercise. This classification may be well incorporated in situations where an individual may have aesthetic goals, a degree of weakness in a specific muscle group (including postinjury rehabilitation), or to supplement a portion of another exercise.

Trunk stability exercises aim to target the musculature of the torso, and a wide variety of exercises exist within this classification. It is important to note that the trunk stabilizing muscles exist in all areas of the torso, including anterior, lateral, and posterior regions.

As part of a comprehensive RT program, one should consider including a variety of multijoint, single-joint, and trunk stability exercises, with the exercises that are most fatiguing — large muscle multijoint, technically demanding, and high intensity — performed before less fatiguing exercises (small muscle multijoint, single-joint, and trunk stability) (Table 5.9). For a thorough overview of RT methods, readers are directed to Suchomel et al. (190).

TABLE 5.9 Order of exercise implementation

Large muscle multijoint → small muscle multijoint ↔ large muscle single joint → small muscle single joint ↔ trunk stability
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Regardless of RT mode, an individual’s ability to safely perform an exercise with *good* movement competency allows the body and joints to pass through a natural ROM, provided there is sound stability and control throughout the movement. As an example, many have propagated the adage “knees don’t go past the toes,” but this alteration of natural mechanics may actually increase negative stress to the body: by keeping

the lower leg more vertical to keep the knees behind the toes, the ankles are not permitted to pass through their full dorsiflexion capacity, the hips remain elevated above knee height, and the torso leans forward, which then shifts the stress of the load to the hips and low back, which then increases the risk of low back injury. For additional reading on this matter, readers are directed to the work by Fry et al. ([191](#)).

Volume of Resistance Training Exercise

RT volume may be quantified by the number of sets and repetitions (sets × repetitions) performed during a given training session or throughout a week. Similar to intensity, the prescribed volume should be determined in conjunction with the individual's RT goals and experience ([Tables 5.10 and 5.11](#)). Untrained individuals may see significant improvements in a variety of fitness characteristics with as few as one set per muscle group per training session. Therefore, low volume RT programs (*e.g.*, <4 weekly sets per muscle group) may be a viable option for untrained or individuals with limited time ([192, 193](#)). In contrast, more advanced individuals may require greater stimuli to achieve specific training goals due to the dose-response relationship. It is important to recognize that altering volume by simply increasing sets may result in a plateau or regression of muscular fitness. Therefore, consideration of the balance between sets, repetitions, and load or intensity should occur to offset the likelihood of an RT plateau and ensure the RT goals are progressed toward and met ([Tables 5.10 and 5.11](#)).

TABLE 5.10 Resistance Training Recommendations for the General Public Individual

Health/Wellness	Untrained	Intermediate	Advanced
Objective	Develop proper foundational technique	Introduce moderately complex exercises	Progress to highly complex exercises
Exercise Types	• Foundational multijoint • Single joint as necessary • Light loads	• Foundational multijoint • Weightlifting derivatives • Single joint as necessary • Increasing loads	• Foundational multijoint • Weightlifting derivatives • Full weightlifting movements • Single joint as necessary • Increase loads and intensity
Frequency	2–3	3–4	4–7
Number of exercises	6–10	5–10	4–5
Sets	1–2	2–5	RT goal dependent
Repetitions	8–12	RT goal dependent	RT goal dependent

Adapted from McKeever and Howard ([180](#)) and Sheppard and Triplett ([168](#)).

TABLE 5.11 Resistance Training Recommendations for the Competitive Athlete

Athletes/Performance	Off-Season	Preseason	In-Season	Postseason
Objective	• Early off-season: Improve work capacity (strength-endurance) • Late off-season: basic strength development	• Maximal strength development • Incorporate velocity-based exercises • Enhance power output	• Maintain strength developed in the late off-season and preseason	• Active rest • Allow the body to recover from the sport season
Exercise Types	• Multijoint full ROM • Foundational strength exercises	• Multijoint full ROM • Foundational strength exercises	• Multijoint full ROM • Foundational strength exercises	• Multijoint full ROM • Foundational strength exercises
Frequency	3–5	2–3	1–2	0–3
Sets	• SE/hypertrophy: 3–6 • Strength: 2–6	3–5	1–5	0–3
Repetitions	• SE/hypertrophy: 6–12 • Strength: ≤6	1–5	3–5	1–5
Inter-set Rest Periods	1.5–3 min	2–5 min	2–5 min	Variable

ROM, range of motion.

Adapted from McKeever and Howard ([180](#)) and Sheppard and Triplett ([168](#)).

While higher volumes of training may promote muscle hypertrophy, they may negatively impact rapid force production and power output ([194, 195](#)); however, untrained individuals may still initially improve these characteristics due to the novel exposure of RT and the associated neural adaptations ([172](#)). Conversely, lower volumes of training may aid in the development of maximal strength and power output due to decreased session and accumulated fatigue and the ability to use heavier loads, yet minor decreases in muscle mass may occur due to the de-emphasis of hypertrophic stimuli. This reversibility of hypertrophy is normal as one advances through the progressive phases of training.

When determining the desired volume per muscle group per week, the sessions may be programmed in such a way that directly or indirectly engages targeted muscle groups. For example, over the course of the 3-RT session week, the quadriceps may be involved during (a) squats, (b) stiff-legged deadlifts (RDLs), and (c) front squats. Though not directly targeted during the stiff-legged deadlift, the quadriceps play a stabilizing role and are stimulated in the coordination of the exercise. Using different exercises to train the same muscle

group may add variety and allow for a training stimulus to remain novel, allowing continual progression to occur (196).

Progression of Resistance Training Exercise

Progressive overload may be defined as the gradual increase of the stress placed on the body (196) and may be necessary to provide a greater training stimulus for improvements in various fitness characteristics. However, as an individual gets closer to their genetic ceiling, the rate and magnitude of further improvements will be limited based on the principle of diminishing returns (197). Therefore, one should consider implementing progressive overload via the progression or regression of RT volumes, loads, intensities, and/or exercises. Multiple methods exist to adjust the RT stimulus, and one should consider the training level of the individual and the applicability toward their goals when determining which aspects to adjust. Additionally, each of these components of overload should be viewed as existing on an overlapping spectrum, where a change in one parameter may be accommodated by a change in another when appropriate, for example, increased load and decreased repetitions per set when transitioning from a strength-endurance to strength block.

Volume alterations include increasing or decreasing the number of sets or repetitions, and while increasing repetitions is a common method of overload, it is not always appropriate to add more volume. It would be appropriate following a high-volume strength endurance/hypertrophy phase with a strength phase, where the number of sets may stay the same or increase and repetitions per set decreases to accommodate the greater load of the strength stimulus, for example, 3×10 to 4×5 . One should exercise caution when increasing the number of repetitions within a given RT phase as additional repetitions may modify the targeted adaptation. Increases in the number of prescribed sets for an exercise may also have exponential effects as the training stimulus would be significantly changed. However, it should be noted that athletes may complete a training phase in which the number of sets is increased for 1 week and decreased during subsequent weeks, termed a planned overreach and taper. Fewer repetitions or sets may promote recovery or may be incorporated to allow the individual to train with heavier loads.

An overload stimulus may also be provided by increasing load during subsequent RT sessions via multiple strategies (171). These include increasing training loads by 2.5%–5% of a 1-RM during subsequent training sessions using the same exercises, typically during the following week; linear loading, where the load is gradually increased; and the 2-for-2 method, whereby loads are increased after individuals can perform two repetitions more than what was prescribed during two consecutive sessions for a given exercise (168). While linear loading and the 2-for-2 methods may be beneficial for untrained individuals or novice athletes, these methods provide little variation and may be detrimental if applied for prolonged periods of time. On the opposite end of the spectrum, set-repetition best and velocity-based training may provide more information about the daily readiness to train using subjective and objective feedback, respectively, and may be more beneficial for individuals with more training experience.

It should be noted that regressing training volume and loads may also provide a unique training stimulus and provide individuals with the opportunity to recover and adapt from previous RT fatigue. For example, if a 4-week training block is prescribed and the load is progressively increased for each of the first 3 weeks, a deload of volume or load during week four may allow individuals to realize the training adaptation provided by the stimulus from the first 3 weeks.

Progressive overload may also be provided by changing the speed or ROM of an exercise. For example, individuals may progress to using a faster or more ballistic exercise that has a similar movement pattern after they have demonstrated adequate competency of the more basic exercise. While there are numerous examples, some include progressing from landing off a box to a drop/depth jump or bench press to a bench press throw. It should be noted that these progressions may also be implemented to promote different training adaptations, in the above example: power output compared to general strength. However, exercise regressions may also take place, especially if an individual has difficulty performing a given exercise with greater speeds.

To adjust the ROM of an exercise, individuals can progress an exercise by performing greater ranges of motion, which has been shown to increase the intensity of the training stimulus (198). For example, some individuals may have difficulty achieving greater squat depths potentially due to inexperience, their stance, or lack of flexibility, mobility, balance, etc.; however, small changes in technique (*e.g.*, modifying stance width and/or slightly turning their toes outward) may help progress to greater ranges of motion (199). Additionally, RT may significantly increase ROM, especially in untrained and sedentary individuals (200). It should be noted that regressions to smaller ROM may be necessary due to potential injuries; however, they may also be used for training specificity and to decrease relative levels of fatigue. For example, partial squats may improve jump and sprint performance to a greater extent than full ROM squats (201) and may also provide an effective force and power output stimulus (202), especially if they are performed with ballistic intent (203).

A unique training stimulus may be provided by implementing different exercise and load combinations. For example, one may program a variety of exercises using heavy or light loads (performed to failure) to achieve similar hypertrophy results (204). However, the combination of bilateral and unilateral exercises may be considered to develop these characteristics as the latter may increase the muscle activation of certain muscle groups (205). Thus, the addition or combination of both bilateral and unilateral exercises may provide a unique overload stimulus to individuals if not already used. From a strength standpoint, a progressive overload may be provided by prescribing exercises that have a greater capacity to produce force. This may include implementing multijoint versus single-joint exercises or exercises that are more ballistic in nature. Finally, one may prescribe exercises that produce greater force and/or velocity characteristics to improve power output. Researchers have shown that the hexagonal barbell jump and jump shrug produced greater power output compared to the jump squat (206, 207). However, it is important to note that differences in power output may be load dependent for each exercise, suggesting that a range of loads should be prescribed to optimize this characteristic (208, 209).

A key facet of exercise progression should be based on movement competence: an individual should not be progressed to a more complex exercise if they fail to demonstrate technical competency and ROM with a simpler exercise. For example, a bodyweight squat should be completed before performing a goblet squat, front squat, or back squat. It should be noted that more complex exercises such as the snatch or clean may not be completed safely without the ability to perform other exercises (*e.g.*, deadlift, stiff-legged deadlift, overhead or front squat, etc.) that serve as the foundation to these movements. RT skill testing batteries have been previously developed and may help determine if an individual is ready to progress to a more complex exercise (210–212). Comfort et al. (213) and Morris et al. (214) also discuss appropriate progressions and regressions of weightlifting movements and their derivatives. Additionally, within an exercise progression, exercises should be taught from less taxing to more taxing. However, it is important that individuals can move through a given ROM without pain. If there are difficulties or limitations during

pain-free movement, offering alternative exercises or modifying the technique of the exercise may be necessary. Regardless, it is important to gradually progress each individual based on their abilities to help them build their confidence and achieve their desired RT goals.

FLEXIBILITY

Flexibility, or the ability to move through a joint's full ROM, has long been considered a component of fitness and therefore can be included as part of any Ex R_x (215). Joint ROM and flexibility can be improved by engaging in a number of different types of exercises that are specific to the target joint (3). These activities can include various types of stretching (active and passive static, proprioceptive neuromuscular facilitation [PNF], dynamic and ballistic) (216), foam rolling (217), vibration (218, 219), and resistance exercises (220) among others. In addition, improved flexibility can be achieved not only with the specific stretched muscle or joint, but moderate magnitude ROM increases can also occur in nonstretched muscles in other parts of the body (221). For example, research has shown that unilateral stretching of the quadriceps will improve ROM of the contralateral quadriceps (222), whereas stretching the hip adductor muscles (groin) can improve the flexibility of the shoulders and stretching the shoulders can improve flexibility of the hamstrings (hip flexion) (223). These findings may be important for individuals who are rehabilitating an injury and cannot stretch particular muscle groups. Stretching major muscle groups seems to have global flexibility effects on the body (221). Joint ROM can be improved immediately after performing stretching exercises and has shown chronic improvement after about 3–4 weeks of regular stretching two to three times per week (3). Postural stability and balance can also be improved by consistently or regularly engaging in flexibility exercises (224). Overall, the goal of a flexibility program should be to develop ROM in the major joints and muscle/tendon groups in accordance with individualized training goals. Moreover, flexibility, along with aerobic capacity and muscular fitness, is a core aspect of minimizing mobility deficits experienced with aging and therefore should be considered as part of an Ex R_x for such purposes (225). With stretch-induced reductions in muscle stiffness, increases in musculotendinous compliance and stretch tolerance, and alterations in muscle fiber fascicle angles and rotations (216, 226–229) increases in mobility will contribute to more efficient movement. Hence, there is less resistance and less energy needed to move through an even greater scope of movement.

Is improved flexibility only important for movement efficiency? Increased flexibility is not only important for improving ROM, decreasing the resistance to movement (226) but can also decrease musculotendinous injury incidence especially with explosive or high-speed movement with change of direction (229, 230). This reduction in musculotendinous injury incidence may be related to the increased forces produced at longer muscle lengths (altered force-length relationship), where most muscle and tendon injuries occur. Chronic stretch training can also improve balance, which may contribute to a decreased incidence of falls and associated injuries, especially with older adults (see [Chapter 6](#)) (230). Chronic stretching training induces consistent pain alleviation from a variety of joints and muscles (231). Pain reduction mechanisms have been primarily related to increased pain (sensory) tolerance, but other possibilities such as neural pain modulation (gate control theory, diffuse noxious inhibitory control), myofascial meridians, and increases in parasympathetic nervous activity may also contribute (231). The cardiovascular system receives benefits from stretching with decreases in arterial stiffness, resting HRs, diastolic BP, and improved endothelial function (232). Furthermore, although RT is very time efficient for improving muscular strength and hypertrophy, recent evidence shows that chronic stretching can also increase muscular strength and hypertrophy (233–238). Muscular strength and hypertrophy were improved with just 10 minutes of daily

plantarflexors static stretching over 8 weeks ([235](#)). Hence, improvements in flexibility through stretching (and other activities to be discussed) have many global positive effects on health and function.

Evidence suggests that the type of stretch ([Box 5.4](#)) and when the stretching is performed may impact exercise performance. Although stretching exercises for the purpose of increasing ROM are encouraged for all populations, performing prolonged static stretching exercises (*e.g.*, >60 seconds per muscle group) for the sole purpose of improving exercise performance or reducing muscle soreness is not recommended ([19](#), [229](#), [239](#), [240](#)). Static stretching, where one slowly stretches a muscle/tendon group to a point of mild discomfort for a period of time, is very common in fitness environments. This type of stretching results in improvements in ROM that can be a result of decreases in neural inhibition, musculotendinous unit stiffness, or tolerance to the stretch ([241-244](#)). However, there is a diminishing rate of return with static stretching, such that stretches performed for more than 60 seconds have a deleterious effect on exercise performance (*i.e.*, sprinting, maximal contractions, etc.) ([229](#)).

Box 5.4 Flexibility Exercise Definitions

Ballistic methods or bouncing stretches use the momentum of the moving body segment to produce the stretch.

Dynamic or slow movement stretching involves a gradual transition from one body position to another and a progressive increase in reach and range of motion as the movement is repeated several times.

Static stretching involves slowly stretching a muscle/tendon group and holding the position for a period of time (*i.e.*, 10–30 s). Static stretches can be active or passive.

Active static stretching involves holding the stretched position using the strength of the agonist muscle as is common in many forms of yoga.

Passive static stretching involves assuming a position while holding a limb or other part of the body with or without the assistance of a partner or device (such as elastic bands or a ballet barre).

Proprioceptive neuromuscular facilitation (PNF) methods take several forms but typically involve an isometric contraction of the selected muscle/tendon group followed by a static stretching of the same group (*i.e.*, contract-relax).

Adapted from (3).

PNF stretching (245) is also used to stimulate the musculotendinous unit and is characterized by an isometric contraction of a target muscle and a concentric contraction of an opposing muscle, along with a controlled approach to the stretch (245). Even though PNF stretching has been previously suggested to yield greater gains than static and ballistic stretching (246), recent evidence suggests that this is not the case (247, 248). While all stretching types can substantially improve ROM, the most recent meta-analysis reports that PNF and static stretching produced greater ROM compared to ballistic/dynamic stretching (216). However, there was no significant difference between static and PNF stretching ROM gains. Therefore, for the purpose of improving ROM, it is reasonable to recommend any type of stretching exercise for individuals engaged in a general fitness program.

Dynamic stretching involves controlled movements through an active ROM, is a recommended component of the warm-up, and mimics the intended exercise or sport activity subsequent to the warm-up. Dynamic stretching can elevate core temperature, which leads to increased neuromuscular conduction and compliance and enzymatic activity, which may accelerate energy production (229). The available evidence suggests that short sessions of dynamic stretches (<30 seconds) do not adversely affect exercise bout performance, and prolonged sessions (>30 seconds) may facilitate performance (229, 240). The augmented performance after dynamic stretching may be related to postactivation performance effects (249, 250) whereby prior dynamic muscle contractions increase myosin light chain phosphorylation and neural potentiation (*e.g.*, decreased motoneuron activation thresholds and increased persistent inward currents) sensitizing the muscle fibers to the calcium release as well as amplify the motoneuron activation and facilitate subsequent performance, respectively (251). Therefore, dynamic stretching is recommended both prior to more vigorous exercise and as a potential adjunct to improving sport performance.

FITT FLEXIBILITY EXERCISE RECOMMENDATION

Flexibility exercises are recommended to improve joint-specific ROM and to improve performance. ROM can be improved with static, ballistic, dynamic, and/or PNF stretching. Joint-specific flexibility exercises are most effective when the muscles are warm, and prolonged static stretching should be avoided prior to an exercise bout ([228, 229, 240, 252](#)). Extensive static, ballistic, and/or PNF stretching should be performed on their own, as part of a specific program to increase ROM, and not preceding any exercise activity. Short duration (<60 seconds per muscle group) static stretching and dynamic stretches in combination with activity-specific dynamic activities (e.g., basketball, soccer, hockey, tennis drills before the sport activity) are encouraged prior to any exercise bout and may also be used to improve performance ([Table 5.12](#)) ([227–229, 240, 252](#)).

TABLE 5.12 Dynamic Stretching Recommendations

FITT	Recommendations
<i>Frequency</i>	• Prior to vigorous activity or sport participation • Daily (≥5 per week provide optimal benefits but improvements are still seen with 2 times per week)
<i>Intensity</i>	• Full range of motion (dynamic stretching: submaximal limb velocity)
<i>Time</i>	• 3–6 sets of 30–90 s with 15-s rest periods between sets
<i>Type</i>	• Dynamic stretching for all major muscle-tendon units is recommended • Dynamic and ballistic (higher velocity movements typically with bouncing actions at the end ROM) flexibility

Types of Flexibility Exercises

Flexibility exercise should target the major muscle tendon units of the shoulder girdle, chest, neck, trunk, lower back, hips, posterior and anterior legs, and ankles ([3](#)). [Box 5.4](#) shows the several types of flexibility exercises that can improve ROM. Properly performed ballistic stretching has been shown in some studies to be equally as effective as static stretching in increasing joint ROM and may be considered for adults or athletes who engage in activities that involve ballistic movements such as basketball ([253–256](#)).

Volume of Flexibility Exercise (Time, Repetitions, and Frequency)

The progression of flexibility exercises is as complex as any other component of the Ex R_x and should be individualized just like the aerobic and muscular fitness components. Evidence suggests that flexibility exercises should be progressed based on individuals' level of discomfort and within their current ROM. Additionally, static and PNF stretching should be performed exclusively as part of a program to improve ROM. For improved performance, prolonged static stretching exercises should be avoided, unless included as part of a dynamic warm-up that are specific to the activity being performed ([227–229, 240, 252](#)). Because static stretching can reduce the incidence of musculotendinous injuries and is effective for increasing ROM for many sport applications, it can still be included in a preevent warm-up if combined with aerobic activities, dynamic stretching, and dynamic sport-specific activities ([227–229, 240, 252](#)). Although a number

of studies have shown no adverse effects of static stretching when incorporated into a full dynamic warm-up ([257, 258](#)), there can be positive psychological effects as well ([259](#)).

If the individual goal is to improve ROM, holding a stretch for 10–30 seconds to the point of tightness or slight discomfort enhances joint ROM. Moreover, evidence suggests that holding a static stretch for over 60–90 seconds without additional dynamic activities will lead to performance decrements ([227, 229, 240](#)). In older adults, stretching for 30–60 seconds may result in greater flexibility gains than shorter duration stretches ([3](#)) (see [Chapter 6](#)). For PNF stretches, it is recommended that individuals of all ages hold a light-to-moderate contraction (*i.e.*, 20%–75% of maximum voluntary contraction) for 3–6 seconds, followed by an assisted stretch for 10–30 seconds ([3](#)). During a flexibility training session, stretching exercises should be repeated two to four times to accumulate a total of 90 seconds of stretching for each flexibility exercise by adjusting time/duration and repetitions according to individual needs ([229](#)). Performing flexibility exercises $\geq 2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$ will improve ROM, but stretching exercises are most effective when performed daily ([3](#)). A stretching routine following these guidelines can be completed by most individuals in ≤ 10 minutes ([3](#)).

FITT FLEXIBILITY VOLUME RECOMMENDATION

A total of 90 seconds of discontinuous flexibility exercise per joint is recommended. Holding a single flexibility exercise for 10–30 seconds to the point of tightness or slight discomfort is most effective. Older adults can benefit from holding the stretch for 30–60 seconds. A 20%–75% maximum voluntary contraction held for 3–6 seconds followed by a 10- to 30-second assisted stretch is recommended for PNF techniques. Performing flexibility exercises $\geq 2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$ is recommended with daily flexibility exercise being most effective. For individuals who are seeking additional or more specific levels of flexibility, the principles of Ex R_x for flexibility training are summarized in [Table 5.13](#) and are presented as *Frequency* ($\text{d} \cdot \text{wk}^{-1}$), *Intensity* (magnitude of loading), and *Type* (RT exercises selected).

TABLE 5.13 Flexibility Exercise Recommendations

FITT	Recommendation
<i>Frequency</i>	• $\geq 2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$ with daily being most effective
<i>Intensity</i>	• Stretch to the point of feeling tightness or slight discomfort
<i>Time</i>	• Holding a static stretch for 10–30 s is recommended for most adults • In older individuals, holding a stretch for 30–60 s may confer greater benefit • For proprioceptive neuromuscular facilitation (PNF) stretching, a 3–6 s light-to-moderate contraction (<i>e.g.</i>, 20%–75% of maximum voluntary contraction) followed by a 10- to 30-s assisted stretch is desirable
<i>Type</i>	• A series of flexibility exercises for each of the major muscle-tendon units is recommended • Static flexibility (<i>i.e.</i>, active or passive), dynamic flexibility, ballistic flexibility, and PNF are each effective

Adapted from [\(3\)](#).

Alternatives to Stretching to Increase ROM

Recent commentaries have questioned whether stretching is actually necessary for improving flexibility ([260, 261](#)). They have suggested stretch training benefits such as improvements in flexibility, balance, cardiovascular measures, pain reduction, and decreased injury incidence are also available with RT. Two meta-analyses have reported no significant differences in ROM gains between resistance and stretch training ([200, 262](#)). While the meta-analysis with 11 studies ([262](#)) demonstrated a small but nonsignificant effect size (Hedges' $g = -0.22$; $p = 0.21$) in favor of stretching, the other meta-analysis analyzed 55 studies ([220](#)) and reported no significant differences with trivial effect size differences ($ES = 0.08$; $p = 0.79$). Thus, free weights, machine-based RT, and Pilates provided similar ROM improvements as static stretching. However, bodyweight calisthenics did not induce similar increases, possibly due to a more restricted ROM. Thus, while additional stretching prior to or after RT may not be necessary, stretch training may still play a role in certain circumstances. For instance, RT may not be appropriate as part of a warm-up prior to competition and therefore stretching could still be an important preparatory activity. Yoga is widely used not only for improving flexibility but also for inducing positive changes in mental (psychological) attitudes (*e.g.*, enhancing calm and peacefulness) ([215](#)). However, yoga encompasses not only stretching, but also breathing techniques, and meditation among other techniques ([215](#)). Stretching is also used for relaxation for many

individuals, and hence, higher intensity activities like RT would not be as useful. Furthermore, more extensive flexibility needs such as with gymnastics, figure skating, certain combat sports (i.e., judo, karate), hurdling, and other sports may necessitate dedicated stretch training routines to achieve the necessary ROM.

Foam rolling is also a very popular preparatory (warm-up) and postexercise (cool-down) activity. Foam rolling can not only increase muscle extensibility improving ROM ([217](#), [263–265](#)) but also reduce pain at the rolled muscle and contralateral muscles ([266](#), [267](#)) and reduce the symptoms of delayed onset muscle soreness ([268](#), [269](#)). A further benefit of foam rolling compared to prolonged static stretching as part of a warm-up is that foam rolling does not induce subsequent performance impairments ([265](#)).

A study questioned whether it was foam rolling or the movements associated with foam rolling that induced ROM increases ([270](#)). They reported that both foam rolling and sham rolling (simulated rolling movements on a padded dolly with wheels but with no rolling pressure on the muscle) provided significant, moderate-to-large magnitude increases in ROM. But there was no significant difference between the foam and sham rolling conditions. Since other studies have demonstrated increased ROM with roller massagers (arms are used to massage the legs, so no full body movement), rolling can increase ROM, but there is no additive effect on ROM of foam rolling compared to just rolling type movements.

In summary, while all types of stretching can increase ROM and provide additional benefits such as reductions in injury incidence, pain, as well as improved balance and cardiovascular functions, it is not the only activity that can accomplish many of these goals. RT, foam rolling, and other techniques such as vibration (conflicting evidence), tissue flossing ([271](#), [272](#)), and others may provide alternatives to stretching dependent on the individual's goals and desires.

FUTURE DIRECTIONS

A considerable body of scientific evidence supports the positive relation between exercise dose and fitness measures that can be achieved by progressing components of the FITT principle. The extent to which exercise intensity modulates physiologic adaptations has generally been a widely debated topic, partly due to inconsistencies with defining and identifying different classification ranges for exercise intensity across studies, scientific statements, and in applied settings. These inconsistencies in the literature requires deeper discussions and a need for concerted investigative efforts. An initial area of focus can be placed on unifying intensity classification terminology and ranges for cardiorespiratory and resistance exercise intensities (i.e., very low, low, moderate, high, very high). Consensus on terminology and intensity classification can facilitate a unified movement that will help investigative efforts to characterize the modulatory effects of intensity and ultimately influence exercise prescriptive practices in applied settings.

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CHAPTER 6

Exercise Prescription for Healthy Populations With Special Considerations

CHILDREN AND ADOLESCENTS

Children and adolescents, defined as individuals aged 6–19 years (also referred to as *youth*), are typically more physically active than adults. The World Health Organization (WHO) Guidelines on Physical Activity and Sedentary Behaviour recommends that children and adolescents accumulate an average of at least $60 \text{ min} \cdot \text{d}^{-1}$ of moderate- to vigorous-intensity physical activity (MVPA) (1). Resistance exercise and bone loading activities are also recommended on at least $3 \text{ d} \cdot \text{wk}^{-1}$ and count toward the $60 \text{ min} \cdot \text{d}^{-1}$ total (1). Globally, only 27%–33% of youth meet the recommended amount of MVPA required for health and well-being (2), with children and adolescents with disabilities less active than the general population (see [Chapter 11](#)) (3). Furthermore, a strong age-related decline in youth physical activity (PA) emerges at an early age, and the relative decline in MVPA is greater in girls than boys (4). Informed strategies are needed to improve movement behaviors in youth to mitigate the health risks associated with newly established levels of physical inactivity.

The intensity of PA can be defined in terms of the rate of energy expenditure required for a given activity or based on a perceived level of effort. Energy expenditure–based intensity metrics, such as metabolic equivalents (METs) or kilocalories, are not translatable to children due to differences in basal metabolic rates, energy expenditures per unit body mass, and movement efficiency for certain activities (5, 6). Although MET-based intensities for PA specific to youth are available (7, 8), youth PA intensity can be estimated using a perceived exertion scale from 0 (sitting) to 10 (highest effort possible), with moderate intensity at 5 or 6 and vigorous intensity starting at 7 or 8, relative to each individual's capacity (9).

In addition to the health benefits associated with PA, there is also evidence that limiting screen time, a marker of sedentary behavior, is independently

related to avoiding health problems in youth, such as increased adiposity and depressive symptoms; decreased fitness; and elevated blood pressure, blood lipids, and glycohemoglobin levels ([10–12](#)). Expert panels from the National Heart, Lung, and Blood Institute and the American Academy of Pediatrics (AAP) have recommended that children and adolescents limit total recreational screen time to less than $2 \text{ h} \cdot \text{d}^{-1}$ ([13, 14](#)). Guidelines for young children are lower, including less than $1 \text{ h} \cdot \text{d}^{-1}$ of screen time for 2- to 5-year-olds and none for infants less than 18 months old ([14, 15](#)). The AAP has updated their screen time guidelines, recommending a more tailored approach, encouraging families to seek a balance in screen time through the development of a Family Media Use Plan that incorporates rules regarding the quantity and quality of media content ([16](#)). Yet nationally, slightly more than half of children aged 6–11 years adhere to these guidelines of viewing less than $2 \text{ h} \cdot \text{d}^{-1}$ of screen time ([17](#)). Globally, about 25% of children younger than 2 years of age and about 35% of children aged 2–5 years meet screen time guidelines ([18](#)). Perhaps most importantly, the PA and sedentary behavior patterns of children can track into adulthood, so it is vital that youth initiate and maintain a physically active lifestyle from an early age ([19–22](#)), while also making a habit of replacing unnecessary sedentary time with healthier alternatives.

Children and adolescents are physiologically adaptive to aerobic exercise training ([23](#)), resistance training ([24](#)), and bone loading exercise (*i.e.*, jumping and lifting) ([25, 26](#)). In fact, evidence suggests that prepubescent children who participate in resistance training can achieve relative strength gains similar to those seen in adolescents ([27–29](#)). Furthermore, there is strong evidence that aerobic and resistance exercise training produces improvements in cardiometabolic risk factors, weight control, bone strength, and well-being ([27, 30–32](#)), and preparatory conditioning that includes strength-building exercises has been found to reduce the risk of sports-related injuries in young athletes ([33–35](#)). Thus, the benefits of exercise are

much greater than the risks (*e.g.*, overuse injuries). Recent evidence also supports the concept that PA and physical fitness are positively associated with cognition and academic achievement ([36–38](#)).

Young, healthy individuals are able to start moderate- to vigorous-intensity exercise training without medical screening, although it is reasonable to begin with less intense physical activities and progress over time. Physiologic responses to acute, graded aerobic exercise are qualitatively similar to those seen in adults. However, there are important quantitative differences, many of which are related to the effects of body mass, muscle mass, and height. In addition, it is notable that children have a lower anaerobic capacity than adults, limiting their ability to perform sustained vigorous-intensity exercise ([39](#)).

Exercise Testing

Generally, the adult guidelines for standard aerobic and resistance exercise testing apply to children and adolescents (see [Chapter 3](#)). However, physiologic responses during exercise differ from those of adults ([Table 6.1](#)) so that the following issues should be considered ([39](#), [42](#)):

TABLE 6.1 Physiologic Responses to Acute Exercise in Children Compared to Adults ([40](#), [41](#))

Variable	Response
Absolute oxygen uptake	Lower
Relative oxygen uptake	Higher
Heart rate	Higher
Cardiac output	Lower
Stroke volume	Lower
Systolic blood pressure	Lower
Diastolic blood pressure	Lower
Respiratory rate	Higher
Tidal volume	Lower
Minute ventilation	Lower
Respiratory exchange ratio	Lower

- Exercise testing for clinical purposes is generally not indicated for children or adolescents unless there is an existing health condition or concern.
- The exercise testing protocol should be based on the reason the test is being performed and the functional capability of the child or adolescent.
- Children and adolescents should be familiarized with the test protocol before testing to minimize stress and maximize the potential for a successful test.
- Treadmill, cycle, and combination (upper and lower body) ergometers should be available for cardiopulmonary testing. Treadmills tend to elicit a higher peak oxygen uptake ($\dot{V}O_{2\text{peak}}$) and maximal heart rate (HR_{max}). Cycle ergometers provide less risk for injury but need to be correctly sized for the child or adolescent. Cycle ergometers also require additional focus

and attention in order for the appropriate self-propelled cadence to be maintained, which may be difficult for some children.

- A handgrip dynamometer can provide an easy, low-skill static test of muscular strength for children and adolescents. Repetition maximum (RM) testing (*e.g.*, 1-RM or 10-RM) can be used to assess muscular strength using free weight or weight machine exercises. RM testing requires proper instruction and technique-driven progression of testing loads.
- Children and adolescents may require extra motivation and support during the test compared to adults.

In addition, health/fitness testing may be performed outside of the clinical setting. In school-based settings, the FITNESSGRAM test battery may be used to assess the components of health-related fitness ([43](#)). The components of the FITNESSGRAM test battery include body composition (*i.e.*, body mass index [BMI], skinfold measurements, or bioelectrical impedance analysis), cardiorespiratory fitness (CRF) (*i.e.*, 1-mi walk/run and progressive aerobic cardiovascular endurance run [PACER]), muscular fitness (*i.e.*, curl-up test, trunk lift test, pull-ups, and push-up test), and flexibility (*i.e.*, back-saver sit-and-reach test and shoulder stretch) ([43](#)). Age- and sex-specific, criterion-referenced standards are available, which allow results to be compared across demographic characteristics ([43](#)). Due to the strong correlation between health and fitness, tests that evaluate aerobic and muscular fitness remain important screening tools.

Exercise Prescription

Children and adolescents should be encouraged to participate in various PAs that are enjoyable and developmentally appropriate. PA in youth should include both unstructured (*e.g.*, active play or bike riding) and structured (*e.g.*, physical education and multidirectional exercise/sport) PA, which

typically consist of sporadic bursts of moderate and vigorous-intensity PA alternating with brief periods of rest. It is important to recognize that fundamental movement skill (FMS) competence can influence the ability to participate in a range of PAs early in life ([44–46](#)), underscoring the importance of exposing young children to simple games and various movement experiences during this developmental stage. FMS include locomotor (*e.g.*, climbing and jumping), object control (*e.g.*, kicking and throwing), and balance/stability (*e.g.*, bending and twisting) skills ([47](#)).

Young athletes may benefit from higher intensity training interventions as part of a long-term approach to performance enhancement and injury risk reduction ([48–51](#)). Regular participation in intentional strength and conditioning programming may be particularly beneficial for developing fitness, enhancing athleticism, and building a functional foundation in children and adolescents ([49, 52, 53](#)). The intensity and complexity of exercise movements should be influenced by the amount of time youth have consistently practiced basic skills and reinforced desired movement patterns.

Aerobic Exercise

Enjoyable and developmentally appropriate activities can take on a wide range of meanings for youth, depending on physical, social, and emotional maturation status ([39](#)). Although aerobic exercise training in adults is commonly performed as steady-state exercise, such prolonged bouts of exercise will likely be neither enjoyable nor appropriate for many children. Rather, children's natural play patterns have an intermittent quality (*e.g.*, tag games, many team sports) that can be emulated in more structured intervention efforts while also contributing to improvements in aerobic fitness. When compared with moderate-intensity continuous exercise, bouts of high-intensity interval exercise have been found to elicit greater improvements in aerobic fitness in youth ([50, 55](#)). If improvements in aerobic fitness are assessed by maximal or $\dot{V}O_{2peak}$, prepubertal children

will demonstrate a blunted response to aerobic exercise training; about 5%–10% increase for children, compared with about a 20% increase for adults ([23](#)). However, measures of aerobic performance (*e.g.*, 1-mi walk/run, PACER), rather than maximal capacity, may be more practical and meaningful in many situations.

Resistance Exercise

In addition to the FITT recommendations above, there are several additional factors that need to be incorporated into any structured youth resistance training program. Additional factors include education on proper form and technique, knowledgeable adult supervision, and appropriately determined initial resistance (weight) plus the timing and magnitude (volume) of resistance progressions ([27](#), [28](#)). Most injuries associated with resistance training in youth are due to a failure in one of these four key areas, not due to the exercise itself ([27](#), [56](#)). As with adults, working larger muscle groups and performing more complex multijoint exercises first are recommended, to avoid excess fatigue when performing the more dynamic exercises. Free weights, weight machines, body weight, medicine balls, and resistance bands have all demonstrated effectiveness in improving strength ([27](#), [39](#)). Although there is no evidence-based minimum age for participation in resistance exercise, if a child has the ability to follow instructions and is ready to begin sport activities being able to participate in beginner strength-building activities that include squatting, pushing, pulling, and rotating seems reasonable ([28](#)). As youth enhance their resistance exercise skill competence, the design of the training program can be progressed to create a more challenging and more effective training stimulus.

FITT FITT RECOMMENDATIONS FOR CHILDREN AND ADOLESCENTS (54)

	Aerobic	Resistance	Bone Strengthening
Frequency	Daily; include vigorous intensity at least $3 \text{ d} \cdot \text{wk}^{-1}$	$\geq 3 \text{ d} \cdot \text{wk}^{-1}$	$\geq 3 \text{ d} \cdot \text{wk}^{-1}$
Intensity	Moderate (noticeable increase in HR and breathing) to vigorous intensity (substantial increases in HR and breathing)	1–2 sets of 8–12 submaximal repetitions ($<60\%$ 1-RM) to lower repetitions and higher loads ($>80\%$ 1-RM) for additional sets with good mechanical form	Variable with emphasis on activities that produce moderate to high bone loading through impact or muscle force production
Time	As part of $\geq 60 \text{ min} \cdot \text{d}^{-1}$ of exercise	As part of $\geq 60 \text{ min} \cdot \text{d}^{-1}$ of exercise	As part of $\geq 60 \text{ min} \cdot \text{d}^{-1}$ of exercise
Type	Enjoyable and developmentally appropriate activities, including tag/running games, hiking/brisk walking, hopping, skipping, jumping rope, swimming, dancing, bicycling, and sports such as soccer, basketball, or tennis	Muscle strengthening physical activities can be unstructured (<i>e.g.</i> , playing on playground equipment, climbing trees, tug-of-war), or structured and appropriately supervised (<i>e.g.</i> , performing body weight exercises such as push-ups and sit-ups, lifting weights, working with medicine balls and resistance bands).	Examples of bone strengthening activities include running, jump rope, basketball, tennis, resistance training, and hopscotch.

HR, heart rate; RM, repetition maximum.

Bone Strengthening Exercise

There is strong evidence supporting the beneficial role of PA on bone health. A review of 22 intervention trials in children aged 3–18 years (57) identified an augmented annual increase in bone accrual of 0.6%–1.7% due to a PA or exercise intervention. However, the precise FITT prescription for improving bone health is not clearly defined (58). Bone responds to PA or exercise stimuli that produce a strain on the bone due to impact (*e.g.*, running, jumping, racket sports) or mechanical load (lifting) and responds optimally when these stimuli are dynamic, relatively short in duration, moderate to

high in intensity, and inconsistent in the type of stimuli or direction of the applied load ([58](#)). These important factors should be considered in youth training programs where an increase in bone health is an important outcome.

Special Considerations

- Children and adolescents may safely participate in strength training activities provided they receive proper instruction and supervision. Generally, adult guidelines for resistance training may be applied (see [Chapter 5](#)), although the additional design and supervision elements mentioned above are crucial for youth resistance training programs.
- Because of immature thermoregulatory systems, youth are at greater risk of heat-related injury. Therefore, youth should avoid sustained, heavy exercise in exceptionally hot humid environments, be properly hydrated before, during, and after activity, and appropriately modify activities. See [Chapter 7](#) and the American College of Sports Medicine (ACSM) position stand on exercising in the heat and fluid replacement for additional information ([59](#)).
- Children and adolescents who have excess weight or are physically inactive may not be able to achieve $60 \text{ min} \cdot \text{d}^{-1}$ of MVPA, initially. Instead, they should start out with moderate-intensity aerobic and resistance exercise using a relative estimate of intensity (*e.g.*, perceived exertion) and gradually increase the frequency, intensity, and duration of PA to achieve the $60\text{-min} \cdot \text{d}^{-1}$ goal.
- Children and adolescents with diseases or disabilities such as asthma (see [Chapter 8](#)), diabetes mellitus (see [Chapter 9](#)), obesity (see [Chapter 9](#)), cystic fibrosis, and cerebral palsy (see [Chapter 11](#)) should be referred to individuals with expertise in these areas.
- Efforts should be made to decrease sedentary activities (*i.e.*, television watching, Web surfing, and playing inactive video games) and increase

activities that promote lifelong activity and fitness (*i.e.*, active play, walking, running, cycling, and resistance training).

- Young athletes should perform strength and conditioning activities to enhance sports performance and reduce injury risk. Consistent participation in this type of training throughout childhood and adolescence is a part of long-term athletic development.

ONLINE RESOURCES

U.S. Department of Health and Human Services. Physical Activity Guidelines for Americans [Internet]. 2nd ed. Washington (DC): U.S. Department of Health and Human Services; 2018 [cited 2020 Mar 30]. 118 p. Available from: <https://health.gov/our-work/physical-activity/current-guidelines>

World Health Organization. WHO guidelines on physical activity and sedentary behaviour. Geneva: World Health organization; 2020. Available from: <https://www.who.int/publications/i/item/9789240015128>

PREGNANCY

The American Congress of Obstetricians and Gynecologists (ACOG) recommends that most pregnant females can exercise ([60–63](#)), and there are few maternal medical conditions in which exercise is absolutely contraindicated. The health benefits of PA are well recognized and include prevention of excessive gestational weight gain and gestational diabetes mellitus and hypertension, decreased risk of preeclampsia and urinary incontinence, improved mood, reduced incidence of low back pain and caesarean section, reduced length of labor, and maintenance or improvement of CRF ([60](#), [64–68](#)). If there is a gestational condition, then a female should consult with relevant specialists and subspecialists (*e.g.*, obstetrician, maternal-fetal medicine, cardiology, pulmonology). In females who have obstetric or medical comorbidities, exercise regimens should be individualized. Due to the plethora of research supporting the benefits of exercise during pregnancy on maternal, fetal, placental, and infant outcomes, ACOG also recommends that bedrest is rarely indicated during pregnancy; typically, some level of PA is warranted throughout pregnancy and to improve outcomes ([60](#)). In contrast, short- and long-term risks associated with sedentary behavior are of concern ([69](#)). It is reasonable to consider that the physical and psychological benefits of PA and negative consequences of sedentary behavior in nonpregnant females generally apply to pregnant females.

In their respective guidelines, the ACOG, the 2018 U.S. Department of Health and Human Services ([54](#)), the 2019 Canadian guideline for PA throughout pregnancy ([62](#)), and the International Olympic Committee (IOC) ([63](#), [67](#), [68](#)) outline the importance of exercise during pregnancy and postpartum and provide evidence-based guidance on exercise prescription (Ex R_x) for the minimization of risk and promotion of health benefits. With

appropriate modifications and progression, pregnancy is an opportunity for sedentary females to adopt PA behavior to improve their health, their pregnancy outcomes, and the health of their unborn baby ([62](#)).

Health Screening

The Canadian Society for Exercise Physiologists prescreening for PA during pregnancy, Get Active Questionnaire for Pregnancy (*GAQ-Pregnancy*) can be used for the health screening of pregnant females before their participation in an exercise program ([Figure 6.1](#)). If a pregnant female responds yes to any of the questions, then she should have the health care provider consultation form for prenatal PA ([70](#)) completed by an obstetric provider prior to continuing or initiating PA.

GET ACTIVE QUESTIONNAIRE FOR PREGNANCY



NAME (+ NAME OF PARENT/GUARDIAN IF APPLICABLE) [PLEASE PRINT]:			
TODAY'S DATE (DD/MM/YYYY):	YOUR DUE DATE (DD/MM/YYYY):	NO. OF WEEKS PREGNANT:	AGE:

Physical activity during pregnancy has many health benefits and is generally not risky for you and your baby. But for some conditions, physical activity is not recommended. This questionnaire is to help decide whether you should speak to your Obstetric Health Care Provider (e.g., your physician or midwife) before you begin or continue to be physically active.

Please answer YES or NO to each question to the best of your ability. **If your health changes as your pregnancy progresses you should fill in this questionnaire again.**

1. In this pregnancy, do you have:		
a. Mild, moderate or severe respiratory or cardiovascular diseases (e.g., chronic bronchitis)?	Y	N
b. Epilepsy that is not stable?	Y	N
c. Type 1 diabetes that is not stable or your blood sugar is outside of target ranges?	Y	N
d. Thyroid disease that is not stable or your thyroid function is outside of target ranges?	Y	N
e. An eating disorder(s) or malnutrition?	Y	N
f. Twins (28 weeks pregnant or later)? Or are you expecting triplets or higher multiple births?	Y	N
g. Low red blood cell number (anemia) with high levels of fatigue and/or light-headedness?	Y	N
h. High blood pressure (preeclampsia, gestational hypertension, or chronic hypertension that is not stable)?	Y	N
i. A baby that is growing slowly (intrauterine growth restriction)?	Y	N
j. Unexplained bleeding, ruptured membranes or labour before 37 weeks?	Y	N
k. A placenta that is partially or completely covering the cervix (placenta previa)?	Y	N
l. Weak cervical tissue (incompetent cervix)?	Y	N
m. A stitch or tape to reinforce your cervix (cerclage)?	Y	N
2. In previous pregnancies, have you had:		
a. Recurrent miscarriages (loss of your baby before 20 weeks gestation two or more times)?	Y	N
b. Early delivery (before 37 weeks gestation)?	Y	N
3. Do you have any other medical condition that may affect your ability to be physically active during pregnancy? What is the condition? Specify:	Y	N
4. Is there any other reason you are concerned about physical activity during pregnancy?		

Go to Page 2 Describe Your Physical Activity Level

Describe Your Physical Activity Level



During a typical week, what types of physical activities do you take part in (e.g., swimming, walking, resistance training, yoga)?

During the same week, please describe ON AVERAGE how often and for how long you engage in physical activity of a light, moderate or vigorous intensity. See definitions for intensity below the box.

ON AVERAGE	FREQUENCY (times per week)	INTENSITY (see below for definitions)	DURATION (minutes per session)
How physically active were you in the six months before pregnancy?	☐ 0 ☐ 3–4 ☐ 1–2 ☐ 5–7	☐ light ☐ moderate ☐ vigorous	☐ <20 ☐ 31–60 ☐ 20–30 ☐ >60
How physically active have you been during this pregnancy?	☐ 0 ☐ 3–4 ☐ 1–2 ☐ 5–7	☐ light ☐ moderate ☐ vigorous	☐ <20 ☐ 31–60 ☐ 20–30 ☐ >60
What are your physical activity goals for the rest of your pregnancy?	☐ 0 ☐ 3–4 ☐ 1–2 ☐ 5–7	☐ light ☐ moderate ☐ vigorous	☐ <20 ☐ 31–60 ☐ 20–30 ☐ >60

Light intensity physical activity: You are moving, but you do not sweat or breathe hard, such as walking to get the mail or light gardening.

Moderate intensity physical activity: Your heart rate goes up and you may sweat or breathe hard. You can talk, but could not sing. Examples include brisk walking.

Vigorous intensity physical activity: Your heart rate goes up substantially, you feel hot and sweaty, and you cannot say more than a few words without pausing to breathe. Examples include fast stationary cycling and running.

General Advice for Being Physically Active During Pregnancy

Follow the advice in the 2019 Canadian Guidelines for Physical Activity throughout Pregnancy: csepguidelines.ca/pregnancy

It recommends that pregnant women get at least 150 minutes of moderate-intensity physical activity (resistance training, brisk walking, swimming, gardening), spread over three or more days of the week. **If you are planning to take part in vigorous-intensity physical activity, or be physically active at elevations above 2,500 m (8,200 ft), then consult with your health care provider.** If you have any questions about physical activity during pregnancy, consult a Qualified Exercise Professional or your health care provider beforehand. This can help ensure that your physical activity is safe and suitable for you.

Declaration

To the best of my knowledge, all of the information I have supplied on this questionnaire is correct. **If my health changes, I will complete this questionnaire again.**

☐ I answered NO to all questions on Page 1.

Sign and date the declaration below.
Physical activity is recommended.

I answered YES to one or more questions on Page 1 and I will speak with my health care provider before beginning or continuing physical activity.
The Health Care Provider Consultation Form for Prenatal Physical Activity can be used to start the conversation (www.csep.ca/getactivequestionnaire-pregnancy).

☐ I have spoken with my health care provider who has recommended that I take part in physical activity during my pregnancy.

Sign and date the declaration below.

NAME (+ NAME OF PARENT/GUARDIAN IF APPLICABLE) [PLEASE PRINT]:		SIGNATURE (OR SIGNATURE OF PARENT/GUARDIAN IF APPLICABLE):
TODAY'S DATE (DD/MM/YYYY):	TELEPHONE (OPTIONAL):	EMAIL (OPTIONAL):

Figure 6.1 The Canadian Society for Exercise Physiologists Pre-Screen for Physical Activity in Pregnancy: Get Active Questionnaire

for Pregnancy. If a pregnant female answers “Yes” to a question, they are encouraged to have their obstetric health care provider complete the *Health Care Provider Consultation Form for Prenatal Physical Activity*. Reprinted with permission from <http://www.csep.ca/cmfiles/publications/parq/parmed-xpreg.pdf>.

Exercise Testing

Although maximal exercise testing should not be performed on females during pregnancy, there might be instances in which this is medically required (71, 72). Therefore, the exercise professional should refer to these articles that performed maximal tests with a cycle ergometer during various time points in pregnancy (73–83). Aside from medical necessity, submaximal testing is the preferred method of exercise testing for pregnant females. The acute physiologic responses observed with exercise in nongravid adults are modified during pregnancy (Table 6.2) (85). Because of the physiologic changes that accompany pregnancy, assumptions of submaximal protocols may be compromised; therefore, it is best to utilize tests that are found valid and reliable in pregnancy in order to personalize training rather than accurately estimating maximal aerobic power (68, 71, 86–89). For physical fitness testing, there are a variety of instruments that have been utilized during pregnancy depending on equipment available, environment of testing, as well as purpose (*e.g.*, cardiorespiratory, muscular endurance, muscular strength, flexibility, balance). For CRF, the most common submaximal protocols use a treadmill or cycle ergometer. There have been 56 different protocols using cycle ergometers and 14 different treadmill protocols during pregnancy (88). The most commonly used test of CRF during pregnancy is the Modified Balke protocol, which is validated for pregnant females (88). If a treadmill or cycle ergometer are not available, then other protocols have been used such as the 6-minute walk test or three different step tests (88); however, these have not been validated for pregnant females (87–89). For muscular strength, the primary test used during pregnancy is the maximal

hand-grip strength test ([87–89](#)); although this test is not validated in pregnant females, it has been associated with infant outcomes. Muscular endurance tests used during pregnancy are the Sit-to-Stand Test and Trendelenburg test; the Sit-to-Stand test is reliable for pregnancy and easy to perform needing only a chair and timer but has not been validated ([87–89](#)). Various tests have been used to assess flexibility, balance, and speed in pregnancy ([87–89](#)). For flexibility, although no tests have been validated for pregnancy, there have been 13 different protocols to assess flexibility during pregnancy, including sidebending, sit-and-reach, back-scratch, motion analysis, and goniometry; the most common tests used are the passive abduction of the left fourth finger, which is reliable, as well as the back-scratch test. Both of the tests are related to delivery or infant outcomes ([87–89](#)). Balance testing for pregnant females has been completed with static and dynamic balance tests and are associated with maternal outcomes ([87–89](#)). Since platforms are not available in all settings, the one-legged standing protocol can be used to assess static balance during pregnancy ([87–90](#)). Lastly, speed has been tested for pregnant females with the 10-m timed walk test, which is reliable and valid in pregnancy ([87–89](#)). If a quick assessment needs to be done with no equipment available, then the International Fitness Scale can be used ([89, 91](#)). All exercise professionals, clinicians, and pregnant females should be educated on the warning signs for when to stop exercise ([Box 6.1](#)).

TABLE 6.2 Physiologic Responses to Acute Exercise During Pregnancy Compared to Nonpregnancy (84)

Oxygen uptake (during weight-dependent exercise)	Increase
Heart rate	Increase
Stroke volume	Increase
Cardiac output	Increase
Tidal volume	Increase
Minute ventilation	Increase
Ventilatory equivalent for oxygen ($\dot{V}E/\dot{V}O_2$)	Increase
Ventilatory equivalent for carbon dioxide ($\dot{V}E/\dot{V}CO_2$)	Increase
Systolic blood pressure	No change/decrease
Diastolic blood pressure	No change/decrease

Box 6.1 Warning Signs to Stop Exercise During Pregnancy ([63](#), [92–94](#))

- Vaginal bleeding
- Abdominal pain
- Regular, painful contractions
- Amniotic fluid leakage
- Dyspnea before exertion
- Dizziness
- Headache
- Chest pain
- Muscle weakness affecting balance
- Calf pain or swelling

Exercise Prescription

The ACOG recommends $20\text{--}30 \text{ min} \cdot \text{d}^{-1}$ of moderate-intensity exercise on most or all days of the week throughout pregnancy ([Table 6.3](#)) ([60](#)). Exercise recommendations for pregnant females should be modified according to the female's prior exercise history as well as symptoms, discomforts, and abilities across the pregnancy time course. In general, international guidelines for exercise during pregnancy agree that $150 \text{ min} \cdot \text{wk}^{-1}$ of moderate-intensity exercise is recommended for improved maternal, placental, birth, and infant outcomes. There may be periods when following exercise recommendations is not possible; females should do what they can and return to following the recommendations when they are able under the health care providers care ([62](#)). Importantly, growing evidence supports that there is a dose-response effect of benefits from maternal exercise ([95](#)). Further guidance specifically for athletes can be found in the evidence summary from the IOC ([63](#)).

TABLE 6.3 Physical Activity Recommendations During Pregnancy From Three Guideline Documents

	Canada 2019 (62)	ACOG 2020 (60)	United States 2018 (54)
Duration	$\geq 150 \text{ min} \cdot \text{wk}^{-1}$	$\geq 20\text{--}30 \text{ min} \cdot \text{d}^{-1}$	$\geq 150 \text{ min} \cdot \text{wk}^{-1}$
Frequency	Minimum of 3 d $\cdot \text{wk}^{-1}$; being active every day is encouraged	Most or all days of the week	Spread throughout the week
Intensity	Moderate intensity defined as physical activity intense enough to noticeably increase heart rate; a person can talk but not sing during activities of this intensity Target heart rate zones for pregnant females based on age, “talk test”	Moderate intensity, RPE 13–14 on a scale of 6–20, “talk test” — can talk while exercising	Light to moderate intensity, RPE 5–6 on a scale of 0–10, “talk test” — can talk while exercising; additionally, females who engage in vigorous-intensity aerobic activity can continue these activities if they remain healthy and discuss with their health care provider.
Type	Aerobic and resistance training activities including brisk walking, stationary cycling (moderate effort), swimming or aquafit, carrying moderate loads, household chores	Aerobic and strength-conditioning exercises including walking, swimming, stationary cycling, low-impact aerobics, modified yoga or Pilates, running, racquet sports	Aerobic and muscle strengthening

	Canada 2019 (62)	ACOG 2020 (60)	United States 2018 (54)
	(<i>e.g.</i> , gardening, washing windows)		

ACOG, American College of Obstetricians and Gynecologists; RPE, rating of perceived exertion.

Exercise Frequency, Intensity, Time, and Type

These points assume the obstetric or medical clearance for exercise during pregnancy.

- Similar to all populations, a warm-up and cool-down is suggested before and after exercise, respectively, in order to lower the chance for injury ([62](#), [96](#)).
- Exercise frequency should be regular, occurring throughout the week, and adjusted based on total exercise volume (*i.e.*, number of days may vary based on intensity and duration of exercise). The typical frequency of exercise recommendation is 3–4 times per week ([95](#)).
- Exercise intensity should be moderate to vigorous intensity, which has been associated with decreased pregnancy complications and excessive gestational weight gain, as well as improved infant outcomes ([95](#)). A talk test can be used to monitor exercise intensity. For the talk test, a female should be able to maintain a conversation during exercise and reduce the intensity if she cannot talk without being out of breath ([62](#)). Higher intensity or prolonged exercise may require caloric intake beforehand ([60](#)).
- For previously inactive females, a progression, from lower intensity and/or duration to at least the recommended level, is recommended rather than reduced or irregular frequency ([62](#)). Exercise goals and progression may vary at different time points during pregnancy, and exercise routines should remain flexible. Importantly, maternal and pregnancy benefits of exercise are dose related ([95](#), [97–99](#)), thus, it is important to aim for at

least the minimum of exercise recommendation and maintain through the duration of pregnancy.

- Exercise time, or duration, for each session should be spread throughout the week, such as in 20–30-minute sessions, although bouts of any duration are beneficial ([54](#), [60](#), [95](#)). In general, pregnant females should aim for at least 150 minutes of moderate intensity, aerobic and/or strength, exercise every week throughout the duration of pregnancy.
- Importantly, the earlier females start exercise during pregnancy, the more beneficial it is for the pregnancy and fetal development.

Exercise Types to Consider

- Examples of safe aerobic exercises include walking, swimming, stationary cycling, cardio machines (*i.e.*, elliptical, treadmill), low- or high-impact aerobics, jogging, running, and aquatic aerobics ([60](#), [100](#)).
- Research has demonstrated the safety of resistance training during pregnancy ([101](#)–[106](#)). Resistance exercise can be done alone or in combination with aerobic exercises. The total goal of exercise is at least 150 minutes of moderate intensity every week. Therefore, depending on whether resistance exercise is done alone or in combination will depend on the length of time for each session. In general, resistance exercise can also be done at light to moderate intensity, 30- to 50-minute sessions, for 1–3 times each week. This mode of exercise is beneficial for maternal and infant health ([97](#), [104](#), [107](#)–[109](#)). Of note, the Valsalva maneuver and excessive pressure toward the pelvic floor should be avoided. The pelvic floor muscles should be contracted before and during heavy lifting to counteract the impact of increased intraabdominal pressure on the pelvic floor from weightlifting ([63](#)).
- Pelvic floor muscle training, such as Kegel exercises, may be performed on a daily basis to decrease the risk of urinary incontinence during and after pregnancy ([62](#), [63](#), [110](#), [111](#)). Pelvic floor contractions should be done

with near-maximal contractions in sets of 8–12, at least 3 times each day on most days ([63](#)); typically these techniques are more effective individually and not in a group setting ([112](#)). Pelvic floor muscle training during pregnancy reduces the length of both the first and second stage of labor without negative impacts on labor and birth outcomes ([113–115](#)). It is important to note that females with diastasis recti abdominis (DRA, discussed later), pelvic floor muscle exercises should be avoided as they can worsen the condition ([116](#)).

- Substitution of certain activities may be necessary given the physiologic changes the female experiences over the time course of pregnancy (see [Table 6.3](#)) ([68](#)).

Exercise Types to Avoid

- Females should avoid sports activities with high risks of falling or loss of balance or with high risk of abdominal trauma. However, experts working under affiliation of the IOC ([63](#)) published a long list of sports to be avoided, at the same time they mentioned that *noncontact training* may be continued in such sports as artistic gymnastics, equestrian, field hockey, freestyle skiing, ski jumping, figure skating, rugby, short track speed skating, snowboard, some track and field events, trampoline, and volleyball. Importantly, medical experts involved in the National Forum on Pregnancy and Sport in 2001 in Sydney advised that “damage to the womb of the kind that could injure an unborn child is usually associated with forces equivalent to those occurring in a car accident. All medical experts agreed that falls and direct contact of the kinds that occur during contact sports were unlikely to cause damage to the womb or the unborn child” ([117](#), p. [6](#)). It is important to note that the risk of falling is higher in females who do not exercise during pregnancy than in those who are active ([118](#)). Females should avoid activities that present an increased risk of falling ([60](#), [119](#)).

- There is insufficient data related to scuba diving during pregnancy. Therefore, scuba diving during pregnancy should be avoided until further research is completed ([60](#), [62](#), [120](#)).
- Hot Pilates and hot yoga should be avoided due to the risk of rising core temperatures ([60](#)).
- Although exertion or prolonged periods in the supine position might reduce venous return and subsequent cardiac output, restricting blood flow to the uterus and fetus, the available research suggests that exercise in the supine position is not associated with negative consequences for the course of pregnancy or the development of the fetus ([121](#)). Caution is still advised if there is light-headedness, most likely due to hypotension (and potential reductions in blood flow to the baby), to modify position. Thus, for females who are uncomfortable in this position (either in physical or mental terms) or experience hypotension, they should modify the exercise, for example, using the semirecumbent or sitting position, or a pillow below the right hip.
- In any PA, avoid using the Valsalva maneuver, prolonged isometric contraction, and motionless standing, which may result in decreased venous return and hypotension ([60](#), [62](#)).
- During pregnancy, some females experience diastasis recti, indicated by a visible separation of their abdominal muscles termed DRA ([62](#)). Pelvic floor and drawing-in exercises increase the interrectus distance (IRD), while head lift, curl-up, and diagonal curl-up decreased the IRD in pregnant females with DRA ([116](#), [122–124](#)). Females with DRA should seek physical therapy advice and focus on curl-up and head lift abdominal exercises only to strengthen the abdominal core and avoid worsening the condition ([116](#), [122–124](#)). Exercise professionals should work collaboratively with a physical therapist to help manage this condition on an individual basis for each client.

Special Considerations

- *Heat* — Females who are pregnant should avoid exercising in a hot humid environment, be well hydrated, and dress appropriately to avoid heat stress ([60](#), [63](#)). On hot humid days, it may be best to exercise indoors.
- *Hydration* — Drink water before, during, and after PA ([62](#)). Further information on fluid replacement is detailed in the *ACSM Exercise and Fluid Replacement* position stand ([59](#)).
- *Low back pain (LBP)* — For females experiencing LBP during pregnancy, an excellent alternative to land-based exercise may be water exercise ([60](#)).
- *Metabolic demand and weight* — Although the metabolic demand increases during pregnancy, some females may not need to increase caloric intake to meet the additional caloric costs of pregnancy and exercise. Appropriate weight gain during pregnancy is based on prepregnancy BMI, with higher BMI needing less weight gain, if any ([125](#), [126](#)). PA may help regulate weight gain during pregnancy ([54](#)). In order to avoid excessive weight gain during pregnancy, females should consult with a qualified nutritionist and their health care provider regarding appropriate weight gain and nutritional intake ([119](#), [125](#)).
- *High altitude* — For lowlanders, PA up to an altitude of 6,000 ft (~1,829 m) is safe ([60](#)). PA at higher elevations should be discussed with an obstetrician with knowledge of the impact of high altitude on maternal and fetal outcomes ([62](#)). Others recommend refraining from high-intensity training at altitudes higher than approximately 5,000 ft (>1,500–2,000 m) during pregnancy for those not acclimated ([63](#)).

Exercise During Postpartum

Females should work with their health care provider to determine when PA can be resumed following delivery; healing times will differ depending on vaginal or cesarean delivery. Resumption may differ based on the type and

intensity of the PA during pregnancy. When returning to activity, PA should be resumed gradually, as soon as medically safe, because of normal deconditioning in the initial postpartum period ([60](#)). Females with higher CRF levels and more rigorous exercise routines prior to and during pregnancy may be able to resume exercise sooner. The health benefits of exercise during postpartum can include improved postpartum recovery, prevention of postpartum weight retention, improvement of mood, and decreased risk of postpartum depressive symptoms ([60](#), [67](#), [68](#)). If females had DRA during pregnancy, then the same Ex R_x applies in the postpartum period. Pelvic floor and drawing-in exercises increase the IRD, while abdominal crunch exercises decreased the IRD in postpartum females with DRA ([122](#), [124](#), [127](#)).

Exercise in the postpartum period is important for return to prepregnancy BMI and does not interfere with breastfeeding, as long as the female takes in appropriate food and fluid ([128](#), [129](#)). For babies that do not breastfeed well immediately after maternal exercise, mothers could feed them or express milk immediately before exercise, which may also make exercise more comfortable ([60](#), [128](#)).

ONLINE RESOURCES

American College of Obstetricians and Gynecologists: guideline (60) and resources: <http://www.acog.org>

Canadian Society of Exercise Physiology: Get Active Questionnaire for Pregnancy: <https://csep.ca/2021/05/27/get-active-questionnaire-for-pregnancy/>

Exercise During Pregnancy and the Postpartum Period: regularly updated literature review (104): <https://www.uptodate.com/contents/exercise-during-pregnancy-and-the-postpartum-period>

2018 Physical Activity Guidelines for Americans, 2nd ed. (54): <https://health.gov/paguidelines/second-edition/>

2018 Scientific Report supporting the Physical Activity Guidelines for Americans: <https://health.gov/paguidelines/second-edition/report/>

2019 Canadian Guideline for Physical Activity Throughout Pregnancy: guideline (62) and resources: <https://csepguidelines.ca/guidelines/pregnancy/>

LOW BACK PAIN

LBP is defined as pain, muscle tension, or stiffness localized below the rib margin and above the inferior gluteal folds, with or without referred or radiating leg pain ([130](#), [131](#)). LBP is considered a major public health problem, with the lifetime prevalence reported as high as 84% ([132](#)). However, the presence of LBP disability is not consistent across cultures ([133](#)). In western countries, between 4% and 33% of the adult population experience LBP at any given point in time ([134](#)), and recurrent episodes of LBP can occur in over 70% of cases ([135](#)). Approximately 20% of LBP cases become chronic, and about 10% of the cases progress to a disability ([132](#)).

Individuals with LBP can be classified into one of three broad categories: (a) LBP potentially associated with another specific spinal cause (*e.g.*, cancer, fracture, infection, ankylosing spondylitis, or cauda equina syndrome), (b) LBP potentially associated with radiculopathy or spinal stenosis, and (c) and nonspecific LBP (LBP with no known pathoanatomical cause), which encompass over 85% of all cases ([136](#)). For prognosis and outcome purposes, LBP can be described as acute (<6 weeks), subacute (6–12 weeks), and chronic (>12 weeks) ([132](#), [137](#)). The best available evidence supports a classification approach that deemphasizes the importance of identifying specific anatomical lesions after red flag screening is completed ([137](#)). The American Physical Therapy Association suggests five classifications of LBP:

- LBP with mobility deficits
- LBP with radiating pain
- LBP with referred lower extremity pain
- LBP with movement coordination impairments
- LBP with related generalized pain

It is frequently stated that approximately 90% of acute LBP episodes resolve within 6 weeks regardless of treatment ([138](#)). However, it is more accurate to state that 90% of individuals with LBP who receive primary care will have stopped consulting with symptoms within 3 months, yet a majority of individuals will experience recurrence of their LBP within 1 year after consultation ([139](#)).

To reduce the probability of disability, individuals with LBP should stay active by continuing ordinary activity within pain limits, avoid bed rest, and return to work as soon as possible ([140](#)). Current research shows that following an acute bout of LBP, early access (within 3 weeks of acute onset) to physical therapy results in dramatic reductions in the need for advanced imaging, administration of opioids, injections, and surgery, as well as decreased disability ([141–143](#)). If disabling pain continues beyond 6 weeks, a multidisciplinary approach that includes addressing psychosocial factors is recommended ([137](#)). Many individuals with LBP experience fear, anxiety, or misinformation regarding their LBP, exacerbating a persistent pain state ([144](#)). A combination of therapeutic and aerobic exercise, in conjunction with pain education, can improve individual attitudes, outcomes, perceptions, and pain thresholds ([145, 146](#)). Psychosocial factors that increase the risk of developing or perpetuating long-term disability and work loss associated with LBP can be found in [Box 6.2](#).

Box 6.2 Psychosocial Factors for Long-Term Disability and Work Loss Associated With Low Back Pain ([147](#))

- A negative attitude that back pain is harmful or potentially severely disabling
- Fear avoidance behavior and reduced activity levels
- An expectation that passive, rather than active, treatment will be beneficial
- A tendency to depression, low morale, and social withdrawal
- Social or financial problems

Current literature does not support a definitive cause for initial bouts of LBP ([137](#)). However, previous LBP is one of the strongest predictors for future back pain episodes ([130](#)); therefore, once an individual suffers an initial episode of LBP, they are in fact more susceptible to future episodes of LBP. Recurrent episodes of LBP tend toward increased severity and duration, higher levels of disability, including work disability, and higher medical and indemnity costs ([84](#), [148](#)). Current guidelines place a heavy emphasis on preventive measures and early interventions to minimize the risk of an acute LBP episode from becoming chronic and/or disabling ([149](#)). Best evidence for treating LBP indicates PA as a key component in managing the condition ([150–153](#)). The biopsychosocial model is the prevailing framework used for understanding, managing, and treating back pain. This approach suggests that in addition to biology, psychological, socioeconomic, environmental, and cultural factors, all contribute to the incidence and persistence of back pain symptoms ([154](#)).

Within this framework, specific considerations must be given to individuals with LBP who are fearful of pain or reinjury and thus avoid PA and equally to those individuals who persist in PA despite worsening symptoms ([155](#), [156](#)). Individuals with LBP who are fearful of pain or

reinjury often misinterpret any aggravation of symptoms as a worsening of their spinal condition and hold the mistaken belief that pain necessarily implies tissue damage ([157](#)). In contrast, those with LBP who persist in PA may not allow injured tissues the time that is needed to heal. Both behaviors, persistent PA during pain and being fearful of pain, are associated with chronic pain ([155](#)). Therefore, when designing and implementing exercise programs, it is imperative that a thorough understanding of the individual's beliefs have been considered. [Box 6.3](#) further highlights the LBP clinical practice guidelines from the Orthopedic Section of the American Physical Therapy Association ([137](#)).

Box 6.3 Low Back Pain: Clinical Practice Guidelines ([137](#))

Clinicians should not utilize patient education and counseling strategies that either directly or indirectly increase the perceived threat or fear associated with low back pain, such as education and counseling strategies that:

- Promote extended bed rest.
- Provide in-depth, pathoanatomical explanations for the specific cause of the patient's low back pain.

Patient education and counseling strategies for patients with low back pain should emphasize the following:

- The promotion of the understanding of the anatomical/structural strength inherent in the human spine
- The neuroscience that explains pain perception
- The overall favorable prognosis of low back pain
- The use of active pain coping strategies that decrease fear and catastrophizing
- The early resumption of normal or vocational activities, even when still experiencing pain
- The importance of improvement in activity levels, not just pain relief

When LBP is a symptom of another serious pathology (*e.g.*, cancer), exercise testing and prescription should be performed in consultation with the health care team managing the serious pathology. For all other causes, and in the absence of a comorbid condition (*e.g.*, cardiovascular disease [CVD] with its associated risk factors), recommendations for exercise testing and prescription are similar as for healthy individuals (see [Chapter 5](#)). Given that the vast majority of LBP cases are nonspecific, the focus of the Ex R_x recommendations presented here will address individuals with LBP that are not associated with trauma or any specific underlying conditions (*e.g.*, cancer or infection).

Exercise Testing

Individuals with acute or subacute LBP appear to vary in their individual levels of PA independent of their pain-related disability. However, chronic LBP with high levels of disability may lead to low levels of PA ([158](#)).

Individual beliefs about back pain will often influence a person's willingness to exercise ([159](#)). As such, exercise testing and subsequent activities may be symptom limited in the first weeks following symptom onset ([156](#), [160](#)).

Cardiorespiratory Fitness

Current evidence has failed to find a clear relationship between CRF and pain ([161](#), [162](#)). Few studies have subjected individuals with LBP to exercise tests to exhaustion ([163](#)). Submaximal exercise tests are considered reliable and valid for individuals with LBP ([156](#)); however, actual or anticipated pain may limit submaximal testing as often as maximal testing ([156](#), [163–166](#)).

Therefore, the choice of maximal versus submaximal testing in individuals with LBP should be guided by the same considerations as for the general population (see [Chapter 3](#)).

Muscular Strength and Endurance

Individuals with LBP frequently have deficits in trunk muscle strength and endurance ([167–169](#)) and neuromuscular imbalance ([170](#), [171](#)); however, the role these play in the development and progression of LBP remains unclear ([168](#), [172](#)). Decreases in muscular strength and endurance may be independent of the period and intensity of LBP ([173](#), [174](#)), but improvements in trunk and core muscle strength and endurance have been linked to improved pain and disability ([175](#)) so testing may be warranted when appropriate. General testing of muscular strength and endurance in individuals with LBP should be guided by the same considerations as for the general population (see [Chapter 3](#)). In addition, tests of the strength and

endurance of the trunk musculature (*e.g.*, isokinetic dynamometers with back attachments, resistance machines with weight stack, and back hyperextension benches) are commonly assessed in individuals with LBP ([149](#)). However, the reliability of these tests is questionable because of the considerable learning effect in particular between the first and second sessions ([176](#), [177](#)). Performance of muscular strength and endurance assessments are often limited by actual or anticipated fear of reinjury in individuals with LBP ([178](#)).

Flexibility

There is no clear relationship between gross spinal flexibility and LBP or associated disability ([150](#)). A range of studies have shown associations between measures of spine flexibility, hip flexibility, and LBP ([179](#)), yet the nature of these associations is likely complex and requires further study. There appears to be some justification, although based on relatively weak evidence, for flexibility testing in the lower limbs, and in particular the hips of individuals with LBP ([137](#), [180](#)). In general, flexibility testing in individuals with LBP should be guided by the same considerations as for the general population (see [Chapter 5](#)). It is essential, however, to identify whether the assessment is limited by stretch tolerance of the target structures or exacerbation of LBP symptoms.

Exercise Prescription

Current guidelines for the management of LBP consistently recommend staying physically active and avoiding bed rest ([132](#), [137](#), [149](#), [153](#), [181](#)). Although it may be best to avoid exercise in the first few days immediately following an acute and severe episode of LBP so as not to exacerbate symptoms ([151](#), [153](#)), individuals with subacute and chronic LBP, as well as recurrent LBP, are encouraged to be physically active within pain limitations ([151](#)). Within 2 weeks of an acute LBP episode, activities can be carefully

introduced. Regular walking is a good way to encourage individuals with LBP to participate in activity that does not exacerbate symptoms ([149](#)). Both progressive aerobic training and progressive resistance training have been shown to be equally effective at decreasing pain intensity in individuals with chronic LBP ([182](#)).

When recommendations are provided, they should follow very closely with the recommendations for the general population, combining resistance, aerobic, and flexibility exercise (see [Chapter 5](#)). In chronic LBP, exercise programs that incorporate individual tailoring, supervision, stretching, mind-body (*e.g.*, Tai Chi), and strengthening, coupled with client preference and practitioner expertise, are associated with the best outcomes ([151](#), [153](#), [183](#)). Utilizing graded exercise exposure focusing on incremental increases in difficulty, intensity, volume, or other standard variables based on the individual's response has been shown to effectively reduce pain and disability ([184](#)). It can be helpful to utilize feedback loops such as the pain monitoring model ([185](#)) to help guide rate of progression. Encouraging activity in a safe zone of symptoms in the 3–5/10 (numeric pain rating scale) range or 9–14 on the Borg RPE Scale when paired with positive feedback and patient input can help yield positive outcomes. Furthermore, the evidence supporting the multidimensional nature of nonspecific chronic LBP shows most favorable outcomes with an individualized approach that addresses psychological distress, fear avoidance beliefs, self-efficacy in controlling pain, and coping strategies ([172](#)). Whereas Ex Rx can play an integral role in helping a client manage LBP, the diagnosis and treatment of LBP falls outside the scope of practice of most exercise professionals and needs to be referred to a licensed health care provider ([186](#)).

Special Considerations

- Exercises that address coordination, endurance, and strengthening of the trunk can be used to reduce LBP and disability in individuals with

subacute and chronic LBP with movement coordination impairments ([187](#)). However, there is insufficient evidence for any benefit of emphasizing single-dimension therapies such as abdominal strengthening ([172](#), [180](#), [188](#)).

- Individual response to back pain symptoms can be improved by providing assurance, encouraging activity, and providing early referral to physical therapy ([141](#), [143](#), [180](#)).
- There is a lack of agreement on the definition, components, and assessment techniques related to core stability. Furthermore, the majority of tests used to assess core stability have not demonstrated validity ([189](#), [190](#)).
- Certain exercises or positions may aggravate symptoms of LBP. Walking, especially downhill, may aggravate symptoms in older adults with LBP ([191](#)). However, walking on an inclined treadmill or cycling with the lumbar spine flexed may be helpful for individuals that find more upright positions bothersome.
- Certain individuals with LBP may experience a “peripheralization” of symptoms, that is, a spread of pain into the lower limbs with certain sustained or repeated movements of the lumbar spine ([192](#)). Limits should be placed on any activity or exercise that causes spread of symptoms ([152](#)).
- Repeated movements and exercises such as prone press-ups or knees to chest in a supine position that promote centralization (*i.e.*, a reduction of pain in the lower limb from distal to proximal) are encouraged to reduce symptoms in patients with acute LBP with related lower extremity pain ([137](#)).
- Flexibility exercises are generally encouraged as part of an overall exercise program. Hip and lower limb flexibility should be promoted ([150](#), [179](#)). However, in individuals with LBP and movement coordination

impairments, strengthening and/or motor control exercises should be emphasized, not flexibility ([137](#)).

- Consider progressive, low-intensity aerobic exercise for individuals with chronic LBP with generalized pain (pain in more than one body area) and moderate- to high-intensity aerobic exercise for individuals with chronic LBP without generalized pain ([137](#)).
- Exercises such as yoga, Pilates, aquatic therapy, and strength training for the posterior chain (*e.g.*, deadlifts) have shown to be effective interventions for LBP; however, the research is not clear on whether any single intervention is superior to another; therefore, the choice of exercise should fundamentally be driven by client preference and practitioner expertise ([193–199](#)).
- Traditional strength training movements such as the deadlift have been shown to improve pain and function in individuals with chronic LBP. Utilization may be most beneficial in those with lower pain scores and higher baseline paraspinal strength ([199](#)). Utilization of this exercise intervention should strongly be considered when it matches an individual's goals or expectations.
- General consensus continues to highlight the superiority of these active interventions in the management of chronic LBP when compared to passive treatments or no treatment, but more evidence is required to determine any specific intervention better than others ([200](#)).

ONLINE RESOURCES

American Physical Therapy Association. *Low Back Pain* [Internet]. Alexandria (VA): American Physical Therapy Association; 2011 [cited 2019 Apr 22].

Available from:

<https://www.choosept.com/symptomsconditionsdetail/physical-therapy-guide-to-low-back-pain>

Go4Life Resources [Internet]. Washington (DC): National Institute on Aging.

Available from: <https://www.nia.nih.gov/health/exercise-physical-activity>

National Council on Aging. *Senior Fitness & Exercise Programs* [Internet].

Arlington (VA): National Council on Aging; 2015 [cited 2019 Apr 22].

Available from: <https://www.ncoa.org/center-for-healthy-aging/basics-of-evidence-based-programs/physical-activity-programs-for-older-adults/>

OLDER ADULTS

The term *older adult* represents a diverse spectrum of ages and physiologic capabilities, typically including individuals aged ≥ 65 years and individuals aged greater than 50 years with clinically significant conditions or physical limitations that affect movement, physical function, physical fitness, or PA ([201](#)). Older adults have the greatest range of physical function of any age group, with elite masters athletes at one end of the continuum and disabled institutionalized elders at the other end. A way to illustrate this is with “aging curves,” which illustrate how physical function may change over the lifespan dependent largely on lifestyle behaviors ([202](#)). These curves show that function improves during growth and development, is fairly well maintained during adult life, and declines in older age and may reach a threshold that puts a person at risk for disease, disability, and/or death. The rate of decline in older age is primarily a function of lifestyle behaviors including exercise/PA, with those choosing healthy behaviors seeing a much slower rate of decline ([202](#)). This general pattern is applicable to most physiological processes and functional capacities, although many are highly correlated (*e.g.*, muscle strength and bone mineral density), each will follow its own trajectory depending on the factors that impact it.

Based on the aging curve concept ([Figure 6.2](#)), it is clear that the effects of chronological aging are generally less impactful on overall physical function than the effects of deconditioning and/or disease ([Table 6.4](#) provides a list of age-related changes in key physiologic variables). Therefore, health and functional status are better indicators of the ability to engage in PA than chronological age.

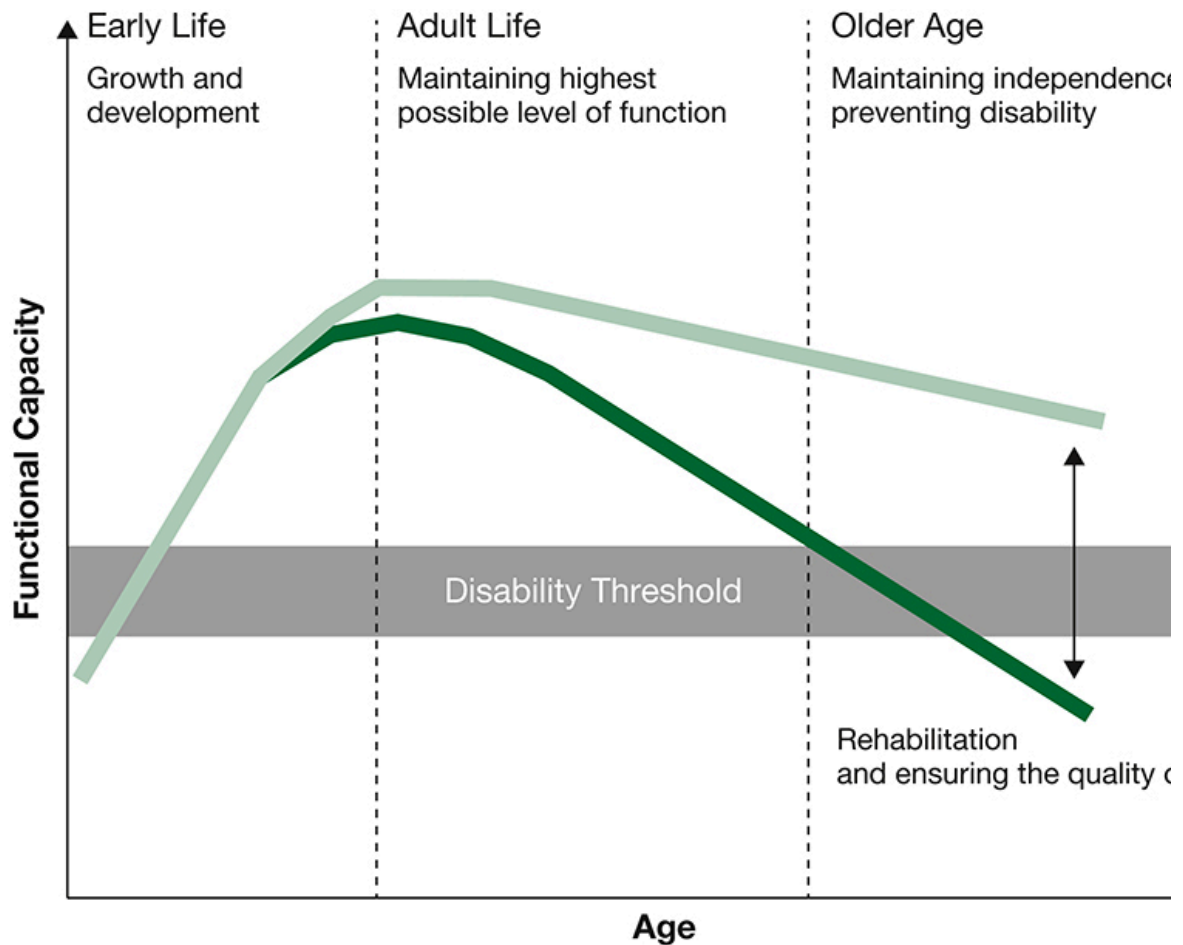


Figure 6.2 Aging curve (202). Originally from Moritz, Inka, Stein, Claudia L'Engle & WHO Ageing and Health Programme. (1999). A life course perspective of maintaining independence in older age prepared for WHO by Inka Moritz and Claudia Stein under the guidance of WHO's Ageing and Health. World Health Organization. <https://link.edgepilot.com/s/b2ebdf2b/ESu40brgwk2zmZeTHkfp2g?u=https://iris.who.int/handle/10665/65576> (figure 1, page 4)

TABLE 6.4 Effects of Aging on Selected Physiologic and Health-Related Variables (201)

Variable	Change
Resting heart rate	Unchanged
Maximum heart rate	Lower
Maximum cardiac output	Lower
Resting and exercise blood pressure	Higher
Absolute and relative maximum oxygen uptake reserve ($\dot{V}O_{2Rmax}$ L · min ⁻¹ and mL · kg ⁻¹ · min ⁻¹)	Lower
Residual volume	Higher
Vital capacity	Lower
Reaction time	Slower
Muscular strength	Lower
Flexibility	Lower
Bone mass	Lower
Fat-free body mass	Lower
% Body fat	Higher
Glucose tolerance	Lower
Recovery time	Longer

Overwhelming evidence supports the benefits of PA in (a) slowing typical age-related changes that impair exercise capacity, (b) optimizing age-related changes in body composition, (c) promoting psychological and cognitive well-being, (d) managing chronic diseases, (e) reducing the risks of physical disability, and (f) increasing longevity (201). Despite these benefits, older adults are the least physically active of all age groups. In fact, only about 16% of individuals aged 65–74 years report engaging in aerobic and muscle strengthening activities that meet the federal guidelines detailed in the 2018

Physical Activity Guidelines Advisory Committee Scientific Report and only 7% of individuals aged 85 years and older meet these same guidelines ([203](#)).

The consequences of low levels of PA and exercise include *sarcopenia*, a loss of muscle function (*i.e.*, strength and power) and a multifactorial loss of physical function known as *frailty*. Sarcopenia is now a diagnosable muscular disease and can be determined by low muscle performance measures rather than measurement of low muscle mass or poor muscle quality. This has improved the ability to screen individuals for sarcopenia due to the wide range of valid assessments of muscle performance including grip strength, lower body strength, muscle power, and gait speed ([204](#)). The SARC-F questionnaire is a five-item self-administered questionnaire that exhibits low to moderate sensitivity and high specificity in predicting low muscle strength and therefore sarcopenia risk ([Table 6.5](#)). A score of 4 or greater on the SARC-F scale is suggestive of sarcopenia and warrants further assessment of muscle function ([205](#)).

TABLE 6.5 SARC-F Screen for Sarcopenia

Component	Question	Scoring
Strength	How much difficulty do you have in lifting and carrying 10 lb?	None = 0 Some = 1 A lot or unable = 2
Assistance in walking	How much difficulty do you have walking across a room?	None = 0 Some = 1 A lot, use aids, or unable = 2
Rise from a chair	How much difficulty do you have transferring from a chair or bed?	None = 0 Some = 1 A lot or unable without help = 2
Climb stairs	How much difficulty do you have climbing a flight of 10 stairs?	None = 0 Some = 1 A lot or unable = 2
Falls	How many times have you fallen in the past year?	None = 0 1–3 falls = 1 4 or more falls = 2

Adapted from Malmstrom et al. ([205](#)).

Additionally, frailty is an umbrella term that describes the modifiable conditions of weight loss, slowness, weakness, sedentariness, and fatigue. Frailty is not a form of chronic disease, but it both exacerbates, and in turn is exacerbated by, underlying diseases ([206](#)). The FRAIL Scale ([Box 6.4](#)) is a questionnaire that assesses one's degree of frailty with high validity ([207](#)).

Box 6.4 Frail Scale ([207](#))

Fatigue: Are you fatigued?

Resistance: Cannot walk up one flight of stairs?

Aerobic: Cannot walk one block?

Illnesses: Do you have more than five illnesses?

Loss of weight: Have you lost more than 5% of your in the past 6 months?

Scoring:

Robust (score = 0), prefrail (score = 1–2), and frail (score = 3–5)

Although the prevalence of frailty and sarcopenia increases with age and chronic disease burden, they are not inevitable, particularly when a physically active lifestyle is maintained.

Exercise Testing

Most older adults do not require an exercise test prior to initiating a moderate-intensity PA program (see [Chapter 2](#)). However, if exercise testing is recommended, it should be noted that the associated electrocardiogram (ECG) has higher sensitivity (*i.e.*, ~84%) and lower specificity (*i.e.*, ~70%) for coronary artery disease than in younger age groups (*i.e.*, <50% sensitivity and >80% specificity), producing a higher proportion of false positive outcomes for older adults. This situation may be related to the greater frequency of left ventricular hypertrophy and the increased presence of conduction disturbances among older as compared to younger adults ([208](#)).

Although there are no specific exercise test termination criteria for older adults beyond those presented for all adults in [Chapter 3](#), the increased prevalence of cardiovascular, metabolic, and orthopedic problems among older adults increases the overall likelihood of early test termination. Therefore, exercise testing in older adults may require subtle differences in both protocol and methodology, and should only be performed when

indicated by a physician or other health care provider. Special considerations when testing older adults include the following:

- Initial workload should be light (*i.e.*, <3 METs) and workload increments should be small (*i.e.*, 0.5–1.0 MET) for those with low work capacities. The modified Naughton treadmill protocol is a good example of such a protocol (see [Figure 4.1](#)) ([201](#)).
- A cycle ergometer may be preferable to a treadmill for those with poor balance, poor neuromotor coordination, impaired vision, impaired gait patterns, weight-bearing limitations, and/or orthopedic problems. However, local muscle fatigue may be a factor for premature test termination when using a cycle ergometer ([201](#)).
- Adding a treadmill handrail support may be required because of reduced balance, decreased muscular strength, poor neuromotor coordination, and fear. However, handrail support for gait abnormalities will reduce the accuracy of estimating peak MET capacity based on the exercise duration or peak workload achieved ([201](#)).
- Treadmill workload may need to be adapted according to walking ability by increasing grade rather than speed ([201](#)).
- Many older adults exceed the age-predicted HR_{max} during a maximal exercise test. The frequently used $(220 - \text{age})$ HR_{max} equation tends to underpredict HR_{max} in older adults ([209](#)); therefore, it is best to use other HR_{max} equations (see [Table 5.3](#)).
- The influence of prescribed medications on the ECG and hemodynamic responses to exercise may differ from usual expectations (see [Appendix A](#)).

Currently, there is little evidence demonstrating increased mortality or cardiovascular event risk during exercise, or exercise testing, in this segment of the population, therefore eliminating the need for exercise testing unless medically indicated (*e.g.*, symptomatic CVD, uncontrolled diabetes).

Otherwise, individuals free from CVD symptoms should be able to initiate a light intensity (<3 METs) exercise program without undue risk ([210](#)).

Physical Performance Testing

Physical performance testing has largely replaced exercise stress testing for the assessment of functional status of older adults. Some test batteries have been developed and validated as correlates of underlying fitness domains, whereas others have been developed and validated as predictors of subsequent disability, institutionalization, and death. Physical performance testing is appealing in that most performance tests require little space, equipment, and cost; can be administered by lay or health/fitness personnel with minimal training; and are considered safe in healthy and clinical populations ([211](#)). Many widely used physical performance tests have either age and sex normative data or threshold values indicative of functional limitations associated with poorer health status that can be targeted for an exercise intervention. Some of the most commonly used physical performance tests are described in [Table 6.6](#). Before performing physical performance assessments: (a) carefully consider the specific population for which each test was developed, (b) be aware of known floor or ceiling effects (*i.e.*, the inability of the least robust participants to score lower and the most robust to score higher on the performance test), and (c) understand the context (*i.e.*, the sample, age, health status, and intervention) in which change scores or predictive capabilities are attributed.

TABLE 6.6 Commonly Used Physical Performance Tests

Measure and Description	Administration Time	Cutpoint Indicative of Lower Function
Senior Fitness Test (212)		
Seven items: 30-s chair stand, 30-s arm curls, 8 ft up and go, 6-min walk, 2-min step test, sit and reach, and back scratch with normative scales for each test	30 min total Individual items range from 2 to 10 min each.	≤25th percentile of age-based norms
Short Physical Performance Battery (213)		
A test of lower extremity functioning that combines scores from usual gait speed and timed tests of balance and chair stands; scores range from 0 to 12 with higher score indicating better functioning.	10 min	10 points
Usual Gait Speed		
Usually assessed as the better of two trials of time to walk a short distance (3–10 m) at a usual pace	<2 min	1 m · s ⁻¹
6-Min Walk Test		
Widely used as an indicator of cardiorespiratory endurance; assessed as the most distance an individual can walk in 6 min. A change of 50 m is considered a substantial change (214).	<10 min	≤25th percentile of age-based norms
Continuous Scale Physical Performance Test (215)		
Two versions — long and short — are available. Each consists of serial performance of daily living tasks, such as carrying a weighted pot of water, donning	60 min	57 points

Measure and Description	Administration Time	Cutpoint Indicative of Lower Function
and removing a jacket, getting down and up from the floor, climbing stairs, carrying groceries, and others, performed within an environmental context that represent underlying physical domains. Scores range from 0 to 100 with higher scores representing better functioning.		

The *Senior Fitness Test* (SFT) was developed using a large, healthy, community-dwelling sample and has published normative data for males and females aged 60–94 years for items representing upper and lower body strength, upper and lower body flexibility, CRF, agility, and dynamic balance ([212](#)). There are now published thresholds for each test battery assessment that defines the level of capacity needed between the ages of 65–85 years to remain independent to age 90 years ([216](#)).

The Short Physical Performance Battery (SPPB) ([213](#)), a test of lower extremity functioning, is best known for its predictive capabilities for disability, institutionalization, and death, but it also has known ceiling effects that limit its use as an outcome for exercise interventions in generally healthy older adults. A change of 0.5 point in the SPPB is considered a small meaningful change, whereas a change of 1.0 point is considered a substantial change ([214](#)).

Usual gait speed, widely considered the simplest test of walking ability, has comparable predictive validity to the SPPB ([215](#)), but its sensitivity to change with exercise interventions has not been consistent. A change in usual gait speed of $0.05 \text{ m} \cdot \text{s}^{-1}$ is considered a small meaningful change, and a change of $0.10 \text{ m} \cdot \text{s}^{-1}$ is considered a substantial change ([217](#)). In fact, predicted 10-year survival at age 75 years varies between 19% and 87% and between 35% and 91% in females across a range of gait speeds, with significant

increases in mortality risk per $0.10 \text{ m} \cdot \text{s}^{-1}$ reduction in gait speed ([218](#)). It has been suggested that gait speed be included as a vital sign associated with clinical evaluation of older adults ([219](#)). The 400-m usual-pace walk test has also been shown to be reliable as an assessment of mobility status in older adults with functional limitations ([220](#)).

Measurement of muscle power is also of interest given the relationship between muscle power and functional capacity in older adults ([221](#)). The Gallon Jug Transfer Test is a muscle power field test for older adults to measure the total time to transfer five full one gallon water jugs from a shelf at knee level to a shelf at shoulder level and has a high degree of sensitivity and specificity ([222](#), [223](#)).

Several assessment batteries have been developed to assess fall risk in older adults ([224](#)). Two assessments that can be combined into a short test battery include the Functional Reach Test which measures the distance reached forward from a stationary standing position and the Timed Up-and-Go Test which is similar to the SFT's 8-Foot Up and Go Test ([212](#)) which measures the time elapsed to get up, walk around a cone 3 m away, and return to the chair to sit down. These assessments provide an accurate overview of the physical capacities that affect fall risk: joint mobility (both range of motion and stability), sensory function, static balance, dynamic balance and gait. It is also useful to pair the objective score of each of these tests with basic movement observations during the performance of the tests that can identify the specific task demands that pose difficulty to the older adult ([224](#)).

See [Table 6.7](#) for additional criterion-referenced fitness standards for maintaining physical independence in older adults.

TABLE 6.7 Criterion-Referenced Fitness Standards for Maintaining Physical Independence in Older Adults

	Age Groups							% of Decline Reflected Over 30 yr
	60- 64	65- 69	70- 74	75- 79	80- 84	85- 89	90- 94	
Lower body strength (number of chair stands in 30 s)								
Female	15	15	14	13	12	11	9	40.0
Male	17	16	15	14	13	11	9	47.1
Upper body strength (number of arm curls in 30 s)								
Female	17	17	16	15	14	13	11	35.3
Male	19	18	17	16	15	13	11	42.1
Aerobic endurance (yards walked in 6 min)								
Female	625	605	580	550	510	460	400	36.0
Male	680	650	620	580	530	470	400	41.2
Alternate aerobic endurance (number of steps in 2 min)								
Female	97	93	89	84	78	70	60	38.1
Male	106	101	95	88	80	71	60	43.4
Agility/dynamic balance (8-foot up-and-go, s)								
Female	5.0	5.3	5.6	6.0	6.5	7.1	8.0	37.5
Male	4.8	5.1	5.5	5.9	6.4	7.1	8.0	40.0
Mean decline = 40.1								

NOTE: The proposed fitness standards were developed for use with the Senior Fitness Test (SFT) battery.

FITT FITT RECOMMENDATIONS FOR OLDER ADULTS ([225](#), [226](#), [228](#), [229](#))

	Aerobic	Resistance	Flexibility
Frequency	$\geq 5 \text{ d} \cdot \text{wk}^{-1}$ for moderate intensity; $\geq 3 \text{ d} \cdot \text{wk}^{-1}$ for vigorous intensity; $3\text{--}5 \text{ d} \cdot \text{wk}^{-1}$ for a combination of moderate and vigorous intensity	$\geq 2 \text{ d} \cdot \text{wk}^{-1}$	$\geq 2 \text{ d} \cdot \text{wk}^{-1}$
Intensity	Moderate, RPE 12–13 and progressing to vigorous, RPE 14–17 using the 6–10 Borg scale, or equivalent ranges of 3–4 and ≥ 5 , respectively, using the CR10 scale (Fig. 4.2)	<i>Progressive weight training:</i> Light intensity (<i>i.e.</i> , 40%–50% 1-RM) for beginners; progress to moderate-to-vigorous intensity (60%–80% 1-RM); alternatively, adjust resistance so that the last two repetitions are challenging to perform.	Stretch to the point of feeling tightness or slight discomfort.
Time	$150 \text{ min} \cdot \text{wk}^{-1}$ of moderate-intensity exercise or $75 \text{ min} \cdot \text{wk}^{-1}$ of vigorous-intensity exercise; or an equivalent combination of moderate and vigorous-intensity exercise; may be accumulated over the course a day	<i>Progressive weight training:</i> 8–10 exercises involving the major muscle groups; ≥ 1 set of 10–15 repetitions for beginners; progress to 1–3 sets of 8–12 repetitions for each exercise <i>Power training:</i> 3 sets of 6–10 repetitions with high velocity or 12 sets of 2 repetitions or 6 sets of 4 repetitions (<i>i.e.</i> , cluster set configuration)	Hold stretch for 30–60 s.
Type	Any modality that does not impose excessive orthopedic stress such as walking. Aquatic exercise and stationary cycle exercise may be advantageous for those with limited tolerance for weight-bearing activity.	Progressive or power weight-training programs or weight-bearing calisthenics, stair climbing, and other strengthening activities that use the major muscle groups	Any physical activities that maintain or increase flexibility using slow movements that terminate in static stretches for each muscle group rather than rapid ballistic movements

1-RM, one repetition maximum.

Exercise Prescription

The general principles of Ex Rx apply to adults of all ages (see [Chapter 5](#)). The relative adaptations to exercise and the percentage of improvement in the components of physical fitness among older adults are comparable with those reported in younger adults and are important for maintaining health and functional ability and attenuating many of the physiologic changes that are associated with aging (see [Table 6.4](#)). Low aerobic capacity, muscle weakness, impaired balance and overall deconditioning are more common in older adults than in any other age group and contribute to loss of independence ([225](#)). Therefore, an appropriate Ex Rx should be a multicomponent program that combines aerobic exercise, resistance training, balance, and flexibility/mobility exercises. The *2018 Physical Activity Guidelines Advisory Committee Scientific Report* highlighted strong evidence from numerous randomized controlled trials and cohort studies that aerobic, muscle-strengthening, balance, and/or multicomponent PA programs improved physical function and reduced risk of age-related loss of physical function in the general aging population, as well as in older people with specific chronic conditions ([54](#)). In fact, the physical function benefits of multicomponent exercise programs in older adults are greater than single-component exercise.

For Ex Rx, an important distinction between older adults and their younger counterparts should be made relative to intensity. For apparently healthy adults, MVPAs are defined relative to METs, with moderate-intensity activities defined as 3–5.9 METs and vigorous-intensity activities as ≥ 6 METs. In contrast for older adults, activities can be defined by a number of methods, including physiological responses such as heart rate (HR) or Rating of Perceived Exertion (RPE) which ranges from 0 (an effort equivalent to sitting) to 10 (an all-out effort), with moderate intensity defined as 3 or 4 and vigorous intensity as ≥ 5 . A moderate-intensity PA should produce a noticeable increase in HR and breathing, whereas a vigorous-intensity PA

should produce a large increase in HR or breathing. Other RPE options include the Borg 6 to 20 scale of self-perceived exertion to assess intensity of activity ([227](#)).

High-Intensity Interval Training for Increasing Aerobic Capacity and Improving Cognitive Function

Research has demonstrated that high-intensity interval training protocols with alternating vigorous and moderate- to low-intensity exercise over the course of a training session elicits more rapid and more substantial training benefits to CRF and aspects of cognitive function than moderate-intensity continuous exercise and is both safe and enjoyable for older adults ([230–232](#)). Use of cardiovascular equipment (*e.g.*, treadmill, cycle ergometer, recumbent stepper, etc.) to elicit 85%–95% HR_{peak} during high-intensity intervals are associated with greater than 80% $\dot{V}O_{2peak}$. It is important to note that there is evidence that bodyweight or externally loaded high-velocity exercises may elicit 85%–95% HR_{peak} but much lower oxygen transport (<60% $\dot{V}O_{2peak}$) and therefore may not improve CRF effectively ([233](#)).

Neuromotor (Balance) Exercises for Older Adults, and/or Individuals With Mobility Limitations

Almost one in four individuals ≥ 65 years falls in the United States every year ([233](#)). Furthermore, falls are the leading cause of fatal injury and the most common cause of nonfatal trauma-related hospital admissions among older adults ([233](#)). Neuromotor exercise training, which combines balance, agility, and proprioceptive training, is effective in reducing and preventing falls, if performed $2\text{--}3\text{ d} \cdot \text{wk}^{-1}$ ([225](#), [229](#)). The 2018 PAGAC Scientific Report ([54](#)) cited strong and consistent evidence demonstrating that multicomponent PA that includes two or more components of strength, balance, endurance, or flexibility significantly reduced the risk of fall-related injuries by about 32%–

40%, including severe falls that result in bone fracture, head trauma, open wound soft tissue injury, or any other injury requiring medical care or admission to hospital ([234–237](#)), with similar benefits observed between older adults who were at high risk of falling versus those who were at an unspecified risk ([234](#)). Also, fall prevention programs using multicomponent activity reduced the risk of fall-related bone fractures by 40%–66% among older adults in community and home settings ([234–237](#)).

At this point, there is no consensus on the dose/response relationship for balance and fall risk reduction training for older adults. General recommendations for balance training include using the following: (a) progressively difficult postures that systematically reduce the base of support (*e.g.*, narrow stance, semitandem stance, tandem stance, one-legged stance), (b) dynamic movements that challenge limits of stability and/or base of support (*e.g.*, reaching from a narrow stance, tandem walking, stepping over obstacles), (c) stressing postural muscle groups (*e.g.*, heel or toe walks), (d) reducing sensory input (*e.g.*, standing with eyes closed, walking while turning the head), and (e) tai chi ([225](#), [238](#)). Many of these exercises do not elicit significant fatigue and therefore should be practiced on a daily basis.

Additional balance improvement and fall risk reduction exercises can be implemented to improve lower body strength, balance, and motor performance, as well as for increasing daily levels of PA ([239](#), [240](#)). Examples of such include postures or walking with gradual reduction in the base of support (*e.g.*, upgrading tandem stand to one-leg stand over time) and dynamic movements that perturb the center of gravity (stepping over obstacles). Balance training interventions examined the effects of four different types of balance training interventions (multidimensional [activities such as functional exercises, tai chi, and ball games], control center of mass, mobility, and reaching) on several different dimensions of balance performance ([241](#)). Overall, balance training interventions report significant

benefits from programs that included any of these four types of balance training versus interventions that did not involve any balance training ([241](#)). Multitask activities with high levels of physical (speed, coordination, balance), mental, and social demands (*e.g.*, dancing, team sports, handball) appeared particularly effective in improving functional performance and balance ([64](#)), whereas Nordic walking is significantly effective in improving dynamic balance, functional balance, muscle strength of lower limbs, and aerobic capacity in healthy older people ([242](#)). Active computer gaming (ACG) also has a significantly beneficial effect on balance and on functional exercise capacity when the volume of the ACG was greater than 120 min · wk⁻¹ ([243](#)).

Muscle Power Training in Older Adults

Older adults should engage in muscle power training (*e.g.*, medicine ball throws, kettlebell swings, etc.) because this element of muscle fitness declines most rapidly with aging and insufficient power has been associated with a greater risk of functional decline, falls, and frailty ([92](#), [221](#), [244](#), [245](#)). Muscle power training has a greater impact on improving functional capacities such as gait speed and Timed-Up-And-Go performance than traditional resistance training in older adults ([246](#)).

Traditionally, muscle power training recommendations for older adults included performing 1–3 sets using light-to-moderate loading (30%–60% of one repetition maximum [1-RM]) for 6–10 repetitions with high velocity ([93](#)). However, recent research has indicated that power training in older adults is more effective with less repetitions per set and shorter recovery times, termed cluster set configurations. Limiting each set to no more than 4 repetitions has been shown to maintain power output throughout the set more effectively and less attenuation of power output during training sets is associated with larger increases in muscle power over a training program.

Additionally, the cluster set approach has been shown to result in higher measures of enjoyment and adherence in older adults ([228](#), [247](#)).

Special Considerations for Exercise Programming

There are numerous considerations to maximize the effective development of an exercise program, including the following:

- Intensity and duration of PA should be light at the beginning, in particular for older adults who are highly deconditioned, functionally limited, or have chronic conditions that affect their ability to perform physical tasks.
- Progression of PA should be individualized and tailored to tolerance and preference; a conservative approach may be necessary for the older adults who are highly deconditioned or physically limited.
- Muscular strength decreases rapidly with age, especially for those aged greater than 50 years. Although resistance training is important across the lifespan, it becomes more important with increasing age ([93](#), [226](#), [229](#)).
- For strength training involving use of selectorized machines or free weights, initial training sessions should be supervised and monitored by personnel who are sensitive to the special needs of older adults.
- If chronic conditions preclude activity at the recommended minimum amount, older adults should perform PA as tolerated to avoid being sedentary.
- Older adults should gradually exceed the recommended minimum amounts of PA and attempt continued progression if they desire to improve and/or maintain their physical fitness.
- Older adults should consider exceeding the recommended minimum amount of PA to improve management of chronic diseases and health conditions for which a higher level of PA is known to confer a therapeutic benefit.

- Moderate-intensity aerobic PA (especially those that combine a cognitive and physical task, such as counting backward while walking) ([54](#)) should be encouraged for individuals with cognitive decline given the known benefits of PA on cognition. Individuals with significant cognitive impairment can engage in PA but may require individualized assistance.
- Structured PA sessions should end with an appropriate cool-down, particularly among individuals with CVD. The cool-down should include a gradual reduction of effort and intensity and, optimally, flexibility exercises.
- Incorporation of behavioral strategies such as social support, self-efficacy, the ability to make healthy choices, and perceived safety all may enhance participation in a regular exercise program (see [Chapter 12](#)).
- The exercise professional should also provide regular feedback, positive reinforcement, and other behavioral/programmatic strategies to enhance adherence.

ONLINE RESOURCES

Alliance for Aging Research: <https://www.agingresearch.org/>

Canadian Society for Exercise Physiology. *Canadian Physical Activity*

Guidelines for Older Adults (65 years and older) [Internet]. Ottawa, Ontario

(Canada): Canadian Society for Exercise Physiology; 2019 [cited 2019 Apr

24]. Available from: [https://csepguidelines.ca/adults-](https://csepguidelines.ca/adults-65/#resourceshttps)

[65/#resourceshttps](https://csepguidelines.ca/adults-65/#resourceshttps)

National Council on Aging; Healthy Living: [https://www.ncoa.org/healthy-](https://www.ncoa.org/healthy-aging/)

[aging/](https://www.ncoa.org/healthy-aging/)

U.S. Department of Health and Human Services; Healthy Aging:

<https://www.hhs.gov/aging/healthy-aging/index.html>

TRANSGENDER AND GENDER DIVERSE INDIVIDUALS

Transgender individuals are people whose gender identities differ or are opposite their sex at birth ([Table 6.8](#)). In the United States, over 1.6 million people identify as transgender, with 0.5% of adults and 1.4% of youth ages 13–17 ([250](#)). Transition refers to a process by which a transgender person begins to live according to their gender identity and may start in childhood, adolescence, and stages of adulthood. Transition may involve one or many processes, social, legal or medical, which aim to align the outward self with one's internal gender identity ([251](#)). Some but not all transgender people will undergo medical transition at some point in their lives, meaning that they start a medication or have surgery under the supervision of a medical provider to align their body with their gender identity. Guidelines for medical and surgical interventions, including eligibility, requirements, and expected effects, are summarized in The World Professional Association for Transgender Health Standards of Care ([252](#)) and the Endocrine Society Guidelines ([253](#)).

TABLE 6.8 Terminology

Sex	A multidimensional biological construct based on anatomy, physiology, genetics, and hormones (248).
Gender	A multidimensional construct that encompasses gender identity and expression, as well as social and cultural expectations about status, characteristics, and behavior as they are associated with certain sex traits (248).
LGBTQ+	Lesbian, Gay, Bisexual, Transgender, Queer/Questioning, + = other identities not encompassed with this acronym
Transgender	A person whose gender identity differs or is opposite their sex at birth.
Cisgender	A person whose gender identity aligns with their sex at birth.
Nonbinary	An umbrella term for gender identities that are outside the gender binary.
Gender-affirming hormone therapy	This may encompass testosterone given to induce masculine secondary sex characteristics or estradiol (typically with a medication to block or suppress endogenous testosterone) to induce feminine secondary sex characteristics.
Gender-affirming surgery	Surgery to align one's body with one's gender identity. Feminizing surgeries may include: breast augmentation, facial feminization, vaginoplasty, orchiectomy, tracheal shave, or others. Masculinizing surgeries may include chest masculinization surgery, facial masculinization, hysterectomy (with or without oophorectomy), phalloplasty/metoidioplasty, or others.

These definitions were adapted from the National Institute of Health ([248](#)) and the National Academies of Sciences ([249](#)).

In the 2022 Trevor Project Survey, LGBTQ+ youth (see [Table 6.8](#) for definition) noted athletics and exercise as a source of joy ([254](#)). However, in the same survey, 3% of transgender and nonbinary youth “worried about

transgender people being denied the ability to play sports due to state or local laws” (254). Legislation that restricts participation based on birth sex is increasingly common and differs by state in the United States (255). Many sporting organizations, especially at the higher levels, have regulations regarding participation of transgender individuals, especially in the women’s competition category. These policies have been shown to shift over time and often differ by sport, with the most comprehensive source being transathlete.com (256).

There are many barriers, external and internal, to PA and sports participation for transgender individuals, and overall transgender individuals are less physically active than cisgender individuals (257–259). External barriers may include lack of inclusive environments, prior negative experiences in sports, lack of suitable changing and showering facilities, gender-specific sports-related clothing, trouble concealing breasts or genitals, and restrictive policies for competitive sports (especially for transgender women competing in the women’s category) (260–262). Additional barriers including fear, stigma, gyms being unwelcoming, or reactions of others, including bullying or hateful rhetoric from teammates, opponents, coaches, referees/umpires, patrons, staff, and others (263–265). These may also be fueled by negative portrayals of transgender athletes in the media (266). Internal barriers may include gender incongruence (outward appearance does not reflect inward gender identity), body dissatisfaction/dysphoria and anxiety about others’ reactions (260). There may be further barriers for nonbinary individuals, as many exercise facilities and sports are divided into binary gender categories. Coaches, athletic trainers, and others should be aware of these factors, as well as other possible medical or mental health comorbidities. Transgender individuals are more likely to be overweight or obese (267, 268) and have lower bone mineral density than their cisgender counterparts (269–272). Forty percent of transgender adults and 35% of transgender youth have attempted suicide

in their lifetime ([273](#), [274](#)) and have a high prevalence of depression, anxiety, and eating disorders ([275](#)). In other populations, regular exercise is associated with lower risk of mental health problems and suicidality, which supports the importance of PA and exercise for transgender people ([276](#)).

Generally, transgender people feel more comfortable engaging in PA after, than before transition, or with improved appearance congruence (appearance aligns with gender identity) ([277](#)). Transgender people who are on gender-affirming hormone therapy, which is related to higher body satisfaction, are more engaged in PA than transgender people not on gender-affirming hormone therapy ([258](#)). Although, for transgender men, many avoid or limit PA and sports participation before chest masculinizing surgery ([278](#)). Factors that facilitate participation in PA and sports include body satisfaction and changes, gender affirming surgery, safe spaces or “trans only environments” ([260](#)).

Medical Interventions

For youth/adolescents, puberty-delaying medications may be prescribed to block or pause development of secondary sex characteristics that do not align with gender identity ([252](#), [253](#)). These medications, gonadotropin-releasing hormone agonists (GnRHa), when prescribed alone, result in decreased bone turnover and bone mineral apparent density Z scores ([272](#), [279](#)). Individuals on GnRHa treatment may benefit from other measures to optimize bone health, such as adequate calcium and vitamin D and adhering to age-based recommendations for weight-bearing exercise. Bone density Z scores improve after discontinuation of the GnRHa or initiation of gender-affirming hormone therapy ([279](#)). Gender-affirming hormone therapy (testosterone or estradiol) may be prescribed to eligible individuals to induce secondary sex characteristics to align the body with one’s gender identity ([252](#), [253](#)). Some adults may undergo gender-affirming surgery, to further align their body with their gender identity (see [Table 6.8](#)). There are limited

data on the impact of gender-affirming medications or surgery on exercise and athletic performance ([251](#)). Until additional data are available, recommendations for exercise should follow age-based guidelines or guidelines based on additional medical needs.

Exercise Testing

Generally, the guidelines for standard exercise testing apply based on the age group of the individual, including those for adults (see [Chapter 3](#)), older adults, or children and adolescents (see [Chapter 6](#)). More data are needed regarding when in medical transition, standards based on gender identity should be applied, particularly at elite levels of participation.

Exercise Prescription

The exercise prescription (Ex Rx) guidelines outlined based on age group apply to this population as well, for adults (see [Chapter 5](#)), older adults, or children and adolescents (see [Chapter 6](#)). A multifaceted and individualized approach to exercise prescription is recommended. Understanding individual sources of gender dysphoria and barriers to PA, medical and mental health comorbidities (if present), and sources of joy are important in planning PA goals. Exercise professionals should be aware that some transgender people may exercise to achieve a certain body shape ([263](#)). Currently, there is limited research on exercise in transgender individuals and limited data on the impact of gender-affirming medications and surgeries on athletic performance or how to tailor an exercise prescription. There are known sex differences in athletic performance (see [Chapter 5](#)) that may impact the training regimen.

Special Considerations

- Ask about name and pronouns, use affirmed name and pronouns.

- Consider signs demonstrating that transgender and/or LGBTQ+ people are welcome in the space.
- Be aware of gendered bathrooms, locker or changing rooms, have gender neutral or single stall bathrooms or locker rooms available.
- Be aware of gendered uniforms or apparel (*e.g.*, swimsuits) and options available to transgender and nonbinary people (*e.g.*, waivers may be available in some sports).
- Be aware of items people may wear to affirm their gender, such as binders to compress the chest (which can limit breathing in aerobic activity) or genital packers.
- Be aware of clothing/uniforms that do not adequately shield genitalia, when that is desired by the individual (*e.g.*, swimsuits, tight-fitting pants, etc.).
- Some, but not all, transgender youth may take medications to block/suppress/pause endogenous puberty. These medications will prevent feminizing or masculinizing secondary sex characteristics and may impact athletic performance.
- Some, but not all transgender people, may take hormones to align their body with their gender identity, and this may impact athletic performance.
- Some, but not all transgender adults, may have surgery to align their body with their gender identity. Some people may be more likely to exercise after surgery (*e.g.*, chest masculinizing surgery for transgender men or nonbinary individuals).
- Professional sports have widely varying regulations regarding participation of transgender women in the women's competition category (primarily regarding duration or degree of testosterone suppression).

ONLINE RESOURCES

TRANSATHLETE.COM, which includes an overview of policies ranging from K-12 to Olympic level sports (256): <https://www.transathlete.com>

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CHAPTER 7

Environmental Considerations for Exercise Prescription

INTRODUCTION

This chapter addresses unique factors associated with environmental effects on exercise, with specific emphasis on altitude, heat, and cold. Aside from describing general considerations for exercising in environmental extremes, this chapter also provides details on common medical environmental illnesses and injuries, prevention strategies, and organizational planning factors that are intended to assist those who train or provide coverage to athletes under these conditions. The exercise professional must be familiar with environmental effects on clients and athletes in order to provide effective and safe counseling for exercising in a variety of environments.

EXERCISE IN HIGH-ALTITUDE ENVIRONMENTS

Altitude, vertical elevation above mean sea level, is broken into the following elevation bands: *low altitude* (0–1,500 m; 0–4,921 ft), *high altitude* (1,500–3,500 m; 4,921–11,483 ft), *very high altitude* (3,500–5,500 m; 11,483–18,045 ft), and *extreme altitude* (5,500–8,850 m; 18,045–29,035 ft) (1). As altitude increases, there is a corresponding reduction in atmospheric pressure, which directly reduces the partial pressure of oxygen inhaled. This drop in partial pressure of oxygen leads to a decrease in arterial oxygen levels and induces physiological compensation. These changes induce several normal physiologic responses as the body attempts to adjust to higher elevations (1). The immediate compensatory responses include increased ventilation and cardiac output, the latter usually through elevated heart rate (HR) (2). These responses then produce a respiratory alkalosis, which leads to renal compensation and occurs through the elimination of bicarbonate in the urine. Practically, as the athlete exhales greater and greater levels of carbon dioxide (CO₂), the kidneys remove bicarbonate to maintain pH of the blood, which can lead to increased risk of dehydration as a function of increased respiration. Therefore, exercising at altitude requires an increased fluid intake beyond the normal athletic requirements (1).

Aerobic performance decreases with increasing altitude in the near term. In general, the physical performance decrement will be greater as elevation, physical activity duration, and muscle mass increases, but this decrement can be ameliorated with proper altitude acclimatization (1). As a rule of thumb, individuals can expect a decrease in exercise performance of 1.5%–3.5% per 300 m of elevation after 1,500 m (3, 4). The most common altitude effect on physical task performance is an increased time for task completion because of reduced pace or the need for more frequent rest breaks. With altitude exposure of ≥1 weeks, significant acclimatization occurs (*i.e.*, increased ventilation and arterial oxygen content and restored acid-base balance), and the time to complete a task is reduced, but still longer relative to sea level (3). Exercise professionals should be aware of the differences in acclimatization versus acclimation with acclimatization happening in the natural or “uncontrolled” environment like regional mountains and acclimation in the controlled environment, such as an indoor training facility. The document here will focus on acclimatization in the natural environment, precautions, considerations, planning, and treatment intervention for altitude illness. The estimated increases in performance time to complete tasks of various durations during altitude exposure are given in [Table 7.1](#) (5).

TABLE 7.1 Estimated Impact of Increasing Altitude on Time to Complete Physical Tasks at Various Altitudes (5)

Percentage Increase in Time to Complete Physical Tasks Relative to Sea Level								
	Tasks Lasting <2 min		Tasks Lasting 2–5 min		Tasks Lasting 10–30 min		Tasks Lasting >3 h	
Altitude	Initial	>1 wk	Initial	>1 wk	Initial	>1 wk	Initial	>1 wk
High	0	0	2–7	0–2	4–11	1–3	7–18	3–10
Very high	0–2	0	12–18	5–9	20–45	9–20	40–65	20–45
Extreme	2	0	50	25	90	60	200	90

For those able to stay long term at altitude, stays of 30 days or longer produce not only acclimatization effects but also physiologic changes to include increased erythropoietin and hemoglobin (6). These produce the classic lay reporting of improved performance when returning to lower altitudes. Living at altitude and then training at a higher altitude, even a modest increase in line with recommendation here such as 350 m, can also provide improvement in maximal $\dot{V}O_2$ (6). The change between living altitude and training altitude causes more physiological changes that benefit the athlete than living and training at the same high altitude (6). The outcome goal of the exercise prescription ($Ex R_x$), length of available time to remain at altitude, and access to altitude changes (such as living in driving distance) should be considered by the provider when developing their $Ex R_x$. Elite or endurance athletes with access to altitude change and a desire to return to peak performance will have a very different profile and altitude requirement from the person seeking to return to weekend sport at altitude such as skiing or biking.

Altitude Acclimatization

Proper acclimatization is typically more effective, when time allows, than premedication for the prevention of acute altitude illnesses. A detailed description of acute altitude illnesses may be found in the “Medical Considerations: Altitude Illnesses and Preexisting Conditions” section below. With proper acclimatization, individuals can achieve optimal physical and cognitive performance for the altitude at which they will be performing. Altitude acclimatization consists of physiologic adaptations that develop in a time-dependent manner during continuous exposures to high altitudes (1). Breathing low concentrations of oxygen or restricting ventilation using masks, hoods, and other equipment (*i.e.*, normobaric hypoxia) is not as effective as being exposed to the natural altitude environment (*i.e.*, hypobaric hypoxia) for inducing functionally useful altitude acclimatization (7).

A graded ascent to altitude of $600 \text{ m} \cdot \text{d}^{-1}$ and a rest day every 600–1,200 m is the most widely advocated method for reduction in risk and prevention of high-altitude illnesses (8). Above 3,000 m, ascent should be limited by a sleeping altitude increase of 500 m per night, with a rest day of every 3–4 days (9). Since this is not always feasible because of terrain, transportation, or accommodations, modifications can be made by averaging the ascent rate for the entire trip (days ascending/total altitude gain) and maintaining it at no more than the $500 \text{ m} \cdot \text{d}^{-1}$ threshold (9). Additional rest days should be included after nights where the altitude gain exceeded 500 m (9).

For individuals ascending from low altitude, at least partial altitude acclimatization can develop by living at a high-altitude elevation temporarily while leading up to the event or ascending to a higher target elevation, which is termed *staging*. The goal of staged ascents is to gradually promote development of altitude acclimatization while averting the adverse consequences (*e.g.*, altitude sickness) of rapid ascent to high altitudes (7). The first stage of all staged ascent protocols should be ≥ 3 days of residence at high altitude. At this altitude, individuals will experience small decrements in physical performance and a low incidence of altitude sickness. At any given altitude, almost all of the acclimatization response is attained between 7 and 12 days of residence at that altitude. Short stays of 3–7 days at high altitudes will decrease susceptibility to altitude illness at higher altitudes. Stays of 6–12 days are required to improve physical work performance (7). The magnitude of the acclimatization response is increased with additional higher staging elevations or a longer duration at a given elevation. The final staging elevation should be as close as possible to the target elevation. It should be noted that this method is effective and designed for when there is ample time available before the event. A return to lower elevation during staging will negate or reduce the accumulated benefits of acclimatization (7).

The general staging guideline is as follows (7):

- For every day spent greater than 1,200 m (3,937 ft), an individual is prepared for a subsequent rapid ascent to a higher altitude equal to the number of days at that altitude times 305 m (1,000 ft).
- For example, an athlete stages at 1,829 m (6,000 ft) for 6 days. Therefore, 305 m (1,000 ft) per staged day multiplied by 6 days equals 6,000 ft additional ascent with reduced risk. In this case, the physical performance and altitude sickness risk will be reduced up to altitudes of 3,657 m (12,000 ft) compared to the athlete who does not stage.
- This guideline applies to altitudes up to 4,267 m (14,000 ft).

Rapid Ascent

Many individuals travel directly to high mountainous areas from sea level for skiing or trekking vacations and are unacclimatized when beginning their activities. During this time, physical

activity should not be excessive, and endurance exercise training should be stopped, or its intensity greatly reduced to minimize the possibility that acute mountain sickness (AMS) will develop or be exacerbated if already present (8). Beginning within hours of a rapid ascent to a given altitude, AMS may begin to present, and physical and cognitive performances will be at their nadir for these unacclimatized individuals (9). As acclimatization occurs, individuals may gradually resume normal activities and exercise training once the altitude symptoms begin to subside over the subsequent hours to days (8). Classically, acute altitude illnesses are associated with travel above 2,500 m, but cases of AMS have been observed as low as 2,000 m (9).

Assessing Individual Altitude Acclimatization Status

Assessing the acclimation process can allow the exercise professional to provide an Ex Rx with informed decisions on activity quantity and intensity in order to reduce risk of altitude illnesses. The best indices of altitude acclimatization over time at a given elevation are improved physical performance, decreased HR (both resting and exercise), an increase in arterial oxygen saturation (SaO_2), and if present, a decrease in AMS symptoms (9). The uncomplicated resolution of AMS or its absence in the first 3–4 days following ascent indicates a normal acclimatization response. After 1–2 weeks of acclimatization, physical performance improves such that most tasks can be performed for longer periods of time and with less perceived effort relative to the initial exposure to the same elevation (9). Another early sign of appropriate adaptation to altitude is increased respiratory rate and subsequently increased urine volume, which generally occurs during the first several days at a given elevation. Urine volume will continue to increase with additional ascent and return to normal with subsequent adaptation after acclimatization is complete (1).

The response to acclimatization and possibility of an acute altitude illness can be roughly predicted by measuring an individual's maximal heart rate (HR_{max}) at their normal altitude. To minimize the risk of AMS, the goal should be to maintain an HR below 85% of the individual's HR_{max} . Doing so reduces the incidence of AMS in unacclimatized/acclimatizing individuals to only 20%. This is in contrast to an incidence of over 60% for those who exceed 85% of max threshold (10). HR_{max} estimation formulas have been found to be less accurate at altitude than at sea level when predicting athletes' capacities (11). Therefore, it is recommended that direct/real-time HR monitoring or pulse oximetry be used during any training while in the acclimatization process and that this should be limited to 85% of previous measured HR_{max} .

Indirect measurement of SaO_2 by noninvasive pulse oximetry (SpO_2) is a very good indicator of acclimatization. Pulse oximetry should be performed under nonstimulating, resting conditions (12). From its nadir on the first day at a given altitude, SpO_2 should progressively increase over the next 3–7 days before stabilizing. For example, with initial exposure to an altitude of 4,300 m (14,107 ft), resting SpO_2 is 81%; after a week of continuous residence at the

same elevations, resting SpO₂ progressively rises to approximately 88% (12). Pulse oximetry often reveals pronounced hypoxia and should be utilized as soon as there are any concerns for acute altitude illness. While continuous pulse oximetry is typically not necessary, frequent checks of asymptomatic personnel may be helpful for situational awareness (13). When using pulse oximetry at altitude in monitoring for altitude illness or measuring Ex R_x performance, additional considerations are required. Vasoconstriction from cold environment can reduce accuracy, increase in ambient sunlight from the environment at altitude (14), and the reduced PaO₂ of altitude itself (15) can all skew readings. These can be controlled for by warming the patient, testing indoors, tent, or other shelter and by having the patient rest and take gentle breaths for several minutes (16).

Use of commercial fitness trackers has exploded in recent years, and heart rate variability (HRV) has become a target of clinical research and grown as a form of fitness measurement (17, 18). Literature review already conducted supports the use of HRV for measuring clinical improvement as well as athletic performance with an increase in HRV being correlated with improved fitness and performance (17, 18). Use of HRV to adjust exercise load and training plan shows promise for improved outcomes during rehabilitation compared with standardized noncustomized training regimes including while at altitude (19, 20). This was achieved by using waking HRV scores to adjust the days training load based on higher or lower HRV scores both as a way to maximize effective training days and to reduce overtraining on less effective days, especially in a high-altitude environment (19).

Medical Considerations: Altitude Illnesses and Preexisting Conditions

Rapid ascent to high and very high altitude increases individual susceptibility to altitude illness. The primary altitude illnesses are AMS, high-altitude cerebral edema (HACE), and high-altitude pulmonary edema (HAPE). Additionally, many individuals may develop a sore throat and bronchospasm from drier air that may produce disabling, severe coughing spasms at high altitudes. Susceptibility to altitude illnesses is increased in individuals who rapidly ascend without acclimatization, by immediate prolonged physical exertion, dehydration early in the altitude exposure, and with a prior history of acute altitude illnesses (13).

AMS is the most common form of altitude sickness. Symptoms include headache, nausea, vomiting, decreased appetite, fatigue or weakness, dizziness or light-headedness, and poor sleep. The Lake Louise Scoring system is used to determine severity of AMS (13), and the assessment should be performed by personnel trained in this scoring system (*e.g.*, Wilderness First Responder). AMS typically develops within the first 24 hours of altitude exposure, and its incidence and severity increase in direct proportion to ascent rate and altitude (21). The estimated incidence of AMS in unacclimatized individuals rapidly ascending directly to high

altitudes is $\leq 15\%$; to very high altitudes, 15%–70%; and to extreme altitudes, 70%–85% (21). In most individuals, if ascent is stopped and physical exertion is limited, AMS symptoms peak at about 18–22 hours, and recovery occurs over the next 24–48 hours (21).

HACE is a potentially fatal, although uncommon illness that occurs in less than 2% of individuals ascending greater than 3,658 m (12,000 ft) (13). HACE may begin with or without AMS; symptoms may start with headache or nausea and progresses to ataxia, altered consciousness (confusion, drowsiness, impaired decisions), and then coma. HACE most often occurs in individuals who have AMS, yet continue to ascend without treatment; deterioration to coma may occur in as little as 12 hours (13). HACE alone, without concomitant HAPE, is more common at very high altitudes such as cases in Tibet. Whereas cases at high altitude often have both HACE and HAPE present (1).

HAPE is a potentially fatal illness that occurs in less than 0.6% of individuals ascending greater than 3,658 m (12,000 ft) (13). Initial symptoms include shortness of breath with exertion, progressing to shortness of breath at rest, and cough which may be productive of blood-tinged sputum. Individuals making repeated ascents and descents greater than 3,658 m (12,000 ft), and who exercise strenuously early in the exposure, have an increased susceptibility to HAPE. Half of those with HAPE have preceding AMS that was untreated. Of those diagnosed with HAPE, 16% have concurrent HACE at presentation, whereas upon autopsy, 50% of those with HAPE also demonstrate HACE. The presence of crackles (*i.e.*, rales) in the lungs, severe dyspnea, and elevated respiratory and HRs may indicate impending HAPE. Most HAPE presentations begin during the night or early morning, between the first and third days of ascent (13).

Altitude can affect preexisting conditions in a variety of ways such as predisposing individuals to acute altitude illnesses, interfering with training regimens, or by exacerbating (or in some cases alleviating) symptoms from preexisting conditions. Examples of this variability include coronary artery disease, where the altitude outcome depends on how well the disease is controlled; sickle cell disease, where the individual should avoid altitude training altogether; and asthma, where the symptoms may actually improve with altitude (13). Individuals with conditions present prior to ascent should discuss their planned ascent with their health care team in order to prepare for altitude and make any necessary accommodations or restrictions (13).

Prevention and Treatment of Altitude Sickness

Altitude acclimatization is the best prevention for all altitude sickness. Minimizing initial sustained physical activity while maintaining adequate hydration and food intake will reduce susceptibility to altitude sickness during the acclimatization process, the duration of which will vary considerably based on individual health and behaviors. Medications may play a role in

reducing altitude sickness, even when acclimatization is possible, and should be prescribed by a licensed health care provider if used. The most common prophylaxis medication used is acetazolamide (*i.e.*, Diamox) and is used at a dosage of 125 mg twice a day to speed the elimination of urine bicarbonate, which hastens acclimatization (9). In situations requiring a rapid ascent above 3,500 m with expectation of immediate moderate-to-strenuous physical activity, acetazolamide should be augmented with 4 mg of dexamethasone (*i.e.*, Decadron, Hexadrol) every 12 hours orally as prescribed by a licensed health care provider (9).

When symptoms of altitude illness present, AMS may be treated with therapeutic use of acetazolamide 250 mg twice a day, which should be prescribed by a licensed health care provider. Headaches are most effectively treated with ibuprofen, which is additionally shown to support overall AMS symptom reduction (9). Oxygen therapy or portable hyperbaric chamber (*i.e.*, Gamow Bag) therapy can relieve AMS symptoms and the associated poor sleep. Dexamethasone is taken orally and may be used for prevention; it can also be delivered intravenously (IV) or intramuscularly (IM) for acute treatment of altitude illness (22). Dexamethasone may be used in combination with or in place of other treatments such as acetazolamide. Note that dexamethasone should not be used in pediatric individuals because of the systemic suppression effects on children (22). When moderate-to-severe symptoms and signs of an altitude-related sickness develop, the most effective treatment is to descend to a lower altitude. Descents of 300–1,000 m (985–3,280 ft) with an overnight stay can be effective in prevention and recovery of all altitude illnesses, and further descent may be necessary (9). If no medical provider is readily available for evaluation, the symptomatic individual should seek out medical evaluation as soon as possible. An important distinction is that while medications are considered first-line treatment for AMS (in conjunction with descent), the most critical treatment for individuals diagnosed with HACE or HAPE is urgent, rapid, controlled descent, oxygen therapy, and/or hyperbaric bag therapy (9).

Exercise Prescription

During the first few days at high altitudes, individuals should minimize their exercise and physical activity to reduce susceptibility to altitude illness. After this period, if their Ex R_x specifies a target heart rate (THR), the individual may utilize the same THR used at sea level. The personalized number of weekly training sessions and the duration of each session at altitude can remain similar to those used at sea level for a given individual. This approach reduces the risk of altitude illness and excessive physiologic strain. For example, at high altitudes, reduced speed, distance, or resistance will achieve the same THR as at lower altitudes. As altitude acclimatization develops, the THR will be achieved at progressively higher exercise intensity (8). Monitoring exercise HR provides a safe, easy, and objective means to quantify exercise intensity at altitude, as it does at sea level. Therefore, using any HR-based Ex R_x model

at altitude will provide a similar training stimulus to sea level as long as the quantity and duration of training sessions are maintained. Be mindful that for the same perceived effort, the individuals jogging or running pace will be reduced at altitude compared to sea level, independent of altitude acclimatization status.

Special Considerations

Active individuals who are acclimatized to altitude, adequately rested, nourished, and hydrated minimize the risk for developing altitude sickness and maximize their physical performance capabilities for the altitude to which they are acclimatized. The following factors should be considered to further minimize the negative effects of high altitude:

- Prepare for the environment: High-altitude regions are often associated with more daily extremes of temperature, humidity, wind, and solar radiation. Follow appropriate guidelines for hot ([23](#)) and cold environments ([24, 25](#)).
- Monitor the weather: A drop in barometric pressure with worsening weather can more directly exacerbate altitude changes with increased elevation. Sudden swings in weather can also radically change the environmental protection needed in accordance with the mentioned guidelines. Remember mountains may generate or change their own weather ([9](#)).
- Modify activity at high altitudes: Activity levels should be based on altitude acclimatization status, physical fitness, nutrition, sleep quality and quantity, age, exercise time and intensity, and availability of fluids. Longer or more frequent rest breaks and shortened activity times should be incorporated to facilitate rest and recovery. Longer duration activities are affected more by high altitude than shorter duration activities ([8, 9](#)).
- Hydration: Reasonably increase hydration beyond the normal expected amount for exertion. This is to offset fluid loss from the acclimatization process and the water loss through respiration due to the lower humidity with increased elevation. Overhydration with hypotonic fluids (*e.g.*, water), while rare, can pose a risk for life-threatening hyponatremia ([26, 27](#)).
- Clothing: Individual clothing and equipment need to provide protection over a greater range of temperature, wind conditions, and solar radiation.
- Education: The training of exercise professionals, coaches, and community emergency response teams enhances the reduction, recognition, and treatment of altitude-related illnesses.

Organizational Planning

When individuals exercise in high-altitude locations, physical fitness facilities and organizations should formulate a standardized management plan that includes the following procedures:

- Screening and surveillance of at-risk individuals.
- Using altitude acclimatization procedures to minimize the risk of altitude sickness and enhance physical performance.
- Consideration of the hazards of mountainous terrain when designing exercise programs and activities.
- Planning proper ascent profiles or staging methods to allow acclimatization.
- Awareness of the signs and symptoms of altitude illness.
- Development of organizational procedures for emergency medical care of altitude illnesses.
- Team physicians should consider maintaining a supply of oxygen and pharmaceuticals for preventing and treating altitude sickness.

EXERCISE IN COLD ENVIRONMENTS

Individuals can successfully exercise and work in many cold weather environments ([28](#)). Several factors, including the environment, clothing, body composition, anthropometry, health status, nutrition, age, sex, ethnicity, fitness, and exercise intensity, interact to determine the physiologic strain and injury risk during cold weather exercise ([24](#), [25](#), [28](#)). In most cases of cold weather exercise, the body is able to balance heat production with environmental heat loss, limiting the risk of cold injury ([25](#)). However, there are scenarios (*i.e.*, immersion, rain, prolonged exposure, and low-ambient temperature with wind) where whole body or local thermal balance cannot be maintained during exercise-related cold stress, which in turn contributes to hypothermia, frostbite, and diminished exercise capability and performance ([24](#), [25](#), [28](#), [29](#)). Furthermore, exercise-related cold stress may increase the risk of morbidity and mortality in at-risk populations such as those with cardiovascular disease ([30](#)) or asthmatic conditions, and inhalation of cold air may also exacerbate these conditions ([31](#)). Individuals with long QT syndrome type 1 are advised to avoid sports that involve diving into cold water as this is associated with increased risk of arrhythmias ([32](#)). It is prudent for the exercise professional to understand the different predisposing factors that may lead to a cold-related injury ([Table 7.2](#)).

TABLE 7.2 Predisposing Factors for Cold Injury

Decreased Heat Production	Increased Heat Loss	Impaired Thermoregulation	Conditions Aggravated by Cold Exposure	Others
Low energy Low caloric intake Inactivity Fatigue Fitness Endocrine Hypopituitarism Hypoadrenalism Hypoglycemia Diabetes Age Children Age >60 yr	Immersion Rain Wind Low body fat Anthropometry Age (young and old) Sex Ethnicity Skin changes Sunburn Dermatitis Psoriasis Open wound	Neuropathies Multiple sclerosis Parkinson disease Stroke Illicit drug abuse Alcohol abuse Vascular Raynaud syndrome Diabetes Peripheral artery disease Inadequate/wet clothing Constrictive clothing/boots	Asthma Exercise-induced bronchoconstriction Coronary artery disease Long QT-syndrome type 1 Raynaud disease Chronic obstructive pulmonary disease	Infection Renal failure Previous cold injury Vasoconstriction and reduced skin blood flow

Adapted from [\(16, 19\)](#).

Medical Considerations: Cold Injuries

Hypothermia develops when heat loss exceeds heat production, causing the core body temperature to drop below 35 °C (<95 °F) [\(31, 33\)](#). The environment, individual characteristics, and clothing all impact the development of hypothermia, with some specific hypothermia risk factors being immersion, rain, wet clothing, low body fat, older age (*i.e.*, ≥60 years), hypoglycemia [\(25\)](#), and accidental injury or exhaustion making individuals unable to exercise [\(24\)](#). Exercise in wet and windy environments markedly increases heat loss and can increase risk of hypothermia [\(31\)](#).

Frostbite occurs when body tissue temperature drops below 0 °C (32 °F) [\(31\)](#). Frostbite is most common in exposed tissue with poor perfusion (*i.e.*, hands, feet, head) [\(24\)](#). Contact

frostbite may occur by touching or gripping cold objects with bare skin, causing rapid heat loss (24). Risk factors for frostbite include inadequate clothing, age, sex, ethnic background, diseases affecting neural, vascular and metabolic function, and prior cold injury (24, 34). Exercise can be protective in preventing frostbite by elevating core and peripheral temperatures through increased heat production and cold-induced peripheral vasodilatation (35).

The principal cold stress determinants for frostbite are air temperature, wind speed, and wetness. Wind exacerbates heat loss by facilitating convective heat loss and reducing the insulative value of clothing (25, 28). The Wind Chill Temperature Index (WCT) (Figure 7.1) integrates wind speed and air temperature, estimating the cooling power of the environment. The WCT estimates the danger of cooling exposed skin of individuals walking at $1.3 \text{ m} \cdot \text{s}^{-1}$ (3 mi $\cdot \text{h}^{-1}$) (25, 28). The following needs to be considered concerning wind and the WCT:

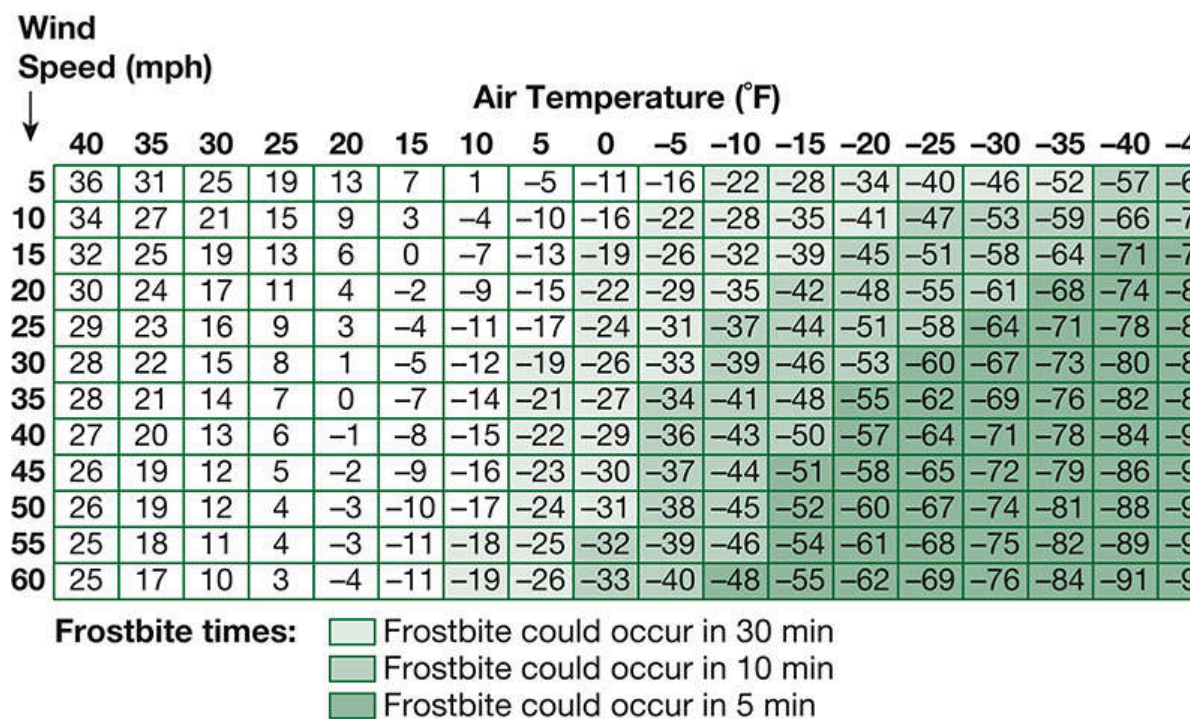


Figure 7.1 Wind Chill Temperature Index and frostbite times for exposed facial skin. (From NWS Windchill Chart [Internet]. Silver Spring (MD): National Oceanic and Atmospheric Administration, National Weather Service; 2009 [cited 2015 Aug 18]. Available from: <https://www.weather.gov/safety/winter>; Canada's Windchill Index: Windchill Hazards and What to Do [Internet]. Gatineau, Quebec (Canada): Environment Canada; 2011 [cited 2015 Aug 18]. Available from: <http://www.ec.gc.ca/meteo-weather/default.asp?lang=En&n=5FBF816A-1>)

- Wind does not cool an exposed object below the ambient temperature (25).
- Wind speeds from weather reports do not take into account self-made wind (e.g., running, skiing) (24). Athletes moving and creating self-made wind must be accounted for in WCT.

- The WCT presents the relative risk of frostbite and predicted times to freezing (see [Figure 7.1](#)) of exposed facial skin. Exposed fingers will freeze at warmer WCT than the cheek ([24](#)).
- Wet skin exposed to the wind cools faster. The ambient temperature used for the WCT table should be 10 °C lower than the actual ambient temperature ([24, 36](#)).
- Frostbite cannot occur if the air temperature is greater than 0 °C (32 °F) ([25](#)).
- Ambient temperature is never adequate to determine frostbite risk except for stationary personnel (*e.g.*, spectators). The risk of frostbite is less than 5% when the ambient temperature is above –15 °C (5 °F). Increased safety surveillance of exercisers is warranted when the WCT falls below –27 °C (–8 °F), when frostbite can occur in 30 minutes or less in exposed skin ([24](#)).

Nonfreezing cold injuries (NFCIs) typically occur when tissues are exposed to cold-wet conditions (0–15 °C [32–59 °F]) for prolonged periods of time ([24, 33](#)), causing tissue temperatures to fall below 15 °C (59 °F) ([24](#)). NFCI may occur due to immersion or by a damp environment inside boots or gloves, as often seen during heavy sweating. Most commonly, NFCI develops when wet socks and shoes are worn continuously over many days ([34](#)). The likelihood of NFCI in most sporting activities is low; however, severe NFCI are reported among ice skaters, cyclists, divers, long-distance polar rowers, hikers, and mountaineers ([24](#)). Intense vasoconstriction, ischemia, and/or reperfusion cause neurovascular tissue damage, and peripheral tissues are most commonly affected (*i.e.*, fingers, toes, nose, ears) ([24](#)). A mild form of NFCI is chilblains that occurs above freezing, following 1–5 hours of cold-wet exposure. Chilblains are small, swollen, itchy, erythematous papules, which may be tender and painful ([24, 37](#)).

Diagnosing NFCI involves observation of clinical symptoms over time as different and distinct stages emerge days to months after the initial injury ([24](#)). During cold exposure, the affected area is pale, cold, and numb. During rewarming, tissues remain pale and numb, and can become cyanotic or swell in severe cases ([24](#)). In some cases, the following hyperemic phase can last between 2 weeks and 3 months. Individuals may experience hot, red, and dry skin with some paresthesia and blistering in severe cases ([24](#)). The exposure temperature and time needed to develop NFCI is quite variable and most likely individual, with estimates ranging from 12 hours to 3–4 days in cold-wet environments ([24, 31, 33, 37](#)).

NFCIs can be prevented by keeping the body warm by remaining active ([24](#)). Appropriate waterproof, windproof and breathable clothing, and protection of hands and feet are important. Waterproof boots and gloves with breathable membranes preventing sweat accumulation during activity, and cooling during rest are required ([24](#)). Keeping feet and hands dry by continually changing socks and gloves throughout the day is highly recommended during long-term exposure to cold-wet environments ([24, 37](#)). In addition, restrictive footwear should be

avoided as reduced blood flow increases cooling rates. With great interindividual NFI susceptibility, individualized risk mitigation is recommended (24).

Cardiac and Respiratory Considerations

Exercise-induced bronchoconstriction (EIB) is a transient narrowing of the airways during or after vigorous exercise. Exercise in cold environments increase the relative risk of EIB in both asthmatic (EIBa) and nonasthmatic individuals (EIB) and can reduce athletic performance (38, 39). Appropriate medications should be provided to cold weather athletes with EIB or EIBa; however, no pharmaceutical therapy completely resolves EIB (38, 39). The World Anti-Doping Agency (WADA) allows competing athletes to use some inhaled corticosteroids and β_2 -agonists, as long as the use is documented to be therapeutic. Currently, the only inhaled β_2 -agonists permitted by the WADA are dose-limited inhaled rapid-acting salbutamol and long-acting formoterol and salmeterol. Combinations of β_2 -agonists or use of oral β_2 -agonists are, with a few exceptions, prohibited (38). Athletes, coaches, and medical staff should refer to the WADA world antidoping code to adhere to antidoping regulations (38, 40).

The negative effects of cold air inhalation in athletes with known EIB or EIBa can be limited by the use of face masks, specific warm-up routines, and continual movement on sidelines, when applicable (38, 39). Masks can promote humidification and warm inspired air and have been shown in small studies to reduce cold weather EIB (41, 42). The reduction in EIB symptoms appears to persist when a mask is utilized only in warm-up and rest periods before the main sport activity (43). Athletes with known EIB/EIBa may benefit from an interval-based high-intensity preexercise warm-up routine in order to trigger a refractory period that reduces hyperresponsiveness. The warm-up strategy should include some exercise near peak oxygen consumption or HR_{max} (*i.e.*, sprints) approximately 20–30 minutes before the event or competition (44, 45). The duration of this protective effect from an induced refractory period should be considered short-term (45).

Exercising in cold water deserves special mention due to two conditions unique to this environment. Swimming-induced pulmonary edema (SIPE) is a rare cause of acute breathlessness in swimmers. SIPE occurs during swimming in the absence of aspiration when fluid accumulates in the lungs causing acute dyspnea and hemoptysis (46). SIPE affects roughly 0.01%–0.75% of competitors in open water-based events (46, 47), with 8.6 times higher odds in females compared with males and 12.8 times higher odds in an older, compared to a younger age group (47). SIPE may occur in tropical waters, but there appears to be a higher incidence in cold water, presumably due to increased central venous pooling and subsequent increase in cardiac preload (46). Of note, recurrences are unpredictable yet fairly common, with recurrence rates between 13% and 30% (46, 48). Swimming in cold water also increases the risk for cardiac arrhythmias, which are seen in about 2% of cold water immersions. Cold water immersion

activates two powerful reflexes in the human body. The diving response when the face is immersed in cold water stimulates the parasympathetic system causing bradycardia, and the cold shock response when the skin is immersed in cold water stimulates the sympathetic system causing tachycardia ([49](#)). Under most normal conditions, these two reflexes coexist without difficulty; however, with breath holding and the addition of other predisposing factors (*e.g.*, ischemic heart disease, channelopathies), this “autonomic conflict” may pose a risk for more serious arrhythmias ([49](#)). While widespread screening for these conditions may not be warranted, it would be prudent to counsel those who have experienced SIPE of the high rate of recurrence. Likewise, it is critical for those providing medical coverage to open water events to be aware of the elevated risk of SIPE and arrhythmias with cold water immersion.

Clothing Considerations

Cold weather clothing protects against hypothermia and frostbite by reducing convective and evaporative heat loss through the insulation provided by wind- and waterproofing and air trapping between clothing layers ([24](#), [25](#)). Typical cold weather clothing consists of three layers: an inner layer in contact with skin that wicks moisture to the outer layers, a middle layer that provides the primary insulation, and an outer layer designed to allow moisture transfer to the air while repelling wind and rain. Recommendations for cold weather clothing include the following considerations ([24](#), [25](#), [35](#)):

- Adjust clothing insulation to minimize sweating.
- Use clothing vents to reduce sweat accumulation.
- Avoid an outer layer unless rainy or very windy.
- Reduce clothing insulation as exercise intensity increases.
- Individualize clothing to physiological response (*e.g.*, sweating) in group exercise settings.
- Avoid constrictive clothing and footwear to minimize blood flow restriction and risk of frostbite.
- Consider using chemical or electric hand/foot warmers to maintain peripheral warmth (avoid direct skin contact or placement constricting blood flow).
- If body heat cannot be maintained during exercise, availability of extra dry and waterproof clothing and buddy supervision is recommended.
- Emollients do not protect exposed skin against cold injuries and may increase risk of frostbite.

Exercise Prescription

For athletes at higher risk for injury with exercise in cold, the following should be considered:

- Individuals with known cardiovascular disease should use caution when exercising in cold environments. Cold exposure increases incidence of cardiovascular events and angina, likely due to increase in arterial pressure, total peripheral resistance, altered myocardial functioning, “autonomic conflict” due to sympathetic and parasympathetic coactivation, and cardiac work/myocardial oxygen demand ([28](#), [30](#)).
- Individuals with long QT syndrome type 1 are advised to avoid sports that involve diving into cold water as this is associated with increased risk of arrhythmias ([32](#)).
- Endocrine conditions should be well controlled prior to cold weather athletics ([37](#)). Hypoglycemia and other endocrine abnormalities can impair the shivering response ([31](#)).
- Athletes should avoid vasoconstrictive medications, smoking, drugs, alcohol, and central nervous system depressant medications. Vasoconstrictive medications and cigarette smoking can impair peripheral circulation increasing risk of cold injury ([37](#)).
- Individuals with EIBa or EIB should follow the WADA world antidoping code and have appropriate documented medical management prior to and during exercise ([38](#)).
- Preexercise warm-up and wearing face masks to warm and humidify inspiratory air can protect against bronchoconstriction. High-intensity intervals and variable intensity during warm-up can provide protection against bronchoconstriction between 2 and 4 hours ([38](#), [39](#), [43](#), [45](#)).

EXERCISE IN HOT ENVIRONMENTS

As homeotherms, humans must maintain a fragile balance of heat dissipation and heat retention in order to stay within a safe and healthy body temperature range. The body's mechanisms for thermoregulation occur through a balance of heat production and heat loss. Significant shifts in this fragile balance can result in heat illness (50). Muscular contractions produce metabolic heat that is transferred from the active muscles to the blood stream, which in turn raises the body's core temperature (50). Subsequent body temperature elevations elicit heat loss responses of increased skin blood flow and increased sweat secretion so that heat can be dissipated to the environment via evaporation (51). As a result of elevated skin blood flow, the cardiovascular system plays an essential role in temperature regulation (51). Heat exchange between skin and environment via sweating and dry heat exchange is governed by biophysical properties dictated by surrounding temperature, humidity and air motion, sky and ground radiation, and clothing (52). There are two main mechanisms by which a body may gain heat during exercise, internal (e.g., working muscles producing heat) or external (e.g., high ambient temperature). There are also mechanisms by which cooling may occur (e.g., conduction or evaporation). The **heat balance equation** describes this balance (53):

$$S=M-(\pm Work)-E\pm R\pm C\pm K$$

S represents overall heat storage, *M* represents metabolic heat production, *E* represents evaporation, *R* represents radiation, *C* represents convection, and *K* represents conduction. Heat gain is indicated by a positive value, whereas heat loss is indicated by a negative value. Options for heat loss include the *E*, *R*, *C*, and *K* pathways, whereas options for heat gain include the *M*, *E*, *R*, *C*, and *K* pathways. Note that for an exercising individual, *M* will always be positive and contribute to heat gain. Radiation is heat gained or lost due to the energy transferred in the infrared electromagnetic energy spectrum independent of air temperature. Radiation, such as that from the sun, depends on the gradient that exists between the surfaces receiving and producing the heat. Convection is heat loss or gain through a transfer layer of either air or water. Wind chill is based on this principle. High winds increase the heat loss and the perceived feeling of "cold" in the air. Water has an even greater ability to transfer heat. It is approximately 40 times more effective than air at removing or releasing heat to or from the body. This is an extremely important consideration when determining course of treatment for an EHS victim. Evaporation occurs as the result of the latent heat of vaporization of water from the skin. The evaporation of water requires energy in the form of heat, which is called the latent heat of vaporization. Given this input, the heat from the body is transferred to sweat. When this liquid is evaporated and converted to a gaseous state, the heat is transferred and removed. This is the main method of cooling that the body relies on in high ambient temperatures. However, high

humidity hinders the ability of the body to cool itself in this fashion. Humidity becomes a large factor because once ambient temperature rises above skin temperature, cooling by radiation and convection is lost and now contributes to heat gain as opposed to loss. This leaves evaporation as the primary means of cooling. Unfortunately, the rate of evaporative cooling decreases as humidity rises. Conduction is the transfer of heat from one object to another that is in direct contact with it—for example, the heat lost when transferred from the body to clothes. This type of cooling makes up a small amount of the total heat exchange (~1%) and does not contribute greatly to cooling of the body.

When the amount of metabolic heat exceeds heat loss, hyperthermia (*i.e.*, elevated internal body temperature) may develop (50). Sweat that drips from the body or clothing provides no cooling benefit. In fact, if secreted sweat drips from the body and is not evaporated, a higher sweating rate will be needed to achieve the evaporative cooling requirements (51). Sweat losses vary widely among individuals and depend on the amount and intensity of exercise, clothing, protective equipment, and environmental conditions (*e.g.*, humidity). Other factors such as hydration state, body composition, and level of aerobic fitness can alter sweat rates and ultimately fluid needs (52). For example, heat acclimatization results in higher and more sustained sweating rates, whereas aerobic exercise training has a modest effect on enhancing sweating rate responses (51). When properly controlled and compared, the differences in thermoregulation (*e.g.*, sweating) between males and females are minimal (54, 55).

During exercise-induced heat stress, dehydration increases physiologic strain as measured by core temperature, HR, and perceived exertion responses (56-58). The greater the body water deficit, the greater the increase in physiologic strain for a given exercise task (59, 60). Dehydration greater than 2% body mass loss can exacerbate core temperature elevations during exercise in temperate (60) as well as in hot environments (59-62), with typical increases of about 0.2 °C (0.4 °F) with each 1% of dehydration (58). The greater heat storage with dehydration is associated with a proportionate decrease in heat loss; thus, decreased sweating rate (*i.e.*, evaporative heat loss) and decreased cutaneous blood flow (*i.e.*, dry heat loss) are responsible for greater heat storage observed during exercise when hypohydrated (50). Both aerobic and anaerobic exercise performance suffer with dehydration as well as exercise in the heat. However, anaerobic performance tends to require greater extremes of dehydration and environmental heat to elicit deficits (58, 63, 64). This is true whether individuals commence exercise in a dehydrated state or accumulate fluid loss during exercise (65).

The critical water deficit (*i.e.*, >2% body mass for most individuals) and magnitude of performance decrement are likely related to environmental temperature, exercise task, and the individual's unique biological characteristics (*e.g.*, tolerance to dehydration) (65). Acute dehydration impairs endurance performance regardless of whole-body hyperthermia or environmental temperature, and endurance capacity (*i.e.*, time to exhaustion) is reduced more in a hot environment than in a temperate or cold one (66).

Mitigating Risk

Individuals have varying sweat rates, and as such, fluid needs for individuals performing similar tasks under identical conditions can be different. Sweat rates within an individual may also need to be periodically reassessed to account for changes from influencing factors, such as heat acclimatization status, fitness, exercise intensity, or exercise environmental conditions.

Determining sweat rate ($\text{L} \cdot \text{h}^{-1}$) by measuring body weight before and after exercise provides a fluid replacement guide. Active individuals should drink 0.5 L (1 pint) of fluid for each pound of body weight lost. Meals can help stimulate thirst resulting in restoration of fluid balance. Snack breaks during longer training sessions can help replenish fluids and be important in replacing sodium and other electrolytes ([67](#)). There is presently no scientific consensus for how to best assess hydration status in a field setting ([65](#)). However, in most field settings, the additive use of first morning body mass measurements in combination with some measure of first morning urine concentration and gross thirst perception provides a simple and inexpensive way to dichotomize euhydration from gross dehydration resulting from sweat loss and poor fluid intakes ([Figure 7.2](#)) ([27](#), [54](#)). When assessing first morning urine, a paler color indicates adequate hydration; a darker yellow/brown color indicates a greater degree of dehydration ([68](#)). [Box 7.1](#) provides recommendations for how to assess sweat rate ([27](#)).

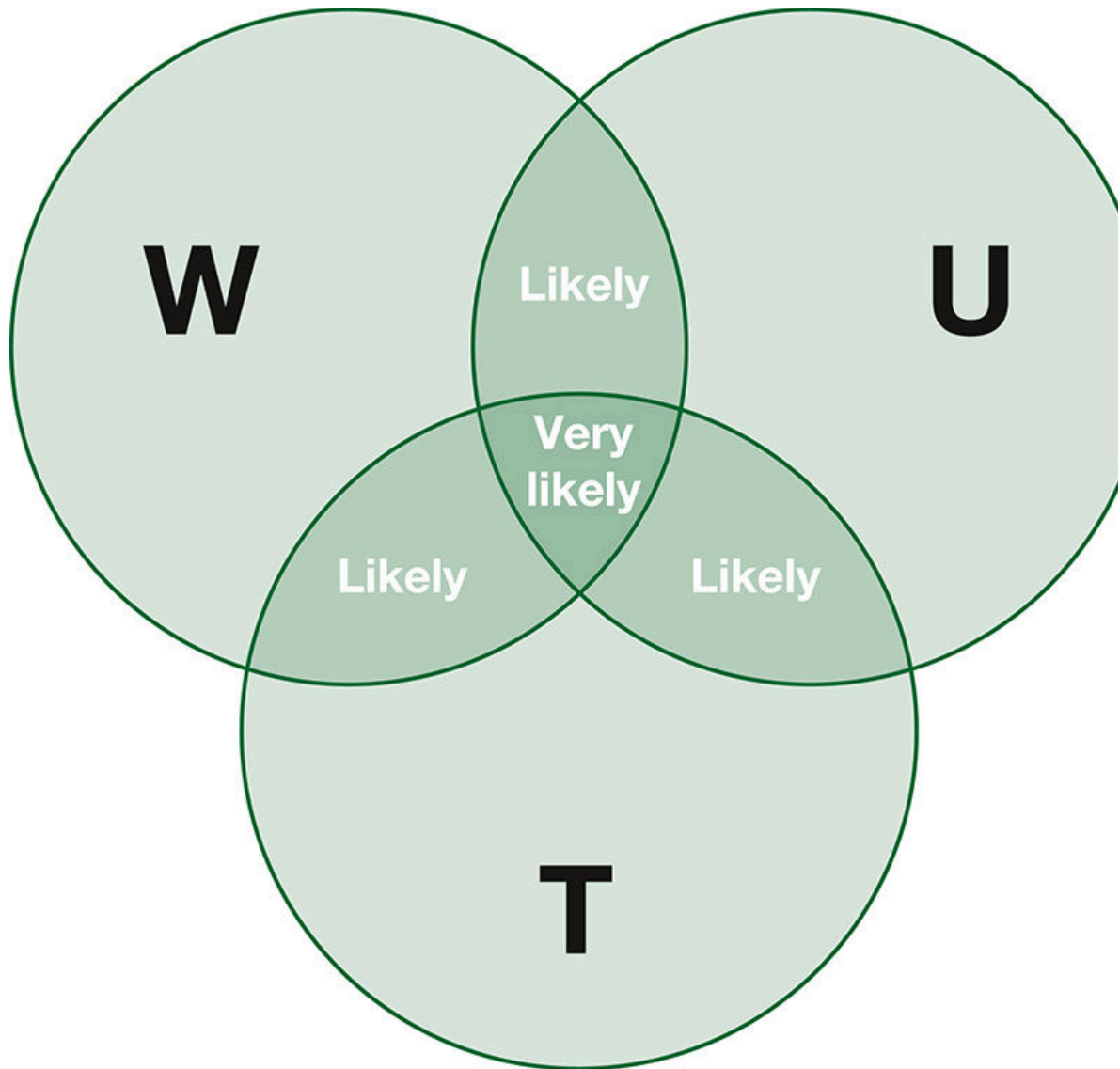


Figure 7.2 W stands for “weight.” U stands for “urine.” T stands for “thirst.” When two or more simple markers are present, dehydration is likely. If all three markers are present, dehydration is very likely. Reprinted with permission from ([70](#)).

Box 7.1 Calculating Sweat Rate

Variable	Equation	Notes
A. Nude bodyweight preexercise (kg)		Void your bladder before completing the preexercise nude bodyweight. Do not consume water until preexercise water bottle weights have been recorded.
B. Water bottle(s) weight preexercise (kg)		Weigh the water bottle(s) before exercise begins.
C. Urine jug weight preexercise (kg)		Any urine loss during exercise should be collected to ensure accuracy. Weigh an empty urine collection jug before urinating during exercise. If you do not have the means to collect urine, avoid urinating during exercise if possible.
D. Nude bodyweight postexercise (kg)		Ensure to dry off all sweat before collecting postexercise weight. If unable to collect urine during exercise, urination should occur <i>after</i> this value is collected to ensure accuracy. Do not consume water after recording postexercise bodyweight until postexercise water bottle weights have been recorded.
E. Water bottle(s) weight postexercise (kg)		Weigh the water bottle(s) after exercise.
F. Urine jug weight postexercise (kg)		This is only necessary if urine was collected during exercise.
G. Change in bodyweight (kg)	$A - D$	
H. Volume of fluid consumed (kg)	$B - E$	
I. Urine volume (kg)	$C - F$	This is only necessary if urine was collected during exercise.
J. Sweat loss (L)	$G + (H - I)$	If you did not urinate during exercise, use $G + H$.
K. Exercise time (h)		Record how long your exercise session was in hours by taking the total amount of time in minutes divided by 60. For example, if you exercised for 1 h and 30 min, you record it as 1.5 h. If you exercised for 45 min, you record it as 0.75 h.
L. Sweat rate (L/h)	J/K	

Using kilograms (kg) for all weights allows for simplification of this equation as 1 kg = ~1 liter (L). To determine sweat rate in milliliters per minute ($\text{mL} \cdot \text{min}^{-1}$), divide J by 1,000 to find sweat loss in mL and then divide sweat loss in mL by total exercise time in minutes. Exercise intensity, hydration status, environmental conditions, heat acclimatization status, and the amount of clothing/equipment worn can influence sweat rate. It is important to calculate sweat rate under the various circumstances listed.

Overdrinking hypotonic fluid (*e.g.*, water) can lead to exercise-associated hyponatremia, a state of lower than normal blood sodium concentration (typically $<135 \text{ mEq} \cdot \text{L}^{-1}$) accompanied by nausea, vomiting, headache, extremity edema, and severe symptoms such as pulmonary edema and altered cognitive status ([27](#)). Hyponatremia tends to be more common in long duration physical activities and is precipitated by consumption of hypotonic fluid in excess of sweat losses (typified by body mass gains) ([27](#), [69](#)). When participating in exercise events that result in many hours of continuous or near-continuous sweating, hyponatremia can be prevented by practices such as having an individualized hydration plan, not drinking in excess of sweat rate, and consuming salt-containing fluids or foods ([27](#)). For additional information, see the American College of Sports Medicine (ACSM) position stand on fluid replacement ([26](#), [27](#)).

Other risk factors largely associated with heat illness are exercise intensity unmatched for physical fitness, hot environments or lack of weather-based exercise modifications, use of protective equipment, and lack of heat acclimatization ([71-74](#)). Monitoring weather conditions as measured by the Wet Bulb Globe Temperature (WBGT) and applying regionally based exercise modifications have had success in reducing EHI incidence by as much as 79% ([75](#)), and adoption of heat acclimatization policies have demonstrated a 55% reduction ([73](#)). In combination with the use of the progressive introduction of exercise, these tools have great potential to reduce risk of EHI.

Medical Considerations: Exertional Heat Illnesses

Heat illnesses vary by severity ranging from mild (*i.e.*, muscle cramps) to life-threatening exertional heat stroke and are described in [Table 7.3](#). The term “heat illness” may be misleading as many of these do not necessitate hot environments to occur and can result from a myriad of risk factors. Dehydration may be either a direct (*i.e.*, exercise-associated muscle cramps [EAMCs] and heat exhaustion) ([62](#)) or indirect (*i.e.*, exertional heat stroke) ([76](#)) factor in heat illnesses.

TABLE 7.3 A Comparison of the Signs and Symptoms of Illnesses That Occur in Hot Environments (80)

Disorder	Prominent Signs and Symptoms	Mental Status Changes	Core Temperature Elevation
Exertional heatstroke	Disorientation, dizziness, irrational behavior, apathy, headache, nausea, vomiting, hyperventilation, wet skin	Marked (disoriented, unresponsive)	Marked (>40 °C [>104 °F])
Exertional heat exhaustion	Low blood pressure, elevated heart rate and respiratory rates, skin is wet and pale, headache, weakness, dizziness, decreased muscle coordination, chills, nausea, vomiting, diarrhea	Little or none, agitated	None to moderate (37–40 °C [98.6–104 °F])
Heat syncope	Heart rate and breathing rates are slow; skin is pale; individual may experience sensations of weakness, tunnel vision, vertigo, or nausea before syncope	Brief fainting episode	Little or none
Exertional heat cramps	Begins as feeble, localized, wandering spasms that may progress to debilitating cramps	None	Moderate (37–40 °C [98.6–104 °F])

EAMCs are painful involuntary muscle contractions or spasms most often in the abdomen, arms, or legs that may occur during or after strenuous activity (77). The term *heat cramps* is often used interchangeably but technically is an inappropriate term because EAMCs present during exercise in hot and cold environments unassociated with elevated core temperatures (23). Some controversy exists regarding the etiology of EAMCs; the cause is likely multifactorial and possibly unique to each athlete (77). Evidence suggests that EAMCs may be more related to muscle fatigue and neuronal excitability compared to hydration status or electrolyte concentrations (78). However, water loss and significant sweat sodium losses have been proposed as contributing factors and may play a role in cramping in individuals identified as “heavy sweaters” (*e.g.*, >2 L · h⁻¹) or those who lose appreciable amounts of body fluid and sodium. One treatment or prevention strategy is not likely to work for every individual (23). However, EAMCs have been shown to respond to rest, prolonged stretching, and dietary sodium chloride (*i.e.*, 1/8–1/4 tsp of table salt or one to two salt tablets added to 300–500 mL of fluid, bullion broth, or salty snacks). There is limited anecdotal evidence for IV fluid use. Oral hydration remains the best practice for most athletes (79).

Heat syncope, or orthostatic dizziness, is a temporary circulatory failure caused by the pooling of blood in the peripheral veins, particularly of the lower extremities. Heat syncope tends to occur more often among physically unfit, sedentary, and nonacclimatized individuals. Heat syncope is caused by standing erect for a long period or at the cessation of strenuous, prolonged, upright exercise, because maximal cutaneous vessel dilation results in a decline of

blood pressure (BP) and insufficient oxygen delivery to the brain. Symptoms range from light-headedness to loss of consciousness; however, recovery is rapid once individuals lay supine and elevate their legs to restore adequate perfusion. Complete recovery of stable BP and HR may take a few hours ([23](#)). See the ACSM and/or National Athletic Trainer's Association position stand on heat illness during exercise for additional information ([23](#), [26](#)).

Heat exhaustion is the most common form of heat illness ([81](#), [82](#)). Heat exhaustion is the incapacity to perform exercise in the heat, due to a combination of factors to include cardiovascular insufficiency, hypotension, energy depletion, and central fatigue ([83](#)). Heat exhaustion manifests with an elevated core body temperature (but usually $<40.5^{\circ}\text{C}$) when the body cannot sustain the level of blood volume needed to support skin blood flow for thermoregulation and blood flow for metabolic requirements of exercise ([67](#)). Heat exhaustion is characterized by prominent fatigue and progressive weakness without end-organ damage (*e.g.*, renal insufficiency, rhabdomyolysis, altered mental status, or liver injury) ([23](#)). Oral fluids are preferred for rehydration in individuals who are conscious, able to swallow, and not losing fluid (*i.e.*, vomiting and diarrhea). IV fluid administration facilitates recovery in those unable to ingest oral fluids or who have severe dehydration ([79](#)). With elite athlete care both in and out of competition, providers are reminded that the use of IV fluid administration outside of a hospital setting is strictly regulated by some governing bodies (*e.g.*, WADA, United States Anti-Doping Agency) and may require a therapeutic use exemption. In addition to fluid replacement, moving the individual to a cool and/or shaded environment, elevating the legs, and providing light cooling (*e.g.*, cold-wet towels, cold water dousing) can further aid in the recovery from heat exhaustion.

Exertional heat stroke is a life-threatening condition caused by hyperthermia and is characterized by elevated body temperature ($>40.5^{\circ}\text{C}$ or 105°F) ([23](#), [50](#)), profound central nervous system dysfunction, and multiple organ system failure that can result in delirium, convulsions, or coma. Exertional heat stroke should not be confused with classical heat stroke. Exertional heat stroke typically affects healthy individuals during physical activity, while classical heat stroke occurs in individuals with underlying health issues due to passive heating of the body. The greatest risk for exertional heat stroke exists during very-high-intensity exercise of short duration or prolonged exercise when the WBGT exceeds 28°C (82°F) ([72](#), [84](#), [85](#)). Exertional heat stroke requires immediate (within 30 minutes) and effective whole-body cooling with cold water and ice water immersion therapy ([23](#)). Cooling should continue until rectal temperature reaches $38.9\text{--}39.4^{\circ}\text{C}$ ($102\text{--}103^{\circ}\text{F}$). Removal prior to complete cooling, or using inadequate cooling modalities, places the athlete at risk for complications or death ([86](#)). Inadequate physical fitness, excess adiposity, improper clothing, protective pads, incomplete heat acclimatization, recent or current illness, punishment workouts, and medications or dietary supplements that contain stimulants (*e.g.*, ephedra; synephrine) also increase the risk of exertional heat stroke ([23](#), [70](#), [72](#), [87](#)).

Exercise Prescription

Exercise professionals may use evidence-based, regionally specific WBGT standards to define levels at which the risk of heat injury is increased. Exercise may be performed, however, if stepwise modifications addressing exercise duration, number of breaks, duration of breaks, and removal of equipment is utilized ([84](#), [85](#)).

If an $Ex R_x$ specifies a THR, it will be achieved at a lower absolute workload when exercising in a warm/hot versus a cooler environment. For example, in hot or humid weather, an individual will achieve their THR with a reduced running speed. Reducing one's workload to maintain the same THR in the heat will help to reduce the risk of heat illness during acclimatization or in hot environments ([88](#)).

Special Considerations

Adults and children who are adequately rested, nourished, hydrated, provided appropriate weather-based modifications, and acclimatized to heat are at less risk for exertional heat illnesses. However, when individuals exercise in fitness or recreational settings while in hot/humid conditions, staff (coaches, personal trainers, educators, etc.) should consult with a medical professional (e.g., ATC, physician) to formulate a standardized heat stress management plan that incorporates the following considerations to minimize the effects of hyperthermia and dehydration, along with the questions in [Box 7.2](#) ([23](#), [54](#)):

- Create a heat stress prevention and management plan, ideally in consultation, coordination, and communication with local medical professionals.
- Monitor the environment: Use the WBGT index to determine appropriate action and based on established criteria for modifying or canceling exercise/events.
- When performing multiple exercise sessions during the day, allow at least 3 hours, and preferably 6 hours, of recovery and rehydration time between sessions.
- Modify activity in extreme environments: Enable access to ample fluid and bathroom facilities, provide longer and/or more rest breaks to facilitate heat dissipation, remove equipment, and shorten or delay playing times. Perform exercise at times of the day when conditions will be cooler compared to midday (early morning, later evening). Children and older adults should modify activities in conditions of high WBGT values.
- Individualize fluid intake that (a) matches the volume of fluid consumed to the volume of sweat lost and (b) limits body weight change to less than 2% of body weight. Do not restrict or limit fluid intake during activity.
- Screen and monitor at-risk individuals and establish specific emergency procedures with local medical professionals.

- Consider heat acclimatization status, physical fitness, sleep deprivation, recent or current illness (especially vomiting and/or diarrhea), age of individuals; intensity, time/duration, WBGT; and availability of fluids.
- Heat acclimatization adaptations include a 10%–20% increase in plasma volume, which allows the body to store more heat with minimal impact to its core temperature. Acclimatization results in the earlier onset of sweating, reduced sodium loss in sweat, decreased skin blood flow, and increased synthesis of heat shock proteins to prevent cellular damage. Additional adaptations include decreased rectal temperature, HR, and RPE; increased exercise tolerance time; and an increased sweating rate.
- Acclimatization results in the following: (a) improved heat transfer from the body's core to the external environment, (b) improved cardiovascular function, (c) more effective sweating, and (d) improved exercise performance and heat tolerance. Seasonal acclimatization will occur gradually during late spring and early summer months with sedentary exposure to the heat. However, this process can be facilitated with a structured program of moderate exercise in the heat across 10–14 days to stimulate adaptations to warmer ambient temperatures.
- Equipment: Athletes should follow the gradual introduction of equipment as outlined in heat acclimatization guidelines. During and following heat acclimatization, WBGT-based modifications for equipment use should be applied as well.
- Education: The training of individuals, fitness specialists, coaches, and community emergency response teams enhances the reduction, recognition, and treatment of heat-related illness. Such programs should emphasize the importance of recognizing signs/symptoms of heat illness, modifying exercise in extreme heat, being hydrated, fed, rested, and acclimatized to heat. Educating individuals about dehydration, assessing hydration state, and using a fluid replacement program can help maintain hydration.

Box 7.2 Questions to Evaluate Readiness to Exercise in a Hot Environment ([72](#))

Adults should ask the following questions to evaluate readiness to exercise in a hot environment. Corrective action should be taken if any question is answered “no.”

- Have I developed a plan to avoid dehydration and hyperthermia?
- Have I acclimatized by gradually increasing exercise duration and intensity for 10–14 days?
- Do I limit intense exercise to the cooler hours of the day (early morning)?
- Do I avoid lengthy warm-up periods on hot, humid days?
- When training outdoors, do I know where fluids are available, or do I carry water bottles in a belt or a backpack?
- Do I know my sweat rate and the amount of fluid that I should drink to replace body weight loss?
- Was my body weight this morning within 1% of my average body weight?
- Is my 24-hour urine volume plentiful?
- Is my urine color “pale yellow” or “straw colored”?
- When heat and humidity are high, do I reduce my expectations, my exercise pace, the distance, and/or duration of my workout or race?
- Do I wear loose-fitting, porous, lightweight clothing?
- Do I know the signs and symptoms of heat exhaustion, exertional heatstroke, heat syncope, and heat cramps (see [Table 7.3](#))?
- Do I exercise with a partner and provide feedback about his or her physical appearance?
- Do I consume adequate salt in my diet?
- Do I avoid or reduce exercise in the heat if I experience sleep loss, infectious illness, fever, diarrhea, vomiting, carbohydrate depletion, some medications, alcohol, or drug abuse?

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CHAPTER 8

Exercise Prescription for Individuals With Cardiovascular and Pulmonary Diseases

INTRODUCTION

In this chapter, we discuss exercise testing and prescription recommendations for individuals with various cardiovascular and pulmonary diseases.

CORONARY HEART DISEASE

In the United States, the estimated prevalence of adults (≥ 20 years old) who have been diagnosed with coronary heart disease (CHD) is 7.1% and is higher in males (8.7%) than females (5.8%) ([1](#)). Among men, the annual incidence of myocardial infarction (MI) is 25 per 100,000 population in those 33–44 years old, increases with age up to 155 per 100,000 population in those 65–74 years old, and is lower among those greater than 74 years old. Annual incidence is lower in females and increases across all age groups (35–44 years old = 10 per 100,000; 75–84 years old = 125 per 100,000) ([1](#)). Ischemic heart disease is the leading cause of death in all regions of the world except for Sub-Saharan Africa (stroke is the leading cause in females) and South Asia (stroke is the leading cause in males and females) ([2](#)).

Exercise Testing

Guidelines for the conduct and interpretation of exercise testing in patients with CHD are consistent with those discussed in [Chapter 4](#). In asymptomatic patients with prior coronary revascularization or MI, the American College of Cardiology (ACC) and the American Heart Association (AHA) categorize an exercise stress test with ECG only as “may be appropriate” to “rarely appropriate” ([3](#)). The exception is patients with CHD prior to initiating an unsupervised exercise program and those participating in cardiac rehabilitation (CR) ([3](#)). Both of these indications are considered “appropriate” use ([3](#)).

Exercise Prescription

Results from meta-analyses of randomized, controlled trials in patients with CHD have consistently supported the benefits of aerobic exercise training. In a 2023 meta-analysis of 85 trials and 23,430 subjects with CHD, Dibben et al. ([4](#))

showed that exercise training is associated with a 26% lower risk of cardiovascular-related mortality, a 23% lower risk of all-cause hospitalization, and an 18% lower risk of MI. In addition, exercise training in patients with CHD increases exercise capacity, improves quality of life, reduces angina pectoris, and facilitates return to work ([5](#), [6](#)). As a result, regular exercise training (aerobic and strength) is a class 1 (highest level) recommendation from the AHA and the ACC ([7](#)). CR is the primary model of clinic-based, supervised exercise training programs for individuals with CHD (see *Cardiac Rehabilitation* below).

The exercise prescription (Ex Rx) for apparently healthy individuals ([8](#)) (see [Chapter 5](#)) is largely appropriate for individuals with CHD, with a few special considerations.

Special Considerations

General Recommendations

The physical activity (PA) goal for all adults to accumulate at least 150 min · wk⁻¹ of moderate (≥3 to <6 metabolic equivalents of task [METs]) intensity and/or 75 min · wk⁻¹ of vigorous (≥6 METs) intensity PA ([9](#)) might not initially be realistic for many patients with CHD due to low exercise capacity. The majority of patients with CHD in CR are not able to exercise at workloads ≥3 METs ([10](#)), which emphasizes the importance of aerobic exercise training to improve cardiorespiratory fitness. Exercise sessions should include a warm-up and cool-down. Exercise programs should focus on aerobic exercise training utilizing exercise modes that engage large muscle groups. Additional benefits can be seen with the addition of resistance training and should be a part of each patient's exercise plan ([11](#)).

Symptoms and Medications

Individuals with CHD should be educated on symptoms of exercise intolerance such as excessive fatigue, dizziness or lightheadedness, angina, and anginal

equivalents ([12](#)). People with CHD should take their prescribed medications at their usual time as recommended by their health care providers. Those on a beta-adrenergic blocking agent (β -blocker) will have lower resting heart rate (HR) and an attenuated HR response during exercise ([13](#)). Individuals who are prescribed a target heart rate (THR) determined from a sign and symptom-limited maximal exercise test and whose β -blocker dose was altered after the test may benefit from a new exercise test ([12](#)). Alternately, a new THR can be estimated by measuring the individual's HR during exercise at the same workload performed before the change in dose and establishing a range that is ± 5 beats of the new HR. It should be noted that while this is based on physiological principles, this method has not been tested in a research study.

Target Heart Rate Range

When possible, an exercise training THR should be established based on a sign and symptom-limited maximal exercise test after the individual's coronary event and after their medications have been optimized ([6](#), [14](#), [15](#)). Estimates of maximal HR, such as $220 - \text{age}$ should NOT be used to determine an exercise training THR in patients with CHD ([13](#), [16](#)).

Myocardial Ischemia

Patients should not exercise at an intensity that causes signs or symptoms of myocardial ischemia (*e.g.*, angina, anginal equivalents, or significant ST segment depression) or symptoms of poor perfusion (*e.g.*, lightheadedness, dizziness). Until determined otherwise, chest discomfort reported by a patient with CHD should be considered angina pectoris, and exercise intensity should be reduced to avoid it. The upper limit of an exercise training THR should be set at 10 beats below the HR at the ischemic threshold ([6](#), [15](#)). A prolonged warm-up may help reduce the occurrence of exercise-induced angina. Individuals who repeatedly experience angina may benefit from prescribed nitroglycerin before exercise.

Resistance Training

Resistance training is safe and an important adjunct to aerobic exercise training in those with CHD ([11](#)). It is prudent to consider adding resistance training after adherence to 2–4 weeks of regular aerobic exercise training ([12](#)).

Contraindications to resistance training reported by the AHA ([11](#)) are shown in [Box 8.1](#). As it relates to relative contraindications, consultation with a physician might be assumed for those individuals who are referred to outpatient CR.

Box 8.1 Contraindications to Resistance Training in Patients With CHD From the American Heart Association ([11](#))

Absolute

- Unstable CHD
- Decompensated heart failure
- Uncontrolled hypertension
- Severe pulmonary hypertension (mean pulmonary arterial pressure >55 mm Hg)
- Severe and symptomatic aortic stenosis
- Acute myocarditis, endocarditis, or pericarditis
- Uncontrolled hypertension (>180/110 mm Hg)
- Aortic dissection
- Marfan syndrome
- High-intensity RT (80%–100% of 1-RM) in patients with active proliferative retinopathy or moderate or worse nonproliferative diabetic retinopathy

Relative (consult a physician before participation)

- Major risk factors for CHD
- Diabetes at any age
- Uncontrolled hypertension (>160/>100 mm Hg)
- Low functional capacity (<4 METs)
- Musculoskeletal limitations
- Individuals who have implanted pacemakers or defibrillators

CHD, coronary heart disease; 1-RM, one repetition maximum; MET, metabolic equivalent of task.

High-Intensity Interval Training

High-intensity interval training (HIIT) may be beneficial for individuals with CHD; however, there are no universally accepted guidelines for HIIT at this time. The evidence on the impact of HIIT training across several clinical populations is still limited ([17](#)). The 4 × 4 method (*i.e.*, 4 minutes at high intensity separated by 3 minutes active recovery periods performed 4 times) is the most widely studied protocol ([18](#)). It is important to note that in these studies, HIIT Ex R_x is often established based on data from a sign and symptom-

limited maximal exercise test. Expert recommendations on the practical application of HIIT have been previously made ([18](#)).

Exercise Prescription Without a Maximal Exercise Test

Although a sign and symptom-limited maximal exercise test to determine an exercise THR is recommended for individuals with CHD who are participating in exercise training ([3](#), [12](#)), just 17% of patients in CR programs in the United States have a recent test ([19](#)). Previous recommendations suggested setting the initial exercise intensity at a HR that is 20–30 beats above a patient’s resting HR ([6](#), [12](#)). However, this results in a suboptimal training intensity in the majority of patients. Self-reported perceived exertion using scales such as the Borg Rating of Perceived Exertion (RPE) Scale have been promoted as a reliable way to guide exercise intensity ([6](#), [8](#), [15](#)). However, the Borg RPE is influenced by multiple factors, including age, sex, exercise experience, and self-efficacy ([20](#)). RPE can be a valid tool to guide exercise intensity, but only after an individual’s perceived exertion is aligned with a prescribed HR or target workload, such as a THR from a sign and symptom-limited maximal exercise test ([20](#)). In the absence of a target exercise HR/workload, patients will exercise at their self-selected pace ([20–22](#)). Several studies have shown the “Talk Test” (see [Chapter 5](#)) to be a valid method to guide exercise intensity in patients with CHD ([23–25](#)).

FITT RECOMMENDATIONS FOR INDIVIDUALS WITH HISTORY OF CORONARY ARTERY DISEASE ([6](#), [11](#), [12](#), [15](#), [17](#)) OR HEART FAILURE ([11](#), [26](#))

	Aerobic	Resistance	Flexibility
Frequency	$\geq 3-5 \text{ d} \cdot \text{wk}^{-1}$	2-3 nonconsecutive d · wk ⁻¹	$\geq 2-3 \text{ d} \cdot \text{wk}^{-1}$ with daily being most effective
Intensity	Consider starting at 40%–60% HRR during first few weeks and progressing to 60%–80% of HRR as tolerated by the patient 10 beats below the HR at ischemic threshold, when present Without an exercise test, titrate intensity based on a combination of the Talk Test, RPE, and clinical judgement	Perform 10–15 repetitions at 60%–70% 1-RM	Stretch to the point of feeling tightness or slight discomfort.
Time	$\geq 20-60 \text{ min}$	1–2 sets	10–30 s hold for static stretching; 2–4 repetitions of each exercise
Type (mode)	Walking (jogging if able), leg ergometer, combination of upper and lower (dual action) extremity cycle ergometer, upright and recumbent cycle ergometer, recumbent stepper, rower, elliptical, stair climber, treadmill	Multijoint and single-joint exercises focusing on major muscle groups using multiple modalities	Exercise for each of the major muscle-tendon units

1-RM, one repetition maximum; HR, heart rate; HRR, heart rate reserve; RPE, rating of perceived exertion; $\dot{V}O_{2R}$, volume of oxygen uptake reserve.

CARDIAC REHABILITATION

CR is used to deliver exercise training and other lifestyle interventions and consists of a coordinated, multifaceted intervention designed to reduce risk, foster and promote healthy behaviors, and reduce disability for individuals with several forms of heart disease ([27](#)). CR is delivered in inpatient (phase 1) and outpatient (phases 2 [early] and 3 [maintenance] CR) settings.

Inpatient Cardiac Rehabilitation

Inpatient CR is driven by the inpatient medical care team. The focus is patient assessment and clinical stability, prevention of secondary events (*e.g.*, embolisms), avoiding bed rest, promoting early mobilization and ambulation, health education, and discharge planning ([12](#)). Health education should include information regarding diagnosis, risk factors, guidance on the resumption of activities of daily living (ADLs), and medications. Early mobilization might start in the cardiothoracic intensive care unit ([12](#)). Ambulation and health education might be provided by a variety of professionals, including clinical exercise physiologists, dietitians, nurses, and occupational/physical therapists ([12](#)). As appropriate, discharge planning should include referral to outpatient CR; preferably with a physician directly encouraging the patient to participate ([28](#)). Over the past several decades, as the mean length of stay of patients hospitalized for a CHD-related event and/or procedure has decreased, there has been a shift from supervised, progressive inpatient exercise programs to earlier preparation of the patient for discharge, including referral to outpatient CR ([12](#)).

Supervised ambulation may be initiated pursuant to the conditions in [Box 8.2](#) being met and accounting for the contraindications for CR in [Box 8.3](#). These recommendations may be superseded by the clinical judgment of the supervising physician. [Box 8.4](#) provides a list of possible adverse responses for which discontinuing an exercise session would be indicated. During the

hospital stay, simple exposure to gravitational stress, such as intermittent sitting or standing, within the initial 12–24 hours after an MI may prevent deterioration in exercise tolerance that often follows an acute cardiac event and subsequent bed rest ([29](#), [30](#)). The optimal dose of exercise for patients during their inpatient stay has not been defined, but PA should progress from self-care activities (*e.g.*, sitting, toileting), arm and leg range of motion, and postural changes, to limited supervised walking short-to-moderate distances with minimal or no assistance 3–4 times per day on the hospital floor ([12](#)). Other activities may include upper body movement exercises and minimal stair climbing in preparation for returning home ([12](#)). During progressive phases of ambulation, the patient should be monitored for appropriate hemodynamic responses (HR, systolic blood pressure [SBP]), rhythm, and ST changes in the electrocardiogram [ECG]) and/or new signs or symptoms (*e.g.*, chest pain, shortness of breath, palpitations, fatigue) ([12](#)). Although all individuals might not be suitable candidates for inpatient exercise, virtually all will benefit from PA counseling and health education.

Box 8.2 American Association of Cardiovascular and Pulmonary Rehabilitation (AACVPR) Parameters for Inpatient Cardiac Rehabilitation Daily Ambulation ([34](#))

- No new or recurrent chest pain in previous 8 hours
- Stable or falling creatine kinase and troponin values
- No indication of decompensated heart failure (*e.g.*, resting dyspnea and bibasilar rales)
- Normal cardiac rhythm and stable electrocardiogram for previous 8 hours

Box 8.3 Indications and Contraindications for Inpatient and Outpatient Cardiac Rehabilitation

Indications

- Medically stable post MI
- Stable angina
- Coronary artery bypass graft surgery
- Percutaneous transluminal coronary angioplasty
- Stable heart failure caused by either systolic or diastolic dysfunction (cardiomyopathy)
- Heart transplantation
- Valvular heart disease/surgery
- Peripheral arterial disease
- At risk for coronary artery disease with diagnoses of diabetes mellitus, dyslipidemia, hypertension, or obesity
- Other individuals who may benefit from structured exercise and/or individual education based on physician referral and consensus of the rehabilitation team

Contraindications

- Unstable angina
- Uncontrolled hypertension (resting SBP > 180 mm Hg and/or resting diastolic blood pressure [DBP] > 110 mm Hg)
- Orthostatic SBP drop of >20 mm Hg with symptoms
- Significant aortic stenosis (aortic valve area <1.0 cm²)
- Uncontrolled atrial or ventricular arrhythmias
- Uncontrolled sinus tachycardia (>120 beats · min⁻¹)
- Uncompensated heart failure
- Third-degree atrioventricular block without pacemaker
- Active pericarditis or myocarditis
- Recent embolism (pulmonary or systemic)
- Acute thrombophlebitis
- Aortic dissection
- Acute systemic illness or fever
- Uncontrolled diabetes mellitus
- Severe orthopedic conditions that would prohibit exercise

- Other metabolic conditions, such as acute thyroiditis, hypokalemia, hyperkalemia, or hypovolemia (until adequately treated)
- Severe psychological disorder

MI, myocardial infarction; SBP, systolic blood pressure.

Information from ([35](#)).

Box 8.4 Adverse Responses to Inpatient Exercise Leading to Exercise Discontinuation

- DBP $\geq$ 110 mm Hg
- Decrease in SBP $>$ 10 mm Hg during exercise with increasing workload
- Significant ventricular or atrial arrhythmias with or without associated signs/symptoms
- Second- or third-degree heart block
- Signs/symptoms of exercise intolerance including angina, marked dyspnea, and electrocardiogram (ECG) changes suggestive of ischemia

DBP, diastolic blood pressure; SBP, systolic blood pressure.

Used with permission from [\(34\)](#).

Health systems that have an opt-out, automatic referral to outpatient CR via electronic medical record system capture the most referrals ensuring continuity of care [\(31\)](#). An exercise plan should be provided before discharge. Until evaluated with a sign and symptom-limited maximal exercise test or entry into a clinically supervised outpatient CR program, the upper limit of HR or RPE noted during exercise should not exceed those levels observed during the inpatient program [\(12\)](#). Individuals should be counseled to identify abnormal signs and symptoms suggesting exercise intolerance and the need for medical evaluation.

All eligible individuals should be strongly encouraged, and if possible, given an appointment, to participate in a clinically supervised outpatient phase 2 CR program prior to discharge. [Table 8.1](#) lists strategies that may be utilized to increase the percentage of individuals with heart disease participating in outpatient CR. Additionally, the Million Hearts initiative has published guidelines and strategies to help CR programs become more efficient in order to achieve a goal of 70% participation of eligible patients [\(33\)](#).

TABLE 8.1 Strategies That Influence Referral and Enrollment to Cardiac Rehabilitation (32)

Strategy	Brief Description
Automatic inpatient CR referral system	CR referral is carried out as an automatic EMR order for all eligible individuals
Inpatient “liaison” to help educate and refer individuals to outpatient CR	A liaison meets with inpatients who are eligible for CR, educating and guiding them in the CR enrollment process
Combination of automatic CR referral system and “liaison”	Combination of the two strategies listed above
Limit or eliminate out-of-pocket expenses to individuals for CR services	Negotiate with insurance companies to limit or eliminate copayments and other out-of-pocket expenses for individuals enrolled in CR
Inclusion of home-based CR option for individuals who are not able to attend a center-based CR program	Protocol-driven, home-based approaches to CR delivery provide CR services to individuals at home for low- to moderate-risk individuals may be nurse-managed
Flexible hours of operation	Increased flexibility of CR center hours to include early morning, noontime, after work, and weekend hours
Early outpatient appointment established before hospital discharge	Inpatient staff members work and EMR set up an outpatient CR enrollment appointment for each eligible individual within 12 d of hospital discharge
Use of CR referral performance measures in a quality improvement system	CR referral is assessed, reported, and acted on in a systematic quality improvement program

CR, cardiac rehabilitation; EMR, electronic medical record.

Outpatient Cardiac Rehabilitation

There is strong evidence that outpatient CR is effective in preventing a secondary cardiac event for individuals diagnosed with acute coronary

syndrome, MI, heart failure with reduced ejection fraction (HFrEF), and after a coronary revascularization (coronary artery bypass graft [CABG] surgery or percutaneous coronary intervention), heart valve surgery, and heart transplant ([5](#), [36](#), [37](#)). Currently, Medicare and most third-party health insurance providers in the United States provide coverage for outpatient CR as a benefit for these diagnoses. Participation in exercise training through CR reduces the rate of mortality and morbidity in individuals with various forms of CHD ([4](#)). The benefits provided by CR are important to the individual and to public health as subsequent health care costs may be reduced following participation ([38](#)), with cost-effectiveness greater in individuals with a higher risk for subsequent cardiac events ([39](#)). CR continues to be underutilized, despite strong evidence demonstrating desirable outcomes ([40](#)). A recent analysis of Medicare beneficiaries in the United States revealed less than 29% of eligible patients participated in ≥ 1 session of early outpatient CR within 1 year of an eligible diagnosis ([40](#)).

Outpatient CR is delivered in a traditional center/facility-based program and/or alternative models such as in the home/community with asynchronous or synchronous monitoring by CR staff. Regardless of the model, CR programs should incorporate the core clinical and operational components for CR as outlined by the ACC and AHA ([5](#)). Alternative delivery models are important to address low participation rates. There is evidence that alternative delivery models of CR (*e.g.*, virtual CR) are effective at improving adherence, overall PA, and functional capacity ([41](#)). In the only randomized clinical trial of facility-based CR versus hybrid CR (combination of facility and supervised home via telehealth), compared to facility-based CR, hybrid CR resulted in similar participation rates and improvements in exercise capacity and self-reported health ([42](#)). Lastly, CR programs provide risk factor and disease management education. The goals of outpatient CR are listed in [Box 8.5](#), and the components of CR are listed in [Box 8.6](#) and include established CR components regardless of the program being delivered in a traditional outpatient or virtual setting.

Box 8.5 Goals for Outpatient Cardiac Rehabilitation

- Develop and assist the individual to implement a safe and effective formal exercise and lifestyle physical activity program.
- Provide appropriate supervision and monitoring to detect change in clinical status.
- Provide ongoing surveillance to the individual's health care providers in order to enhance medical management.
- Return the individual to vocational and recreational activities or modify these activities based on the individual's clinical status.
- Provide individual and spouse/partner/family education to optimize secondary prevention (*e.g.*, risk factor modification) through aggressive lifestyle management and judicious use of cardioprotective medications.

Box 8.6 Components of Outpatient Cardiac Rehabilitation

- Cardiovascular risk factor assessment and counseling on aggressive lifestyle management
- Education and support to make healthy lifestyle changes to reduce the risk of a secondary cardiac event
- Development and implementation/supervision of a safe and effective personalized exercise plan
- Monitoring with a goal of improving BP, lipids/cholesterol, and diabetes mellitus
- Psychological/stress assessment and counseling
- Communication with each individual's physician and other health care providers regarding progress and relevant medical management issues
- Return to appropriate vocational and recreational activities

BP, blood pressure.

At the time of CR program entry, the following assessments should be performed ([12](#)):

- Medical and surgical history including the most recent cardiovascular event, comorbidities, and other pertinent medical history.
- Review relevant medical history including risk factors for CHD, medications, 12-lead ECG, coronary angiogram, echocardiogram, stress test (exercise or pharmacologic studies), cardiac surgeries or percutaneous interventions, and settings for pacemakers and implantable cardiac defibrillators.

While exercise training is safe for most individuals with CHD, participants should be monitored on a regular basis for signs and symptoms of disease progression. Important items to monitor during CR include HR, cardiac rhythm, blood pressure (BP), body weight, symptoms of ischemia, symptoms of decompensating heart failure (HF), excessive fatigue, and medication changes.

Continuous Electrocardiographic

Monitoring

While monitoring the ECG during exercise might be considered the defining characteristic of outpatient CR, it is not required. The inclusion and extent of ECG monitoring varies between programs and may, in part, be determined based on the patient's risk, whether they have had a sign and symptom-limited maximal exercise test, and requirements of third-party health insurance providers ([12](#)).

INDIVIDUALS WITH HISTORY OF SPONTANEOUS CORONARY ARTERY DISSECTION

Spontaneous coronary artery dissection (SCAD) is a cause of acute coronary syndrome including MI and cardiac arrest secondary to dissection or hematoma formation within the wall of a coronary artery thereby limiting blood flow to the heart. Patients with SCAD are a unique subset of patients. Most are females, mean age of 40–50 years at time of first MI, with a lower burden of atherosclerotic risk factors than patients with conventional atherosclerotic MI. Some patients, more often men, report the onset of SCAD with physical exertion and a subset of patients describe a precipitating emotional stressor ([1](#), [2](#)). Fibromuscular dysplasia (FMD) is a noninflammatory, nonatherosclerotic arteriopathy that can cause arterial stenosis, aneurysm, dissection, occlusion, and tortuosity ([43](#)). Arterial FMD, aneurysm, ectasia, and dissection are common and diagnosed in 25%–86% of SCAD patients ([44](#)).

SCAD is also a cause of MI associated with pregnancy and can occur in a small subset of patients with inheritable connective tissue disease such as Marfan, Ehlers-Danlos, or Loeys-Dietz syndrome ([44](#)). Other reported comorbidities include hyperlipidemia (30%), hypertension (30%), and obesity (20%) ([45](#)). SCAD patients are commonly treated with β -blockers and antihypertensive medications as there is some observational data to suggest that these may be factors that influence risk for recurrent SCAD ([46](#)).

Antiplatelet therapy depends on whether coronary stenting was performed, although a conservative strategy allowing for coronary artery healing is preferred ([44](#)). CR and moderate exercise are considered safe and beneficial for SCAD patients ([47–49](#)).

Exercise Testing

Many patients with a history of SCAD have persistent chest pain. Often, this is multifactorial, nonexertional, may be nonischemic, and generally improves over time for many patients. If there are concerns about ischemia, pharmacologic or exercise testing coupled with cardiac imaging can be helpful. Data regarding the diagnostic yield of pharmacologic or exercise testing with cardiac imaging in this patient population is minimal. If exercise testing is performed, it may give insight regarding arrhythmia, chronotropic incompetence, cardiopulmonary limitations, and overall exertional capacity, which can be used to guide patients during their future exercise regimens.

Exercise Prescription

The general approach to an exercise prescription is similar to individuals with other types of CVD with special considerations. While not true for all SCAD patients ([45](#)), a subset of patients may already be at an above average or high level of fitness. Therefore, many SCAD patients require a program tailored to specific needs, physical capabilities, and preferences with individualized progression ([50](#), [51](#)). Patients should exercise to a moderate, not extreme, level of exhaustion ([52](#)). In the absence of sufficient clinical evidence, arbitrary HR and BP limits with exercise are not recommended and may have unintended or negative consequences on patient well-being.

Special Considerations

- Moderate exercise to a 12–14 rating of perceived exertion (RPE, Borg scale 6–20) is encouraged. Patients should not exercise to a state of complete exhaustion.
- Resistance training is encouraged, with a preference for lighter weights and higher number of repetitions.

- Avoid the Valsalva maneuver during resistance training to minimize excessive changes in intrathoracic pressure, BP, and HR.
- Recommend against stretches or exercises, which require hyperextension or extreme positions of the head and neck to reduce risk of spontaneous arterial dissection in the carotid and vertebral arteries especially since FMD is common.
- Chiropractic treatments (*e.g.*, spinal manipulation or adjustment), and other activities involving rapid motions of the head and neck, should be avoided due to concern for iatrogenic dissection of vulnerable vasculature.
- The patient demographic includes postpartum females, parents of young children, and individuals who return to full-time work after SCAD. Therefore, childcare and work schedules may need to be considered when arranging CR sessions.
- Anxiety, depression, and distress are common after SCAD such that psychosocial support is an important component to incorporate into CR programs whenever possible ([53](#), [54](#)).
- Patients with a history of SCAD may also have chronic HF, sternotomy, pacemaker/implantable cardioverter defibrillator (ICD), or cardiac transplant as a result of their MI. See the relevant sections in this chapter.

CHRONIC HEART FAILURE

Chronic HF is characterized by exertional dyspnea and fatigue in the setting of HFrEF (*i.e.*, systolic dysfunction), a preserved left ventricular ejection fraction (heart failure with preserved ejection fraction [HFpEF], *i.e.*, diastolic dysfunction), or a combination of both systolic and diastolic dysfunction. Due partly to the aging of the population and to improved survival with cardiovascular disease, the prevalence of HF is increasing. HF currently affects an estimated 6.5 million American adults and is estimated to increase 46% by 2030 ([1](#)). Despite this increasing prevalence, hospitalization for acute decompensated HF has decreased ([1](#)). Among individuals admitted for acute HF, 53% have HFrEF and 47% have HFpEF, and 1-year mortality is nearly 30% ([1](#)).

Exercise Testing

Sign and symptom-limited maximal exercise testing is safe in individuals with HFrEF, and when combined with the measurement of expired gases (cardiopulmonary exercise testing [CPET]) provides useful information pertaining to ECG, hemodynamic responses to exercise, and prognosis ([6](#), [55](#)).

- Compared to age-matched healthy individuals, people with HFrEF and HFpEF exhibit a lower peak HR, peak stroke volume, and peak cardiac output response to exercise.
- Vasodilation of the large vessels (*e.g.*, brachial artery) and resistance vasculature are attenuated, limiting regional and local blood flow ([56](#)).
- Abnormalities in skeletal muscle histochemistry limit oxidative capacity ([56](#)).
- Exercise tolerance in individuals with HF being considered for cardiac transplant is often less than 50% of age-predicted normal or a peak volume of oxygen uptake ($\dot{V}O_2$) less than $12 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ([57](#), [58](#)). Because of this, an exercise protocol that starts at a lower work rate and imposes

smaller increases in work rate per stage, such as the modified Naughton treadmill protocol or a $10 \text{ W} \cdot \text{min}^{-1}$ ramp ergometer protocol (see [Chapter 5](#)), is commonly used.

Exercise Prescription

Exercise training is broadly recognized as a valuable adjunct in the therapeutic approach to the care of individuals with stable chronic HF and is recommended by the ACC and the AHA ([59](#)). The benefits of exercise training in individuals with HFrEF have been well described and include improved clinical outcomes (*e.g.*, hospitalizations) and health-related quality of life ([26](#), [60–62](#)). Exercise training also improves exercise capacity (10%–30%), central hemodynamic function, autonomic nervous system function, and peripheral vascular and skeletal muscle function in individuals with HFrEF ([32](#)). In total, these adaptations allow individuals to exercise to a higher peak work rate or exercise at a submaximal level with a lower HR, less perceived effort, and less dyspnea and fatigue. A meta-analysis of 57 studies that directly measured peak $\dot{V}O_{2\text{peak}}$ reported an average 17% improvement following an exercise training program ([63](#)). Emerging data indicate that individuals with HFpEF also benefit from exercise training, as evidenced by improved skeletal muscle function, quality of life, and exercise capacity ([64](#)); however, many insurance providers do not provide coverage for CR for HFpEF alone.

Special Considerations

- A THR range should be determined based on peak HR measured during a sign and symptom-limited maximal exercise test. There are no data to support the use of estimated peak HR in individuals with HF. See section “*Exercise Prescription without a Maximal Exercise Test*” above.
- For those individuals who have completed a sign and symptom-limited maximal exercise test, higher intensity aerobic interval training may be considered. Compared to moderate continuous training, HIIT has been

shown to provide similar to larger improvements in $\dot{V}O_{2peak}$ in stable individuals with HFrEF ([65–69](#)). See section “*High-Intensity Interval Training*” above.

- The exercise program should be designed to gradually increase the volume of exercise performed over time, with duration and frequency of effort increased before intensity. Among individuals with HF, there is an inverse relationship between volume of exercise and risk for all-cause mortality and hospitalization up to $7 \text{ MET} \cdot \text{h} \cdot \text{wk}^{-1}$ ([70](#)).
- After individuals have adjusted to and are tolerating aerobic exercise training, which usually requires at least 4 weeks, resistance training activities can be considered.
- Approximately 40% of individuals with HF are compliant with a prescribed exercise program at the end of 1 year, which is not different than the long-term adherence reported among individuals with established CHD ([62](#), [71](#), [72](#)).

Left Ventricular Assist Device

- Regular exercise training improves exercise tolerance and quality of life in patients with a left ventricular assist device (LVAD) ([73](#)).
- Exercise training and testing of individuals that received an LVAD for either bridge-to-transplant or as a destination therapy for end-stage disease is becoming increasingly more common. These individuals have a low functional capacity with a $\dot{V}O_{2peak}$ in the range of $7\text{--}23 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ([74](#)).
- Due to the continuous flow of the LVAD and the resultant lack of pulsatile flow, BP must be measured by Doppler instead of auscultation with a stethoscope. In the absence of pulsatile flow, Doppler-derived BP provides an estimate of the mean arterial pressure (MAP). Resting mean pressure should be controlled to between 70 and 80 mm Hg ([75](#)). In general, MAP should increase with increasing work rates. Studies have shown safe performance of

exercise in inpatient settings with MAP maintained between 70 and 90 mm Hg ([76](#)).

- In patients with an LVAD, HR during exercise increases in a manner that is generally linear with an increase in work rate. However, until more definitive information describing the relationship between HR and exercise intensity are available, using subjective methods (RPE, Talk Test) to guide exercise intensity in these patients is appropriate.
- Individuals with an LVAD typically have modest increases in blood flow rate (as high as $10 \text{ L} \cdot \text{min}^{-1}$) during progressive intensity exercise.
- Early-onset fatigue is common with exercise in patients with an LVAD. When starting an exercise training program, fatigue later in the day may be reported. If fatigue occurs, intermittent exercise may reduce the level of fatigue experienced from subsequent exercise training sessions.

HEART TRANSPLANTATION

In individuals with end-stage HF in whom prognosis is poor and standard medical therapy fails to control symptoms, heart transplantation may be an option. In 2016, 3,209 heart transplants were performed in the United States and 30,622 people were living with a heart transplant. Survival after a heart transplant varies by age and race ([77](#)). Between 2009 and 2011, the 3-year posttransplant survival rate was 83.5% in the United States ([77](#)). Following surgery, both aerobic and resistance training are strongly recommended to improve exercise capacity and quality of life, help restore bone mineral density, reverse sarcopenia, and help modify CHD risk factors such as obesity, hypertension, and glucose intolerance ([78](#)). According to the International Society for Heart and Lung Transplantation, strong evidence exists to support the efficacy of aerobic and resistance exercise training after a heart transplant ([78](#)).

Exercise Testing

Although there is some evidence of at least partial reinnervation of cardiac autonomic function a year or more after heart transplantation surgery, in the absence of direct cardiac sympathetic efferent innervation, peak cardiac output is 20%–35% below normal ([79](#)). The skeletal muscle and peripheral abnormalities (*e.g.*, endothelial dysfunction) present before surgery are not normalized by the surgery per se and, therefore, contribute to a low-exercise capacity observed in patients with heart transplants when compared to age-matched healthy individuals ([79](#)).

Special Considerations

- Resting HR is often elevated, and the HR response to exercise is attenuated. In the absence of parasympathetic innervation, increases in HR are

dependent on circulating catecholamines. As a result, increases in HR following increases in workload are delayed. The highest HRs might occur after the exercise test or training session and recovery HR is slow to return to preexercise levels.

- BP is often elevated at rest, with a slightly attenuated response to peak exercise.
- Given the HR and BP responses and the previously mentioned reduction in exercise capacity, a more gradual exercise testing protocol should be employed, similar to or more gradual than those recommended for individuals with HF.
- Other exercise testing considerations such as test endpoints remain the same as those used for individuals with other forms of heart disease except for detection of angina, which is unlikely due to the denervated heart.

Exercise Prescription

Randomized, controlled trials of aerobic exercise training in patients postheart transplant have shown an 8%–48% improvement in exercise capacity ([80](#)). These improvements are due to improvements in chronotropic response, oxidative capacity of skeletal muscle, cardiac function, and autonomic regulation ([80](#)). Resistance training is an important adjunct to aerobic exercise training ([81](#)). Following heart transplantation, individuals are at risk for several complications including cardiac allograft vasculopathy, graft failure, cancer, hyperlipidemia, hypertension, and diabetes mellitus ([80](#)).

Prescribing exercise in individuals after heart transplantation is, for the most part, similar to that of other individuals with CHD disease. However, because of the denervated myocardium, setting an HR-based training range is not appropriate, and subjective methods should be used to guide exercise intensity. Initial target workloads can be identified from a sign and symptom-limited maximal exercise test. Because of the negative effects of immunosuppressive drugs on the musculoskeletal system, resistance training should be performed regularly and engage all major muscle groups.

Special Considerations

- Due to the delayed HR response, longer warm-up and cool-down periods should be considered.

FITT RECOMMENDATIONS FOR INDIVIDUALS WITH CARDIAC TRANSPLANT ([61](#), [81](#), [82](#))

	Aerobic	Resistance	Flexibility
Frequency	Minimally 3 d · wk ⁻¹ ; preferably up to 5 d · wk ⁻¹	2–3 nonconsecutive d · wk ⁻¹	≥2–3 d · wk ⁻¹ with daily being most effective
Intensity	With an exercise test, use 50%–80% of exercise capacity ($\dot{V}O_{2R}$) based on target workload Without an exercise test, titrate intensity based on a combination of the Talk Test, RPE, and clinical judgement	Perform 10–15 repetitions at 60%–70% 1-RM	Stretch to the point of feeling tightness or slight discomfort
Time	20–60 min · d ⁻¹	1–2 sets	10–30 s hold for static stretching; 2–4 repetitions of each exercise
Type	Aerobic exercise, focusing on treadmill- or free-walking and stationary cycling as capable	Multijoint and single-joint exercises focusing on major muscle groups using multiple modalities	Exercise for each of the major muscle- tendon units

1-RM, one repetition maximum; RPE, rating of perceived exertion.

- Immunosuppression therapy used to prevent graft rejection can lead to bone loss, diabetes mellitus, and hypertension, and both regular aerobic and resistance training exercise can play an important role in helping manage these metabolic disorders.

- HIIT has been used in individuals with cardiac transplantation with positive results. Work/rest intervals are similar to those recommended for individuals with HF with the exception that HR is not as useful a guide of intensity ([80](#)).
- Due to median sternotomy, range of motion and the work rate of activities and exercises involving the upper limbs should be modified for up to 12 weeks from time of transplant. See “Sternotomy” section in this chapter.

STERNOTOMY

The median sternotomy is the standard incision to provide optimal access for cardiovascular surgeries such as CABG, heart valve replacement, LVAD implantation, and heart transplantation. Although most individuals heal without complications and achieve adequate sternal stability in approximately 8–10 weeks, sternal instability has been observed in up to 16% of cases ([11](#), [83](#)). Several factors such as diabetes mellitus, age, certain drugs, and obesity can predispose an individual to sternal instability and limit exercise training.

Special Considerations

For individuals who have had a sternotomy, there are no evidence-based precautions or restrictions on arm movement, and the vast majority of individuals can initiate light arm movement immediately after surgery with little or no risk of dehiscence (*e.g.*, wound separation) ([84](#)). The rate of dehiscence is 1.5%–3% of all individuals with sternotomy. The highest risks of dehiscence are from excess coughing, osteoporosis, or infection ([84](#)). Other considerations include the following:

- Individuals should be encouraged to move their arms freely immediately after surgery, keeping their arms close to the body to reduce stress on the sternum ([85](#)). This has been phrased “Keep Your Move in the Tube” ([85](#)).
- Individuals poststernotomy should be taught how to use the upper body to perform activities that avoid excessive stress on the sternum and provided an individualized activity progression strategy ([85](#), [86](#)).
- Patients poststernotomy are often instructed to limit the use of their upper body and not lift more than 5–10 lb for the 8–12 weeks ([12](#)). Instructions provided to patients can vary between surgeons. Rhythmic upper limb activities (*e.g.*, arm ergometry, dual-action ergometer) that do not strain the sternum can be performed during the first 12 weeks poststernotomy. It may

be prudent to postpone resistance training with the upper body for 12 weeks. The time between surgery and initiating resistance training should be individualized based on age, prior activity and strength levels, presence of comorbidities, and/or surgical complications ([85](#)). Applying the “Keep Your Move in the Tube” strategy can allow for starting resistance training earlier ([85](#)).

PACEMAKER AND IMPLANTABLE CARDIOVERTER DEFIBRILLATOR

Implantable cardiac pacemakers are used to restore an optimal HR at rest and during exercise, to synchronize atrial and ventricular filling and contraction in the setting of abnormal rhythms, and to synchronize right and left ventricular contraction in the setting of left bundle branch block. Specific indications for pacemakers include sick sinus syndrome with symptomatic bradycardia, acquired atrioventricular (AV) block, and persistent advanced AV block. The different types of pacemakers include the following:

- Rate-responsive (*i.e.*, rate-adaptive or rate-modulated) pacemakers that are programmed to increase or decrease HR to match the level of PA (*e.g.*, seated rest or walking).
- Single-chambered pacemakers have one lead placed into the right atrium or the right ventricle. There are also leadless pacemakers that are implanted into the right ventricle that do not require an external battery source. Both of these may be indicated for individuals with chronic atrial fibrillation with concomitant symptomatic bradycardia such as seen with a high-grade AV block or after creation of complete heart block for definitive rate control.
- Dual-chambered pacemakers that have two leads: one placed in the right atrium and one in the right ventricle; indicated for physiologic pacing to reestablish a normal sequence and timing of contractions between the upper and lower chambers of the heart.
- Biventricular pacemakers for cardiac resynchronization therapy have three leads: one in the right atrium, one in the right ventricle, and one in the coronary sinus or, less commonly, the left ventricular myocardium and are indicated in select individuals with HFrEF.

The type of pacemaker, regardless of manufacturer, is identified by a four-letter code:

- The first letter of the code describes the chamber paced (*e.g.*, atria [A], ventricle [V], dual chamber [D]).
- The second letter of the code describes the chamber sensed.
- The third letter of the code describes the pacemaker's response to a sensed event (*e.g.*, triggered [T], inhibited [I], dual (triggered + inhibited) [D], no response [O]).
- The fourth letter of the code describes the rate modulation availability of the pacemaker (*e.g.*, rate modulation/responsive available [R], rate modulation not available or disabled [O]).

For example, a VVIR code pacemaker means (a) the ventricle is paced (V) and sensed (V); (b) when the pacemaker senses a normal ventricular contraction, it is inhibited (I); and (c) the pulse generator is rate responsive (R).

Exercise testing is strongly recommended for the assessment of rate-responsive pacemakers in those contemplating increased PA or competitive sports ([6](#)). In these cases, exercise testing can help to optimize the HR response and thus may increase the exercise capacity of an individual.

The ICD is a device that monitors the heart rhythm and delivers an electrical shock if life-threatening rhythms are detected. ICDs are used for high-rate ventricular tachycardia or ventricular fibrillation in individuals who are at risk for these conditions as a result of previous cardiac arrest, cardiomyopathy, or ineffective drug therapy for abnormal heart rhythms. When ICDs detect an excessively rapid or irregular rhythm, they might initially attempt to pace the heart into a normal rate and rhythm (*i.e.*, antitachycardia pacing). If unsuccessful, they can then deliver an electrical shock (*i.e.*, electrical cardioversion, defibrillation) in an attempt to reset the heart to a normal rhythm and rate. ICDs protect against sudden cardiac death from ventricular fibrillation. Exercise is safe for individuals with an ICD ([87](#)).

Special Considerations

- Programmed pacemaker modes, HR limits, and ICD rhythm detection algorithms should be obtained prior to exercise testing or training.
- Exercise testing should be used to evaluate HR and rhythm responses prior to beginning an exercise program.
- When an ICD is present, the peak HR during the exercise test and exercise training program should be maintained at least 10–15 beats $\cdot$ min⁻¹ below the programmed HR threshold for defibrillation ([6](#)).
- After the first 24 hours following the device implantation, mild upper extremity range of motion activities can be performed and may be useful to avoid subsequent joint complications.
- Activities involving rigorous upper extremity movement such as swimming, bowling, lifting weights, elliptical machines, and golfing should be avoided for at least 3–4 weeks after device implant. However, lower extremity activities are allowable.
- In the United States, isolated pacemaker and ICD implantation are not indications for CR. However, supervised exercise can be important for these individuals.

PERIPHERAL ARTERIAL DISEASE

Peripheral arterial disease (PAD) is atherosclerosis of the arteries outside of those that supply blood to the heart and brain, particularly the lower extremities. Among adults ≥ 40 years, approximately 8.5 million Americans and 110 million people worldwide have PAD of their lower extremities, including more than 14% of people over 70 years old and 22% of people over 80 years old ([1](#)). PAD represents an advanced burden of cardiovascular disease, and patients with PAD have a high risk of morbidity and mortality. Among individuals with PAD, the risk of stroke, MI ([88](#)), and mortality ([89](#)) is equivalent to or greater than the risk in those with CHD. The 10-year mortality among individuals with PAD ranges from 22% to 70% depending on age and severity of risk factors ([90](#), [91](#)). The major risk factors for PAD include hypertension, smoking, diabetes mellitus, and chronic kidney disease ([1](#)).

In addition to a high risk of cardiovascular events, PAD leads to an accelerated decline in physical function and increased disability in both symptomatic and asymptomatic patients ([92–94](#)). For symptomatic patients, the greatest limitation to activity is ischemic pain in the legs known as intermittent claudication. Claudication originating from PAD typically presents as a cramping or tightness sensation in the lower extremities caused by reduced arterial blood flow that predictably waxes and wanes with exercise and rest. Claudication most often presents itself in one or both calves but can also occur in the buttocks or thighs. PAD severity and chronic limb-threatening ischemia can be ranked based on the presence of signs and symptoms ([Table 8.2](#)) ([95](#), [96](#)) and/or by the ankle/brachial pressure index (ABI) ([Table 8.3](#)) ([97](#)).

TABLE 8.2 Classification of Peripheral Artery Disease

Fountaine Classification			
Stage	Symptoms		
1	Asymptomatic		
2	Intermittent claudication		
2a	Distance to pain onset >200 m		
2b	Distance to pain onset <200 m		
3	Pain at rest		
4	Gangrene, tissue loss		
Rutherford Classification			
Grade	Category	Clinical Description	Objective Criteria
0	0	Asymptomatic no hemodynamically significant occlusive disease	Normal treadmill or reactive hyperemia test
	1	Mild claudication	Completes treadmill exercise; AP after exercise >50 mm Hg but at least 20 mm Hg lower than resting value
I	2	Moderate claudication	Between categories 1 and 3
	3	Severe claudication	Cannot complete standard treadmill exercise and AP after exercise <50 mm Hg
II	4	Ischemic rest pain	Resting AP < 40 mm Hg, flat or barely pulsatile ankle or metatarsal PVR; TP < 30 mm Hg
III	5	Minor tissue loss — nonhealing ulcer, focal gangrene with diffuse pedal ischemia	Resting AP < 60 mm Hg, ankle or metatarsal PVR flat or barely pulsatile; TP < 40 mm Hg

Fountaine Classification			
Stage	Symptoms		
	6	Major tissue loss — extending above TM level, functional foot no longer salvageable	Same as category 5

AP, ankle pressure; PVR, pulse volume recording; TP, toe pressure; TM, transmetatarsal.

TABLE 8.3 Ankle/Brachial Pressure Index Scale for Peripheral Arterial Disease ([97](#))

Supine, Resting ABI	Interpretation
>0.90	Normal
≤0.90	Threshold for PAD confirmation
Decrease of >0.15 over time	Significant PAD progression
Postexercise ABI	Interpretation
No change	Normal
Decrease of >30 mm Hg or >20% from resting ABI	Reasonable to consider as threshold for PAD confirmation, whether ABI is normal or abnormal at rest
Decrease of >0.15 over time	Significant PAD progression

ABI, ankle/brachial pressure index; PAD, peripheral arterial disease.

Exercise Testing

Exercise testing can be performed in individuals with PAD to determine functional capacity, although individuals with claudication will likely be limited by their pain. Exercise testing is most useful to determine the time of onset of claudication pain and total walking time before and following therapeutic intervention and to diagnose the presence of cardiovascular disease and assess for other exercise safety factors ([97](#), [98](#)):

- Medication dose and timing should be noted and repeated in an identical manner in subsequent exercise tests assessing potential therapeutic changes.
- Ankle and brachial artery SBP should be measured bilaterally after 5–10 minutes of rest in the supine position following standardized ABI procedures ([99](#)). The ABI is calculated by dividing the highest ankle SBP reading by the highest brachial artery reading. Note there are multiple diagnostic criteria using the ABI at rest and during exercise ([100](#)).
- A standardized motorized treadmill protocol should be used to ensure reproducibility of pain-free maximal walking time ([97](#)). Claudication pain perception can be monitored using a numerical rating scale (see [Figure 4.3](#)) ([101](#)).
- The exercise test should begin with a slow speed and have gradual increments in grade ([102](#)) (see [Chapter 4](#)).
- Following the completion of the exercise test, individuals should recover in the seated position.
- The 6-minute walk test can be used to objectively assess ambulatory functional limitations in those not amenable to treadmill testing ([97](#)).

Exercise Prescription

The AHA and ACC strongly recommend supervised exercise training for patients with claudication and as a treatment option before revascularization ([103](#), [104](#)). Data support supervised exercise over community-based programs, and the primary exercise should be walking to moderate-to-maximum claudication alternating with periods of no exercise rest ([103](#), [104](#)). Revascularization might be considered when lifestyle-limiting claudication persists despite guideline-directed medical therapy and to avoid chronic limb ischemia ([103](#)).

Multiple review articles and position statements support walking as beneficial and as the primary mode of exercise for patients with PAD ([102](#)–[105](#)). A meta-analysis looking at exercise in patients with PAD concluded that the optimal exercise program included intermittent walking at near-maximal

claudication pain ([106](#)). In these studies, the distance to claudication onset increased an average of 179% and the distance to maximal claudication increased an average of 122% ([106](#)).

Although exercise training for patients with PAD should focus on weight-bearing exercise (walking is preferred), they should be individualized relative to intensity, work-to-rest ratios, duration, and frequency ([104](#)). An exercise workload should be selected that elicits mild claudication pain within 5 minutes and moderate to moderately severe claudication within 10 minutes followed by rest ([104](#)). Exercise should resume when claudication has fully subsided ([104](#)). Analogous to outpatient CR for patients with heart disease, supervised exercise therapy for symptomatic PAD (aka, SET PAD) is the primary model of clinic-based, supervised exercise training programs for patients with PAD ([107](#)). In the United States, it is a covered benefit by Medicare and many third-party health insurance providers for patients with symptomatic PAD ([107](#)). The reader is referred to the review by Ehrman et al. ([107](#)) for more information on SET PAD.

Special Considerations

- Studies have shown that improvement in PAD with exercise training might be most noticeable in the initial 2–3 months ([108](#)). However, improvement in PAD continues to increase in programs lasting 6 months or longer ([109](#)).
- Exercise training protocols emphasizing lower intensity (*i.e.*, below claudication threshold) and longer duration walking might be as beneficial as protocols that emphasize higher intensity and shorter duration walking ([110](#)); however, guidelines recommend exercise to moderate-severe pain ([97](#), [102](#), [104](#), [108](#)).
- Some individuals may need to begin exercise by accumulating only 15 min · d⁻¹, gradually increasing time by 5 min · d⁻¹ biweekly.
- Although the focus of an exercise training program in individuals with PAD should be walking, non-weight-bearing exercise may provide additional benefits ([104](#), [108](#)).

- Resistance training has not consistently been shown to improve pain-free walking ability in individuals with PAD ([104](#), [108](#)).
- Cycling or other non-weight-bearing exercise modalities may be used as a warm-up but should not be the primary type of activity ([104](#)).
- The optimal work-to-rest ratio has not been determined for individuals with PAD and may need to be adjusted for each individual ([104](#)).

FITT RECOMMENDATIONS FOR INDIVIDUALS WITH LOWER EXTREMITY, SYMPTOMATIC PERIPHERAL ARTERIAL DISEASE ([104](#), [107](#))

	Aerobic	Resistance	Flexibility
Frequency	Minimally 3 d · wk ⁻¹ ; preferably up to 5 d · wk ⁻¹	2–3 nonconsecutive d · wk ⁻¹	≥2–3 d · wk ⁻¹ with daily being most effective
Intensity	Exercise workload that elicits moderate to moderate-severe claudication (<i>i.e.</i> , 3–4 out of 4 on the claudication pain scale) within 10 min, followed by rest with resumption of exercise when pain is <i>completely</i> alleviated	Perform 10–15 repetitions at 60%–70% 1-RM	Stretch to the point of feeling tightness or slight discomfort
Time	30–45 min · d ⁻¹ (excluding rest periods) for a minimum of 12 wk; may progress to 60 min · d ⁻¹	1–2 sets	10–30 s hold for static stretching; 2–4 repetitions of each exercise
Type (mode)	Weight-bearing is preferred (free or treadmill walking is best). Consider non-weight-bearing as an adjunct or if weight-bearing exercise is not tolerated	Multijoint and single-joint exercises focusing on major muscle groups using multiple modalities	Exercise for each of the major muscle-tendon units

1-RM, one repetition maximum.

- A cold environment may aggravate the symptoms of intermittent claudication; therefore, a longer warm-up may be necessary ([111](#)).

- Increasing functional capacity in patients with PAD may unmask symptoms of underlying CHD.

POSTURAL ORTHOSTATIC TACHYCARDIA SYNDROME

Postural orthostatic tachycardia syndrome (POTS) is a clinical syndrome most common in premenopausal females between the ages of 15–50 years, characterized by chronic orthostatic intolerance with an inordinate increase in HR ($\geq 30 \text{ beats} \cdot \text{min}^{-1}$) from lying to standing over 10 minutes without orthostatic hypotension ([112](#)). The pathophysiology of POTS remains unclear; however, a variety of subtypes contribute to this abnormal cardiovascular and hemodynamic response. POTS is often associated with migraine disorders, myalgic encephalomyelitis/chronic fatigue syndrome, joint hypermobility, gastrointestinal, and autoimmune disorders.

Physiologic changes suggest that the orthostatic tachycardia in POTS is an appropriate compensatory response to reduced cardiac size and mass, blood volume, stroke volume, and $\dot{V}O_{2\text{peak}}$. Additional root cause pathophysiology has been described as POTS subtypes in which neurological, autoimmunity, volume regulation, adrenergic, or histamine responses are unique drivers of symptomology ([112](#)).

Treatment of POTS should be overseen by a medical doctor or practitioner with specialized training in human pathophysiology and exercise physiology. Treatment should focus on nonpharmacological intervention with the use of pharmacological intervention only as required to support rehabilitation. Care should be taken to address conditions which mimic POTS such as those listed in *Special Considerations*.

Preexercise Assessment

The 10-minute stand test can be performed with minimal equipment. The individual is placed in the supine position quietly for 10 minutes before measuring supine BP and HR. The individual then stands quietly unassisted for

10 minutes, during which HR and BP are measured at 1, 3, 5, and 10 minutes ([112](#)). These measurements are used to calculate HR zones for the exercise prescription in [Table 8.4](#).

Table 8.4 Summary of Exercise Program for POTS

Exercise Session	Intensity	Month 1	Month 2	Month 3
Base Pace	RPE 13–15 HRR 60%	12 × 30 min	6 × 30 min 3 × 30–45 min	5 × 35 min 4 × 45–60 min
Max Steady State	RPE 16–18 HRR 75%	1 × 20 min 1 × 25 min	1 × 25 min 1 × 30 min 1 × 35 min	1 × 30 min 1 × 35 min 1 × 40 min
Recovery	RPE 9–12 HRR < 60%	2 × 40 min	1 × 40 min 1 × 30 min	3 × 25 min
Resistance training	RPE 13–15	8 × 30 min	8 × 30 min	8 × 30 min
Exercise Modality		Recumbent bike Rowing (preferred) Core, lower body	Month 1 modes plus upright bike Core, lower body	Month 1 and 2 modes plus elliptical and treadmill walking Core, lower body

Adapted from ([114](#)).

If conducting a maximal exercise test, individuals may demonstrate higher HRs during submaximal and maximal exercise due to a lower stroke volume compared with sedentary individuals but no difference in HR at relative workloads (*i.e.*, % $\dot{V}O_{2peak}$) ([113](#)). Depending on the individual's level of

deconditioning, consider lower initial workloads with smaller increments using recumbent position. Assess for use of medications that impact exercise responses and consider their effects when interpreting exercise testing.

Exercise Prescription

Short-term progressive exercise training (summary [Table 8.4](#)) is well tolerated by individuals with POTS and leads to improvement or remission of POTS symptoms ([114](#), [115](#)).

It is recommended to use the HR reserve method and provide patients with a HR range (*e.g.*, 5 beats either side of 75% HRR) rather than a specific target. For individuals using HR limiting medications, the rate of perceived exertion (RPE) scale is recommended instead of HR zones. A chest-based strap or wrist worn HR monitor is strongly recommended to ensure patients receive the optimal dose of exercise.

It is strongly recommended that aerobic exercise is performed in the recumbent position to reduce gravitational stress and allow progressive adaptation of the cardiovascular system. Recommended modalities include recumbent biking, rowing ergometry, or swimming ([115](#)). It is imperative that individuals perform a light warm-up before and cool-down after each exercise session. Aerobic sessions should not be performed consecutively. Resistance exercise is recommended to support cardiovascular conditioning with a focus on strengthening the core and lower body such as Pilates or yoga (nonheated).

To further enhance the efficacy of rehabilitation, the following modifications are recommended while implementing [Table 8.4](#)

- Plasma volume expansion: increase daily sodium intake to 7,000–10,000 mg (salt tablets are not recommended) with 2–3 L of noncaffeinated fluids ([114](#)).
- Remain upright throughout the day (rest in semirecumbent position if necessary).
- Place bed in reverse Trendelenburg position (4–6 in elevation from supine).

Adherence to this program is associated with improvement in symptoms and quality of life, increases in cardiorespiratory fitness ([115](#)), physiological cardiac remodeling ([113](#)), and improvements in autonomic function ([116](#)). Individuals that complete the exercise prescription should maintain an active lifestyle that includes regular exercise 4–5 days per week to maintain cardiovascular health. Individuals who experience a remission of symptoms may repeat [Table 8.4](#).

For further details to guide exercise prescription for POTS, the reader is encouraged to refer to Bryarly et al. ([112](#)).

Special Considerations

The following categories should be assessed for and distinguished before prescribing exercise as they may mimic POTS-like presentations.

Iron deficiency (ID) — muscle function and oxygen transport depend on iron availability. ID can result in orthostatic intolerance, increased HRs and shortness of breath during exercise, fatigue, headaches, and reduced cognitive function ([117–119](#)).

- Ferritin less than $50 \text{ ng} \cdot \text{dL}^{-1}$
- %SAT (transferrin saturation) less than 20%

ID with anemia (above markers plus) ([120](#), [121](#))

- Hgb (hemoglobin) less than $12 \text{ g} \cdot \text{dL}^{-1}$
- MCV (mean cell volume) less than 80 fL

Medications — commonly prescribed, but not a complete list

- SNRI (serotonin and norepinephrine reuptake inhibitor) — increased noradrenergic effects
- NDRI (norepinephrine-dopamine reuptake inhibitor) — increased sympathomimetic effects
- Drospirenone — diuretic effects leading to decreased fluid volume

- Stimulants — increased sympathomimetic effects

Ingested substances — increased sympathomimetic effects

- Cocaine
- Caffeine
- Workout supplements
- Energy drinks

Hyperthyroidism — excessive thyroid hormone production is associated with palpitations, increased HR, exercise intolerance, and widened pulse pressure. (↑HR, ↑cardiac output, ↓SVR, ↓diastolic blood pressure [DBP], ↓afterload) ([122](#), [123](#)).

- Thyroiditis
- Overmedicated
- Thyroid/pituitary tumor

Inappropriate sinus tachycardia — tachycardia (>100 bpm) unrelated to rest or positioning ([112](#)).

PEDIATRIC CARDIAC REHABILITATION

Many pediatric heart diseases are identified in utero and often require prolonged hospitalizations and surgical interventions early in life, resulting in chronic impairments in physical function, cardiorespiratory fitness, and frailty ([124–126](#)). Pediatric CR is like adult CR in that it consists of a coordinated interdisciplinary approach to improve physical function, foster an active lifestyle, and reduce disability. Pediatric CR services may span from infancy through early adulthood and include a spectrum of disciplines such as physical and occupational therapy, speech-language therapy, psychological care, dietary and feeding therapies, and exercise training ([127](#)). Thus, many professionals believe the terms “rehabilitation” and “prehabilitation” may not be completely appropriate in this context, as many of these youth have deficits in physiological and neurodevelopmental function since birth.

Delivering CR to a pediatric population requires modifications for developmental appropriateness and consideration of the unique needs and/or constraints associated with congenital or acquired pediatric heart diseases. Pediatric CR is delivered in both inpatient and outpatient settings. Inpatient CR occurs after cardiac surgery or with prolonged hospitalization. Inpatient exercise prescriptions are often highly individualized and include nursing, physical or occupational therapists, neurodevelopmental specialists, and clinical exercise physiologists. Studies have demonstrated that inpatient CR delivered by a clinical exercise physiologist is safe and feasible for patients with prolonged hospitalization for HF awaiting transplant and/or equipped with ventricular assist device ([128–130](#)).

Initiation of outpatient CR may be within weeks of surgical repair or several years after surgical repair of their congenital heart disease as certain functional difficulties or disabilities may not be recognized until later in childhood/adolescence, or patients may be too young to participate in exercise

therapies ([127](#)). Although the CR exercise prescription (FITT principle box) for pediatrics is like the exercise prescription for adult CR, there are some unique differences ([131](#)). For example, extremely deconditioned patients with complex congenital heart disease may require intervention to improve movement literacy and motor proficiency prior to implementation of an aerobic and resistance exercise program ([132](#), [133](#)). Other individualized modifications in training volume and intensity may be needed for those with neurodevelopmental delay and decreased attention span, and younger children may benefit from the gamification of exercise ([134](#)). Certain exercises may need to be modified based on risk, with the most common concerns being arrhythmia triggered by high-intensity training and mitigating hemodynamic load related to high-intensity isometric exercises ([127](#)). The effect of exercise training for improving cardiac output may be limited due to the unique hemodynamic features of some congenital heart diseases. Thus, exercises focused on peripheral/skeletal muscle adaptations may be a specific target in this population ([131](#)). For example, lower-body resistance training has been found to improve ventricular filling and stroke volume in adolescents with univentricular cardiac physiology following Fontan palliation ([135](#)). Outpatient CR often requires patients to travel to a children's hospital facility due to the lack of CR facilities designed to accommodate children and adolescents or with expertise in congenital heart disease. The use of eHealth technologies for in-home delivery of developmentally appropriate pediatric CR is growing in popularity and may result in improved adherence and fewer school absences ([136](#), [137](#)).

FITT RECOMMENDATIONS FOR PEDIATRIC CR ([131](#))

	Aerobic	Resistance	Flexibility
Frequency	Minimally 2–3 d · wk ⁻¹	Minimally 2–3 nonconsecutive d · wk ⁻¹	Up to daily but as often as tolerated
Intensity	55%–80% of measured peak HR OR RPE 11–13 (most patients) and 15–16 (low-risk patients)	30%–70% of 1-RM Perform 10–15 repetitions of each exercise using a resistance that is 40%–70% of 1-RM Avoid Valsalva maneuver	To the point of feeling tightness or slight discomfort
Time	30 min per session gradually working up to 60 min per session	2–3 sets; 5–7 exercises per session; most complex movements first and a greater focus on lower-body exercises Rest periods between sets >60 s	20–30 s for static stretching
Type	Developmentally appropriate exercises such as walking, running, biking, swimming, and active play Greater focus on locomotor movement proficiency, balance, and coordination in younger children High-intensity interval training can be used in patients who are hemodynamically stable (active rest ratio of 1:2)	Developmentally and body size appropriate equipment or body weight exercises Tempo: 2 s eccentric/0 s isometric/2 s concentric Avoid excessive hand gripping and exercises with a high static component to mitigate exaggerated hemodynamic load	Static and dynamic stretches focused on major joints in limbs and lower back; focus on upper body Avoid breath holding during stretch Modifications if recent cardiac surgery (once cleared sternal precautions) or presence of epicardial pacemaker

- This exercise prescription represents general recommendations for youth with congenital heart disease. Considering the heterogeneity in pediatric and congenital heart diagnoses and age, exercise prescriptions for pediatric CR should be individualized based on the child's cardiac anatomy, relative risk, and developmental function.
- Exercise prescriptions should only be provided by exercise physiologists or healthcare providers with knowledge and experience in pediatric/congenital heart disease and exercise medicine. Exercise

prescription may require consultation with a pediatric cardiologist with expertise in exercise medicine.

INDIVIDUALS WITH A CEREBROVASCULAR ACCIDENT (STROKE)

When blood flow to a region of the brain is obstructed (*i.e.*, cerebrovascular accident, CVA, or stroke), brain function deteriorates quickly and leads to neuronal cell death ([138](#), [139](#)). This can result in motor, sensory, cognitive, and psychosocial impairments, the extent of which are greatly influenced by the size and location of the affected area and presence or absence of collateral blood flow ([140–142](#)). Eighty-seven percent of strokes are ischemic and 13% hemorrhagic ([1](#)). Each year, nearly 800,000 U.S. residents experience stroke, with females having a higher lifetime risk of stroke than males ([1](#), [143](#)). Average age of stroke in males and females is 67.5 and 71.8 years of age, respectively, with age of incident stroke declining ([144](#), [145](#)).

Rehabilitation services, such as physical and occupational therapy, are utilized poststroke to improve physical function such as mobility, balance, and engagement in ADLs ([146](#)). Exercise is recommended by the AHA/ASA to improve cardiovascular health and aerobic fitness and to optimize recovery ([146](#)). Poststroke, decreased cardiorespiratory fitness, fatigue, and sedentary behavior are common and exacerbate the functional decline observed in stroke ([146](#), [147](#)). Exercise engagement improves aerobic capacity and physical function such as walking endurance, walking speed, ADL engagement, reduces risk of a secondary cardiovascular event, and may improve fatigue and quality of life ([146–150](#)).

Exercise Testing

Compared to age- and sex-matched peers, cardiorespiratory fitness is lower in individuals poststroke ([151–153](#)). Factors such as motor ability, chronotropic

incompetence, medication use, and early-onset fatigue should be considered with exercise testing. Submaximal exercise testing may be preferred to maximal testing when motor or cognitive limitations are present.

- Exercise testing should employ a mode of testing that accommodates an individual's motor performance and cognitive ability ([146](#)).
- Cycle ergometry and dual action semirecumbent seated steppers and combo (upper and lower body) ergometers may be preferred to mitigate balance demand ([146](#), [154](#)). Modifications of the device (*e.g.*, pedal type, swivel seated, seated back, leg attachments for hemiparesis, hand foot straps) may be needed to facilitate safety and ease of use. The modified Total Body Recumbent Stepper Maximal (mTBRS) and TBRS Submaximal Exercise tests are valid methods for assessing cardiorespiratory fitness in stroke ([155](#), [156](#)).
- Standard walking treadmill test protocols, such as the Modified Bruce Protocol, may be used if an individual demonstrates sufficient balance and ambulates with minimal to no assist ([146](#)). Handrails and/or a body weight support system (*e.g.*, walking harness) may be used as necessary, including an exercise testing protocol with reduced incremental increases in incline and speed ([146](#), [157](#)). Gait belts, walkers, and other assistive devices can also be employed for overground gait assessments.

Exercise Prescription

Strong evidence exists to support exercise for individuals with history of stroke, as reported in the AHA/ASA Physical Activity and Exercise Recommendations for Stroke Survivors ([146](#)). The majority of individuals who experience a stroke also have multiple comorbidities including CVDs, arthritis, and metabolic disorders ([146](#), [158](#)). These comorbidities and their concomitant medications should be considered when prescribing exercise. Older adults who experience stroke may require additional testing and programming considerations (see [Chapter 5](#) and [Chapter 6](#)).

Exercise should occur across all phases of stroke to promote functional recovery and enhance cardiovascular health and physical function ([146](#)). Stroke phases include: acute (<1 week), subacute (1 week to 6 months), and chronic (≥ 6 months) ([159](#)). During the acute phase, out of bed activity and mobility are recommended, especially after 24 hours poststroke, and then a goal is to increase activity to mitigate decreases in physical function, strength, and muscle mass ([146](#), [160](#)). In subacute stroke, the focus is on furthering functional return and initiating an exercise program to regain or exceed prestroke fitness levels ([146](#)). Exercise programs should then be continued into the chronic phase of stroke recovery and progressed as necessary ([146](#)). A primary focus of exercise prescription poststroke should be to promote exercise self-efficacy for long-term continuation of exercise engagement outside of traditional rehabilitation ([146](#), [161](#)). The following FITT guidelines for Ex Rx are general recommendations for individuals with cerebrovascular disease.

Exercise Training Considerations

- HR monitoring and RPE can be used to ensure safety and reaching prescribed exercise intensity ([162](#)).
- If unable to conduct an exercise test, ACSM HR equations (see [Chapter 5](#) and [6](#)) can be used to guide the exercise prescription with the understanding that much variability in maximal HR exists among individuals.
- Avoid the Valsalva maneuver during resistance training to avoid excessive elevations in BP.
- Exercise modality should be chosen based on motor and cognitive ability and individual preference to promote long-term exercise engagement ([146](#), [161](#)).
- Consider adding neuromuscular training 2–3 times per week (*i.e.*, balance and coordination activities, yoga, and tai chi) to exercise programs to reduce fall risk, improve balance, physical function, and mental health ([146](#)).

All components of exercise training (aerobic, resistance, and neuromuscular training) are important in stroke recovery.

FITT RECOMMENDATIONS FOR INDIVIDUALS WITH A CEREBROVASCULAR ACCIDENT ([146](#))

	Aerobic	Resistance	Flexibility
Frequency	Minimally 3 d · wk ⁻¹ ; preferably up to 5 d · wk ⁻¹	2–3 d · wk ⁻¹ ; performed on nonconsecutive days	≥2–3 d · wk ⁻¹ with daily being most effective
Intensity	If using HR data from a recent GXT or HR _{max} prediction equation, use 40%–70% of HRR/ $\dot{V}O_2R$ or 55%–80% HR _{max} . If atrial fibrillation or chronotropic incompetence are present and/or continuous HR monitoring is not available, use RPE of 11–14 on a 6–20 scale	50%–80% of 1-RM	Stretch to the point of feeling tightness or slight discomfort
Time	Progressively increase from 20 to 60 min · d ⁻¹ . Consider multiple 10-min sessions	1–3 sets of 10–15 repetitions of 8–10 exercises	10–30 s hold for static stretching; 2–4 repetitions of each exercise
Type	Cycle ergometry and semirecumbent seated steppers; may need modification based on functional and cognitive demands. Treadmill and overground walking can be considered if individual has sufficient balance and ambulation with minimal to no assist	Use equipment and exercises that consider and improve safety in those with deficits: machine vs. free-weight, bar vs. hand weights; elastic bands; weight-bearing or partial weight-bearing; seated vs. standing as indicated	Static and/or dynamic

1-RM, one repetition maximum; GXT, graded exercise test; HR, heart rate; HRR, heart rate reserve; $\dot{V}O_2R$, $\dot{V}O_2$ reserve; HR_{max}, maximal HR; RPE, rating of perceived exertion.

Other Considerations

- Exercise should be initiated only after the individual is medically stable. Individuals at risk for exertion-related cardiovascular events should undergo exercise testing ([146](#)).
- Emerging evidence suggests that high-intensity interval training (HIIT) in chronic stroke may be effective for improving physical function, walking, and

cardiorespiratory fitness. However, specifics regarding optimal intensity are unknown, and more research is needed to provide exercise guidelines for HIIT ([150](#), [163](#), [164](#)).

- CVD risk factor reduction should be a focus of exercise prescription in stroke ([146](#)).
- Early-onset local muscle and general fatigue are common and should be considered when setting work rates and progression.
- Comprehensive stroke care involves being attentive to affective issues such as mood, motivation, frustration, and confusion. Correctly managing affective issues can favorably influence adherence and response to exercise. Strategies to minimize negative influences include close supervision, individualized instruction until independence is established, and involvement of family members.

Exercise Training for Return to Work

For individuals desiring to return to their previous vocation, the exercise plan should consider musculature used and workload required to perform the occupational tasks. A list of MET levels associated with a wide range of occupational tasks has been published and can be used to estimate required workload ([165](#)). Specificity of training should be employed with aerobic, resistance, and neuromuscular exercise to mimic and optimize movements required for a given vocation. Factors such as cardiovascular demand, muscle groups used, and work environment (altitude, temperature, noise level, etc.) should be considered.

PULMONARY DISEASES

Chronic pulmonary diseases are a significant cause of morbidity and mortality. Currently, more than 500 million people globally have chronic pulmonary disease, an increase of approximately 40% from 1990 ([166](#)). Characterized by debilitating symptoms of dyspnea and fatigue, chronic pulmonary diseases cause exertional intolerance, reduced health-related quality of life, and increased risk of hospitalization and mortality ([166](#)). Pulmonary rehabilitation (PR) is essential in the continuum of care in people with chronic pulmonary disease. There is strong evidence that PR meaningfully improves exercise tolerance, reduces symptoms, and improves quality of life ([167–171](#)). For adults with chronic obstructive pulmonary disease (COPD), PR is strongly recommended in those with stable disease and in those hospitalized for an exacerbation of the disease ([167](#)). Moreover, evidence-based recommendations ([170–172](#)) and clinical practice guidelines ([169, 173–176](#)) indicate that exercise training should be a mandatory component of PR for adults with COPD. Scientific rationale strongly supports exercise training in people with non-COPD respiratory diseases (*i.e.*, cystic fibrosis [CF], pulmonary hypertension [PH], idiopathic pulmonary fibrosis [IPF]) and confirms similar benefits as those seen in COPD ([167, 168, 177](#)). A key focus of exercise in PR is long-term behavior change to achieve continued participation in PA and exercise. Long-term participation in hospital- or home-based PR (*i.e.*, >3 months) may provide greater and longer lasting physiologic and behavioral benefits than traditional, shorter duration PR programs ([178–183](#)). Adjuncts that have the potential to improve exercise performance, symptoms, and quality of life in appropriate individuals include supplemental oxygen, bronchodilation techniques, breathing retraining techniques such as pursed lips breathing, specific inspiratory muscle training (IMT) (see *Respiratory Muscle Training* section), and use of assistive ambulatory devices (*i.e.*, rolling walkers) ([184](#)). Individuals whose physical limitations prevent them from

exercising at higher intensities can achieve significant improvement in aerobic conditioning with lower-intensity endurance training ([185](#)). Resistance exercise training has been shown to be essential to improve upper and lower body strength and muscle mass, ADLs, and health-related quality of life in individuals with chronic obstructive lung diseases ([170](#), [186–189](#)). More research is needed on the effect of both aerobic and resistive exercise training in individuals with restrictive and interstitial lung diseases (ILDs) such as pulmonary fibrosis, although recent trials have shown encouraging results ([190–192](#)). A list of respiratory diseases in which exercise is of potential benefit is shown in [Box 8.8](#).

Asthma

Asthma, which is a heterogeneous chronic inflammatory disorder of the airways, is estimated to affect 262 million people worldwide. Asthma is characterized by a history of episodic bronchial hyperresponsiveness; variable airflow limitation; and recurring wheeze, dyspnea, chest tightness, and coughing that occur particularly at night or early morning. These symptoms are variable and often reversible. Despite significant advances in the pharmacological management of asthma, even when pharmacotherapy is optimized, people with asthma still experience a high disease burden that limits PA and ADLs ([193](#), [194](#)). Compared to people without asthma, those with a diagnosis of asthma are less likely to achieve recommended PA levels ([195](#), [196](#)). While structured exercise training may provide some therapeutic benefit for people with asthma, it is clear that asthma symptoms can be provoked or worsened by exercise, which may contribute to reduced participation in PA or sports, ultimately leading to deconditioning and lower CRF. With deconditioning, the downward cycle or “spiral” continues with asthma symptoms being triggered by less intense PA and subsequent worsening of exercise tolerance.

Conclusive evidence for exercise training as an effective therapy for asthma is lacking, and at present, there are no specific evidence-based guidelines for

exercise training in these individuals. However, findings from randomized controlled trials have suggested that a step-based exercise prescription using a pedometer, a combined weight loss plus aerobic and resistance exercise training program, and high-intensity interval training can significantly improve PA in people with asthma ([197–199](#)). In addition, strong evidence is available for recommending regular PA because of its general health benefits and reduced incidence of exacerbations ([200](#)). In general, exercise training is well tolerated and safe in people with stable asthma ([201, 202](#)). A significant improvement in CRF from before to after exercise training is a somewhat consistent finding in people with asthma ([201, 202](#)), and it has been suggested that exercise training may positively affect health-related quality of life in these patients ([201, 203](#)). Improved lung function and increased days without asthma symptoms have been reported after exercise training, but this finding is not universal across previous findings ([201–203](#)). The data examined from these reviews are limited by small numbers of randomized controlled trials and heterogeneity of trial methods and individuals. Nevertheless, based on the available evidence, exercise training should be encouraged in people with stable asthma.

Exercise-induced bronchoconstriction (EIB), defined as airway narrowing that occurs because of exercise, is experienced in a substantial proportion of people with asthma ([204](#)), but people without a diagnosis of asthma may also experience EIB. For athletes, environmental triggers such as cold or dry air and air pollution including particulate matter, allergens, and trichloramines in swimming pool areas may stimulate a bout of EIB. EIB can be successfully managed with pharmacotherapy ([204](#)). Strong recommendations have also been made for 10–15 minutes of vigorous intensity or variable intensity (combination of light and vigorous intensity) warm-up exercise to induce a “refractory period” in which EIB occurrence is attenuated ([204, 205](#)).

Exercise Testing

- Assessment of physiologic function should include evaluations of cardiopulmonary capacity, pulmonary function (before and after exercise), and oxyhemoglobin saturation via noninvasive methods (*i.e.*, pulse oximetry).
- Administration of an inhaled bronchodilator (*i.e.*, β_2 -agonists) (see [Appendix A](#)) prior to testing may be indicated to prevent EIB, thus providing optimal assessment of cardiopulmonary capacity.
- Exercise testing is typically performed on a treadmill, or cycle ergometer. Targets for high ventilation and HRs are better achieved using the treadmill. For athletes, a sports-specific mode may be more relevant.
- The degree of EIB should be assessed using vigorous-intensity exercise achieved within 2–4 minutes and lasting 4–6 minutes with the individual breathing relatively dry air. The testing should be accompanied by a spirometric evaluation of the change in forced expiratory volume in one second (FEV_{1,0}) from baseline and the value measured at 5, 10, 15, and 30 minutes following the exercise test ([204](#)). The criterion for a diagnosis of EIB varies, but many laboratories use a decrease in FEV_{1,0} from baseline of $\geq 15\%$ because of its greater specificity ([204](#)). As it is more effort dependent for the individual, peak expiratory flow rate is *not* a reliable marker of EIB.
- Appropriately trained staff should supervise exercise tests for EIB, and physician supervision may be warranted when testing higher-risk individuals because severe bronchoconstriction is a potential hazard following testing. Immediate administration of nebulized bronchodilators with oxygen is usually successful for relief of bronchoconstriction ([206](#), [207](#)).
- Although exercise testing is considered highly specific for detecting EIB, when it is unavailable or unfeasible, surrogate tests to evaluate airway's hyperresponsiveness include eucapnic voluntary hyperventilation of dry air, inhalation of hyperosmolar aerosols of 4.5% saline, dry powder mannitol, or methacholine ([206](#)). These tests should be administered by appropriately trained individuals with medical supervision.

- Procedural details for EIB diagnostic testing have been described ([204](#), [206](#)). Although none of these surrogate tests are 100% sensitive or specific for EIB, they are useful in identifying airway hyperresponsiveness.
- Evidence of oxyhemoglobin (SpO_2) desaturation $\leq 80\%$ should be used as test termination criteria in addition to standard criteria, although individual laboratories may choose a lower threshold as an indication to stop the test (*e.g.*, SpO_2 85%) ([208](#)).
- The 6-minute walk test (6-MWT), incremental (ISWT), or endurance (ESWT) shuttle walk tests may be used in individuals with moderate-to-severe persistent asthma when other testing equipment is not available ([175](#), [209](#)).

Exercise Prescription

Exercise training is generally well tolerated in asthmatic individuals successfully managed with pharmacotherapy when triggers to bronchoconstriction (*e.g.*, cold; dry, dusty air; inhaled pollutants) are removed to bring about symptom relief ([201](#)). As such, the general FITT recommendations for comprehensive exercise in healthy adults, adjusted to individual capabilities, are suitable (see [Chapter 5](#)). Position statements on exercise in asthma ([210](#)) and systematic reviews ([201](#)) support this recommendation.

Special Considerations

- Caution is suggested in using target HR based on prediction of maximal heart rate (HR_{max}) because of the wide variability in its association with ventilation and the potential HR effects of asthma control medications.
- Individuals experiencing exacerbations of their asthma should not exercise until symptoms and airway function have improved.
- Use of short-acting bronchodilators may be necessary before or after exercise to prevent or treat EIB (see [Appendix A](#)).
- Individuals on prolonged treatment with oral corticosteroids may experience cachexia of peripheral muscle and may therefore benefit from resistance

training.

FITT RECOMMENDATIONS FOR INDIVIDUALS WITH ASTHMA (201, 210)

	Aerobic	Resistance	Flexibility
Frequency	Minimally $3 \text{ d} \cdot \text{wk}^{-1}$; preferably up to $5 \text{ d} \cdot \text{wk}^{-1}$	At least $2 \text{ d} \cdot \text{wk}^{-1}$ performed on nonconsecutive days	$\geq 2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$ with daily being most effective
Intensity	Begin with moderate intensity (40%–59% HRR or $\dot{V}\text{O}_2\text{R}$). If well tolerated, progress to 60%–70% HRR or $\dot{V}\text{O}_2\text{R}$ after 1 mo	Strength: 60%–70% of 1-RM for beginners; $\geq 80\%$ for experienced weight trainers Increase workload by 2%–10% if one to two repetitions over the desired number are well tolerated on two consecutive training sessions Endurance: $<50\%$ of 1-RM	Stretch to the point of feeling tightness or slight discomfort
Time	Progressively increase to at least $30\text{--}40 \text{ min} \cdot \text{d}^{-1}$	Strength: 2–4 sets, 8–12 repetitions Endurance: ≤ 2 sets, 15–20 repetitions	10–30 s hold for static stretching; 2–4 repetitions of each exercise
Type	Aerobic activities using large muscle groups such as walking, running, cycling, swimming, or pool exercises	Weight machines, free weight, or body weight exercises	Static, dynamic, and/or PNF stretching

1-RM, one repetition maximum; HRR, heart rate reserve; PNF, proprioceptive neuromuscular facilitation; $\dot{V}\text{O}_2\text{R}$, oxygen uptake reserve.

- Exercise in cold environments or in the presence of airborne allergens or pollutants should be limited to avoid triggering bronchoconstriction in

susceptible individuals. EIB can also be triggered by prolonged exercise durations or high-intensity exercise sessions.

- IMT has been shown to facilitate significant increases in inspiratory muscle strength; however, whether IMT confers other clinical benefits (*e.g.*, improved exercise capacity, improved health-related quality of life, decreased health care use) remains unclear ([211](#), [212](#)).
- Use of a nonchlorinated pool is preferable because this will be less likely to trigger an asthma event.
- Be aware of the possibility of asthma exacerbation shortly after exercise, particularly in a high-allergen environment.
- Exercise prescription combined with education centered on the benefits and safety of PA and exercise, patient self-monitoring during exercise, and how to best prevent debilitating asthma-related symptoms during PA and exercise may maximize uptake and adherence to exercise programming in people with asthma ([213](#)).

Chronic Obstructive Pulmonary Disease

Globally, almost 12% of people have COPD, which is a major cause of chronic morbidity and is the third leading cause of death worldwide, accounting for approximately 6% of total deaths. COPD is preventable and treatable. It is characterized by predisposing risk factors resulting in chronic airway inflammation chiefly due to exposure to noxious gases and particles, especially tobacco smoke and various environmental and occupational exposures. Dyspnea, chronic cough, and sputum production are common symptoms. Significant systemic effects such as weight loss, nutritional abnormalities, sarcopenia, skeletal muscle dysfunction, vascular dysfunction, PAD, and pulmonary vascular disease often accompany COPD ([169](#), [214–218](#)). COPD encompasses chronic bronchitis and/or emphysema. The severity of lung function impairment, and specifically the severity of airflow obstruction, can be graded based on either percent predicted or z-score values for forced expiratory volume in 1 second (FEV₁) ([Table 8.5](#)) ([219](#), [220](#)). Individuals may

be classified according to disease severity based on pulmonary function tests and Global Initiative for Chronic Obstructive Lung Disease (GOLD) criteria, including level of symptoms and exacerbation frequency ([219](#)).

TABLE 8.5 Global Initiative for Chronic Obstructive Lung Disease (GOLD) and American Thoracic Society (ATS)/European Respiratory Society (ERS) Classification of Airflow Obstruction in Individuals With Chronic Obstructive Pulmonary Disease Based on the FEV_{1.0} Obtained From Pulmonary Function Tests ([219](#), [220](#))

Severity of Airflow Obstruction	GOLD ^a	ATS/ERS
	FEV _{1.0} (% predicted)	FEV _{1.0} (z-score)
Mild	≥80% predicted	−1.65 to −2.5
Moderate	50% ≤ FEV _{1.0} < 80% of predicted	−2.51 to −4.0
Severe	30% ≤ FEV _{1.0} < 50% of predicted	<−4.0
Very Severe	FEV _{1.0} < 30% of predicted	

^a Based on postbronchodilator FEV_{1.0} when postbronchodilator FEV_{1.0}/FVC is <0.70. FEV_{1.0}, forced expiratory volume in 1 s; FVC, forced vital capacity.

Dyspnea or shortness of breath (SOB) with exertion is a cardinal symptom of COPD resulting in PA limitations and deconditioning. Disuse muscle atrophy is common in individuals with COPD because of the adverse downward spiral of increasing ventilatory limitations, SOB, and further decreases in PA. This contributes to the loss of muscle strength, power, and endurance and decrements in the performance of everyday functional activities. Exercise is an effective and potent intervention that can improve symptoms, lessen the development of functional impairment and disability, and increase quality of life in all individuals with COPD regardless of disease severity ([169](#), [171](#)). In

adults with stable COPD, meta-analysis of the available data from randomized controlled trials suggests that, compared to control, PR increases exercise capacity (*e.g.*, 6-MWT and ISWT distance, peak work capacity), decreases dyspnea, and improves health-related quality of life to an extent approaching or exceeding the minimally important clinical difference for each of the outcome variables ([167](#)). Similar improvements have been noted following PR in people after hospitalization for a COPD-related exacerbation, along with consistent evidence that PR reduces health care subsequent use in such individuals ([167](#)). It has also been noted that early adoption of PR after an acute COPD exacerbation (*within 90 days of hospital discharge*) may be associated with a lower risk of mortality at 1 year ([221](#)). The beneficial effects of exercise occur primarily through adaptations in the musculoskeletal and cardiovascular systems that in turn reduce stress on the pulmonary system during exercise ([222](#)).

Exercise Testing

- Exercise testing (*e.g.*, treadmill, cycle ergometer, 6-MWT) has multiple purposes in the assessment of individuals with chronic pulmonary disease, including COPD. These purposes include quantifying exercise capacity prior to PR entry ([170](#), [175](#), [208](#), [223](#)), establishing a baseline for outcome documentation, evaluating drug treatment efficacy, assisting in the development of the Ex R_x ([170](#), [185](#)), evaluating unexplained dyspnea and exercise intolerance, and prognostic evaluation for individual risk stratification. Ideally, all individuals who enter a PR program should undergo at least one exercise assessment evaluation prior to PR entry (*e.g.*, cardiopulmonary exercise test, 6-MWT, or shuttle walk test) ([175](#), [176](#), [208](#)).
- Evidence-based guidelines confirm the utility of CPET in adults with COPD as well as other chronic lung diseases (*i.e.*, ILD, primary PH, and CF) in providing an objective measure of exercise capacity, mechanisms of exercise intolerance, prognosis, disease progression, and treatment response ([207](#), [208](#), [224](#)).

- Incremental exercise tests (*e.g.*, graded exercise test [GXT] on a treadmill or cycle ergometer) may be used to assess cardiopulmonary function and CRF. Modifications of traditional protocols (*e.g.*, smaller work rate increments) may be warranted depending on functional limitations and the onset of dyspnea. A test duration of 8–12 minutes is optimal in those with mild-to-moderate COPD ([225](#)), whereas a test duration of 5–9 minutes is recommended for individuals with severe and very severe disease ([207](#)).
- Individuals with moderate-to-severe COPD may exhibit oxyhemoglobin desaturation with exercise. Therefore, a measure of blood oxygenation, either the partial pressure of arterial oxygen (P_aO_2) or percent saturation of arterial oxygen (SaO_2), should be made periodically during the initial GXT and during any follow-up GXTs to help determine degree of improvement or decline in peripheral blood oxygenation. Treadmill-based exercise testing may be favorable over cycle ergometer-based exercise testing for identifying and quantifying the degree of exercise-induced oxyhemoglobin desaturation in people with COPD ([226](#)).
- Resting SpO_2 of $\leq 85\%$ (*breathing room air*) should be considered a relative contraindication to exercise testing ([208](#)). A $\geq 5\%$ decrease in SpO_2 during exercise suggests hypoxemia. Evidence of oxyhemoglobin desaturation $\leq 80\%$ should be used as test termination criteria in addition to standard criteria ([223](#)), although individual laboratories may choose a lower threshold as an indication to stop the test (*e.g.*, SpO_2 85%) ([208](#)).
- Submaximal exercise testing may be used depending on the reason for the test and the clinical status of the individual. However, individuals with pulmonary disease may have ventilatory limitations to exercise; thus, prediction of $\dot{V}O_{2peak}$ based on age-predicted HR_{max} may not be appropriate as criteria for terminating the submaximal GXT.
- A constant work rate (CWR) test using 80%–90% of peak work rate achieved from the GXT is appealing, as it assesses the type of work-related activity levels likely to be encountered in everyday life ([227](#)) particularly when performed on a treadmill.

- The measurement of flow volume loops during the GXT using commercially available instruments may help identify individuals with expiratory airflow limitation and/or dynamic lung hyperinflation and associated increased dyspnea. Use of bronchodilator therapy may be beneficial for such individuals ([170](#)).
- Exertional dyspnea is a common consequence of several different pulmonary diseases. The modified Borg Category-Ratio 0–10 (CR10) scale ([Figure 8.1](#)) ([228](#)) has been used extensively to measure dyspnea before, during, and after exercise ([229](#)). Individuals should be given specific, standardized instructions on how to relate the wording on the scale to their level of breathlessness ([209](#), [230](#)). Because dyspnea scales are subjective, some caution is advised in their interpretation as exercise intolerance may be accompanied by exaggerated dyspnea scores without corresponding physiological confirmation.

0	No breathlessness at all
0.5	Very, very slight (just noticeable)
1	Very slight
2	Slight breathlessness
3	Moderate
4	Somewhat severe
5	Severe breathlessness
6	
7	Very severe breathlessness
8	
9	Very, very severe (almost maximal)
10	Maximal

Figure 8.1 Borg Category-Ratio 0–10 scale modified for dyspnea.

- The 6-MWT is a widely used exercise assessment tool for cardiorespiratory function in PR ([175](#), [209](#), [230](#)). The test is safe, easy to administer, involves the use of minimal technical resources, is well tolerated, and accurately reflects walking abilities. To obtain valid and reliable results, it is essential to standardize the test procedure (*i.e.*, staff, exercise track or hallway

distance/configuration, individual instructions, verbal reinforcement used during testing, type, and flow rate of supplemental oxygen, walking aides, and number of trials) ([175](#), [209](#), [231](#)). The minimal clinically important difference in 6-MWT distance has been reported to be a mean of 30 m (98.42 ft) in people with COPD ([175](#), [209](#)).

- ISWT and ESWT are also used to assess cardiopulmonary function in individuals with chronic lung disease ([232–235](#)). The ISWT is an incremental, symptom-limited walk test that simulates a symptom-limited CPET ([236](#)). This test measures a symptom-limited walking distance over a marked walking course of 10 m (33 ft). This distance correlates well with $\dot{V}O_{2peak}$ in individuals with chronic lung disease ([175](#), [237](#)) and has been shown to be a reliable, valid, and responsive measure of estimated functional capacity in ILD ([238](#)) and asthma ([239](#)). The ISWT utilizes an audible pacing timer to incrementally increase the pacing frequency. The individual walks according to the pacing timer frequency until they are too breathless to continue or cannot keep pace with the external pacing signal. Like the 6-MWT, the primary test result of the ISWT is the total distance walked. The ESWT is a derivative of the ISWT, where the individual walks as long as possible at a predetermined percentage of maximum walking performance assessed by the ISWT on a subsequent day of testing ([235](#)). For this test, the outcome measure is total time walked (minute).
- The exercise testing mode is typically either walking or stationary cycling. Walking protocols may be more suitable for individuals with severe disease who lack the muscle strength to overcome the increasing resistance of cycle leg ergometers.
- Although arm ergometry is a good adjunct to aerobic weight-bearing exercises, it should be used with caution in individuals with COPD because it can result in increased dyspnea that may limit the intensity and duration of the activity.

Exercise Prescription

Presently, there are no evidence-based guidelines that describe the specific application of the FITT principle for individuals with COPD, although expert reviews, official statements, and clinical practice guidelines for the components of the FITT principle have been published ([169–171](#), [173](#)) and tend to be in general agreement.

Aerobic exercise training is recommended for individuals in all stages of COPD who can exercise ([170](#), [173](#)). Pulmonary diseases and their treatments affect both the lungs and skeletal muscles (*i.e.*, limb muscle dysfunction due to atrophy and weakness) ([240–242](#)). Resistance training is the most potent intervention to address the muscle dysfunction seen in COPD and should be an integral part of the Ex R_x ([169](#), [170](#), [173](#), [243](#), [244](#)). The effects of resistance training on disease outcome and pulmonary function are not yet well understood in individuals with chronic lung diseases. However, limited evidence from a systematic review and meta-analysis on resistance training outcomes in individuals with COPD demonstrated improvements in forced vital capacity (FVC) and peak minute ventilation ($\dot{V}E_{peak}$) but not FEV_{1.0} ([245](#)).

Of paramount concern is the common observation of falls in people with COPD ([246](#), [247](#)). Because muscle weakness, gait, and balance abnormalities are among the risk factors for falling, lower extremity muscle strengthening and balance training are effective countermeasures. As such, functional exercises for balance, posture, and proper gait should be incorporated into PR exercise training sessions.

Exercise Training Considerations

- Higher intensities yield greater physiologic benefits (*e.g.*, reduced $\dot{V}E$ and HR at a given workload) and should be encouraged when appropriate ([171](#), [173](#)). Although limited, there is evidence that the increase in exercise capacity (*i.e.*, 6-MWT distance) after PR is greater in people with COPD who are capable of developing quadriceps muscle fatigue during a single exercise session compared to those who are not ([248](#)).

- For individuals with mild COPD, intensity guidelines for healthy older adults are appropriate (see [Chapter 6](#)). For those with moderate-to-severe COPD, intensities representing less than 60% peak work rate have been recommended ([170](#)).
- Light-intensity aerobic exercise is appropriate for those with severe COPD or very deconditioned individuals. Intensity may be increased as tolerated within the target time window.
- Interval training may be an alternative to standard continuous endurance training for those who have difficulty in achieving their target exercise intensity due to dyspnea, fatigue, or other symptoms ([240](#), [249](#)). Interval training is a modification of endurance training in which higher-intensity exercise is interspersed with periods of rest or lower-intensity exercise ([170](#)). Several randomized, controlled trials ([249–252](#)) and systematic reviews ([253](#), [254](#)) have found no clinically important differences between interval and continuous training protocols in exercise capacity, health-related quality of life, and skeletal muscle adaptations following training. Thus, individual characteristics will warrant the use of either interval or continuous exercise training protocols.
- Supervision at the outset of training allows guidance in correct execution of the exercise program, enhanced safety, and optimized benefit ([255](#)).
- Ventilatory limitation at peak exercise in individuals with severe COPD coincides with significant metabolic reserves during whole body exercise ([256](#)). This may allow these individuals to tolerate relatively high work rates that approach peak levels ([169](#)) and achieve significant training effects.
- As an alternative to using peak work rate or $\dot{V}O_{2peak}$ to determine exercise intensity, dyspnea ratings of between 3 and 6 on the Borg CR10 scale may be used (see [Figure 8.1](#)) ([170](#), [257](#)). A dyspnea rating between 3 and 6 on the Borg CR10 scale has been shown to correspond with 53% and 80% of $\dot{V}O_{2peak}$, respectively. Most individuals with COPD can accurately and reliably produce a dyspnea rating obtained from an incremental exercise test as a target to regulate/monitor exercise intensity ([257](#)).

- Intensity targets based on percentage of estimated HR_{max} or HRR may be inappropriate ([258](#)). Particularly in individuals with severe COPD, resting heart rate (HR_{rest}) is often elevated, and ventilatory limitations as well as the effects of some medications prohibit attainment of the predicted HR_{max} and thus its use in exercise intensity calculations.
- The use of oximetry is recommended for the initial exercise training sessions to evaluate possible exercise-induced oxyhemoglobin desaturation and to identify the workload at which desaturation occurred.
- Flexibility exercises may help overcome the effects of postural impairments that limit thoracic mobility and therefore lung function ([170](#)).
- Regardless of the prescribed exercise intensity, the exercise professional should closely monitor initial exercise sessions and adjust exercise intensity and duration according to individual responses and tolerance. In many cases, the presence of symptoms, particularly dyspnea/breathlessness, supersedes objective methods of Ex R_x .

Special Considerations

- Peripheral muscle dysfunction contributes to exercise intolerance ([240](#)) and is significantly and independently related to increased use of health care resources ([261](#)), poorer prognosis ([262](#)), and mortality ([263](#)), which emphasizes the importance of strength training in these individuals.
- Maximizing pulmonary function using bronchodilators before exercise training in those with airflow limitation can reduce dyspnea and improve exercise tolerance ([170](#)).
- Because individuals with COPD may experience greater dyspnea while performing ADLs involving the upper extremities, include resistance exercises for the muscles of the upper body.
- Supplemental oxygen is indicated for individuals with a $P_aO_2 \leq 55$ mm Hg or an $SpO_2 \leq 88\%$ while breathing room air ([264](#)). This recommendation applies when considering supplemental oxygen during exercise. In individuals using ambulatory supplemental oxygen, flow rates will likely

need to be increased during exercise to maintain SpO₂ levels greater than 88% ([169](#), [170](#), [265](#)).

FITT RECOMMENDATIONS FOR INDIVIDUALS WITH CHRONIC

OBSTRUCTIVE PULMONARY DISEASE

([169](#)–[171](#), [259](#), [260](#))

	Aerobic	Resistance	Flexibility
Frequency	Minimally 3 d · wk ⁻¹ ; preferably up to 5 d · wk ⁻¹	At least 2 d · wk ⁻¹ performed on nonconsecutive days	≥2–3 d · wk ⁻¹ with daily being most effective
Intensity	Moderate-to-vigorous intensity (50%–80% peak work rate as tolerated or 3–6 on the Borg CR10 scale)	Strength: 60%–70% of 1-RM for beginners; ≥80% for experienced weight trainers. Increase workload by 2%–10% if one to two repetitions over the desired number are well tolerated on two consecutive training sessions Endurance: <50% of 1-RM	Stretch to the point of feeling tightness or slight discomfort
Time	Initially 10–15 min · d ⁻¹ for first 3–5 exercise sessionsProgressively increase to 20–60 min · d ⁻¹ at moderate-to-high intensities as tolerated. If the 20- to 60-min durations are not achievable, accumulate ≥20 min of exercise interspersed with intermittent exercise rest periods of lower-intensity work or rest	Strength: 2–4 sets, 8–12 repetitions Endurance: ≤2 sets, 15–20 repetitions	10–30 s hold for static stretching; 2–4 repetitions of each exercise
Type	Common aerobic modes including walking (free or treadmill), stationary cycling, and upper body ergometry	Weight machines, free weight, resistance bands, or body weight exercises	Static, dynamic,

	Aerobic	Resistance	Flexibility
			and/or PNF stretching

1-RM, one repetition maximum; PNF, proprioceptive neuromuscular facilitation.

- Individuals suffering from acute exacerbations of their pulmonary disease should limit exercise until symptoms have subsided.

Exercise Training for Pulmonary Diseases Other Than Chronic Obstructive Pulmonary Disease

Despite substantially less investigation into the benefits of exercise training in non-COPD chronic lung diseases, strong scientific evidence supports the inclusion of exercise training for many lung diseases and chronic conditions other than COPD with demonstrated clinical and physiologic benefits ([168](#), [174](#), [177](#), [190](#), [191](#), [266](#), [267](#)). Methods for adapting exercise training in individuals with non-COPD chronic lung disease have been published ([174](#)), and each program should be modified to include disease-specific strategies based on currently available evidenced-based guidelines.

Pulmonary Arterial Hypertension

PH is a global term for elevated pulmonary arterial pressure. Established guidelines define PH as invasively derived mean pulmonary arterial pressure (mPAP) at rest of ≥ 25 mm Hg, although recently reviewed criteria suggest a diagnostic cut-off of mPAP greater than 20 mm Hg combined with a pulmonary vascular resistance (PVR) ≥ 3 Wood Units ([268](#)). Clinically, PH is classified according to five groups: Group 1, pulmonary arterial hypertension (PAH); Group 2, PH associated with left heart disease; Group 3, PH associated with lung diseases and/or hypoxia; Group 4, PH associated with pulmonary artery obstructions; and Group 5, PH with unclear and/or multifactorial mechanisms

([269](#)). PAH is a complex, progressive, and often devastating disease of the pulmonary vasculature. It negatively affects pulmonary perfusion and causes excessive right ventricular (RV) afterload, which can lead to RV dysfunction and failure. A consequent reduction in exercise, and eventually resting, cardiac output combined with arterial hypoxemia, sympathetic hyperactivity, endothelial dysregulation, and skeletal muscle dysfunction with impaired aerobic capacity causes progressive fatigue, dyspnea, syncope, and exertional intolerance, even to low-intensity activities. In individuals with PAH, pulmonary arterial stiffness and vasodilatory dysfunction increase vascular resistance during acute exercise, resulting in impaired RV stroke volume and cardiac output, as well as chronotropic incompetence in older individuals ([270](#)).

Despite optimized medical treatment, most people with PAH still experience reduced health-related quality of life and disease progression ([271](#)), with pharmacotherapy largely unable to normalize pulmonary vascular pressures or reverse the degradation in RV function. In people with PAH, pulmonary pressures can increase suddenly and dramatically during exercise, predisposing them to RV decompensation and cardiovascular collapse ([272](#)). Additionally, there has been concern that increased cardiac output and pulmonary blood flow during exercise may engender pulmonary vascular wall shear stresses that trigger pulmonary vascular remodeling and disease worsening. Due to these “risks,” people with PAH have been historically discouraged from exercise training.

Despite these concerns, it was shown that low-dose exercise training significantly improved 6-MWT distance, WHO functional class, health-related quality of life, and cardiorespiratory fitness (*i.e.*, $\dot{V}O_{2peak}$) in people with severe chronic PH and right HF ([273](#)). A continually growing body of evidence suggests that supervised exercise training consistently and meaningfully improves exercise capacity and health-related quality of life, may reduce pulmonary vascular pressures, and appears to do so without increased risk of serious adverse events in PAH patients who are clinically stable and receiving optimal pharmacological treatment ([167](#), [266](#), [269](#), [274](#)). Updated clinical

guidelines advise that a supervised and closely monitored exercise program in specialized clinics (*i.e.*, *PR*) should be recommended for stable PAH patients ([167](#), [269](#), [275](#)). However, the optimal exercise prescription for those with PAH remains unknown. In general, the following exercise guidelines may be applicable or appropriate for *all* major PH classifications, except where noted ([270](#), [276](#), [277](#)):

- Due to cardiac remodeling and RV failure often associated with PAH, continuous telemetry ECG monitoring may be useful for detection of high-grade ventricular ectopy or inappropriate bradyarrhythmias.
- Oxygen supplementation may be warranted in many individuals to keep exercise SpO₂ levels greater than 90% due to the potential for hypoxemia, which can further increase pulmonary artery pressures, potentially leading to complex arrhythmias or circulatory collapse.
- Low-intensity aerobic exercise consisting of treadmill, level surface walking, recumbent stepper, and arm or cycle ergometry should be employed. The optimal frequency for aerobic exercise sessions is 5 or more d · wk⁻¹ (including home exercise) to produce significant gains in overall functional capacity. There is no clear guidance on how aerobic exercise intensity should be progressed. It has been suggested that aerobic exercise training targeting 40%–70% of HRR (RPE 11–14 on the Borg 6–20 scale) is achievable in people with PH ([278](#)). However, it should be noted that exercise intensity was limited according to an absolute measure of HR (not more than 120 beats · min⁻¹) in some prior investigations of exercise training in people with PH ([273](#), [279](#), [280](#)). Also, a decreased chronotropic response to exercise has been reported in people with PAH ([281](#), [282](#)), which may need to be considered when prescribing exercise intensity based on THR.
- At first, patients may be unwilling to exert themselves to target exercise intensity due to fear that it will be harmful. One-on-one coaching into an exercise program may help overcome learnt fear of activity. Fear avoidance and anxiety about activities that bring on breathlessness may be barriers

that need to be addressed when prescribing exercise in people with PAH ([278](#)).

- BP, HR, and SpO₂ should be closely monitored during each exercise session.
- Pacing and energy conservation are vitally important for the individual with PAH during PR and home exercise sessions.
- Resistive exercise that utilizes body weight and/or light dumbbells may be sufficient to complement aerobic training regimens. However, machine weights, bands, and tubes have been shown to be safe in those with PAH under proper supervision and may be incorporated on a person-by-person basis.
- Although the 6-MWT is the most common functional capacity assessment technique for those with PAH, CPET provides a wealth of information for individual prognosis and exercise prescription.
- Submaximal CPET can help identify the presence and severity of PAH ([283](#)) and may also have utility in Ex R_x.
- IMT may be an effective adjunct to PR in people with PH, decreasing ambulatory BP and dyspnea, and enhancing inspiratory muscle strength and exercise capacity ([284](#)).

Interstitial Lung Disease

ILD encompasses a group of disorders characterized by variable degrees of fibrosis and inflammation (or both) in the alveolar compartments of the lung parenchyma. These disorders include IPF, asbestosis, sarcoidosis, and drug-induced pneumonitis, among other related fibrotic lung conditions. Common symptoms of ILD include dry cough, exertional dyspnea, hypoxemia, and exercise intolerance. In contrast to COPD, individuals with ILD typically have low lung volumes and reduced diffusing capacity, resulting in a rapid, shallow breathing pattern. These individuals oftentimes require oxygen supplementation due to impaired gas exchange, as their diffusing capacity diminishes, particularly during exercise. The combination of these detrimental cardiopulmonary changes results in increased dead space ventilation and

respiratory rate, peripheral muscle dysfunction, exertional hypoxemia, and an overall diminished health-related quality of life ([285](#), [286](#)).

Exercise guidelines have been presented for those with ILD ([190–192](#)):

- The FITT guidelines are similar to those for COPD, although moderate-intensity aerobic exercise should comprise the core component of the exercise program.
- Exercise intensity should be below that which provokes severe dyspnea, oxygen desaturation, or hypertension.
- A high fraction of inspired oxygen (FIO₂) may be required during exercise training due to exertional hypoxemia.
- A strong focus on pacing and energy conservation techniques should be incorporated into the exercise prescription as well as preexercise use of bronchodilators and gradual warm-up.
- As with most chronic lung conditions, CPET is valuable in these individuals due to the complex, multifactorial basis of exercise limitations of ILD.

Cystic Fibrosis

CF is a hereditary disease that affects the lungs and digestive system where the body produces excessively thick, viscous mucus that clogs the lungs and obstructs the pancreas. CF leads to major morbidity with symptoms of cough, sputum production, dyspnea, intermittent hemoptysis, exercise intolerance, functional impairment, and decreased quality of life. Although there is no cure for CF, regular exercise and increased levels of PA have been shown to be beneficial for individual with CF. Higher levels of physical fitness have been associated with better survival rates in individuals with CF ([287](#)). Exercise programming used in individuals with COPD is applicable to those with CF before and after lung transplantation ([288](#)) when modifications are adapted to the individual's exercise tolerance. Maintenance of adequate nutrition, regular use of airway clearance techniques, pacing/energy conservation, and infection control should be a priority for the individual with CF. Specific CF exercise

prescription guidelines for children, adolescents, and adults have been presented ([289](#)).

Lung Transplantation

Lung transplantation is an effective treatment option for select patients with end-stage pulmonary disease, generally facilitating improvements in lung function, exercise capacity, and health-related quality of life that often exceed the effects of other treatments for end-stage lung disease ([290](#), [291](#)). Exercise training is increasingly recognized as a key component in the continuum of care for lung transplant candidates and recipients. Both before and after lung transplant, exercise training can maintain or improve functional exercise capacity ([292](#)), with improvements in 6-MWT distance often greater than the minimally important clinical difference reported in patients with chronic lung diseases ([293–295](#)). For example, Ochman et al. reported that 12 weeks of Nordic walking-based exercise resulted in approximately 63-m increase in 6-MWT distance in patients referred for lung transplantation; this contrasted the *decrease* in 6-MWT distance (~57 m) observed in the comparator control group who did not take part in exercise training ([296](#)). Similarly, Langer et al. found a significantly greater increase in 6-MWT distance (*expressed as % of predicted*) in lung transplant recipients who undertook 3 months of exercise training compared to a control group ([297](#)). Given that exercise capacity is an important predictor of pretransplant waiting list survival and is associated with posttransplant outcomes including length of hospital stay and survival ([288](#), [298–301](#)), exercise training before (*i.e., prehabilitation*) and/or after (*i.e., rehabilitation*) surgery can help individuals to better tolerate the rigors of the transplant process and enhance postoperative recovery and outcomes ([170](#), [292](#), [300](#), [302](#), [303](#)).

Currently, there are no standardized, evidence-based guidelines for exercise-based rehabilitation for lung transplant candidates and recipients. However, evidence-informed recommendations for exercise-based prehabilitation and rehabilitation in such individuals have been presented ([170](#), [302](#)). The stability

and clinical course of the underlying lung disease along with factors such as respiratory and cardiovascular reserve and muscle strength and endurance can have substantial impact on the frequency, intensity, type, duration, and rate of progression of prescribed exercise ([302](#)). Thus, comprehensive assessment of exercise capacity (*e.g.*, CPX, 6-MWT), peripheral and respiratory muscle strength and endurance, and physical performance and mobility (*e.g.*, gait speed, short physical performance battery, balance tests) is important for optimizing aerobic and resistance exercise programming in lung transplant candidates and recipients. Evidence-informed recommendations for exercise training include the following ([302](#)):

Lung Transplant Candidates (Prehabilitation)

- Aerobic exercise should be performed at least $2 \text{ d} \cdot \text{wk}^{-1}$, preferably up to $5 \text{ d} \cdot \text{wk}^{-1}$.
- Aerobic exercise intensity of 50%–80% HRR, 3–5 on the modified Borg Category-Ratio 0–10 (CR10) scale can be targeted. 60%–80% of 6-MWT speed can also be used.
- Supplemental O_2 may be needed to maintain SpO_2 greater than 85%–88%. Guidelines for O_2 supplementation for exercise are not clearly defined. As such, institutional policies/guidelines should be followed.
- Walking and/or leg or arm cycling exercise modalities can be used.
- Aerobic exercise can be performed for 15–30 minutes (*continuous*) or for 5–10 minutes $\times$ 2–3 bouts (*intermittent*).
- Resistance exercise should be performed on $2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$ on nonconsecutive days; 1–2 sets of 8–15 repetitions at 30%–80% of 1-RM and targeting the major muscle groups of the upper and lower body is advised.

Lung Transplant Recipients (Rehabilitation)

- Aerobic exercise should be performed at least $3 \text{ d} \cdot \text{wk}^{-1}$, preferably up to $5 \text{ d} \cdot \text{wk}^{-1}$.

- Aerobic exercise intensity of 50%–80% HRR, 3–5 on the modified Borg Category-Ratio 0–10 (CR10) scale can be targeted (75%–100% of 6-MWT speed can also be used). Intensive aerobic or resistive exercise should be avoided for 4–6 weeks posttransplant.
- Walking and/or leg cycling exercise modalities can be used. Arm ergometry may need to be avoided for up to 3 months to allow incision healing.
- Aerobic exercise can be performed for 20–30 minutes (*continuous*).
- Resistance exercise should be performed on 2–3 d · wk⁻¹ on nonconsecutive days: 1–3 sets of 8–15 repetitions at 50%–80% of 1-RM.
- In many settings, it is suggested that upper extremity lifting/pushing/pulling greater than 10 lb should be avoided in the first approximately 10–12 weeks after transplant, especially in the setting of sternal instability ([302](#)); however, accumulating evidence suggests that general poststernotomy movement strategies including modified sternal precautions and Keep Your Movement in the Tube allow safe, increased levels of lifting and activity that promotes improved functional status and confidence ([85](#), [304](#)). Reference and adherence to local institutional policy and surgeon guidance is advised.

Additional Considerations

- Pretransplant, individuals should exercise close to the highest workload that they can tolerate from the standpoint of dyspnea and fatigue.
- Exercise should be closely supervised to ensure that the prescribed workload can be safely tolerated to the level of intensity that will have a beneficial effect.
- Exercise should be continued up to the time of surgery in a PR program, complimented by home exercise.
- Institutional infection control procedures should be followed. This is essential for preventing the spread of certain infections (*e.g.*, methicillin-resistant staphylococcus aureus, mycobacterium abscesses) during group exercise programs.

- Individual education is crucial for the lung transplant person, with topics that include the surgical procedure, posttransplant wound care, potential complications of transplantation, management of anxiety/depression, methods of assisted ventilation, and strategies to optimize nutrition.
- Interval training has been associated with lower levels of dyspnea and fewer unintended breaks in lung transplant candidates compared to continuous training while achieving similar improvements in overall exercise capacity (305). Thus, interval training may be a preferred mode of exercise for the PR individual.
- Exercise intolerance and functional disability often persist after lung transplant despite restoration of near-normal lung function dynamics and pulmonary gas exchange.
- Skeletal muscle dysfunction plays a major role in exercise impairment for the lung transplant individual. Postoperative rehabilitation can begin as early as 24 hours after surgery to minimize the detrimental effect of immunosuppressants and bed rest on skeletal muscle.
- Rehabilitation in the early postoperative period should include ROM exercises, transfer activities (*e.g.*, sit to stand), breathing pattern efficiency training, and airway clearance education.
- Poor posture and gait may aggravate incisional discomfort. Balance, posture, and gait should be monitored closely postsurgery, and exercises should be incorporated to improve each, when indicated.
- Musculoskeletal problems may arise once an individual is exercising at a higher-intensity or duration-level posttransplant.

Other Tests of Muscular Fitness for Individuals With Chronic Lung Disease

There are several validated physical function tests in senior adults that may be used to assess upper and lower muscular strength and endurance in PR populations. Testing is performed at program entry and periodically thereafter for assessment of individual improvement or decline. Upper and lower body

muscular power (*e.g.*, watts) may be assessed from some of these tests. These types of tests include, but are not limited to, the following:

- Timed Up and Go test ([306](#))
- Five-Times-Sit-To-Stand test ([307](#))
- 30-Second Chair Stand test ([308](#))
- 30-Second Arm Curl test ([309](#))
- 6-Minute Pegboard and Ring test ([310](#))
- Handgrip dynamometer test ([311](#), [312](#))
- The seated medicine ball throw test ([313](#))
- The gallon jug shelf transfer test ([305](#))
- The Short Physical Performance Battery ([314](#))
- 4-meter gait speed ([315](#))

Although normative data currently do not exist for many of these tests for individuals with chronic lung disease, published normative values may provide a reasonable reference point to determine rate of improvement (or decline) over time. These data may also show the individual how they compare to individuals in their particular age category ([309](#)). Consideration must be given to the severity and type of lung disease, comorbidities, musculoskeletal abnormalities, and hypoxemia with regular subjective and objective assessment of these types of tests. Until these (and other) muscular fitness tests have been validated in pulmonary populations, test results should be viewed with caution when comparing normative values to geriatric populations.

Telerehabilitation

Structured exercise training as a therapeutic intervention for people with chronic pulmonary disease is typically associated with patient engagement in outpatient PR programs. Traditionally center-based PR programs involve supervised and monitored engagement in exercise in a multiple person class setting. However, despite its proven efficacy in enhancing physical function, clinical status, and health-related quality of life in people with chronic

pulmonary disease, center-based PR remains underutilized; poor referral rates, issues associated with program availability, accessibility, and transportation, and poor awareness of its effectiveness are all barriers to adoption and adherence of center-based PR ([316–318](#)). Telerehabilitation, which makes use of communication mediums such as telephone calls, videoconferencing, and dedicated websites to deliver the essential components of PR, has received increasing attention as an alternative method to deliver exercise training to people with chronic pulmonary disease ([167](#)). Programmatically, challenges related to delivering *all* the key components of PR (*e.g.*, comprehensive physical assessment, patient-specific prescription, and monitoring of exercise) via telerehabilitation need to be overcome. Despite this, meta-analysis of available randomized controlled trial data suggests that center-based PR and telerehabilitation strategies elicit similar improvements in functional exercise capacity (*i.e.*, 6-MWT distance), health-related quality of life, and dyspnea ([167](#)). Moreover, while there may be justifiable fear of adverse events, including cardiovascular events, falls, or musculoskeletal injuries, during remote, telerehabilitation-based exercise, the rate of exercise-related adverse events appears to be not different with telerehabilitation compared to center-based PR ([167](#)). These findings have led to clinical practice guidelines suggesting that telerehabilitation should be offered as an alternative to traditional center-based PR in people with chronic pulmonary disease ([167](#)). However, while the available evidence suggests that telerehabilitation may be an effective alternative to traditional center-based PR programs, several implementation considerations and potential barriers remain; these include but are not limited to the need for patient access to mobile technology (*e.g.*, smart phone, tablet) and internet access, a baseline level of technology literacy, and the potential need to develop smartphone applications to facilitate delivery ([167](#)).

Respiratory Muscle Testing and Training

Respiratory Muscle Strength Testing

Respiratory muscle testing is an important component of assessing the health of an individual's respiratory system ([319](#), [320](#)). Impaired respiratory muscle strength is associated with dyspnea and exercise intolerance ([321](#)). The most widely utilized assessments of respiratory muscle performance are maximal inspiratory pressure (MIP) and maximal expiratory pressure (MEP), which are measures of peak respiratory muscle strength ([319](#), [320](#)).

Both in clinical practice and research, noninvasive handheld manometers with flanged mouthpieces are most frequently used to measure MIP and MEP ([319](#), [320](#)). These devices possess excellent reliability and are simple to operate and affordable ([319](#), [320](#), [322](#)). For individuals without use of their mouth, measurements can be taken using a nasal probe for sniff pressure ([320](#)).

To assess MIP, individuals are instructed to fully expire and then perform a maximal inspiratory effort through the manometer for 3–5 seconds, at minimum 1.5 seconds. MEP can be assessed by having an individual fully inspire and then perform a maximal expiratory effort into the manometer for 3–5 seconds at minimum 1.5 seconds. The maximum sustained pressure of at least 1 second detected by the transducer in the manometer for either maneuver is recorded. A minimum of three trials and a maximum of eight trials are performed with 1 minute of rest between trials ([320](#)). Each measurement value should be within 10% of each other to be considered acceptable, and the highest value recorded during testing is used ([319](#), [320](#)). Since the results of these tests are effort dependent and influenced by technique, using a familiarization period, encouragement and cuing during test administration are recommended. For safety, patients should be seated in a chair with back support during training. Nose clips should be used to help create a hermetic seal between the individual's mouth and the device.

To help with interpretation, normative values matched to sex and age for MIP have been produced ([Table 8.6](#)) ([320](#)). While an individual may have an MIP or

MEP below age-sex matched normative values, the general threshold for respiratory muscle weakness is much lower; males = MIP: 83 cmH₂O and MEP: 109 cmH₂O; females = MIP: 62 cmH₂O and MEP: 81 cmH₂O (323). Age-specific thresholds matched to sex have also been provided for inspiratory muscle weakness (Table 8.7) (320). Prediction equations are also available (Box 8.7) (323).

TABLE 8.6 Normative Values for MIP (321)

Age (years)	Male	Female
18–29	128 (116.3–139.5)	97 (88.6–105.4)
30–39	128.5 (118.3–138.7)	89 (84.5–93.5)
40–49	117.1 (104.9–129.2)	92.9 (78.4–107.4)
50–59	108.1 (98.7–117.6)	79.7 (74.9–84.9)
60–69	92.7 (84.6–100.8)	75.1 (67.3–82.9)
70–83	76.2 (66.1–86.4)	65.3 (57.8–72.7)

Units in cmH₂O.

TABLE 8.7 Age- and Sex-Matched Thresholds for Inspiratory Muscle Weakness (321)

Age (years)	Male	Female
<40	63	58
40–60	55	50
61–80	47	43
>80	42	38

Units in cmH₂O. Reproduced with permission of the © ERS 2024: Laveneziana P, Albuquerque A, Aliverti A, et al. ERS statement on respiratory muscle testing at rest and during exercise. Eur Respir J. 2019;53(6):1801214. doi: 10.1183/13993003.01214-2018

Box 8.7 Prediction Equations for MIP and MEP ([323](#))

$$PI_{\max} (\text{females}) = 61.48 + 0.66 * \text{age} + 1.55 * \text{BMI} - 0.01 * \text{age}^2$$

$$PI_{\max} (\text{males}) = 98.60 + 1.18 * \text{age} + 0.76 * \text{BMI} - 0.02 * \text{age}^2$$

$$PE_{\max} (\text{females}) = 74.75 + 1.67 * \text{age} + 1.75 * \text{BMI} - 0.02 * \text{age}^2$$

$$PE_{\max} (\text{males}) = 58.11 + 3.71 * \text{age} + 2.64 * \text{BMI} - 0.04 * \text{age}^2$$

Key: Age in years, BMI in $\text{kg} \cdot \text{m}^{-2}$

Respiratory Muscle Endurance Testing

One of the most common and simplest methods to assess respiratory muscle endurance is a constant load endurance test. This test requires an individual to breathe against a respiratory muscle training (RMT) device set at a constant submaximal resistance (*e.g.*, 25% or 50% of MIP or MEP) until task failure ([320](#)). Task failure is indicated by volitional fatigue or two consecutive failed repetitions. Testing performance is measured either by the total time until task failure (seconds) or work expressed in pressure time units (*i.e.*, pressure $\times$ time; *e.g.*, $\text{cmH}_2\text{O} \times \text{seconds}$) ([319](#), [320](#)). To keep the work rate constant, breathing frequency should be controlled using a metronome, and breathing depth should be monitored with feedback from the examiner. To avoid ceiling effects, it is recommended that the testing load should result in task failure within 5–10 minutes of testing ([320](#)).

Respiratory Muscle Training

RMT has been shown to improve respiratory muscle strength, dyspnea, exercise capacity, and quality of life in numerous clinical populations including but not limited to asthma, HF, COPD, obesity, and patients with neuromuscular disorders ([212](#), [324–328](#)). This is accomplished by improving respiratory muscle strength, which reduces the relative work of the respiratory muscle during exercise and PA ([329](#), [330](#)). Patients with baseline respiratory muscle

weakness tend to receive the greatest benefit from RMT ([322](#)). In healthy individuals without dyspnea and athletes, RMT may only provide modest benefits on exercise performance ([331](#)).

There are several different device options available for RMT; however, the most widely used and researched are pressure threshold devices. These devices require an individual to breathe through a conduit with valve set to a pressure threshold. Once enough pressure is generated, the valve opens and air flows through the device. A percentage of the MIP or MEP is used as the training load based on the type of protocol. A key advantage of pressure threshold RMT is that the training resistance set is flow-independent, which means irrespective of the speed or depth of breathing, the resistance remains the same.

Protocols to improve respiratory muscle strength typically use higher resistance ranging from 40% to 60% of MIP or MEP and at frequencies of 20–30 breaths per session and performed twice per day. Protocols to improve RM endurance typically use lower resistance 15%–20% of MIP or MEP for 5–10 minutes per session and performed twice per day. For safety, patients should be seated in a chair with back support during training, and sessions should take place at least 6 hours apart. Respiratory muscle testing and training should not be performed in the presence of contraindications ([Box 8.8](#)). Training parameters may require adjustments according to an individual's tolerance and goals.

Box 8.8 Contraindications for Respiratory Muscle Testing and Training

Recent surgery (thoracic, abdominal, brain, ear, nose, or throat)

Pneumothorax

Myocardial infarction

Aortic aneurysm

Hemoptysis

Pulmonary embolism

Acute diarrhea

Stress incontinence

Hypertensive crisis (systolic >180 mm Hg, diastolic >120 mm Hg)

Confusion and dementia

Patient discomfort

Vomiting

Contagious infections

Retinopathy, glaucoma, and cataracts

ONLINE RESOURCES

American Association for Cardiovascular and Pulmonary Rehabilitation:

<http://www.aacvpr.org>

American Heart Association: <https://www.heart.org/en/health-topics/cardiac-rehab>

American Lung Association: <http://www.lungusa.org/lung-disease/copd/>

EPR3: Guidelines for the Diagnosis and Management of Asthma (Expert Panel Report 3): <http://www.nhlbi.nih.gov/guidelines/asthma/asthgdln.htm>

CTSNet. Goodbye Sternal Precautions, Hello Move in the Tube?

<https://www.ctsnet.org/article/goodbye-sternal-precautions-hello-move-tube>

Global Initiative for Asthma: <http://www.ginasthma.org>

Global Initiative for Chronic Obstructive Lung Disease:

<https://goldcopd.org/gold-reports/>

Society for Vascular Medicine: <http://www.vascularmed.org>

American Stroke Association: <http://www.strokeassociation.org>

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CHAPTER 9

Exercise Prescription for Individuals With Metabolic Disease and Cardiovascular Disease Risk Factors

INTRODUCTION

This chapter contains the exercise prescription (Ex R_x) guidelines and recommendations for individuals with metabolic and cardiovascular disease (CVD) risk factors. The Ex R_x guidelines and recommendations are presented using the *Frequency, Intensity, Time, and Type (FITT)* principle of Ex R_x based on the available literature. Information is often lacking regarding volume and progression for the chronic diseases and health conditions presented in this chapter. In these instances, the guidelines and recommendations provided in [Chapter 5](#) for apparently healthy populations should be adapted with sound clinical judgment for the chronic disease(s) and health condition(s) being targeted.

DIABETES MELLITUS

Diabetes mellitus (DM) is a group of metabolic diseases characterized by elevated blood glucose levels as a result of defects in insulin secretion and/or an inability to use insulin. Sustained elevated blood glucose levels place individuals at risk for long-term complications such as microvascular diseases (*e.g.*, retinopathy, nephropathy), macrovascular comorbidities (*e.g.*, coronary artery disease [CAD], peripheral artery disease), as well as neuropathies (peripheral and autonomic). Furthermore, those with DM are more likely to have a higher prevalence of other elevated CVD risk factors (*e.g.*, dyslipidemia, inflammatory markers) compared to those without DM. According to the Centers for Disease Control and Prevention, 37.3 million people, or 11.3% of the U.S. population, have diabetes, with 23.0% of those undiagnosed ([1](#)). Four types of diabetes are recognized based on etiologic origin: Type 1 diabetes mellitus (T1DM), Type 2 diabetes mellitus (T2DM), gestational (*i.e.*, diagnosed during pregnancy), and other specific origins (*i.e.*, genetic defects and drug induced); however, most individuals have T2DM (90%–95% of all cases) followed by T1DM (5%–10% of all cases) ([1](#)).

T1DM, now classified in three distinct phases, is most often caused by the autoimmune destruction of the insulin producing β cells of the pancreas and includes latent autoimmune diabetes of adulthood (LADA), although some cases of T1DM are idiopathic in origin ([2](#)). The primary characteristics of individuals with T1DM are nearly absolute insulin deficiency and a high tendency for ketoacidosis. T2DM is driven by insulin-resistant skeletal muscle, adipose tissue, and liver combined with an insulin secretory defect, best described as nonautoimmune progressive loss of adequate β -cell insulin secretion frequently associated with insulin resistance and metabolic syndrome (Metsyn) ([2](#)). A common feature of T2DM is excess body fat with fat distributed in the upper body (*i.e.*, central obesity) ([2](#)).

Assigning the type of diabetes frequently depends on the circumstances present at the time of diagnosis, with some individuals not necessarily fitting clearly into a single category (such as having T1DM or T2DM). Both are heterogeneous diseases and clinical presentation and disease progression may vary considerably ([2](#)); for example, those with T2DM who are on insulin therapy may have similar complication risks (*e.g.*, hypoglycemia or ketoacidosis) as those with T1DM, and adults with LADA have a long duration of marginal insulin secretory capacity and may be initially misdiagnosed with T2DM.

Central obesity and insulin resistance often progress to prediabetes, a condition characterized by (a) elevated blood glucose in response to dietary carbohydrate, termed *impaired glucose tolerance* (IGT), and/or (b) elevated blood glucose in the fasting state, termed *impaired fasting glucose* (IFG) ([Table 9.1](#)). Individuals with prediabetes are at very high risk of developing diabetes because the capacity of the β -cells to hypersecrete insulin diminishes over time and becomes insufficient to restrain elevations in blood glucose. The American Diabetes Association (ADA) posits that future classification schemes for diabetes will likely focus on the pathophysiology of the underlying β -cell dysfunction.

TABLE 9.1 Diagnostic Criteria for Prediabetes and Diabetes Mellitus (3)

Normal	Prediabetes	Diabetes Mellitus
A1C < 5.7% ($\leq 38 \text{ mmol} \cdot \text{mol}^{-1}$)	A1C = 5.7%–6.4% ($39\text{--}47 \text{ mmol} \cdot \text{mol}^{-1}$)	A1C $\geq 6.5\%$ ($\geq 48 \text{ mmol} \cdot \text{mol}^{-1}$)
FPG < 100 mg $\cdot \text{dL}^{-1}$ (5.6 mmol $\cdot \text{L}^{-1}$)	FPG = 100–125 mg $\cdot \text{dL}^{-1}$ (5.6–6.9 mmol $\cdot \text{L}^{-1}$) (IFG)	FPG $\geq 126 \text{ mg} \cdot \text{dL}^{-1}$ (7.0 mmol $\cdot \text{L}^{-1}$)
2-h PG < 140 mg $\cdot \text{dL}^{-1}$ (7.8 mmol $\cdot \text{L}^{-1}$) during an OGTT	2-h PG = 140–199 mg $\cdot \text{dL}^{-1}$ (7.8–11.0 mmol $\cdot \text{L}^{-1}$) during an OGTT (IGT)	2-h PG $\geq 200 \text{ mg} \cdot \text{dL}^{-1}$ (11.1 mmol $\cdot \text{L}^{-1}$) during an OGTT
		In an individual with classic symptoms of hyperglycemia or hyperglycemic crisis, a random PG $\geq 200 \text{ mg} \cdot \text{dL}^{-1}$ (11.1 mmol $\cdot \text{L}^{-1}$)

FPG, fasting plasma glucose (at least 8-h fasting); A1C, glycated hemoglobin; IFG, impaired fasting glucose; IGT, impaired glucose tolerance; OGTT, oral glucose tolerance test (75 g glucose); PG, plasma glucose.

The fundamental goal for the management of DM is glycemic control using diet, exercise, and, in many cases, medications such as insulin or other antidiabetic agents. Intensive treatment to manage blood glucose levels reduces the risk of progression of diabetes complications in anyone with the condition (2). Criteria for diagnosis of DM and prediabetes are presented in [Table 9.1](#). The ADA and World Health Organization (WHO) now endorse using hemoglobin A1C $\geq 6.5\%$, elevated fasting plasma glucose (FPG) ($\geq 126 \text{ mg} \cdot \text{dL}^{-1}$ or $7.0 \text{ mmol} \cdot \text{L}^{-1}$), and 2 hours plasma glucose following a 75-g oral glucose tolerance test (OGTT; $\geq 200 \text{ mg} \cdot \text{dL}^{-1}$ or $11.1 \text{ mmol} \cdot \text{L}^{-1}$) as being equally appropriate as diagnostic methodologies for diabetes (2, 4).

Further details on the handling of A1C, FPG, and OGTT diagnostic tests are highlighted in the ADA Classification and Diagnosis of Diabetes recommendations, which are updated annually ([2](#)). Beyond glycemic control, management of other comorbidities (*e.g.*, obesity, coronary heart disease) and CVD risk factors (*e.g.*, hypertension) are of importance. The presence of these other factors will affect the approach to both exercise testing and Ex R_x.

Benefits of Regular Physical Activity for Diabetes

Physical activity (PA) is a key management tool for any type of diabetes and can assist in preventing multiple diabetes-related health complications, insulin resistance, prediabetes, and T2DM progression. The most recent consensus statement from the ACSM on exercise and PA in people with T2DM highlights that regular exercise undertaken by individuals with T2DM results in less daily time in hyperglycemia and lower overall glycemia (0.5%–0.7% measured by A1C) ([5](#)). Other important benefits for individuals with T1DM, T2DM, or prediabetes include improvements in multiple CVD risk factors and overall quality of life ([5–10](#)). Regular exercise participation may also prevent or delay the transition to T2DM for individuals with prediabetes at high risk for developing the disease ([11](#)). In those with T1DM, exercise may improve insulin sensitivity, thus lowering requirements for exogenous insulin despite having little to no impact on pancreatic function ([8](#)). A decreased insulin dose has been linked to lower risk of CAD in T1DM ([9](#)). Further benefits of regular PA include improved all-cause and CVD mortality risk profile as well as decreased risk for stroke ([10, 12](#)). Those with diabetes also experience an increased risk for psychological conditions ([13](#)), which are associated with impaired glucose control among other risk factors. Regular PA has been shown to improve psychological

profiles in those with diabetes, which may improve biological markers such as glucose homeostasis ([13](#)).

Moderate-intensity exercise totaling $150\text{--}300\text{ min} \cdot \text{wk}^{-1}$ is associated with reduced morbidity and mortality in observational studies in all populations, including those with DM ([14](#)). Prolonged sedentary time has been found to be independently associated with deleterious health outcomes, such as T2DM and all-cause mortality; however, the deleterious outcome effects associated with sedentary time generally decrease with higher levels of PA ([14](#), [15](#)). Moreover, the interruption of sedentary time, via increasing overall PA as well as through multiple bouts of activity, is beneficial for glycemic control. Thus, all individuals with DM or prediabetes should be encouraged to be regularly physically active, including more daily physical movement, structured exercise, and decreased overall sedentary time, to improve their health and longevity ([5](#)).

Exercise Testing

The following are special considerations for exercise testing in individuals with DM:

- When beginning an exercise program of light to moderate intensity, exercise testing is generally not necessary for individuals with DM or prediabetes who are asymptomatic for CVD ([5](#), [6](#), [16](#), [17](#)). ACSM does encourage those with DM who are sedentary, even if asymptomatic for CVD, to receive medical clearance before beginning an exercise program independent of desired intensity. In contrast, the ADA concludes that medical clearance is not warranted in those with DM and asymptomatic for CVD wishing to begin an exercise program of light-to-moderate intensity ([6](#)).
- Electrocardiogram (ECG) stress testing may be indicated for individuals with DM ([5](#), [6](#), [18](#)), especially anyone who has been sedentary and desires

to participate specifically in vigorous-intensity activities.

- Follow-up diagnostic testing may be performed if positive or nonspecific ECG changes in response to exercise are noted or nonspecific ST and T wave changes at rest are observed. The Detection of Ischemia in Asymptomatic Diabetes trial, involving 1,123 individuals with T2DM and no symptoms of CAD, found that screening with adenosine stress radionuclide myocardial perfusion imaging for myocardial ischemia over a 4.8-year follow-up period did not alter rates of cardiac events ([19](#)). Results from the DYNAMIT (Do You Need to Assess Myocardial Ischemia in Type-2 diabetes) trial found similar results in which detection of silent myocardial ischemia (reduced blood flow to the myocardium in the absence of anginal symptoms) in asymptomatic individuals provided no benefit in predicting risk of future cardiac events ([20](#)). Thus, the cost-effectiveness and diagnostic value of more intensive testing remains in question. Furthermore, the most recent statement by the U.S. Preventive Services Task Force suggests against the use of resting or exercise ECG to predict cardiac events in asymptomatic adults ([21](#)).
- Silent myocardial ischemia in individuals with DM often goes undetected ([22](#)); therefore, annual CVD risk factor assessments should be conducted ([5](#), [6](#)).

Exercise Prescription

The FITT principle of Ex R_x for healthy younger and older adults generally applies to individuals of similar ages with DM (see [Chapter 5](#)) without complications. In those with other complications/comorbidities, Ex R_x should be modified as appropriate to these other conditions. Participating in an exercise program confers benefits that are extremely important to individuals with T1DM and T2DM. Maximizing the cardiovascular and metabolic benefits resulting from exercise is a key outcome for both types of

diabetes. Healthy weight loss and maintenance of appropriate body weight are often issues for those with T2DM and prediabetes, but excess body weight and fat can be present in those with T1DM as well, and an exercise program can be useful for either (see “Overweight and Obesity” and “Metabolic Syndrome” sections).

A systematic review and meta-analysis found no evidence that resistance exercise differs from aerobic exercise in the beneficial impact on multiple cardiovascular risk factors or safety in individuals with T2DM with the exception of A1C. Resistance exercise decreases A1C to a greater magnitude although it was not clinically significant ([23](#)). It is important to also note that aerobic exercise significantly increased cardiorespiratory fitness (CRF) in greater magnitude compared to resistance exercise. CRF has been shown to be one of the strongest independent predictors of mortality among those with T2DM ([24–26](#)). Thus, it is encouraged that those with T2DM, and those with T1DM, participate in sufficient volumes of both aerobic exercise and resistance exercise. Several studies have provided evidence to suggest a combination of aerobic exercise and resistance exercise is superior to only aerobic or only resistance exercise in the management of glucose homeostasis as well as other risk factors ([27, 28](#)). Whether the added benefits are caused by a greater overall caloric expenditure ([29](#)) or by the combination of aerobic and resistance training ([8, 28, 30](#)) has not yet been fully elucidated.

Exercise Training Considerations

- Many people with DM have comorbid conditions; tailor the Ex Rx accordingly. Many individuals with prediabetes or DM are at high risk for or have CVD (see [Chapter 8](#)).
- Most individuals with T2DM and prediabetes and many with T1DM are overweight or obese (see “Overweight and Obesity” section and the relevant ACSM position stand ([33](#))).

- Due to low initial fitness levels, most individuals with T2DM will require at least $150 \text{ min} \cdot \text{wk}^{-1}$ of moderate-to-vigorous aerobic exercise to achieve optimal CVD risk reduction and increased CRF ([5](#), [6](#)).

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH DIABETES ([5–7](#), [31](#), [32](#))

	Aerobic	Resistance	Flexibility and Balance
Frequency	3–7 d · wk ⁻¹ No more than 2 consecutive days without activity	A minimum of 2 nonconsecutive d · wk ⁻¹ , but preferably 3 d	≥2–3 d · wk ⁻¹ for both
Intensity	40%–59% of $\dot{V}O_2R$ or HRR (moderate), RPE 11–12; or 60%–89% of $\dot{V}O_2R$ or HRR (vigorous), RPE 14–17	Moderate (50%–69% of 1-RM) to vigorous (70%–85% of 1-RM) to improve strength	Stretch to the point of tightness or slight discomfort Balance exercises: light-to-moderate intensity
Time	T1DM and T2DM: 150–300 min · wk ⁻¹ at moderate or 75–150 min · wk ⁻¹ at vigorous intensity, or a combination thereof	At least 8–10 exercises with 1–3 sets of 10–15 repetitions to near fatigue per set early in training	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise. Balance for any duration
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming) Continuous activity or HIIT	Resistance machines, free weights, resistance bands, and/or body weight	Static, dynamic, other stretching, yoga

Unstructured activity (errands, housework, yard work) and small doses of physical activity throughout the day to break up sedentary time are encouraged in those with diabetes to aid in reducing sedentary time, increasing daily energy expenditure, and assisting with weight management ([6](#)).

1-RM, one repetition maximum; HIIT, high-intensity interval training; T1DM; Type 1 diabetes mellitus; T2DM, Type 2 diabetes mellitus.

- Interspersing very short, high-intensity intervals during moderate-intensity aerobic exercise may be useful to increase CRF and to lessen the decline in blood glucose levels during the early postexercise recovery period ([34](#), [35](#)).
- Given the established importance of CRF, a greater emphasis should eventually be placed on vigorous-intensity aerobic exercise if not contraindicated by diabetes- or other health-related complications. Better overall blood glucose control may be achieved by engaging in vigorous-intensity exercise training as well. Both high-intensity interval training (HIIT) and continuous training are recommended forms of vigorous-intensity exercise for individuals with DM ([6](#), [7](#), [36](#)), although not appropriate initially for most individuals starting out with low levels of CRF or at all for others.
- For T2DM, allow no more than two consecutive days without aerobic exercise to prevent a period of excessive decline of insulin action/insulin sensitivity; regular PA is also recommended for individuals with T1DM.
- Resistance training should be encouraged for individuals with DM or prediabetes in the absence of contraindications, such as uncontrolled hypertension or unstable severe proliferative retinopathy. Higher resistance (*i.e.*, heavier weight) may be beneficial for optimization of skeletal muscle strength, insulin action, and blood glucose ([32](#), [37](#), [38](#)) control, although moderate resistance may be equally effective in previously sedentary individuals ([39](#)).
- Appropriate progression of resistance exercise is important to prevent injury because those with T2DM often have an increased risk for tendinopathy ([40](#)) and may have limited joint mobility due to the process of glycation of collagen, especially older adults ([41](#)). Initial training intensity should be moderate, involving 10–15 repetitions per set, with

increases in weight or resistance undertaken with a lower number of repetitions ([5](#), [6](#)) only after the target number of repetitions per set can consistently be exceeded ([6](#), [41](#)). This increase in resistance can be followed by a greater number of sets and lastly by increased training frequency ([42](#)).

- During combined training, completing resistance training prior to aerobic training may lower the risk of postexercise hypoglycemia in individuals with T1DM ([35](#), [43](#), [44](#)). HIIT training, combining both anaerobic and aerobic exercise, may have a similar effect ([35](#), [44](#)). Timing of exercise sessions may also impact blood glucose responses ([45](#)).
- Although flexibility training may be desirable for individuals with all types of diabetes, it should not substitute for other recommended activities (*e.g.*, aerobic and resistance training). However, yoga, tai chi, and other alternative forms of exercise have been shown to improve glycemic control in T2DM and potentially prevent its onset ([46–48](#)).
- Potential complications/contraindications may affect the appropriateness of some types of activities (*e.g.*, diabetic retinopathy, autonomic and peripheral neuropathy, nephropathy). See the following references for more information ([6](#), [13](#)).
 - Individuals with DM and unstable proliferative retinopathy are at risk for vitreous hemorrhage. However, risk may be minimized by avoiding activities that dramatically elevate blood pressure (BP). Anyone with severe nonproliferative and unstable proliferative diabetic retinopathy should avoid vigorous-intensity aerobic and resistance exercise, jumping, jarring, head-down activities and the Valsalva maneuver ([6](#)).
- During exercise, autonomic neuropathy may cause chronotropic incompetence (*i.e.*, a blunted heart rate [HR] response), attenuated volume of oxygen consumed per unit of time ($\dot{V}O_{2\max}$) kinetics, and anhidrosis (*i.e.*, water deprivation) ([6](#)). In the presence of autonomic neuropathy, the following should be considered:

- Monitor for signs and symptoms of silent ischemia, such as unusual shortness of breath or back pain, because of the inability to perceive angina.
- Monitor BP before and after exercise to manage hypotension and hypertension associated with vigorous-intensity exercise (see “Hypertension” section).
- HR and BP responses to exercise may be blunted secondary to autonomic dysfunction. Rating of perceived exertion (RPE) should be used to assess exercise intensity ([30](#)).
- For individuals with peripheral neuropathy, proper care of the feet is needed to prevent foot ulcers and lower the risk of amputation ([6](#)). Special precautions should be taken to prevent blisters on the feet. Feet should be kept dry, and silica gel or air midsoles as well as polyester or blend socks should be used. All individuals should closely examine their feet on a daily basis to detect and treat sores or ulcers early. Consideration of more non-weight-bearing activities is encouraged.
- For individuals with diabetic kidney disease, exercise does not appear to accelerate its progression even though protein excretion acutely increases after exercise ([6](#), [13](#), [49](#)). Both aerobic and resistance training improve physical function and quality of life in individuals with advanced kidney disease, and individuals should be encouraged to be active. Exercise should begin at a low intensity and volume if aerobic capacity and muscle function are substantially reduced ([6](#), [50](#)) (see [Chapter 10](#)).

Special Considerations

- Hypoglycemia is the most common, acute concern for individuals taking insulin or certain oral hypoglycemic agents that increase insulin secretion ([6](#)).

- Hypoglycemia, defined as a blood glucose level less than $70 \text{ mg} \cdot \text{dL}^{-1}$ ($<3.9 \text{ mmol} \cdot \text{L}^{-1}$) ([51](#)), is a relative contraindication to beginning an acute bout of exercise ([5](#)).
- Rapid decreases in blood glucose may occur with exercise and render individuals symptomatic even if blood glucose is well above $70 \text{ mg} \cdot \text{dL}^{-1}$. Conversely, blood glucose levels may decrease in some individuals without generating noticeable symptoms (*e.g., hypoglycemic unawareness*).
- Common adrenergic symptoms associated with hypoglycemia include shakiness, weakness, abnormal sweating, nervousness, anxiety, tingling of the mouth and fingers, and hunger. More severe neuroglycopenic symptoms may include headache, visual disturbances, mental dullness, confusion, amnesia, seizures, and coma.
- Individuals with DM who take insulin or medications that increase insulin secretion should monitor blood glucose levels before, occasionally during (if needed), and after exercise and compensate with appropriate dietary and/or medication regimen changes (in consultation with their health care provider) as needed to maintain euglycemia ([5](#), [6](#), [35](#), [52](#)). (See ([5](#)), p. 2069, *Table 2*, for dose reduction recommendations.)
- For those with diabetes, preexercise optimal blood glucose levels are between 90 and $250 \text{ mg} \cdot \text{dL}^{-1}$ (5.0 and $13.9 \text{ mmol} \cdot \text{L}^{-1}$). The ADA provides guidelines on carbohydrate ingestion based on preexercise blood glucose levels ([6](#)). An international consensus statement provides exercise management guidelines for individuals with T1DM ([53](#)).
- Hypoglycemia risk is higher during and immediately following primarily moderate-intensity aerobic exercise but can occur up to 12 hours or more postexercise, making food and/or medication adjustments necessary, mostly in insulin users ([35](#), [54](#)). Risk for hypoglycemia also presents with the daily accumulation of low-intensity, unstructured PA ([55](#)).

- However, aerobic exercise at a vigorous intensity has been shown to decrease/lessen the speed at which blood glucose declines following exercise ([56](#)). Also, performing resistance exercise before aerobic exercise may elicit similar effects ([35](#)). Exercise timing and prandial state also impact blood glucose responses to various types and intensities of activities. Nonetheless, frequent blood glucose monitoring is the key to detecting and preventing later-onset hypoglycemia.
- Sulfonylurea drugs and other compounds that enhance insulin secretion (*e.g.*, glyburide, glipizide, glimepiride, nateglinide, and repaglinide) increase the risk of hypoglycemia because the effects of insulin and muscle contraction on blood glucose uptake are additive ([57](#), [58](#)). Blood glucose monitoring is recommended when beginning a program of regular exercise to assess whether changes in doses of these medications are necessary.
- The timing of exercise is particularly important in individuals taking insulin. Changing insulin timing, reducing insulin doses, and/or increasing carbohydrate intake are effective strategies to prevent hypoglycemia and hyperglycemia during and after exercise ([59](#)). Early morning exercise, in particular, may result in elevations in blood glucose levels instead of the usual decrease with moderate activity ([60](#)).
- Prior to planned exercise, rapid- or short-acting insulin doses will likely have to be reduced to prevent hypoglycemia, particularly if exercise occurs during peak insulin times (usually within 2–3 hours). Rapid-acting insulin analogs (*e.g.*, lispro, aspart, and glulisine) induce more rapid decreases in blood glucose than regular human insulin.
- Longer-acting basal insulins (*e.g.*, glargine, detemir, and neutral protamine Hagedorn) are less likely to cause exercise-induced hypoglycemia ([61](#)), although overall doses may need to be reduced to accommodate regular training or activities of extended duration.

- For individuals with T1DM using insulin pumps, insulin delivery during exercise can be markedly reduced by decreasing the basal rate or disconnecting the pump for short durations, depending on the intensity and duration of exercise. Reducing basal insulin delivery rates for up to 12 hours postexercise may be necessary to avoid later-onset hypoglycemia.
- Continuous glucose monitoring (CGM) devices can be very useful in detecting patterns in blood glucose across multiple days and evaluating both the immediate and delayed effects of exercise ([6](#), [35](#), [62](#)). Guidelines are now available for glucose time in range for individuals using CGM ([63](#)).
- Individuals using hybrid closed-loop systems (*e.g.*, insulin pump, CGM, and algorithm control of insulin delivery) may need to make adjustments to target glucose levels and/or insulin delivery prior to, during, or following physical activities ([64](#), [65](#)).
- Risk mitigation for individuals with DM who experience exercise-induced hypoglycemia may include carrying a medical ID identifying diabetes, a cell phone, and glucose tablets or other rapid carbohydrate treatment for hypoglycemia when active.
- Diabetic autonomic neuropathy, long-duration T1DM, and recent antecedent hypoglycemia or exercise contribute to impaired epinephrine, glucagon, and other hormonal responses and hypoglycemia unawareness ([66](#)), so frequent blood glucose monitoring is warranted. In older individuals with T2DM, the joint occurrence of hypoglycemia unawareness and deteriorated cognitive function is a critical factor that needs to be considered in their exercise blood glucose management ([67](#)).
- Hyperglycemia with or without ketosis is a concern for individuals with T1DM. Common symptoms associated with hyperglycemia include polyuria, fatigue, weakness, increased thirst, and acetone breath. Individuals who present with hyperglycemia (*i.e.*, blood glucose $\geq 250 \text{ mg} \cdot \text{dL}^{-1}$ or $16.7 \text{ mmol} \cdot \text{L}^{-1}$), provided they feel well and have no ketones

present when testing either blood or urine, may exercise up to a moderate intensity; however, they should test blood glucose frequently, refrain from vigorous-intensity exercise until glucose levels are declining, and ensure adequate hydration ([6](#)).

- It is recommended that individuals with T1DM check for elevated urine or blood ketones when blood glucose levels are $\geq 250 \text{ mg} \cdot \text{dL}^{-1}$ ($13.9 \text{ mmol} \cdot \text{L}^{-1}$) before starting to exercise ([6](#), [68](#)). Exercise should be postponed when both hyperglycemia and ketones are evident.
- If blood glucose levels are $\geq 350 \text{ mg} \cdot \text{dL}^{-1}$ ($\geq 19.4 \text{ mmol} \cdot \text{L}^{-1}$), even without ketones present, conservative corrective insulin therapy is recommended before exercise ([6](#)).
- If blood glucose has been elevated for less than 2–3 hours following a meal, individuals with T2DM will likely experience a reduction in blood glucose during aerobic exercise because endogenous insulin levels will be high ([57](#), [69](#)). Those with T1DM may experience similar declines in blood glucose levels if injected or pumped levels of insulin are higher during postprandial exercise.
- Regardless of initial blood glucose levels, vigorous activity of any type may cause acute elevations in glucose due to an exaggerated release of counterregulatory hormones like epinephrine and glucagon ([70](#)). In such cases, individuals with T1DM may need small doses of supplemental insulin to lower postexercise hyperglycemia.
- Dehydration resulting from polyuria secondary to hyperglycemia may contribute to a compromised thermoregulatory response ([6](#), [71](#)). Dehydration may also contribute to elevations in blood glucose levels. Anyone with hyperglycemia has an elevated risk for heat illness and should frequently monitor for signs and symptoms (see [Chapter 7](#) and other relevant ACSM position stands ([72](#), [73](#))).
- Given the likelihood that thermoregulation in hot and cold environments is impaired ([6](#), p. 2074) in those with DM, additional precautions for heat

and cold illness are warranted (see [Chapter 7](#) and other relevant ACSM positions stands ([72-74](#))).

ONLINE RESOURCES

American College of Sports Medicine position stand on exercise and Type 2 diabetes mellitus (2010): <https://www.acsm.org/acsm-positions-policy/official-positions>

American College of Sport Medicine consensus statement on exercise/physical activity and in individuals with type 2 diabetes (2022): <https://www.acsm.org/education-resources/pronouncements-scientific-communications/official-statements>

American Diabetes Association: <http://www.diabetes.org>

Diabetes Motion (for information about exercising safely with diabetes): <http://www.diabetesmotion.com>

National Institute of Diabetes and Digestive and Kidney Diseases: <https://www.niddk.nih.gov/>

DYSLIPIDEMIA

Dyslipidemia is an abnormal concentration of lipids (*e.g.*, cholesterol, triglycerides [TGs]) in the blood. It is further defined by the presence of elevated levels of total cholesterol (TC) or low-density lipoprotein cholesterol (LDL-C), elevated levels of TGs, or low levels of high-density lipoprotein cholesterol (HDL-C). LDL-C is the primary lipoprotein for CV risk assessment and is typically calculated using TC and TG. If TG are elevated, the accuracy of calculated LDL-C versus directly measured LDL-C is compromised, and non-HDL-C (TC minus HDL-C) is the preferred lipoprotein for risk assessment ([75](#)). Current definitions for dyslipidemia are found in [Table 2.4](#). Over 50% of adults in the United States have dyslipidemia ([76](#)), a major risk factor for atherosclerotic CVD.

There are many causes of dyslipidemia. The most common contributing factors are poor dietary and lifestyle choices; however, genetics may play a prominent contributing role, and very high levels of cholesterol often cluster within families. Familial hypercholesterolemia (LDL-C > 190 mg · dL⁻¹ in adults) is a genetic disorder occurring in 1 in 300–500 people ([77](#)). Various disease states can also increase blood lipids. LDL-C levels are often increased in individuals with hypothyroidism or nephrotic syndrome. Very high levels of TG may be present in individuals with obesity, insulin resistance, or DM ([78](#)). MetSyn is partially defined by the presence of high TG levels. Additionally, the use of oral anabolic steroids has been associated with a 20%–70% reduction in HDL-C levels ([79](#)).

Lifestyle improvements are the first-line treatment of dyslipidemia even for individuals who may require medications to control their dyslipidemia. Exercise may improve dyslipidemia, although the magnitude of effect is often small. Aerobic and/or dynamic resistance exercise training may reduce LDL-C by 3–6 mg · dL⁻¹ (0.08–0.16 mmol · L⁻¹), TGs by 4–12 mg ·

dL^{-1} ($0.05\text{--}0.14 \text{ mmol} \cdot \text{L}^{-1}$), and increase HDL-C by $1\text{--}2 \text{ mg} \cdot \text{dL}^{-1}$ ($0.03\text{--}0.05 \text{ mmol} \cdot \text{L}^{-1}$) (7). Additionally, weight loss may result in decreases in TC, LDL-C, TG and should be encouraged (80, 81).

Several types of medications are used to treat dyslipidemia (75, 82, 83). Statins are first-line drugs, which improve multiple outcomes for patients with dyslipidemia including better survival. Even with maximal dose statins and improved lifestyle, some patients may require a second nonstatin medication to optimally control dyslipidemia. Ezetimibe, PCSK9 inhibitors (evolocumab, alirocumab, inclisiran), or bempedoic acid when combined with statins result in substantially greater reductions in LDL-C than with statins alone. Statin intolerance, the most common side effect being mild-to-moderate muscle pain and weakness, may necessitate using only nonstatin medications. For high or very high levels of TG, a fibric acid derivative (gemfibrozil) or omega 3 fatty acids are effective treatment options. The four most important groups of people who benefit from blood lipid improving medications are (a) individuals with established CVD, (b) individuals with LDL-C levels greater than $190 \text{ mg} \cdot \text{dL}^{-1}$, (c) individuals with DM who are ≥ 40 years, and (d) individuals with an estimated 10-year risk for CVD of $\geq 7.5\%$. The 10-year risk score is based on the presence and severity of risk markers for heart disease and can be calculated using readily available online calculators (see “Online Resources” at the end of this section).

For adults in the United States, TC, TG, and LDL-C levels improved between 1999 and 2018 (84). This improvement is attributed to greater cholesterol awareness, changes in dietary eating patterns, and increased use of medications (84). However, substantial numbers of people throughout the United States have uncontrolled dyslipidemia (84).

The ACSM makes the following recommendations regarding exercise testing and training of individuals with dyslipidemia.

Exercise Testing

- In general, an exercise test is not required for asymptomatic individuals prior to beginning an exercise training program at a light-to-moderate intensity and then gradually progressing to vigorous-intensity exercise (see [Chapter 5](#)).
- Standard exercise testing methods and protocols are appropriate for use with individuals with dyslipidemia cleared for exercise testing (see [Chapters 3](#) and [4](#)).
- Use caution when testing individuals with dyslipidemia because undetected underlying CVD may be present.
- Special consideration should be given to the presence of other chronic diseases and health conditions (*e.g.*, Metsyn, DM, obesity, hypertension) that may require modifications to standard exercise testing protocols and modalities (see the sections of this chapter and other relevant ACSM position stands on these chronic diseases and health conditions ([33](#), [85](#))).

Exercise Prescription

The FITT principle of Ex R_x for individuals with dyslipidemia without comorbidities is very similar to the Ex R_x for healthy adults (see [Chapter 5](#)) ([42](#), [86](#)). However, an important difference in the FITT principle of Ex R_x for individuals with dyslipidemia and overweight/obesity compared to healthy adults is that healthy weight maintenance should be highly emphasized (see “Overweight and Obesity” section and the relevant ACSM position stand ([33](#))).

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH DYSLIPIDEMIA ([42](#), [87](#), [88](#))

	Aerobic	Resistance	Flexibility
Frequency	$\geq 5 \text{ d} \cdot \text{wk}^{-1}$ to maximize caloric expenditure	$2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$	$\geq 2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$
Intensity	40%–75% $\dot{V}\text{O}_2\text{R}$ or HRR	Moderate (50%–69% of 1-RM) to vigorous (70%–85% of 1-RM) to improve strength	Stretch to the point of tightness or slight discomfort
Time	30–60 min $\cdot \text{d}^{-1}$. To promote or maintain weight loss, 50–60 min $\cdot \text{d}^{-1}$ or more of daily exercise is recommended	2–4 sets, 8–12 repetitions for strength; ≤ 2 sets, 12–20 repetitions for muscular endurance	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming)	Resistance machines, free weights, and/or body weight	Static, dynamic, and/or PNF stretching

1-RM, one repetition maximum; HRR, heart rate reserve; $\dot{V}\text{O}_2\text{R}$, oxygen uptake reserve; PNF, proprioceptive neuromuscular facilitation.

Exercise Training Considerations

- The FITT principle of Ex R_x may need to be modified should the individuals with dyslipidemia present with other chronic diseases and health conditions such as Metsyn, DM, obesity, and hypertension (see “Metabolic Syndrome,” “Type 2 Diabetes,” “Overweight and Obesity,” and “Hypertension” sections and other relevant ACSM position stands on these chronic diseases and health conditions ([33](#), [85](#), [89](#))).
- Adults older than age 65 years and with dyslipidemia should follow the ACSM exercise guidelines for older adults ([90](#)).

Special Consideration

- Individuals taking statins, particularly at higher doses or in combination with fibric acid derivatives, rarely may experience severe muscle pain and weakness due to rhabdomyolysis ([91](#)). Rhabdomyolysis is a potentially life-threatening event, and a health care provider must be promptly consulted if an individual experiences unusual or persistent muscle soreness, especially if accompanied by brown- or red-tinged urine when exercising while taking these medications.

ONLINE RESOURCES

American Heart Association: 2018 Prevention Guidelines Tool CV Risk

Calculator (<https://static.heart.org/riskcalc/app/index.html#!/baseline-risk>)

HYPERTENSION

Hypertension or high BP is the most common, costly, and modifiable major risk factor for the development of CVD (including CAD and stroke) and premature CVD morbidity and mortality in the United States and world ([92–94](#)). Hypertension is defined by the 2017 ACC/AHA Task Force on Clinical Practice Guidelines as having a resting systolic blood pressure (SBP) ≥ 130 mm Hg or a resting diastolic blood pressure (DBP) ≥ 80 mm Hg or both, confirmed by a minimum of two measurements taken on at least 2 separate days, taking antihypertensive medication for BP control, or any combination of these criteria ([95](#)). Using this definition, an estimated 122.4 million (or $\sim 47\%$) of U.S. adults ≥ 20 years of age have hypertension ([94](#)).

These BP thresholds to define hypertension represent a departure from the long-standing BP criteria to define hypertension by the 2003 Seventh Report of the Joint National Committee (JNC7) on the Prevention, Detection, Evaluation, and Treatment of High Blood Pressure as having a resting SBP ≥ 140 mm Hg and/or a resting DBP ≥ 90 mm Hg ([96](#)). Of note, the JNC7 BP thresholds to define hypertension remain consistent with the hypertension clinical practice of professional societies across the world, including The Task Force for the management of arterial hypertension of the European Society of Cardiology (ESC) and the European Society of Hypertension (ESH) ([97, 98](#)) and the International Society of Hypertension (ISH) ([99](#)).

It is important to be familiar with the ACC/AHA, ESC/ESH, ISH, and JNC7 BP thresholds and classifications as these differences result in slight variations in prevalence and control rates, treatment of, and patient education about hypertension. See [Table 2.3](#) for both sets of criteria.

The underlying causes of hypertension are often unknown given how complex and multifaceted BP regulation is. However, chronic primary (or essential) hypertension accounts for 95% of all cases. There are a myriad of

established risk factors for primary hypertension including, but not limited to: older age, race/ethnicity, sex/gender, genetics/family history, social determinants of health, pregnancy complications, physical inactivity, suboptimal sleep, overweight/obesity, tobacco use, excessive alcohol use, and poor diet quality (*e.g.*, higher sodium intake and dietary fats, lower potassium intake) ([94](#), [95](#), [100](#)). Secondary hypertension accounts for a small percentage (~5%) of hypertension cases; however, recognizing the causes is important as medical intervention can often improve BP levels and delay disease progression ([95](#)). The principal causes of secondary hypertension include chronic kidney disease, primary aldosteronism, renal disease or stenosis, and pheochromocytoma ([101](#)).

The central goal in the treatment, control, and management of hypertension is to reduce the risk of cardiovascular mortality and morbidity. The most current clinical practice guidelines provide specific guidance for hypertension prevention, treatment, and management ([95](#), [97–99](#)). In general, target BP levels can be achieved through a variety of pharmacologic and/or nonpharmacologic lifestyle interventions; largely dependent on baseline risk ([95](#)). Lifestyle modifications, such as regular PA, are essential for the prevention, treatment, and control of hypertension and all major clinical practice guidelines for hypertension-recommend exercise as antihypertensive lifestyle therapy ([95](#), [97](#), [98](#), [102](#)). Indeed, a recent AHA scientific statement reinforces lifestyle modification as a critical component of first-line treatment for individuals with mild-to-moderately elevated BP to prevent the development of hypertension ([103](#)). Other recommended lifestyle changes to prevent, treat, and manage hypertension include nicotine cessation, weight management, reduced sodium intake, moderation of alcohol consumption, and an overall healthy dietary pattern consistent with the Dietary Approaches to Stop Hypertension diet ([95](#)).

In addition to lifestyle behavioral changes (see [Chapter 12](#)), medications may be required to treat and manage hypertension. Recent meta-analyses

continue to reinforce the importance of exercise and antihypertensive medications to treat and manage hypertension ([104](#)). Of note is the new finding that the BP-lowering effects of antihypertensive medications when combined with exercise are stronger than with either intervention alone ([104](#)). Current guidelines for the management of hypertension provide specific guidance on the implementation of pharmacologic therapies ([95](#)). Most individuals treated with antihypertensive medication require more than one medication class to achieve their targeted BP ([95](#), [105](#)). Some antihypertensive medications may affect the physiological response to exercise, and these side effects should be taken into consideration during exercise testing and when prescribing exercise (see [Appendix A](#)) ([95](#)).

Exercise Testing

Although hypertension is not an indication for exercise testing *per se*, the test may be used to evaluate the BP response to exercise, which can be useful with guiding Ex Rx ([17](#), [18](#), [95](#)). Individuals with hypertension may have an exaggerated BP response to exercise, even if resting BP is controlled ([106](#)). Some individuals with mildly elevated BP may also have an exaggerated BP response to exercise ([107](#)).

Recommendations regarding medical clearance (*i.e.*, medical examination or exercise test) for individuals with hypertension vary depending on their BP level, ASCVD risk, current PA levels, desired exercise intensity, and the presence of signs or symptoms and/or known cardiovascular, metabolic, renal or target organ disease, or clinical CVD ([17](#), [95](#)). Individuals with hypertension whose BP is not controlled (*i.e.*, resting SBP $\geq$ 130 mm Hg and/or DBP $\geq$ 80 mm Hg according to the ACC/AHA guidelines) should consult with their health care provider but does not necessarily preclude participation in light- to moderate-intensity PA ([17](#), [95](#)).

Recommendations include the following:

- Individuals with SBP $\geq$ 160 mm Hg and/or DBP $\geq$ 100 mm Hg or presence of signs or symptoms and/or known cardiovascular, metabolic, renal or target organ disease (*e.g.*, left ventricular hypertrophy, retinopathy) must not engage in any exercise, including exercise testing, prior to a medical evaluation and adequate BP management ([17](#), [95](#)).
- When exercise testing is performed for the specific purpose of designing the Ex R_x, it is preferred that individuals take their usual antihypertensive medications as recommended ([18](#)).
- Individuals on β -blocker therapy are likely to have an attenuated HR response to exercise and reduced maximal exercise capacity. Individuals on diuretic therapy may experience hypokalemia and other electrolyte imbalances, cardiac dysrhythmias, or potentially a false-positive exercise test (see [Appendix A](#)).

Exercise Prescription

- Chronic exercise of any intensity, duration (time), type, and volume leads to reductions in resting SBP and DBP among those with normal and elevated BP and those with hypertension ([108](#), [109](#)). An ACSM Pronouncement concluded the magnitude of the BP reductions to exercise training is directly proportional to resting BP, with the greatest BP benefits occurring among those with hypertension (5–8 mm Hg) with reductions on the order of 2–4 and 1–2 mm Hg for individuals with elevated and normal BP, respectively ([108](#)). In addition, regular aerobic exercise training has been shown to translate to favorable structural changes, including regression of cardiac wall thickness and left ventricular mass among individuals with hypertension ([107](#), [110](#)) and reduced left ventricular mass among individuals with elevated BP and moderate-to-high physical fitness ([111](#)). It has been a commonly held notion that aerobic exercise elicits BP reductions that are superior to

other types of exercise; however, recent research has demonstrated that other exercise modalities are also effective in lowering BP, particularly dynamic resistance exercise with emerging evidence for isometric resistance and neuromotor exercise ([102](#), [109](#), [112](#)). Therefore, the Ex Rx for individuals with hypertension no longer places an emphasis on aerobic exercise alone but rather encourages participation in a variety of multimodal exercises that reflect personal preference.

- Individuals with hypertension are recommended to engage in aerobic or resistance exercise alone or combined (*i.e.*, concurrent exercise) on most, preferably all, days of the week for at least $20\text{--}30\text{ min} \cdot \text{d}^{-1}$ to total at least $90\text{--}150\text{ min} \cdot \text{wk}^{-1}$ of continuous or accumulated exercise of any duration.
- In addition, neuromotor and flexibility exercise may be added, depending on personal preference to improve balance and range of motion, reduce the risk of musculoskeletal injury, decrease the risk of falls, and lower BP. Neuromotor exercise should be performed $\geq 2\text{--}3\text{ d} \cdot \text{wk}^{-1}$ at low-to-moderate intensity for $\geq 20\text{--}30$ minutes per session and include exercise involving motor skills and/or functional body weight and flexibility exercise such as yoga, Pilates, and tai chi ([108](#)). Flexibility exercise should be performed after a thorough warm-up or during the cool-down period following the guidelines for healthy adults (see [Chapter 5](#)). Neuromotor functional body weight exercise can be substituted for resistance exercise, and depending on the amount of flexibility exercise integrated into a session, neuromotor flexibility exercise can be substituted for flexibility exercise depending on patient preference.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH HYPERTENSION

	Aerobic	Resistance	Flexibility
Frequency	$\geq 2-3 \text{ d} \cdot \text{wk}^{-1}$	$\geq 2-3 \text{ d} \cdot \text{wk}^{-1}$	$\geq 2-3 \text{ d} \cdot \text{wk}^{-1}$
Intensity	Moderate (40%–59% $\dot{V}\text{O}_2\text{R}$ or HRR; RPE 12–13 on a 6–20 scale) to vigorous (60%–80% $\dot{V}\text{O}_2\text{R}$ or HRR; RPE 14–16 on a 6–20 scale)	Moderate (60%–70% 1-RM; may progress to 80% 1-RM; for older individuals and novice exercisers, begin with 40%–50% 1-RM)	Stretch to the point of feeling tightness or slight discomfort
Time	$\geq 20-30 \text{ min} \cdot \text{d}^{-1}$ of continuous or accumulated exercise of any duration	2–4 sets of 8–12 repetitions of 8–10 resistance exercises of each of the major muscle groups per session to total $\geq 20 \text{ min}$ per session with rest days interspersed depending on the muscle groups being exercised	Hold static stretch for 10–30 s with 2–4 repetitions of each exercise targeting the major muscle tendon units to total 60 s of total stretching time for each exercise; $\leq 10 \text{ min}$ per session
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming)	Resistance machines, free weights, resistance bands, and/or functional body weight exercise	Static, dynamic, and/or proprioceptive neuromuscular facilitation

1-RM, one repetition maximum; HRR, heart rate reserve; $\dot{V}\text{O}_2\text{R}$, oxygen uptake reserve; RPE, rating of perceived exertion.

Exercise Training Considerations

- Consideration should be given to the level of BP control, recent changes in antihypertensive drug therapy, medication-related adverse effects, the presence of target organ disease, other comorbidities, and age. Adjustments to the Ex R_x should be made accordingly. In general, progression should be gradual, avoiding large increases in any of the FITT components of the Ex R_x , especially for intensity in most individuals with hypertension.

- An exaggerated BP response to relatively low exercise intensities and at HR levels less than 85% of the age-predicted maximal heart rate (HR_{max}) is likely to occur in some individuals, even after resting BP is controlled with antihypertensive medication. In some cases, an exercise test may be beneficial to establish the exercise HR corresponding to the exaggerated BP in these individuals.
- It is prudent to maintain $SBP \leq 220$ mm Hg and/or $DBP \leq 105$ mm Hg when exercising ([85](#)).
- Moderate-intensity aerobic exercise (40%–59% $\dot{V}O_2R$) is generally recommended to optimize the benefit-to-risk ratio. However, the magnitude of the BP reductions resulting from aerobic exercise are directly proportional to intensity such that the greatest BP reductions occur after vigorous-intensity exercise if the patient is willing and able to perform vigorous-intensity exercise ([102](#), [112](#), [113](#)).
- Individuals with hypertension are often overweight or obese. Ex Rx should focus on increasing caloric expenditure coupled with reducing caloric intake to facilitate weight reduction (see “Overweight and Obesity” section and the relevant ACSM position stand ([33](#))).
- Mounting evidence suggests that nontraditional mind-body types of exercise such as yoga and tai chi are effective in reducing BP ([112](#), [114](#), [115](#)). Additionally, emerging research suggests that isometric resistance exercise training is effective at reducing BP among patients with high normal BP and hypertension ([109](#), [112](#)). For these reasons, individuals with hypertension should be encouraged to participate in a variety of multimodal exercises that reflect personal preference.
- Inhaling and breath-holding while engaging in the actual lifting of a weight (*i.e.*, Valsalva maneuver) can result in extremely high BP responses, dizziness, and even fainting. Thus, such practice should be avoided during resistance training.

Special Considerations

- Exercise testing and vigorous-intensity exercise training for individuals with hypertension at moderate-to-high risk for cardiac complications should be medically supervised until the safety of the prescribed activity has been established ([18](#)).
- β -Blockers and diuretics may adversely affect thermoregulatory function. β -Blockers may also increase the predisposition to hypoglycemia in certain individuals (especially individuals with DM who take insulin or insulin secretagogue medication that increases pancreas insulin secretion) and mask some of the manifestations of hypoglycemia (particularly tachycardia). In these situations, educate individuals about the signs and symptoms of heat intolerance and hypoglycemia and the precautions that should be taken to avoid these situations (see [Appendix A](#)).
- β -Blockers, particularly the nonselective types, may reduce submaximal and maximal exercise capacity primarily in individuals without myocardial ischemia (see [Appendix A](#)). The peak exercise HR achieved during a standardized exercise stress test should then be used to establish the exercise training intensity. If peak exercise HR is not available, RPE should be used.
- Antihypertensive medications such as α -blockers, calcium channel blockers, and vasodilators may lead to sudden excessive reductions in postexercise BP. Therefore, termination of the exercise should be gradual, and the cool-down period should be extended and carefully monitored until BP and HR return to near resting levels.
- Many older individuals are likely to have hypertension. The exercise-related BP reduction is independent of age. Therefore, older individuals experience similar exercise-induced BP reductions as younger individuals (see [Chapter 6](#) and the relevant ACSM/AHA recommendations ([116](#))).

- The BP-lowering effects of aerobic exercise are immediate and clinically meaningful, a physiologic response referred to as postexercise hypotension. Recent research suggests that the antihypertensive benefits of more long-term exercise training may be in large part due to postexercise hypotension. For these reasons, monitoring BP before and after exercise should be encouraged as a behavioral strategy to increase exercise adherence ([117](#)).
- Individuals with hypertension should be encouraged to engage in greater frequencies of exercise to optimize benefit from the acute effects of exercise on BP ([103](#)).
- If an individual with hypertension has ischemia during exercise, the Ex R_x recommendations for those with CVD with ischemia should be utilized. See [Chapter 8](#) for more information.

ONLINE RESOURCES

American College of Sports Medicine Special Communications. Physical activity to prevent and treat hypertension: a systematic review:

https://journals.lww.com/acsm-msse/fulltext/2019/06000/physical_activity_to_prevent_and_treat.26.aspx

American College of Sports Medicine. Exercise to treat hypertension: late breaking news on exercise prescriptions that FITT:

https://journals.lww.com/acsm-csmr/fulltext/2022/08000/exercise_to_treat_hypertension_late_breaking_news.6.aspx

American College of Sports Medicine. What's new in the ACSM pronouncement on exercise and hypertension?:

<https://www.acsm.org/home/featured-blogs---homepage/acsm-blog/2019/06/11/new-acsm-pronouncement-exercise-hypertension>

American College of Sports Medicine. FITT recommendations for

hypertension infographic: https://www.acsm.org/read-research/resource-library/resource_detail?id=708f04d2-c8f7-49a1-b314-d9c200d3aa26

Alves AJ, Wu Y, Lopes S, et al. Exercise for hypertension: new recommendation strategies for blood pressure control. J Clin Exerc Physiol. 2024;12(4):120–128.

OVERWEIGHT AND OBESITY

The prevalence of *overweight* and *obesity* has been increasing in the United States and in developed countries around the world. Notably, nonjudgmental person-first language (*e.g.*, adults with obesity) rather than disease-first language (*e.g.*, obese adult) is recommended ([118](#), [119](#)). Recent estimates indicate that approximately 70% of the U.S. population have either overweight or obesity ($\text{BMI} \geq 25 \text{ kg} \cdot \text{m}^{-2}$), with approximately 41.9% with obesity alone ($\text{BMI} \geq 30 \text{ kg} \cdot \text{m}^{-2}$), including 7% with severe obesity ($\text{BMI} \geq 40 \text{ kg} \cdot \text{m}^{-2}$) ([120](#)). Moreover, obesity is disproportionately more prevalent in individuals that identify as Black and Hispanic adults, evident in 49.9% and 45.6% of these cohorts, respectively ([121](#)). Importantly, 22 states in the United States had a prevalence of obesity of 35% or higher in 2022, while approximately 10 years ago, no states had a prevalence of obesity greater than 35%.

Statistics relating to youth indicate that 32% of children and adolescents are classified as overweight or obese ([122](#)). In the United States, the percentage of children 6–11 years of age who have obesity is 19.3%; the percentage of adolescents (12–19 years of age) who have obesity is 20.9% ([122](#)). The troubling data on overweight/obesity prevalence among the adult and pediatric populations and its health implications have precipitated an increased awareness in the value of identifying and treating individuals with excess body weight ([33](#), [123–125](#)). In fact, obesity is associated with an elevated risk for T2DM, CVD, CVD-related risk factors, musculoskeletal problems, and some cancers. It is estimated that obesity-related conditions account for more than 7% of total health care costs in the United States, and the direct and indirect costs of obesity are in excess of \$190 billion annually ([126](#)).

Obesity is multifactorial and its management is complex. Lifestyle remains the cornerstone therapy for patients with obesity. Specifically, a sustained chronic negative energy balance (energy expenditure [EE] > energy intake [EI]) results in weight loss. Sustained weight loss of 3%–5% of initial body weight results in clinically meaningful reductions in several CVD risk factors, including TGs, blood glucose, and HbA1C levels, and the risk of developing T2DM ([127](#)). There is evidence that as little as 2%–3% weight loss can result in similar CVD risk factor improvement ([33](#)). Weight loss >10% within the first year of starting a structured lifestyle intervention has been associated with lower risk for major CVD ([128](#)). These benefits are more likely to be sustained through the maintenance of weight loss, but maintenance is challenging with weight regain averaging approximately 33%–50% of initial weight loss within 1 year of terminating treatment. This is concerning as weight loss is usually characterized by a concomitant loss of fat mass as well as lean mass, while weight regain is typically resulting from an almost exclusive fat mass gain. Lean mass preservation in the setting of weight loss remains, in fact, a major focus of investigation especially considering the growing prevalence of sarcopenic obesity, which describes the concomitant presence of sarcopenia (see [Chapter 6](#)) and obesity ([129](#), [130](#)).

Lifestyle interventions for weight loss that combine reductions in EI with increases in EE through exercise and other forms of PA often result in an initial 5%–10% reduction in body weight ([131](#)). PA appears to have a modest impact on the magnitude of weight loss observed across the initial weight loss intervention compared with reductions in EI ([132](#)). A review of weight loss interventions found that programs, which combined diet and exercise, resulted in a 20% (~3 kg) greater weight loss versus diet restriction alone ([131](#)); however, this effect is lost when EI is severely reduced ([33](#)). PA and diet restriction will provide comparable weight loss if they provide similar levels of negative energy balance ([33](#)). Due to low levels

of CRF characteristic of overweight/obesity, it may be difficult for these individuals to perform a volume of PA required to achieve clinically meaningful weight loss. Therefore, the combination of moderate reductions in EI with adequate levels of PA maximizes the likelihood of achieving a clinically meaningful weight loss in individuals with overweight and obesity. The addition of PA to dietary energy restriction interventions in individuals with obesity is important for additional beneficial metabolic adaptations that cannot be achieved by diet alone, including improvements in insulin sensitivity and up-regulated expression of genes involved in mitochondrial health and angiogenesis ([133](#)). Importantly, even in the absence of weight loss, increased PA and exercise training should be recommended in patients to improve CRF. CRF mediates many of the detrimental effects of obesity; therefore, improving CRF, with or without weight loss, should remain one of the primary goals of obesity treatment ([134](#)).

There is a dose-response relationship between PA levels and the magnitude of weight loss. The ACSM's position stand on PA and weight loss concluded that (a) less than $150 \text{ min} \cdot \text{wk}^{-1}$ of PA promotes minimal weight loss, (b) greater than $150 \text{ min} \cdot \text{wk}^{-1}$ of PA results in modest weight loss of approximately 2–3 kg, and (c) greater than $225\text{--}420 \text{ min} \cdot \text{wk}^{-1}$ of PA results in a 5–7.5-kg weight loss ([33](#)).

PA appears necessary for most individuals to prevent or slow down the weight regain after achieving the targeted weight loss, but there are no adequately powered energy balance studies to provide evidence for the amount of PA needed to prevent weight regain after weight loss ([33](#)). However, there is literature that suggests it may require more than the consensus public health recommendation for PA of $150 \text{ min} \cdot \text{wk}^{-1}$ or 30 minutes of PA on most days of the week ([33](#), [86](#)). Some studies support the value of approximately $200\text{--}300 \text{ min} \cdot \text{wk}^{-1}$ of PA during weight maintenance to reduce weight regain after weight loss, and it seems that “more is better” ([33](#)).

Based on the existing scientific evidence and practical clinical guidelines ([33](#)), the ACSM makes the following recommendations regarding exercise testing and training for individuals with overweight and obesity.

Exercise Testing

- An exercise test is often not necessary in individuals with overweight and obesity prior to beginning a low- to moderate-intensity exercise program.
- Individuals with overweight and obesity are at risk for other comorbidities (*e.g.*, dyslipidemia, hypertension, hyperinsulinemia, hyperglycemia), which are associated with CVD risk.
- The timing of medications to treat comorbidities relative to exercise testing should be considered, particularly in those who take β -blockers and glucose-lowering medications.
- The presence of musculoskeletal and/or orthopedic conditions may necessitate the need for using leg or arm ergometry.
- The potential for low exercise capacity in individuals with overweight and obesity may necessitate a low initial workload (*e.g.*, 2–3 METs) and small increments per testing stage of 0.5–1.0 MET (see examples in [Chapters 3](#) and [4](#)).
- Exercise equipment must be adequate to meet the weight specification of individuals for safety and calibration purposes.
- The appropriate cuff size should be used to measure BP to minimize the potential for inaccurate measurement.
- Because of the potential presence of elevated BP, strict adherence to protocols for assessing BP before and during exercise testing should be followed (see [Chapter 2](#)) ([109](#)).

Exercise Prescription

The goals of exercise during the active weight loss phase are to (a) maximize the amount of caloric expenditure to enhance the amount of weight loss and (b) integrate exercise into the individual's lifestyle to prepare them for a successful weight loss maintenance phase.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH OVERWEIGHT AND OBESITY ([33](#), [135](#))

	Aerobic	Resistance	Flexibility
Frequency	$\geq 5 \text{ d} \cdot \text{wk}^{-1}$	$2-3 \text{ d} \cdot \text{wk}^{-1}$	$\geq 2-3 \text{ d} \cdot \text{wk}^{-1}$
Intensity	Initial intensity should be moderate (40%–59% $\dot{V}O_{2R}$ or HRR); progress to vigorous ($\geq 60\%$ $\dot{V}O_{2R}$ or HRR) for greater health benefits	60%–70% of 1-RM; gradually increase to enhance strength and muscle mass	Stretch to the point of feeling tightness or slight discomfort
Time	$30 \text{ min} \cdot \text{d}^{-1}$ ($150 \text{ min} \cdot \text{wk}^{-1}$); increase to $60 \text{ min} \cdot \text{d}^{-1}$ or more ($250-300 \text{ min} \cdot \text{wk}^{-1}$)	2–4 sets of 8–12 repetitions for each of the major muscle groups	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming)	Resistance machines and/or free weights	Static, dynamic, and/or PNF

1-RM, one repetition maximum; HRR, heart rate reserve; $\dot{V}O_{2R}$, oxygen uptake reserve; PNF, proprioceptive neuromuscular facilitation.

Exercise Training Considerations

- The duration of moderate-to-vigorous intensity PA should initially progress to at least $30 \text{ min} \cdot \text{d}^{-1}$ ([86](#)).
- To promote long-term weight loss maintenance, individuals should progress to at least $250 \text{ min} \cdot \text{wk}^{-1}$ ($\geq 2,000 \text{ kcal} \cdot \text{wk}^{-1}$) of moderate to vigorous exercise. To achieve the weekly maintenance activity goal of $\geq 250 \text{ min} \cdot \text{wk}^{-1}$, exercise and PA should be performed on $5-7 \text{ d} \cdot \text{wk}^{-1}$.
- Individuals with overweight and obesity may accumulate this amount of PA in multiple daily bouts of at least 10 minutes in duration or through increases in other forms of moderate-intensity lifestyle PA. Accumulation of intermittent exercise may increase the volume of PA achieved by

previously sedentary individuals and may enhance the likelihood of adoption and maintenance of PA ([135](#)).

- Resistance training does not result in clinically significant weight loss ([33](#), [136](#)). The addition of resistance exercise to energy restriction does not appear to prevent the loss of fat-free mass or the observed reduction in resting EE ([135](#)). However, resistance exercise coupled with adequate protein consumption (at least $1.2 \text{ g} \cdot \text{kg}^{-1}$) is recommended to minimize the loss of lean mass associated with very-low-calorie diets, particularly in individuals with overweight and obesity that are susceptible to sarcopenia and frailty ([137](#)).
- Resistance exercise may enhance muscular strength and physical function in individuals who are overweight or obese. Moreover, there may be additional health benefits of participating in resistance exercise such as improvements in CVD and DM risk factors and other chronic disease risk factors ([135](#)).

Special Considerations ([33](#), [138](#))

- Utilize goal setting to target short- and long-term weight loss. Target a minimal reduction in body weight of at least 3%–10% of initial body weight over 3–6 months.
- Target reducing current EI to achieve weight loss. A reduction of $500\text{--}1,000 \text{ kcal} \cdot \text{d}^{-1}$ is adequate to elicit a weight loss of $1\text{--}2 \text{ lb} \cdot \text{wk}^{-1}$ ($0.5\text{--}0.9 \text{ kg} \cdot \text{wk}^{-1}$). This reduced EI should be combined with an improvement in diet quality, characterized by the reduction in added sugars and saturated fatty acids and an increase in unsaturated fatty acids (*i.e.*, poly- and monounsaturated fatty acids) associated with a reduction in CVD in both primary ([139](#)) and secondary ([140](#)) prevention.
- Weight loss beyond 5%–10% may require more aggressive nutrition, exercise, and behavioral intervention (see [Chapter 12](#)). For those who do

not respond to any degree of lifestyle intervention, medical treatments such as medications or surgery may be appropriate.

- Medically indicated very-low-calorie diets with energy restrictions of up to $1,500 \text{ kcal} \cdot \text{d}^{-1}$ can result in greater initial weight loss amounts compared to more conservative EI reductions. These medically managed meal plans are typically only used for selected individuals and for short periods of time.
- Incorporate opportunities to enhance communication between health care professionals such as registered dietitian nutritionists, and exercise professionals and individuals with overweight and obesity following the initial weight loss period.
- Target changing eating and exercise behaviors because sustained changes in both behaviors result in significant long-term weight loss and maintenance. Assist individuals with achieving evidence-based recommendations for aerobic exercise during the weight loss and maintenance phases.
- For patients treated with weight loss medications, such as glucagon-like peptide (GLP)-1 receptor agonists among others, participation in structured moderate- to vigorous-intensity exercise programs is safe, effective, and recommended. This combination approach prevents (a) the lean mass loss associated with weight loss achieved with the pharmacologic therapy alone and (b) the increase in resting HR characteristic of some of these agents ([141](#)). Moreover, compared to pharmacological intervention or exercise alone, a combined intervention results in a greater reduction in body fat and sustained weight loss. In addition, combined intervention has shown greater improvements in insulin sensitivity, glycated hemoglobin, and CRF ([141](#)).
- Accommodations should be made to provide privacy during weighing ([118](#)).

- Patients with obesity are likely to present with comorbidities; therefore, special consideration related to those comorbidities such as diabetes and hypertension, among others, should also be considered.

Bariatric Surgery

Surgery for weight loss may be indicated for individuals with a BMI $\geq 40 \text{ kg} \cdot \text{m}^{-2}$ or those with comorbid risk factors and BMI $\geq 35 \text{ kg} \cdot \text{m}^{-2}$.

Comprehensive treatment following surgery includes exercise, as there is evidence for enhanced weight and fat mass loss ([142](#), [143](#)). Moreover, exercise training after bariatric surgery may improve CRF and muscular strength, while reducing bone loss and ultimately preventing weight regain ([134](#)). Once individuals are cleared for exercise training by their physician after surgery, a progressive exercise program for all individuals including aerobic, resistance, and flexibility training should follow the FITT principle of Ex R_x for weight loss and maintenance for individuals with overweight and obesity as listed previously in this section. Those with a prior history of orthopedic injuries should be assessed to reduce the risk of aggravation by weight-bearing exercise. In those for whom excess body weight may limit the ability to engage in weight-bearing exercise or continuous exercise, intermittent exercise and non-weight-bearing alternatives should be considered. Subsequently, continuous exercise and weight-bearing exercise such as walking may be slowly introduced to make up a greater portion of the exercise program.

ONLINE RESOURCES

American College of Sports Medicine position stand on overweight and obesity: <https://www.acsm.org/acsm-positions-policy/official-positions>

National Heart, Lung, and Blood Institute. Clinical guidelines on the identification, evaluation, and treatment of overweight and obesity in adults: the evidence report: <http://www.nhlbi.nih.gov/health-pro/guidelines/archive/clinical-guidelines-obesity-adults-evidence-report>

METABOLIC SYNDROME

The Metsyn is characterized by a clustering of risk factors associated with an increased incidence of CVD, DM, and stroke ([144](#)). Uncertainty exists as to whether Metsyn is a distinct pathophysiological entity or simply a clinical marker of future untoward events, particularly CVD mortality. Observational research shows a greater risk of CVD death in individuals with Metsyn compared to those without Metsyn ([145](#)), yet no evidence exists from prospective studies to confirm these findings.

Initial criteria for defining Metsyn varied by organization ([146](#)) and yielded a prevalence rate of 34% and 39% in U.S. adults ([147](#), [148](#)). A consensus definition now exists ([147](#)), in which each organization includes hyperglycemia (or currently taking blood glucose medication), elevated BP (or currently taking hypertension medication), dyslipidemia (or currently taking lipid-lowering medication), and national or regional cut points for central adiposity based on waist circumference; however, differences in specific value within these criteria remain ([Table 9.2](#)). It is further agreed that an individual is categorized as having Metsyn when he or she displays at least three of the defining risk factors.

TABLE 9.2 Metabolic Syndrome Criteria: NCEP/ATP III, IDF, and WHO

Criteria	NCEP/ATP III (149)	IDF (150)	WHO (151) ^a
Body weight	Waist circumference ^{a,b}	Waist circumference ^c	Waist-to-hip ratio (males > 0.90; females > 0.85) and/or BMI of > 30 kg · m ⁻²
Males	> 102 cm (> 40 in)	≥ 94 cm (≥ 37 in)	> 0.9 ratio
Females	> 88 cm (> 35 in)	≥ 80 cm (≥ 31.5 in)	> 0.85 ratio
Insulin resistance/glucose	≥ 100 mg · dL ^{-1d} or on drug treatment for elevated blood glucose	≥ 100 mg · dL ⁻¹ or previously diagnosed Type 2 diabetes	See footnote ^e
Dyslipidemia			
HDL	Males: < 40 mg · dL ⁻¹	Males: < 40 mg · dL ⁻¹	Males: < 35 mg · dL ⁻¹
	Females: < 50 mg · dL ⁻¹	Females: < 50 mg · dL ⁻¹	Females: < 39 mg · dL ^{-1f}
	Anyone on drug treatment for reduced HDL-C	Anyone on drug treatment for reduced HDL-C	
Triglycerides	≥ 150 mg · dL ⁻¹ or on drug treatment for	≥ 150 mg · dL ⁻¹ or on drug treatment for	≥ 150 mg · dL ⁻¹ or on drug treatment for high triglycerides

Criteria	NCEP/ATP III (149)	IDF (150)	WHO (151) ^a
	high triglycerides	high triglycerides	
Elevated blood pressure	≥ 130 or ≥ 85 mm Hg or on drug treatment for hypertension	≥ 130 or ≥ 85 mm Hg or treatment of previously diagnosed hypertension	Antihypertensive medication and/or a BP of ≥ 140 or ≥ 90 mm Hg
Other	N/A	N/A	Urinary albumin excretion rate ≥ 20 $\mu\text{g} \cdot \text{min}^{-1}$ or albumin:creatinine ratio ≥ 30 $\text{mg} \cdot \text{g}^{-1}$

NOTE: To convert glucose from $\text{mg} \cdot \text{dL}^{-1}$ to $\text{mmol} \cdot \text{L}^{-1}$, multiply by 0.0555. To convert HDL from $\text{mg} \cdot \text{dL}^{-1}$ to $\text{mmol} \cdot \text{L}^{-1}$, multiply by 0.0259. To convert triglycerides from $\text{mg} \cdot \text{dL}^{-1}$ to $\text{mmol} \cdot \text{L}^{-1}$, multiply by 0.0113.

^a Overweight and obesity are associated with insulin resistance and the metabolic syndrome (Metsyn). However, the presence of abdominal obesity is more highly correlated with these metabolic risk factors than is elevated BMI. Therefore, the simple measure of waist circumference is recommended to identify the body weight component of the Metsyn.

^b Some males develop multiple metabolic risk factors when the waist circumference is only marginally increased (94–102 cm [37–39 in]). Such individuals may have a strong genetic contribution to insulin resistance. They should benefit from changes in life habits, similarly to males with categorical increases in waist circumference.

^c A required criterion defined as waist circumference ≥94 cm (≥37 in) for Europid males and ≥80 cm (≥31.2 in) for Europid females, with ethnicity-specific values for other groups.

^d The American Diabetes Association has established a cut point of $\geq 100 \text{ mg} \cdot \text{dL}^{-1}$, above which individuals have either prediabetes (impaired fasting glucose) or diabetes mellitus ([152](#)). This cut point should be applicable for identifying the lower boundary to define an elevated glucose as one criterion for metabolic syndrome.

^e A required criterion is one of the following: Type 2 diabetes mellitus, impaired fasting glucose, impaired glucose tolerance, or for individuals with normal fasting glucose levels ($< 100 \text{ mg} \cdot \text{dL}^{-1}$), glucose uptake below the lowest quartile for background populations under investigation under hyperinsulinemic and

euglycemic conditions. Note this value has been updated to be consistent with current ADA recommendations regarding normal fasting plasma glucose levels ([152](#)).

^f These values have been updated from those originally presented to ensure consistence with ATP III cut points.

ATP III, Adult Treatment Panel III; BMI, body mass index; BP, blood pressure; HDL-C, high-density lipoprotein cholesterol; IDF, International Diabetes Federation; NCEP, National Cholesterol Education Program; WHO, World Health Organization.

The treatment guidelines for Metsyn recommended by National Cholesterol Education Program Adult Treatment Panel III focus on three interventions including weight control, PA, and treatment of the associated CVD risk factors that may include pharmacotherapy ([153](#)). The International Diabetes Federation (IDF) guidelines for primary intervention include (a) moderate restriction in EI to achieve a 5%–10% weight loss within 1 year, (b) moderate increases in PA consistent with the consensus public health recommendations of 30 minutes of moderate-intensity PA on most days of the week ([86](#)), and (c) change in dietary intake composition consistent with modifying specified CVD risk factors (*e.g.*, decreased simple carbohydrates, increased lean-sources of protein, reduced saturated and trans fatty acids, and increased unsaturated fatty acids) ([154](#)). The IDF secondary intervention includes pharmacotherapy for associated CVD risk factors ([154](#), [155](#)).

Exercise Testing

- The presence of Metsyn does not result in the requirement of an exercise test prior to beginning a low- to moderate-intensity exercise program.
- If an exercise test is performed, the general recommendations can be followed (see [Chapter 3](#)), with particular consideration for dyslipidemia, hypertension, or hyperglycemia when present.
- Because many individuals with the Metsyn have either overweight or obesity, exercise testing considerations specific to those individuals

should be followed (see “Overweight and Obesity” section and the relevant ACSM position stand (32)).

- The potential for low exercise capacity in individuals who have overweight or obesity may necessitate a low initial workload (*e.g.*, 2–3 metabolic equivalents [METs]) and small increments per testing stage (0.5–1.0 MET) (see [Chapter 4](#)).
- Because of the potential presence of elevated BP, strict adherence to protocols for assessing BP before and during exercise testing should be followed (see [Chapter 2](#)) (156).
- Exercise training reduces the long-term need for increased medications use in individuals with Metsyn (157).

Exercise Prescription/Special Considerations

The FITT principle of Ex R_x in Metsyn is generally consistent with the recommendations for healthy adults regarding aerobic, resistance, and flexibility exercise (see [Chapter 5](#)). Similarly, the minimal dose of PA to improve health/fitness outcomes is consistent with the consensus public health recommendations of 150 min · wk⁻¹ or 30 minutes of moderate-intensity PA on most days of the week (86, 102). However, due to the clustering of CVD and DM risk factors, along with the likely presence of chronic diseases and health conditions that accompany Metsyn, the following Ex R_x special considerations are suggested:

- When developing the Ex R_x for Metsyn, attention should be given to each risk factor/condition present, with the most conservative criteria used to set initial workloads (see other sections of this chapter on the FITT principle Ex R_x for these other chronic diseases and health conditions as well as relevant ACSM position stands (32,84,88)). Over time and as

tolerated, longer duration and higher intensities may be necessary to achieve significant health and fitness outcomes.

- To reduce the impact of the Metsyn, variables that are considered risk factors for CVD and DM, initial exercise training should be performed at a moderate intensity (40%–59% $\dot{V}O_2R$ or HRR) totaling a minimum of 150 min $\cdot$ wk⁻¹ or 30 min $\cdot$ d⁻¹ most days of the week to allow for optimal health/fitness improvements. When appropriate, progress to a more vigorous intensity ($\geq 60\%$ $\dot{V}O_2R$ or HRR).
- Reduction of body weight is an important goal for individuals with Metsyn ([149](#)); therefore, gradually increasing PA levels to approximately 250–300 min $\cdot$ wk⁻¹ or 50–60 min on 5 d $\cdot$ wk⁻¹ may be necessary when appropriate ([33](#)). Daily and weekly amounts of PA may be accumulated intermittently. Accumulation of exercise in multiple shorter bouts (≥ 10 min in duration) may increase the volume of PA achieved and can include various forms of moderate-intensity lifestyle PAs. For some individuals, progression to 60–90 min $\cdot$ d⁻¹ of PA may be necessary to promote or maintain weight loss (see “Overweight and Obesity” section and ACSM position stand ([33](#))).
- Resistance training, when combined with aerobic training, can produce greater decreases in Metsyn prevalence than that of aerobic training alone ([158](#), [159](#)). Reported participation in ≥ 2 d $\cdot$ wk⁻¹ of muscle strengthening activity reduces the risk of acquiring dyslipidemia, IFG, prehypertension, and increased waist circumference, all part of the Metsyn cluster ([160](#)) (see [Chapter 5](#)).
- HIIT is also an effective strategy to improve Metsyn and its components at both high-volume and low-volume ([161–163](#)), in addition to showing consistent improvement in fitness. If HIIT is chosen, selecting low-impact activities may mitigate the risk for injuries ([164](#), [165](#)).

ONLINE RESOURCES

American College of Sports Medicine position stand on exercise and hypertension: <https://www.acsm.org/acsm-positions-policy/official-positions>

American College of Sports Medicine position stand on weight loss and prevention of weight gain: https://journals.lww.com/acsm-msse/Fulltext/2009/02000/Appropriate_Physical_Activity_Intervention.26.aspx

American Heart Association, metabolic syndrome:
http://www.heart.org/HEARTORG/Conditions/More/MetabolicSyndrome/Metabolic-Syndrome_UCM_002080_SubHomePage.jsp

Mayo Clinic Diseases and Conditions, metabolic syndrome:
<https://www.mayoclinic.org/diseases-conditions/metabolic-syndrome/symptoms-causes/syc-20351916>

National Heart Lung and Blood Institute. What is metabolic syndrome?:
<http://www.nhlbi.nih.gov/health/health-topics/topics/ms>

METABOLIC DYSFUNCTION- ASSOCIATED STEATOTIC LIVER DISEASE

Characterized by an abnormal amount of fat in the liver, metabolic dysfunction–associated steatotic liver disease (MASLD) is the leading cause of liver disease worldwide, affecting a third of the global population ([166](#)). Previously known as nonalcoholic fatty liver disease (NAFLD), the term MASLD was adopted instead following a June 2023 update to the nomenclature used to discuss fatty liver disease by multiple leading global liver societies ([167](#)). This update occurred in response to patient feedback regarding the stigmatizing nature of the term NAFLD as well as to better reflect disease pathophysiology. Importantly, many factors contribute to the development of MASLD, and physical inactivity plays a key role in MASLD development ([168](#)). If uncorrected, MASLD can progress and individuals with this common condition can develop liver fibrosis (scarring), cirrhosis, or primary liver cancer. Individuals with MASLD also have greater risk of CVD and cancer outside their liver.

MASLD can be initially detected by elevated liver enzymes as found in blood tests including alkaline phosphatase, alanine transaminase, and aspartate transaminase. Further tests including liver ultrasound and/or biopsy are needed to officially diagnose this disease. Individuals with MASLD do not consume alcohol in excess, defined as more than two drinks/day for males and one drink/day for biological females. Though many MASLD patients do not have symptoms, some may experience fatigue, abdominal pain, and poor health-related quality of life. It is also common that these patients have other metabolic comorbidities including obesity, DM, hypertension, and hyperlipidemia.

At this point, there is no regulatory agency-approved drug therapy or cure for MASLD; therefore, lifestyle modification with Mediterranean-informed dietary change and increased PA is the foundation of treatment.

Unfortunately, most individuals with MASLD are physically inactive ([168](#), [169](#)). In fact, only 10%–20% meet guideline-based amounts of weekly activity. MASLD patients, on average, walk approximately 10 km less per week than those without MASLD and each additional hour of sedentary behavior is linked to 1.2% higher liver fat content. Common barriers to PA include low rates of counseling from health care providers, a lack of time, no access to exercise resources (*e.g.*, exercise professionals, fitness facilities), and pain with activity ([170](#)).

Individuals with MASLD are recommended to perform at least 150 min · wk⁻¹ of moderate-intensity aerobic activity or 75 min · wk⁻¹ of vigorous-intensity PA ([168](#), [169](#)). This amount of aerobic activity should be included in a formal Ex R_x and should be supplemented with 2 days of resistance training. When guideline-based amounts of exercise are completed, many health benefits can be expected ([170](#)). These include improvement in liver fat and inflammation, body composition with loss of body fat and gain of lean muscle mass, CRF, metabolic disease control (*e.g.*, blood sugar, blood pressure, cholesterol), health-related quality of life, and a reduction in cardiovascular risk. Importantly, many of these benefits occur without clinically significant body weight loss, defined as 5% or more. Whether exercise training improves liver fibrosis without weight loss remains controversial. Although emerging evidence does suggest this may be the case, most experts agree that at least 7% body weight loss is necessary to improve liver fibrosis ([171](#)), and this is still recommended for all individuals with MASLD who are overweight or obese.

Exercise Testing

- Exercise testing is generally not necessary for MASLD patients beginning an exercise program, especially for low- to moderate-intensity programs.
- Prior to starting a moderate to vigorous exercise training program, individuals with MASLD who are inactive are encouraged to seek medical clearance. For those who are actively engaged in a moderate exercise training program, medical clearance is recommended if seeking to begin a vigorous exercise training program.
- In individuals with signs or symptoms suggestive of cardiovascular, metabolic, and/or renal disease, starting with low-intensity exercise and progressing intensity as physiologic acclimation occurs may be considered.
- If an exercise test is performed, the general recommendations can be followed (see [Chapter 3](#)), with consideration for dyslipidemia, hypertension, or hyperglycemia present.
- Because many individuals with MASLD are overweight or have obesity, exercise testing considerations specific to those individuals should be followed. See “Overweight and Obesity” section and the relevant ACSM position stand ([33](#)).
- Patients with an abnormal ECG or signs/symptoms of CVD may need further testing and should seek consultation with their health care provider prior to starting.
- The timing of medications to treat comorbidities relative to exercise testing should be considered, particularly in those who take β -blockers and glucose-lowering medications.

Exercise Prescription

- When designing an Ex R_x for an individual with MASLD, exercise professionals should consider current level of fitness, comorbidities, and personal preferences to increase adherence.
- Exercise can be completed in intermittent bouts during multiple sessions to accumulate the FITT recommendations, if longer sessions are not well tolerated.
- The duration of moderate- to vigorous-intensity PA should initially progress to at least $30 \text{ min} \cdot \text{d}^{-1}$ ([86](#)).
- Exercise training can be completed by a variety of modalities including moderate-intensity aerobic exercise and HIIT.
- If HIIT is chosen, low-impact activities can mitigate injury risk.
- Resistance training should be considered 2 days a week in addition to prescribed aerobic training.
- Yoga or Pilates may be beneficial for sedentary individuals as these types of exercise programs can improve flexibility and balance.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH MASLD

	Aerobic	Resistance	Flexibility and Balance
Frequency	3–5 d · wk ⁻¹	2 d · wk ⁻¹	≥2–3 d · wk ⁻¹
Intensity	Moderate (40%–59% HRR) or vigorous (> 60% HRR)	Moderate (50%–85% 1-RM to improve strength, < 50% 1-RM to improve muscle endurance)	Stretch to the point of tightness or slight discomfort Balance exercises: light-to-moderate intensity
Time	150 min · wk ⁻¹ for moderate intensity (<i>e.g.</i> , 30 min · d ⁻¹ , 5 d · wk ⁻¹). 75 min · wk ⁻¹ for vigorous intensity	2–4 sets of 8–12 repetitions for each of the major muscle groups, 12–20 repetitions for muscular endurance	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise Balance for any duration
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming)	Resistance machines, free weights, and/or body weight	Static, dynamic, other stretching, yoga, Pilates

1-RM, one repetition maximum; HRR, heart rate reserve

Special Considerations

- Exercise testing and exercise at vigorous intensities in individuals with MASLD should be medically supervised until the safety of the activities are well established.
- Individuals with MASLD have more exercise-limiting conditions, and they experience greater pain sensitivity (*e.g.*, higher ratings of perceived exertion), fatigue, and osteoarthritis than those without MASLD.
- A graded, “staircase” approach is encouraged for patients with MASLD to progress and achieve moderate- to vigorous-intensity activity.

- Individuals with overweight and who are obese may accumulate this amount of PA in multiple daily bouts or through increases in other forms of moderate-intensity lifestyle PA. Accumulation of intermittent exercise may increase the volume of PA achieved by previously sedentary individuals and may enhance the likelihood of adoption and maintenance of PA ([135](#)).
- Focusing on short-term gains, such as body composition (“Are your clothes fitting differently?”), fitness (“Do you have more stamina?”), or quality of life (“Are you feeling better in general?”), can be an effective way to encourage exercise adherence rather than directing attention to weight loss.

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CHAPTER 10

Exercise Testing and Prescription for Populations With Other Chronic Diseases and Health Conditions

INTRODUCTION

This chapter contains the exercise testing and exercise prescription (Ex Rx) guidelines and recommendations for individuals with noncommunicable diseases (NCDs) and other health conditions not addressed in [Chapters 8](#) (cardiovascular and pulmonary), [2](#) (metabolic), and [11](#) (neurologic diseases and conditions). As with the other chapters, the Ex Rx guidelines and recommendations are presented using the *Frequency, Intensity, Time, and Type (FITT)* principle of Ex Rx based on the available professional society position papers and scientific statements or using other literature. Note that information is often lacking regarding volume and progression of exercise training for the NCDs presented in this chapter. In these instances, the guidelines and recommendations provided in [Chapter 5](#) for apparently healthy populations should be adapted with good clinical judgment for the health condition(s) being targeted.

ARTHRITIS

Arthritis is an umbrella term for over 100 rheumatic conditions (1), which together, when measured in years lived with disability (YLD), provides the second most common cause of disability in the United States (2) and worldwide (3). And the disability burden of arthritis is rapidly escalating, with the YLD specifically attributable to osteoarthritis (OA) alone having increased globally by 75% from 1990 to 2013 (3). Among adults in the United States, approximately 23% (54.4 million) report having a doctor's diagnosis of arthritis (4), and of these, 44% (23.7 million) complain of arthritis-related physical activity (PA) limitations (2).

The two most common forms of arthritis are OA and rheumatoid arthritis (RA), with the latter being the most well-investigated inflammatory joint disease. OA is a progressive local degenerative joint disease affecting one or more synovial joints (*e.g.*, hands, hips, spine, and knees) and is associated with risk factors including the following: having overweight or obesity, history of joint injury or surgery, genetic predisposition, aging, female sex, and certain occupations. RA is a chronic, systemic, inflammatory autoimmune disease of unknown etiology, in which the local inflammatory response, causes swelling (inflammation) of the joint lining (synovitis), damage to articular cartilage and supporting ligaments, and may lead to bony erosions; and, systemically, results in significant fatigue, muscle loss, fat gain, increased risk of osteoporosis, and significantly exacerbated cardiovascular disease (CVD) risk primarily due to accelerated atherosclerosis (5, 6). Other common rheumatic diseases include gout, spondyloarthropathies (*e.g.*, ankylosing spondylitis, psoriatic arthritis, reactive arthritis, and enteropathic arthritis), and specific connective tissue diseases (*e.g.*, systemic lupus erythematosus, systemic sclerosis [scleroderma], and dermatomyositis).

Regardless of type, arthritis is generally characterized by pain, impaired physical function, fatigue, and muscle loss and increased adiposity, with (a) 66% of individuals with OA being either overweight or obese at the time of disease onset (7) and (b) approximately two-thirds of RA patients experiencing rheumatoid cachexia, a condition characterized by loss of muscle mass and increased adiposity and metabolism, while body weight remains unchanged (8). Due to the aging population and burgeoning obesity epidemic, the prevalence of physician-diagnosed arthritis is expected to increase to an estimated 78 million Americans (26%) by 2040 (9), with the same trend anticipated globally (3).

Pharmaceutical treatment of OA primarily involves analgesics (*e.g.*, topical and oral nonsteroidal anti-inflammatory drugs), and for RA, disease-modifying antirheumatic drugs, biologic therapies, and glucocorticoids. Optimal treatment of arthritis features a multidisciplinary approach, which includes drug treatment, individual education in self-management, physical therapy, occupational therapy, and exercise (10–14). This treatment approach is supported by the European Alliance of Associations for Rheumatology, the leading association for developing health care recommendations in rheumatic and musculoskeletal diseases (RMD) (11, 15). When joint damage and loss of mobility are severe, and restoration of a reasonable level of function and control of pain is no longer achievable by pharmacological and conservative management (*i.e.*, “end-stage” disease), total joint replacement and other surgeries are the remaining therapeutic options.

Although pain and functional limitations can present challenges to individuals with arthritis in terms of performing PA, regular exercise is essential for managing these conditions and hence are a core component of international guidelines and recommendations. For instance, the disease process in those with inflammatory arthropathies may lead to more muscle wasting and excess adiposity compared to same age and sex matched healthy individuals ([4](#), [16](#)), and this is a phenomenon that cannot be reversed through medication ([17](#)). Thus, regular exercise plays an important role in weight control and achieving a healthy body composition both through the anabolic and lipolytic effects of exercise itself and via the anti-inflammatory effects of regular PA ([18](#)). Regular exercise provides numerous additional benefits to individuals with arthritis, including minimizing functional decline or improving functional capacity in the deconditioned; reducing fall risk; attenuating pain and joint stiffness; reducing comorbidities such as CVD, Type 2 diabetes mellitus (T2DM), metabolic syndrome, and osteoporosis; and improving mental health and quality of life ([11](#), [19–24](#)).

Exercise Testing

Most individuals with arthritis tolerate symptom-limited exercise testing consistent with recommendations for apparently healthy adults (see [Chapters 3](#) and [4](#)). The following are special considerations for individuals with arthritis:

- High-intensity exercise, as during a maximal stress test, is contraindicated when there is acute inflammation (*i.e.*, hot, swollen, and painful joints; “disease flare”). If individuals are experiencing acute inflammation, exercise testing should be postponed until the flare has subsided.
- Although most individuals with arthritis tolerate treadmill walking, the use of leg or arm cycle ergometry may be less painful for some, thereby providing a more accurate assessment or estimation of cardiorespiratory fitness (CRF) and/or capacity. The mode of exercise chosen should be based on what is least painful for the individual being tested.
- Allow time for individuals to warm-up (at a very-light- or light-intensity level) according to each individual’s functional status prior to beginning the graded exercise test.
- Monitor pain levels and perceived exertion during testing using a validated exercise intensity scale, such as the Borg CR-10 Scale (see [Figure 4.2](#)) ([25](#)) and a validated pain scale such as the visual numeric pain scale ([Figure 10.1](#)) ([26](#)).

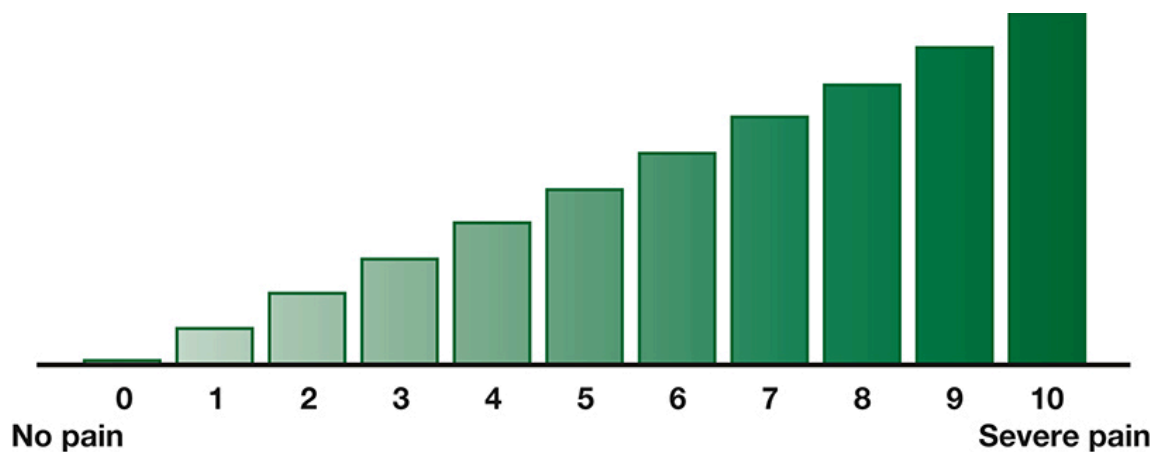


Figure 10.1 Visual numeric pain scale. Reprinted from (26).

- Muscle strength and endurance can be measured using standard protocols (see [Chapter 3](#)). However, the tester should be aware that pain and swelling may impair maximum voluntary muscle contraction via neural inhibition of muscle fiber recruitment in affected joints.

Exercise Prescription

A major barrier to individuals with arthritis starting an exercise program is the belief that exercise, particularly weight-bearing exercise, will exacerbate joint damage and symptoms such as pain and fatigue (27, 28). This fear is prevalent not only among individuals with arthritis but also among some physicians and allied health professionals (29). Thus, individuals with arthritis need to be reassured that exercise is not only safe but also widely and consistently reported to reduce pain, fatigue, inflammation, and disease activity (11, 20–24, 30–34). Those with arthritis, particularly those with pain and those who are deconditioned, should gradually progress to exercise intensities and volumes that provide clinically significant health benefits. However, pain medication is not recommended for patients with RMD to have relief and thus participate in exercise programs (35). In general, recommendations for Ex Rx are consistent with those for apparently healthy adults (see [Chapter 5](#)) with observance of FITT recommendations and additional consideration of an individual's disease activity, pain, joint integrity, functional limitations, and personal exercise/PA preferences (1). Although these recommendations will likely be appropriate for most individuals with arthritis for both aerobic and resistance training, an individual's personal exercise mode and intensity preference needs to be considered to optimize adoption and adherence to exercise.

Exercise Training Considerations

- The goal of aerobic exercise training is to improve CRF without exacerbating joint pain or damage. There is no clear evidence that supports avoiding or discouraging high-impact activities, such as running, stair climbing, and those with stop and go actions, in individuals with arthritis unless they present with obvious biomechanical or joint stability issues. However, because individuals with arthritis often have lower levels of CRF (36), muscle strength, and neuromuscular protective

reflexes, they should approach high-impact exercise cautiously in order to minimize the risk of incurring injury or aggravating joint symptoms.

- Long, continuous bouts of aerobic exercise may initially be difficult for those who are very deconditioned and restricted by pain and joint mobility, so breaking up sedentary behavior by encouraging movement throughout the day is beneficial and should be encouraged ([37](#)). It is appropriate to start with short bouts of as little as 5 minutes, or what can be initially tolerated.
- In addition to improving muscular strength and endurance, research evidence indicates that resistance training improves physical function and is likely to reduce chronic pain through local (*i.e.*, enhanced dynamic stability, attenuated joint forces) and systemic changes (*i.e.*, decreased inflammation, elevated endogenous opioids) ([38](#), [39](#)).
- Flexibility training is important to enhance range of motion (ROM) and to counteract the negative effects of arthritis on joint mobility. Additionally, active ROM can shorten episodes of “morning stiffness” in individuals with RA.
- Balance and agility training are important for individuals with lower limb involvement as pain and impaired coordination, protective reflexes, and proprioception put them at greater risk of falling. Both static (*e.g.*, standing in tandem foot position or on one leg) and dynamic (*e.g.*, walking, changing directions, stepping over obstacles) balance activities are recommended.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH ARTHRITIS ([11](#), [14](#), [23](#), [27–32](#))

	Aerobic	Resistance	Flexibility
Frequency	3–5 d · wk ⁻¹	2–3 d · wk ⁻¹	Daily
Intensity	Moderate (40%–59% $\dot{V}O_2R$ or HRR) to vigorous ($\geq 60\%$ $\dot{V}O_2R$ or HRR).	60%–80% 1-RM. Initial intensity should be lower (<i>i.e.</i> , 50%–60% 1-RM) for those unaccustomed to resistance training.	Move through ROM feeling tightness/stretch without pain. Progress ROM of each exercise only when there is little or no joint pain.
Time	Accumulate 150 min · wk ⁻¹ of moderate intensity, or 75 min · wk ⁻¹ of vigorous intensity, or an equivalent combination of the two.	Use healthy adult values and adjust accordingly (<i>i.e.</i> , 8–12 repetitions for 1–3 sets); include all major muscle groups.	Up to 10 repetitions for dynamic movements; hold static stretches for 10–30 s and repeat two to four times.
Type	Activities with low joint stress, such as walking, cycling, swimming, or aquatic exercise ¹	Machine, free weights, resistance bands, tubing. Body weight exercises are also appropriate for most individuals with arthritis.	A combination of active, static, and proprioceptive neuromuscular facilitation stretching (see Box 5.4) of all major joints with a focus on affected joints and muscles crossing these joints.

1-RM, one-repetition maximum; HRR, heart rate reserve; ROM, range of motion; $\dot{V}O_2R$, oxygen uptake reserve.

- Adequate warm-up and cool-down periods (at least 5–10 minutes) are critical for minimizing pain. Warm-up and cool-down activities should involve controlled movement of joints through their full ROM along with light-intensity aerobic exercise.
- Individuals with significant pain and functional limitation may need interim goals that are shorter than the recommended FITT and should be encouraged to undertake and maintain any amount of PA that is safely tolerated. In the absence of specific recommendations for people with arthritis, the general population recommendation of increasing aerobic exercise duration by 5–10 minutes every 1–2 weeks over the first 4–6 weeks of an exercise training program can be applied.
- Regular progression of resistance training exercises is critical for achieving gains in muscular strength, endurance, and function. Following the ACSM guidelines ([40](#)) for progression of resistance training in healthy adults is a good starting point while recognizing individuals with arthritis may need to increase loads at slower rates and in smaller increments to minimize localized joint reaction and discomfort.

Special Considerations

- Based on data from ([29](#), [30](#), [41](#)).

- Avoid strenuous exercise during acute flare-ups. However, it is appropriate to gently move joints through their full ROM and break up sedentary behavior with light activity during these episodes.
- Inform individuals with arthritis that a small amount of discomfort in the muscles or joints during or immediately after exercise is common following performance of unfamiliar exercise and hence does not necessarily mean joints are being further damaged. Higher pain ratings 48–72 hours after exercise may be due to delayed-onset muscle soreness, especially in those who are new to exercise; individuals should be informed that this is a normal response to unaccustomed exercise, which will progressively diminish as their training progresses and they adapt to the demands of exercise.
- If specific exercises exacerbate joint pain, alternative exercises that work the same muscle groups and energy systems should be considered.
- Encourage individuals with arthritis to exercise during the time of day when pain is typically least severe and/or in conjunction with peak activity of pain medications. In some cases of inflammatory arthritis, such as RA, this may be after the 2-hour morning stiffness episodes.
- Appropriate shoes that provide good shock absorption and stability are particularly important for individuals with arthritis. Shoe specialists can provide recommendations appropriate for an individual's biomechanics.
- Incorporate functional exercises such as the sit-to-stand, step-ups, stair climbing, and carrying to improve neuromuscular control, balance, and enhance the ability to perform activities of daily living (ADL).
- For pool-based exercise, a water temperature of 83 °F–88 °F (28 °C–31° C) aids in relaxing and increasing ROM, compliance of muscles, and reducing pain.

ONLINE RESOURCES

American College of Rheumatology: <http://www.rheumatology.org>; <https://www.rheumatology.org/L-Am-A/Patient-Caregiver/Diseases-Conditions/Living-Well-with-Rheumatic-Disease/Exercise-and-Arthritis>

Arthritis Foundation: <http://www.arthritis.org>. Individuals can click on “Your Local Area” to locate appropriate walking, group exercise, and aquatic classes in their community.

Exercise is Medicine’s Rx for Health Series:

https://www.exerciseismedicine.org/support_page.php/rx-for-health-series/

CANCER

Cancer is the second leading cause of mortality in the world after heart disease. In 2022, there were approximately 20 million new cancer cases and 9.7 million deaths from cancer. Further, there were 53.5 million people alive within 5 years following their cancer diagnosis ([42](#)). Cancer is increasingly being recognized as not one, but many diseases, defined not only by different anatomic locations but also by cell of origin, etiologic factors, and susceptibility to treatment. Thus, cancer treatment has become increasingly individualized, and this is being recognized in the latest guidelines for exercise assessment and prescription for people with cancer ([43](#)).

Evidence is well established supporting the role of PA and exercise in both cancer prevention and control. Higher volumes of PA are associated with lower risk of developing 13 types of cancer ([44, 45](#)) and a reduction of cancer-related mortality in individuals with several common cancers ([46](#)). Exercise, a subset of PA, has also been shown to mitigate the side effects of cancer therapy, enhance effectiveness of primary treatments, and improve physical function and quality of life for individuals with cancer ([47–50](#)). To understand the relationship between exercise and reduced cancer-specific mortality, there is also growing research into how exercise affects tumor and cancer cell biology with a range of mechanisms being explored ([51, 52](#)). The benefits of PA for primary cancer prevention are briefly outlined in [Chapter 1](#). This section describes the benefits of PA and exercise for individuals with a history of cancer, known as cancer survivors ([53](#)), and offers considerations for exercise testing and prescription in this population.

Overview of Importance of Physical Activity in Cancer Survivors

Cancer-Related Mortality

Findings from observational studies show that participation in regular and sufficient amounts of PA after diagnosis of early-stage, potentially curable cancer is associated with a lower risk of cancer-specific mortality in breast, prostate, and colon cancers ([45](#)). Emerging evidence is that sedentary behavior and screen-based activities are independent risk factors for cancer-specific mortality ([54](#)). Currently, there are no definitive data from randomized controlled trials (RCTs) evaluating the effects of exercise interventions with cancer recurrence or mortality as a primary endpoint. However, several multicenter randomized trials will determine if targeted exercise reduces the risk of cancer progression, reduces recurrence, and prolongs survival in people who have had a cancer diagnosis ([55–57](#)).

Physiological and Quality of Life Outcomes

Cancer survivors derive a variety of physiological and quality of life benefits from exercise. Meta-analyses and systematic reviews of exercise intervention trials in individuals with cancer during and after treatment demonstrate that aerobic exercise during and after cancer treatment increases CRF

(58), and resistance exercise during and after cancer treatment increases upper and lower extremity muscular strength, and increases lean body mass (59), and it is likely that supervised exercise improves these outcomes more than unsupervised exercise (60). It is also apparent that the specificity of the Ex R_x may be high in people with cancer. For example, combined resistance and impact loading exercise (e.g., jumping, hopping, skipping) prevents or slows loss of bone mineral density (BMD) of the lumbar spine and hip (61, 62). Further, the *interference effect* of aerobic exercise blunting muscle hypertrophy from resistance training may be more pronounced in people with cancer due to compromised immune capacity or hormone therapy (62).

Many studies have also evaluated the effect of exercise on quality of life and related outcomes in cancer survivors. Findings from meta-analyses are that exercise interventions reduce fatigue and depression during and after cancer treatment and also improve quality of life and reduce sleep disturbance after cancer treatment (63–65). Again, studies suggest that both supervised and unsupervised interventions can be effective; however, supervised tend to yield greater improvements (66, 67) and also show that higher levels of baseline fatigue and other symptoms predict larger benefits from exercise interventions (60, 66, 68).

Physical Activity Patterns in Cancer Survivors

Exercise volume often decreases during cancer treatment and may not return to prediagnosis volume after completing treatment (69–71). In a nationally representative sample of cancer survivors, only 8% engaged in 150 min · wk⁻¹ of moderate- to vigorous-intensity exercise (72). A similar study demonstrated that breast cancer survivors engaged in a daily average of 1 minute of moderate- to vigorous-intensity exercise, spending most of their day in sedentary (66%) or light-intensity activities (33%) (73). Consequently, there are significant opportunities to utilize exercise as a therapeutic modality to improve numerous outcomes in cancer survivors (58). Also, more than 60% of cancer survivors are ≥65 years (74) and will often have other preexisting health conditions, such as CVD, T2DM, arthritis, sarcopenia, and obesity (75) (see [Chapters 6](#) and [9](#)). The combined result of cancer-related side effects, aging, and other health conditions often manifest as impaired CRF, functional limitations, and reduced quality of life in cancer survivors (76–79). Further, low muscle mass and high fat mass are associated with higher mortality post cancer diagnosis (80). Therefore, promoting exercise without creating unnecessary barriers to participation is of critical importance in cancer survivors (81, 82). Exercise is safe for almost everyone, including most cancer survivors, even those with advanced disease and undergoing difficult treatments, and the health benefits of exercise outweigh the risks for most people (83, 84).

Preparticipation Evaluation

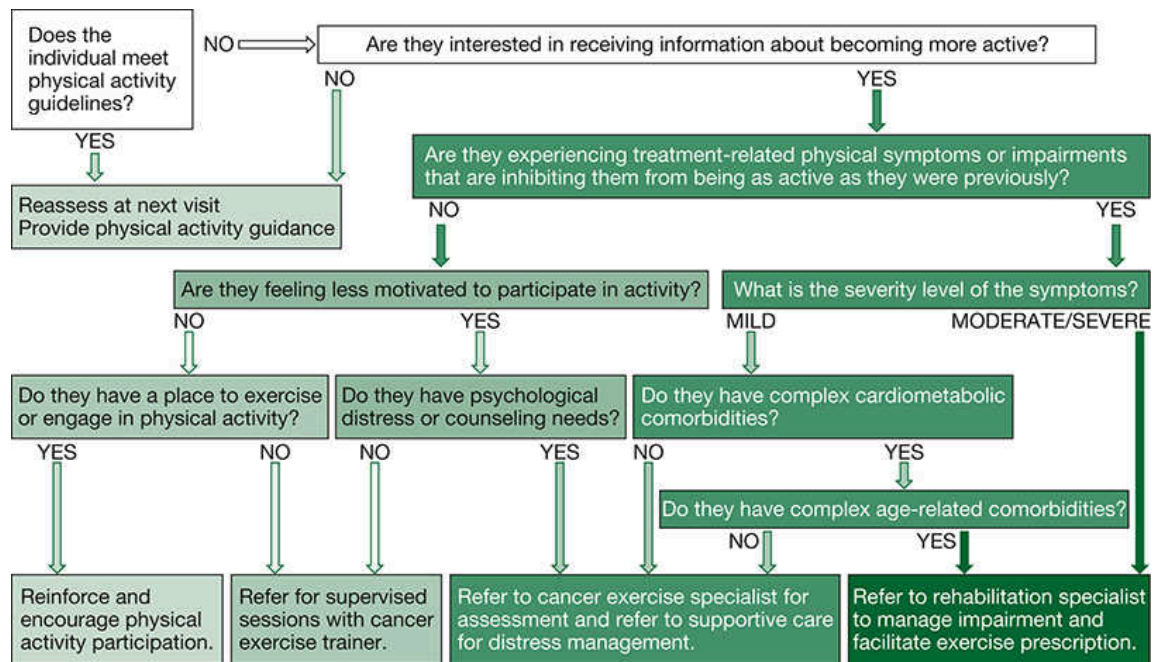
Preexercise Assessments

Given the known benefits of exercise in cancer survivors and the current low adherence to exercise guidelines, it is important to not create barriers to exercise. Given the low absolute risk of serious adverse events that occur with exercise, most screening methods in asymptomatic individuals will

produce high false-positive rates (85). However, cancer survivors often experience a variety of acute, chronic, and late side effects from cancer and the treatments (43) that may influence the approach to exercise testing and prescription (86). A preexercise assessment based on self-reported instruments such as the Physical Activity Readiness Questionnaire for Everyone (PAR-Q+) (*e.g.*, [Chapter 2](#)) can identify cancer survivors with overt cardiopulmonary symptoms (*e.g.*, chest discomfort at rest) who may benefit from a medical evaluation or exercise testing prior to engaging in moderate- to vigorous-intensity exercise. Exercise professionals can administer a brief cancer history and symptom inventory to inform the design of the Ex Rx ([Box 10.1](#)) in addition to a screening algorithm ([Figure 10.2](#)) that helps determine the recommended level of exercise supervision, exercise setting, and exercise or rehabilitation professional relative to the complexity of the individual's condition.

Box 10.1 Sample Cancer History Questions

- What kind of cancer?
- Whether the individual is currently receiving cancer treatment (and if so, what specifically)?
- Whether the cancer was removed or is still present?
- If the individual has any considerations, symptoms, or side effects attributed to cancer treatment? Including:
 - Neuropathy
 - Lymphedema
 - Sarcopenia
 - Osteopenia or osteoporosis
 - Ostomy
 - Bone metastases
 - Any other issues the individual believes may influence their ability to exercise



A

	What	Who	Where
High complexity	Physician clearance Medical workup Supervised exercise program Impaired management Caregiver guidance	Physiatrist Rehabilitation providers (PT/OT/RN)	Acute rehabilitation setting Multidisciplinary outpatient rehabilitation setting
Moderate complexity	Physician clearance Supervised exercise program Impaired management Caregiver integration	Rehabilitation providers Cancer exercise physiologist	Outpatient rehabilitation setting Medically oriented gym or fitness center
Low complexity	Group or supervised exercise program Lifestyle coaching Impairment monitoring	Exercise physiologist Cancer exercise trainer	Gym or fitness center Wellness center
Independent	Group exercise program Home-based independent Community-based facility	Self-guided Coach Technology assisted	Any setting of individual preference

B

Figure 10.2 **A**. Algorithm for an exercise referral clinical pathway. The pathway is intended to stratify individuals to higher (*dark green*) or lower (*light green*) condition complexity, which provides insight into the level of supervision and guidance that they may need to successfully engage in exercise and informs referrals to the settings outlined in panel (B). **B**. Suggested exercise settings and supervision based on individual condition complexity and risk for decline. OT, occupational therapist; PT, physical therapist; RN, registered nurse. Reprinted from [87].

Medical Assessment and Exercise Testing

The ACSM preparticipation screening algorithm can be used to determine whether exercise testing is needed for cancer survivors prior to participation in moderate- to vigorous-intensity exercise. As described earlier, exercise testing is recommended but not required for preparticipation assessment for most cancer survivors ([82](#)), and the 2019 American College of Sports Medicine Roundtable on Exercise Guidelines for Cancer Survivors concluded that exercise testing is not required before walking, resistance, or flexibility activities ([65](#)). Specific cancer survivor populations for whom medical evaluation and/or exercise testing should be considered include those with metastatic disease, those with persistent and significant cancer treatment–related side effects (*e.g.*, sarcopenia, cardiac toxicities), or those with significant comorbidities ([88](#)). Given the lack of precision regarding the definition of “significant” side effects and comorbidities, collaboration between exercise professionals and the oncology team and/or primary care provider is strongly encouraged. In addition, a preexercise medical assessment is suggested ([Table 10.1](#)).

TABLE 10.1 Preexercise Medical Assessments for Individuals With Cancer

Cancer Site	Breast	Prostate	Colon	Adult Hematologic (No. HSCT)	Adult HSCT	Gynecologic
General medical assessments recommended prior to exercise	Recommend evaluation for peripheral neuropathies and musculoskeletal morbidities secondary to treatment regardless of time since treatment. If there has been hormonal therapy, recommend evaluation of fracture risk. Individuals with known metastatic disease to the bone will require evaluation to discern what is safe prior to starting exercise. Individuals with known cardiac conditions (secondary to cancer or not) require medical assessment of the safety of exercise prior to starting. There is always a risk that metastasis to the bone or cardiac toxicity secondary to cancer treatments will be undetected. This risk will vary widely across the population of survivors. Fitness professionals may want to consult with the patient's medical team to discern this likelihood. However, requiring medical assessment for metastatic disease and cardiotoxicity for all survivors prior to exercise is not recommended, as this would create an unnecessary barrier to obtaining the well-established health benefits of exercise for the majority of survivors, for whom metastasis and cardiotoxicity are unlikely to occur.					
Cancer site specific medical assessments recommended prior to starting an exercise program	Recommend evaluation for arm/shoulder morbidity prior to upper body exercise.	Evaluation of muscle strength and wasting.	Patient should be evaluated as having established consistent and proactive infection prevention behaviors for an existing ostomy prior to engaging in exercise training more vigorous than a walking program.	None	None	Patients with morbid obesity may require additional medical assessment for the safety of activity beyond cancer-specific risk. Recommend evaluation for lower extremity lymphedema prior to vigorous aerobic exercise or resistance training.

NOTE: To be applied in addition to preexercise screening for adults.

HSCT, hematopoietic stem cell transplantation.

Reprinted with permission from [\(89\)](#).

There is no evidence that the level of medical supervision required for symptom-limited or maximal exercise testing needs to be different for cancer survivors than for other populations. Exercise testing techniques and contraindications for the general population are appropriate for cancer survivors, with the following cancer-specific considerations:

- *Upper or lower body morbidity and lymphedema:* Cancer survivors with upper or lower body morbidity including lymphedema that makes it unsafe or not possible for resistance exercise testing should be referred to physical therapy for rehabilitation (90). Resistance exercise with one-repetition maximum testing is safe among breast cancer survivors with and at risk for upper extremity lymphedema (65).
- *Bone metastases:* Cancer survivors with bone metastases are at an increased risk of skeletal fracture, spinal compression, and exacerbation of bone pain. Selected modalities for exercise testing should avoid direct musculoskeletal loading to metastatic lesions or loading of muscles that are proximal to metastatic lesions (91–93).
- *Neuropathy:* Cancer survivors with peripheral neuropathy may have instability, balance difficulty, and altered gait biomechanics that increase the risk of falls (94). Assessment of stability, balance, and gait biomechanics may be useful to refine selection of exercise testing modality (e.g., stationary cycle vs. treadmill).
- *Ostomy:* During resistance exercise testing, survivors should be reminded to avoid inducing excessive intra-abdominal pressure (e.g., Valsalva maneuver). There is no empirical evidence to support this recommendation, and it is based on expert opinion (65).

Exercise Prescription

General Recommendations

The *2018 Physical Activity Guidelines for Americans* forms the basis from which adaptations are made for cancer survivors (83). Important recommendations from these guidelines that are applicable to cancer survivors include avoiding inactivity, accumulating at least 150–300 min · wk⁻¹ of moderate-intensity, or 75–150 min · wk⁻¹ of vigorous-intensity aerobic exercise when possible, engaging in resistance exercise of all major muscle groups on 2 or more days each week. Balance and flexibility exercises should also be integrated into the Ex R_x. Multiple organizations including the ACSM (65) and the American Cancer Society (92) have endorsed similar guidelines for Ex R_x in cancer survivors.

Cancer survivors range from the otherwise apparently healthy to those severely impacted by the disease and treatments. Associated with cancer are levels of low neuromuscular and CRF; poor function; low muscle mass; compromised metabolic, pulmonary, and cardiovascular health; and/or low bone density. It is recommended that cancer survivors undergo health and fitness assessment appropriate to their health status and the issues or deficits placing them at great risk of morbidity and mortality identified and prioritized (43). A tailored Ex R_x must then be formulated to minimize risk while maximizing benefit while considering patient capacity in terms of access, finances, time constraints, and motivations (43). Providing the most safe and effective care is very important, and this should incorporate a framework for triaging patients with cancer into the optimal exercise program with appropriate level of supervision, monitoring, and setting (95).

As part of the general recommendations for exercise testing and prescription, the fitness professionals should understand the relevant contraindications and considerations ([Table 10.2](#)).

TABLE 10.2 Considerations for Starting Exercise, Stopping Exercise, Progression, Regression, and Injury Risk for Cancer Survivors

	Breast	Prostate	Colon	Adult Hematologic (No. HSCT)	Adult HSCT	Gynecologic
General considerations for starting an exercise program common across all cancer sites	Allow adequate time to heal after surgery. The number of weeks required for surgical recovery may be as high as 8. Do not exercise individuals who are experiencing fever, extreme fatigue, significant anemia, or ataxia. Follow <i>ACSM Guidelines</i> for exercise prescription with regard to cardiovascular and pulmonary contraindications for starting an exercise program. However, the potential for an adverse cardiopulmonary event might be higher among cancer survivors than age-matched comparisons given the toxicity of radiotherapy and chemotherapy and long-term/late effects of cancer surgery.					
Cancer-specific considerations for starting an exercise program	Patients with acute arm or shoulder problems secondary to breast cancer treatment should seek medical care to resolve those issues prior to exercise training with the upper body.	None	Physician opinion recommended for patients with an ostomy prior to participation in contact sports (risk of blow), weight training (risk of hernia).	None	None	Females with swelling or inflammation in the abdomen, groin, or lower extremity should seek medical care to resolve these issues prior to exercise training with the lower body.
Cancer-specific reasons for stopping an exercise program. (Note: General <i>ACSM Guidelines</i> for stopping exercise remain in place for this population)	Changes in arm/shoulder symptoms or swelling should result in regression or avoidance of upper body exercise until appropriate medical evaluation and treatment resolves the issue.	None	Hernia, ostomy-related systemic infection.	None	None	Changes in swelling or inflammation of the abdomen, groin, or lower extremities should result in regression or avoidance of lower body exercise until appropriate medical evaluation and treatment

	Breast	Prostate	Colon	Adult Hematologic (No. HSCT)	Adult HSCT	Gynecologic
						resolves the issue.
General injury risk issues in common across cancer sites	Patients with bone metastases require tailored exercise prescription regarding intensity, duration, and mode given increased risk for skeletal fractures. Infection risk is higher for patients who are currently undergoing chemotherapy or radiation treatment or have compromised immune function after treatment. Care should be taken to reduce infection risk in fitness and exercise facilities and equipment frequented by cancer survivors. Patients currently in treatment and immediately following treatment may vary from exercise session to exercise session with regard to exercise tolerance, depending on their treatment schedule. Individuals with cardiac conditions (secondary to cancer or not) will require modifications and may require increased supervision for safety.					
Cancer-specific risk of injury, emergency procedures	The arms/shoulders should be exercised, but proactive injury prevention approaches are encouraged, given the high incidence of arm/shoulder morbidity in breast cancer survivors. Monitor for lymphedema exacerbation. Be aware of risk for fracture among those treated with hormone therapy, a diagnosis of osteoporosis, or bony metastases.	Be aware of risk for fracture among patients treated with ADT, a diagnosis of osteoporosis or bony metastases.	Advisable to avoid excessive intra-abdominal pressures for patients with an ostomy.	Multiple myeloma patients should be treated as if they are osteoporotic.	None	The lower body should be exercised, but proactive injury prevention approaches are encouraged, given the potential for lower extremity swelling or inflammation in this population. Monitor for lymphedema exacerbation. Be aware of risk for fractures among those treated with hormonal therapies, with diagnosed osteoporosis, or with bony metastases.

ACSM, American College of Sports Medicine; ADT, androgen deprivation therapy; HSCT, hematopoietic stem cell transplantation.
Information from (65).

FITT Principle

Exercise training is safe during and after cancer treatment, and cancer survivors should avoid physical inactivity and engage in exercise on a regular basis. Overall recommendations for cancer survivors are consistent with the guidelines presented in [Chapter 5](#) and the following publications ([96–100](#)); however, emphasis on mode and dosage must be tailored for cancer survivors with significant health conditions ([43](#)).

Exercise professionals may wish to implement these recommendations sequentially, first prescribing a small volume of activity, then incrementally increasing the frequency, intensity, and time of exercise, as tolerated ([88](#)). As stated above, Ex R_x for cancer survivors must be tailored to address the health and fitness issues causing greatest morbidity and risk of mortality ([43](#)). In some instances, other diseases such as sarcopenia, cachexia, and CVD may take priority for exercise management. An extensive discussion of health issues and recommended Ex R_x considerations is provided elsewhere ([43](#)); however, some key cancer-specific considerations include the following:

- *Upper or lower extremity lymphedema:* Survivors with established lymphedema may prefer to wear a compression garment during resistance exercise; however, this does not increase or decrease lymphedema after a resistance training session ([102](#)). Resistance intensity should progress slowly, monitoring for possible exacerbation, and it is preferable to be under the supervision of a certified exercise professional. It has been established that it is unlikely that light or heavy resistance training exacerbates lymphedema and there is no upper limit on the resistance training intensity for breast cancer survivors with or at risk for lymphedema ([90](#), [103](#), [104](#)). The safety of exercise for lower extremity lymphedema remains unknown ([105](#)), although it is likely reflective of the upper extremity research.

FITT FITT RECOMMENDATIONS FOR CANCER SURVIVORS

	Aerobic	Resistance	Flexibility
Frequency	3–5 d · wk ⁻¹ .	2–5 d · wk ⁻¹ with a minimum of 48 h between training a given muscle group.	2–3 d · wk ⁻¹ up to daily.
Intensity	40%–<60% $\dot{V}O_2R$ or HRR. Survivors may find RPE useful to gauge exercise intensity.	60%–80% 1-RM or allow for 6–15 repetitions. Increase weight as tolerated and when repetitions >15. RPE is correlated with % 1-RM in cancer survivors (101). Repetitions to failure is not necessary and sets can cease 1–2 repetitions prior, <i>i.e.</i> , 1–2 RIR.	Stretch within limits of pain to the point of tightness or slight discomfort.
Time	≥30 min · d ⁻¹ . No lower limit on bout length. During treatment, exercise length may need to be modified due to chemotherapy or radiation-related toxicities.	≥1 set, ≥6 repetitions per set; ≥60-s rest between sets.	Hold each stretch for 10–30 s.
Type	Walking, rowing, cycling, swimming, exercise circuits, etc., with MICT or HIIT formats. Swimming should not be prescribed for survivors with central lines, those with ostomies, those in an immunocompromised state, or who are currently receiving radiation therapy.	8–10 exercises of major muscle groups; bodyweight, machines, or free weights.	Static stretches (passive and/or active), for all major muscle tendon groups. May be supplemented with tai chi and yoga.

1-RM, one-repetition maximum; HRR, heart rate reserve; RPE, rating of perceived exertion; $\dot{V}O_2R$, oxygen uptake reserve; RIR, repetitions in reserve; MICT, moderate-intensity continuous training; HIIT, high-intensity interval training.

- **Bone metastases:** Selected modalities for exercise should avoid direct musculoskeletal loading to metastatic lesions or loading of muscles that are proximal to metastatic lesions ([92](#), [93](#), [106](#)), and the modular multimodal exercise program is recommended as a guide ([47](#)). Most recently, controlled isometric and even slow dynamic resistance training exercise have been implemented at the sites of metastatic lesions, but this should be applied with caution and under close supervision ([107](#), [108](#)). Bone pain should be monitored during and after exercise as a key sign ([92](#), [93](#)). If bone pain worsens, the exercise should be regressed or ceased; if pain does not improve with cessation of exercise, referral to the medical team is encouraged.

- *Neuropathy*: Systematic assessment of falls may be informative in older cancer survivors ([109](#)) or those with a history of falls and/or significant neuropathy of the lower extremities ([94](#), [110](#)). Weight-bearing activities should be carefully selected to reduce risk of falls. Neuropathy symptoms should be monitored during and after exercise. If neuropathy worsens, exercise should be ceased or alternative exercises considered; if neuropathy symptoms do not improve with cessation of exercise, referral to medical provider is encouraged.
- *Ostomy*: Cancer survivors with an ostomy should adhere to infection risk reduction practices. Resistance exercise should start with low resistance and progress slowly. Avoid contact sports and exercises that cause excessive intra-abdominal pressure (*e.g.*, Valsalva maneuver). There is no empirical evidence to support this recommendation, and it is based on expert opinion ([65](#)).
- *Sarcopenia or cachexia*: Sarcopenia (low muscle mass) (see [Chapter 4](#)) and cachexia (wasting syndrome) require special consideration for Ex R_x. Emphasis should be on resistance training to drive muscle hypertrophy so higher volume load (exercises × sets × reps × load) with emphasis on the largest muscle groups. Aerobic exercise should be minimal or not included as this may exacerbate the sarcopenia or cachexia ([43](#)).
- *Osteopenia or osteoporosis*: Cancer and some treatments such as hormone and chemotherapy cause loss of bone content and some patients rapidly become osteoporotic. Because this is driven by disease and/or treatments, the decline is often more rapid and may not respond to normal exercise interventions such as resistance training. In this case, the addition of impact loading is recommended, which must be programmed after 2–3 months of resistance training with gradual increase in intensity and with progression or regression as appropriate ([43](#)). Impact exercises include heel drops, hopping, skipping, bounding, and drop jumps.

Among all cancer survivors, the presence of ataxia, severe fatigue, severe anemia (<8 g/dL), low platelet count (<50,000), low white blood cells (<3,000), profound weakness, or any other worsening or changing physical condition will require more extensive patient assessment (*e.g.*, cardiopulmonary exercise test [CPET], DEXA, pulmonary function) and precise Ex R_x and monitoring that is best provided by exercise specialists in oncology such as clinical exercise physiologists, physical therapists, or exercise medicine doctors with training in cancer. If in doubt, referral to more intensive care is required. Health/fitness professionals should use prudent judgment in deciding the level of exercise supervision as needed on an individual basis.

Summary

- All cancer survivors should be encouraged to avoid inactivity and be as physically active as possible on most if not every day regardless of cancer type, stage, or even when undergoing difficult treatments.
- Exercise is generally safe for cancer survivors during and after cancer treatment.
- General Ex R_x for most [cancer survivors](#):
 - At least 150 min · wk⁻¹ of moderate intensity or 75 min · wk⁻¹ of vigorous intensity or an equivalent combination of moderate- and vigorous-intensity aerobic activity. Preferably, aerobic

activity should be spread throughout the week.

- Resistance training of moderate-to-vigorous intensity that involves all major muscle groups on 2 or more days each week, as this mode provides different and specific health benefits.
- If resistance training more than $3 \text{ d} \cdot \text{wk}^{-1}$, a split program is required to provide at least 48 hours between exercise on the same muscle group.
- Stretch major muscle groups and engage in balance activities on as many days as tolerable.
- Health and fitness assessment should be conducted to prioritize key issues and the Ex Rx tailored to specifically ameliorate those issues causing the greatest morbidity and risk of mortality while minimizing risk of adverse events or exacerbating illness.
- Tailoring should also incorporate an individual's abilities, preferences, financial capacity, and access to supervision and facilities.
- For individuals undergoing active cancer treatment, living with metastatic cancer, or any other major health issues, exercise professional collaboration with clinical exercise physiologists, physical therapists, and oncology providers in a team care arrangement is recommended.

ONLINE RESOURCES

American Cancer Society: <http://www.cancer.org>

American College of Sports Medicine/American Cancer Society Certified Cancer Exercise Trainer:

<https://www.acsm.org/get-stay-certified/get-certified/specialization/cet>

Exercise is Medicine Australia: <https://exerciseismedicine.org.au/resources/factsheets/>

Livestrong at the YMCA: <https://www.livestrong.org/what-we-do/program/livestrong-at-the-ymca>

Cancer Exercise Toolkit: <https://cancerexercisetoolkit.trekeeducation.org/>

Exercise is Medicine: Moving Through Cancer: <https://www.exerciseismedicine.org/eim-in-action/moving-through-cancer/>

American Society of Clinical Oncology (ASCO): <https://www.asco.org/>

National Cancer Care Network (NCCN): <https://www.nccn.org/>

FIBROMYALGIA

Fibromyalgia syndrome, the most common chronic widespread pain condition in rheumatology, is primarily characterized by chronic widespread pain ([111](#)). In addition to pain, common symptoms include fatigue, sleep disturbances, and mood disorders such as anxiety and depression, cognitive dysfunction, autonomic disturbances, regional pain syndromes, hypersensitivity to external stimuli, and stiffness ([112–117](#)) ([Table 10.3](#) for a listing of signs and symptoms). Fibromyalgia is often associated with comorbidities including infections, diabetes, rheumatic diseases, and psychiatric or neurological disorders, musculoskeletal conditions, cardiovascular or endocrinology disorders, interstitial cystitis bladder syndrome, chronic pelvic pain, and temporomandibular joint disorder ([116, 118–121](#)). Fibromyalgia pain and other symptoms fluctuate both in location and severity between and even within single days ([116, 122](#)). Fibromyalgia significantly influences quality of life and the ability to carry out normal daily activities and carries significant health care and economic burdens ([111, 123](#)). The complexity and heterogeneity of this disease highlights the need for personalized management and treatment strategies ([116, 122](#)).

TABLE 10.3 Clinical Characteristics and Associated Symptoms of Fibromyalgia ([112](#)–[117](#))

Symptom Type	Examples of Specific Symptoms	Percentage of Individuals Who Experience Symptoms
Pivotal Symptoms for Diagnosis		
Pain	• Generalized head to toe • Neuropathic pain	100%
Fatigue	• Physical fatigue • Mental fatigue	More than 90%
Sleep disturbances	• Insomnia • Frequent awakening • Nonrestorative sleep	80%
Common Symptoms and Features		
Mood disorders	• Anxiety • Depression • Posttraumatic stress disorder	30%–60%
Cognitive impairment	• Concentration difficulties • Memory deficits	42%–52%
Autonomic disturbances	• Blurred vision • Orthostatic hypotension	More than 30%
Regional pain syndromes	• Migraine or headache • Irritable bowel syndrome or abdominal pain	More than 50%
Hypersensitivity to external stimuli	• Chemical sensitivity • Hypersensitivity to light, odors, and sounds	42%–94%
Stiffness	• Morning stiffness	39%–88%

Current evidence recognizes fibromyalgia as a central nervous system condition causing disordered pain regulation, often classified as central sensitization or central nervous system–driven pain amplification syndrome ([112](#), [116](#)). These hypersensitive pain responses include hyperalgesia (increased pain responses to stimuli that normally provoke pain) and allodynia (pain responses to stimuli that do not normally provoke pain) ([116](#), [124](#)). Fibromyalgia also seems to have a genetic component, as fibromyalgia and other chronic pain conditions cluster in families, and more than 100

genes have been identified as influencing pain experiences ([116](#), [122](#), [123](#)). Recent discoveries have identified noninflammatory immunoglobulin G serum antibodies associated with fibromyalgia, suggesting fibromyalgia could be an autoimmune condition ([111](#), [125](#), [126](#)). There are several risk factors for fibromyalgia, such as female sex, traumatic events (*e.g.*, car accident), advanced age, family history, lupus, RA, and viral infection ([127](#)). The pathophysiology of fibromyalgia remains controversial, reflecting the lack of specific validated biological and instrumental biomarkers, with multiple hypothetical pathogenic mechanisms identified ([122](#), [125](#)).

The first diagnostic criteria for fibromyalgia were published by the American College of Rheumatology (ACR) in 1990, requiring widespread pain to be present for longer than 3 months and at least 11 of 18 tender points test positive for sensitivity to pain ([128](#)). The ACR later published a symptom-based alternative method for diagnosis in 2010, replacing the tender points count with a symptom severity scale and self-reported pain index ([129](#)), with revisions proposed in 2011 ([130](#)). Further revisions were proposed by the ACR in 2016, requiring generalized pain in at least four of five regions for at least 3 months ([131](#)). In 2019, the Analgesic, Anesthetic, and Addiction Clinical Trial Translations Innovations Opportunities and Networks — American Pain Society Pain Taxonomy published an alternative framework for fibromyalgia diagnosis, requiring pain in at least six anatomic regions and moderate to severe sleep problems or fatigue for the past 3 months ([132](#)). At present, any of the 2010, 2011, 2016, and 2019 criteria are appropriate for clinical diagnosis ([121](#)).

Globally, fibromyalgia is experienced by 2%–6% of the population, approximately 80% of whom are females, although prevalence ranges vary considerably depending on the diagnostic criteria used ([111](#), [116](#), [122](#)). While fibromyalgia generally appears between 30 and 35 years of age, it is most common among females aged 50–60 years, those residing in rural areas and individuals of low socioeconomic and educational status ([116](#), [133](#)). Fibromyalgia is also more common among individuals who are overweight or obese and who experience inflammatory conditions ([114](#), [123](#)).

Understanding and identifying management strategies for fibromyalgia has improved vastly in recent decades. Evidence-based guidelines for fibromyalgia management emphasize individualized strategies developed in partnership with patients, engaging in symptom-based and stepwise progressions toward shared goals ([121–123](#)). Further, the treatment and management of fibromyalgia should be multimodal, incorporating education, exercise training, pharmacotherapy, and psychotherapy ([121–123](#)). The overall goals for the management and treatment of fibromyalgia focus on maintaining or improving function and quality of life, and managing symptoms, with exercise training highlighted as an important component of management ([122](#), [123](#), [134](#)).

Individuals with fibromyalgia may have a higher risk of deconditioning. Therefore, it is important to focus on both cardiovascular and muscular fitness to enhance physical functioning and contribute to various aspects of mental well-being ([135](#), [136](#)). In particular, individuals with fibromyalgia experience reduced quality of life and difficulties in carrying out daily activities ([116](#), [122](#)). While exercise is strongly recommended, individuals with fibromyalgia are much less likely to achieve adequate levels of exercise ([123](#), [134](#)). They may be concerned that exercise will worsen their pain and fatigue ([113](#), [123](#), [137](#)). Decreased physical functioning and increased sedentary behavior can also make exercise more challenging, and individuals with lower physical fitness are less likely to engage in

PA, including daily/weekly exercise and daily activities that require energy expenditure ([113](#), [137–139](#)). Lower levels of PA contribute to reduced quality of life and higher levels of depression among individuals with fibromyalgia ([137](#)). Further, individuals with fibromyalgia experiencing higher pain intensity, physical and mental fatigue, and more severe disease impact are more likely to remain sedentary ([113](#), [140](#)). Conversely, individuals with fibromyalgia who experienced better social and physical function, and had more vitality, were better able to reduce their sedentary behavior ([113](#), [140](#)). Engaging in regular PA also reduces risks of many chronic diseases, including CVD and diabetes ([113](#), [141](#), [142](#)). There is also evidence that regular exercise improves some symptoms of fibromyalgia and maintains or improves physical fitness ([136](#), [143–148](#)). It is important to recognize, however, that these findings are based on studies that have focused primarily on middle-aged females from high-income countries.

Pharmacological Treatment Options

Importantly, clinicians and practitioners who work with individuals managing fibromyalgia must be sensitive to the individual's expression of pain and understand that the pharmacological treatments for fibromyalgia may have side effects (see [Table 10.3](#)) that can affect physical performance and that pain and performance may not be consistent session to session ([Table 10.4](#) for a listing of signs and symptoms) ([149](#)). Medications used for the management of fibromyalgia may interfere with exercise capacity, tolerability, exercise responses, and recovery from exercise.

TABLE 10.4 Pharmacological Treatment Options of Fibromyalgia

Drug	Common/Serious Side Effects	Major Precautions
Labeled Drugs		
Duloxetine (SNRI)	Abdominal pain, constipation, drowsiness, ED, suicidal ideation, SS	Platelet aggregation may be impaired due to platelet serotonin depletion. Additional serotonergic agents may increase SS risk.
Milnacipran (SNRI)	Constipation, headache, insomnia, ED, suicidal ideation, SS	Platelet aggregation may be impaired. Additional serotonergic agents may increase SS risk.
Pregabalin (anticonvulsant, analgesic, anxiolytic properties)	Ataxia, constipation, drowsiness, weight gain, angioedema, confusion, anaphylaxis, GI bleed, suicidal ideation	Concomitant use with other CNS and/or respiratory depressants may exacerbate these symptoms.
Off-Labeled Drugs		
Amitriptyline (TCA)	Blurred vision, constipation, memory impairment, weight gain, ED, suicidal ideation, SS, torsade de pointes	Concomitant use with anticonvulsants may increase CNS depression and lower seizure threshold. Concomitant use with CNS and/or respiratory depressants may exacerbate these symptoms. Additional serotonergic agents may increase SS risk.
Cyclobenzaprine (skeletal muscle relaxant)	Sedation, dry mouth	Concomitant use with CNS depressants may exacerbate these symptoms. Additional serotonergic agents may increase SS risk.
Fluoxetine (SSRI)	Transient nausea, anorexia, drowsiness, insomnia, ED, suicidal ideation, SS, QT prolongation	Platelet aggregation may be impaired due to platelet serotonin depletion. Additional serotonergic agents may increase SS risk. Fluoxetine is a strong inhibitor of CYP2D6, CYP3A4.
Gabapentin (anticonvulsant, analgesic)	Headache, dizziness, sedation, nausea	Concomitant use with CNS and/or respiratory depressants may exacerbate these symptoms.
Venlafaxine (SNRI)	Headache, insomnia, drowsiness, ED, suicidal ideation, SS	Platelet aggregation may be impaired due to platelet serotonin depletion. Additional serotonergic agents may increase SS risk.

CNS, central nervous system; ED, erectile dysfunction; GI, gastrointestinal; SNRI, serotonin-norepinephrine reuptake inhibitor; SS, serotonin syndrome; SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant.

Adapted from (149).

Exercise Testing

Research on the effects of exercise training for adults with fibromyalgia has used different evaluation methods to test various components of physical fitness (150–156). Understanding the fibromyalgia symptoms the individual is experiencing, while also considering their exercise history, will help

clinicians to make informed decisions when choosing which tests and equipment are best suited to the individual.

Cardiopulmonary exercise testing is the gold standard approach to determine an individual's level of fitness, in which the main variable of maximal oxygen uptake ($\dot{V}O_{2\max}$) reflects the best measure of exercise capacity (157). However, as there seems to be no evidence about the reliability and validity of maximal exercise tests in individuals with fibromyalgia, we suggest using submaximal aerobic tests (158). A systematic review (155) identified physical capacity tests (including field tests) that are valid and reliable for individuals with fibromyalgia. The following submaximal tests can be used to predict maximal aerobic capacity levels and changes over time (155):

- The Åstrand bicycle test (159)
- The modified Åstrand test (160)
- The lean body mass–based Åstrand test (161)
- Submaximal bicycle ergometer test following protocols other than the Åstrand tests (162)
- 5-minute (160, 163), 6-minute (164), and 10-minute walk tests (165)
- Shuttle walk test (166)

In contrast, researchers have advised against the use of the aerobic 2-km walk test, mainly because it is not possible to control effort during the walk (155, 167).

There is no preferred protocol for testing strength and flexibility in individuals with fibromyalgia, and clinicians should use the tests used in the general population (see [Chapter 3](#)). The tests should be as specific as possible to the planned exercise.

One potential fitness assessment option that may be appropriate for many individuals with fibromyalgia is the Senior Fitness Test battery and handgrip strength measurements (156, 168, 169). The Senior Fitness Test battery comprises field tests that evaluate cardiorespiratory endurance, muscular strength, speed/agility, and flexibility (see [Chapter 6](#)). This was originally developed for older adults living in the community (169), and subsequent research has established criterion-referenced fitness standards for older adults that predict the level of capacity needed to maintain physical independence into later life (169).

In summary, individuals with fibromyalgia can safely participate in exercise testing, but clinicians should always be aware of the individual's symptoms and how the individual feels (*i.e.*, before, during, and after the test period).

Before Testing

- Prior to testing, the practitioner should consider the individual's medical history, fibromyalgia symptoms, current lifestyle, level of PA and sedentary behavior (170), exercise history, and attitudes toward exercise.
- The assessment should include information on the times of day during which symptoms (including morning stiffness, pain, fatigue) are usually less severe, and all tests and training sessions should be scheduled for those times.

- Understanding the individual's current fibromyalgia symptoms in conjunction with their exercise history will help the practitioner to select tests and equipment that are best suited to that individual. For example, if the individual has painful gluteal tender points, consider a walking test instead of a cycle ergometer test. In contrast, if the individual has leg pain before testing, consider a weight-supported test using a leg or recumbent cycle ergometer. This information will help the practitioner to be mindful of symptom-limited tests.
- People with fibromyalgia may have symptoms that make exercise testing difficult to complete. The disease-specific Fibromyalgia Impact Questionnaire (FIQ) ([171](#)) or FIQ-Revised ([172](#)) may help in assessing physical function, overall impact of disease, and fibromyalgia symptoms.
- A daily record of symptoms, symptom fluctuations during the day, exercise, and PA may help individuals with fibromyalgia and clinicians clarify potential and ongoing symptom impacts on and benefits of exercise.
- If the individual presents with cognitive dysfunction, the clinician needs to determine the individual's level of understanding ([117](#), [173](#)) and adapt the instructions accordingly. This assessment should inform the presentation and delivery of verbal and written instructions in ways that are most helpful to the individual with fibromyalgia.
- The clinician should discuss with the individual about the differences between postexercise soreness and fatigue, and the normal fluctuations in pain, fatigue, and other symptoms that occur with fibromyalgia.

During Testing

- Clinicians should establish how best to encourage individuals (verbally or otherwise for individuals hard of hearing and intellectually impaired) to perform well during an exercise test while being consistent across individuals and testing sessions.
- Clinicians should ensure that individuals get enough rest between tests as well as consider the order of testing to allow for adequate rest and recovery of different physiologic systems and/or muscle groups.
- The limits of painful movements experienced by an individual should guide their positioning on the exercise testing equipment and inform the choice of test. The clinician should modify the exercise test equipment to reflect any limits of painful movement.
- The practitioner should monitor how the individual is feeling during the test. The Borg CR-10 scale (see [Figure 4.2](#)) and visual analog scales for pain and fatigue can help with monitoring the rate of perceived exertion (RPE) and how the individual is feeling. The practitioner should ensure that the individual knows that they can ask for the test to stop or they can agree on a signal for stopping the test at any time ([174](#)).
- The practitioner should measure the individual's heart rate (HR) and blood pressure at rest and during the test.

Exercise Prescription

Evidence-based guidelines and international scientific papers recognize exercise as an important component in fibromyalgia symptom management and overall well-being ([175–183](#)). Exercise training may improve health-related quality of life and physical function; lead to decreases in pain, stiffness, anxiety, fatigue, and depression; and maintain or improve physical fitness and sleep ([143, 144, 184](#)). These benefits are reported for land- and water-based exercise training and for training using one type (aerobic or resistance) or a combination (aerobic, resistance, and flexibility) of exercise training types ([136, 185–189](#)).

Flexibility exercise as a stand-alone form of training does not appear to improve symptoms of fibromyalgia ([190, 191](#)), but when added to aerobic or resistance training programs, research suggests a flexibility component will contribute to overall individual physical functioning ([192, 193](#)). Meditative exercise programs with a flexibility component such as tai chi and yoga may improve some fibromyalgia symptoms, independent of aerobic training ([194–197](#)). There is emerging literature on dance as exercise for individuals with fibromyalgia, although the benefits of this type of exercise are not yet clear ([198–203](#)).

The FITT recommendations summarize current fibromyalgia and exercise training research from RCTs that have been included in the cited systematic reviews of RCTs. The reporting of adverse events (safety data) in the studies is poor. The dose-response curves for intensity and frequency of each type of exercise versus fibromyalgia symptoms and whether the dose-response curves differ among subgroups of individuals with fibromyalgia are unclear based on the current body of evidence.

The FITT principles of Ex R_x are based on the current fibromyalgia and exercise training research literature.

Exercise Training Considerations

- It is crucial to recognize that individuals with fibromyalgia are part of a heterogeneous group. Ex R_x must be individualized, based on the individual's baseline and ongoing physical function, severity and fluctuation of pain, fatigue and other fibromyalgia symptoms, and tolerance to exercise and exercise-induced pain ([210](#)).
- Evidence-based guidelines advocate shared decision-making ([100, 211](#)) and active participation of individuals with fibromyalgia alongside social support ([212](#)) in symptom management. Clinicians should develop individualized prescriptions for exercise and PA in collaboration with the individual, to achieve prescriptions that best fit the individual's symptoms, potential symptom flares, and exercise preferences. This approach has potential to promote regular long-term adherence to exercise to maximize function and well-being.
- The rate of progression of the FITT principle Ex R_x should be tailored to the individual's fibromyalgia symptoms and level of functioning.
- Recovery times should be individualized to minimize worsening or exacerbation of fibromyalgia symptoms.

- Research has shown that exercise intervention dropout rates are lower when delivered by professionals who have an expertise in Ex Rx ([213](#)).
- Although aerobic, resistance, or combined exercises undertaken at a frequency of 1–2 d · wk⁻¹ has demonstrated symptom reduction, the reduction is greater when the frequency is increased to 3 d · wk⁻¹ ([143](#)).
- Individuals with fibromyalgia seem to tolerate the combination of aerobic and resistance training, with low exercise dropout. Aerobic exercises are essential for the transport and use of oxygen, and strength training is essential to gain muscle mass, generating strength for daily tasks ([208](#)). Hence, mixed exercise programs are recommended to promote long-term physical function and health ([214](#)).
- A strengthening exercise program of light-to-moderate intensity brings benefits through gaining muscle mass and generating the strength necessary for daily life tasks ([208](#)).
- Aerobic exercise training can begin at very light intensities (<30% of oxygen uptake reserve [$\dot{V}O_{2R}$] or heart rate reserve [HRR]) but should be progressed to light ($\dot{V}O_{2R}$ or HRR 30%–39%) and then moderate intensities ($\dot{V}O_{2R}$ or HRR 40%–59%) as tolerated. As recommended in [Chapter 5](#) of this book, when initiating an exercise program for deconditioned individuals, the principle of “start low and go slow,” increasing exercise time/duration per session before increasing intensity, may help to avoid adverse effects. Light-to-moderate intensities are more widely tolerated, with lower dropout rates ([213](#)), but studies have shown that some individuals can tolerate progression to vigorous intensities ($\dot{V}O_{2R}$ or HRR 60%–89%) ([138](#), [143](#), [146](#), [185–187](#), [190](#), [215](#)).

FITT EVIDENCE-BASED FITT RECOMMENDATIONS FOR INDIVIDUALS WITH FIBROMYALGIA ([136](#), [148](#), [185–190](#), [204–209](#))

	Aerobic	Resistance	Flexibility
Frequency	Begin with 1–2 d · wk ⁻¹ and gradually progress to 2–3 d · wk ⁻¹ .	2–3 d · wk ⁻¹ with a minimum of 48 h between sessions.	2–3 d · wk ⁻¹ .
Intensity	Begin with light (30%–39% $\dot{V}O_{2R}$ or HRR). Gradually progress to moderate intensity (40%–59% $\dot{V}O_{2R}$ or HRR; RPE 9–12 on a 6–20 scale).	40%–80% 1-RM. As tolerated, progress from 40% to 60% to 60%–80% concentric 1-RM for strength. For muscle endurance, use ≤50% 1-RM. OMNI-RES 1–4 on a 0–10 scale.	Stretch within limits of pain to the point of tightness or slight discomfort.
Time	Begin with 10 min · d ⁻¹ and progress to a total of 30–60 min · d ⁻¹ as tolerated.	Strength: Gradually progress, as tolerated, from 4–5 to 8–12 initially to 15 repetitions, increasing from 1 to 2–4 sets per muscle group with at least 2–3 min of rest between sets; endurance: 15–20 repetitions, increasing from 1 to 2 sets with a shorter rest interval. Intensity progression should be dependent upon postexercise symptomatology.	10–30 s hold for static stretching; 2–4 repetitions of each exercise; 8–10 dynamic, low-intensity, and low-amplitude mobility and stability drills (5–10 min per session).
Type	Low impact (<i>e.g.</i> , aquatic exercise, walking, dance and other aerobic movement to music, swimming, cycling).	Body weight, elastic bands, dumbbells, cuff/ankle weights, weight machines. For resistance in water, use devices to manipulate turbulence (velocity, surface area).	Static stretches (passive and/or active), for all major muscle tendon groups. Dynamic stretches may also be used.

1-RM, one-repetition maximum; HRR, heart rate reserve; $\dot{V}O_{2R}$, oxygen uptake reserve. OMNI-RES, OMNI-resistance exercise scale of perceived exertion; RPE, rating of perceived exertion.

- Mixed aquatic ([189](#), [205](#), [206](#), [216](#)) and land-mixed exercise ([186](#), [208](#)) that include two or all three types of training (aerobic, resistance, and flexibility) in each session or within each week of training, as well as training using a single type of exercise, are equally beneficial in fibromyalgia symptom management ([186](#), [208](#)).
- Teach individuals with fibromyalgia to adjust exercise intensity according to their symptoms (self-regulation).
- RPE can be used to prescribe exercise intensity for both aerobic and muscular strength and endurance training. RPE may be particularly useful during flares of pain and/or fatigue ([174](#)).
- Exercise professionals should teach individuals breathing control to avoid the Valsalva maneuver.

- Individuals should include stretching exercises, breathing exercises, and relaxation techniques at the end of exercise sessions.
- Exercise professionals should teach and demonstrate or offer access to videos and leaflets about the correct mechanics for performing each exercise to reduce the potential for injury and pain.

Special Considerations

- Monitor fatigue and pain ([171](#), [172](#)) intermittently to assess the impact of exercise on fibromyalgia and its symptoms.
- Individuals should avoid prolonged sitting and inactivity as much as their symptoms allow and should be encouraged to use functional activities (*e.g.*, walking, stair climbing, rising from a chair, dancing) to maintain light- to moderate-intensity PA. The clinician should strategize with the individual to identify ways to translate this idea into daily life. For example, to identify times of day for exercise when symptoms are the least intrusive, as individuals who experience morning stiffness should avoid early morning exercise.
- It is important to ensure that the individual has information about the potential benefits of exercise for symptoms of fibromyalgia and for physical fitness ([217](#)) and health ([218](#)).
- Exercise professional should work with the individual to identify strategies to maintain inclusion of some PA during flares of symptoms such as pain and fatigue. It may be better to decrease intensity or duration prior to reducing frequency to maintain a pattern of regular PA ([219](#)).
- It is important to set realistic short- and long-term goals because improvements in symptoms with exercise training are modest and may take more than 7 weeks after initiating an exercise program to be clinically relevant and noticeable to the individual ([143](#)).
- Individuals may experience an increase in symptoms during the first days or weeks, until exercise adaptation occurs, so it is important to prioritize low intensities at the beginning to promote adherence to activities and maintenance of frequency throughout the program ([204](#)).
- For aquatic testing and exercise, use pools with water temperatures of 91 °F–97 °F (33 °C–36 °C) for comfort and to maximize performance ([143](#), [216](#)).
- To minimize barriers to adherence, focus on the individual's exercise experiences and preferences in applying the principle of specificity when choosing exercise tests and prescribing exercise.
- Although group settings are not for everyone, supervised or group exercise early in a training program can provide a social support system for reducing physical and emotional stress and promoting adherence ([220](#), [221](#)). Clinicians should, however, discuss ways to foster exercise independence to increase long-term adherence and for individuals to be able to exercise on their own in the long term.

MYALGIC ENCEPHALOMYELITIS/CHRONIC FATIGUE SYNDROME

Background

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) involves a constellation of severely disabling symptoms and signs involving multiple body systems. The prevalence of ME/CFS is 1.4% of the general population, suggesting this condition is relatively common. Females are more frequently affected than males ([222](#)). ME/CFS is encountered in both adults and children ([222](#)).

ME/CFS often follows an identifiable precipitating event ([223](#), [224](#)). Precipitating events may include the following:

- Viral infection (*e.g.*, Epstein-Barr virus ([225–231](#)), human herpesvirus 6 ([232](#)), West Nile virus ([233](#)), Ebolavirus ([234](#), [235](#)), coxsackie B enteroviruses ([236](#), [237](#)), and SARS-CoV-2 ([238](#)))
- Bacterial infection (*e.g.*, *Coxiella* ([225](#), [239](#)) and *Borrelia* ([240](#)))
- Exposure to environmental chemicals ([241](#), [242](#))
- Physical trauma (*e.g.*, motor vehicle accidents and surgery)
- Psychosocial trauma
- Adverse vaccine event

Postexertional malaise (PEM) is a diagnostic hallmark of ME/CFS ([223](#), [224](#)). PEM involves signs and symptoms following physical or cognitive exertion, resulting in an inappropriate loss of physical and/or mental stamina, rapid physical and/or mental fatigability, or tendency for other associated symptoms to worsen ([Box 10.2](#)) ([223](#), [243](#), [244](#)). There also must be a pathological length of time to recover of often 24 hours or more ([223](#), [245](#)). Other conditions potentially explaining the pattern of symptoms must be excluded ([223](#), [224](#), [226](#), [246](#), [247](#)). Some common symptoms and signs and symptoms of PEM are listed in [Box 10.2](#) ([223](#), [224](#), [246](#), [248–252](#)).

Box 10.2 Common Symptoms and Signs of Postexertional Malaise

- Profound physical and/or mental fatigue
- Cognitive dysfunction (*e.g.*, impairment in attention and short-term memory)
- Dizziness or vertigo
- Generalized body pains
- Headaches
- Palpitations
- Signs and symptoms of viral reactivation (*e.g.*, fever, sore throat, swollen lymph nodes)
- Sleep disturbance
- Weakness

ME/CFS results in substantially impaired health-related quality of life for individuals compared to normative data ([253–257](#)), healthy control participants matched to age and body mass index (BMI) ([242, 258](#)) and individuals with other fatigue-related health conditions ([255, 258](#)). The severity of ME/CFS ranges from none to mild (*i.e.*, “high functioning”) to severe (*i.e.*, housebound/bedbound) ([Table 10.5](#)). Low quality of life is related to the breadth and severity of signs and symptoms ([248, 249, 251, 261, 262](#)). Only about 7% of people living with ME/CFS report full improvement to premorbid status ([263](#)), and 70%–92% of people living with ME/CFS had not returned to work within 2–5 years following diagnosis ([263](#)).

TABLE 10.5 Guidelines for Staging the Severity of ME/CFS

Severity	Exercise Testing (Weber Classification (259))	Function (United Kingdom Institute for Health and Care Excellence (260))
Mild	$\dot{V}O_2 \text{ max (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } >20$ $\dot{V}O_{2\text{max (METs): } >5.7$ $\dot{V}O_{2\text{VAT (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } >14$ $\dot{V}O_{2\text{VAT (METs): } >4$	• Care for themselves and do some light domestic tasks (sometimes needing support) but may have difficulties with mobility • May still be working or in education but probably have stopped all leisure and social pursuits • Often have reduced hours, take days off, and use the weekend to cope with the rest of the week
Moderate	$\dot{V}O_2 \text{ max (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } 16\text{--}20$ $\dot{V}O_{2\text{max (METs): } 4.6\text{--}5.7$ $\dot{V}O_{2\text{VAT (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } 11\text{--}14$ $\dot{V}O_{2\text{VAT (METs): } 3\text{--}4$	• Reduced mobility and are restricted in all activities of daily living, although they may have peaks and troughs in their level of symptoms and ability to do activities • Usually have stopped work or education • Need rest periods, often resting in the afternoon for 1 or 2 h • Sleep at night is generally poor quality and disturbed
Severe	$\dot{V}O_2 \text{ max (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } 10\text{--}16$ $\dot{V}O_{2\text{max (METs): } 2.9\text{--}4.5$ $\dot{V}O_{2\text{VAT (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } 8\text{--}11$ $\dot{V}O_{2\text{VAT (METs): } 2.3\text{--}3.1$	• Unable to do any activity for themselves or can carry out minimal daily tasks only (such as face washing or cleaning teeth) • Severe cognitive difficulties and may depend on a wheelchair for mobility • Often are unable to leave the house or have a severe and prolonged after-effect if they do so • May also spend most of their time in bed and are often extremely sensitive to light and sound
Very severe	$\dot{V}O_2 \text{ max (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } <10$ $\dot{V}O_{2\text{max (METs): } <2.9$ $\dot{V}O_{2\text{VAT (mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}\text{): } <8$ $\dot{V}O_{2\text{VAT (METs): } <2.3$	• Are in bed all day and dependent on care • Need help with personal hygiene and eating, and are very sensitive to sensory stimuli • Some may not be able to swallow and may need to be tube fed

NOTE: The potential correlation between CPET and functional observation criteria has not yet been established and these criteria should not be considered related.

mL, milliliters; kg, kilograms; min, minute(s); METs, metabolic equivalents; VAT, ventilatory anaerobic threshold.

Diagnosis and Comorbidities

Conducting a thorough preparticipation health screening is necessary to determine disease severity, pattern of symptoms, known triggers, history of PEM, previous and current PA history, how PA is being tolerated, and past medical history. The diagnosis of ME/CFS is not mutually exclusive with the presence of other chronic health conditions. PEM symptoms characteristic of ME/CFS have been documented in people with post COVID-19 condition (also known as Long COVID-19) ([264–269](#)), postural orthostatic tachycardia syndrome ([270–272](#)), posttreatment Lyme disease ([273](#)), chronic effects of mycotoxicity ([274](#)), Gulf War illness ([275–277](#)), and cancer ([278, 279](#)). Prevalent comorbidities with ME/CFS are anxiety, depressive mood conditions, fibromyalgia, irritable bowel disease and syndrome, mast cell activation syndrome, migraine, and small fiber neuropathy ([249, 280](#)).

Altered cardiac autonomic regulation is common in those with ME/CFS ([281](#)). People with ME/CFS typically have higher resting HR and higher ratio of low-frequency power to high-frequency power from heart rate variability (HRV) at rest compared to healthy people. During exercise, peak HR, HR at ventilatory anaerobic threshold (VAT), and high-frequency power HRV are significantly lower in people with ME/CFS versus healthy people ([281](#)).

Exercise Testing

For individuals with ME/CFS, serial exercise tests are most useful to explore the unique postexertional physiology associated with the illness, which underlies the waxing and waning nature of symptoms, signs, and functioning ([282](#)). The first CPET is used to establish a baseline for metabolic, cardiac, and ventilatory functioning, while also serving as a standardized stressor to elicit PEM. A second exercise test conducted 24 hours after the first test provides a measurement of metabolic, cardiac, and ventilatory functioning in the postexertional state ([282](#)). Use of the 2-day CPET methodology in ME/CFS assesses a patient's ability to recover normally following exertion ([282](#)).

Research evidence from CPET suggests the systems-level pathophysiology of ME/CFS. Individuals with ME/CFS demonstrate significantly lower volume of oxygen consumption ($\dot{V}O_2$) and work rate on the second test of a 2-day CPET than deconditioned individuals, despite meeting criteria for a maximal test ([283–287](#)). Reductions in $\dot{V}O_2$ at both peak exercise and at the VAT suggest impaired aerobic functioning in the postexertional state ([284](#)). In addition, individuals with ME/CFS demonstrate impaired chronotropic responses at VAT and peak exertion compared to deconditioned individuals ([288](#)), which is worsened in the postexertional states. It takes an average of 2 weeks for individuals with ME/CFS to recover from 2-day CPET, while almost all deconditioned individuals recover within 24 hours ([245, 289](#)). Individuals with ME/CFS also have impaired cerebral blood flow following orthostatic challenge ([270, 290, 291](#)), which results in PEM ([271](#)). These data suggest that PEM is associated with diminished exercise responses and impaired recovery responses from even light PA, which is distinct from deconditioned individuals.

Not every individual with ME/CFS needs an exercise test. However, exercise testing may be useful for differential diagnosis, to characterize exertional and postexertional responses, and to determine

disability status as appropriate given the individual's risk to benefit analysis ([282](#)). Every effort should be made to avoid prefatiguing the individual prior to initial exercise testing. Tests based on age-predicted maximal HR that are considered submaximal for other clinical populations may be maximal in those with ME/CFS secondary to chronotropic incompetence. A 2-day maximal CPET should be considered to conduct standardized physiological measurement of the postexertional state. An electronically braked cycle ergometer should be used to obtain accurate measurements of work rate and to maintain the individual's safety. Abnormal exercise responses may implicate specific aspects of energy production and physiological systems affected by ME/CFS, which may contribute to PEM. Immediate posttest recovery measures (*e.g.*, HR, blood pressure, electrocardiography, oxygen saturation, and recovery time) should be closely monitored. Disrupted posttest recovery dynamics, particularly following the second CPET, are not unusual in this population. Signs and symptoms including pain and dyspnea should be documented before, during, and after exercise.

Pacing

Pacing is the intentional act of balancing activity with rest. Although it is not curative, pacing assists the individual with ME/CFS in symptom management and to make better informed decisions about how to allocate their energy. Several guidelines exist for pacing ([292](#)). The simplest advice involves coaching the individual to maintain the intensity of their ADLs within the fund of energy they have in each moment, also called the “energy envelope” ([293](#)). This approach may involve asking the individual to maintain a journal of activities and symptoms to help identify activities or circumstances that trigger PEM. A challenge of this approach is that symptoms may not present immediately, which reduces the ability to identify cause-and-effect relationships between symptoms and functioning.

Another popular pacing strategy among individuals with ME/CFS involves HR biofeedback ([292](#)). VAT is considered a boundary between sustainable and unsustainable activities because functioning above VAT is only practical for short periods of time. Research evidence suggests that individuals with ME/CFS consistently function above VAT during even light ADLs ([294](#)). Therefore, HR at VAT may provide an appropriate threshold to establish the intensity of acceptable exertion for individuals living with ME/CFS. HR at VAT obtained during the second of a 2-day CPET is preferred to set the threshold for pacing through HR biofeedback. HR thresholds for HR biofeedback also may be established with journaling to determine the relationship between activities triggering PEM, activity duration and HR, and symptoms.

Exercise Prescription

PA is a commonly reported trigger for PEM in those with ME/CFS. Therefore, recommending an increase in PA only should take place after the individual has achieved a satisfactory and stable functional baseline through pacing. Abnormal exercise recovery responses in ME/CFS suggest an inability to physiologically adapt to chronic exercise. Individuals with ME/CFS often worsen with standard exercise dosages that may be appropriate for other illness ([260](#)).

Abnormal exercise recovery responses observed during PEM challenges the FITT principle, because physiological capacity may be limited to adapt to fixed incremental increases in PA ([283](#), [288](#), [295](#)). As such, specific cautions against standard Ex R_x appear as part of contemporary clinical guidelines for ME/CFS ([260](#), [296](#)). A highly personalized approach to Ex R_x is advocated when appropriate, which is initiated when the individual feels ready to progress their PA beyond their paced ADLs and/or desires to incorporate exercise/activity into managing their ME/CFS or comorbidities. The exercise practitioner should carefully explain the risks and benefits of exercise to the individual, and exercise should not be presented as a curative intervention ([260](#), [296](#)).

For individuals who do progress into a formal exercise program, the goal of exercise in ME/CFS should be to maintain functional capacity to the extent possible and to manage comorbidities within the individual's goals, preferences, and available fund of energy. Appropriate exercise/activity will be that which does not cause immediate or delayed PEM. Clinicians may begin with diaphragmatic breathing and passive ROM. If those can be tolerated, then the clinician may consider progression with HR biofeedback and monitoring closely for PEM.

Frequency

Sessions initially should be spaced at least 1–2 weeks apart to assess for latent effects of PEM ([248](#)).

Intensity

HR biofeedback is recommended to be in place during exercise sessions to minimize time spent above VAT.

Time

Exercise duration should not exceed 90 seconds without significant recovery interval. Individuals with ME/CFS must fully recover between sets, exercises, and sessions, including no increase in PEM. Time lags between precipitating events and corresponding increase in PEM ([248](#)) indicate the need for a long duration between steps in an exercise progression.

Type

It is the fundamental responsibility of the exercise practitioner to maximize benefit and minimize risk for individuals in exercise selection. Standard Ex R_x may be contraindicated even in those with ME/CFS that have mild impairments because the aerobic energy system may not recover normally between bouts. Supine exercise involving gravity eliminated limb movements is preferred to minimize physiological strain and maximize venous return because dysautonomia is prevalent in those with ME/CFS ([271](#), [288](#), [290](#)).

Special Considerations

The potential for increased PEM from factors outside the exercise program should be monitored (*i.e.*, triggers from confounding physical, cognitive, emotional, and environmental stimulation), as these

factors may affect the individuals ability to participate with the exercise program. A strong therapeutic alliance with the client is necessary to ensure that the exercise program does not take place at the expense of ADLs.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH ME/CFS

	Symptom Severity	Aerobic Exercise	Strengthening	Stretching
Frequency	None-mild Mild-moderate Moderate-severe Severe	Practitioner must balance risk and benefit for people with mild to moderate disease. Aerobic exercise cannot be recommended for those with severe disease.	Begin with once per week and monitor for PEM and progress to no more than 3 times per week Contraindicated	Begin with once per week and monitor for PEM, progressing to 2–3 d · wk ⁻¹ as tolerated Occasionally as tolerated based on PEM and hypersensitivity to stimulation
Intensity	None-mild Mild-moderate Moderate-severe Severe	Practitioner must balance risk and benefit for people with mild to moderate disease. Aerobic exercise cannot be recommended for those with severe disease.	Start with one set of 2–3 repetitions and progress to two sets of 4–6 repetitions A HR monitor must be in place to ensure intensities below HR at VAT Contraindicated	Begin with passive ROM and progress to active ROM as tolerated without PEM A HR monitor must be in place to ensure intensities below HR at VAT Passive pain-free ROM in mid-ranges only
Time	None-mild Mild-moderate Moderate-severe Severe	Practitioner must balance risk and benefit for people with mild to moderate disease. Aerobic exercise cannot be recommended for those with severe disease.	Duration less than 90 s with complete recovery between exercises, longer than 1:3 work-to-rest ratio Contraindicated	To individual tolerance without pain, PEM, and below HR at VAT
Type	None-mild Mild-moderate Moderate-severe Severe	Practitioner must balance risk and benefit for people with mild to moderate disease. Aerobic exercise cannot be recommended for those with severe disease.	Begin with gravity eliminated limb movements and progress to against gravity as tolerated without PEM Contraindicated	Passive and active ROM may be appropriate based on pain, PEM, and below HR at VAT Pain-free passive ROM only

Summary

- Pacing is a prudent first step when ME/CFS is suspected.
- Inappropriate training stimuli, pathological responses to exercise, and the lack of recognition of PEM complicates standard Ex Rx.
- Once precipitating factors that create PEM and recognition of PEM symptoms are understood to avoid deleterious effects, exercise training may be recommended in some individuals.
- Exercise training should not elicit PEM or cause diminishment of ADLs.

ONLINE RESOURCES

U.S. Centers for Disease Control and Prevention | Myalgic Encephalomyelitis/Chronic Fatigue Syndrome: <https://www.cdc.gov/me-cfs/index.html>

National Institute of Health and Care Excellence | Myalgic encephalomyelitis (or encephalopathy)/chronic fatigue syndrome — diagnosis and management: <https://www.nice.org.uk/guidance/ng206>

MEAction Network | MEPedia: [https://me-pedia.org/wiki/Welcome to MEPedia](https://me-pedia.org/wiki/Welcome_to_MEpedia)

Workwell Foundation | Educational Materials: <https://workwellfoundation.org/resources/>

HUMAN IMMUNODEFICIENCY VIRUS

The success of antiretroviral therapy (ART) in the treatment of human immunodeficiency virus (HIV) infection has dramatically improved the health and lifespan of people with HIV (PWH) ([297](#)). ART is now initiated as soon as possible after diagnosis, leading to a marked reduction in the opportunistic infections and wasting seen with acquired immunodeficiency syndrome (AIDS). HIV is now considered more of a chronic comorbidity that can be successfully treated with long-term ART ([298, 299](#)). However, prolonged exposure to unsuppressed levels of HIV-1 prior to treatment and exposure to some ART regimens may be associated with an accelerated aging process and the premature development of age-related comorbidities ([300–303](#)). Both HIV and ART may contribute to dyslipidemia, abnormal distribution of body fat (*i.e.*, abdominal obesity with subcutaneous fat loss), insulin resistance ([304](#)), and sarcopenia (loss of muscle mass and function). With earlier initiation and less weight-suppressive ART, PWH are now experiencing an increasing prevalence of obesity, emphasizing the importance of lifestyle changes ([305](#)). PWH may also experience a low level of ongoing inflammation, despite suppression of HIV-1 with effective ART. Additional factors including substance use, sociodemographic risks, or other stressors (*e.g.*, trauma, homelessness) may occur at a higher frequency among PWH ([306, 307](#)). In combination, these factors contribute to an increased risk of cardiometabolic diseases and physical function decline. It is imperative that cardiovascular health and maintenance of physical function are emphasized throughout the disease process to attenuate the declines often seen with aging ([308](#)).

Exercise training is a safe and effective way for PWH to maintain physical function and improve multiple health domains, such as cardiovascular health, muscle strength and mass, bone density, quality of life, and body composition ([308–312](#)). Furthermore, increased minutes of PA is associated with reduced fatigue in PWH, one of the most common and severe symptoms reported in this population ([313, 314](#)). PWH may face unique barriers to initiating or maintaining regular PA. Understanding and addressing individual or system barriers is crucial in supporting safe, effective, and consistent PA for PWH ([315–317](#)).

It is important to note that there is no evidence to suggest that regular participation in exercise will suppress immune function in either asymptomatic or symptomatic individuals, but rather exercise appears to improve immune function both among PWH and the general population. Overall, the benefits of PA for PWH far outweigh the risks.

Exercise Testing

Consideration as to whether a preparticipation exercise test is needed should be guided by the general recommendations for any adult looking to begin an exercise program. As such, seeking medical clearance prior to beginning a new exercise program is recommended for adults, including PWH, with known history of cardiovascular event (*i.e.*, heart attack or stroke), metabolic syndrome, or renal disorder according to ACSM PAR-Q+ guidelines ([318](#)). HIV may be considered as an additional CVD

risk factor, and therefore, providers may have a lower threshold for obtaining cardiac testing for PWH ([303](#), [319](#), [320](#)).

In addition, the following issues should be considered:

- Ensure that potential precautions to exercise are assessed from medical history prior to initiating a maximal stress test. For example, if a PWH reports peripheral neuropathy, testing mode may need to be altered (*i.e.*, using cycle ergometer vs. treadmill) to accommodate any functional limitations, such as impaired sensation or balance.
- Variability of exercise test results may be higher among PWH than among people without HIV. PWH may have a significantly lower peak oxygen consumption ($\dot{V}O_{2peak}$) and greater evidence of chronotropic incompetence when compared to age-matched healthy individuals ([321–324](#)).
- PWH and advanced HIV (AIDS) may have significant limitations in exercise time and ability to reach ventilatory threshold or greater than 85% of age-predicted HR_{max} ([325](#), [326](#)).
- For PWH who are very deconditioned, alternative protocols may be applied to help the patient reach the requirements for a maximal stress test (see [Chapter 3](#)).
- PWH have an increased risk of cardiac arrhythmias and may benefit from having ECG monitoring during a maximal exercise test ([327](#), [328](#)).
- As HIV is a bloodborne pathogen, no additional precautions beyond standard cleaning protocols between patients are required. Consistent with the management of all cardiopulmonary exercise testing, attention to compliance with established disinfecting protocols and universal precautions should always be employed for the protection of patients and examiners.

Exercise Prescription

The chronic disease and health conditions associated with HIV suggest that health benefits would be gained by regular participation in a program of combined aerobic, resistance, balance, and flexibility exercises. Indeed, numerous clinical studies have shown that participation in habitual PA results in physical and mental health benefits ([312](#), [329–333](#)). Furthermore, data indicate that PWH adapt readily to exercise training, with some studies showing more robust responses than would be expected in a healthy population ([332](#), [334–336](#)) and tolerance of vigorous-intensity aerobic exercise despite comorbidities ([337](#)).

Although there are no HIV-specific contraindications to exercise, PWH are more likely to have comorbidities. As many as 83% of PWH who are ≥ 50 years and 63% of PWH between 18 and 49 years old have at least one comorbidity ([299](#)). Exercise prescriptions should be individualized, and comorbidities, functional limitations, or medications identified through the medical history should be taken into account. Potential side effects from ART will be listed under *Special Considerations* below. The general FITT principle for this population is consistent with guidelines for adults without HIV (see [Chapter 5](#)) or older adults (see [Chapter 6](#)), with appropriate modifications made by exercise professionals for more medically complex patients when necessary.

Exercise Training Considerations

- Adaptation to exercise, particularly cardiovascular exercise, may take longer compared to people without HIV ([339](#)). Higher-intensities or interval training may be required to achieve similar cardiovascular benefits, after the patient is appropriately acclimated to exercise.
- PWH should be encouraged to decrease sedentary time throughout their day ([340](#)). Depending on lifestyle factors, time may be a significant barrier to exercise. Smaller bouts of exercise may be more feasible than one bout of continuous exercise and will still provide health benefits ([341](#)).
- Due to the higher prevalence of osteopenia and osteoporosis, incorporation of weight-bearing resistance training should be encouraged to improve bone density and reduce risk of fractures ([311](#)). For those with a known diagnosis of osteoporosis, particularly with a history of nontraumatic fracture, high-impact and plyometric activities should be used under close supervision to ensure safety.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH HUMAN IMMUNODEFICIENCY VIRUS

	Aerobic	Resistance	Flexibility
Frequency	3–5 d · wk ⁻¹ .	2–3 d · wk ⁻¹ .	≥2–3 d · wk ⁻¹ .
Intensity	Begin at low intensity (30%–39% $\dot{V}O_{2R}$ or HRR) to allow for exercise acclimation. Gradually progress to moderate intensity (40%–59% $\dot{V}O_{2R}$ or HRR) or high intensity (60%–70% $\dot{V}O_{2R}$ or HRR).	Begin at light intensity with goal of gradual progression to 60% 1-RM.	Stretch to the point of tightness or slight discomfort.
Time	Begin with 10-min increments and progress to 30–60 min · d ⁻¹ . Perform at least 150 min · wk ⁻¹ (338).	1–2 sets, with gradual progression to 3 sets of 8–10 repetitions.	Hold static stretch for 30–60 s; 2–4 repetitions of each exercise.
Type	Modality will vary with the health status and interests of the individual. High-impact or plyometric activities should be carefully monitored in the presence of osteoporosis.	Machine weights are safe and effective without supervision; free weights or resistance bands can be used after initial supervision or demonstration. Aim to strengthen all major muscle groups. Activities that incorporate balance training are encouraged.	Static stretching may be used following exercise, dynamic stretching may be used prior to exercise.

1-RM, one-repetition maximum; HRR, heart rate reserve; PNF, proprioceptive neuromuscular facilitation; $\dot{V}O_{2R}$, oxygen uptake reserve.

- PWH also tend to have lower skeletal muscle mass and greater fatty infiltration of skeletal muscle and consequently lower strength (342). Response to resistance training does not appear to be impaired, but resistance exercise should be carefully matched to the abilities of each person initially and then titrated slowly to avoid injuries.
- The Centers for Disease Control and Prevention Physical Activity Guidelines recommend balance training for adults aged ≥65 years. With the high prevalence of balance impairments, peripheral neuropathy, and falls among PWH aged ≥50 years, we recommend that exercises incorporating balance training are included for PWH ≥50 years or older and considered for younger PWH depending upon additional risks (338, 343, 344). Balance exercises should be tailored to the person's ability to ensure safety.

Special Considerations

- There are no established guidelines regarding contraindications for exercise for PWH.
- Exercise is strongly recommended for all PWH at any stage of their diagnosis (308, 345). Those with multiple comorbidities or who are very deconditioned are encouraged to seek guidance from an

exercise professional to ensure safe participation in an exercise program.

- PWH with a high medication burden may report increased side effects during or after exercise.
- Although modern ART are generally well tolerated, PWH who have a history of AIDS, exposure to older nucleoside reverse transcriptase inhibitors, or prolonged ART use may have an increased prevalence of peripheral neuropathy, mitochondrial impairment, and exercise intolerance ([339](#), [346](#), [347](#)). These conditions may require adjustment of exercise type, intensity, time (duration), and ROM.
- Regular monitoring of the health/fitness benefits related to PA and CVD risk factors is critical for clinical management and continued exercise participation of PWH.

ONLINE RESOURCES

Centers for Disease Control and Prevention: <http://www.cdc.gov/hiv/>

KIDNEY DISEASE

Most recent estimates indicate that more than 35 million adults in the United States (*i.e.*, ~14% of the adult population) have chronic kidney disease (CKD) (348), and the incidence is expected to increase due to the increasing prevalence of diabetes mellitus and obesity. Hypertension, diabetes mellitus, and CVD are very common in the CKD population, with the prevalence of these comorbidities rising incrementally with the severity of CKD (349). CKD is diagnosed in individuals who have either kidney damage or have poor kidney function. Kidney damage is evidenced by moderately increased urine levels of albumin, whereas poor kidney function is noted with a ≥ 3 -month estimated glomerular filtration rate (eGFR) of less than $60 \text{ mL} \cdot \text{min}^{-1} \cdot 1.73 \text{ m}^{-2}$ (350). CKD is categorized into five distinct stages, based on the eGFR and the amount of albumin present in the urine. The level of kidney function and evidence of damage are used to identify the risk of disease progression and the likelihood of poor outcomes (Figure 10.3) (350). Stage 1 CKD indicates normal or high eGFR, but this is associated with some evidence of kidney disease or damage. Stage 5 individuals have an eGFR less than $15 \text{ mL} \cdot \text{min}^{-1} \cdot 1.73 \text{ m}^{-2}$ and are approaching the need for renal replacement therapy such as hemodialysis (in-center or at home), peritoneal dialysis, kidney transplantation, or conservative management (*e.g.*, no dialysis or transplant). The latter option is often chosen by frail, older individuals. The symptoms and complications of late-stage CKD dictate the timing and initiation of renal replacement therapy.

				Albuminuria Categories		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30–299 mg/g 3–29 mg/mmol	≥300 mg/g ≥30 mg/mmol
GFR Stages	G1	Normal or high	≥90			
	G2	Mildly decreased	60–90			
	G3a	Mildly to moderately decreased	45–59			
	G3b	Moderately to severely decreased	30–44			
	G4	Severely decreased	15–29			
	G5	Kidney failure	<15			

Key to Figure:

Colors: Represent the risk for progression, morbidity and mortality by color from best to worst.

- Low Risk (if no other markers of kidney disease, no CKD)
- Moderately Increased Risk
- High Risk
- Very High Risk
- Highest Risk

Figure 10.3 Staging of chronic kidney disease. Reprinted with permission from the National Kidney Foundation. <https://www.kidney.org/atoz/content/stages-chronic-kidney-disease-ckd>

Exercise Testing

Those who have not participated in regular exercise training in the previous 3 months should be referred for medical clearance prior to beginning exercise (see [Chapter 2](#)). CVD is highly prevalent in individuals with CKD, so when CVD is diagnosed or symptoms are present, exercise testing may be indicated as part of the medical clearance process prior to beginning an exercise program of moderate-to-vigorous intensity. In some cases, exercise testing may also be included in the workup for possible kidney transplantation. However, exercise testing for individuals with end-stage kidney disease (*i.e.*,

stage 5 CKD) and/or frailty can be problematic because their performance may be affected by muscle fatigue. As such, exercise testing may be an unnecessary barrier to their participation in an exercise training program. If performed, exercise testing of individuals with CKD should use standard test termination criteria and test termination methods (see [Chapter 4](#)).

Individuals with CKD have low functional capacities across all stages of the disease, or approximately 50%–80% of healthy age- and sex-matched controls, with $\dot{V}O_{2\text{peak}}$ ranges between 15 and 25 mL · kg⁻¹ · min⁻¹ ([351](#), [352](#)). $\dot{V}O_{2\text{peak}}$ values can increase with training by approximately 17%–23% but in general will never reach the values achieved by age- and sex-matched controls ([351](#), [352](#)). This reduced functional capacity is thought to be related to several factors including a sedentary lifestyle, cardiac dysfunction, anemia, and musculoskeletal dysfunction. In those referred for exercise testing, the following considerations should be noted:

- Medical clearance should be obtained.
- Individuals with CKD are likely to be on multiple medications, many of which could impact exercise test capacity or results (see [Appendix A](#)).
- When performing an exercise test on individuals with CKD, standard exercise testing procedures should be followed (see [Chapters 3](#) and [4](#)).
- In individuals receiving maintenance hemodialysis, testing should be scheduled for nondialysis days, and BP should be monitored in the arm that does not contain the arteriovenous fistula or graft.
- For comfort purposes, individuals receiving continuous ambulatory peritoneal dialysis should be tested with little dialysate fluid in their abdomen.
- Standard exercise testing procedures are used with individuals who are transplant recipients.
- Both treadmill and cycle leg ergometry protocols can be used to test individuals with kidney diseases. Because of the low functional capacity in this population, more conservative treadmill protocols such as the modified Balke or Naughton are appropriate ([353](#)) (see [Chapter 4](#)). If cycle leg ergometry is used, initial warm-up work rates should be 20–25 W with the work rate increased by 10–30 W increments every 1–3 minutes.
- In individuals receiving maintenance hemodialysis, the peak heart rate (HR_{peak}) is often attenuated and may not surpass 75% of age-predicted maximum ([354](#)); therefore, RPE should always be monitored (see [Chapter 4](#)).
- As a result of the very low functional capacity, traditional exercise tests may not always yield the most valuable information for $\text{Ex } R_x$ and the assessment of exercise training adaptations ([355](#)). Consequently, a variety of physical performance tests that have been used in other populations (*e.g.*, older adults) can be used (see [Chapter 6](#)). The short physical performance battery has been identified as a useful functional test in low-functioning individuals with CKD ([356](#)). Tests can be chosen to assess CRF, muscular strength, balance, and flexibility ([355](#), [356](#)).
- Isotonic strength testing should be done using a 3-RM or higher load (*e.g.*, 5- to 10-RM) as 1-RM testing is generally thought to be contraindicated in individuals with advanced CKD.
- Muscular strength and endurance can be safely assessed using isokinetic dynamometers employing angular velocities ranging from 60 to 180 degrees · s⁻¹ ([353](#)).

- Muscle power should be assessed using a computerized dynamometer because power appears to be more related to functional ability than either muscular strength or endurance (355). To assess power, individuals should be asked to perform a repetition at a specific percentage of their estimated maximum as quickly as possible.

Exercise Prescription

Exercise training in those with CKD leads to improvements in cardiovascular and muscular fitness and overall quality of life (357). The ideal FITT principle of Ex Rx for individuals with CKD has not been fully developed, but based on the available research, programs for these individuals should consist of a combination of aerobic and resistance training (358). The Kidney Diseases: Improving Global Outcomes clinical practice guidelines recommend that those with CKD aim for PA of an aerobic nature $5 \text{ d} \cdot \text{wk}^{-1}$ for at least 30 minutes but do not provide more specific guidance regarding the Ex Rx (350). Recently, the UK Kidney Association has published the clinical practice guideline exercise and lifestyle in CKD summarizing the evidence and providing specific recommendations for nondialysis, hemodialysis, and transplant patients (358), while the International Society for Peritoneal Dialysis and the Global Renal Exercise Network have provided practice recommendations on PA and exercise for peritoneal dialysis patients (359). Overall, the FITT recommendations for CKD follow those for the general population, but specific comorbidities and advanced stages of CKD might require significant adjustments, so it is prudent to initially use light-to-moderate intensities and gradually progress over time based on individual tolerance.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH KIDNEY DISEASE

	Aerobic	Resistance	Flexibility
Frequency	3–5 d · wk ⁻¹ Perform aerobic exercise spread out at least 3 d during the week, with no more than 2 consecutive days between bouts of activity	2–3 d · wk ⁻¹ Undertake resistance exercise at least twice weekly on nonconsecutive days, but more ideally 3 times a week, along with regular aerobic exercise	5–7 d · wk ⁻¹ Flexibility exercises should be performed 5–7 d · wk ⁻¹
Intensity	At least moderate intensity (40%–59% $\dot{V}O_{2R}$ RPE 12–13 on a scale of 6–20) Additional benefits may be gained from vigorous-intensity aerobic exercise (<i>i.e.</i> , >60% of $\dot{V}O_{2max}$)	65%–75 % 1-RM Performance of 1-RM is not recommended unless medically cleared for such effort; instead, estimate 1-RM from a ≥ 3 -RM test	Static: stretch to the point of tightness or slight discomfort; PNF: 20%–75% of maximum voluntary contraction
Time	Target of 30 min of continuous activity (up to 60 min) In deconditioned patients unable to maintain sustained exercise, intervals of several minutes could be used	1–3 sets of 8–12 repetitions Choose 8–12 different exercises targeting the major muscle groups	60 s per joint for static (10–30 s hold per stretch); 3–6 s contraction followed by 10–30 s assisted stretch for PNF
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming)	Machines, free weights, or bands Each session of resistance exercise should involve a combination of the major upper and lower body muscle groups (legs, hips, chest, back, abdomen, shoulders, and arms)	Static or PNF

1-RM, one-repetition maximum; 3-RM, three-repetition maximum; PNF, proprioceptive neuromuscular facilitation; RPE, rating of perceived exertion; $\dot{V}O_{2R}$, oxygen uptake reserve.

Exercise Training Considerations

- Some individuals with CKD are unable to do continuous exercise and therefore should perform intermittent exercise with intervals as short as 3 minutes interspersed with 3 minutes of rest (*i.e.*, 1:1 work-to-rest ratio). As the individual adapts to training, the duration of the work interval can be gradually increased, whereas the rest interval can be decreased. Initially, a total exercise time of 15 minutes can be used, and this can be increased within tolerance to achieve the target of 30 minutes of continuous activity (up to 60 minutes).
- All individuals with CKD, including those with kidney failure, should be gradually progressed to a greater exercise volume over time, allowing sufficient time for adaptations to the increased volume, with the potential to lower the risk of injury and to promote adherence.
- The rate at which a program is progressed should be individualized based on the responses to training and personal goals.
- Progression may consist of increasing any of the components of the basic FITT principles.
- For aerobic training, the duration should be increased first (*e.g.*, by 5–10 minutes), followed by the frequency and then intensity of the session.

- For resistance training, an initial increase in the number of repetitions is recommended before an increase in load.
- The clinical status of the individual is important to consider, and progression may need to be slowed if the individual has a medical setback.
- Because individuals with CKD tend to be sedentary, in addition to stressing the need to increase their levels of PA to comply with current recommendations, they should also be encouraged to decrease the daily time spent engaged in sedentary behavior.
- Intradialytic exercise training can be performed at any time during dialysis, as long as the patient is stable.
- RPE is typically used to monitor intensity during intradialytic exercise because the removal of fluid during dialysis can cause both resting and exercise heart rate to vary unpredictably.
- There is no evidence that resistance exercise has adverse effects on dialysis patient's vascular access once the arteriovenous fistula is mature, so resistance training can be safely prescribed in dialysis patients.

Special Considerations

- Nondialysis CKD
 - Patients with CKD who are not on dialysis have been shown to benefit from exercise training programs using standard exercise prescriptions for the general population, so long as the individuals are screened properly, start at a low to moderate intensity, and progress gradually.
 - Nondialysis patients are likely able to tolerate greater intensity and volume of exercise compared to those on dialysis; however, every patient needs to be considered individually.
 - Extremely deconditioned patients may need to start with flexibility exercises and a resistance program of lower workloads and higher repetitions before beginning any cardiorespiratory activity.
- Hemodialysis
 - Both intra- and interdialytic exercise are important to consider, though exercise adherence tends to be greater in the former, due to increased supervision.
 - Hemodialysis patients typically have extremely low fitness levels, so gradual progression is critical.
 - Immediately postdialysis, most individuals do not feel energetic enough to engage in PA. However, for those who feel capable, there are no contraindications to doing so. It is recommended that patients have a snack at least 1 hour prior to being active and/or during the PA to support the exercise.
 - Patients should avoid upper limb activity with temporary or healing arteriovenous fistulas. Individuals may exercise the arm with permanent arteriovenous access but should always avoid placing weight or pressure on the access device ([360](#)).
 - Measure BP in the arm that does not contain the fistula.
 - During dialysis, individuals should not exercise the arm with permanent arteriovenous access.

- 1-RM testing for strength is not recommended because of secondary hyperparathyroidism-related bone and joint problems.
- Peritoneal dialysis
 - Patients with a peritoneal catheter should avoid *full* sit-ups and activities that involve full flexion at the hip. They can accomplish abdominal strengthening by performing isometric contractions and partial sit-ups or crunches.
 - Swimming may be a challenge in those with peritoneal catheters because of infection risk. Patients should cover the catheter with protective tape and clean around the catheter after swimming; lake swimming is *not* recommended.
 - Patients perform exercise more comfortably with abdominal cavities emptied of dialysis fluid, reducing diaphragmatic pressure, breathlessness, and chest discomfort.
- Recipients of kidney transplants.
 - Weight management often becomes an issue following transplant.
 - Patients are often fearful of overexertion; thus, gradual progression should be stressed.
 - Exercise should be decreased in intensity and duration during episodes of rejection, not eliminated completely.
 - Patients may experience frequent hospitalizations during the first year after the transplant. Because patients are immunosuppressed, effort must be made to avoid infectious situations (*e.g.*, strict sterilization procedures must be followed for exercise testing and training equipment).

ONLINE RESOURCES

Kidney Disease: Improving Global Outcomes: <http://kdigo.org/home/>

National Institute of Diabetes and Digestive and Kidney Diseases: <http://www2.niddk.nih.gov/>

National Kidney Foundation: <http://www.kidney.org/>

United States Renal Data System: <http://www.usrds.org/atlas.htm>

Global Renal Exercise Network: <https://grexercise.arizona.edu/>

Global Renal Exercise Network Exercise in CKD course: <https://grexercise.arizona.edu/renal-exercise-education-resources/exercise-ckd-course>

MULTIPLE SCLEROSIS

Multiple sclerosis (MS) is an immune-mediated and neurodegenerative disease of the CNS with a prevalence of 1 million adults in the United States and 2.8 million adults worldwide (361, 362). The cause of MS is unknown but may include a combination of environmental (*e.g.*, vitamin D deficiency and cigarette smoking) and genetic factors, and/or exposure to infectious agents (*e.g.*, Epstein-Barr virus) (361). The underlying pathogenesis of MS involves autoreactive cells from both the adaptive and innate immune systems (*e.g.*, B cells and T cells) (361) initiating a cascade of cellular events resulting in demyelination and transection of axons and eventual loss of neurons. The extent and location of damage in the CNS results in the many symptoms of MS (*e.g.*, optic neuritis, ataxia, and cognitive and bladder dysfunction) (361). The early phase of MS is often characterized by transient, intermittent episodes of inflammation in the CNS resulting in acute neurological deficits, known as relapses, and this typically transitions into a progressive disease over time. The diagnosis of MS is based on a combination of results from a clinical examination, neuroimaging, and laboratory assessments, and most people who develop MS experience a single episode, known as a clinically isolated syndrome, which resolves over time, and a formal diagnosis of MS occurring with a second relapse.

The onset of MS usually occurs between the ages of 20 and 50 years, and MS occurs in females two times more often than males (362). The disease course of MS is highly variable between individuals and within a given individual over time (Table 10.6). People with MS who have intermittent episodes of relapses followed by silent periods are described as having relapsing remitting MS (361). This is the most common type of MS occurring in 85% of cases (361). Of those with the relapsing remitting clinical course, 15%–30% will develop progressive disability with or without relapses (*i.e.*, secondary progressive MS) over time (361). Approximately 15% of people with MS have progressive disability from the onset of MS, and this disease course is described as primary progressive MS (361). MS can be further subcategorized as active or nonactive, where active MS is defined as “the occurrence of clinical relapse or the presence of new T2 or gadolinium-enhancing lesions over a specified period of time, preferably at least one year” (363). Table 10.7 is a summary of the expanded disability status scale (EDSS; range 0–10), and the EDSS is based on a neurological examine for measuring disability and the progression of MS over time (364).

TABLE 10.6 Disease Courses of Multiple Sclerosis

Type	Characteristic
Relapsing-remitting	Periodic exacerbations followed by full or partial recovery of deficits
Primary progressive	Continuous disease progression from onset with little or no plateaus or improvements
Secondary progressive	Slow and steady disease progression that transitioned from the relapsing-remitting type
Progressive-relapsing	Progression from onset with distinct relapses superimposed on the steady progression with or without full recovery

TABLE 10.7 Summary of Kurtzke Expanded Disability Status Scale

Rating	Disability
0–2.5	None to minimal disability
3–5.5	Moderate disability but still ambulatory without assistive device
6–7	Severe disability but still ambulatory with assistive device
7.5–9	Essentially wheelchair-bound or bedbound
10	Death attributable to multiple sclerosis

Common symptoms of MS include spasticity; fatigue; pain; mobility impairment; ataxia and tremor; bladder, bowel, and sexual dysfunction; depression and anxiety; cognitive impairment; and visual disturbances ([361](#)) ([Box 10.3](#)). Those symptoms often limit ADLs and compromise quality of life. Fatigue is one of the most common symptoms of MS ([365](#)), as well as mobility impairment, which interfere with PA participation among people with MS. Fatigue can be both primary (*i.e.*, directly related to disease pathology) and secondary based on reduced physical fitness. People with MS may experience heat sensitivity and impaired temperature regulation, which may temporarily result in worsening of symptoms including fatigue and physical and cognitive function during and after PA ([366](#)). Avoidance of PA because of fatigue and impaired thermoregulation may lead to reduced aerobic capacity ([367](#)), which decreases with increasing levels of disability ([368](#)). This can result in a negative cycle of deconditioning, reduced participation in PA ([369](#)), and worsening of symptoms including fatigue and mobility impairment.

Box 10.3 Common Signs and Symptoms of Multiple Sclerosis

Symptoms

- Muscle weakness
- Symptomatic fatigue
- Numbness
- Visual disturbances
- Walking, balance, and coordination problems
- Bladder dysfunction
- Bowel dysfunction
- Cognitive dysfunction
- Dizziness and vertigo
- Depression
- Emotional changes
- Sexual dysfunction
- Pain

Signs

- Optic neuritis
- Nystagmus
- Paresthesia
- Spasticity

Reprinted from [\(370\)](#).

The decline in muscle performance is commonly observed in MS. This includes a reduction in upper and lower limb isometric muscle strength, lower limb muscle power, and lower limb rate of force development in people with MS compared to those without MS [\(371\)](#). There may be a decrease in muscle flexibility in people with MS, particularly among those with muscle spasticity and rigidity. The changes in muscle performance may be associated with reduced muscle mass among people with MS [\(372–374\)](#), although this is not consistent across all studies [\(375–377\)](#). For example, the reduction in maximal voluntary contraction, in the presence of no changes in cross-sectional area, suggests that impaired central activation in MS contributes to decreased muscle performance [\(377\)](#). Physical inactivity may further contribute to reductions in muscle size and muscle strength and therefore manifest in a negative cycle of deconditioning and physical inactivity.

Participation in regular PA is associated with improvements in cardiovascular risk factors (*e.g.*, waist circumference, cholesterol levels, glucose levels) [\(378\)](#), longevity [\(379\)](#), and improvements in health-related quality of life [\(380\)](#). The reduction in PA participation may be associated with worsening of MS symptoms [\(381\)](#). There is evidence that exercise interventions, including aerobic exercise training, resistance training, and a combination of aerobic and resistance training improve health-related quality of life [\(382\)](#), walking speed and walking endurance [\(383\)](#), balance [\(384\)](#), depressive symptoms [\(385\)](#), muscle strength [\(386\)](#), and CRF [\(367, 387\)](#) in people with mild-to-moderate disability.

Interventions that incorporate gait, balance, and functional training demonstrate the greatest improvement in balance, but these improvements may not carry over to reduced falls ([384](#)). There is low-quality evidence (according to the Grading of Recommendations, Assessment, Development and Evaluation or GRADE approach) that PA programs (including physiotherapy and structured exercise programs) improve spasticity in people with MS ([388](#)). Among people with progressive MS and moderate disability, there is weak evidence that aerobic exercise training improves CRF, mobility, and cognitive function and that resistance training improves muscle strength ([389](#)).

Fatigue is one of the most common and debilitating symptoms of MS, and managing fatigue is often an important objective for people with MS. There is moderate-quality evidence that exercise interventions can improve fatigue among people with MS compared with no exercise ([390](#)). There specifically is moderate-quality evidence that aerobic exercise training and mixed modes of training have a positive effect on fatigue in comparison with no exercise ([390](#)). However, studies are limited to people with minimal-to-moderate disability ([390–392](#)), and there is large variability between studies in terms of the type of exercise intervention and type of comparator or control condition ([390, 391](#)). Studies further assess fatigue using different outcome measures, such as the Fatigue Severity Scale and the Modified Fatigue Impact Scale, which may not measure the same construct, and therefore, the results may not be comparable ([390](#)). Fatigue is not typically the primary outcome of interest in studies of exercise interventions, and therefore, the majority of studies to date did not include persons with MS who are not prescreened for the presence of fatigue or calculate the sample size required to detect a difference in fatigue.

As indicated previously, the majority of research regarding exercise for MS includes individuals with minimal-to-moderate disability. Although exercise has benefits for this group, similar exercise training approaches may not be accessible or feasible for people with severe disability. One review of exercise for individuals with an EDSS score of ≥ 6.0 concluded that there was unclear evidence regarding the benefits of conventional aerobic exercise programs but that conventional progressive resistance training may improve muscular fitness, balance, fatigue, and quality of life ([393](#)). Although benefits of aerobic exercise were not clear, included studies reported that it was safe and feasible for people with severe disability ([393](#)). Among those who are unable to perform conventional exercise, adapted exercise training may be particularly useful, and this includes bodyweight support treadmill training and recumbent arm-leg stepping, which may improve disability, strength, fatigue, and quality of life for these individuals ([393](#)).

Exercise Testing

Exercise testing is useful for determining the fitness level, physiological response, and the effectiveness of exercise training in individuals with MS. Prior to exercise testing, appropriate screening should be undertaken and medical clearance should be sought if there are risk factors (see [Chapter 2](#)), and it is recommended to review an individual's medical history, list of medications, and functional capacity. An expert group recommended the following core set of outcome measures for use within exercise studies in MS ([394](#)): quality of life assessed using the Multiple Sclerosis Impact Scale (MSIS-29) or MSQoL54, fatigue assessed using the Modified Fatigue Impact Scale or Fatigue Severity

Scale, exercise tolerance assessed using the 6-minute walk test, muscle function assessed using the Timed Up and Go, body composition assessed using waist-to-hip ratio or BMI. Additionally, a clinical practice guideline for outcome measures among adults with neurologic conditions recommended the use of the Berg Balance Scale to assess static and dynamic sitting and standing balance, the Functional Gait Assessment to assess walking balance, the 10-m walk test to assess walking speed, and the five times sit-to-stand to assess transfer ability ([395](#)).

Exercise Testing Considerations

- Avoid testing during a relapse or period of acute exacerbation of MS symptoms.
- Closely monitor for any signs of fatigue, overheating, or general worsening of symptoms as exercise intensity increases.
- Perform exercise testing earlier in the day because fatigue generally worsens throughout the day in individuals with MS.
- Conduct exercise testing in a climate-controlled room (72 °F–74 °F [22.2 °C–24.4 °C], low humidity) and use electric fans or cold neck packs as appropriate.
- Use RPE in addition to HR to evaluate exercise intensity. Individuals with MS may experience cardiovascular dysfunction as a result of autonomic dysfunction ([396](#)). HR responses may be blunted during exercise, and therefore, HR may not be a valid indicator of exercise intensity ([397](#)).
- Cycle ergometry is the recommended mode of testing aerobic fitness in most people with MS because this modality requires less balance and coordination compared with walking on a treadmill ([398](#)). Individuals with balance and coordination problems may require the use of an upright or recumbent cycle leg ergometer with foot straps.
- In select individuals, a recumbent arm-leg stepping ergometer (*i.e.*, device that allows for the use of upper and lower extremities) may be advantageous because it distributes work to all extremities, thus minimizing the potential influence of local muscle fatigue or weakness in one limb on maximal exercise testing.
- Individuals who are nonambulatory with sufficient upper body function can be assessed using an arm ergometer.
- Assessment of $\dot{V}O_{2\text{peak}}$ is a valid measure of CRF in individuals with mild disability (EDSS score ≤ 4.0). However, $\dot{V}O_{2\text{peak}}$ in individuals with moderate disability (EDSS score > 4.0) may be symptom limited and therefore indicate functional ability rather than CRF ([399](#)).
- Depending on the disability and physical fitness level of the individual, the use of a continuous or discontinuous protocol of 3–5 minutes stages increasing work rate for each stage from 12 to 25 W for leg ergometry and 8 to 12 W for arm ergometry may be recommended, although shorter stages of 1–2 minutes have been applied with success.
- In general, muscle strength and endurance can be determined using standard protocols in individuals with MS. Each large muscle group and all limbs should be tested because weakness may present itself in a particular muscle group or limb due to the heterogeneity of lesion location and impact in MS. Isokinetic dynamometry can be used to reliably evaluate muscle performance ([371](#)).

In a clinical or community setting, an 8- to 10-RM or functional testing (*e.g.*, 30-second sit-to-stand test) can be used to measure muscular strength and endurance.

Exercise Prescription

The National MS Society recommends that individuals with MS should strive toward guidelines for PA of 150 minutes of moderate intensity per week or undertake a prescriptive exercise training program consisting of aerobic, resistance, and flexibility modalities. Indeed, individuals with minimal disability (EDSS 0–2.5) should undertake an exercise program based on the FITT principle of Ex Rx that is generally consistent with those outlined in [Chapter 5](#) for healthy adults. As MS symptoms and level of disability increase, the following modifications outlined may be required.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH MULTIPLE SCLEROSIS

	Aerobic	Resistance	Flexibility
Frequency	2–5 d · wk ⁻¹	2 d · wk ⁻¹	5–7 d · wk ⁻¹ , one to two times · d ⁻¹
Intensity	40%–70% $\dot{V}O_2R$ or HRR; RPE 12–15	60%–80% 1-RM	Stretch to the point of feeling tightness or mild discomfort
Time	Increase time initially to a minimum of 10 min before increasing intensity. Progress to 30–60 min as tolerated	Begin with 1 and gradually work up to 2 sets of 10–15 repetitions	Hold 30–60 s, 2–4 repetitions
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming)	Multijoint and single-joint exercises using machines, free weights, resistance bands, or body weight	Static stretching

1-RM, one-repetition maximum; HRR, heart rate reserve; RPE, rating of perceived exertion; $\dot{V}O_2R$, oxygen uptake reserve.
Based on data from (400).

Exercise Training Considerations

- With individuals who have significant paresis, consider assessing RPE of the extremities separately using the 0–10 OMNI scale (Figure 10.4) (401) to evaluate effects of local muscle fatigue on exercise tolerance.

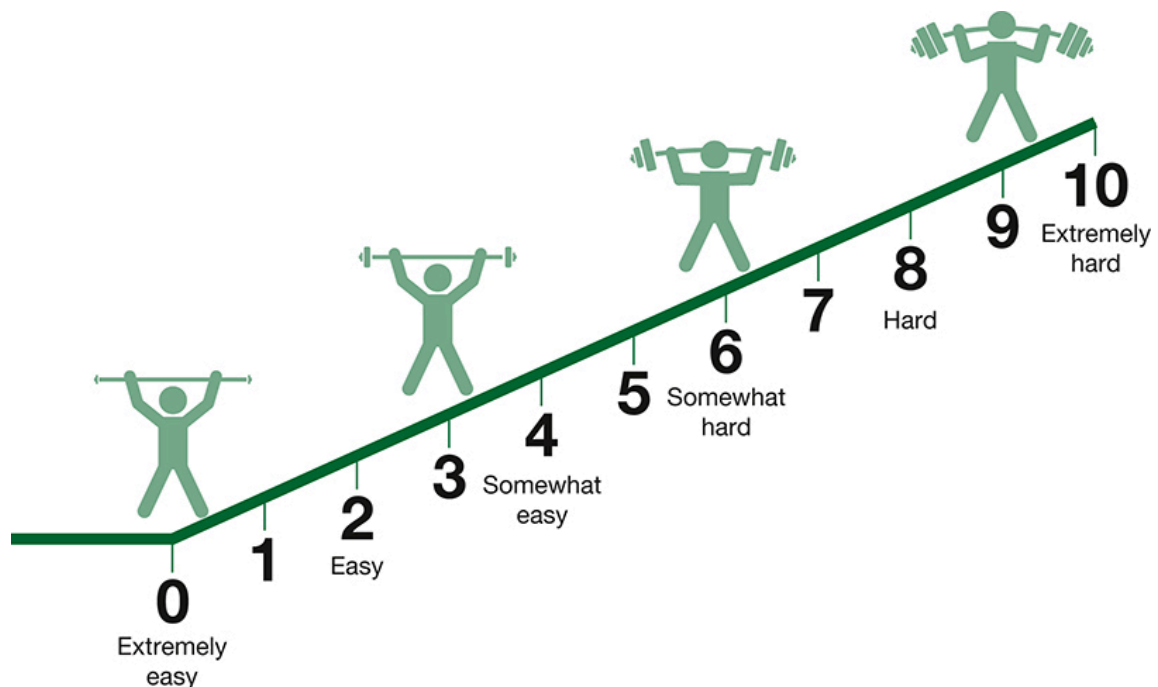


Figure 10.4 OMNI Resistance Exercise Scale of perceived exertion. Used with permission from (400).

- During a relapse or exacerbation of MS symptoms, decrease the FITT of the Ex R_x to the level of tolerance. If the exacerbation is severe, focus on maintaining functional mobility and/or focus on aerobic exercise and flexibility. Recognize that in times of severe relapse, any exercise may be too difficult to perform.
- When strengthening weaker muscle groups or working with easily fatigued individuals, increase rest time (*e.g.*, 2–5 minutes) between sets and exercises as needed to allow for full muscle recovery. Focus on large muscle groups and minimize total number of exercises performed.
- To eliminate balance concerns during flexibility exercises, slow and gentle passive ROM exercise should be performed while seated or lying down.
- Muscles and joints with significant tightness or contracture may require longer duration (several minutes to several hours) and lower load positional stretching to achieve increases in joint ROM. Very low-intensity, low-speed, or no-load cycling may be beneficial in those with frequent spasticity.

Special Considerations

- Commonly used disease-modifying medications such as interferon β -1a and glatiramer acetate have common side effects including altered mood, flulike symptoms, liver failure, and localized irritation at the injection site. Take medication side effects into consideration with exercise testing and scheduling.
- The individual should be helped to understand the difference between more general centrally mediated MS fatigue and temporary peripheral exercise-related fatigue.
- Some individuals may restrict daily fluid intake because of bladder control problems. If present, these persons should be counseled to increase fluid intake with increased PA levels to prevent dehydration and hyperthermia, secondary to impaired thermoregulation.
- Many individuals with MS have some level of cognitive deficit, and this may affect the understanding of testing and training instructions. Some persons may also have short-term memory loss that requires written instructions and frequent verbal cueing and reinforcement.
- Watch for transient worsening of sensory and motor symptoms, most commonly, visual impairment, associated with exercise and elevation of body temperature. Symptoms can be minimized by using cooling strategies and adjusting exercise time and intensity.

ONLINE RESOURCES

MS Canada Physical Activity Guidelines: <https://mscanada.ca/physical-activity>

National Institute of Neurological Disorders and Stroke: <https://www.ninds.nih.gov/Disorders/All-Disorders/Multiple-Sclerosis-Information-Page>

National Multiple Sclerosis Society: <http://www.nationalmssociety.org>

Physical Activity Guidelines for Americans: https://health.gov/sites/default/files/2019-09/16_F-10_Individuals_with_Chronic_Conditions.pdf

OSTEOPOROSIS

Osteoporosis is a systemic skeletal disease that is characterized by low BMD and detrimental changes in the microarchitecture of bone that increase susceptibility to fracture. Clinically, osteoporosis is defined in postmenopausal females and in males ≥ 50 years as a BMD T-score of the lumbar spine, total hip, or femoral neck of ≤ -2.5 (402). The National Bone Health Alliance Working Group proposes additional diagnostic criteria for osteoporosis to include those with diagnosed low bone mass (previously osteopenia) who have sustained a low-trauma vertebral, proximal humerus, pelvis, or distal forearm fracture, or who have an elevated fracture risk per the Fracture Risk Algorithm (403). Using the defined T-score criteria, approximately 13% of U.S. adults aged 50 years and over have osteoporosis and 43% have low bone mass (404).

The burden of osteoporosis on the individual and society is significant. More than 54 million individuals in the United States have osteoporosis or low bone mass, including over 18 million males (405). Mortality at 1 year postfracture is 20% and 25% in males and females, respectively (406). Further, approximately 50% of those who suffer a hip fracture are permanently incapacitated in some way (407). Economically, the direct cost of osteoporotic fracture in the United States was \$17 billion in 2005 and was projected to increase by 50% by 2025 (408). These estimates do not account for indirect costs, such as time off work, reduced physical function, or increased morbidity related to the fracture.

Evidence suggests that exercise can slow age-associated bone loss and potentially delay the onset of osteoporosis and/or reduce fracture risk (409–417). The benefits of exercise on bone health occur throughout the lifespan due primarily to increases in bone density, volume, and strength (411, 413). Exercise also improves balance in both young and older populations, which can reduce falls and subsequent osteoporotic fracture risk (414, 418). Thus, exercise can generally be regarded as the primary nonpharmacological treatment for the prevention and management of osteoporosis. Nevertheless, many investigators have concluded that large RCTs are still needed in both females and males to determine the ideal $Ex R_x$ for preventing both osteoporosis and associated fracture (411, 412, 416).

Exercise Testing

In general, when an exercise test is clinically indicated for those with osteoporosis, normal testing procedures should be followed (see [Chapter 3](#)) with some additional considerations:

- Use of cycle leg ergometry as an alternative to treadmill exercise testing to assess cardiorespiratory function may be indicated in individuals with severe vertebral osteoporosis for whom walking is painful or risky.
- Vertebral compression fractures contributing to a loss of height and spinal deformation can compromise ventilatory capacity and result in a forward shift in the center of gravity. The latter may affect balance during treadmill walking.

- Maximal muscle strength testing may be contraindicated in individuals with severe osteoporosis due to the risks of skeletal injury. There are no established guidelines for contraindications for maximal muscle strength testing.
- Balance testing or fall risk assessment should be considered in individuals with osteoporosis or low bone density. Available assessments include the Performance-Oriented Mobility Assessment and the Modified Falls Efficacy Scale ([419](#)).

Exercise Prescription

Currently, there is no consensus on the optimal Ex R_x for individuals with osteoporosis, and general recommendations are similar to the Ex R_x for older adults (see [Chapter 6](#)) ([97](#), [100](#)). In general, aerobic exercise is primarily for overall health benefits; weight-bearing aerobic exercise (*e.g.*, walking) is preferred to weight-supported (*e.g.*, cycling) ([420](#)), and all Ex R_x should include some form of higher-impact, higher-velocity, and/or higher-intensity resistance training ([97](#), [411](#), [421](#), [422](#)). Supervised training appears to be superior to unsupervised training with regard to most outcomes (bone mass, balance, fall prevention), although there is limited evidence on unsupervised training ([423](#)).

Special Considerations

- Aerobic activity primarily benefits individuals with osteoporosis or low bone mass through improved cardiovascular and metabolic health. Depending upon the type of aerobic activity, impact (ground reaction forces) and speed associated with weight-bearing aerobic exercise may also contribute to bone loading.
- It is difficult to quantify exercise intensity in terms of bone loading forces. However, the magnitude of bone loading force generally increases in parallel with exercise intensity quantified by conventional methods (*e.g.*, %HRR for aerobic training or %1-RM for resistance training), and bone strengthening only occurs in the involved area. Weight-bearing aerobic and high-velocity resistance training modes are recommended, as tolerated. Proper form and alignment are more important than intensity, especially for those with a history of fractures ([411](#), [421](#), [422](#)).

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH OSTEOPOROSIS ([97](#), [411](#), [421](#), [422](#))

	Aerobic	Resistance	Flexibility
Frequency	4–5 d · wk ⁻¹ .	Start with 1–2 nonconsecutive d · wk ⁻¹ ; may progress to 2–3 d · wk ⁻¹ .	5–7 d · wk ⁻¹ .
Intensity	Moderate intensity (40%–59% $\dot{V}O_{2R}$ or HRR). Use of the CR-10 scale with ratings of 3–4 might be a more appropriate method of establishing intensity.	Adjust resistance so that last 2 repetitions are challenging to perform. High-intensity and high-velocity training can be beneficial in those who can tolerate it.	Stretch to the point of tightness or slight discomfort.
Time	Begin with 20 min; gradually progress to a minimum of 30 min (with a maximum of 45–60 min).	Begin with 1 set of 8–12 repetitions; increase to 2 sets after ~2 wk; no more than 8–10 exercises per session.	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise.
Type	Walking, cycling, or other individually appropriate aerobic activity (weight bearing preferred). Impact loading exercises such as jumping or bench stepping can be used in those with low or moderate risk for fracture.	Standard equipment can be used with adequate instruction and safety considerations. Compound movement exercises are best.	Static stretching of all major joints.

HRR, heart rate reserve; $\dot{V}O_{2R}$, oxygen uptake reserve.

- There are currently no established guidelines regarding contraindications to exercise for individuals with osteoporosis. The general recommendation is to prescribe moderate-intensity weight-bearing exercise that does not cause or exacerbate pain. Exercises that involve explosive movements or high-impact loading should be avoided, especially in those at high risk for fracture ([411](#)). Specific exercises or portions of group-led routines (*e.g.*, yoga, Pilates) that require excessive twisting, bending, or compression of the spine should also be carefully assessed and avoided, particularly in those with very low spinal BMD values or prior history of vertebral fracture ([97](#), [411](#)).
- Falls in those with osteoporosis increase the likelihood of a bone fracture. For older females and males at increased risk for falls, the Ex R_x should also include activities that improve balance (see [Chapters 5](#) and [6](#) and the relevant ACSM position stand ([97](#))). Primary considerations should be exercises that strengthen the quadriceps, hamstrings, and gluteal and trunk muscles because these are the primary balance muscles ([418](#)). Tasks done with the eyes closed should also be considered for individuals with low or moderate (but not high) risk for fracture ([411](#)).
- In light of the rapid and profound effects of immobilization and bed rest on bone loss, and poor prognosis for recovery of BMD after remobilization, even the frailest older adults should remain as physically active as health permits because this will best preserve musculoskeletal integrity. Even short bouts of standing or walking are desirable during prolonged illnesses.

- The recommendations in this section are generalized for exercise in individuals with, or at risk for, osteoporosis. Modifications based on clinical judgment may be necessary for some individuals ([411](#), [421](#), [422](#)). Goals and preferences of the individual should also be considered to help with compliance ([411](#), [422](#)).

ONLINE RESOURCES

International Society of Clinical Densitometry: <http://www.iscd.org/official-positions/>

National Institutes of Health Osteoporosis and Related Bone Diseases:

http://www.niams.nih.gov/Health_Info/Bone/default.asp

Bone Health and Osteoporosis Foundation (previously National Osteoporosis Foundation):

<https://www.bonehealthandosteoporosis.org/>

SPINAL CORD INJURY

Spinal cord injury (SCI) results in a loss of somatic, sensory, and autonomic functions below the lesion level. Lesions in the cervical region typically result in tetraplegia (also known as quadriplegia, resulting in the partial or total loss of function below the cervical level of lesion), whereas lesions in the thoracic, lumbar, and sacral regions lead to paraplegia (which does not influence arm function). SCI level and neurological completeness are currently determined by the International Standards for Neurological Classification of SCI (ISNCSCI) exam or the American Spinal Cord Injury Impairment Scale (AIS) Classification ([424](#)). This classification primarily considers motor and sensory impairment at or below the level of injury. SCI may lead to either motor-complete (AIS A and B) or motor-incomplete (AIS C and D) paralysis. An individual with a loss of sacral sparing (*i.e.*, loss of sensory and motor response at S4–S5) is considered AIS A (*i.e.*, complete loss in both motor and sensory function). Individuals with sensory function below the neurological lesion level, but no motor function, are classified as AIS B. Motor-incomplete injuries are distinguished by functional ability below the level of injury (AIS C; more than half the key muscle groups having a muscle testing grade less than 3: AIS D; at least half the key muscle groups having a muscle testing grade ≥ 3).

Incomplete injuries are the most frequent neurological classification (45.8% and 20.9% for incomplete tetraplegia and paraplegia, respectively) in which some somatosensory functions below the lesion are still maintained ([425](#)). Approximately 19.7% of individuals have motor-complete paraplegia, with the remaining 13.2% being classified as motor-complete tetraplegia. The most common causes of SCI can be attributed to motor vehicle accidents or falls, with males accounting for approximately 81% of new SCI cases ([425](#)). SCI of traumatic origin often occurs at an early age (18–40 years old) followed by another peak that occurs at a later age greater than 60 years due to cervical myelopathy or loss of balance that results in falls. Individuals with SCI have a high risk for the development of secondary conditions (*e.g.*, shoulder pain, urinary tract infections, skin pressure ulcers, osteopenia, neuropathic pain, problematic spasticity, joint contractures, depression, anxiety, fatigue, obesity, dyslipidemia, T2DM, and CVD).

The SCI level and completeness of the lesion directly impact physical function, along with metabolic and autonomic cardiorespiratory responses to exercise. For example, individuals with SCI at or above T6 exhibit a loss of autonomic supraspinal control of visceral organs (including the heart and blood vessels), which blunts maximal HR and impairs blood redistribution and BP regulation during exercise ([426](#)). Such alterations result in reduced venous return to the heart, which limits stroke volume and thus cardiac output, often leading to premature fatigue. It is therefore crucial to consider the SCI lesion level when performing exercise testing and prescribing exercise for those with motor-complete SCI. The following points describe specific motor and autonomic considerations, dependent on the level of SCI lesion:

- L2–S2 lack voluntary and autonomic control of the bladder, bowel, and sexual function; however, the upper extremities and trunk usually have normal function.

- T6–L2 have respiratory and motor control that depends on the functional capacity of the abdominal muscles (*i.e.*, minimal at T6 to maximal at L2).
- T1–T6 can experience impaired thermoregulation, orthostatic and postexercise hypotension (with debilitating symptoms, *i.e.*, light-headedness, dizziness, fatigue), and autonomic dysreflexia (AD). AD is an unregulated, spinally mediated reflex response called the *mass reflex* that can be a life-threatening medical emergency with sudden hypertension, bradycardia, pounding headache, piloerection, flushing, goose bumps, shivering, sweating above the level of injury, nasal congestion, and blotching of the skin. When there is no supraspinal sympathetic input to the heart (T1–T5), resting HR may be bradycardic due to cardiac vagal dominance, and HR_{peak} is limited to approximately $115\text{--}130 \text{ beats} \cdot \text{min}^{-1}$. Breathing capacity is further diminished by intercostal muscle paralysis; however, arm function is normal.
- C5–C8 are tetraplegic. Those with C8 lesions have voluntary control of the scapula, shoulder, elbow, and wrist but decreased hand function, whereas those with C5 lesions rely on the biceps brachii and shoulder muscles for self-care and mobility.
- Above C4 requires ventilator support for breathing. These individuals are commonly recommended to receive an implanted phrenic nerve stimulator for direct activation of the diaphragm muscle in order to function as a diaphragmatic pacing technique. AD, orthostatic and postexercise hypotension, along with thermoregulatory issues, can also occur in individuals with C4–C8 injuries.

Exercise Testing

It should be noted that exercise-testing options following SCI have advanced to include functional electrical stimulation (FES) of paralytic muscles, locomotor modalities (*i.e.*, body weight-supported treadmill training [BWSTT], robotic exoskeletons), and other hybrid forms including brain-computer interface (BCI), neuromodulation (*i.e.*, epidural or transcutaneous spinal cord stimulation), and virtual reality strategies, which can be used in conjunction with traditional locomotor modalities. This is in addition to commonly available forms of exercise that primarily focus on the voluntary activation of spared muscle above the level of injury (*i.e.*, arm-crank ergometer [ACE], circuit resistance training, wheelchair propulsion, pulleys, and resistance bands).

When exercise testing individuals with SCI, consider the following issues:

- Level of functional independence or spinal cord injury independence measures (SCIM) should be assessed including ROM, strength, using manual muscle testing for both lower and upper extremities, sitting and standing balance (if the individual is able to ambulate), independence in transfer, wheelchair mobility, and upper and lower extremity motor involvement. This assessment will facilitate the choice of exercise testing equipment, protocols, and adaptations.
- Consider the purpose of the exercise test, the level and completeness of SCI, and the physical fitness level of the individual to optimize equipment and protocol selection.
- Choose an exercise mode that allows the individual to engage the largest possible muscle mass. If substantial trunk and lower limb function is intact, consider combined voluntary arm and leg cycle ergometry (hybrid exercise) or recumbent stepping. If injuries are motor-complete, voluntary arm-

crank ergometry is the easiest to perform and is norm-referenced for the assessment of CRF ([427](#)). Other norm-referenced data exist for wheelchair exercise ([428](#), [429](#)).

- The paradigms of FES necessary to evoke muscle activation may vary from a single-trained muscle, using neuromuscular electrical stimulation (NMES), or multiple muscle activation using functional electrical stimulation lower extremity cycling (FES-LEC; *i.e.*, up to six muscle groups). With either paradigm, it is recommended that adequate recovery is provided between sessions (at least 48 hours) to avoid exercise-induced muscle damage that typically occurs following activation of paralyzed skeletal muscles ([430](#)). Recent evidence may suggest that the addition of FES-LEC to NMES may influence musculoskeletal and neuromuscular systems differently compared to applications of FES-LEC ([431](#)).
- Individuals with SCI at or above the T6 level are likely to experience AD (sudden increase in systolic BP 20–40 mm Hg above baseline) during FES-LEC and sometimes NMES. Individuals prone to AD should have access to a fast-acting antihypertensive agent (*i.e.*, nifedipine, captopril, or nitroglycerin), which can be administered should systolic BP remain above 150 mm Hg. It is important for exercise practitioners to periodically monitor BP throughout each exercise session (ideally every 3–5 minutes) and modulate the stimulation parameters to reduce the charge density developed under the electrodes during FES-LEC. When exercising, individuals should always be in a seated position (to ensure an orthostatic hypotension response protecting the cerebrovasculature) and exercise may be paused or terminated based on the individual's BP responses. Anecdotal findings suggest that preceding FES-LEC with NMES may attenuate the effects on AD in persons with tetraplegia ([432](#)).
- Additionally, a high-intensity interval training (HIIT) FES cycling program may benefit cardiometabolic health in persons with chronic SCI ([433](#)).
- If available, a stationary wheelchair roller system or motor-driven treadmill should be used with the individual's wheelchair adjusted as necessary. Motor-driven treadmill protocols allow for realistic simulation of external conditions such as slope and speed alterations ([434](#)).
- Incremental exercise tests for the assessment of CRF in the laboratory should begin at 0 W with incremental increases of 5–10 W per stage for individuals with tetraplegia. Depending on function and fitness, individuals with paraplegia can use increments of 10–25 W per stage.
- For sport-specific CRF assessments in the field, an incremental test adapted from the Léger and Boucher shuttle test around a predetermined rectangular court is recommended. However, floor surface characteristics and wheelchair-user interface should be standardized ([434](#), [435](#)).
- After maximal-effort exercise in individuals with tetraplegia, it may be necessary to treat postexercise hypotension and exhaustion with rest, recumbency, leg elevation, and fluid ingestion. Individuals should consider wearing elastic leg stockings and/or an abdominal binder to prevent venous blood pooling during exercise. Maximal exercise testing, particularly in individuals with high-level SCI, should be accompanied with concurrent ECG monitoring for cardiac arrhythmias. There are no special considerations for the assessment of muscular strength regarding the exercise-testing mode beyond those for the general population except for the lesion level, which will

determine residual motor function and thus the need for trunk stabilization, hand-wraps (*i.e.*, Active Hands), and accessibility of testing equipment.

- Individuals with SCI requiring a wheelchair for mobility may develop joint contractures because of muscle spasticity, strength imbalance, flexed joint position in the wheelchair (*i.e.*, hip flexion, hip adduction, and knee flexion), excessive wheelchair pushing, and manual transfers (*i.e.*, anterior chest and shoulder). Therefore, intensive sport-specific training must be complemented with an upper extremity stretching of prime movers and strengthening antagonists program to promote muscular balance around the joints.
- During locomotor exercise (*i.e.*, BWSTT, robotic exoskeleton), individuals can experience orthostatic hypotension (a drop in systolic or diastolic BP of 20 or 10 mm Hg, respectively) and associated debilitating symptoms when transitioning to standing. This is particularly common in individuals with injuries at or above T6. BP should be routinely monitored and individuals returned to the seated or supine position if symptoms persist.

Exercise Prescription

The goals of exercise training include the prevention of deconditioning; improved health (*i.e.*, weight management, glucose homeostasis, lower cardiovascular risk); improved muscular strength, muscular endurance, and flexibility for functional independence (wheelchair mobility, transfers, ADL); prevention of falls and sports injuries; and improved performance (safety and success in adaptive sports and recreational activities). Currently, there are no published consensus recommendations for developing an Ex R_x in the SCI population, and further research is warranted ([436](#), [437](#)). Thus, the specific FITT principle of Ex R_x recommendations provided in the box is based on several systematic reviews and consensus documents as listed in the Ex R_x box.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH SPINAL CORD INJURY ([438](#)–[441](#))

	Aerobic	Resistance	Flexibility
Frequency	Minimum of 2 d · wk ⁻¹ ; progress to 3 d · wk ⁻¹ . Athletes can increase to 3–5 d · wk ⁻¹ .	Minimum of 2 d · wk ⁻¹ .	Daily, especially in the presence of joint contracture, spasticity, or frequent wheelchair propulsion and manual transfers.
Intensity	Beginners: moderate intensity (40%–59% HRR) progressing to vigorous intensity (75%–90% HRR).	Initially, use 50% of 1-RM progressing to 80% 1-RM for all large muscle groups for each exercise.	Do not allow stretching discomfort >2 on the 0–10 pain scale.
Time	Initially, bouts of 5–10 min alternating with 5-min active recovery periods. Gradually increase to at least 20–40 min per session ^a or 30–44 min per session, ^b depending on health variable of interest. Aim to decrease or eliminate rest periods altogether with progression.	Initially, 1–2 sets of 8 repetitions each exercise per session. Gradually progress to 3 sets of 10 repetitions.	Stretch each muscle group repeatedly for 3–4 min · d ⁻¹ , preferably after warm-up or following training/competition.
Type	Engage the largest possible muscle mass: voluntary arm + leg ergometry or combined FES-LEC and voluntary arm ergometry or rowing, recumbent stepping, arm ergometry, wheelchair ergometry/rollers, or wheeling.	Accessible resistance exercise machines are convenient and safe. If not available, use dumbbells, cuff weights, or elastic bands/tubing. Surface neuromuscular electrical stimulation–resistance training can also be performed for paralytic muscles.	Active stretching is preferred, but if this is not possible, low-intensity passive stretching may be used by the individual or assistant. A standing frame can also be utilized. ^c

^a Minimal guideline recommendation if aerobic exercise is combined with resistance training to improve cardiorespiratory fitness, power output, and muscle strength.

^b Minimal guideline recommendation if solely aerobic exercise is performed (without resistance training) to improve muscle strength, body composition, and cardiometabolic disease risk.

^c Those with spinal cord injury (SCI) and limited or no recent standing history may be at an increased risk for fracture due to osteoporosis. Full weight-bearing activities should be limited to those individuals with uncomplicated history of standing or for whom prior medical clearance for full weight bearing has been obtained. Preferably, individuals should have a dual-energy x-ray absorptiometry scan to assess BMD, with T-score less than –2.5 SD for total body or BMD less than 0.6 g · cm⁻² for distal femur or proximal tibia, considered high risk for standing.

1-RM, one-repetition maximum; FES-LEC; functional electrical stimulation lower extremity cycling; HRR, heart rate reserve; BMD, bone mineral density.

The recent systematic review, which informed the aforementioned FITT recommendations for individuals with SCI, mentioned that high-quality RCT evidence to inform population-specific exercise guidelines are still lacking ([438](#)). The low prevalence and considerable heterogeneity of the SCI population negatively impacts the quality and quantity of the existing exercise training literature. Other authors have advocated for volumes of moderate- to vigorous-intensity exercise more akin to able-bodied guidelines, such as those proposed by ACSM (*i.e.*, >150 min · wk⁻¹) ([442](#)). Such recommendations may be purely aspirational for profoundly inactive individuals or those with significant motor impairments. Therefore, it has been suggested that stratifying individuals into “*beginning*” (inactive, unlikely to reach sufficient exercise volume within 3 months), “*intermediate*” (currently not meeting recommendations but possible in the next 3 months), and “*advanced*” (currently meeting guidelines, maintenance, and management to prevent overuse injuries) can assist

with appropriate Ex R_x and the application of able-bodied guidelines into clinical practice for individuals with SCI. The aforementioned FITT recommendations are consistent with the recent International Spinal Cord Society (ISCoS) exercise guidelines (439) (*i.e.*, 20–30 minutes of moderate-to-vigorous intensity two to three times per week). However, it is important to note that this volume of exercise is well below that advised by the World Health Organization (WHO) and ACSM to improve CRF and cardiometabolic disease risk factors (443). Thus, while the SCI-specific exercise guidelines are more or less universally accepted and feasible, based on the totality of the available scientific evidence in this population (level 3–4 studies), this should be considered a minimal dose and starting point for individuals classified as “*beginning*.” There are no clear guidelines on progression as this is highly individualized due to the heterogenous nature of the injury. However, benefits can be derived from small, progressive increases in exercise volume (*e.g.*, $5 \text{ min} \cdot \text{wk}^{-1}$), thereby transitioning individuals to “*advanced*” ($>150 \text{ min} \cdot \text{wk}^{-1}$) over time.

A recent overview of systematic reviews in the SCI literature suggests that ACE training with/without additional therapeutic interventions (20 minutes, moderate-to-vigorous intensity, twice weekly for 6 weeks) may improve CRF; similarly, resistance training with/without additional therapeutic interventions (three sets of 8–10 repetitions, moderate-to-vigorous intensity, twice weekly for 6 weeks) may improve muscle strength (444). In addition to improving CRF (445), upper body aerobic exercise training ($>75\%$ maximum heart rate) appears to improve waist circumference and hepatic insulin sensitivity but appears insufficient for improving fasting glucose, lipid profile, or resting blood pressure (446).

The aforementioned FITT recommendations are primarily for forms of exercise that focus on the voluntary activation of spared muscle above the level of injury. The guidelines for FES exercise are currently not well established (447). However, preliminary evidence supports that two to three times per week may be a reasonable frequency to evoke skeletal muscle hypertrophy, improve cardiometabolic health, and attenuate bone loss. Recent recommendations have been published for different paradigms of electrical stimulation exercise for persons with SCI (448).

It is encouraged that applications of FES start as soon as a few weeks postinjury to attenuate injury-induced loss in both muscle and bone tissues (449). Muscle is a highly plastic tissue, and individuals with SCI are likely to experience substantial disuse muscle atrophy, accompanied with the infiltration of intramuscular fat, just a few weeks postinjury (450). Supporting evidence in individuals with acute SCI indicated that FES two to three times per week (for 30–60 minutes per session) ranging from 8 weeks up to 6 months is likely to attenuate muscle atrophy (447).

It is interesting to note that two systematic reviews confirmed the robust effects of FES/NMES on muscle size or lean mass (451). However, there is still inconclusive evidence on the effects of electrical stimulation exercise on reducing fat mass. Surprisingly, Bekhet et al. (452) showed that the progressed dose of NMES training may differently influence muscle size compared to FES-LEC.

For bone health with FES, studies have been inconclusive. At least 6 months of FES training are necessary to induce changes in trabecular bone as well as decreasing biomarkers of bone resorption (453). Loads of between 1 and 1.5 times body weight, applied via electrical stimulation of the soleus muscle for 3 years, can result in higher distal tibia trabecular BMD compared to the untrained

contralateral limb ([454](#)). Therefore, it may be that more intensive, long-term training regimes are necessary to improve bone health with FES in individuals with SCI.

Other forms of exercise that involve locomotor training to improve CRF have been recommended (*i.e.*, BWSTT and exoskeletal-assisted walking). For BWSTT, the general consensus has been to conduct exercise training three to five times per week for 60 minutes, starting with 50% body weight support or enough support that individuals would not buckle at the knees. RCTs have been unable to establish the superiority of BWSTT compared to over ground ambulation, especially in individuals with acute incomplete SCI ([455](#)). It appears that the dosing and intensity of training is highly proportional to the extent of recovery in individuals with incomplete SCI. Based on the available evidence, to optimize spontaneous neurological recovery, training should be scheduled for five times per week during the first 12–18 months postinjury and then gradually limited to three times per week.

The consensus regarding exoskeletal-assisted walking is also still not well established. Currently, the available evidence supports that two to three times per week for 30–60 minutes may be sufficient to improve quality of life and promote CRF in individuals with chronic SCI. However, a major caveat is that the speed of walking is restricted by the individual's ability to achieve postural control and balance. Walking below $0.5 \text{ m} \cdot \text{s}^{-1}$ does not provide a perceived exertion greater than 12–13, which may not elicit an exercise intensity above that recommended by the ACSM guidelines to improve CRF. Therefore, similar to BWSTT, it is possible that health improvements achieved with exoskeletal-assisted walking is dependent on the level and severity of SCI. It should also be considered that these approaches are resource heavy, that is, they rely on trained staff and expensive equipment that might not be available to all individuals with SCI or exercise practitioners.

Evidence for exercise and various health outcomes specific to individuals with SCI is weak for individuals with acute (<1 year) SCI, motor-incomplete injuries (AIS C–D), and older adults (those aged >65 years are neglected in the wider exercise and SCI literature). The strength of existing evidence is also higher for young and middle-aged adults ([438](#), [439](#)). The evidence is currently insufficient to comment on inverse dose-response associations for exercise and health-related variables. If changes in individuals' exercise behaviors are accurately tracked and recorded (using validated methods) ([456](#)), future research may permit a deeper understanding of the optimal dose of exercise, specifically in this population. Importantly, few adverse events have been documented with exercise in this population, besides the occasional musculoskeletal complaints, and the risk seems to be comparable to those observed in the general population ([290](#)).

Exercise Training Considerations

- Include resistance exercises for all innervated muscle groups, typically involving the upper body but not ignoring paretic arm, trunk, or leg muscles. Paralyzed muscle groups can now be exercised using surface NMES-resistance training ([457](#)). A recent video publication has provided the opportunity for clinicians and researchers to learn step-by-step procedures on how to simply activate paralyzed muscles, using both surface NMES-resistance training and FES-LEC in individuals with SCI ([458](#)). Based on this established protocol, electrically evoked resistance training is offered in the forms of 4×10 repetitions per leg (40 repetitions). While seated in a wheelchair, individuals with SCI start

moving their legs against gravity without ankle weights for 1–2 weeks. This is followed by gradually loading the legs, using ankle weights with 2-lb incremental progression on a weekly basis. Failure to achieve 40 repetitions precludes the progression of adding more weight.

- Despite this strategy being safe to apply in individuals with SCI, certain precautions need to be considered, including conducting regional bone scans for hip and knee joints using dual-energy x-ray absorptiometry. A T-score of less than 3.5% SD of the femoral necks or BMD of less than $0.6 \text{ g} \cdot \text{cm}^{-2}$ may require medical clearance to ensure safe participation in a protocol aiming to electrically evoke resistance training of the paralyzed muscles.
- Resist overemphasis of “pushing” motions such as bench/chest press or rickshaw dips that develop the anterior shoulder/pectoral muscles, scapular protractors, and internal rotators (prime movers for functional skills such as wheelchair propulsion and transfers).
- Balance the “pushing” exercises with “pulling” exercises such as rowing and lat pulldowns that develop the scapular retractors and depressors, posterior deltoids, external rotator cuff muscles, and latissimus dorsi.
- If strength is the goal and arm overuse syndromes do not develop, increase resistance to 5- to 10-RM. As exercise tolerance increases and if arm overuse syndromes do not develop, increase volume to 3–4 sets per session.
- Advanced exercises for athletes may include ballistic medicine ball exercises, battle rope training, and sport-specific skills requiring power and speed.
- Therapeutic exercises may be indicated for joints with muscle imbalance and spasticity. The primary goal is the prevention/correction of joint contractures and loss of ROM.
- All muscles should be stretched slowly, especially spastic muscles, to minimize elicitation of stretch reflexes (spasticity), which can aggravate muscle imbalance and contracture. Emphasize prime mover muscles of the chest, anterior shoulders, and shoulder internal rotators. Adjacent joints must be stabilized to stretch the intended muscles/tendons.
- Stretch spastic muscles that may cause joint contractures (*e.g.*, elbow flexors, hip/knee flexors, hip adductors, and ankle plantar flexors). Passive/active standing can also stretch hip and plantar flexors.
- Progression (increased ROM) should be slow and based on pain tolerance, especially in advanced age, arthritis, permanent joint contracture, periodic immobilization (bed rest, hospitalizations), heterotopic ossification (HO), and chronic overuse syndromes and pain.

Special Considerations

Autonomic

- Individuals with complete SCI at or above T6, particularly those with complete tetraplegia, may exhibit lower exercise performance due to motor and cardiovascular dysfunction. These individuals may reach their HR_{peak} , peak cardiac output (Q), and oxygen consumption ($\dot{V}\text{O}_2$) at lower exercise levels than those with paraplegia with lesion levels below T5–T6.

- Individuals with higher SCI levels may benefit from the use of lower body positive pressure by applying compressive antithromboembolic stockings, an elastic abdominal binder, electrical stimulation to leg muscles, and/or exercise in a recumbent posture. Beneficial hemodynamic effects may include maintenance of BP, lower HR, and higher stroke volume during arm work to compensate for blood pooling below the lesion. Anecdotal evidence has demonstrated that spinal cord stimulation can increase blood pressure and oxygen pulse (a surrogate for stroke volume) in an individual with tetraplegia, leading to an increase in peak $\dot{V}O_2$ ([459](#)).
- During exercise, the occurrence of AD results in an increased release of catecholamines that increases HR, $\dot{V}O_2$, BP, and exercise capacity ([460](#)). BP may be elevated to excessively high levels (*i.e.*, systolic BP 250–300 mm Hg and/or diastolic BP 200–220 mm Hg). In these situations, immediate emergency responses to decrease BP are needed (*i.e.*, stopping exercise, sitting upright, and identifying and removing the irritating stimulus such as an obstructed catheter/urinary collection device, tight clothing, or braces). Emergency medical attention should be sought immediately if the symptoms persist.
- The practice of self-inducing AD is termed *boosting* and has been used to induce a competitive advantage in athletes with SCI ([461](#), [462](#)). However, the International Paralympic Committee (IPC) prevents athletes from competing with AD due to the potential catastrophic health consequences including cerebral hemorrhage, seizure, myocardial infarction, or even death ([463](#)). The IPC recommends measuring resting BP before competition to detect boosting. If SBP is found to be above 160 mm Hg, then it is reexamined 10 minutes later. If SBP remains elevated above 160 mm Hg, then the athlete is withdrawn from competition. Pharmaceutical intervention is advocated when SBP remains above 150 mm Hg. Changes in SBP greater than 20 mm Hg also are considered indicative of AD. This is relevant when you consider that low resting BP (*i.e.*, 90/60 mm Hg) is common in individuals with higher lesion levels, with an increase to or above 160 mm Hg ($\Delta \geq 70$ mm Hg) considered quite dangerous. The practice of self-inducing AD should be widely discouraged by practitioners.
- Individuals should empty their bowels and bladder or urinary bag before exercising. The most common causes of AD are a full bladder or bowel distension.
- Individuals with SCI tend to endure higher core temperatures during endurance exercise than their able-bodied counterparts. Despite this enhanced thermoregulatory drive, they generally have lower sweat rates. The following factors reduce heat tolerance and should be avoided: lack of acclimatization, dehydration, glycogen depletion, sleep loss, alcohol, and infectious disease. During training and competition, the use of light clothing, ice vests, protective sunscreen cream, and mist spray are recommended ([464](#)).

Musculoskeletal

- Novice, unfit but healthy individuals with SCI will probably experience muscular fatigue before achieving substantial central cardiovascular stimulus. Individuals with tetraplegia who have a very small active musculature will also experience muscular fatigue before exhausting central cardiorespiratory capacity.

- Muscular strength training sessions from a seated position in the wheelchair should be complemented with nonwheelchair exercise bouts to involve all trunk-stabilizing muscles. However, transfers (*e.g.*, from wheelchair to the exercise apparatus) should be limited because they increase the glenohumeral contact forces and the risk of repetitive strain injuries such as shoulder impingement syndrome and rotator cuff strain/tear, especially in individuals with tetraplegia ([465](#)). Special attention should be given to shoulder muscle imbalance and the prevention of repetitive strain injuries. The prime movers of wheelchair propulsion should be lengthened (*i.e.*, muscles of the anterior shoulder and chest), and antagonists should be strengthened (*i.e.*, muscles of the posterior shoulder, scapula, and upper back) ([466](#)).
- *Tenodesis* (*i.e.*, active wrist extensor-driven finger flexion) allows functional grasp in individuals with tetraplegia who do not have use of the hand muscles. To retain the tenodesis effect, these individuals should never stretch the finger flexor muscles (*i.e.*, maximal and simultaneous extension of wrist and fingers).
- HO is considered as an ectopic bone growth that commonly occurs around the hip joints in individuals with SCI. HO may cause nerve pain, ischemic pressure, and limitations in the ROM. Radiological examination is recommended, especially for individuals likely to be involved in FES-LEC or exoskeleton training to prevent additional growth. This condition should be considered as a relative and not absolute contraindication for exercise. Medical clearance should be sought based on the individual's condition.

Skin

- Skin pressure injuries should be avoided at all costs, and potential risk areas should be checked on a regular basis. It is commonly recommended that individuals with sacral or pelvic pressure injuries greater than grade II only exercise following medical clearance. This protects the skin against shear stress and allows essential healing time. Pressure mapping testing is now available, which can provide an indication of problem areas for individuals when seated that are likely to progress into future pressure injuries.

Exercise Options

- In individuals with spastic paralysis above T12 who have substantial sensory loss and respond to stimulation with sustainable static or dynamic contractions, hybrid exercise may provide higher-intensity cardiovascular exercise than voluntary arm exercise alone. Hybrid exercise activates a larger muscle mass and elicits higher peak and submaximal training values of $\dot{V}O_2$, stroke volume, and Q than either arm ergometry or FES-LEC alone, especially for individuals with tetraplegia ([467](#), [468](#)). However, there is evidence that there may not be additional benefit of hybrid cycling versus hand cycling in this population ([469](#)).
- Prescribing exercise intensity based on HR data can be difficult for individuals with autonomic cardiovascular dysregulation (>T6). It has been suggested that ratings of perceived exertion or a talk test are viable alternatives to prescribe exercise at a given intensity in this population ([470–472](#)).

- HIIT may be a viable alternative exercise strategy to promote vigorous-intensity exercise in individuals with SCI ([473](#)). Whereas the optimal protocol for upper body exercise specifically for this population remains to be elucidated ([474](#)), this approach offers a relatively simple, readily available, and time-efficient exercise solution. By nature, HIIT also facilitates rest periods that may reduce peripheral fatigue, and recent evidence suggests that this form of exercise is more enjoyable than traditional moderate-intensity exercise for individuals with SCI ([475](#)).

ONLINE RESOURCES

American Spinal Injury Association Learning Center: <https://asia-spinalinjury.org/learning/>

National Center on Health, Physical Activity and Disability:

<http://www.nchpad.org/Articles/9/Exercise~and~Fitness>

SCI Action Canada: <https://sciactioncanada.ok.ubc.ca/>

Spinal Cord Injury Rehabilitation Evidence: <https://scireproject.com/evidence/rehabilitation-evidence/>

MULTIPLE CHRONIC DISEASES AND HEALTH CONDITIONS

The Centers for Disease Control and Prevention estimates that half of the U.S. adult population (129 million) has at least one of the top 10 chronic disease conditions and that one in four has more than one of these conditions ([476](#)). For those over 65 years of age, 88% have at least one and 73% have at least two chronic conditions ([477](#)).

This makes it increasingly likely exercise professionals will be designing Ex R_x for individuals with multiple chronic diseases and health conditions. [Chapters 8, 9](#), and this chapter present Ex R_x guidelines for many individual chronic diseases and conditions. This section considers guidelines for individuals with more than one chronic disease or condition. In general, recommendations should follow the disease or condition with the most conservative guidelines. Exercise is generally safe for the majority of individuals with multiple diseases and chronic conditions who are medically stable and wish to participate in a light- to moderate-intensity exercise program (see [Chapter 1](#)). However, exercise professionals are encouraged to consult with medical providers when there are questions about an individual's known disease and/or health conditions that may limit their participation in an exercise program.

Exercise Testing

Follow the preparticipation screening process in [Chapter 2](#) to determine if medical clearance is warranted for any single individual. An exercise test is often not warranted to begin a regular exercise program. However, if such a test is recommended by a health care provider or requested by the individual to establish a baseline fitness level, refer to the information for the disease or condition that dictates the most conservative approach.

Exercise Prescription

In general, the FITT principle of Ex R_x for individuals with multiple diseases and/or health conditions will follow the recommendations for healthy adults (see [Chapter 5](#)) except when a disease or condition dictates a more conservative approach. Review the Ex R_x recommendations for each disease and condition to make this determination. The primary challenge is determining the specifics of the Ex R_x that should be recommended for the individual who presents with multiple chronic diseases, because there is some variability in the exercise dose that most favorably impacts a particular disease, health condition, or CVD risk factor (*e.g.*, BP requires lower doses of exercise to improve than does high-density lipoprotein, abdominal adiposity, or bone density).

Special Considerations

- In those with multiple chronic diseases or conditions, it is important to make sure that all are stable prior to initiating an exercise training program.
- In some instances, exercise training adaptations may allow exercise intensity increases to elicit symptoms of a disease. For instance, in the individual with intermittent claudication, regular walking may allow an increase in exercise intensity that may subsequently uncover angina or dyspnea symptoms that were not present at lower-intensity levels.
- A large body of scientific evidence supports the role of PA in delaying premature mortality and reducing the risks of many chronic diseases and health conditions. There is also clear evidence for a dose-response relationship between PA and health. Thus, any amount of PA should be encouraged, even if the level is very low due to a chronic disease or condition.
- Begin with an Ex R_x for the single disease and health condition that confers the greatest risk and/or is the most limiting regarding ADL, quality of life, and/or starting or maintaining an exercise program. Also consider individual preference and goals.
- Alternatively, begin with the most conservative Ex R_x for the multiple diseases, health conditions, and/or CVD risk factors with which the individual presents.
- Know the magnitude and time course of response of the various health outcome(s) that can be expected as a result of the Ex R_x that is prescribed in order to progress the individual safely and appropriately.
- Frequently monitor signs and symptoms to ensure safety and proper adaptation and progression.

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CHAPTER 11

Neurologic Diseases, Conditions, and Disorders

INTRODUCTION

Brain health can be broadly defined as the optimal or maximal functioning of behavioral and biological measures of the brain and the subjective experiences arising from brain function (*e.g.*, mood). The *2018 Physical Activity Guidelines Scientific Report* ([1](#)) concluded that there is unequivocal evidence that exercise influences brain health and that individuals with conditions that affect brain health could greatly benefit from engaging in exercise. There is recent evidence suggesting that exercise, especially with higher intensities, may result in neurogenesis ([2](#), [3](#)). Neurogenesis is defined as the growth and development of nervous tissue that may be associated with improved learning, memory, cognition, and mood states ([4](#), [5](#)). When exposed to moderate- or vigorous-intensity exercise, it is believed that increased stimulation in certain areas of the brain (*e.g.*, hippocampus) may reduce neuroinflammation and O₂ stress of mitochondria, resulting in an adequate cellular environment for the repair and formation of new neurons ([2](#), [3](#)). Although these findings are promising, it must be recognized that most neurodegenerative diseases are not reversible or curable, and future research is needed to better understand the effects of exercise on neurogenesis.

In addition to the term brain health, several other terms are important to understand with respect to exercise prescription (Ex Rx). The terms “neurologic” and “psychiatric” refer to conditions and diseases that have been traditionally defined as having a clear source of pathology affecting the nervous system (neurologic) or a condition affecting behavioral or mental health features of an individual (psychiatric). For example, Alzheimer’s disease (AD) is often referred to as a neurological condition because of the pathological features of the disease, whereas clinical depression is often referred to as a psychiatric condition because of its behavioral symptoms and manifestation. Yet, despite the traditional labeling of these diseases and conditions, the boundaries between the terms and fields are becoming less distinct and blurrier because of the clear neurologic basis for psychiatric conditions and because most neurologic diseases have behavioral or

psychiatric consequences. The term “neurodegenerative disease” more specifically refers to diseases that are associated with atrophy and deterioration of the brain (*e.g.*, AD and Parkinson disease [PD]).

This chapter contains the exercise testing and Ex R_x guidelines and recommendations for individuals with health conditions related to the brain. As with the other chapters, the Ex R_x guidelines and recommendations are presented using the Frequency, Intensity, Time, and Type (FITT) principle of Ex R_x based on the available evidence from professional society position papers and scientific literature. For some neurologic diseases, conditions, or disorders, there is insufficient information regarding appropriate volume and progression of exercise training. In these instances, guidelines and recommendations provided in other chapters of the *American College of Sports Medicine (ACSM) Guidelines* should be adapted with good clinical judgment for the condition being targeted. In many instances, exercise training can be performed without a prior clinical exercise test. However, if an exercise test is to be performed, this chapter presents specific recommendations for individuals with various neurologic diseases, conditions, or disorders.

One area of brain health that is of high public interest is concussion. However, when this 12th edition of the ACSM Guidelines was published, there was limited evidence on the role of exercise or physical activity (PA) in the mitigation of, or recovery from, concussion. Future editions of the *ACSM Guidelines* and other ACSM publications will contain concussion information as it relates to exercise and PA, as the supporting evidence emerges.

ANXIETY AND DEPRESSION

Anxiety and depression are disorders that interfere with social, occupational, or other important aspects of daily functioning. Diagnostic criteria have been outlined in the *Diagnostic and Statistical Manual of Mental Disorders* (6). Anxiety and depression share a negative affective tone. The prevailing subjective experiences that might indicate anxiety or depression involve the anticipation of threat or experience of sadness, respectively, on more days than not for an extended period of time; however, individual presentations can vary widely.

Anxiety and depression are also often comorbid, meaning that individuals with one disorder frequently experience symptoms of the other (sometimes enough to qualify for a dual diagnosis). A critical risk with depression is suicidality (*i.e.*, suicidal ideation, plans, or attempts). Exercise professionals should not attempt to diagnose anxiety or depressive disorders but should instead refer individuals of concern to licensed mental health professionals for assessment, diagnosis, and treatment planning.

The World Health Organization (WHO) (7) estimates that over 300 million people suffer from depression and over 250 million suffer from anxiety disorders worldwide, with prevalence for both increasing over the past decade. Lifetime prevalence is such that over 33% of the U.S. adult population will experience an anxiety disorder and 21% will experience a mood disorder (13% for recurrent major depressive disorder) (8). Every year, over 21% of American adults experience an anxiety disorder and nearly 10% experience a mood disorder (over 7% experiencing recurrent major depressive episodes) (8). Rates of depression (4.4%) and anxiety (9.4%) are lower for children and youth ages 3–17 (9) as well as for older adults ages 55+ (anxiety: 11.4%, depression: 5.6%, mood disorder: 6.8%) (10). Unfortunately, 45%–80% of people with anxiety or depressive disorders do not receive treatment and, of those who do, many do not respond to first-line treatments (11–13).

Mental disorders are among the costliest health conditions in the United States. More money is spent on the treatment and management of mental health

annually (\$201 billion · yr⁻¹) when compared to cancer, heart conditions, diabetes, pulmonary conditions, kidney disease, or hypertension ([14](#)). Although scaling up treatments to treat anxiety and depressive disorders would be costly, the long-term economic benefits from increasing healthy life years and improving productivity are estimated to be 3.3–5.7 times greater ([15](#)).

Exercise Testing

For those with anxiety and depression, submaximal exercise tests are highly recommended because these individuals often have poor physical health and fitness levels, low physical self-worth, and limited aerobic training experience, energy, and motivation for maximal physical effort ([16](#), [17](#)). Pretest anxiety can also influence testing results due to fears of physiological reactions, which mimic physical symptoms of anxiety and depression (*e.g.*, shortness of breath, chest pain) ([18](#)). Frequently used submaximal tests include the 6-minute walk test (6-MWT) ([19](#)) and the Franz ergocycle test ([17](#)).

- All individuals with anxiety and/or depression should be screened for medication use prior to exercise testing for potential contraindications. In particular, benzodiazepines may cause drowsiness and poor coordination as well as reduce the plasma catecholamine response to exercise ([20](#)).
- Individuals with anxiety disorders have had mildly impaired blood pressure (BP) responses to cardiovascular exercise and should be screened prior to exercise testing ([18](#), [21](#)). In particular, a transdiagnostic biomarker of psychopathology for individuals with generalized anxiety disorder is vagal modulation of cardiac activity as estimated by heart rate (HR) variability, which manifests as lower HR variability during stress ([22](#)), which could negatively impact their exercise testing ([23](#)).
- Females with anxiety disorders and without coronary artery disease history have an increased ischemia risk during exercise testing ([24](#)).

Exercise Prescription

Anxiety

Exercise is effective for reducing anxious symptoms, and this applies to people with and without anxiety disorders, and those with and without other medical diagnoses ([16](#), [25–29](#)). The number of high-quality randomized controlled trials is limited, and many trials with clinically anxious participants have combined exercise with other treatments. Nevertheless, although limited, it is possible to make preliminary recommendations on how to prescribe exercise based on the available evidence.

In individuals with clinical anxiety disorders, exercise is effective for reducing symptoms both in combination with other treatments and in comparison to nonactive control groups ([25](#), [27](#)). Symptom severity does not alter these effects ([25](#)). For individuals with other medical problems, multimodal treatments do not appear to be more effective than exercise by itself ([28](#)).

In general, meeting the 2018 Physical Activity Guidelines for Americans recommendations is appropriate for reducing anxiety ([1](#)). These guidelines recommend that adults accumulate at least $150 \text{ min} \cdot \text{wk}^{-1}$ of moderate-intensity PA or $75 \text{ min} \cdot \text{wk}^{-1}$ of vigorous-intensity PA or the equivalent combination thereof. They also recommend that adults engage in muscle-strengthening exercise for all major muscle groups at least two times per week. Beyond those general guidelines, systematic reviews of exercise effects on anxiety symptoms provide specific guidance to inform Ex Rx.

Depression

Exercise is effective for both reducing depressive symptoms in people with and without clinical depression and for reducing the odds of a clinical diagnosis in those who started with a clinical diagnosis of depression. The effects of aerobic exercise are more profound among individuals who are clinically depressed ([35](#), [36](#)). Antidepressant effects of exercise have been observed in trials with older adults with and without mild cognitive impairment (MCI) ([37](#)). In individuals with depression, aerobic exercise has proven to be as effective as psychotherapy

or pharmacotherapy for reducing depressive symptoms and appears to have an additive benefit when combined with standard treatments ([35](#), [36](#), [38](#)). Exercise is also more effective for reducing depression than bright light therapy and other controls ([36](#)). Any level of PA reduces risk of depression relative to inactivity ([39](#)).

Exercise Considerations

- Exercise can induce physiological changes similar to a panic attack (*e.g.*, increased HR, shortness of breath); therefore, individuals with known panic disorders should be advised to expect these symptoms as a normal result of exercise.
- Some exercise appears to be better than none for reducing anxiety and depression, although meeting recommended levels of PA has had the best results.
- For depressed individuals, it is important to find PAs that will be maintained, and they should include a mix of aerobic and resistance training activities.

FITT FITT EXERCISE PRESCRIPTION FOR ANXIETY

	Aerobic	Resistance	Flexibility
Frequency	3–5 d · wk ⁻¹ (30)	2–3 d · wk ⁻¹ (31)	3–5 d · wk ⁻¹ with daily being most effective (30)
Intensity	High intensity such as walking, jogging, or running (60%–90% of HR _{max} or 70% VO _{2max}) is more effective in reducing anxiety; individually tailored programs that progress from walking to running are recommended along with supervision (30)	50%–70% one repetition maximum (1-RM) or 5–8 on the resistance training-specific RPE scale for beginners; 60%–85% 1-RM or 6–10 on the resistance training-specific RPE scale for more advanced exercisers (31 , 32)	Full body, full extension, flexion, rotation, or stretch to the point of slight discomfort
Time	20–60 min; working up to longer durations of 60–90 min has greater effects than 1–30 and 30–60 min (33)	Beginners should do 2–3 sets of 8–20 repetitions and rest between sets 90–150 s; 3–4 sets of 6–15 repetitions with self-selected rest for more advanced exercisers (31)	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise for at least 10 min · d ⁻¹ (30)
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, running) (30)	Machines are best for beginners with more upper than lower body exercises; more experienced exercises can use machines or free weights; both should progress from small to large muscle groups (31)	Slow static stretches for all major muscle groups; yoga; Pilates (34)

HR, heart rate; RPE, rating of perceived exertion.

FITT FITT EXERCISE PRESCRIPTION FOR DEPRESSION

	Aerobic	Resistance	Flexibility
Frequency	Antidepressant effects are more likely from programs lasting 13+ days. The cumulative frequency of exercise matters more for individuals with depressive disorders than those without (35).	Insufficient evidence on resistance training frequency for optimizing antidepressant effects. Recommend following 2 or more days per week based on Physical Activity Guidelines for Americans.	Insufficient evidence to recommend frequency.
Intensity	PA at any intensity level appears to be effective for reducing depressive symptoms (35 , 36). More evidence has been collected on moderate-to-vigorous than light physical activity, but it appears that exercise at all intensities is beneficial for reducing depressive symptoms. Aerobic activity of any intensity is better than no activity.	Insufficient evidence on resistance training intensity for optimizing antidepressant effects. Recommend following standard ACSM guidance recommendations for safe muscle-strengthening exercise progression.	Insufficient evidence to recommend intensity.
Time	Exercise has acute or immediate effects on core affective states that can be useful for temporarily alleviating depressive symptoms after exercise (40 , 41). Bouts as brief as 20 min appear to be sufficient to reduce depressive symptoms in individuals without depressive disorders (35). For individuals with depressive disorders, 45 min is the recommended bout length (16 , 35).	Insufficient evidence on resistance training duration for optimizing antidepressant effects.	Insufficient evidence to recommend duration.
Type	Both aerobic and resistance training can be effective in reducing depressive symptoms, with some evidence suggesting that combination training may have added benefits (35 , 36 , 42 , 43).		Stretching, meditation, and relaxation techniques, are effective on symptoms of depression (36), but not to the same magnitude as resistance and aerobic training.

Special Considerations

- Suicidality involves ideation, plans, or actions intended to end one's life. Suicidality is a risk with depressive disorders, and the consequences can be life-threatening. Exercise interventions do not appear to alter suicidal ideation but have been linked with 77% lower risk for suicide attempts ([44](#)). If exercise professionals are working with an individual who is describing or enacting a plan to harm themselves, they should call 911 immediately to get emergency help.
- Exercise professionals should refer individuals they suspect of depressive disorders to licensed mental health professionals for assessment, diagnosis, and comprehensive treatment planning. Listening supportively and sharing your concern for their well-being can help individuals to accept referrals to mental health professionals. The 988 Suicide and Crisis Lifeline also provides 24/7 access to trained crisis workers for individuals in distress.
- For individuals with depressive disorders, reduced sensitivity to rewards can create additional challenges for adhering to a PA program.
- Individuals can benefit from supplementing exercise with support for self-regulation (*e.g.*, digital tools, training on evidence-based strategies) to manage these chronic disease states.

Future Directions

- Dose-response relations between exercise and mental health outcomes need to be characterized in greater detail using comparative effectiveness research designs. These studies can indicate the optimal prescription in terms of frequency, intensity, timing, and type of activity. Limited information exists on the effects of varying exercise intensities for individuals with depression and anxiety. Additional research is needed for the effects of resistance training with detailed information on total volume, as well as combined or mixed aerobic, resistance training, and flexibility exercises for anxiety and depression. Anxiety

and depression are often comorbid, and the effects of exercise on comorbid anxiety and depression are not well understood.

ONLINE RESOURCES

Anxiety

Anxiety: <https://www.nimh.nih.gov/health/topics/anxiety-disorders/index.shtml>

Anxiety Disorders: <https://www.psychiatry.org/patients-families/anxiety-disorders/what-are-anxiety-disorders>

Generalized Anxiety Disorder: <https://www.nimh.nih.gov/health/publications/generalized-anxiety-disorder-gad/index.shtml>

Panic Disorder: <https://www.nimh.nih.gov/health/publications/panic-disorder-when-fear-overwhelms/index.shtml>

Social Anxiety Disorder: <https://www.nimh.nih.gov/health/publications/social-anxiety-disorder-more-than-just-shyness/index.shtml>

Depression

Depression: <https://www.nimh.nih.gov/health/topics/depression/index.shtml>
<https://www.nimh.nih.gov/health/publications/depression-listing>

Overview and Diagnostic Criteria: <https://www.psychiatry.org/patients-families/depression/what-is-depression>

Depression in Teens: <https://www.nimh.nih.gov/health/publications/teen-depression/index.shtml>

Depression in Older Adults: <https://www.nia.nih.gov/health/mental-and-emotional-health/depression-and-older-adults>

Postpartum Depression: <https://www.nimh.nih.gov/health/publications/postpartum-depression-facts/index.shtml>

Suicide Prevention & Crisis Lifeline: <https://988lifeline.org/>

Suicide Prevention Information:

<https://www.nimh.nih.gov/health/topics/suicide-prevention>

ATTENTION- DEFICIT/HYPERACTIVITY DISORDER

Attention-deficit/hyperactivity disorder (ADHD) is a common neurodevelopmental disorder characterized by inattention, hyperactivity-impulsivity, or both ([45](#)). The prevalence of ADHD worldwide is approximately 5% in children-adolescents and an average of around 2.5%–3.4% for adults ([46](#)). However, estimates in children and adolescents vary across sex, being markedly more prevalent in boys than in girls with a ratio of 2–3:1 ([47](#), [48](#)). Existing data support that ADHD prevalence has not increased over the last three decades ([47](#)). Despite the popular belief that ADHD is mainly a pediatric disorder, meta-analyses of follow-up studies demonstrate around 65% of children with ADHD will continue managing this condition into their adulthood years ([49](#)). Problems related to ADHD include psychiatric comorbidities (*e.g.*, major depression, anxiety, bipolar disorder), health problems, psychological dysfunction, academic and occupational failure, social disability, and risky behaviors (*e.g.*, lying, stealing, and substance abuse) ([45](#)).

ADHD is a complex disorder, and its etiology is not yet completely understood. Although there is evidence that environmental factors play an important role, ADHD has a strong genetic component, with heritability estimates averaging approximately 75% ([49–51](#)). Both pharmacological and nonpharmacological treatments are used to treat ADHD. Although nonstimulants (*e.g.*, selective noradrenaline reuptake inhibitor atomoxetine and long-acting formulations of two α 2-adrenergic agonist drugs clonidine and guanfacine) are sometimes used based on contraindications or personal preferences, stimulants (such as amphetamine or methylphenidate) are mostly used to treat ADHD in individuals of all ages ([45](#)). Additionally, nonpharmacological treatments, such as dietary, neurocognitive, and behavioral therapies, are also used as an alternative or complement to pharmacological treatments.

According to the WHO 2020 guidelines, regular PA improves multiple dimensions of cognition, including two of which are of utmost importance for individuals with ADHD: attention and inhibition ([52](#)). Inattentiveness is one of the core symptoms of ADHD, and the most updated evidence from WHO guidelines strongly supports the use of PA to improve attention ([53](#)). Impulsivity is another core symptom of ADHD. Existing evidence supports a link between engagement in PA and improvements in cognitive inhibition ([54](#), [55](#)). Cognitive inhibition is a major component of executive function dealing with the ability of people to inhibit responses in order to better respond to a specific stimulus. In this context, exercise has been shown to improve inhibition in the general population ([56](#)) and children with ADHD ([54](#)). Other cognitive functions, such as a better ability to plan and organize daily life activities, considered to be part of executive function, are also positively related to exercise in the general population and can provide additional benefits to those with ADHD ([1](#), [57](#), [58](#)). Moreover, sleep duration and quality are often impaired in those with ADHD ([59](#)); see reference for a review ([57](#)). The WHO 2020 guidelines state that physically active people have better sleep quality in terms of the time in bed to sleep onset, the number and duration of times that a person wakes up at night after having fallen asleep, and sleep efficiency, among others ([52](#)). This would, therefore, be another mechanism by which exercise can improve ADHD symptomatology.

Major comorbidities in ADHD include obesity, hypertension (see [Chapter 9](#)), and depression/anxiety (as discussed in this chapter) ([45](#), [46](#), [60](#), [61](#)), and exercise can play a key role in mitigating each of these conditions ([1](#)). In most cases, individuals with ADHD can start a moderate-intensity exercise program without previous medical screening, considering exercise testing for clinical purposes is not necessary unless there is a health concern that warrants exercise testing ([62–64](#)). Given the higher prevalence of ADHD in childhood/adolescence than in adulthood, the considerations about exercise testing for individuals with ADHD will be mainly those for children and adolescents.

Exercise Testing

Exercise testing both in pediatric and adult populations is always informative as a health indicator and for monitoring improvements in fitness as a consequence of exercise ([65](#)). When doing so, the recommendations for exercise testing in the general population (see [Chapter 3](#)) will apply to ADHD ([66](#)). In children and adolescents, the most updated and evidence-based fitness test battery is the European Union–funded ALPHA battery ([67–71](#)), also supported by the Institute of Medicine in the United States ([72, 73](#)). The tests selected for being the most valid, reliable, and related to future health are (a) the 20-m shuttle run test to assess cardiorespiratory fitness (CRF); (b) the handgrip strength and (c) standing broad jump to assess musculoskeletal fitness; and (d) body mass index (BMI), (e) skinfold thickness, and waist circumference to assess body composition ([68](#)). International reference values for correct sex- and age-specific interpretation of physical fitness assessment are available elsewhere ([74–77](#)). If ADHD is present with comorbidities, exercise professionals should review relevant exercise testing options as listed elsewhere in the *ACSM Guidelines*.

Exercise Prescription

Because ADHD is typically diagnosed early in life, the Ex R_x principles for healthy children and adolescents apply to ADHD (see [Chapter 6](#)) ([78, 79](#)). Given that ADHD continues into adulthood for nearly two-thirds of children, adult Ex R_x principles also apply (see [Chapter 5](#)).

Exercise Considerations

- Attention should be paid to potentially coexisting comorbidities, such as overweight/obesity, hypertension, and depression/anxiety. Professionals are recommended to screen for subjacent conditions before exercise programming, and if comorbidity coexists special considerations outlined in other chapters should be consulted (see [Chapter 9](#)).

- Emerging evidence suggests that low physical fitness is common in ADHD ([62–64, 80](#)). Care should be taken to start slow and to set realistic goals for fitness in this population.
- Developmental coordination disorder is frequently present in ADHD individuals ([81–83](#)). Therefore, complex exercises that require specific motor skills with high neuromuscular control (such as dancing) can be incorporated but should be done in a more progressive way ([84, 85](#)).
- To increase exercise adherence in those with ADHD, it is important to choose fun and stimulating activities that provide motivation and positive feedback and, whenever possible, to train in small groups. These can all contribute to enhancing social skills and improving compliance.
- High-intensity interval training (HIIT) may be a good choice for those with ADHD once a moderate fitness level is achieved. The shorter duration and higher intensity tend to require greater levels of concentration and focus, which may be beneficial ([86](#)).
- Either open-skill (such as football, tennis, badminton) or closed-skill activities (such as running, cycling, swimming) might be advantageous for ADHD symptomatology improvement in youth ([87](#)).

Special Considerations

- For most individuals with ADHD, PA, and/or exercise will be complementary to their pharmacological treatment. It has been demonstrated that exercise can enhance the effects of stimulants (*i.e.*, methylphenidate) on clinical symptoms, cognitive function, and brain activity of individuals with ADHD ([88–91](#)). Therefore, those with ADHD are advised to discuss with their primary care provider information about medication doses and how they may interact with an exercise program. Dosage adjustments may be necessary.
- The use of behavior change techniques (see [Chapter 12](#)) to target and create short- and long-term behavior goals might also be appropriate in this population ([92, 93](#)).
- Features to be considered when prescribing exercise for individuals with ADHD:

- Exercise sessions that are structured, previously planned, and part of a routine, done with a sports specialist or in a group setting.
- Program variety, including functional movement patterns.
- A defined, specific, measurable goal for the day and short term.

Future Directions

Overall, there is a need for more rigorous and well-designed randomized controlled trials testing the effects of exercise for children, adolescents, and adults with ADHD. To date, most studies on exercise and ADHD are conducted in youth and most exercise interventions in ADHD focused exclusively on aerobic exercise. Given the multiple benefits of resistance training for health in the general population, future studies should include strength training among individuals with ADHD to explore its effects on overall health and specific ADHD symptomatology. Additionally, future studies should investigate how exercise interacts with ADHD medication (*e.g.*, whether incorporating PA into the daily life of a medicated person with ADHD could contribute to reducing the drug dose).

ONLINE RESOURCES

Attention Deficit Disorder Association (ADDA): <https://add.org/>

ADHD Europe: <https://www.adhdeurope.eu/>

ADHD in Adults: <http://www.adhdinadults.com>

American Academy of Pediatrics ADHD Toolkit:

<https://publications.aap.org/toolkits/pages/ADHD-Toolkit?autologincheck=redirected>

American Professional Society for ADHD and Related Disorders:

<http://www.apsard.org>

Australian NHMRC: <https://www.nhmrc.gov.au/>

Children and Adults With ADHD: <http://www.chadd.org>

European Network on Adult ADHD: <http://www.eunetworkadultadhd.com/>

International Collaboration on ADHD and Substance Abuse:

<http://www.adhdandsubstanceabuse.org>

Mayo Clinic Diseases and Conditions. Attention-deficit/hyperactivity disorder:

<https://www.mayoclinic.org/diseases-conditions/adult-adhd/symptoms-causes/syc-20350878>

National Institute of Mental Health:

<https://www.nimh.nih.gov/health/topics/attention-deficit-hyperactivity-disorder-adhd/index.shtml>

The Canadian ADHD Resource Alliance: <http://www.caddra.ca>

UK Adult ADHD Network: <http://www.ukaan.org>

World Federation of ADHD: <http://www.adhd-federation.org>

AUTISM SPECTRUM DISORDER

Autism spectrum disorder (ASD) is a complex neurological and developmental condition with estimated prevalence rates of 1 case per 36 children in the United States ([94](#)). Diagnosis is based on observation of behaviors, signifying persistent differences in social communication and social interaction, and restricted, repetitive patterns of behavior, interests, or activities ([95](#)). Moreover, co-occurring conditions are also prevalent, including physical health conditions (*e.g.*, seizures, gastrointestinal disorders, metabolic conditions, sleep problems, autoimmune disorders) ([96](#)), psychiatric disorders (*e.g.*, ADHD, anxiety disorders, depressive disorders, bipolar and mood disorders) ([97](#)) and motor deficits ([98](#), [99](#)). ASD is a highly heterogeneous condition with multiple developmental trajectories. Etiology is not completely understood; however, genetic and nongenetic factors are known to be at play. The genetic contribution to ASD is supported by family and twin studies that report heritability estimates from 50% to 95% ([100](#)). The study of nongenetic contributory factors has identified multiple environmental risk factors. Among these, the most convincing evidence points to an association between maternal factors, including advanced age, chronic hypertension, preeclampsia, gestational hypertension, and overweight before or during pregnancy ([101](#)). It should be noted that preferences regarding identifying language vary by individual, with some preferring the terms “person on the autism spectrum” and “autistic person” while others opt for the term “person with autism” ([102](#), [103](#)). Accordingly, practitioners are encouraged to consult with the individual to understand which language they prefer.

Specific supports and services to address individualized needs with daily functioning and quality of life include educational practices, developmental therapies, and behavioral interventions ([104](#)). Specific supports and services include applied behavioral therapy, speech therapy, occupational therapy, and physical therapy. An array of other supports exist that incorporate evidence-based practices, as identified by the National Clearinghouse on Autism Evidence and Practice. These evidence-based practices include antecedent-based

interventions, prompting, modeling, social narratives, and visual supports ([105](#)). Pharmacological treatment options, such as psychotropic medications, are increasingly used by people with ASD ([104](#)). Significant adverse effects of medical treatments for individuals on the autism spectrum are often reported, including weight gain, fatigue, sedation, and somnolence — effects often associated with a discontinuation of treatment.

Exercise and movement is included as one of the 28 evidence-based practices identified for individuals with ASD, cited as positively impacting outcomes related to social and communication skills, joint attention, cognition, challenging/interfering behavior, and motor skills ([105](#)). Meta-analyses confirm these effects to be moderate to large ([106–108](#)). It is evident that exercise addresses ASD traits and cooccurring conditions that interfere with daily functioning and provides health-enhancing benefits. However, people with ASD tend to be less active and less likely to meet the PA guidelines compared to people without ASD ([109](#), [110](#)). This is unsurprising when we consider that the demands of PA, typically a physical, social, sensory, and dynamic experience, may directly conflict with the characteristics of ASD. Therefore, special attention is warranted for exercise testing and prescription among this population.

Exercise Testing

Aside from possible risks associated with PA and exercise that exist for the general population (see [Chapter 1](#)), exercise testing is safe for populations with ASD. Therefore, preparticipation health screening for exercise and exercise testing should follow the general recommendations (see [Chapter 2](#)). However, one should ensure that the testing procedures and environments are understood, motivating, and comfortable. There exists tremendous heterogeneity within this population; for example, individuals vary within cognitive, social, behavioral, sensory, and motor domains. This heterogeneity means that providing recommendations for exercise testing is not a “one size fits all” task. The exercise test administrator should, however, consider how the evidence-based practices identified for individuals with ASD, including visual support, social narratives, task analysis, prompting, and reinforcement, may be used to assist with testing

procedures ([*Table 11.1*](#)). Evidence-based practices are frequently used in combination with one another.

TABLE 11.1 Exercise Testing Recommendations for Individuals With Autism Spectrum Disorder (ASD)

Recommendation	Elaboration/Example
1. Understand the individual's strengths, abilities, and preferences.	• Does the individual... ▪ Experience specific challenges or support needs the exercise professional should be aware of (<i>e.g.</i>, challenges with sensory processing)? ▪ Require specific accommodations or individualized supports (<i>e.g.</i>, communication supports)? ▪ Take medications with side effects such as fatigue or sedation?
2. Choose an appropriate test depending on the individual's attributes.	• A cycle ergometer may be preferred over a treadmill for an individual with poor balance or coordination. • Tests that are familiar to the individual may be preferred.
3. Provide routine and predictability.	• Many individuals with ASD do well with repetition and perform best with daily routines in place. • Be consistent with routines, equipment, and organization. • Visual schedules may help to reinforce routine by providing a graphic representation of scheduled tasks and activities.
4. Prepare the individual for exercise testing.	• Let the individual know what they can expect or how to prepare for exercise testing.

Recommendation	Elaboration/Example
	Consider using a social narrative (111). Social narratives describe new social situations for individuals with ASD by providing relevant cues and descriptions of appropriate behavior expectations.
5. Use task analysis (112), the process in which an activity or behavior is divided into small, manageable steps in order to assess and teach the skill.	- Task analysis of an exercise test, and the subsequent teaching of its “parts,” may be necessary before testing an autistic individual who is not familiar with the activity and/or may find the activity challenging. - Gradually expose the individual to the testing procedures and environment.
6. Use prompting (113) to assist the individual in the testing procedures.	- Prompting procedures include any help given to learners that assist them in using a specific skill. - Prompting procedures may include verbal (keep concise and concrete), gestural, and physical. - Prompts are typically provided in conjunction with other evidence-based practices. A variety of prompting procedures exist, including using least-to-most prompts, simultaneous prompting, and graduated guidance. See this reference for an overview of these procedures (113).
7. Use modeling (114) and/or video modeling	Modeling involves the demonstration of an activity or skill by the instructor to initiate

Recommendation	Elaboration/Example
(115) to provide a demonstration of the testing procedures.	imitation of the behavior/skill by the learner/exerciser. • Video modeling involves using video recordings and display equipment to provide a visual model of the behavior or skill (115) to the learner/exerciser. • Types of video modeling include basic video modeling, video self-modeling, point-of-view video modeling, and video prompting (115).
8. Use reinforcement to increase the probability that the individual will complete the movement (116, 117).	• Provide positive reinforcement to promote a positive experience and increase likelihood of continuing the exercise program. • If working with a minor, ask the parent/guardian or the autistic individual about their preferred reinforcers.

Exercise Prescription

It is recommended that people with ASD are prescribed exercise to gradually progress to a frequency, intensity, and time of that recommended for people without ASD. As various types of PA have shown to be beneficial (106–108), the exercise type prescribed should depend on factors such as the individual's interests and the goal of the exercise program (*e.g.*, a goal of social skill development vs. improved fitness). Activities of type A (*e.g.*, walking, slow cycling) or B (*e.g.*, jogging, running, rowing, elliptical exercise) may be preferable for autistic individuals with motor deficits. Furthermore, due to the social differences associated with ASD, some individuals may prefer to focus on “type A, B, or C (*e.g.*, swimming, skiing)” individual activities. See Activity Types, [Table 5.4](#).

However, type D activities (*e.g.*, soccer, basketball, racquet sports) should not be neglected to ensure that the autistic individual can choose to participate in a range of activities, including activities that offer opportunities for socialization. Importantly, participation in socially demanding physical activities can be anxiety inducing for some participants. Individualized support should therefore be provided when organizing group activities, such as the provision of regular breaks.

Making specific FITT recommendations for exercise interventions is hampered by the large variability that exists within the interventions studied among autistic individuals ([106](#)). For example, a host of interventions have been demonstrated to positively affect social skills among children on the autism spectrum, using exercise types ranging from yoga to multisport camps, at varying doses ranging from 6.5 to 150 h ([118](#)). Similarly, the muscular strength and endurance of children with ASD has been shown to be positively impacted by a host of exercise types ranging from exergaming to equine therapy, at varying doses ranging from 20 to 60 h ([106](#)). A recent meta-analysis does, however, lend support to increased effectiveness of PA programs of longer duration (≥ 2 weeks), longer PA sessions (≥ 90 minutes), and more frequent sessions (≥ 3 per week) for interventions focused on autistic characteristics and traits. Notable, some outcomes, such as self-regulation, have been shown to be positively impacted by short exercise bouts. For example, 10 minutes of low- to moderate-intensity exercise was shown to enhance self-regulation in those with ASD ([119](#)). Although the evidence does not lend itself to making specific exercise programming decisions for this population, it does strongly support the benefit of exercise of varying types and doses for individuals with ASD. Therefore, for inactive autistic people, starting small (*e.g.*, regular 10-minute bouts of light to moderate intensity) in a type of activity enjoyed by the individual, and gradually progressing the frequency, intensity, and time, is recommended.

Exercise Training Considerations

[Table 11.2](#) highlights considerations regarding the exercise environment and exercise programming for individuals with ASD. It should be noted that

approximately 40% of individuals with ASD have an intellectual disability (ID) ([94](#)); therefore, readers may also find the information in this chapter related to those *with intellectual disability* useful.

TABLE 11.2 Exercise Training Recommendations for Individuals With Autism Spectrum Disorder (ASD)

Exercise Training Components	Special Considerations
The exercise environment	• Hypersensitivity issue. Learn about and prepare for sensory processing issues that the individual may have. For example, consider the noise and light in the gym, free weight area, group exercise studio, etc. and modify the environment if necessary. • Predictability. Organize the space in a predictable manner for each training/testing session. • People. Consider the social environment: The individual with ASD may prefer individual, parallel, or small group activities. • Anxiety. New environments may cause anxiety in some individuals with ASD; slowly transition to new environments and give ample time for the individual to become accustomed to the new space.
Exercise programming	• Desire for “sameness.” Maintain a predictable structure in the exercise program and communicate it via a visual schedule. • Motor deficits. Adapt equipment and activities to cater for possible motor deficits (<i>e.g.</i>, coordination, balance, muscular strength). • Fatigue. Offer breaks throughout the exercise period. Be aware that the individual may require breaks from sensory input and social interaction, in addition to physical fatigue. • Apply evidence-based practices (individually or in combination with other practices) for the promotion of exercise behaviors and skills: See Table 11.1 for a summary of evidence-based practices such as task analysis,

Exercise Training Components	Special Considerations
	modeling, video modeling, prompting, and reinforcements. See Wong et al. (116) for a comprehensive overview of evidence-based practices for individuals with ASD.

Future Directions

To understand how to conduct exercise testing and prescription most effectively for individuals with ASD, research is needed with larger, more homogeneous samples using rigorous research designs to determine what, if any, modifications are needed to the FITT principles for this population. The FITT principles that correspond to particular outcomes of relevance to individuals with ASD (*e.g.*, social skills, repetitive behaviors) also require investigation. Moreover, two populations vastly understudied include younger children (<4 years) and adults on the spectrum. Due to the importance of early intervention for this population ([120](#)), how exercise can be used to benefit young children, including being incorporated into other interventions, is an avenue ripe for research. A focus on adults is also warranted; despite the increasing number of adults with ASD and the greater health disparities they experience ([121](#)), exercise interventions for this population are sparse. Finally, research exploring the impact of muscle-strengthening exercises on individuals with ASD is notably limited and urgently requires further investigation.

ONLINE RESOURCES

ACSM/Exercise Connection Autism Exercise Specialist Course:

<https://members.acsm.org/ItemDetail?iProductCode=AUTISM-CE-2018&Category=SPECIALTY&WebsiteKey=e6756b65-2c71-47cb-b31b-cb7858e166e3>

Autism Society: <http://www.autism-society.org>

Autism Speaks: <https://www.autismspeaks.org>

Evidence-Based Practices: <https://autismpdc.fpg.unc.edu/evidence-based-practices>

Exercise Connection: <https://www.exerciseconnection.com>

National Autism Association: <http://nationalautismassociation.org>

The National Professional Development Center on Autism Spectrum Disorder:
<https://autismpdc.fpg.unc.edu/national-professional-development-center-autism-spectrum-disorder>

INTELLECTUAL DISABILITY AND DOWN SYNDROME

ID or intellectual disability is defined in the Diagnostic and Statistical Manual of Mental Disorders, 5th Edition (DSM-5) as neurodevelopmental disorders that begin in childhood and are characterized by intellectual difficulties as well as difficulties in conceptual, social, and practical areas of living ([122](#)). ID has an estimated global prevalence of 1% ([123](#)). ID includes a wide range of intellectual and adaptive functioning, with individuals divided into mild (can live independently with minimum levels of support), moderate (independent living may be achieved with moderate levels of support), severe (individual requires daily assistance with self-care activities and safety supervision), or profound (individual requires 24-hour care) ([122](#)). ID encompasses a broad number of conditions and is thought to be driven by genetic factors (*i.e.*, phenylketonuria, Down syndrome [DS], Lesch-Nyhan syndrome, Fragile X syndrome, Rett syndrome) and/or environmental exposures (*i.e.*, exposure to toxins such as alcohol in utero, maternal rubella infection in the initial trimester of pregnancy, maternal HIV); in some cases, ID can be acquired during childhood, such as encephalitis or meningitis infection, asphyxia, cranial tumor, or head trauma ([124](#)).

Life expectancy for individuals with ID can be up to 12.8 years less than that of the general population ([125](#)). Heart disease was the leading cause of death in individuals with ID in the United States without any other comorbid developmental disability in one population level study of 22,512 individuals with ID. Additionally, individuals with ID were found to be at an increased risk of mortality from pneumonitis, influenza/pneumonia, and choking across all levels of disability severity. Those with mild or moderate ID were also at an increased risk for mortality due to diabetes and malignant neoplasms ([125](#)). Individuals with DS are at an increased risk of mortality at a young age due to congenital heart anomalies and increased risk of mortality across the lifespan from

influenza, pneumonia, respiratory failure, pneumonitis, and choking compared to individuals without DS ([126](#)). However, the life expectancy rates for individuals with ID and DS has increased in recent decades, leading to an increased number of individuals with ID at risk for conditions associated with unhealthy aging ([127](#)). Individuals with ID or DS are at a heightened risk for adverse health conditions, such as obesity ([128](#)), metabolic syndrome ([129](#), [130](#)), epilepsy ([131](#)), cardiac atrophy ([132](#)), and visual impairment ([131](#)), which are accompanied by increased health care cost and health disparity ([133](#)). Individuals with ID have been found to have suboptimal PA levels ([134](#)) and lower fitness ([134](#)), which exacerbates existing adverse health risks. Given that physical fitness is independently associated with 5-year survival in older adults with intellectual disabilities, performing PA to avoid functional decline and unhealthy aging is imperative ([135](#)).

ID, as well as its related comorbidities, provides unique challenges for implementing exercise programs. However, the rewards of exercise programming for people with ID and DS are numerous. Acute exercise has demonstrated positive effects on mood and self-confidence in this population ([136](#)). Prolonged exercise programming has demonstrated increases in self-efficacy ([137](#)) and life satisfaction ([138](#)) and reductions in anxiety ([139](#)) and depression ([140](#)) in individuals with ID and DS. The effects of exercise in this population could even extend further into caregiver and family networks by decreasing caregiver burden as a consequence of increasing physical fitness and independence ([141](#)).

Exercise Testing

A thorough medical history should be obtained prior to engagement in exercise testing, as is standard with healthy individuals. Special attention should be given to cardiovascular risk factors within the medical history despite age, as individuals with ID can face increased risk of early onset cardiovascular disease (CVD) ([142](#)) and congenital heart disease occurs in 40%–50% of individuals with DS ([143](#)). CVD risk factor assessment is recommended for use in healthy adults (see section in [Chapter 2](#) on “Medical History and Cardiovascular Disease Risk Factor Assessment”) and can be employed with caregivers of individuals with ID

or DS to identify participants at heightened cardiovascular risk. Documented medical clearance may be necessary prior to exercise participation. While this could present as a barrier to PA, working with the medical community can offer a link between exercise efforts and medical care.

Familiarity of any individual with testing protocol should be considered prior to testing, as test-retest reliability studies of walking and running protocols show that there is an improvement in performance on the second trial among individuals with ID ([144–146](#)). Each testing protocol should have its own familiarization session ([147](#)). Selecting a strength testing protocol for individuals with DS should be carefully considered to avoid exercise-induced injury (*e.g.*, assuring the exercise equipment fits the participant's body appropriately) considering the potential of limb length differences among individuals with DS ([148](#)) ([Table 11.3](#)).

TABLE 11.3 Recommended Exercise Testing Protocols for Intellectual Disability ([144](#), [146](#), [149–165](#))

Type of Exercise	Recommended Test Protocol(s)
Cardiorespiratory	• 6-min walk test (6-MWT) (with familiarization) • Shuttle run test • Rockport One-Mile Fitness Walking Test • Dual-action ergometer testing • Maximal treadmill protocol
Muscular strength/endurance	• 1-RM (if appropriate) • 6-RM or 12-RM using machines (if appropriate) • Functional Strength Measurement in children with intellectual disability • Isometric maximal voluntary contraction • Isokinetic testing • Grip strength testing (older adults) • 30-s chair stand (older adults) • 5 times chair stand (older adults)
Flexibility	• Goniometry at specific joints
Balance and motor skill	• Bruininks-Oseretsky Test of Motor Proficiency-Second Edition • Modified Berg Balance Scale • Test of Gross Motor Development for youth (all variations) • Comfortable gait speed (older adults) • Static balance protocols
Anthropometrics and body composition	• Body mass index • Waist circumference

Type of Exercise	Recommended Test Protocol(s)
	• Skinfold measurements (use of Slaughter equation or equation from Bertapelli et al. (162)) • Air displacement plethysmography • Dual-energy x-ray absorptiometry

1-RM, one repetition maximum; 6-RM, six repetition maximum; 12-RM, twelve repetition maximum.

Evidence in the area of perceived exertion has indicated that children with ID and DS are able to accurately assess perceived exertion when compared to HR on the Borg scale and a modified version of the Children's OMNI 1–10 walk/run scale ([166](#)); however, further studies are needed to validate these scales and alternatives to these scales in this population ([167](#)). When assessing perceived exertion during exercise in this population, it is important to be cognizant of any distraction that the questioning could cause from the testing; obtaining rating of perceived exertion (RPE) should never come at the cost of participant safety during testing. Practitioner's subjective evaluation of participant effort or relative biometrics such as HR may be able to replace RPE in monitoring exercise effort. Determining test endpoints based on HR may be difficult, as general population estimates could overestimate peak HR in this population; peak HR in individuals with ID/DS are typically 30 bpm below expected ([168](#)). Peak HR calculation for ID and DS could be estimated using the formula $[HR_{peak} = 210 - 0.56(\text{age}) - 15.5(\text{DS})]$, where DS = 2 and all other IDs = 1 ([168](#)).

Both traditional progressive exercise testing and various forms of field testing are feasible for individuals with ID or DS. Laboratory testing for aerobic fitness has been shown to be valid and reliable for children and adolescents ([149](#)) and adults ([150](#)) with ID and DS, including the use of maximal aerobic treadmill testing. Traditionally used alternatives to treadmill testing include cycle ergometry and dual-action cycle ergometry (*e.g.*, airdyne) testing. However, there is not a validated traditional cycle protocol for children or adults in this population, although these protocols have been utilized in studies of adults with DS ([169](#)). Dual-action ergometer testing has been validated for use in adults in

this population ([151](#)). Development and validation of cycle-based protocols for use in children and adults with ID or DS is needed in order to provide alternatives to traditional treadmill testing. Discontinuous maximal treadmill protocols have been utilized in adults with ID and demonstrate efficacious elicitation of plateau in $\dot{V}O_{2peak}$ ([149](#)). Discontinuous tests might also be useful for increasing adherence to aerobic testing in this population ([149](#), [170](#)), as they could provide relief from fatigue and adjustment of facial equipment. Further testing and validation of this method and other adaptations to discontinuous protocols are needed ([171](#)).

Field tests can be useful for those without laboratory testing equipment or for participants who are unsteady or unfamiliar with treadmill testing; however, there are several issues to keep in mind with the use of field tests to evaluate CRF. The Rockport One-Mile Fitness Walking Test could be better suited for children or adolescents with mild-to-moderate ID ([152](#)). Additionally, population-specific formulas have been developed predicting $\dot{V}O_{2max}$ using the Rockport One-Mile Fitness Walking Test for individuals with ID; however, these formulas have both overestimated and underestimated performance in ID ([172](#), [173](#)). Thus, estimates should be used with caution. The 20-m shuttle run test has also been reported as a reliable tool to estimate CRF in adolescents and children ([153](#)) with mild-to-moderate ID. The 16- and 10-m adaptations of this test have also been shown to be reliable in children with ID ([153](#)) and adults with severe ID. However, limitations exist with regard to accuracy ([174](#)). Historically, the 6-MWT has shown inconsistent results with relation to reliability, reproducibility, and validity in adults and adolescents with ID and DS ([144](#), [154](#), [155](#)); however, studies that included familiarization sessions with the 6-MWT reported more favorable results ([144](#), [154](#)). Those interested in utilizing the 6-MWT as a field test of CRF in this population should consider familiarization.

The Functional Strength Measurement in children with ID (FSM-ID) is an option for strength testing. While shown to be a reliable and valid tool, familiarization sessions or practice sessions of each task within the FSM-ID might be required in this population group ([156](#)). In adults, use of a 6, 10, or 12-RM for strength testing could be considered for individuals with little strength training

experience. While a one repetition maximum (1-RM) is typically used, the 6–12 RM may provide inexperienced individuals with less risk of injury and less soreness ([157](#), [175](#)), although this type of protocol has not been validated in ID or DS. Further studies are needed to evaluate the 6–12 RM method in this population. Weight machines or closely supervised lifts with free weights are recommended for all strength protocols to minimize risk of injury that may arise with joint laxity in individuals with DS. Population-level data collected in older adults with ID (aged 50–64 years) indicated the use of grip strength to have acceptable test-retest reliability at both same day (ICC = 0.94) and 2-week intervals (ICC = 0.90). Additionally, the same trial found a 30-second sit-to-stand and alternatively a 5 repetition sit-to-stand to have acceptable test-retest reliability at same day (ICC = 0.72) and 2-week (ICC = 0.65) intervals ([158](#)). There is evidence that individuals with ID have reduced muscle activation ([176](#)) and sarcopenia ([177](#)), which may result in resistance and load that appears to be less than expected as compared to similarly fit nondisabled individuals.

Individuals with ID are at an increased risk for fall and injury due to difficulty integrating sensory information along with motor issues ([178](#)). Therefore, addition of balance, posture, and flexibility testing to an exercise testing battery could be particularly useful in this population group. Flexibility testing could be particularly useful due to its importance in preventing injury and maintaining range of motion (ROM) to allow for activities of daily living (ADL) in healthy adults, although this has not been specifically shown in ID/DS ([179](#)). Extra care should be taken to ensure that participants are comfortable and are able to perform any balance test without increasing fall risk ([180](#)). Level of ID could impact overall performance in exercise testing, and particularly in tests that require higher cognitive or perceptual function, such as balance assessments ([159](#)). Increased age can also impact testing performance because of decreased muscular fitness associated with age ([159](#)). During testing, careful observation by the test administrator is imperative.

Valid and reliable assessment of body composition and anthropometrics is important in ID to track obesity outcomes. Air displacement plethysmography and dual-energy x-ray absorptiometry (DXA) ([160](#), [181](#)) are recommended

methods for use when available; however, these methods can present challenges with this population due to the need for complete stillness for periods of time and the breathing pattern required to complete plethysmography. Skinfold measurements could present accuracy issues because of participant noncompliance and tester error ([161](#)), which could present issues for tracking changes over time; however, a skinfold body fat estimation calculation for children with ID has been developed and validated utilizing DXA ($\%BF = -15.416 + (1.394 \times BMI) + (4.538 \times \text{age}) - (0.262 \times \text{age}^2) + (5.489 \times \text{sex})$, where male = 0 and female = 1) ([162](#)). The skinfold measurement prediction equation for individuals with DS provided by Slaughter et al. ([182](#)) has been shown to agree with air displacement plethysmography results ([Table 11.4](#)). BMI and waist circumference have been shown to be reliable measures in ID and DS populations. While these methods might not demonstrate the same amount of accuracy as DXA or plethysmography, they may provide less stressful alternatives for those with ID.

TABLE 11.4 Skinfold Equations in Youth and Adolescents ([182](#))

Males	$\text{PFDWB} = 0.735 (\text{triceps} + \text{calf}) + 1.0$
Females	$\text{PFDWB} = 0.610 (\text{triceps} + \text{calf}) + 5.1$
Prepubescent White males	$\text{PFDWB} = 1.21 (\text{triceps} + \text{subscapular}) - 0.008 (\text{triceps} + \text{subscapular})^2 - 1.7$
Prepubescent Black males	$\text{PFDWB} = 1.21 (\text{triceps} + \text{subscapular}) - 0.008 (\text{triceps} + \text{subscapular})^2 - 3.2$
Pubescent White males	$\text{PFDWB} = 1.21 (\text{triceps} + \text{subscapular}) - 0.008 (\text{triceps} + \text{subscapular})^2 - 3.4$
Pubescent Black males	$\text{PFDWB} = 1.21 (\text{triceps} + \text{subscapular}) - 0.008 (\text{triceps} + \text{subscapular})^2 - 5.2$
Postpubescent White males	$\text{PFDWB} = 1.21 (\text{triceps} + \text{subscapular}) - 0.008 (\text{triceps} + \text{subscapular})^2 - 5.5$
Postpubescent Black males	$\text{PFDWB} = 1.21 (\text{triceps} + \text{subscapular}) - 0.008 (\text{triceps} + \text{subscapular})^2 - 6.8$
All females	$\text{PFDWB} = 1.33 (\text{triceps} + \text{subscapular}) - 0.013 (\text{triceps} + \text{subscapular})^2 - 2.5$
For a sum of triceps and subscapular >35 mm, the following equation should be applied:	All males $\text{PFDWB} = 0.783 (\text{triceps} + \text{subscapular}) + 1.6$ All females $\text{PFDWB} = 0.546 (\text{triceps} + \text{subscapular})$

PFDWB, percent fat body density, water, and bone mineral.

Exercise Prescription

In general, the Ex R_x for an individual with ID is uncomplicated, and recommending daily PA as encouragement toward healthy behavior and behavior

change could be beneficial to efforts for weight loss and reduction of comorbidities for individuals with IDs ([183](#)). Comorbidities, such as hypertension, diabetes, or joint pain, can affect exercise behavior and practitioners should be cognizant of these factors. Energy expenditure can vary between individuals with ID with more severe versus less severe physical disability, as well as between sex ([184](#)). Slower paced walking ($3.0 \text{ km} \cdot \text{h}^{-1}$) has been shown to mimic energy expenditure of moderate paced walking in individuals with IDs, potentially because of differences in biomechanical efficiency ([184](#)).

FITT FITT EXERCISE PRESCRIPTION FOR INTELLECTUAL DISABILITY ([185](#), [186](#))

	Aerobic	Resistance	Flexibility
Frequency	Most days of the week ($\geq 3 \text{ d} \cdot \text{wk}^{-1}$)	Preferred most days of the week but at least $2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$	Preferred most days of the week but at least $2\text{--}3 \text{ d} \cdot \text{wk}^{-1}$
Intensity	Moderate-to-vigorous: $40\%\text{--}80\% \dot{V}O_{2\text{peak}}$ progressive pattern	Begin with 6–12 reps of $60\%\text{--}70\%$ of one repetition maximum (1-RM); progress to 10–12 reps of $70\%\text{--}80\%$ 1-RM	To the point of slight discomfort
Time	$30\text{--}60 \text{ min} \cdot \text{d}^{-1}$, intermittent bouts of 10–15 min alternative	2–3 sets	Static stretches 10–30 s; repeat two to four times per exercise
Type	Walking-based activities, swimming, ergometry (arm and leg), dance	Machine/cable-controlled protocols preferred	Static stretching

Studies evaluating the CRF of individuals with ID have shown significant deficits in CRF, notably lower HR_{peak} and impairments in ventilatory function as compared to individuals without ID ([163](#), [187](#)). Individuals with DS also have deficits, greater than those with ID ([163](#)). One explored mechanism for reduction in CRF in this population is autonomic dysfunction, specifically chronotropic incompetence (CI), or the inability to increase HR ([168](#)). In DS, autonomic dysfunction is demonstrated in both sympathetic (low catecholamine response to maximal exercise) and parasympathetic (low basal vagal function when compared to controls) dysfunction ([188](#)). This dysfunction results in reduced HR response to exercise ([188](#), [189](#)), typically $25\text{--}30 \text{ beats} \cdot \text{min}^{-1}$ below age-matched maximum HR in those without ID ([188](#)). Evidence of CI has also been found in individuals with ID with submaximal exercise ([190](#)).

Integrative and innovative Ex R_x that feature novel modalities have been tested and demonstrated to be effective for individuals with ID or DS. Combined exercise regimens that use aerobic, balance, and strength training have been shown to be effective for improving CRF ([191](#)), strength, body weight ([192](#)), and flexibility ([192](#)) in adults with ID. Single modality regimens are also effective, with one such

intervention utilizing strength training in a functional paradigm (ascending and descending stairs) to improve performance of functional tasks of daily living in adults with DS ([191–193](#)). Integration of new technologies could also be an area to expand within Ex R_x for this population. Common virtual reality devices combined with exercise programming and videogames focused on engaging players in PA have also been shown to enhance CRF, muscular strength and power, motor function, and speed and agility in individuals with ID ([194](#)). In addition, interventions focused on increasing PA through sport participation have seen success, with sport modalities of swimming ([195](#)), dance ([196, 197](#)), Baduanjin ([198](#)), soccer ([199](#)), and taekwondo ([200](#)). However, these interventions were not consistently effective at modifying fitness parameters.

Both aerobic interval training ([201](#)) and continuous aerobic training at higher intensities ([202](#)) have demonstrated positive effects on body weight and BMI in adults with DS, with a slightly greater effect in interval training. In addition, there is evidence that HIIT can be used to increase aerobic capacity and cardiovascular responsiveness in individuals with DS ([201, 203](#)). Given these findings, interval training could be used in DS Ex R_x as an effective method of weight management. The interval training protocols that elicited these benefits consisted of: (a) 1:3 work:rest ratio with intensity of the work being “all out” through the exercise session ([201](#)) and (b) 10-minute warm-up walk at 50% HR reserve before escalating to 4–2 minute intervals at 75% and 85% HR reserve separated by a 2-minute return to warm-up intensity walking for an average of 29 minutes ([203](#)). As is the case with healthy adults, participants with DS should be trained on proper exercise form before engaging in interval training to reduce risk of injury.

Special Considerations for Individuals With Intellectual Disability

- Individuals with ID could be prescribed medications that present difficulties with weight gain or loss, such as antihypertensives, anticonvulsants, antidepressants, or diabetes medications ([204](#)).

- Balance and gait capabilities can be affected in individuals with ID, as compared to age-matched controls ([180](#)); therefore, fall risk must be considered when prescribing and performing exercise.
- Correlates of symptoms of ADHD have been found in children with ID, including attention deficit ([205](#)). Care should be taken to create an exercise environment with minimal distraction, particularly during more skilled movements such as resistance or balance training.
- Individuals with ID and DS have deficits in short-term memory ([206](#)), so simple and concise language and repetition of direction is recommended. Demonstrations of movements could be necessary.
- Organized sport participation, such as Special Olympics, in this population has demonstrated positive effects on quality of life and self-efficacy ([207](#)), although participation in Special Olympics does not guarantee changes in fitness ([208](#)). However, these types of programs are tools for jumpstarting behavior change that increase exercise self-efficacy, and the group participation aspect could increase an individual's sense of community.
- Exercising as a group, or with familiar people, could be beneficial to facilitate enjoyment and promotion of exercise ([209](#)). Additionally, peer mentoring in exercise could be helpful in increasing self-efficacy through modeling behavior in both the demonstrator and observer roles ([210](#), [211](#)).
- Individuals with ID need higher levels of encouragement to exercise, potentially because of characteristics of personality that predispose them to less than average enjoyment from problem solving and lower levels of expected success when presented with a new task ([212](#)). This encouragement could come in many forms, such as verbal affirmation, goal setting, rewards, or relationship building. However, care should be taken to avoid food-based reward systems, as these rewards are often nutritionally empty and compete with efforts toward weight loss.
- Physical accessibility and financial accessibility are large barriers to PA in adults with ID ([213–215](#)). Exercise professionals should be aware of environmental factors, such as built environment inside and around the individual's living area that could impact an individual's PA choice and

behavior ([216](#)). Engaging with this population and having conversations about what is or is not available to the individual in the built environment and with regard to financial resources could aid in creating better long-term exercise outcomes and behavior.

- Individuals with ID will lose access to educational daytime exercise programming as they enter young- and mid- adulthood and could develop sedentary habits as they continue to age. Given the importance of exercise in aging in individuals with ID ([217](#)), efforts should be made to incorporate PA throughout the lifespan.
- To encourage individuals with ID to exercise outside of the regularly scheduled training, Ex R_x should include exercises to be performed independently (*e.g.*, familiar exercises that require minimal equipment or resources to complete). Additionally, encouragement to integrate PA into occupational time or workday time could greatly impact overall sedentary time in this population ([218](#)).
- Family members and caregivers can be facilitators to exercise in people with ID or DS outside of regularly scheduled training and provide valuable insight into exercise behavior of their loved one ([207](#)). However, family members and caregivers can also provide barriers to exercise in people with ID or DS, such as lack of provision resources necessary to complete home-based activity ([215](#), [219](#)). Understanding the home structure of individuals with ID and DS and integrating caregivers into Ex R_x development could aid exercise professionals in creation of a more holistic Ex R_x. Education for family members, caregivers, and/or supervisors of group home settings on how to encourage exercise in people with ID or DS is needed.
- Opening a dialogue on participants' definition and implementation of a comfortable setting for exercise, including factors such as controlling the number of people in a given space, the playing of preferred music, exercise equipment to select from, etc. could play a role in creating initial participation and maintaining participation over time ([220](#)).
- Individuals with ID or DS could have personality dispositions that lead to uneasiness around unfamiliar people ([212](#)). For this reason, participants should be familiar and comfortable with the personnel that they will be testing

with, including effective channels of communication during exercise testing and training. Additionally, there is evidence to suggest that trainees in health care settings could be avoidant of or infantilizing of disability-related issues in care ([221](#)); further training of young professionals in interactions with individuals with ID could aid in shifting patient-practitioner interactions.

Special Considerations for Individuals With Down Syndrome

- Individuals with DS often have other comorbidities, such as heart disease, obesity, leukemia, AD, dementia, and other infections, such as fungal skin infection, severe bacterial infection, and ear infection ([190](#), [222](#), [223](#)). These conditions can impact one's ability to exercise and motivation to do so.
- Facial features that impact ability to breathe, such as smaller nose, flat nasal bridge, and smaller mouth, should be taken into consideration to maintain an open airway. Taking breaks, maintaining open communication with the participant, and watching for nonverbal cues of distress are encouraged during exercise sessions at high intensity.
- Joint laxity and skeletal muscle hypotonia could also affect the exercise choice for individuals with DS. Individuals with DS can experience atlanto-occipital instability (movement between the C1 vertebrae and the occiput), atlantoaxial instability (excessive movement between the C1 and C2 vertebrae), and cervical instability (general laxity between cervical vertebrae) ([224](#)). Exercises that place pressure on the neck (such as abdominal crunches) are cautioned.
- Knee and hip joints can be in increased strain due to overweight or obese status as well as several gait and biological factors ([225](#), [226](#)). Providing alternative exercises, particularly for weight-bearing exercises, could be helpful in reducing joint stress.
- Lower cardiovascular capacity is consistently seen in individuals with DS when compared to healthy control individuals ([188](#)). Careful consideration of HR should be taken during exercise as well as careful monitoring of participant's perceived exertion.

- Evidence of mitochondrial dysfunction has been observed in individuals with DS, including reduced ATP production, decreased respiratory capacity, and alterations in mitochondrial structure ([227](#)). Inefficient mitochondrial metabolism and increased oxidative stress could contribute to lower levels of PA in this population.

Future Directions

Large-scale exercise trials in this population are scarce, which presents a significant limitation to the field. Additional large-scale interventional trials that examine the effects of varying intensities, training volumes, and exercise modalities are necessary to optimize exercise for both prevention and reduction of chronic disease associated with ID and DS. In particular, evaluating the value of exercise training levels below current guidelines are needed, as evidence in the general population suggests health benefits from performing step counts below recommended levels ([228](#)). Additionally, future trials could continue to examine novel exercise modalities especially related to the impact of varying levels of behavioral coaching and guidance in maintaining exercise behavior, with the aim of understanding the optimal amount of behavioral engagement to produce long-term effects. Observational studies are needed to understand the importance of reduction in sedentary time in people with ID, in conjunction with studies on neurotypical people. In addition to the impact of exercise training on physical abilities, continued evaluation of exercise training for people with ID of both prevention and reduction of chronic disease associated with ID and DS is needed. This information could be used to improve exercise guidelines for people with ID and DS as well as inform the choices of exercise professionals to create optimal Ex Rx. The overarching goal is to guide people with ID and DS into effective, evidence-based exercise programs that are designed to produce meaningful changes in activity-induced health benefits.

ONLINE RESOURCES

American Association on Intellectual and Development Disabilities:

<http://www.aamr.org>

Center for Parent Information & Resources.

Intellectual disability: <http://www.parentcenterhub.org/intellectual>

National Association for Down Syndrome: <http://www.nads.org>

TASH: <http://www.tash.org>

The Arc of the United States: <http://www.thearc.org>

CEREBRAL PALSY

Cerebral palsy (CP) is defined as “a group of permanent disorders of the development of movement and posture, causing activity limitation, that are attributed to nonprogressive disturbances that occurred in the developing fetal or infant brain” ([229](#)). As such, CP encompasses a variety of motor disorders, which may be accompanied by atypical sensation, other neurodevelopmental disorders (*e.g.*, epilepsy, autism, intellectual impairment), and communication ([229](#)). CP is the most common pediatric-onset motor disability and severity of motor impairment varies considerably among those who are diagnosed. Function can range from complete dependence on a motorized wheelchair to imperceptible disturbances in speed, balance, and coordination, which eliminate the need for orthotic or assistive devices. Because CP results from a neurological insult either before, during, or shortly after birth, individuals with CP have increased risk of associated comorbidities, including epilepsy, intellectual impairment, and autism ([230](#)). Typically, CP is diagnosed before age 5 and may be as early as 12 months of age using a combination of standardized motor assessments, neuroimaging, and a medical history ([231](#)). Risk factors for CP include low birth weight, children born in multiple births, placental abnormalities, birth asphyxia, maternal obesity ([232](#)), and neonatal infections ([233](#)). However, the causal pathway to CP is poorly understood, as many children that eventually receive a diagnosis of CP are born at full term and have unremarkable imaging. The prevalence rate of CP is 1.5–3.8 per 1,000 live births in Europe, Australia, and the United States ([234–236](#)) but is starting to increase due to advances in premature birth interventions. Moreover, advances in pediatric health care means that virtually all kids with CP survive into adulthood, and life expectancy for those with mild impairments is similar to the general population ([237](#), [238](#)).

The core features of CP are atypical fine and gross motor function ([229](#)), which manifests as impaired muscular tone, motor control, and coordination ([239](#)). CP is nonprogressive so the underlying impairment does not change over time;

however, ambulation, balance, and function may all change over the lifespan due to an individual's lifestyle and therapeutic interventions. There are different models used to describe the motor characteristics resulting from CP including the type of involvement (*e.g.*, spasticity), capacity for motor function (*i.e.*, walking capacity), or topographical location (*e.g.*, hemiplegia). Type of involvement is the most common layperson description model, and categories include spasticity, dyskinesia, and ataxia with individuals who demonstrate multiple characteristics categorized as mixed. Spasticity is the most common type of manifestation and indicates excessive muscle tone, which is demonstrated in the hallmark scissor gait and is characterized by an increased resistance that is velocity dependent ([240](#)). Spasticity may also present with pathological reflexes (*e.g.*, hyperreflexia). People with dyskinesia have involuntary movements and fluctuating muscle tone, which may contribute to drooling and posture difficulty. Dyskinesia may be further divided as dystonia and choreoathetosis ([229](#)). Ambulatory persons with CP who are ataxic experience a loss of muscular coordination and typically present with low muscle tone and generalized weakness, resulting in balance difficulties, and possible tremors. Regardless of classification description, persons with CP experience reduced muscle strength, impaired muscle agonist-antagonist synchronicity, and reduced joint ROM ([241–243](#)), which result in difficulties with participation in ADL ([244, 245](#)).

Clinically, the classification of individuals with CP according to their motor function is more useful for describing, comparing, and predicting future needs. The Gross Motor Function Classification System (GMFCS) is the predominant instrument used to categorize persons by motor function based on their ambulatory ability ([246](#)) ([Table 11.5](#)). Levels I and II identify children with the ability to walk, whereas levels III through V reflect individuals who use assistive technology, including handheld mobility devices (walkers, crutches, or canes) or wheeled mobility. A full description of each GMFCS level is in [Table 11.5](#). CP was historically categorized according to the distribution of motor impairment, including diplegia, hemiplegia, quadriplegia, and tetraplegia. However, it is now recommended that the terms unilateral (one side of the body) and bilateral CP be used to describe the location (or topography) of involvement. Despite the

shortage of research to track lifelong health and chronic disease trajectories in this population, there is ample evidence demonstrating that individuals with CP have significant and progressive motor impairment, excessive sedentary behavior profiles, inadequate muscle and bone development, increased visceral and musculoskeletal adiposity, impaired glucose tolerance, and insulin resistance ([247–252](#)). Consequently, individuals with CP are at a heightened risk for developing age-related secondary chronic conditions such as osteoporosis, diabetes, and primary cardiovascular disease ([253–259](#)), thereby reducing life expectancy ([260](#)). The chronic disease burden that individuals with CP face may develop or be influenced by the presence of impairments or acceleration of the aging process, suggesting that individuals with CP are prone to “premature aging” ([261](#)). Specifically, having CP may increase the risk of developing a health condition that is directly linked to impairments (*e.g.*, pain, joint contractures), or occurs as an indirect consequence of the impairment itself (*e.g.*, increases in sedentary behaviors and obesity-related conditions such as diabetes). Consequently, there is a need for approaching health care delivery for individuals with CP within the context of a life course health development model ([262](#)). Recent PA recommendations for people with CP have been developed ([248](#)); however, more research and awareness are needed to inform exercise prescription development for this population.

TABLE 11.5 Gross Motor Function Classification System

Level I	Walks without restrictions, with limitations in more advanced gross motor skills
Level II	Walks without assistance, with slight community ambulation limitations
Level III	Walks with the assistive mobility devices, with community ambulation difficulties
Level IV	Self-mobility with limitations; wheeled mobility used in most settings but may transfer with physical assistance and walk short distances with physical assistance or in a body support walker
Level V	Severely limited mobility, even with appliances and adaptations; transported in wheelchair in all settings, but powered mobility may be used with extensive adaptations

Exercise Testing

Exercise testing may be conducted in individuals with CP to determine the functional capacity (aerobic and anaerobic) of the individual, and to identify the appropriate exercise dose for aerobic and strengthening exercises. However, a symptom-limited graded exercise test (GXT) is not required for those with CP to begin an exercise training program. Prior to testing, it is recommended to review an individual's medical history, list of medications, and potential existing comorbidities that may impact test administration (*e.g.*, epilepsy). A high proportion of people with CP experience pain and fatigue, which may impact their performance on exercise testing ([263](#)). Assessment of functional capacity, including ambulatory ability, will facilitate the choice of exercise testing equipment, protocols, and adaptations. In the case of athletes, testing mode will also be guided by the desired sport. Common exercise tests for children and adolescents include the muscle power sprint test, the 20/30-s Wingate cycle or arm cranking test, a 10-m shuttle run for GMFCS levels I and II, and a 7.5-m shuttle walk/run test for GMFCS level III ([264](#)). Further information on these tests is provided in [Table 11.6](#).

TABLE 11.6 Exercise Tests for Children With Cerebral Palsy (CP)

Exercise Test	Physical Fitness Component Assessed	Further Comments
30-s Wingate arm cranking test	Anaerobic fitness	Reliable measure in children with CP in Gross Motor Function Classification System (GMFCS) levels III and IV (265)
20-s Wingate cycle test	Anaerobic fitness	Feasible and reliable measure in children with CP in GMFCS levels I–III (266)
Muscle power sprint test	Anaerobic fitness	Has been adapted to assess anaerobic fitness during self-propelling in a wheelchair in children in GMFCS levels III and IV and found to be valid and reliable (267)
		Walking/running muscle power sprint test has been found to be valid, reliable, and feasible in children in GMFCS levels I and II (265 , 268).
		Normative values for children with CP have been published (269).
10-m shuttle run tests (SRT-I and SRT-II)	Cardiorespiratory fitness	Two shuttle run tests have been developed specifically for children in GMFCS levels I and II, respectively.
		Feasible, reliable, and valid in comparison to treadmill testing for assessing cardiorespiratory fitness among children with CP (270).
		Normative values for children with CP have been published for these tests (269).
7.5-m shuttle run/walk test	Cardiorespiratory fitness	Developed for children who require a mobility aid (GMFCS level III) and is a reliable measure of cardiorespiratory fitness (264).

Exercise Test	Physical Fitness Component Assessed	Further Comments
10-m shuttle ride test	Cardiorespiratory fitness	Feasible, valid, and reliable test of cardiorespiratory fitness for children who use a manual wheelchair (271).

Exercise Testing Considerations

Choice of exercise mode and use of adaptive equipment is important when conducting exercise tests on people with CP. When necessary, testing should be conducted using appropriate adaptive equipment, such as straps and holding gloves, to guarantee safety and optimal testing conditions for mechanical efficiency. Treadmill testing may be used to assess CRF among high-functioning ambulatory individuals (GMFCS levels I and II). Although graded arm ergometry tests are often used to assess CRF in people with mobility impairments, the cardiorespiratory demand during an arm ergometry test has been shown to be significantly lower compared to a wheelchair shuttle test in individuals with CP. Therefore, a 10-m wheelchair shuttle test may provide a more accurate indication of CRF ([265](#)). Submaximal single-stage exercise tests, such as the 6-MWT, provide information about walking capacity but are not validated methods of measuring CRF in persons with CP ([272](#)). However, these tests are appropriate to document functional movement through measures such as distance walked, HR, gait efficiency, or RPE, where true maximal CRF testing is not appropriate or accurate.

Field tests of muscular fitness are also good exercise test options. Tests that consist of functional tasks such as the timed stairs test, the sit-to-stand test, and lateral step-ups are important in examining motor capacity over time. Clinical assessment of muscular fitness typically includes 6–10 RM for multijoint machine-based exercises (*e.g.*, leg press) or the use of isokinetic dynamometry to assess hip, knee, and ankle flexors and extensors ([230](#)). Handheld dynamometers may be used to assess isometric muscle strength in persons with CP (especially GMFCS levels I–III), but reliability of this method may be less than ideal, particularly for assessing hip extension, knee extension, and ankle plantarflexion and dorsiflexion ([273](#)). Moreover, 1-RM testing may be difficult to perform in

individuals with CP because of difficulties identifying appropriate methods of adding resistance, difficulties with coordination and balance, and lack of experience of working to maximal exertion.

Special Considerations

- Joint pain is prevalent among people with CP ([259](#)) and may be exacerbated by joint loading. A mode of exercise test that minimizes joint pain should be chosen.
- Positioning and level of comfort during exercise testing (in a chair or on an exercise device) need to be considered to avoid unintended increases in muscle tone or facilitation of primitive reflexes, which can also cause pain.
- Fatigue in the latter stages of exercise testing may reduce coordination and balance, and care should be given to prevent falls and injuries.
- People with CP often have reduced aerobic and anaerobic exercise responses as compared to those with typical development ([274](#)).
- Prediction of maximal HR using standard equations may not be accurate for individuals with CP.
- Difficulty with balance, bilateral muscular symmetry, and grip coordination makes the use of gloves, resistance bands, and machine-weights warranted.

Exercise Prescription

Individuals with CP have decreased physical fitness levels compared to peers without impairment ([243](#), [269](#)). Interventions associated with CP, such as orthopedic surgery, can have a detrimental short-term effect on muscle strength and CRF ([275](#), [276](#)); further deterioration in physical fitness often occurs with age. Although CP is a nonprogressive condition, between 20% and 50% of adults with CP report a deterioration in mobility in young- to mid-adulthood ([277](#)), which may result from loss of ROM, reduced balance, reduced muscle strength, reduced aerobic capacity, or pain ([278](#)). The combination of low muscle mass, low muscle strength, and loss of physical function has prompted comparison with sarcopenia and dynapenia in older adults with typical development ([279](#)).

People with CP should maintain a lifestyle of PA to offset the decline in function associated with aging and the effects of CP. Exercise participation should be promoted for people with CP before motor patterns are fixed (<7 years) in order to optimize function, promote quality of life, and prevent chronic disease risk. Training to develop muscular strength and aerobic capacity are valuable for people with CP, especially as a means of offsetting the functional deterioration and associated physical dependence that adults with CP experience ([280](#)).

Individuals with CP who are not able to meet the PA guidelines recommendation of $150 \text{ min} \cdot \text{wk}^{-1}$ of PA should engage in regular exercise as frequently as possible. Generally, the FITT principle of Ex R_x recommendations for the general population should be applied to individuals with CP (see [Chapter 5](#)) ([248](#)). However, it is important to note that Ex R_x recommendations for individuals with CP are based largely on expert opinion because research focuses primarily on children and adolescents with minimal or moderate involvement (*i.e.*, those who are ambulatory) ([281](#)). It is also important to note that atypical agonist-antagonist contraction can dramatically increase intensity of an activity and degrade movement economy. Hence, activity intensities may be higher in persons with CP than ranges reported in the compendium of physical activities ([282](#)). The combination of reduced CRF and increased walking energy cost also result in a higher physical strain of walking, and nearly half of ambulatory people may walk at or above their anaerobic threshold ([274](#)).

Even though the design of exercise training programs to enhance health/fitness benefits should be based on the same principles as the general population, modifications to the training protocol may be required based on the individual's functional mobility level, number and type of associated conditions, and degree of involvement of each limb ([248](#)). For this reason, recommendations regarding the FITT principle of Ex R_x are included in the following "Special Considerations" section.

Special Considerations

- Individuals with CP have a higher risk of osteoporosis and fractures compared to individuals without CP ([257](#), [283](#)). Leg and arm ergometry may be used to

minimize risk of falls and fractures, particularly among individuals with balance deficits.

- Training sessions may be more effective for individuals with high muscle tone, if (a) short interval training sessions are conducted rather than one longer session, (b) relaxation and stretching routines are included in the session, and (c) new skills are introduced early in the session ([284](#), [285](#)).
- Resistance exercises should be performed over the full ROM, involving concentric and eccentric muscle contraction, at slow contraction speeds. Eccentric training may decrease cocontraction and improve net torque development ([236](#)).
- Single-joint and unilateral exercises may be more effective than multijoint and bilateral exercises for individuals who are very weak or with poor selective motor control, as they minimize synchronization needs with other muscle groups.
- Different resistance exercises and modes should be tried to activate the appropriate muscle group and appropriate resistance to overload the muscle.
- Before initiating open kinetic chain strengthening exercises using free weights, always check the impact of primitive reflexes on performance (*i.e.*, position of head, trunk, and proximal joints of the extremities) and whether the individual has adequate neuromotor control to exercise with free weights. Machine-weights are always recommended as a precedent because of the enhanced safety and balance supports.
- Good positioning of the head, trunk, and proximal joints of extremities to control persistent reflexes is preferred to strapping. Inexpensive modifications that enable good position such as Velcro gloves to attach the hands to the equipment should be used whenever needed.
- Strong support for sport participation is suggested because elite athletes with CP do not show associated lower neuromuscular fatigue ([286](#)).

Future Directions

The evidence supporting exercise in persons with CP is limited based on access to participants. RCTs and experimental interventions are typically conducted on

ambulatory children, and research is far more limited in adults with CP who have limited mobility. The variability of function makes extrapolation of outcomes and fidelity of an intervention very difficult in this population. Individuals with CP may not be able to perform all exercises as prescribed, and understanding fidelity to the intervention is important for identifying if exercise interventions are ineffective for people with CP or simply not feasible to perform as prescribed. There is very little research focusing on the effect of exercise for adults with CP or people with moderate-to-severe CP. These subgroups need to be included in future studies as the effects of exercise for CP may be dependent on age and motor function. Finally, the effect of exercise for individuals with CP has also only been examined on a relatively small number of outcomes. Fewer studies have explored the effect of exercise on participation in life activities or quality of life. Future studies are needed to investigate the long-term effects of exercise on function and health for people with CP, including pain, fatigue, falls, CVD, osteoarthritis, and depression. Exercise may be particularly beneficial to maintain function and prevent development of chronic disease in this population, and longitudinal studies are definitely needed to better capture the impact of exercise on function in persons with CP.

ONLINE RESOURCES

American Academy for Cerebral Palsy and Developmental Medicine Fact Sheets:

<https://www.aacpdm.org/>

“Learn more about Cerebral Palsy (CP)” from the Centers for Disease Control and

Prevention: <https://www.cdc.gov/ncbddd/cp/index.html>

National Institute of Neurological Disorders and Stroke. Cerebral palsy:

<https://www.ninds.nih.gov/health-information/disorders/cerebral-palsy?search-term=cerebral%20palsy>

Physical Activity Guidelines for Americans. Chapter 10:

https://health.gov/sites/default/files/2019-09/16_F-10_Individuals_with_Chronic_Conditions.pdf

ALZHEIMER'S DISEASE

AD is the most common cause of dementia, comprising 60%–80% of all dementia cases ([287](#)). AD is characterized by early and progressive declines in learning and memory, as well as other cognitive processes (*e.g.*, executive functions), followed by more severe declines in cognition, mood, and motor abilities as the disease progresses ([288](#)). AD is also characterized by increased depressive symptoms, behavioral problems such as wandering or agitation, and significant sleep disturbances, all of which may mediate or exacerbate memory impairments ([289](#)). Over time, the impairments become severe enough to interfere with the ability to complete instrumental activities of daily living (I-ADL). Although AD is most common in individuals over the age of 65 years, its early-onset form commonly affects those younger than 65 years. Additionally, the neurodegenerative and neuropathological processes associated with the disease begin one to two decades before the onset of any cognitive symptoms, making a search for prevention and early treatment of the disease a major priority for public health ([290](#), [291](#)).

AD is currently the sixth leading cause of death in the United States, with nearly 46 million people managing from AD or a related dementia. Given the projected increase in the number of adults over the age of 65 years, the *World Alzheimer's Report* and the Alzheimer's Association both indicate that without the discovery of effective prevention and treatment measures, the number of cases may double by 2050 ([287](#)). Age is the greatest risk factor for AD, with 81% of cases being 75 years or older. The prevalence rate exponentially increases with age such that about 3% of individuals 65–74 years old have the disease, whereas 32% of individuals over the age of 85 years have AD ([292](#)). An individual diagnosed with AD survives for 7–12 years on average ([293](#), [294](#)). Unfortunately, at present, AD is incurable.

AD is primarily characterized by two hallmark pathological features: amyloid- β plaques and neurofibrillary tangles. In the preclinical phase of AD, plaques and tangles accumulate in the brain up to two decades before the development of

cognitive symptoms ([291](#), [295](#)). In fact, 20%–40% of individuals over the age of 65 years without evidence of memory loss or cognitive decline have detectable pathological features of the disease, and the presence of this pathology is thought to lead to other downstream neuropathological processes such as atrophy of the hippocampus and other brain regions ([296](#)). Subsequent to the preclinical stage is a second stage characterized by both presence of AD pathology and neurodegeneration (*e.g.*, hippocampal atrophy) along with signs of MCI. The final stage (AD related dementia) is characterized by more significant objectively measured cognitive impairments as well as an inability to independently perform many I-ADLs and subjective reports of cognitive or memory problems. To diagnose AD, providers use a combination of assessments including a neurological examination, tests of cognitive and functional abilities, and brain imaging (MRI, CT, and/or PET). Additional information from cerebrospinal fluid or blood tests can also be used to aid in disease diagnosis.

There have been numerous studies examining the potential for pharmaceutical and nonpharmaceutical treatments to prevent, delay, or reverse the course of AD. These treatments include pharmaceuticals that target the cholinergic system or enzymatic pathways in the amyloid or tau cascade, nonpharmaceuticals that target oxidative stress, and behavioral interventions such as exercise. Unfortunately, the majority of these treatments have met with limited clinical success. Because of this, it is suggested that the greatest chance of success for altering the trajectory of the disease course is by targeting the earliest possible stages, when amyloid- β first begins to show increased accumulation.

Observational studies often consider leisure activities, PA, and exercise behaviors together when assessing longitudinal risk for dementia or cognitive impairment. In contrast, structured exercise programs are often used in randomized clinical trials that assess whether exercise could be used as an effective treatment for symptoms of the disease. A significant body of research demonstrates that greater amounts of PA are associated with a reduced risk of experiencing cognitive decline in late adulthood ([297](#)) as well as a reduced risk of developing AD ([298](#), [299](#)). There is also preliminary evidence that exercise may improve physical and cognitive function in individuals with AD ([300](#)), but these

findings are far from conclusive (see references [\(301\)](#) and [\(302\)](#)) and more research is necessary to determine the magnitude and consistency of such an effect. Data suggest that higher CRF is related to less brain atrophy in early AD [\(303\)](#) and that increasing fitness levels through exercise interventions may slow memory loss and brain atrophy [\(304\)](#). In cognitively healthy adults, there is limited evidence that exercise might be capable of positively altering the size and function of brain areas that are affected in the disease course [\(303\)](#), which improve as a function of exercise-induced changes in cellular and molecular pathways that could mitigate the effect of neuropathology or reduce the accumulation of amyloid- β plaques. One of the brain areas that may be impacted by exercise is the hippocampus, which supports memory consolidation and retrieval and is among the structures most susceptible to neurodegeneration in AD pathogenesis. Recent studies in animal models and cognitively healthy older adults indicate a relationship between exercise and neurogenesis of the hippocampus [\(305, 306\)](#), though further work is needed to characterize this relationship in individuals with AD.

Based on these data, the 2018 Physical Activity Guidelines Advisory Committee reported that there is strong evidence that greater amounts of PA reduce the risk for dementia [\(307\)](#). Although there is much to learn about the potential for PA and exercise to influence cognitive and brain health in AD, there is emerging evidence of increased functional ability in early stages of AD [\(304\)](#) and strong and unequivocal evidence that exercise reduces risk of falls and injuries as well as improves physiological pathways associated with conditions that often cooccur with AD (*e.g.*, Type 2 diabetes mellitus).

Exercise Testing

Recommendations for exercise testing are dependent on the stage and severity of the disease, such that individuals that are in the preclinical and early stages are much more likely to understand and tolerate exercise testing procedures more so than individuals later in the disease course [\(308, 309\)](#). Thus, clinical judgment is always needed to determine a person's safety for conducting a test, and it is recommended that all exercise testing be conducted in consultation with a

physician and/or neuropsychologist who understands the individual's level of impairment. Because dementia often co-occurs with cardiovascular and cardiometabolic conditions, exercise testing should also take into consideration the presence of these conditions. Maximal exercise testing is safe unless there are balance, muscle, or coronary contraindications, as outlined in [Chapter 4](#). Cycle ergometry may be a more effective and less painful procedure for some individuals with comorbid conditions that increase pain during exercise testing (*e.g.*, significant arthritis).

For all procedures, it is recommended to allow individuals to warm up at a light-intensity level according to the individual's level of impairment and functional status and to provide a sufficient period for cool-down while closely monitoring vital signs. Individuals with more severe cognitive impairments may have difficulty remembering the goals of the test, instructions, or remembering what they are there to do. These memory impairments may fluctuate from one moment to the next, making exercise testing contraindicated in conditions of severe memory loss. Standard measures of exertion such as the Borg scale (see [Figure 4.2](#)) may also be invalid in cases of severe AD but, in preclinical and early stages, could be considered a valid assessment tool. Strength testing is recommended in this population, especially in cases of preclinical, MCI, or early dementia stages, given the widespread physiological and behavioral consequences of strength training (*i.e.*, risk of falls). In cases of more severe memory impairments, strength testing may become more hazardous and should always be conducted in consultation with the physician and/or neuropsychologist who understands the severity of the impairment.

Exercise Prescription

Depending on the severity of the cognitive impairment, health professionals and caregivers are sometimes hesitant to start an exercise program with the fear that cognitive losses might result in a greater likelihood of distraction, forgetting what one is doing, or overestimation of one's current abilities to perform exercise. Despite these fears, individuals with AD are safe to exercise as long as the Ex R_x is progressed and monitored in a similar fashion to that of cognitively normal older

adults (see [Chapter 6](#)). There is evidence that cognitively impaired individuals, including those with AD, might also benefit from lower-intensity activities, thereby allowing them to engage in a variety of exercises with a range of intensities ([297](#)). Individuals with AD may also benefit from targeting multiple exercise modes, aerobic, strength, coordination, flexibility, and balance training ([310](#)), as the usual health benefits associated with these are all applicable in the individual with AD. Individuals with AD should avoid inactivity and sedentariness. As is the case with exercise testing, all Ex R_x should be done in consultation with the physician and/or neuropsychologist who understands the severity of the impairment as well as other comorbid conditions that could affect exercise safety or prescription (see [Chapters 8, 9, and 10](#)). All exercise, including aerobic, resistance, and flexibility training ([311](#)), for individuals with AD should be conducted based on a relative intensity of the person's symptoms and physical status and progress at a rate that would optimize adoption and adherence. Finally, anyone with AD should engage in levels of PA as their abilities allow.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH ALZHEIMER'S DISEASE

	Aerobic	Resistance	Flexibility
Frequency	3 d · wk ⁻¹	2–3 d · wk ⁻¹	≥2–3 d · wk ⁻¹ with daily being most effective
Intensity	Dependent on disease severity, start with light intensity and progress to moderate intensity based on the performance of the individual (40%–59% $\dot{V}O_{2R}$ or HRR; RPE of 12–13 on a scale of 6–20)	40%–50% of one repetition maximum (1-RM) for individuals beginning to improve strength; 60%–70% 1-RM for more advanced exercisers. Always consider the severity of the cognitive impairments as well as the presence of any co-occurring conditions that could modify these levels	Full extension, flexion, rotation, or stretch to the point of slight discomfort
Time	Depending on disease severity, it may be necessary to start with bouts <10 min and progress at a rate comfortable with the individual. Exercise could be performed up to 30–60 min of continuous or accumulated exercise	≥1 set of 8–12 repetitions; 10–15 repetitions in adults with Alzheimer disease starting an exercise program	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, cycling, swimming, dancing)	For safety, avoid free weights; focus on weight machines and other resistance devices (<i>e.g.</i> , bands, body weight)	Slow static stretches for all major muscle groups

HRR, heart rate reserve; RPE, rating of perceived exertion.

Exercise Considerations

- There is growing evidence that individuals with AD can realize benefits of exercise through improved aspects of physiological and brain health; whether an individual can perform exercise activities on their own depends on the severity of the disease.
- Long, continuous bouts of exercise are more likely to be helpful and safe in preclinical and early stages of the disease and less likely to be feasible during later stages of the disease.
- Moderate intensity interval training and HIIT training is feasible and might influence enjoyment and adherence. It is considered safe if all other safety

measures described above are followed.

- Metabolic, cardiovascular, joint, and muscle atrophy comorbidities may restrict the frequency and duration of exercise. Therefore, it may be appropriate to start an exercise regimen with short bouts of 10 minutes or less.
- Targeting multiple modes of activity might be the most effective for enhancing balance, flexibility, strength, and endurance.
- Adequate warm-up and cool-down periods with monitoring of vital signs are critical for minimizing safety concerns.

Special Considerations

- In the earliest stages of AD, including MCI, individuals are still capable of performing exercise independently and in the community.
- Ex Rx for individuals with a diagnosis of AD should always be performed with the consultation of the individual's physician and/or neuropsychologist.
- There may be benefits of training and exercising with the caregiver to help provide support, motivation, and monitoring of safety.
- For individuals with AD exercising in the community, caregivers or other supporting individuals should be present at the end of the session and available to provide support in shared changing areas/locker rooms.
- Inform individuals with AD, as well as caregivers, that a small amount of musculoskeletal pain during or immediately following an exercise bout is a normal consequence of starting an exercise regimen.
- Exercising in the morning hours might be easiest and most beneficial, as morning is often when an individual with AD is demonstrating the lowest severity of symptoms.
- Exercise in nursing homes, memory clinics, or senior care facilities is encouraged as long as there are properly trained staff to monitor individual progress and safety.
- Repeating instructions and providing multiple cues to individuals with AD will help to facilitate understanding and application of instructions.

Future Directions

Future research should be conducted to examine parameters and modes of activity that are optimal for both preventing and treating the symptoms of AD. In addition, we have a poor understanding of whether exercise could reduce the incidence of the disease or the optimal window in the disease course to begin an exercise regimen. Future research should examine whether exercise is more effective at treating symptoms in preclinical stages before cognitive decline is evident or whether it is equally effective at later stages of the disease course.

ONLINE RESOURCES

Alzheimer's Association: <https://www.alz.org>

Mayo Clinic: <https://www.mayoclinic.org/diseases-conditions/alzheimers-disease/symptoms-causes/syc-20350447>

National Institute of Aging: <https://www.nia.nih.gov/health/alzheimers>

World Alzheimer's Report: <https://www.alz.co.uk/research/world-report>

PARKINSON'S DISEASE

Parkinson's disease (PD) is the second most common neurodegenerative disease after AD, and it is estimated that at least 700,000 individuals in the United States aged 45 years and older are living with PD, with this number projected to double by 2030 ([312](#)). It is uncommon for PD to be diagnosed before 50 years of age, but the incidence increases 5- to 10-fold from ages 60 to 90 years ([313](#)). The global estimate of PD prevalence is currently in excess of 7,000,000 individuals ([314](#)). It is projected that between 12.9 and 14.2 million people will have PD by 2040 ([314](#), [315](#)). PD is a chronic, progressive neurological disorder characterized by signs of bradykinesia, resting tremor, rigidity, postural instability, and gait abnormalities ([Box 11.1](#)). PD is a form of parkinsonism, a clinical syndrome including other neurodegenerative parkinsonian disorders, such as multiple systems atrophy, progressive supranuclear palsy, and corticobasal degeneration. They are collectively referred to as atypical parkinsonian disorders and have similar core features yet varying clinical signs that lead to differential diagnoses ([316](#)). The motor features of PD are the result of degeneration of the dopaminergic nigrostriatal pathway of the midbrain, which results in a reduction in the neurotransmitter dopamine in the striatum. The cause of PD is unknown; however, aging, genetic susceptibility, and environmental factors likely all play a role. Inflammation and mitochondrial dysfunction may also contribute to the disease process ([317](#)–[321](#)).

Box 11.1 Common Movement Disorders in Individuals With Parkinson's Disease ([322](#))

Bradykinesia	Reduced movement speed and amplitude; at the extreme, it is known as <i>hypokinesia</i> , which refers to “poverty” of movement.
Akinesia	Difficulty initiating movements.
Episodes of freezing	Motor blocks/sudden inability to move during the execution of a movement sequence.
Impaired balance and postural instability	Difficulty maintaining upright stance with narrow base of support in response to a perturbation to the center of mass or with eyes closed; difficulty maintaining stability in sitting or when transferring from one position to another; can manifest as frequent falling.
Dyskinesia	Overreactivity of muscles; can manifest as dystonia; wriggling/writhing movements; chorea or rarely athetosis.
Tremor	Usually resting tremor; more rarely postural or action tremor.
Rigidity	Hypertonicity and hyperreflexia in agonist and antagonist muscle groups in a given limb.
Adaptive responses	Reduced activity, muscle weakness, reduced muscle length, contractures, deformity, reduced aerobic capacity.

The progression of the stages of the disease is described by the Hoehn and Yahr (HY) scale ([323](#)) ([Box 11.2](#)). The key point to notice in the HY scale is that people in stages 1–2 do not have postural instability.

Box 11.2 The Hoehn and Yahr Staging Scale of Parkinson Disease's ([323](#))

Stage 1 = Unilateral involvement only, usually with minimal or no functional impairment.

Stage 2 = Bilateral or midline involvement, without impairment of balance.

Stage 3 = First sign of impaired righting reflexes. This is evident by unsteadiness as the individual turns or is demonstrated when pushed from standing equilibrium with the feet together and eyes closed.

Functionally, this person is somewhat restricted in activities but may have some work potential depending on the type of employment. Individuals are physically capable of leading independent lives, and their disability is mild to moderate.

Stage 4 = Fully developed, severely disabling disease; the person is still able to walk and stand unassisted but is markedly incapacitated.

Stage 5 = Confinement to bed or wheelchair unless aided.

The standard clinical scale for assessing PD is the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS). The MDS-UPDRS is a valid and reliable comprehensive clinical rating scale used to monitor the burden and extent of PD ([324–326](#)). The MDS-UPDRS consists of 65 items and covers four different domains: Part I assesses the nonmotor experiences of daily living such as cognition, depression, sleep, fatigue, and hallucinations; part II assesses the individual's perception of their ability to engage in ADLs such as eating, dressing, hobbies, and walking; part III covers the motor evaluation, which includes ratings for rigidity (stiffness), bradykinesia (slowness), gait, postural stability, and tremor; and part IV assesses the motor complications including ratings for dyskinesias (involuntary movements), dystonia (painful cramps), and motor fluctuations (irregular responses to PD medication). Each item is rated on a 0–4 scale (0 = normal, 1 = slight, 2 = mild, 3 = moderate, 4 = severe), with higher scores indicating a greater impact of PD signs ([325](#)). Individuals with PD may have difficulty performing ADLs and suffer from reduced quality of life ([327](#)). The primary feature of PD is bradykinesia, which refers to slowness in movement and reduced movement amplitude. Clinically, it causes a reduction in dexterity manifest by micrographia, reduced arm swing, and difficulty with fine motor tasks. Resting tremor is present in some individuals with PD ([328](#)) and is typically a larger amplitude, rhythmic shaking of the distal limbs, most commonly

occurring in the hand or arm. Usually, it is asymmetric and can be suppressed by voluntary activity, sleep, and complete relaxation of axial muscles. Stress and anxiety increase resting tremor ([329](#)). Rigidity is an increase in tone, often with a ratchetlike quality called cogwheeling, which can lead to an increase in the individual's perception of movement effort and may be related to feelings of fatigue ([330](#)). These three signs of the disease (micrographia, reduced arm swing, difficulty with fine motor tasks) can occur in HY stages 1–2.

Postural instability refers to “the impairment in balance that compromises the ability to maintain or change posture such as standing and walking” ([331](#)). This is a feature of more advanced stages of the disease (HY stages 3–5) and can lead to falls. Individuals with more advanced PD may have a stooped posture (rounded shoulders, forward head, increased flexion of the trunk and knees) and a reduced stride length when walking such that the gait appears to be shuffling, with decreased arm swing, and may have poorer walking economy when compared to persons without PD ([332](#), [333](#)). Difficulty and slowness in performing turns, getting up, transfers, and ADL are common as PD advances ([334](#)). Other problems include excessive salivation or drooling; soft, slurred speech; and a variety of nonmotor features, including cognitive impairment, mood disorders, and sleep disorders that impact quality of life. Individuals with PD also suffer from autonomic nervous system dysfunction including cardiovascular dysfunction, especially in advanced stages ([335](#), [336](#)). Autonomic dysfunction in PD may result in orthostatic hypotension, cardiac arrhythmias, blunted HR response during exercise and exercise testing, sweating disturbances, and urinary problems ([335–337](#)).

It is important to note that people almost certainly have had PD for several years before they are given the diagnosis of the disease. This is referred to as the prodromal stage of the disease and is characterized by rapid eye movement sleep disorder, constipation, depression, loss of smell (hyposmia), anxiety, and excessive daytime sleepiness ([313](#)). Exercise can help all of these symptoms, which can precede the disease by many years.

Treatment of PD is complex due to the progressive nature of the disease, the vast range of motor and nonmotor symptoms, and the different side effects

associated with therapeutic interventions ([338](#)). One key point about the disease is that it is relentlessly progressive. Different signs and symptoms progress at different rates, and progression is often fastest early in the disease. The progression of the motor signs is between 2 and 3 points per year in the PPMI data set when measured by the MDS-UPDRS part 3 ([339](#)). The progression of the disease has considerable variability from person to person. Treatments include but are not limited to exercise, drug therapy, surgery, and physical therapy ([340–344](#)). With respect to drug therapy, levodopa is a dopamine precursor that is converted to dopamine in the brain. Carbidopa is added to levodopa to prevent the breakdown of levodopa in the blood, allowing the maximum amount of medication to get to the brain while limiting potential adverse effects caused by dopamine in the body, such as nausea and vomiting ([341](#)). Levodopa + carbidopa is the most prescribed drug for the management of PD and is the single most effective drug available to treat all cardinal features of PD, with the exception of rest tremor ([338](#)). As PD progresses, the effect of levodopa may wane. Long-term use is associated with motor complications including motor fluctuations and dyskinesias in about 50% of individuals within 5 years ([345](#)). Other side effects include nausea, sedation, orthostatic hypotension, and psychiatric symptoms such as hallucinations, confusion, and paranoia ([346](#)). Other medications that are used to treat PD symptoms are illustrated in [Table 11.7](#). These drugs may be used as a monotherapy or adjunct therapy to provide symptomatic relief in PD and may have side effects that are important to consider when prescribing exercise to those with PD. Exercise professionals working with individuals with PD are advised to become familiar with these medications (see [Table 11.7](#)). It is important for exercise professionals to be aware of the effects of these medications for two reasons. First, it is important to optimize medication timing and dosage to help obtain the best exercise session possible. Second, it is important to choose exercise interventions that are the least compromised by side effects.

TABLE 11.7 Common Medications for Treatment of Motor Symptoms of Parkinson's Disease

Medication Class	Medication Name	Adverse Effects	Indication
Levodopa-PDDI	Levodopacarbido- Levodopabenserazide	Nausea, orthostatic hypotension, dyskinesia, hallucinations	All motor symptoms
Dopamine agonists	PramipexoleRopiniroleRotigotine	Nausea, orthostatic hypotension, hallucinations, ICDs, edema, increased sleepiness	All motor symptoms
MAOBIs	Selegiline	Stimulant effect, dizziness, headache, confusion, exacerbation of levodopa adverse effects	Early, mild symptoms, and motor fluctuations
	Rasagiline	Headache, arthralgia, dyspepsia, depression, flulike syndrome, exacerbation of levodopa adverse effects, constipation	
COMTIs	Entacapone	Dark-colored urine, exacerbation	Motor fluctuations

Medication Class	Medication Name	Adverse Effects	Indication
		of levodopa adverse effects	
	Tolcapone	Dark-colored urine, exacerbation of levodopa adverse effects, hepatotoxicity	
Unspecified	Amantadine	Hallucinations, confusion, blurred vision, ankle edema, livedo reticularis, nausea, dry mouth, constipation	Gait dysfunction, dyskinesia
Anticholinergic	TrihexyphenidylBenzotropine	Hallucinations, cognitive impairment, nausea, dry mouth, blurred vision, urinary retention, constipation	Tremor

COMTIs, catechol-*O*-methyltransferase inhibitors; ICDs, impulse control disorders; MAOBIs, monoamine oxidase type B inhibitors; PDDI, peripheral dopa decarboxylase inhibitor.

Table adapted from ([347](#)).

Individuals with advanced PD whose motor complications are inadequately managed with medication may choose to undergo deep brain stimulation (DBS). DBS stimulates the brain at high frequencies and thus replaces abnormal high and variable neuronal firing with a consistent pattern ([348](#)). It is very effective for treating bradykinesia, tremor, and rigidity ([349–351](#)). DBS is more effective

than drug therapy in advanced PD in reducing dyskinesias, improving motor function, and increasing quality of life ([352](#)).

Alongside drug therapy and surgery in the treatment and management of PD is exercise, which is a vital component of managing the signs and symptoms of the disease. Regular exercise will decrease or delay secondary sequelae affecting musculoskeletal and cardiorespiratory systems that occur because of reduced PA. Because PD is a chronic progressive disease, sustained exercise is necessary to maintain benefits. Evidence suggests that exercise can reduce disease severity ([353](#), [354](#)) and slow down the progression of the signs of the disease ([355–357](#)). Exercise also improves strength ([353](#), [358–360](#)), aerobic capacity ([355](#), [361](#), [362](#)), gait performance ([362–364](#)), cognition ([365](#)), and quality of life ([363](#)) in individuals with PD. While the precise mechanisms are not known, there is evidence of neuroprotection in preclinical studies ([366–369](#)) as well as neuroplasticity in human studies ([370](#)). Inflammation is increased in PD and can be reduced by exercise ([320](#)).

Exercise Testing

Exercise testing can be used to determine current fitness levels, physiological response to a given exercise bout, and any functional limitations prior to prescribing exercise so that the program can be specified to the individual's particular needs. Individuals with PD can have impaired mobility and problems with gait, balance, and functional ability, which vary from individual to individual. Many individuals with PD experience fluctuations of their motor symptoms from day to day, even from moment to moment ([341](#)). These fluctuations are sometimes attributed to the timing and dosage of their medication and can vary within the same individual and among different individuals. This is important to consider during exercise testing and programming, as the variability of motor fluctuations may influence testing outcomes ([371](#)) and daily exercise performance. As an attempt to control for the variability in testing outcomes due to symptom fluctuations and to help document accurate changes in performance, it may be helpful to utilize a graded symptom checklist on days of pre- and posttesting ([371](#)). The impairments of PD are often accompanied by low levels of

physical fitness (*e.g.*, CRF, muscular strength and endurance, core stability, and flexibility).

There are several special considerations that should be considered prior to performing exercise testing for individuals with PD. Because many individuals with PD are older and have reduced PA levels, assessment of cardiovascular risk (see [Chapter 2](#)) may be warranted prior to performing an exercise test. Autonomic nervous system dysfunction can occur with these individuals ([372](#)), thereby increasing the risk of developing BP abnormalities ([373, 374](#)), which can be further affected by medications ([374](#)). Additionally, individuals with PD may experience orthostatic hypotension. The occurrence of orthostatic hypotension is directly related to the duration and severity of the disease and can also be induced by any of the dopaminergic drugs that are used to manage PD symptoms ([375](#)). Tests of balance, gait, general mobility, ROM, flexibility, muscular strength, core stability, and aerobic capacity are recommended before exercise testing is performed. Results of these tests can guide how to safely exercise test the individual with PD. Static and dynamic balance evaluation and physical limitations of the individual should be used in making decisions regarding testing modes for test validity and safety.

Several studies have suggested that some people with PD have a blunted HR response also known as CI or chronotropic insufficiency ([376, 377](#)). CI is defined as failure to achieve 85% of age-predicted maximal HR during exercise testing. It is also the case that people with PD may take beta blockers to either reduce BP and in some cases, tremor. Beta blockers can reduce resting HR ([378, 379](#)). As such, it is recommended that people with PD have a GXT to determine peak HR so the exercise prescription can be implemented accurately. The actual peak HR of a person with PD can deviate by as much as 30 beats from the age-predicted maximal HR ([380](#)).

For individuals with PD, clinical balance tests include the functional reach test ([381](#)), Berg Balance Scale ([382](#)), Mini-BESTest ([383](#)), tandem stance ([384](#)), single limb stance ([384](#)), and pull tests ([384–386](#)). Functional mobility can be assessed with the Timed Up and Go test ([387, 388](#)) and chair sit-to-stand test ([389, 390](#)). Gait observation can be done during the 10-m walk test at a comfortable walking

speed ([391](#)). The clinimetric properties of all tests used in the study of balance in PD have been summarized by a task force commissioned by the Movement Disorders Society ([391](#)). The sensitivity of gait observations to detect change in gait can be improved by a comprehensive assessment of gait parameters ([392–394](#)). Meanwhile, strength can be evaluated by manual muscle testing, arm curl tests, repetition maximum assessment using weight machines, dynamometers, and chair rise tests just as it is in older adults ([395](#)). Hand-held dynamometers can be used to measure force production of both the upper and lower extremities in individuals with PD but only if performed by the same examiner at each timepoint ([396](#)). Flexibility assessments should focus on truncal posture (C7 to wall test) ([397](#)), hamstring flexibility (90–90 test) ([398](#)), and hip flexor length (Thomas test) ([399](#)). Schenkman et al. provided a summary of impairment measures that can be tested for ([400](#)). Aerobic capacity can be assessed submaximally with the 6-MWT ([401](#)).

Decisions regarding submaximal and maximal exercise testing protocols may be influenced by the stage of PD and the impact this has on posture and balance (see [Box 11.2](#)) or physical limitations of the individual. Use of a cycle ergometer alone or combined with arm ergometry may be more suitable for individuals with severe gait and balance impairment or with a history of falls. However, use of leg/arm ergometers may preclude individuals with PD from achieving a maximum cardiorespiratory response because of early muscular fatigue before the maximal cardiorespiratory levels are attained ([402](#)). Treadmill protocols can be used safely in individuals with early-stage PD or HY stages 1–2 ([355](#), [403](#)). Submaximal tests may be most appropriate in advanced cases (HY stage ≥ 3) or with severe mobility impairment. For individuals with very advanced PD (HY stage ≥ 4) and those unable to perform a GXT for various reasons, such as inability to stand without falling, severe stooped posture, and deconditioning, may require a pharmacologic stress test or dobutamine stress echocardiogram. Moreover, for an individual who is deconditioned, demonstrates lower extremity weakness, or has a history of falling, care and precautions should be taken, especially at the final stages of the treadmill protocol when fatigue occurs and the individual's walking may deteriorate. A gait belt should be worn, and a technician should

stand by close to the subject to guard during the treadmill test. For people with advanced symptoms, symptom-limited exercise testing should be considered as an alternative to heart-rate limited exercise testing, although this recommendation may be modified in certain situations ([404](#)). Symptoms include but are not limited to fatigue, shortness of breath, abnormal BP responses, and signs of discomfort. The Borg RPE scale is a valid tool that can and should be utilized as an aid during exercise testing to monitor physical exertion levels and assess fatigue ([403](#), [405](#)). The RPE scale should be thoroughly explained prior to beginning the exercise test to ensure accurate representations of intensity. Antiparkinsonian medication intake should be noted prior to performing the exercise test due to the individual's susceptibility to experiencing orthostatic hypotension as a side effect of dopaminergic medication. If possible, GXT should be conducted when antiparkinsonian medication is at its peak effect. In addition, it is important to practice walking on a treadmill prior to testing and to use the modified Bruce protocol (see [Figure 4.1](#)) or a modified Gardner protocol ([406](#)) where the speed stays constant and the incline increases at regular intervals until volitional fatigue. If utilizing a modified Gardner protocol, it is important to identify and utilize the fastest speed an individual can safely and comfortably walk at to avoid tests that are too short in duration, too long in duration, or cause the individual to stop the test due to muscular fatigue caused by steep incline as opposed to fatigue of the cardiorespiratory system ([407](#), [408](#)). The Bruce and modified Bruce protocols are the most used protocols for exercise testing on a treadmill ([407](#)), and works well at least through HY Stage 3 ([408](#)).

For individuals with DBS, the signal from the DBS pulse generator interferes with the ECG recording. Therefore, clinicians should consult with a neurologist prior to performing the exercise test in these individuals, and deactivation of the DBS should be done by a trained clinician or neurologist. HR monitoring can be used when DBS is not activated. RPE should be used to monitor physical exertion levels during exercise testing.

In addition to the previously mentioned concerns, standard procedures, contraindications to exercise testing, recommended monitoring intervals, and standard termination criteria are used to exercise test individuals with PD (see

[Chapters 3](#) and [4](#)). There have been no known serious adverse effects exacerbated by the interaction of PD medications and exercise. A few episodes of systolic blood pressure (SBP) drops of greater than 20 mm Hg during treadmill training sessions have been reported ([409](#)). However, no association between medication usage and drop in SBP during exercise was found. Cognitive impairment is frequently observed in individuals with PD, although not all individuals with PD will experience cognitive deficits. This may present as feeling distracted, forgetful, slower thinking and information processing, and difficulty concentrating or managing complex tasks. It is recommended that all testing instructions be explained slowly, concisely, and repeated as necessary.

Exercise Programming

The main goal of the Ex R_x for individuals with PD should ultimately be to slow down the rate at which the signs of the disease progress, reduce the signs and symptoms of the disease, reduce comorbidities, prevent secondary complications from muscle disuse, and improve functional ability, independence, and quality of life. The FITT principle of Ex R_x should address CRF, muscular strength and endurance, flexibility, neuromotor training, and motor control. The ability to rigorously quantify, measure, and prescribe aerobic, resistance, and flexibility exercises has resulted in FITT principles of exercise design that focus predominantly on these three domains. However, the importance of incorporating neuromotor training and exercises that enhance motor control should not be overlooked or undervalued, regardless of the difficulty associated with determining a precise prescription for these types of exercises. As a rule, neuromotor training should progress exercise motor complexity (the coordinative and control requirements of the motor activity) and quantitative training parameters (*i.e.*, FITT principles).

Also, the importance of identifying exercise modalities that an individual enjoys should not be underestimated because adherence is a key ingredient to gaining maximal benefit from exercise. Because PD is a chronic and progressive disorder, an exercise program should be prescribed early when the individual is first diagnosed and continue for a regular, lifetime basis. The Ex R_x should be

reviewed and revised as PD progresses because different physical problems occur at different stages of the disease.

The primary key health outcomes of an exercise program designed for individuals with PD are improved (a) aerobic capacity, (b) muscle strength and endurance, (c) gait mechanics, (d) balance, (e) physical function, and (f) flexibility. Because the FITT principle of Ex R_x recommendations for individuals with PD is based on a smaller literature, the Ex R_x for healthy adults generally applies to those with PD ([410](#)); however, the limitations imposed by the disease process should be assessed, and the Ex R_x should be individually tailored accordingly.

FITT FITT RECOMMENDATIONS FOR INDIVIDUALS WITH PARKINSON DISEASE's ([353](#), [355](#), [358](#), [359](#))

	Aerobic	Resistance	Flexibility	Neuromotor
Frequency	3–4 d · wk ⁻¹	2–3 d · wk ⁻¹	≥2–3 d · wk ⁻¹ with daily being most effective	2–3 d · wk ⁻¹
Intensity	High intensity (80%–85% maximum heart rate [HR _{max}]) for mild-to-moderate PD; moderate intensity (60%–65% HR _{max}) for deconditioned individuals or those with more advanced PD; attempt to progress to 80%–85% HR _{max} if possible.	30%–60% of one repetition maximum (1-RM) for individuals beginning to improve strength; 60%–80% 1-RM for more advanced exercisers	Full extension, flexion, rotation, or stretch to the point of slight discomfort	N/A
Time	30 min of continuous or accumulated exercise	1–3 sets of 8–12 repetitions, beginning with 1 set and working up to 3 sets	Hold static stretch for 10–30 s; 2–4 repetitions of each exercise	30–60 min
Type	Prolonged, rhythmic activities using large muscle groups (<i>e.g.</i> , walking, running, cycling, swimming, rowing, elliptical)	Major muscle groups of upper and lower body on nonconsecutive days. Avoid free weights for individuals in more advanced stages of the disease; focus on weight machines and other resistance devices (<i>e.g.</i> , bands, body weight).	Slow static stretches for all major muscle groups working on increasing range of motion	Exercises involving motor skills (<i>e.g.</i> , balance, agility, coordination, gait, dual tasks) such as tai chi, yoga, multidirectional step training and instability training

Accumulating evidence suggests that long-term aerobic training may attenuate PD progression and does reduce the signs of PD ([411](#)–[413](#)). General aerobic training at a moderate intensity may improve aerobic fitness, fatigue, mood, executive function, and quality of life in those with mild-to-moderate PD ([414](#), [415](#)). High-intensity endurance exercise (80%–85% HR_{max}) can be safely

prescribed to individuals with early-stage PD (HY stages 1–2) and has been shown to attenuate the worsening of motor signs ([355](#)). Individuals should be encouraged to exercise at high intensity to the extent that they are willing to do this. Individuals at HY stages ≥ 3 should also consider high-intensity aerobic exercise but check with their physician if they have autonomic dysfunction.

Resistance training is well tolerated in individuals with mild-to-moderate PD ([416](#)) and should be progressive ([353](#)). Physical parameters, such as muscle strength and power ([359](#)), movement speed ([417](#)), and dynamic balance, along with quality of life parameters such as fatigue, are improved with resistance training in individuals with PD ([416](#), [418](#)), with strength improvements being similar compared to neurologically normal controls ([364](#)). Therefore, recommendations for resistance exercise in neurologically healthy older adults ([419](#)) may be applied to individuals with PD ([364](#)). A recent modification to progressive resistance training that has proven beneficial for individuals with PD is the incorporation of unstable devices into the resistance exercises ([420](#)). Results from this type of training have shown improved mobility, motor signs, neuromuscular outcomes, balance, reduced cognitive impairment, reduced fear of falling, and improved quality of life, which may be attributed to the progression of exercise motor complexity (*i.e.*, degree of instability) and quantitative training parameters (*i.e.*, frequency, intensity, and time) ([420–422](#)).

Recommendations for Neuromotor Exercise for Individuals With Parkinson Disease's

Endurance exercise, resistance exercise, and stretching will benefit posture and balance ([423](#), [424](#)), but there are also benefits to performing exercises specifically for posture, balance, and mobility. Balance impairment and falls are major problems in individuals with PD, with approximately 61% of people with PD experiencing at least one fall and 39% of people suffering from recurring falls ([425](#)). Balance training is a crucial exercise in all individuals with PD. Postural instability and balance performance in individuals with mild-to-moderate PD can be improved with PA and exercise ([426](#)). Static, dynamic, and balance training during functional activities should be included. Clinicians should take steps to

ensure the individual's safety (*e.g.*, using a gait belt and nearby rails or parallel bars and removing clutter on the floor) when using PAs that challenge balance. Training programs may include a variety of challenging PAs (*e.g.*, multidirectional step training, step-up and step-down, reaching forward and sideways, obstacles, turning around, walking with suitable step length, standing up, and sitting down) ([322](#), [427](#), [428](#)). When external cueing in the form of rhythmic auditory stimulation is utilized during multidirectional step training, individuals with PD show improvements in functional gait parameters, including balance, and maintain these improvements longer than when external cueing is not utilized ([322](#)). Tai chi, Tango, and Waltz are other activities to improve balance in PD ([429–431](#)). Incorporating unstable devices, such as balance pads, dyna discs, balance discs, BOSU balls, or Swiss balls, into a resistance training regimen has also been shown to improve balance in PD ([420](#)). Osbourne and colleagues have provided a comprehensive set of recommendations consistent with suggestions made in this section ([344](#)).

Exercise Training Modalities and Considerations

- The selection of the exercise modality should be dependent on the individual's clinical presentation of PD severity and personal preference.
- Treadmill exercise, stationary cycles, recumbent cycles, ellipticals, rowers, and arm ergometers are safe and effective modalities for aerobic training.
- Water exercises, robotic gait training, virtual reality training, mental practice, boxing, and Nordic walking are effective for some people living with PD ([432](#)).
- Dance programs, especially ones that include visual and auditory cues and rhythmic tasks, have been shown to improve some of the motor characteristics of the disease and improve functional mobility ([433](#)).
- Tai chi has been shown to be effective in improving motor function, balance, and quality of life in individuals with PD, with limited evidence also showing improvements in fall risk and depression ([434](#)). Free weights may be utilized for individuals with mild-to-moderate PD but may become unsafe at more advanced stages and in those with increased severity of tremor, especially during exercises that involve overhead lifting.

- Weight machines and other resistive devices such as resistance bands or body weight are safe alternatives to free weights. It may be necessary to modify certain exercises due to decreased ROM associated with PD. During resistance training, emphasize extensor muscles of the trunk and hip to prevent faulty posture. Train all major muscles of the lower extremities to maintain mobility.
- Flexibility and ROM exercises should include slow static stretches and passive ROM exercises for all major muscle groups and joints, with an emphasis on the upper extremities and trunk. Spinal mobility and axial rotation exercises are recommended for all severity stages ([435](#)). Neck flexibility exercises should be emphasized because neck rigidity is correlated with posture, gait, balance, and functional mobility ([436](#)).
- Functional exercises such as the sit-to-stand, step-ups, turning over, and getting out of bed as tolerated should be incorporated in an exercise program to improve neuromotor control, balance, and maintenance of ADL.
- Incorporating the concepts of the Lee Silverman Voice Training BIG program (specific exercises involving large amplitude, exaggerated movement patterns) into functional exercise has demonstrated benefit ([437](#), [438](#)).

Special Considerations

- A blunted HR response during exercise testing and training may be observed in those with CI. It is important to take this into consideration when prescribing exercise based on age-predicted maximal HR, especially for those who have not undergone maximal exercise testing to determine peak HR ([377](#)).
- Some medications used to treat PD further impair autonomic nervous system functions ([374](#)). Levodopa/carbidopa may produce exercise bradycardia and transient peak dose tachycardia and dyskinesia. Caution should be used in testing and training an individual who has had a recent change in medications because the response may be unpredictable. Several nonmotor symptoms may burden exercise performance ([Box 11.3](#)) ([439](#), [440](#)).
- The outcome of exercise training varies significantly among individuals with PD because of the complexity and progressive nature of the disease.

- Cognitive decline and dementia are common nonmotor symptoms in PD ([342](#)) and may burden the training and progression of the disease. It is recommended that instructions be explained slowly, clearly, concisely, and repeated as necessary. Exercises should be demonstrated and broken down into a series of short, simple steps. Utilize verbal, visual, and tactile cues while instructing the individual.
- Fall history should be recorded and fall prevention/reduction and education should be incorporated into the exercise program. Precautions should be taken to prevent falls whenever possible, such as avoiding narrow and/or uneven walkways, avoiding sharp turns and pivots, removing any obstacles on the floor, and avoiding complex dual task exercises or long complex movements ([441](#), [442](#)).
- Balance training should be emphasized for all individuals with PD ([443](#)).
- Use dual tasking or multitasking with novice exercisers with great care. Individuals with PD have difficulty paying full attention to multiple tasks. Dual task performance during gait has been correlated with increased risk of falling and diminished quality of life ([443](#)). Dual task training has been shown to significantly increase stride length and cadence in individuals with PD ([444](#)) and may also better prepare an individual with PD for responding to a balance perturbation ([445](#)). Dual task training can be incorporated into training once the individual is able to perform well in a single task.
- Visual and auditory cueing can be used to improve gait in persons with PD during exercise ([446](#)).
- Some individuals with PD experience freezing of gait (FOG), which is an intermittent feeling that their feet are “frozen” or stuck to the floor when trying to walk. While resistance and balance training do not seem to improve FOG in individuals with PD ([447](#)), utilizing both visual and auditory cues will help during, but will not necessarily alleviate freezing episodes ([448](#)). Utilizing exercise regimens that limit the opportunity for freezing episodes such as stationary cycling and resistance exercises alongside auditory cues are additional ways to handle FOG.

- Although no reports exist suggesting resistive exercise may exacerbate symptoms of PD, considerable attention must be paid to the development and management of fatigue ([449](#)).

Box 11.3 Nonmotor Symptoms in Parkinson Disease's ([439](#))

Domains	Symptoms
Cardiovascular	Symptomatic orthostasis, fainting, light-headedness
Sleep/fatigue	Sleep disorders, excessive daytime sleepiness, insomnia, fatigue, lack of energy, restless legs
Mood/cognition	Apathy, depression, loss of motivation, loss of interest, anxiety syndromes and panic attacks, cognitive decline
Perceptual problems/hallucinations	Hallucinations, delusion, double vision
Attention/memory	Difficulty in concentration, forgetfulness, memory loss
Gastrointestinal	Drooling, swallowing, choking, constipation
Urinary	Incontinence, excessive urination at night, increased frequency of urination
Sexual function	Altered interest in sex, problems having sex
Miscellaneous	Pain, loss of smell/taste and appetite/weight, excessive sweating, fluctuating response to medication

Future Directions

Exercise might play a neuroprotective role in individuals with PD; however, the direct evidence is limited to animal models ([450–454](#)). The data from animal models is very encouraging both in terms of deficits directly linked to dopamine depletion such as slowness of movement ([452, 455](#)) and in terms of deficits not linked to dopamine depletion but which affect those with PD such as hyposmia, anhedonia, lack of novelty-seeking behavior, depression, and anxiety ([451](#)).

Although some studies do suggest promising findings with respect to changes in the brain of those with PD ([456, 457](#)), and exercise-induced increases in blood brain-derived neurotrophic factor levels in human with PD ([458](#)), none of these changes relate to the growth of new neurons or the protection of neurons.

Therefore, there are grounds for cautious optimism that exercise will be shown to be neuroprotective once techniques are developed to address this question in humans. Future studies should be carried out to elucidate whether exercise is in

fact neuroprotective, determine the influence of different types of exercise on neurogenesis and neuroplasticity, and establish the combination of interventions that have the most beneficial effect on both the motor and nonmotor symptoms of PD. These studies will provide clarity on the most advantageous ways to modify disease progression and the biological mechanisms behind it. In addition, we think we are getting closer to a point we will be able to prescribe a personalized Ex Rx for people with PD. We will also be able to monitor blood chemistry to understand how to reduce inflammation [320] and restore the balance of the neuroendocrine system.

ONLINE RESOURCES

American Parkinson Disease Association: <http://www.apdaparkinson.org>

Davis Phinney Foundation: <http://www.davisphinneyfoundation.org>

European Parkinson's Disease Association: <http://www.epda.eu.com>

National Institute of Neurological Disorders and Stroke:

<https://www.ninds.nih.gov/health-information/disorders/parkinsons-disease?search-term=parkinsons%20disease>

National Parkinson Foundation: <http://www.parkinson.org/>

The Michael J. Fox Foundation for Parkinson's Research:

<http://www.michaeljfox.org>

The Parkinson Alliance: <http://www.parkinsonalliance.org/>

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CHAPTER 12

Behavior-Based Strategy to Promote Physical Activity and Exercise

INTRODUCTION

The purpose of this chapter is to provide exercise professionals a basic understanding of how to assist individuals with adopting and adhering to the Ex R_x recommendations that are made throughout the *Guidelines*. [Chapter 1](#) of the *Guidelines* focuses on the public health recommendations for a physically active lifestyle; yet most adults in the United States do not engage in the recommended amounts of physical activity (PA) ([1–3](#)). This highlights that simply providing knowledge and promoting awareness of the importance of Ex R_x does not always lead to behavior change or consistent engagement in recommended amounts of PA and exercise ([4](#)).

Behavioral theories help to explain an individual's behavior and help to provide a framework and guide an exercise programming strategy. However, it is often the key targets or constructs of these theories that exercise and health professionals will use to facilitate engagement in PA and exercise. Behavior change requires a multifaceted understanding of individuals and/or population groups (*e.g.*, age, gender, socioeconomic status, education, ethnicity, identity) and their current behavior. Additionally, knowledge of what may explain their behavior (engaging or not engaging in exercise and why), future goals, and environment is also important throughout the behavior change process.

This chapter begins by establishing a basic understanding of behavioral theories and how to promote engagement in exercise by using key constructs from these theories. [Table 12.1](#) highlights key terms that will assist with navigating the content in these sections. The application portion of the chapter will focus on behavioral targets that are frequently included in Ex R_x and program design along with techniques to effectively promote PA behavior change and an overview for developing client-centered conversations.

TABLE 12.1 Key Terminology

Key Term	Definition
Theory	A structured approach for understanding and predicting behavior such as physical activity.
Model	A visual presentation that illustrates a phenomenon or theory.
Behavior	Anything that a living being does. This includes not only actions such as engaging in physical activity but also thoughts and expressions of emotion.
Construct	A set of concepts or variables that are included within theories.

From ([5, 6](#)).

Additional key terms and definitions are included throughout the *Theoretical Foundations for Understanding Exercise Behavior* and *Relevant Targets to Promote Physical Activity Behavior Change* sections below.

THEORETICAL FOUNDATIONS FOR UNDERSTANDING EXERCISE BEHAVIOR

When designing exercise prescriptions and programs, various behavior change theories can be valuable for guiding effective and adaptive Ex R_x. Theories and models provide a foundation for understanding what drives Ex R_x like the factors that may promote or deter an individual from being physically active. Different theories offer unique insights into human behavior and behavior change processes. The choice of which theoretical approach to use will depend on the person or population, and their specific goals and characteristics. The purpose of this section is to provide a general overview of the most widely used theories and models that may promote and explain PA behavior.

Social Cognitive Theory

Social cognitive theory (SCT) is a theoretical framework that emphasizes the interaction between personal factors (*e.g.*, emotion, personality, cognition, biology), behavior (*e.g.*, past and current experiences and achievement), and environment (physical, social, and cultural). It is important to recognize that these three factors are dynamic, they influence each other, and can change over time ([7](#)).

For example, an individual who begins an exercise program may feel a sense of accomplishment, encouraging even more exercise, which leads to making their environment better for encouraging exercise (*e.g.*, buying home exercise equipment). Conversely, another individual may start an exercise program, work too hard and feel fatigued, lose motivation, and move the exercise equipment to the basement to make the environment less optimal for encouraging exercise.

Central to SCT are three key concepts, (a) self-efficacy, (b) outcome expectations and expectancies, and (c) self-regulation or self-control. Self-efficacy refers to one's beliefs that they are capable of successfully completing a course of action such as exercise ([8](#), [9](#)). Outcome expectations (what is *anticipated* to occur) or expectancies (what one *hopes* will occur) are anticipatory results of a behavior (*e.g.*, PA) and the value placed on the outcomes or results. If specific outcomes are seen as likely to occur and are valued, then it is more likely that the behavior will occur ([10](#)). Self-regulation/self-control is an individual's ability to set goals, monitor progress toward those goals (or self-monitor), problem solve when faced with barriers, and engage in self-reward. Additionally, SCT incorporates the concept of reinforcement (positive or negative), which can influence whether a behavior is repeated in the future ([7](#)). The *Relevant Targets to Promote Physical Activity Behavior Change* section breaks down key strategies to use these constructs in exercise programming. The theory and strategies derived from SCT have been successfully applied in exercise interventions across diverse populations ([11–13](#)).

Transtheoretical Model

The transtheoretical model (TTM) suggests that individuals go through distinct stages when adopting a behavior. The TTM includes five stages of change: precontemplation (*i.e.*, no intention to be regularly active in the next 6 months), contemplation (*i.e.*, intending to be regularly active in the next 6 months),

preparation (*i.e.*, intending to be regularly active in the next 30 days), action (*i.e.*, regularly active for <6 months), and maintenance (*i.e.*, regularly active for ≥6 months) ([14](#), [15](#)).

Key distinctions between each stage are intention (early stages) and action (later stages). Prior to moving to the next stage, an individual must have a certain degree of readiness, and it may be influenced by key constructs such as (a) processes of change, (b) decisional balance, and (c) self-efficacy ([15](#), [16](#)). The 10 processes of change, both cognitive and behavioral in nature, consider strategies that can be used to move an individual through the five stages of change. Decisional balance involves weighing the pros and cons of changing exercise behavior. At each stage, the balance between the pros and cons of being more physically active may influence whether or not a person moves to the next stage. Self-efficacy is lowest in the earliest stages of change and highest in the latest stages of change ([17](#), [18](#)). [Table 12.2](#) provides a combined overview of how processes of change, decisional balance, and self-efficacy are overlaid across the five stages.

TABLE 12.2 Transtheoretical Model Constructs and Strategies

Stage of Change	Precontemplation	Contemplation	Preparation	Action	Maintenance	General Example
	<i>Consciousness Raising</i> Raising awareness of a behavior and its consequences					Reads cont that focus health ben of physic activity (F
	<i>Dramatic Relief</i> Emotionally responding to a behavior and its consequences					Express feelings related t results c engaging not engag in PA
	<i>Environmental Reevaluation</i> Assessing how a behavior impacts personal environment (physical, social)					Considers l being activ inactive impacts otl
Process of Change	<i>Social Liberation</i> Seeing opportunities to allow for the behavior change					Realizes sc trends th promote deter P2
		<i>Self-Reevaluation</i> Connecting the individual to the behavior and consequences				Ask self ab how the might fe about bei active o inactive
			<i>Self-Liberation</i> Committing to behavior change			Sets goal verbalize intent, ar makes plan be activ
				<i>Reinforcement Management</i> Rewarding and reevaluating behavior		Works tow activity gc and rewal
				<i>Helping Relationships</i> Building a support system		Identifies social supp system aro being acti
				<i>Counter Conditioning</i> Discovering positive alternatives to current		Identifie active behaviors

				behaviors	replace inactive behavior
				<i>Stimulus Control</i> Avoiding triggers and relapses	Manages environm to promo activity a avoid inactivity c
Stage of Change	Precontemplation	Contemplation	Preparation	Action	Maintenance
Decisional Balance	Con > Pros	Cons > Pros	Cons = Pros	Cons < Pros	Cons < Pros
Self- Efficacy	Lowest Highest				

[Box 12.1](#) provides strategies for how an exercise professional might help individuals progress to the next stage. The *Relevant Targets to Promote Physical Activity Behavior Change* section of the chapter provides greater details on how to develop many of the strategies mentioned.

Box 12.1 Example Strategies to Facilitate Stage Transitions

Precontemplation → Contemplation

- Provide information about the benefits of regular PA.
- Discuss strategies for overcoming perceived barriers such as “Activity can be done in shorter and accumulated bouts if they don’t have the time for longer exercise sessions.”
- Have them visualize what they would feel like if they were physically active with an emphasis on short-term, easily achievable benefits of activity such as sleeping better, reducing stress, and having more energy.
- Explore how their inactivity impacts individuals other than themselves such as their spouse and children.

Contemplation → Preparation

- Explore potential solutions to their PA barriers.
- Assess level of self-efficacy and begin techniques to build efficacy.
- Emphasize the importance of even small steps in progressing self-toward being regularly active.
- Encourage viewing oneself as a healthy, physically active individual.

Preparation → Action

- Help develop an appropriate plan of activity to meet their PA goals and use a goal setting worksheet or contract to make it a formal commitment.
- Use reinforcement to reward steps toward being active.
- Teach self-monitoring techniques such as tracking time and distance.
- Continue discussion of how to overcome any obstacles they feel are in their way of being active.

Action → Maintenance

- Provide positive and contingent feedback on goal progress.
- Explore different types of activities they can do to avoid burnout.
- Encourage them to work with and even help others become more active.
- Discuss potential rewards that can be used to maintain motivation.

Movement through these stages is not always linear. For example, when a person attempts to increase PA, there may be repeated episodes of relapse (decrease or stopping exercise). However, this does not mean that the process completely restarts (*e.g.*, return to precontemplation). An individual may reenter the process at any of the stages. The exercise professional is key for facilitating a positive restart so the client can successfully increase PA even after several unsuccessful attempts. Stage-based interventions, across various groups and populations, have been effective in helping individuals make progress toward and/or becoming regularly physically active ([17–19](#)).

Health Belief Model

The health belief model (HBM) suggests that behavior change depends on an individual’s beliefs about whether or not they are susceptible to disease and their individual perceptions of the benefits of trying to avoid the disease. In order to evoke readiness to change, an individual needs to believe that they are

susceptible to a disease *and* believe that the benefits of taking action outweigh the perceived barriers (20).

The HBM suggests six constructs that may be useful for motivating individuals to change their exercise behavior because of health issues (Table 12.3).

TABLE 12.3 Health Belief Model Constructs and Strategies

Construct	Exercise-Specific Definition	Change Strategy
Perceived susceptibility	Beliefs about the chances of getting a disease/condition if they do not exercise	• Explain risk information based on current activity, family history, other behaviors, etc.
Perceived severity	Beliefs about the seriousness/consequences of disease/condition as a result of inactivity	• Refer individual to medically valid information about disease. • Discuss different treatment options, outcomes, and costs.
Perceived benefits	Beliefs about the effectiveness of exercising to reduce susceptibility and/or severity	• Provide information on benefits of exercise to preventing/treating condition or disease. • Provide information regarding all of the other potential benefits of exercise (<i>e.g.</i>, quality of life, mental health).
Perceived barriers	Beliefs about the direct and indirect costs associated with exercise	• Discuss Ex R_x options to minimize burden. • Provide information on different low-cost activity choices.
Cues to action	Factors that activate the change process and get someone to start exercising	• Help individuals look for potential cues. • Ask the individual what it would take for them to get started.
Self-efficacy	Confidence in ability to exercise	• Assess level of confidence for different types of activity. • Use self-efficacy building techniques to enhance exercise confidence.

While not as widely studied as other theories for PA, the HBM may be most suitable for understanding and intervening with populations that are motivated to be physically active primarily for health reasons (21). Constructs from this theory may play an important role as the field moves toward implementing more medical referrals to certified exercise professionals and has been applied in cardiac rehabilitation and diabetes mellitus prevention and management (22, 23).

Social Ecological Models

The defining feature of social ecological behavior models is emphasis on relations between individuals and their physical environments ([24](#), [25](#)). Ecological models suggest that behavior results from influences at multiple levels ([Table 12.4](#)). Importantly, environmental factors influence behavior directly and indirectly through an individual's perceptions. Targeting aspects of the individual are important, but if a physical environment is not conducive to changing one's lifestyle, then the exercise intervention will not be successful. A key belief is that interventions are most likely to be effective when they target multiple levels ([24](#), [25](#)). For example, adding a walking path in a park is most likely to be effective in increasing PA when there is a campaign to promote awareness of the path, perhaps combined with an intervention that targets individual beliefs and motivations about walking. Although research examining the impact of interventions based on ecological models is limited in part because of the complexity of delivering these multilevel interventions, results examining the impact of interventions based on ecological models is promising ([25](#)). Barriers at multiple levels need to be considered when developing exercise programming (see "Barriers and Problem Solving" section in *Relevant Targets to Promote Physical Activity Behavior Change*).

TABLE 12.4 Levels of Social Ecological Model and Potential Physical Activity Intervention Strategies

Social Ecological Level	Components	Potential Change Strategies
Intrapersonal factors	• Knowledge, attitudes, behaviors, beliefs, perceived barriers, motivation, enjoyment • Skills and self-efficacy • Demographics (age, sex, education, and socioeconomic and employment status)	• Focus on changing individual's knowledge, skills, and attitudes for physical activity. • Target exercise enjoyment. • Promote skill learning and confidence building for physical activity. • Use demographic information to identify subgroups at risk or subgroups that require different approaches to programming.
Interpersonal factors/social environment	• Family, spouse, or partner • Peers • Coworkers • Access to social support • Influence of health professionals • Community norms • Cultural background	• Use community education, support groups, and peer programs. • Social marketing campaigns may promote positive community attitudes and awareness toward participation in physical activity. • Use consistent, accurate, and encouraging messages to promote physical activity.
Organizational factors	• Schools, workplaces, faith-based settings, and community organizations	• Create opportunities for organizations, at both the individual and group level, to adopt or increase physical activity.
Physical environment	• Natural factors such as weather or geography • Availability and access to exercise facilities • Aesthetics or perceived qualities of facilities or the natural environment • Safety such as crime rates and traffic • Community design • Public transportation options	• Create walking trails or parks. • Enhance existing environments (<i>e.g.</i>, park/neighborhood cleanups). • Help individuals become more aware of opportunities for physical activity in their communities (<i>e.g.</i>, parks, trails, community centers).
Policy	• Urban planning policies	• Align physical activity participation with priorities such as reducing reliance on fossil

Social Ecological Level	Components	Potential Change Strategies
	• Education policies such as physical education classes • Health policies • Environmental policies • Workplace and other organizational policies	- fuels and the reduction of greenhouse gas emissions. • Emphasize the importance of regular physical education. • Encourage workplaces to provide support for physical activity. • Vote for politicians who advocate for increased physical activity.

Self-Determination Theory

Self-determination theory (SDT) is primarily focused on understanding motivation ([26–28](#)). The theory proposes that motivation exists on a continuum from amotivation (not motivated) to levels of extrinsic motivation (externally motivated) then to intrinsic motivation (internally motivated). [Table 12.5](#) helps break down the various stages and sources of motivation. The key is understanding the difference between extrinsic and intrinsic motivation. Extrinsic motivation in all its forms refers to doing something for reasons that are external to the individual such as being physically active to make oneself more attractive to others, out of sense of obligation, to pursue rewards, or out of fear of punishment. This is the opposite of intrinsic motivation that refers to doing something, like exercise, because it is inherently enjoyable or interesting to the individual ([27, 28](#)).

TABLE 12.5 Self-Determination Continuum and Constructs

	Becoming More Self-Determined					
Motivation Level	Amotivation	Extrinsic Motivation				Intrinsic Motivation
Type of Motivation	Lack of motivation — No intention Known as “going through the motions”	<i>External:</i> Externally driven (<i>e.g.</i> , rewards, punishment)	<i>Introjected:</i> Somewhat externally driven (<i>e.g.</i> , compulsion, guilt, self-esteem)	<i>Identified:</i> Somewhat internally driven (<i>e.g.</i> , sets goals)	<i>Integrated:</i> Internally driven (<i>e.g.</i> , participates because they find it to be important but not enjoyable)	Purely internally driven (<i>e.g.</i> engaging in exercise for enjoyment, curiosity, interest, a sense of accomplishment)
Example	Does not exercise or when in exercise situations gives minimal effort	Exercises to get a reward for going to the gym multiple times	Exercises to get compliments from others at an upcoming event	Exercises to prepare for a leg of a relay race	Exercises to improve blood pressure because the physician recommended it	Exercises daily because of how it makes a person feel
Influences	Relatedness		Competence		Autonomy	

Important to SDT is that motivation is influenced by three primary psychological needs individuals attempt to satisfy across the motivation continuum: (a) relatedness or the ability to experience meaningful social interactions with others, (b) demonstration of competence (understanding) or mastery of skill, and (c) autonomy or self-determination, which is having the ability to make one’s own decisions. Individuals with intrinsic motivation have the highest degree of autonomy and are interested in engaging in exercise simply for the satisfaction, challenge, or pleasure it brings, whereas extrinsic motivation appears to be more driven by the constructs of relatedness and competence (26) (see [Table 12.5](#)).

Theory of Planned Behavior

The theory of planned behavior (TPB) helps to explain exercise intentions (29, 30). Intention to perform a behavior is the primary factor of actual behavior (31). However, there is less evidence that interventions based on the TPB are effective at increasing PA levels (32, 33). Intentions reflect an individual’s perceived probability or likelihood that they will exercise but do not always translate directly into behavior.

The TPB includes the following constructs that help explain the gap between intentions moving to action (engaging in exercise) (34). These include (a) attitudes, (b) subjective norms, and (c) perceived behavioral control. Attitudes are the degree to which an individual has a favorable or unfavorable evaluation of behavioral outcomes. For example, if a person has a negative attitude about going to the gym, that may weaken the intention and reduce the likelihood of going regularly. Subjective norms are

about whether an individual believes important people in their life value a behavior. For example, if a family regularly goes for a walk after dinner, an individual may be more likely to continue this behavior because it is the norm. Finally, perceived behavioral control is the perception of how easy or difficult it is to engage in a behavior, which may be related to barriers. For example, if an individual perceives low control over their ability to engage in exercise when the weather is poor, they are more likely to skip an exercise session if it is raining.

The way that all three of these constructs impact each other needs to be considered when trying to understand how intentions can or cannot turn into action. An individual's subjective norms may lead them toward healthier behavior, and a more positive attitude, but powerful barriers outside of the individual's control may limit exercise participation. An individual is more likely to move from intention to being more physically active (taking action) when all three constructs align and (a) they believe that exercise would lead to desired outcomes, (b) is valued by someone whose opinion they value, and (c) is within the individual's own control. Strategies for targeting intentions can be found in the "Intention" section of *Relevant Targets to Promote Physical Activity Behavior Change*.

Dual-Processing Theories

Dual-processing theories refer to a class of theories that focus on both the conscious and nonconscious aspects of behavior ([35](#), [36](#)). For example, many people will state a desired motivation or intention to exercise (conscious motivation) but at the same time that they experience dread of exercise (nonconscious motivation). The dread component is part of hedonic motivation, an automatic association between a behavior and previous affective responses to the behavior ([37](#)). Affective response describes how an individual's affect changes as a result of engaging in a behavior. Affect is a neurobiological state that encompasses (often involuntary) physiological reactions and subjective evaluations of experiences, to characterize how an individual feels about a given situation, experience, or behavior. For example, how an individual feels during a single exercise session — the affective response — is predictive of exercise behavior 6 and 12 months later, and negative affective responses to exercise are typically associated with exercise avoidance ([38](#), [39](#)).

Hedonic motivation suggests that people will tend to seek out experiences that are pleasurable and enjoyable and avoid those that bring displeasure. The key distinction between this type of motivation and conscious motivation is that it is automatic, it happens without thinking, and it does not consider the pros and cons, goals, or intentions. The more positive the experiences surrounding a behavior, the greater the hedonic desire becomes for the behavior. Likewise, the more negative the experiences surrounding a behavior, the greater the hedonic dread becomes for a behavior ([37](#)). When people struggle to start or continue an exercise program, it can be a result of their hedonic dread of exercise influencing their behavior despite their conscious desire or intention to exercise. See "Affect" section in *Relevant Targets to Promote Physical Activity Behavior Change* for more strategies on how to incorporate affect responses into Ex R_x.

RELEVANT TARGETS TO PROMOTE PHYSICAL ACTIVITY BEHAVIOR CHANGE

Many theories share similar concepts and ideas about how to understand PA behavior but use different combinations of constructs or behavioral targets to help explain why people are or are not active.

[Figure 12.1](#) provides a visual representation of the key targets that provide an opportunity for exercise and health professionals to intervene.

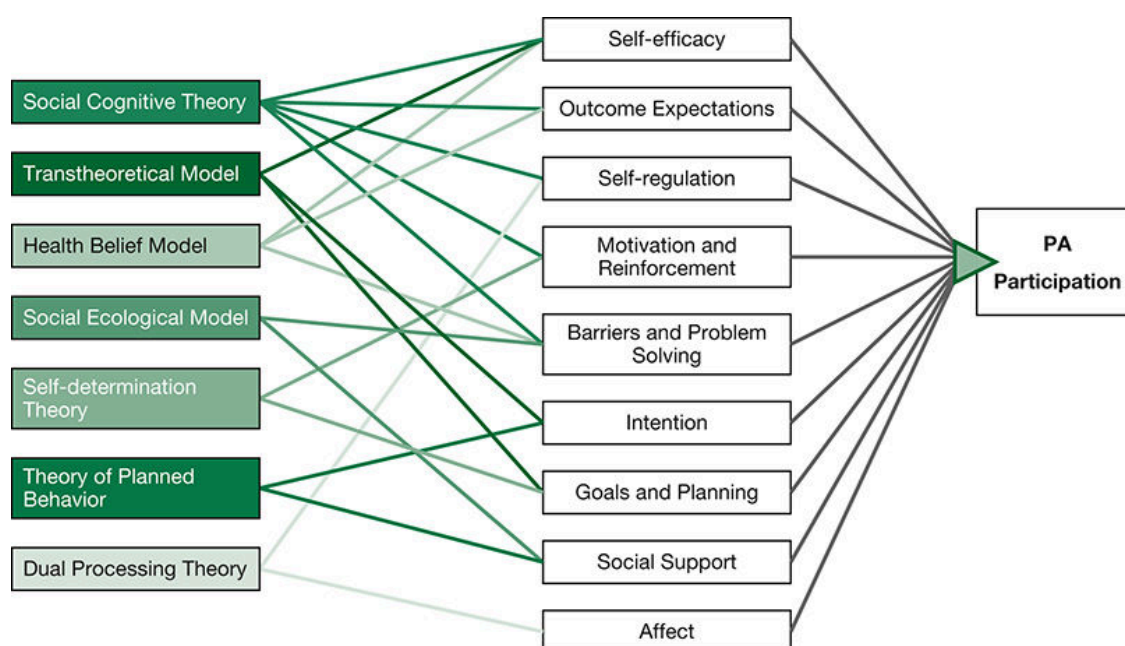


Figure 12.1 Theory-based targets for modifying physical activity behavior.

PA adoption can be strengthened by focusing these key behavior-change targets and skills (see [Figure 12.1](#)). These targets are important parts of many of the theories that were reviewed, but often they are not used in isolation. Rather, multiple targets are used when developing exercise programs or working with an individual depending on the topic and need at the time of intervening. These targets can assist with initiating, progressing, and maintaining exercise programs and should be used to inform the specific components of the Ex Rx (FITT). This section provides a further understanding of these constructs along with the application of strategies and skills exercise professionals can apply to promote PA behavior.

Self-Efficacy

Self-efficacy is an individual's confidence in their ability to carry out actions necessary to perform certain behaviors ([9](#)). Increased self-efficacy is associated with greater levels of PA ([11](#)). Self-efficacy is situation specific; for example, an individual may have high self-efficacy for walking but a low self-efficacy for resistance training. Exercise and health care professionals can promote being physically

active through various sources of task self-efficacy. Task self-efficacy refers to an individual's belief that they can actually do the specific activity or exercise ([40, 41](#)). Sources of task self-efficacy ([Table 12.6](#)) include (a) mastery experiences, (b) vicarious experiences, (c) verbal persuasion, (d) physiological feedback, and (e) emotional states.

TABLE 12.6 Source of Self-Efficacy and Strategies for Enhancing Physical Activity (PA)

Source of Self-Efficacy	Description	Example	Strategies for the Exercise Professional
Mastery experiences	Successfully engaging in a behavior. Once an individual participates in a behavior that can impact future engagement in the behavior.	Successfully attending a personal training session influences an individual's belief that they can attend the next session.	<i>Promoting Mastery:</i> • Set realistic goals that can be achieved. • Progress Ex Rx gradually over time. • Provide proper PA instruction and demonstration. • Use logs to track PA progress for accountability.
Vicarious experiences	Learning about a behavior from others who are similar. Seeing others succeed at a behavior strengthens an individual's belief in their ability to engage in the behavior.	Seeing other gym attendees that are similar to the individual who have successfully engaged in personal training sessions.	<i>Promoting Vicarious Experiences:</i> • Have appropriate exercise leaders that individuals can identify with. • Discuss/show "success" stories of individuals with similar backgrounds and characteristics. • Use materials (video, educational content, etc.) that are tailored to the individual/population. • Build support groups of individuals with similar goals and characteristics.
Verbal persuasion	Verbal praises and support for a behavior. Receiving positive feedback and encouragement about PA behaviors strengthens an individual's belief in their ability to continue engaging in the behavior.	At each personal training session, the exercise professional reviews progress and provides encouraging feedback.	<i>Executing on Verbal Persuasion:</i> • Express confidence in the individual's abilities to engage in PA. • Prompt discussion of previous successful attempts at PA. • Discuss existing skills and knowledge, which can help to support PA participation. • Highlight progress toward goals that has been made, even if small.

Source of Self-Efficacy	Description	Example	Strategies for the Exercise Professional
Physiological feedback	Assessment of physiological symptoms associated with the behavior change. Examining and understanding physiological changes experienced during PA such as an increased heart rate or changes in energy levels.	During personal training sessions, physiological responses to exercise (heart rate, rating of perceived exertion, pain, etc.) are monitored and the meaning is explained related to the Ex Rx.	<i>Enhancing Physiological Feedback:</i> • Provide appropriate PA instruction and regular reassurance about physiological responses experienced during PA bouts. • Ask how each exercise makes the individual physically feel. • Provide education about the possible discomfort associated with PA. • Encourage use of wearable technology (when appropriate, feasible) for continuous monitoring of physiological responses during exercise.
Emotional states	Positive or negative feelings associated with the behavior change. Discuss the connection between positive feelings and positive PA experiences.	During the personal training session, the exercise professional uses a playlist that enhances the mood of the individual they are working with.	<i>Enhancing Emotional States:</i> • Discuss past PA successes and failures and the feelings linked to those experiences. • Utilize positive imagery and relaxation techniques to maximize opportunities for positive feelings and mood. • Ask how each exercise makes the individual emotionally feel. • Encourage using music, scenery, etc., to make PA pleasurable.

An additional type is barrier self-efficacy, an individual's belief that they can regularly be active in the face of common barriers, such as their ability to engage in PA when there is a lack of time or poor weather. Refer to section "Barriers and Problem Solving" in *Relevant Targets to Promote Physical Activity Behavior Change* for strategies to overcome and evaluate barriers to PA engagement.

The higher the sense of self-efficacy, the greater the effort, persistence, and resilience an individual will exhibit, particularly when faced with barriers or challenges. For example, an older adult who does

not believe that they can lift weights would not even consider enrolling in a program that included resistance training. This individual would have to work on increasing their confidence in their capability to perform resistance training (task self-efficacy) prior to focusing on their ability to resistance train in the face of challenges (barriers self-efficacy).

Outcome Expectations and Expectancies

Outcome expectations and expectancies may help determine whether or not an individual will engage in PA. If specific outcomes of being physically active are anticipated and desired, and are also valued by the individual, then increased PA is more likely to occur ([10](#)). For example, an older adult who wants to improve their fitness level and believes that walking will help with this outcome is more likely to start and maintain a walking program. Motivational interviewing is a communication strategy that can be used to learn about an individual's anticipated outcomes, desired outcomes, and values related to PA. To learn more about motivational interviewing, see *Client-Centered Conversations and Motivational Interviewing* sections.

Self-Regulation

Self-regulation is an individual's ability to set goals, monitor progress toward those goals, solve problems when faced with barriers, and engage in self-reward ([13](#)). An important self-regulation skill to promote PA is self-monitoring.

Self-monitoring involves observing and recording behavior and has been shown to be important in changing PA engagement ([11](#), [13](#)) by helping individuals track progress toward PA goals, identify barriers to becoming more active, and as a cue or prompt to be active. Self-monitoring of PA can be in the form of a paper-and-pencil log, a heart rate monitor, pedometer, or wearables such as a smart watch. Wearable technology and apps can provide the individual with real-time feedback such as minutes of exercise, intensity, distance traveled, or step counts. Self-monitoring using wearable technology or apps provides a platform for tracking PA as well as generalized feedback on progress, but by itself may only have limited, short-term effects ([42](#)).

The exercise professional is a critical part of the feedback loop, providing client support with any self-monitoring technology platform. This can include connecting clients with others who are also using the technology ([43](#), [44](#)), helping set realistic goals, and reviewing progress in the technology or app to provide personalized feedback. Modern advances in technology often provide automated feedback, but it appears that the addition of reinforcement and explanation from the exercise professional is most effective. See section on "Motivation and Reinforcement" in *Relevant Targets to Promote Physical Activity Behavior Change*.

A meta-analysis found that exercise interventions were most effective when self-monitoring was combined with at least one other self-regulation technique, such as prompting goal setting, providing feedback on behavior or outcomes, and reviewing goal progress ([13](#)).

Autonomy

Programs should be designed to enhance autonomy by promoting individual choice, not driven solely by the exercise professional. Autonomy can also be supported by incorporating basic and feasible exercises initially to enhance feelings of competence and enjoyment. Levels of PA have been shown to increase when programs target strategies to promote autonomy (45–47). Exercise professionals can increase PA autonomy by utilizing client-centered communication strategies to identify exercises that the client would enjoy, and collaboratively develop goals and a plan for completing an exercise program.

Goal Setting

Goal setting is a powerful tool that can lead to positive changes in exercise behavior when used as part of process that involves developing, monitoring, and adjusting goals (11). Goal setting is a collaborative approach that includes the individual and the exercise professional.

The exercise professional can assist with the implementation, measurement, and revision of goals on a consistent basis to provide direction to the individual's intentions and efforts; provide regular feedback to enhance their motivation; and identify new strategies. However, the goal is ultimately identified by the individual based on their desired results (self-determination). The exercise professional's role is to support the individual's autonomy of choice and help guide the development and details required to set strong and achievable goals. The SMARTS principle can be used to guide effective goal setting:

- **Specific:** Goals should be precise and clear.
- **Measurable:** Goals should be quantifiable.
- **Action-oriented:** Goals should indicate what needs to be done.
- **Realistic:** Goals should be achievable.
- **Timely:** Goals should have a specific and realistic time frame.
- **Self-determined:** Goals should be developed primarily by the individual.

Examples of SMARTS goals are as follows:

- “My goal is to walk on the treadmill for $20 \text{ min} \cdot \text{d}^{-1}$, $3 \text{ d} \cdot \text{wk}^{-1}$, for the next month.”
- “My goal is to take the stairs instead of the elevator to class each day this semester.”
- “My goal is to swim 10 laps each Saturday for the next 3 months.”

It is important for individuals to set both short- and long-term goals that allow for measurement and feedback on a regular basis. Individuals often focus on long-term goals; however, when attempting to start a new behavior, like an exercise program, setting short-term achievable goals (*i.e.*, daily or weekly) is important for increasing self-efficacy (48). The exercise professional should regularly monitor progress, provide individualized feedback, and discuss successes and struggles with the individual. Goal setting is guided by the fitness professional, but the goals should ultimately be identified by the client. Both short- and long-term goals need to be person-centered, which includes evaluating for readiness, willingness/importance, and ableness/confidence. The *Client-Centered Conversations and Motivational*

Interviewing section provides communication strategies for supporting the client's autonomy and eliciting client goals and motivation.

Intention

Intentions do not always lead to action. For example, setting a goal to be physically active does not always result in increases in PA. Intentions can be challenging targets because of the many factors that can influence them. The exercise professional can address intention by focusing on the implementation plan for goals. An individual's specific and detailed plan to be active bridges the gap between intentions and action.

Implementation of intentions includes where an individual will exercise, when they will exercise, and with whom they will exercise. This target expands upon intentions/goals and outlines specifically how that goal will be reached (49). Evidence has supported the addition of these specific targets to improve exercise behavior (50, 51). This strategy has been applied most frequently to clinical populations, including cancer patients (52), cardiac rehabilitation participants (53), and pregnant females (54). [Table 12.7](#) provides an example of how goals can be expanded into action plans.

TABLE 12.7 Examples of Moving Goals to Action

SMARTS Goal	Action Plan
"My goal is to walk on the treadmill for 20 min · d ⁻¹ , 3 d · wk ⁻¹ , for the next month."	"I will walk on the treadmill at my workplace wellness center at 3.0 m · h ⁻¹ on Monday, Wednesday, and Friday each week this month. I will walk at 12 pm, the start of my lunch hour, on each of those days and will schedule this activity on my calendar."
"My goal is to take the stairs instead of the elevator to class each day this semester."	"I will take the main staircase in the building instead of the elevator each Monday through Friday when I attend my classes. This will include taking the stairs to my morning class and my afternoon class."
"My goal is to swim 10 laps each Saturday for the next 3 months."	"I will swim 10 laps at the local YMCA each Saturday in September, October, and November. I will swim in the morning between 7:30 and 8:00 am with my neighbor who also swims each morning."

To further promote that intentions/goals move into action plans for PA, it is important to also work with individuals to understand barriers that might get in the way of their PA plan. In addition, social support and affect need to be assessed and considered throughout exercise programming. The following sections highlight strategies to target barriers, social support, and affect.

Barriers and Problem Solving

Common barriers to PA include lack of time, lack of motivation, fear of injury, lack of skill, lack of resources, and lack of social support. Example strategies the exercise professional can promote for overcoming these common barriers are listed in [Table 12.8](#).

TABLE 12.8 Common Physical Activity (PA) Barriers

PA Barrier	Strategy for Overcoming the PA Barrier
"I don't have <i>enough time</i> to be physically active"	• Spend time reviewing the individual's schedule and commitments to identify feasible opportunities to incorporate PA. • Schedule short bouts (10 min, 3 times) of PA throughout the day instead of long bouts (30 min, 1 time). • Add PA to daily activities. For example, take the stairs instead of the elevator.
"I'm just <i>not motivated</i> to be physically active"	• Identify forms of PA that are enjoyable. • Determine SMARTS goals and rewards for PA that include activities the individual looks forward to. • Build in social support to PA (see "Social Support").
"I don't have <i>the skill</i> to be physically active"	• Start with activity that is familiar and feasible such as walking. • Ask individuals what they are comfortable starting with (promote autonomy). • Start slow, give detailed instruction, and guide the individual through each exercise successfully.
"It <i>costs too much</i> to be physically active"	• Identify PA that does not require any resources such as walking (indoor or outdoor), taking the stairs, or body weight exercises. • Look for low cost or no cost opportunities for activity such as free community group exercise classes or worksite wellness programming. • Recommend online videos and exercise apps that allows the individual to stay active outside of sessions with the exercise professional (to lower costs).
"There is <i>no one</i> to be active with me"	• Ask the individual if there are opportunities to be active with family, friends, and/or coworkers. • Create support groups and social activities in facilities that bring people together around the theme of being active. • Encourage the use of technology and apps that include social challenge components and rewards.

Individuals face a number of personal, social, and environmental-related barriers in both the adoption and maintenance of PA (55). Problem solving can assist with identifying strategies to reduce or eliminate barriers and includes four main steps that the exercise professional guide the individual through (56):

1. Identify the barrier for PA.
2. Brainstorm strategies with the individual to overcome the barrier.

3. Select a strategy that the individual views as most likely to be successful.
4. Analyze how well the plan worked and revise as necessary.

For example, if lack of time is a barrier to engaging in exercise, the individual, in conjunction with the exercise professional, can identify possible solutions for overcoming this barrier (*e.g.*, schedule exercise appointments before work, incorporate PA into existing activities with the family).

Strategies for overcoming barriers should ideally be generated by the individual and not by the exercise professional. Client-centered conversations and motivational interviewing should be used to identify possible solutions to barriers that are suggested by the client (see “Client-Centered Conversations and Motivational Interviewing”).

Social Support

Social support is a powerful motivator for PA for many individuals. Social support can come from family, friends, coworkers, exercise professionals, and groups of people with similar activity goals. Social support can be provided to individuals in various ways that are (a) instrumental, (b) emotional, (c) informational, and promote (d) companionship ([57, 58](#)) ([Table 12.9](#)).

TABLE 12.9 Social Support Type and Actions and Statements for Physical Activity

Type of Social Support	Exercise-Specific Definition and Supportive Action/Statement
Instrumental	This type of social support is provided when someone has “done something to help you be physically active.” For example, “I’ll make dinner so you can go for a walk.”
Emotional	This type of social support is provided when someone has “encouraged you to do physical activity.” For example, “I’m proud of you for taking the time to exercise each day.”
Informational	This type of social support is provided when someone has “discussed physical activity with you.” For example, “I just saw this step challenge program that you might enjoy.”
Companionship	This type of social support is provided when someone has “done or offered to do physical activity with you.” For example, “I wanted to invite you to take this exercise class with me.”

Social support can be provided in one-on-one and group settings. In group settings, such as group exercise classes and small group training, exercisers can benefit from watching others complete their routines and from instructors and fellow exercisers giving encouragement and input on proper technique. Creating supportive exercise groups has been linked with greater levels of exercise ([4](#)). However, some individuals do not feel comfortable exercising in front of others, which can be a barrier that deters participation and motivation. It is important for the exercise professional to consider this and explore each individual’s comfort level for group exercise.

Aspects of social support can also be present in PA apps and wearable devices (59). Technology allows for a level of support through social rewards, such as praise and for individuals to share their activity progress. It is important that the social support provided through technology is matched to the individual's needs and is not the only form of social support for that individual. Furthermore, not all clients will feel supported through technology. The exercise professional should explore for whom and how to appropriately incorporate technology into programming.

Affect

Individuals are often advised to pick an exercise activity they enjoy. By picking an enjoyable form of PA, people are more likely to adopt and maintain PA (60). Engaging in enjoyable forms of PA is also a key component to establishing intrinsic motivation (61), which is predictive of long-term adherence to recommended levels of PA (28). Affective response varies between individuals, and some exercise intensities and types may be more enjoyable for certain individuals and not for others.

Because cumulative positive experiences can shape automatic connections with engagement in exercise, it is important to understand which exercise scenarios promote positive affective responses. The exercise professional should encourage individuals to use self-ratings of affect to understand how they feel about different intensities, modalities, and specific exercises. This information is useful for continuously adjusting the Ex R_x using the FITT principle to promote better enjoyment and long-term adherence. It has been demonstrated that at low-to-moderate intensities affective response to exercise is typically positive; however, newer exercisers may not experience a positive affect at moderate levels. Once individuals reach high intensities that approach their maximum threshold, affective response during exercise is typically aversive (62). For example, if an exerciser reports increased feelings of displeasure after a high-intensity activity, the exercise professional should likely reduce intensity to modify the Ex R_x.

Evidence on affective response to high-intensity interval training (HIIT) suggests that affect varies by the individual; some individuals will respond positively, whereas others may find this type of exercise aversive. See [Box 12.2](#) for additional information on affect and HIT.

Box 12.2 Affect and HIIT Training

HIIT consists of short bouts of high-intensity anaerobic exercise alternating with very short bouts of less intense recovery. HIIT has gained popularity, both in the fitness industry and in research. The *2018 Physical Activity Guidelines Advisory Committee Scientific Report* found that HIIT is an efficient method for increasing cardiorespiratory fitness, and there is moderate evidence that it can effectively improve insulin sensitivity, blood pressure, and body composition in adults (63). However, the scientific report found that the unpleasant affective response associated with increased intensity is greatest above the lactate and ventilatory thresholds (63-66). Thus, caution should be used in implementing HIIT among people with little exercise experience and those who have been sedentary (67).

To promote a positive affect for PA, exercise professionals can develop an Ex R_x that includes the following:

- Exercises that can be completed in an environment that the individual enjoys (*e.g.*, with friends or in a location that is pleasing to the individual) (68)
- Promotes engagement in a variety of exercises, trying new activities (69)
- Establishes a reward system for exercise (70)
- Regular evaluation of affective response to Ex R_x

Motivation and Reinforcement

[Table 12.5](#) illustrates that motivation exists on a continuum from amotivation (not motivated to engage in PA) to levels of extrinsic motivation (externally motivated to engage in PA) then to intrinsic motivation (internally motivated to engage in PA). Exercise professionals can promote motivation for PA and guide individuals from being less self-determined to more self-determined. Strategies outlined previously to promote PA self-efficacy, self-regulation, and outcome expectations also promote self-determination and enhanced motivation. Additionally, an exercise professional can target motivation through reinforcement.

Reinforcement can be positive (rewards) or negative (punishment); however, positive reinforcement is the focus in many behavior change theories and models. Exercise professionals should encourage individuals to reward themselves for meeting long- and short-term behavioral goals. Rewards can be extrinsic or intrinsic but should be important to the individual to be most effective at increasing the likelihood of engaging in PA behavior again in the future.

Extrinsic rewards include tangible, physical rewards such as money or a new pair of shoes, as well as social reinforcement such as verbal praise and acknowledgment. For example, an individual may reward themselves with new exercise clothes after accomplishing their first short-term goal of exercising for 100 minutes, 2 weeks in a row.

Extrinsic rewards are often used to initiate PA behavior change (71), and evidence suggests that this can be effective, at least in the short term (72). However, individuals are more likely to adhere to regular exercise over the long term if they are doing the activity for intrinsic reasons (28, 73).

Intrinsic rewards are intangible rewards that come from within, such as a feeling of achievement, confidence, or enjoyment. For example, an individual who feels accomplished after reaching their PA goals is more likely to continue to engage in PA. It may be difficult for the exercise professional to give

intrinsic reinforcers to individuals, but it is possible to develop an environment that can promote intrinsic motivation. These environments focus on the autonomy of the individual and have been shown to lead to higher levels of PA (45). For example, an exercise professional can create an autonomous environment by assisting an individual with outlining a SMARTS goal and using personal rulers to assess their importance, willingness, commitment, confidence, and readiness (74), but then allows the individual to take ownership over deciding how to best achieve that goal. The action plan and Ex R_x should incorporate this important information.

Relapse Prevention

Regularly active individuals will occasionally encounter situations that make adhering to their exercise program difficult or nearly impossible. Therefore, exercise professionals play an important part in helping individuals maintain their PA levels through the development of strategies to overcome setbacks or lapses in PA (75). Although it is not unusual to have a brief lapse from PA (*e.g.*, missing an exercise session), preparing individuals for situations that may result in an elongated lapse (*e.g.*, missing a week of exercise sessions), or relapse (*e.g.*, stopping exercise completely), is critical.

Relapse prevention strategies include being aware of and anticipating high-risk situations (*e.g.*, travel, vacation, holidays, illness, competing family obligations, and poor weather) and having a plan to ensure that a short-term lapse does not become a long-term relapse (75). For some individuals, it may be important to vary exercise routines and create new goals to avoid boredom as this can lead to a relapse if enjoyment, excitement, and curiosity are reduced. It is also important that individuals do not get discouraged when they miss a PA session, as missing a planned session is unavoidable. Relapse prevention planning can assist so individuals can recover and stay on track.

For example, an individual has a plan to go for a 30-minute walk during their lunch break at work but gets called into a meeting. An all-or-nothing response that can lead to a lapse might be not exercising for the rest of the day because the planned session was missed. However, a relapse prevention strategy that has some flexibility can allow space to gracefully recover when unanticipated things impact PA plans. Since the individual often gets called into lunch meetings, they have a secondary plan to take a shorter walk after dinner with their family. While they did not get in as many PA minutes as originally planned, this feasible back-up strategy allows them to still complete some PA rather than no PA.

POPULATION CONSIDERATIONS

An important area of exercise promotion is the proper tailoring of programs and Ex Rx to promote PA across diverse populations. This can present unique challenges. Proper personalization requires obtaining an understanding of beliefs, values, environments, and obstacles within a population or individual. Although every individual is clearly unique, the following sections discuss population considerations broken into the distinct areas of: (a) age and lifespan, (b) medical status or health condition, and (c) cultural diversity. This section will also highlight the unique behavioral dynamics when people exercise in groups (*e.g.*, group fitness classes, small group training).

Age and Lifespan

The strategy for promoting PA engagement may differ with distinct age groups or at different time points across the lifespan. The behavioral strategies provided previously generally apply to most adults. However, there are behavior change strategies that have been identified specifically for children and older adults. Of note, Ex Rx recommendations should not be assumed based on age category alone without including appropriate exercise assessments and considering the individual needs of the person or group.

Youth

When working with children (see [Chapter 6](#)), it is important to recognize whether they are engaging in an exercise program because their parents wish them to, implying an extrinsic motivator. This scenario likely requires tangible forms of social support (*e.g.*, transportation, payment of fees) ([55](#)). However, to help children maintain exercise behavior over their lifetime, they need help shifting toward a sense of autonomy, or choice over the decision to be physically active and what activities to participate in ([76](#)). Establishing a sense of autonomy, self-efficacy, and intrinsic motivation through the creation of a supportive environment should be a high priority when fostering PA among children and youth ([45](#)). An appropriately engaged family can be important for PA promotion as well ([77](#)).

Schools are an appealing setting for implementing PA programs, as they reach a majority of youth. Simple modifications to physical education classes ([76](#), [78](#)), small changes during recess ([79](#)), and promoting structured PA within the classroom can increase PA ([80](#)). Physical education, recess, and classroom-based PA programs either enhance or do not take away from academic achievement, academic behavior, and cognitive skills and attitudes ([81](#)).

Older Adults

Older adults (see [Chapter 6](#)) have their own set of distinct behavioral considerations exercise professionals need to understand in order to promote PA adoption and adherence ([82](#), [83](#)). Older adults may lack knowledge about the benefits of PA or how to set up a safe and effective exercise program; therefore, exercise professionals may need to provide some initial education ([84](#)). Although typically viewed as beneficial, certain forms of social support are not necessarily positive ([85](#)). Family and

friends may exert negative influences by telling them to “take it easy” and “let me do it.” The implicit message is that they are too old or frail to be physically active.

Although older adults experience many of the commonly reported barriers to PA (*e.g.*, lack of time, motivation) ([55](#), [83](#)), there may be several unique barriers, including lack of or indifferent social support; increased social isolation; fear of falling/safety; and physical ailments such as injury, chronic illness, and poor health ([55](#), [86](#)). Possibly, the largest barrier to exercise participation in older adults is the fear that exercise will cause injury, pain, and discomfort, or make existing conditions worse ([55](#)). In addition, older females in particular may have had little early-life exposure to PA due to social norms that were less accepting of females being physically active. These unique barriers can be significant and require careful consideration when promoting PA and developing programs for this population. Recommendations include finding enjoyable activities, starting low and going slow if the individual lacks a history of exercise, monitoring affect responses to new and changing activities, and being aware of chronic conditions that might be present.

Medical Status and Health Conditions

PA improves symptoms associated with a number of chronic diseases and health conditions. Despite these benefits (see *Health Belief Model*), this may not be enough to motivate an individual to begin or sustain an exercise program. For example, increasing PA may improve the management of Type 2 diabetes. However, beyond this knowledge, if a person does not believe that that exercise will help regulate glucose levels, and they possess barriers that make it difficult to exercise (*e.g.*, busy work schedule, family commitments), it is less likely that they will see a benefit to engaging in PA. Individuals with health conditions may contemplate the value of adding in PA while balancing the other requirements of managing their health care (*e.g.*, medical visits, medication management, stress, etc.).

An additional concern when working with individuals with health conditions is their ability to physically perform exercises. This requires the exercise professional to focus on enhancing task self-efficacy to ensure that individuals believe that they can do what is being asked of them. Often, this may require appropriate modifications to ensure that it is safe and appropriate for the individual’s current medical status and capabilities.

While the individual health beliefs and self-efficacy of a person are important, there are also condition-specific behavioral considerations and barriers ([55](#)). For example, individuals with arthritis report pain, fatigue, and mobility limitations as barriers to PA participation ([87](#), [88](#)). Those with neurological conditions (*i.e.*, muscular dystrophy, multiple sclerosis, motor neuron disease, and Parkinson’s disease) cite fatigue, fear of falling or losing balance, and safety due to the progression of disease as barriers ([89](#)). Being aware of the unique barriers and fears of individuals with health conditions can help exercise professionals choose appropriate exercises and promote self-efficacy.

Additional behavioral considerations are presented for persons with obesity and those diagnosed with some form of mental health disorder given the recent increases in prevalence of both conditions in the U.S. adult population ([90](#), [91](#)).

Obesity

Individuals with obesity may face unique, weight-related barriers to engaging in PA, such as feeling physically uncomfortable while exercising, being uncomfortable with their appearance, and not wanting to exercise in front of others (social physique anxiety) ([92](#)). Individuals with obesity may have had negative experiences with exercise in the past and will need to enhance their self-efficacy so they believe that they can successfully exercise ([93](#), [94](#)). Furthermore, they may perceive moderate-intensity exercise as challenging, so keeping activities fun and at an intensity that feels good may be particularly important to promote positive perceptions of PA ([95](#)).

While PA decreases across body mass index categories, with individuals with obesity being the least active group ([96](#)), concerns about excess weight is often the primary reason why many individuals with obesity adopt an exercise program ([97](#)). While goals should remain self-determined, individuals with obesity may need help from the exercise professional to set realistic weight loss goals and identify appropriate levels of PA to help them reach those goals (see [Chapter 9](#)) ([98](#)). Other recommendations include starting the Ex Rx low and progressing frequency, time, and intensity slowly, utilizing multiple shorter bouts of activity ([99](#)), and evaluating enjoyment and affect ([95](#)) regularly.

Mental Health Disorders

Although adults with diagnosed mental health disorders are not at increased risk of harm with exercise participation (see [Chapter 11](#)), many cite similar barriers to exercise as the general population, but depending on the disorder, may experience additional barriers as a result of their psychological and psychosomatic health. The barriers, which may be intensified in this population, include perceived fear or actual lack of resources, social support, self-efficacy, and motivation. These barriers exist in addition to side effects associated with psychiatric medications ([100](#)). Individuals with a mental health disorder often have trouble working up the energy or motivation to exercise, and therefore, the exercise professional can help identify enjoyable activities and guide setting small, realistic goals.

Other recommendations include being active with others for motivation and support when appropriate, which can improve mood and reduce sadness or anxiety. Additionally, exercising outside has been shown to have positive effects on mood.

Strong evidence exists to support the role of exercise in reducing anxiety, depression, depressive symptoms in adults, and moderate evidence exists to support the benefits of exercise for adults with schizophrenia and attention deficit hyperactivity disorder (see [Chapter 11](#)) ([63](#)). In adults with severe mental illnesses like major depressive disorder and schizophrenia, evidence suggests that exercise improves symptoms, cognition, and quality of life ([101](#)).

Cultural Diversity

In order to provide culturally competent care to exercisers, it is necessary to be exposed to and understand the cultural beliefs, values, and practices of the desired population. This includes but is not limited to housing, neighborhood characteristics, religion, access to resources, crime, race, ethnicity, age, gender identity, ability level, and social class. Although there is homogeneity (similarities) among

groups, there is heterogeneity (differences) as well. For example, people of the same race may have cultural ties to another country, which may impact their perception of and participation in PA.

Higher levels of physical inactivity may not only be influenced by environmental constraints but also by cultural beliefs (102). For example, African American females have cited barriers to PA participation, which may include a lack of PA exposure and role models, family responsibilities (*i.e.*, need to be the caregiver), issues related to body size (*i.e.*, larger body sizes and curves tend to be appreciated, and thus lower perceived need for PA), and hairstyles (103). Various groups may share barriers and facilitators for PA; thus, it is important for the exercise professional to know about the individual and not make generalized assumptions. PA participation can be linked to neighborhood characteristics, access to resources, and can be influenced by religious and other sociodemographic factors.

Perhaps the most important characteristic of exercise programs that target different cultures is being culturally sensitive and tailored. One common, incorrect assumption made when culturally tailoring exercise materials is to only translate programming resources, such as educational handouts or available video and media content, into another language (102). Such superficial tailoring is not adequate. There should be an in-depth knowledge and understanding of the beliefs, values, and barriers of both the individual and their culture. This deeper understanding can lead to more culturally relevant exercise recommendations along with better adherence to Ex Rx.

Group Exercise Behavioral Dynamics

Separate from attempts to implement individual behavior change is the concept of group-based programming (group exercise classes, small group training, PA support groups) to improve exercise adoption and adherence.

Exercise leaders have an influence on PA participation and the psychological benefits that may occur (104). An exercise leader with a socially supportive leadership style is one that provides encouragement, verbal reinforcement, praise, and interest in all individuals (105). When an exercise leader has a socially supportive leadership style, individuals report greater self-efficacy, more energy, more enjoyment, stronger intentions to exercise, less fatigue, and less concern about embarrassment (106).

In addition to the exercise leader, aspects of the exercise group may also influence PA participation. One such aspect is that of group cohesion, that is, a dynamic process reflected in the tendency of a group to stick together and remain united in the pursuit of its objectives (increasing PA levels) (107). Five principles have been successfully used to improve cohesion and lower dropout rates among exercise groups (108, 109):

- Distinctiveness — creating a group identity (*e.g.*, group name)
- Positions — giving members of the class responsibilities and roles for the group
- Group norms — adopt common goals for the group to achieve
- Sacrifice — individuals in the group giving up something for the greater good of the group
- Interaction and communication — the belief that the more social interactions that are made possible for the group, the greater the cohesion

CLIENT-CENTERED CONVERSATIONS AND MOTIVATIONAL INTERVIEWING

Changing any behavior, including PA, requires a multifaceted approach to prompt behavior change. In order to obtain the relevant information that has been previously outlined in this chapter, the exercise and health care professional must understand how to effectively elicit information from individuals. This information will provide a roadmap for where to start and/or how to tailor the Ex Rx and programs to meet the individual/population where they are on their PA journey.

The Role of Client-Centered Counseling

A proven technique for increasing exercise adoption and helping clients to re-adopt PA after lapse or relapse is through the use of client-centered counseling ([110](#)). Client-centered counseling approaches can be based on any of the constructs or targets discussed previously; however, it is imperative that they be more thorough than simply asking an individual about their PA levels and advising them to increase PA engagement. The following example differentiates the differences between general queries and creating client-centered conversation:

General Question: “How did your exercise go this week?”

Client-Centered Question: “I noticed from your activity tracker that you did 3 exercise sessions this week. Great job! I would love to hear more about how it went and how you felt.”

The differences between the two questions links back to incorporating behavioral strategies like providing feedback from self-monitoring, positive reinforcement, and asking the individual to report on affect, feelings, and potential barriers. A client-centered conversation can be used to explore why individuals do or do not engage in PA through the use of open-ended questions, listening to and learning from the individual, and recognizing that individuals may be resistant to change. When using client-centered counseling approaches, it is important to always keep the communication focused on the individual and avoid providing unsolicited advice.

Motivational Interviewing

Motivational interviewing is one specific client-centered counseling approach that has been successfully applied to many health behaviors including PA ([111](#), [112](#)) and has been studied in a variety of settings ([113](#)).

Motivational interviewing is a method of communication where the exercise professional and the individual work together to explore motivations and plans for PA and resolve resistance to engaging in PA. The exercise professional’s approach should be nonjudgmental, empathic, and encouraging. Motivational interviewing techniques allow the exercise professional to respect and promote individual autonomy and view the individual as responsible for changing their behaviors. Motivational interviewing does not involve persuading or guiltning individuals to exercise or any form of arguing.

A major focus of motivational interviewing is to help an individual who is ambivalent. Ambivalence about behavior change occurs when an individual has conflicting viewpoints about changing their behavior. For example, “I know I should be more active to reduce my risk of developing a health condition, but I don’t like the way I feel when I’m exercising.” Addressing ambivalence involves helping guide the individual to realize the different personal opportunities and motivators that can lead to making a change. This can be achieved by eliciting change talk.

Change talk refers to an individual’s mention or discussion of a desire or reason to change. Change is more likely to occur whenever change talk is elicited ([114](#)). [Table 12.10](#) provides different approaches and examples of ways to elicit change talk.

TABLE 12.10 Methods to Elicit Change Talk

Approach	Description	Examples
Ask evocative questions.	Ask the individual questions regarding: • Disadvantages of the status quo • Advantages of change • Optimism about change • Intention to change • Explore and resolve ambivalence	“What do you think will happen if you don’t change anything?” “What are some benefits of becoming more physically active?” “What changes would work best for you?” “What do you intend to do?”
Use the importance ruler.	Ask simple questions to assess how important physical activity is to the individual and what might make it more important.	“How important would you say it is for you to be physically active? (After response) “Why do you believe that?” “What would it take for you to increase the importance of exercise?”
Use the confidence ruler.	Ask simple questions to assess the individual’s confidence and what might increase their confidence.	“How confident are you that you can engage in regular physical activity? (After response) “What makes you feel that way?” “What would it take for you to feel more confident about this?”
Explore pros and cons.	Encourage individual to discuss the positive and negative aspects of their present behavior. Help explore and resolve ambivalence.	“Are there things that you like about being physically inactive?” “Are there disadvantages of being physically inactive?”
Elaborate.	When health professional hears any arguments for change, encourage the individual to elaborate to reinforce change talk.	“You said exercise might make you feel better. Can you tell me more about that?”
Query extremes.	When the individual has little desire for change, encourage them to consider extreme consequences of not changing and best consequences of changing.	“Suppose you continue on as you have, with no physical activity in your life. What do you imagine are the worst things that might happen to you?” “What might be the best results you could imagine if you make a change?”
Look back.	Help the individual remember a time in their life when they were physically active.	“You mentioned that you used to walk regularly. What was that like?”
Look forward.	Help the individual envision a changed future.	“If you don’t like what you see in the future about yourself, how would you like things to be different?”
Explore values and	Ask the individual to tell you what things are most important in their life and then ask	“What in life is most important to you?”

Approach	Description	Examples
goals.	if being inactive fits with this picture.	(After a response) “Does being physically active or inactive matter to this?”

Adapted from ([107](#), [108](#)).

A tool to help elicit change talk that can be used to build effective SMARTS goals is the personal ruler ([74](#), [115](#)). The personal ruler that can range from 0 to 10 that asks participants to rank their perceived level of confidence and importance. This can be used for any behavior change but can be tailored to specifically build PA and exercise goals, both short and long term. Fitness professionals can use this as a simple assessment to get at a client’s present level of motivation.

The following are examples of motivational interviewing-driven conversations (adapted from ACSM Resource: The Value of Targeting Motivation and Relevance) between an exercise professional and a client. Of note, every conversation does not look and feel the same way. In fact, there may be differences between which behavioral targets are incorporated when an individual is adopting an exercise program compared to when an exercise professional is working with an individual after a relapse in PA.

Example 1: Motivational Interviewing for a Client Adopting PA

Exercise Professional: “I see on your forms that you want to start an exercise program. Tell me a little more about why you decided to take this first step.”

Counseling Prompt/Strategy: Eliciting change talk, using open-ended questions

Client: “I sit for most of the day and feel out of shape. I have noticed that activities such as taking the stairs or carrying groceries into the house are more challenging than they used to be. My doctor has also recommended that I be more physically active to help manage my blood pressure.”

Key Behavior Change Information: Client provided an intention for increasing PA through change talk, stated their personal reasons for wanting to increase PA, indicated a desire to feel better (moving towards intrinsic motivation), and has some extrinsic motivation to increase PA from their doctor.

Exercise Professional: “It sounds like you see a connection between regular inactivity and your ability to engage in activities of daily living, and view PA as an important health promoting behavior. You mentioned that you sit for most of the day, is this because of your job?”

Counseling Prompt/Strategy: Reflective listening and learning, communication back/acknowledging what was heard, using open-ended questions prompting the discussion between the client and the exercise professional to continue.

Example 2: Motivational Interviewing for a Client After Relapse

Exercise Professional: “I see on your form that you want to get back to the exercise program that you were engaged in 8 months ago. Tell me a little more about why you took a break from the exercise program and why now is the right time to start again.”

Counseling Prompt/Strategy: Using open-ended questions based on the information provided to better understand the reasons why they stopped exercising. Note the intention of the exercise professional to understand previous barriers and new motivations.

Client: “I was feeling unmotivated to do the same exercises each day and always felt like I had to sacrifice time with my family to be active, so I took a break from exercising.”

Key Behavior Change Information: Client reported losing motivation which may indicate the need to explore ways to enhance enjoyment and excitement (to become more intrinsically motivated) through autonomy and self-determined goal setting and they stated that they value time with family.

Exercise Professional: “It sounds like you have experienced some common barriers to exercise such as feeling unmotivated and short on time to be active. Why do you want to start being active again now? Are there any exercises that you enjoy doing or that you think your family would be willing to do with you?”

Counseling Prompt/Strategy: Reflective listening and learning, communication back/acknowledging what was heard, using open-ended questions prompting the discussion between the client and the exercise professional to continue. Note the intention of the exercise professional to ask again why they are motivated to re-start now, and connect back to the key information (enjoyment, family) to dive deeper so previous barriers and de-motivators are considered and new motivations are incorporated into the Ex Rx.

While motivational interviewing is a common client-centered communication strategy, it is not the only approach that can be utilized. Another approach is to use the 5A's (Ask, Advise, Assess, Assist, Arrange), which also promotes a collaborative conversation between an exercise professional and an individual wanting to increase PA ([116](#)). These five targets can provide a checklist for starting conversations. From here, the exercise professional can dive deeper into the key mentions provided by an individual and use that information to enhance the program and personalize the overall experience.

Understanding the behavioral-based reasons discussed in this chapter for why an individual chooses to engage or not engage in PA can enhance the overall success of an Ex Rx. This requires proper tailoring based on key information learned during client-centered conversations that consider both individual- and population-specific beliefs and behavioral needs. Balancing these behavioral considerations with a physiologically appropriate Ex Rx can further enhance long-term PA adoption and adherence.

ONLINE RESOURCES

Exercise is Medicine: <http://exerciseismedicine.org>

National Cancer Institute Behavioral Research Program Theories Project:

http://cancercontrol.cancer.gov/brp/research/theories_project/index.html

National Physical Activity Plan: <http://www.physicalactivityplan.org/>

2018 Physical Activity Guidelines Scientific Report: <https://health.gov/our-work/nutrition-physical-activity/physical-activity-guidelines/current-guidelines/scientific-report>

ACSM Resource: The Value of Targeting Motivation and Relevance:

https://www.acsm.org/docs/default-source/member-hub-documents/member-resources/acsm-bpr-the-value-of-targeting-motivation-and-relevance-pp1.pdf?sfvrsn=c7fde8be_2

Readiness Ruler: <https://case.edu/socialwork/centerforebp/resources/readiness-ruler>

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APPENDIX A

Common Medications

LIST OF COMMON MEDICATIONS

[Appendix A](#) lists common medication categories that exercise and health care professionals are likely to encounter among individuals who are soon to be, or are, physically active. This table is not intended to be exhaustive or all-inclusive and is not designed for the determination of pharmacotherapy/medication prescription for individuals by clinicians/physicians. Rather, this listing should be viewed as a resource for exercise professionals to assist in understanding how typical hemodynamic responses to exercise may be impacted by certain medications. For a more detailed informational listing, the reader is referred to the online resources of the American Hospital Formulary Service (AHFS) Drug Information or the U.S. Food and Drug Administration and U.S. Department of Health and Human Services.

[Table A.1](#) lists the common categories of medications with available published data regarding their influence on the response to exercise, specifically hemodynamics, the electrocardiogram (ECG), and exercise capacity. Exercise data are presented by drug category. The influence of common medications during rest and/or exercise is presented with the directional relationships when specified in the literature. *Exercise capacity* is a generic term that often was used and not defined by a specific measure in the literature. In cases where measures of exercise capacity were reported, they are listed as maximal oxygen consumption ($\dot{V}O_{2\max}$), endurance, performance, and tolerance, often without clear distinctions provided by the author.

Table A.1 Effects of Medications on Hemodynamics, the Electrocardiogram (ECG), and Exercise Capacity (1-8)

Medications	Cardiac Output	Heart Rate (HR)	Blood Pressure (BP)	ECG Changes	Exercise Capacity
I. Cardiovascular medications					
β-Blockers (BB)	↓ or ↔ Exercise	↓ Rest and exercise ↓ Restless by intrinsic sympathomimetic activity (ISA) + BB ↓ Exercise less by cardioselective BB	↓ Rest and exercise	↓ Rest ↓ Ischemia during exercise	↑ In those with myocardial ischemia
Angiotensin-converting enzyme inhibitors (ACE-I)	↔ Exercise	↔ Exercise	↓ Rest and exercise		↔ Performance; ↑ tolerance in individuals with heart failure (HF)
Angiotensin-neprilysin inhibitors (ARNI)	↑		↓ Rest and exercise	↓ Ventricular arrhythmias	↑ Performance
Angiotensin II receptor blockers (ARB)		↓ or ↔ Rest and exercise	↓ Rest and exercise		↔
Calcium channel blockers (CCB)					
Nondihydropyridines (non-DHP)		↓ Rest and exercise	↓ Exercise		↔ Performance and endurance; responses can be variable
Dihydropyridine	Nifedipine: ↔ Exercise ↓ Stroke volume	↔ Exercise	↓ Exercise (greater vs. non-DHP)		↔ Performance and endurance; responses can be variable
II. Vasodilating agents					
Nitrates		↑ Rest ↑ or ↔ Exercise	↓ Rest ↓ or ↔ Exercise	↑ Rest HR ↑ or ↔ Exercise HR	↑ Individuals with angina ↔ Individuals without angina

Medications	Cardiac Output	Heart Rate (HR)	Blood Pressure (BP)	ECG Changes	Exercise Capacity
				↓ Exercise ischemia	↑ or ↔ Individuals with HF
α-Blockers	↔ Exercise Doxazosin: ↑ exercise at 50% $\dot{V}O_{2max}$	↔ Exercise Doxazosin: ↑ exercise at 75% $\dot{V}O_{2max}$	↓ Exercise systolic BP (SBP) (not diastolic BP [DBP])	↓ Exercise ischemia	↔ Performance
Central α-agonist	↔ Exercise	↓ Exercise except guanabenz	↓ Exercise		Clonidine: blunts the sympathetic response to exercise; consider avoiding if exercising.

III. Antiarrhythmic agents

				All antiarrhythmic agents may cause new or worsened arrhythmias (<i>i.e.</i> , proarrhythmic effect).	
Class I					
Quinidine		↑ or ↔ Rest and exercise	↓ or ↔ Rest	↑ or ↔ Rest HR Exercise may cause false-negative test results.	↔
Disopyramide			↔ Exercise	Rest may prolong QRS and QT intervals.	
Procainamide		↔ Rest and exercise	↔ Rest and exercise	Rest may prolong QRS and QT intervals. Exercise may result in false-	↔

Medications	Cardiac Output	Heart Rate (HR)	Blood Pressure (BP)	ECG Changes	Exercise Capacity
				positive test results.	
Propafenone		↓ Rest ↓ or ↔ Exercise	↔ Rest and exercise	↓ Rest HR ↓ or ↔ Exercise HR	↔
Class II					
BB (see Cardiovascular medications)					
Class III					
Amiodarone	↔	↓ Rest and exercise	↔ Rest ↑ Exercise	↓ Rest HR	↑ or ↔
Sotalol				↔ Exercise	
Class IV					
CCB (see Cardiovascular medications)					
Others					
Digitalis		↓ Individuals with atrial fibrillation and possibly HF Not significantly altered in individuals with sinus rhythm	↔ Rest and exercise	Rest may produce nonspecific ST-T wave changes. During exercise, may produce ST-segment depression	↑ Individuals with atrial fibrillation or HF
IV. Respiratory					
Inhaled corticosteroids		↔ Rest and exercise	↔ Exercise	↔ Exercise	↔
Bronchodilators and anticholinergics		↔ Rest and exercise	↔ Rest and exercise	Bronchodilators: ↔ rest and exercise Anticholinergics: ↑ or ↔ HR	↑ or ↔ In individuals with chronic obstructive pulmonary disease (COPD)

Medications	Cardiac Output	Heart Rate (HR)	Blood Pressure (BP)	ECG Changes	Exercise Capacity
					Bronchodilators: ↔ $\dot{V}O_{2\max}$
Sympathomimetics (β_2-receptor agonists)		↔ Rest and exercise	↔		↔ Performance or $\dot{V}O_{2\max}$; ↑ or ↔ In individuals with COPD
Albuterol		May ↑ exercise			↔ Performance and $\dot{V}O_{2\max}$
Pseudoephedrine		↔ Rest and exercise May ↑ exercise	↔ Exercise May ↑ exercise SBP	May produce premature ventricular contractions (PVC)	↔ Performance
Xanthine derivatives				Rest and exercise may produce PVCs.	
Theophylline		↑ Rest ↔ Exercise			↔ Performance and $\dot{V}O_{2\max}$
Caffeine	↔	↑ Resting ↑ or ↔ exercise	↑ Exercise		↑ Endurance
Antihistamines		↑ Rest ↔ Exercise			↔ Performance and endurance
V. Hormonal					
Human growth hormone		↔ Rest and exercise	↔	↔	↑ Performance and $\dot{V}O_{2\max}$
Androgenic-anabolic		↔ Rest and exercise	↑ DBP		↔ or ↑ Performance and $\dot{V}O_{2\max}$
Thyroid agents		↑ Rest and exercise	↑ Rest and exercise	↑ HR May provoke arrhythmias	↔ Unless angina worsens during exercise
Levothyroxine					↑ Cardiopulmonary reserve ↔ Recovery and performance

Medications	Cardiac Output	Heart Rate (HR)	Blood Pressure (BP)	ECG Changes	Exercise Capacity
VI. Central nervous system					
Antidepressants		↑ or ↔ Rest and exercise	↓ or ↔ Rest and exercise	Variable rest	
Antipsychotics					
Lithium		↔ Rest and exercise	↔ Rest and exercise		
Antianxiety		↑ or ↔ Rest and exercise	↓ or ↔ Rest and exercise	Variable rest	
Stimulants		↑	↑		↑ or ↔ Endurance and performance
Nicotine replacement therapy		↑	↑		↔ or ↓
Nonsteroidal anti-inflammatory drugs (NSAIDs)					↔ Performance; combined with dehydration may cause acute renal failure

↓, decreased; ↑, increased; ↔, not changed; $\dot{V}O_{2\max}$, maximal volume of oxygen consumed per unit time.

It is important to note that exercise may impact the pharmacokinetic (*i.e.*, what the body does to the medication) and pharmacodynamic (*i.e.*, what the medication does to the body) properties of a medication, necessitating a change in (a) dose, (b) dosing interval, (c) length of time the individual takes the medication, and/or (d) the exercise prescription.

[Table A.2](#) lists common categories of medications prescribed to manage diabetes (9), their associated hypoglycemia risk, effects on weight, and commonly reported side effects. Having a general understanding of this information, in addition to the guidelines provided in the *Diabetes Mellitus* section, can help aid the exercise professional in safely delivering exercise interventions.

Table A.2 Diabetes Medication Classes and Clinical Considerations

Medication	Hypoglycemia Risk	Weight Change	Clinical Considerations
Metformin	No	Neutral (potential for modest loss)	• GI side effects common • Potential for vitamin B₁₂ deficiency
SGLT2 inhibitors	No	Loss (intermediate)	• DKA risk, rare in T2DM • Increased risk of genital mycotic infections • Rare reports of necrotizing fasciitis of the perineum (Fournier gangrene) • Attention to volume status, blood pressure
GLP-1 RAs	No	Loss (intermediate to very high)	• Risk of thyroid C-cell tumors in rodents; human relevance not determined • Potential for GI side effects; typically temporary • Potential for ileus • Pancreatitis has been reported in clinical trials but causality has not been established
Dual GIP and GLP-1 RA	No	Loss (very high)	• Risk of thyroid C-cell tumors in rodents; human relevance not determined • Potential for GI side effects which are temporary • Not recommended for individuals with a history of gastroparesis • Pancreatitis has been reported in clinical trials but causality has not been established
DPP-4 inhibitors	No	Neutral	• Pancreatitis has been reported in clinical trials but causality has not been established • Joint pain • Bullous pemphigoid

Medication	Hypoglycemia Risk	Weight Change	Clinical Considerations
Thiazolidinediones	No	Gain	• Congestive HF (pioglitazone, rosiglitazone) • Fluid retention (edema; heart failure) • Benefit in MASH • Risk of bone fractures
Sulfonylureas (second generation)	Yes	Gain	• FDA Special Warning on increased risk of CV mortality based on studies on an older sulfonylurea (tolbutamide); glimepiride shown to be CV safe
Insulin (human or analog)	Yes	Gain	• Injection site reactions • Higher risk of hypoglycemia with human insulin vs. analogs

For agent-specific dosing recommendations, please refer to manufacturers' prescribing information.

CV, cardiovascular; DKA, diabetic ketoacidosis; DPP-4, dipeptidyl peptidase 4; GI, gastrointestinal; GIP, glucose-dependent insulintropic polypeptide; GLP-1 RA, glucagonlike peptide 1 receptor agonist; HF, heart failure; MASH, metabolic dysfunction-associated steatohepatitis; SGLT2, sodium-glucose cotransporter 2; T2DM, Type 2 diabetes mellitus.

Table adapted from the Standards of Care in Diabetes—2024 (9).

[Table A.3](#) lists categories of medications prescribed to manage cancer including treatment types, common medications within treatment type, and the associated side effects.

Table A.3 Common Cancer Drugs and Side Effects

Common Cancer Drugs		
Treatment Type	Drug Class	Common Side Effects
Systemic chemotherapy	Anthracycline (A) — “rubicin” (Doxorubicin, Epirubicin)	Low blood cell count, hair loss, nausea or vomiting, fatigue, cardiotoxicity
	Platinum-based agent (PA) — “platin” (Cisplatin, Carboplatin)	Low blood cell count, nausea or vomiting, taste changes, fatigue, peripheral neuropathy
	Taxane (T) — “taxel” (Paclitaxel, Docetaxel)	Low blood cell count, arthralgias, myalgias, fatigue, peripheral neuropathy
	Antimetabolite (AM) (5-FU or Fluorouracil, Capecitabine)	Low blood cell counts, fatigue, nausea and vomiting, poor appetite
Targeted chemotherapy/immunotherapy	Monoclonal antibody (MA) — “mab” (Trastuzumab, Pembrolizumab)	Muscle/joint pain, nausea, generalized weakness, fatigue, abnormal labs (sodium, glucose)
	Aromatase inhibitor (AI) (Letrozole, Anastrozole)	Hot flashes, muscle/joint pain, weight gain, high cholesterol, nausea
	Antiestrogen (AE) (Tamoxifen)	Hot flashes, nausea, weight loss, menstrual irregularities, mood changes
For list of commonly prescribed drugs and their associated side effects. See NIH for exhaustive list https://www.cancer.gov/about-cancer/treatment/drugs		
Common Chemotherapy Treatment Acronyms		
AC	Doxorubicin (A), Cyclophosphamide (AA)	
CAF	Cyclophosphamide (AA), Doxorubicin (A), Fluorouracil (AM)	
CAV	Cyclophosphamide (AA), Doxorubicin (A), Vincristine (PA)	
CHOP	Cyclophosphamide (AA), Hydroxydaunorubicin (A), Vincristine (VA), Prednisone (C)	
FOLFOX	Fluorouracil (AM), Folinic Acid (FAA), Oxaliplatin (PA)	
FOLFIRINOX	Fluorouracil (AM), Folinic Acid (FAA), Irinotecan (TI), Oxaliplatin (PA)	
TCH	Docetaxel (T), Carboplatin (PA), Trastuzumab (MA)	
TAC or ACT	Docetaxel (T), Paclitaxel (T), Doxorubicin (A), Cyclophosphamide (AA)	
VAD	Vincristine (VA), Doxorubicin (A), Dexamethasone (G)	

A, anthracycline; AA, alkylating agent; AE, antiestrogen; AI, aromatase inhibitor; AM, antimetabolite; C, corticosteroid; FAA, folic acid analog; G, glucocorticoid; MA, monoclonal antibody; PA, platinum agent; T, taxane; TI, topoisomerase; VA, vinca alkaloid.

ONLINE RESOURCES

American Hospital Formulary Service Drug Information: <http://www.ahfsdruginformation.com>

U.S. Food and Drug Administration, U.S. Department of Health and Human Services:

[http://www.accessdata.fda.gov/scripts/cder/drugsatfda/index.cfm?
fuseaction=Search.Search_Drug_Name](http://www.accessdata.fda.gov/scripts/cder/drugsatfda/index.cfm?fuseaction=Search.Search_Drug_Name)

National Cancer Institute: <https://www.cancer.gov/about-cancer/treatment/drugs>

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APPENDIX B

Electrocardiogram Interpretation

[Appendix B](#) provides a quick reference for electrocardiogram (ECG) recording and interpretation, with a focus on normal expectations of the ECG. [Figures B.1](#) and [B.2](#) serve as visual references for appropriate ECG waves, intervals, and segments as well as appropriate ECG lead placement, respectively. Each of the tables ([Tables B.1–B.4](#)) should be used as part of the overall clinical profile when making diagnostic decisions about an individual.

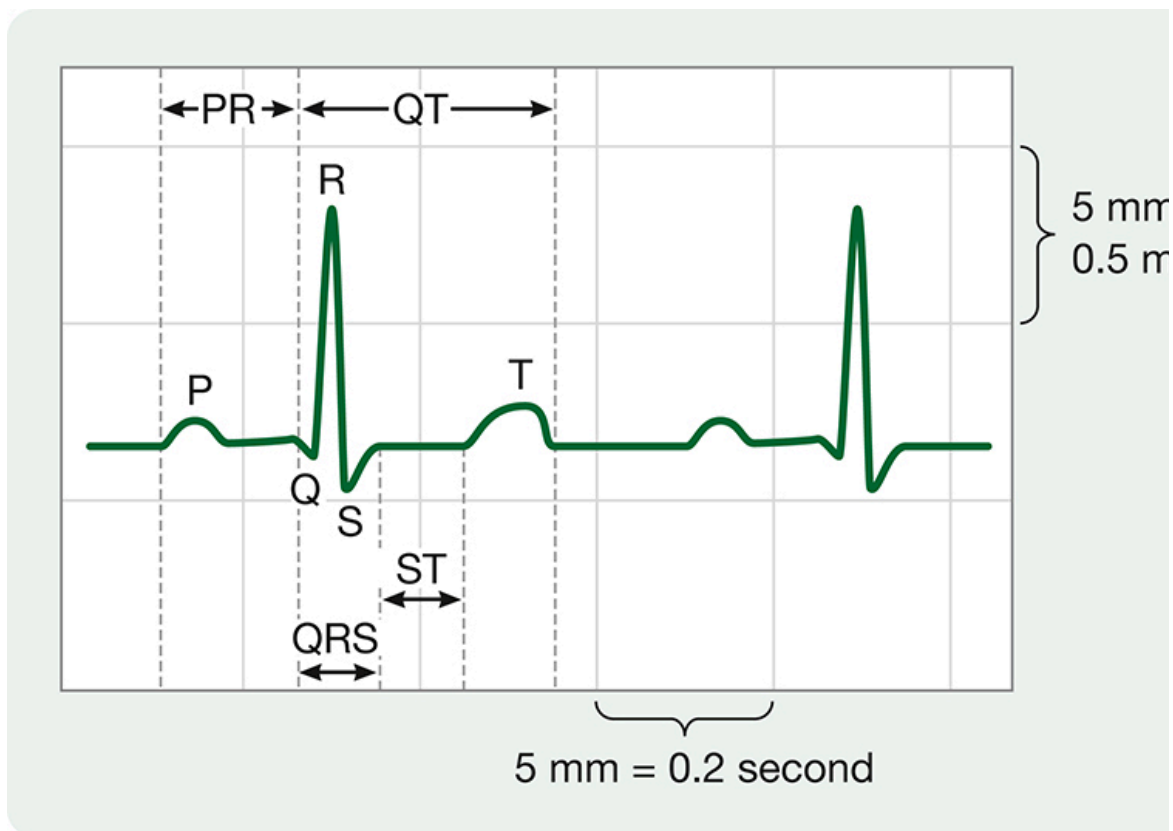


Figure B.1 ECG waves, intervals, and segments. (Reprinted from Figure 9.9 Feito Y, Magal M. ACSM's fitness assessment manual. 6th ed. Philadelphia (PA): Wolters Kluwer Health; 2021. 208 p.)

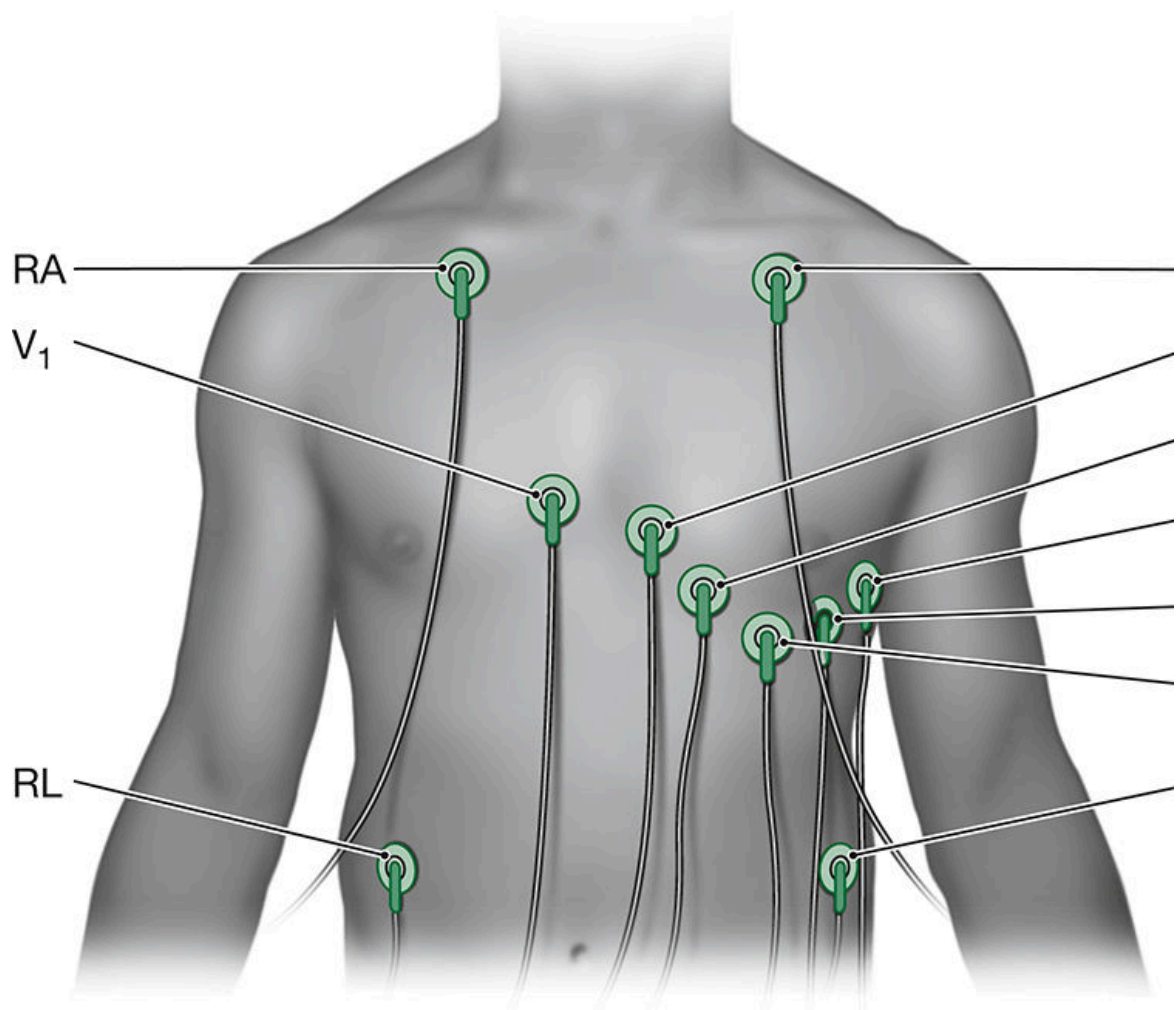


Figure B.2 Modified 12-lead ECG placement. (Reprinted from Figure 24.4 Golan DE. Principles of pharmacology: the pathophysiologic basis of drug therapy. 4th ed. Philadelphia (PA): Wolters Kluwer Health; 2017. 1024 p.)

Table B.1 Limb and Augmented Lead Electrode Placement^{[a](#)}

Lead	Electrode Location and Polarity	Heart Surface Viewed
Lead I	Left arm (+), right arm (-)	Lateral
Lead II	Left leg (+), right arm (-)	Inferior
Lead III	Left leg (+), left arm (-)	Inferior
aVR	Right arm (+)	None
aVL	Left arm (+)	Lateral
aVF	Left leg (+)	Inferior

^a Exercise modifications: The limb leads are positioned over the left and right infraclavicular region for the arm leads and over the left and right lower quadrants of the abdomen for the leg leads. This electrocardiogram (ECG) configuration minimizes motion artifacts during exercise. However, torso-placed limb leads should be noted for all ECG tracings to avoid misdiagnosis of an ECG tracing. The most common changes observed are produced by right axis deviation and standing that may obscure or produce Q waves inferiorly or anteriorly and T wave or frontal QRS axis changes even in normal people ([1](#), [2](#)).

Table B.2 Precordial (Chest Lead) Electrode Placement

Lead	Electrode Placement	Heart Surface Viewed
V ₁	Fourth intercostal space just to the right of the sternal border	Septum
V ₂	Fourth intercostal space just to the left of the sternal border	Septum
V ₃	At the midpoint of a straight line between V ₂ and V ₄	Anterior
V ₄	On the midclavicular line in the fifth intercostal space	Anterior
V ₅	On the anterior axillary line and on a horizontal plane through V ₄	Lateral
V ₆	On the midaxillary line and on a horizontal plane through V ₄ and V ₅	Lateral

Adapted from (3).

Table B.3 Electrocardiogram (ECG) Interpretation Steps

1. Check for correct calibration (1 mV = 10 mm) and paper speed (25 mm · s⁻¹). 2. Verify the heart rate and determine if heart rhythm is regular. 3. Measure intervals (PR, QRS, QT). 4. Determine the mean QRS axis and mean T-wave axis in the limb leads. 5. Look for morphologic abnormalities of the P wave, QRS complex, ST segment, T wave, and U wave (<i>e.g.</i>, chamber enlargement, conduction delays, infarction, repolarization changes). 6. Interpret the present ECG. 7. Compare the present ECG with previous available ECGs. 8. Offer conclusion, clinical correlation, and recommendations.
--

Table B.4 Resting Electrocardiogram (ECG): Normal Limits (4–6)

Parameter	Normal Limits	Considered Abnormal when:	Possible Interpretation(s) ^a
Heart rate	60–100 beats · min ⁻¹	<60 beats · min ⁻¹	Bradycardia
		>100 beats · min ⁻¹	Tachycardia
P wave	< 0.12 s	Broad and notched (> 0.12 s) in leads I, II, aVL, and V ₄ –V ₆ and inverted in V ₁	Left atrial hypertrophy
	< 2.5 mm tall	Peaked (> 2.5 mm tall) in leads II, III, and aVF and upright in V ₁	Right atrial hypertrophy or enlargement
		Peaked and broad in leads I, II, III, aVL, aVF, and V ₄ –V ₆ and biphasic in V ₁	Combined atrial hypertrophy
PR interval	0.12–0.20 s	<0.12 s	Preexcitation (<i>e.g.</i> , W-P-W or L-G-L)
		>0.20 s	Sinus delay (<i>e.g.</i> , AV block)
QRS duration	0.06–0.10 s	If ≥0.11 s	Conduction abnormality (<i>e.g.</i> , incomplete or complete bundle branch block, W-P-W, IVCD, or electronic pacer)
QT interval	Rate dependent	QTc long	Drug effects, electrolyte abnormalities, or ischemia

Parameter	Normal Limits	Considered Abnormal when:	Possible Interpretation(s) ^a
		QTc short	Digitalis effect, hypercalcemia, or hypermagnesemia
QRS axis	-30° to +110°	<-30°	Left axis deviation (<i>e.g.</i> , hemiblock, MI)
		>+110°	Right axis deviation (<i>e.g.</i> , RVH, COPD, PE, hemiblock, MI)
		Indeterminate	All limb leads transitional
T wave	Upright in leads I, II, and V ₃ -V ₆ ; inverted in aVR; flat, inverted, or biphasic in III and V ₁ -V ₂	Upright, inverted, flattened, or biphasic alone or with ST-segment changes	Can be a normal variant; ischemia, LVH, or caused by physiologic conditions (posture changes, respiration, drugs)
T axis	Generally same direction as QRS axis	The T axis (vector) is typically deviated away from the area of "mischief" (<i>e.g.</i> , ischemia, bundle branch block, or hypertrophy).	Chamber enlargement, ischemia, drug effects, bundle branch blocks, or electrolyte disturbances
ST segment	Generally at isoelectric line (PR segment) or within 1 mm	Elevation of ST segment	Normal variant (early repolarization), injury, ischemia, pericarditis, or electrolyte abnormality

Parameter	Normal Limits	Considered Abnormal when:	Possible Interpretation(s) ^a
	The ST segment may be elevated up to 1–2 mm in leads V ₁ –V ₄ .	Depression of ST segment 80 ms after the J-point	Injury, ischemia, electrolyte abnormality, drug effects, or normal variant
Q wave	<0.04 s and <25% of R wave amplitude (exceptions: leads III and V ₁)	>0.04 s and/or >25% of R wave amplitude except leads III and V ₁	MI or pseudoinfarction (as from chamber enlargement, conduction abnormalities, W-P-W, COPD, or cardiomyopathy)
Transition zone	Usually between V ₂ and V ₄	Before V ₂	Counterclockwise rotation (early transition)
		After V ₄	Clockwise rotation (late transition)

^a If supported by other electrocardiograms and related clinical criteria.

AV, atrioventricular; COPD, chronic obstructive pulmonary disease; IVCD, intraventricular conduction delay; L-G-L, Lown-Ganong-Levine syndrome; LVH, left ventricular hypertrophy; MI, myocardial infarction; PE, pulmonary embolism; QTc, QT corrected for heart rate; RVH, right ventricular hypertrophy; W-P-W, Wolff-Parkinson-White syndrome.

ONLINE RESOURCES

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APPENDIX C

Metabolic Calculations and Methods for Prescribing Aerobic Exercise Intensity

METABOLIC CALCULATIONS

Measuring oxygen consumption ($\dot{V}O_2$) requires expensive laboratory equipment managed by trained professional staff who can perform the test as well as interpret the data. This equipment does not lend itself to large numbers of clients, participants, or patients. When it is not possible or feasible to measure $\dot{V}O_2$, reasonable estimates of the $\dot{V}O_2$ during exercise can be made from regression equations derived from measured $\dot{V}O_2$ during exercise on ergometric devices and while walking and running. In this regard, the American College of Sports Medicine (ACSM) developed equations for estimating $\dot{V}O_2$ during *steady-state, submaximal* aerobic exercise as described in this Appendix ([1](#)). More recently, the Fitness Registry and the Importance of Exercise National Database (FRIEND) was created to meet the clinical need to establish normal standards for *maximal exercise capacity* (maximal volume of oxygen consumed per unit time [$\dot{V}O_{2\max}$]) for a healthy (without known coronary vascular or pulmonary diseases) and diverse (age, sex, race, body mass, geographic distribution) population, composed of thousands of participants ([2](#)). Furthermore, this registry has been utilized to develop equations that provide estimates of $\dot{V}O_2$ at *maximal exercise capacity* ($\dot{V}O_{2\max}$) ([3–7](#)). See [Table C.2](#).

ESTIMATION OF ENERGY EXPENDITURE: THE ACSM METABOLIC CALCULATIONS

[Table C.1](#) presents the ACSM metabolic equations for the *gross* or total oxygen cost, expressed in $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, of walking, running, leg ergometry, arm ergometry, and stepping. For each prediction equation, there are essential known physiologic constants, such as how much oxygen is required to move the body horizontally (walking on the flat) and vertically (walking up a grade or hill), or the oxygen cost of pedaling at no resistance ([1](#)). Detailed explanations, examples, and applications are reviewed elsewhere ([Figure C.1](#)) ([1, 8](#)). Moreover, for exercise prescription purposes, these equations may be used to determine the required exercise intensity associated with a desired level of energy expenditure ([1](#)). When using these equations to determine caloric expenditure, $\dot{V}\text{O}_2$ should be used by subtracting the resting $\dot{V}\text{O}_2$, or $3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. This value is commonly used for the resting metabolic rate (RMR) ([9, 10](#)) and has been termed a metabolic equivalent (MET). METs have been used as a tool in epidemiologic studies that easily reflect a standard intensity of a wide variety of activities by dividing the measured energy cost ($\dot{V}\text{O}_2$ [$\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$]) of the activities by $3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ ([9](#)). There has been concern that $3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ overestimates measured RMRs and is influenced by age, body mass, and sex ([10](#)). Furthermore, an equation has been developed resulting in a “corrected MET” value ([11](#)). The corrected MET may be appropriate for exercise prescription and to estimate one’s energy expenditure ([9, 11](#)). However, it is unlikely that the value of $3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ introduces meaningful error as used in the metabolic equations. The ACSM metabolic equations should *not* be used to predict $\dot{V}\text{O}_{2\text{max}}$, especially in a clinical setting where an accurate estimate of maximal exercise

capacity is necessary for therapeutic or prognostic applications. These equations were not developed to equate to a value of $\dot{V}O_2$ measured during progressive, non-steady-state, maximal exercise tests ([4](#), [5](#)). Additionally, these equations only utilized a limited number of rather homogeneous subjects that are not representative of the population undergoing cardiopulmonary exercise testing (CPET) ([4](#), [5](#)).

Table C.1 Metabolic Calculations for the Estimation of Gross Energy Expenditure ($\dot{V}O_2$ [$\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$]) During Common Physical Activities

Sum of Resting + Horizontal + Vertical/Resistance Components				
Mode	Resting Component	Horizontal Component	Vertical Component/Resistance Component	Limitations
Walking	3.5	$0.1 \times \text{speed}_a$	$1.8 \times \text{speed}_a \times \text{grade}_b$	Most accurate for speeds of $1.9\text{--}3.7 \text{ mi} \cdot \text{h}^{-1}$ ($50\text{--}100 \text{ m} \cdot \text{min}^{-1}$)
Running	3.5	$0.2 \times \text{speed}_a$	$0.9 \times \text{speed}_a \times \text{grade}_b$	Most accurate for speeds $>5 \text{ mi} \cdot \text{h}^{-1}$ ($134 \text{ m} \cdot \text{min}^{-1}$)
Stepping	3.5	$0.2 \times \text{steps} \cdot \text{min}^{-1}$	$1.33 \times (1.8 \times \text{step height}_c \times \text{steps} \cdot \text{min}^{-1})$	Most accurate for stepping rates of $12\text{--}30 \text{ steps} \cdot \text{min}^{-1}$
Leg cycling	3.5	3.5	$(1.8 \times \text{work rate}_d) / \text{body mass}_e$	Most accurate for work rates of $300\text{--}1,200 \text{ kg} \cdot \text{m} \cdot \text{min}^{-1}$ ($50\text{--}200 \text{ W}$)
Arm cycling	3.5	—	$(3 \times \text{work rate}_d) / \text{body mass}_e$	Most accurate for work rates

Sum of Resting + Horizontal + Vertical/Resistance Components				
Mode	Resting Component	Horizontal Component	Vertical Component/Resistance Component	Limitations
				between 150 and 750 kg · m · min ⁻¹ (25–125 W)

^a Speed in m · min⁻¹.

^b Grade is grade percentage expressed in decimal format (*e.g.*, 10% = 0.10).

^c Step height in m.

Multiply by the following conversion factors:

lb to kg: 0.454; in to cm: 2.54; ft to m: 0.3048; mi to km: 1.609; mi · h⁻¹ to m · min⁻¹: 26.8; kg · m · min⁻¹ to W: 0.164; W to kg · m · min⁻¹: 6.12; $\dot{V}O_{2\max}$ L · min⁻¹ to kcal · min⁻¹: 4.9; $\dot{V}O_2$ MET to mL · kg⁻¹ · min⁻¹: 3.5.

^d Work rate in kilogram meters per minute (kg · m · min⁻¹) is calculated as resistance (kg) × distance per revolution of flywheel × pedal frequency per minute. Note: Distance per revolution is 6 m for Monark leg ergometer, 3 m for the Tunturi and BodyGuard ergometers, and 2.4 m for Monark arm ergometer.

^e Body mass in kg.

MET, metabolic equivalent; $\dot{V}O_2$, volume of oxygen consumed per unit of time; $\dot{V}O_{2\max}$, maximal volume of oxygen consumed per unit time.

Adapted from ([1](#), [8](#)).

Using metabolic calculations (see Table D.1) to determine target work rate ($\text{kg} \cdot \text{m}^{-1} \cdot \text{min}^{-1}$) on a Monark leg cycle ergometer

Available data:

A 42 yr old female

Weight: 190 lb (86.4 kg)

Height: 70 inches (177.8 cm)

Desired $\dot{V}\text{O}_2$: $18 \text{ kg} \cdot \text{m}^{-1} \cdot \text{min}^{-1}$; determined from the exercise prescription

Formula: $\dot{V}\text{O}_2 = 3.5 + 3.5 + (1.8 \times \text{work rate}) / \text{body mass}$

1. Calculate work rate on cycle ergometer:

$\dot{V}\text{O}_2 = 7.0 + (1.8 \times \text{work rate}) / \text{body mass}$

$$18 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 7.0 + (1.8 \times \text{work rate}) / 86.4 \text{ kg}$$

$$11 = (1.8 \times \text{work rate}) / 86.4$$

$$950.4 = 1.8 \times \text{work rate}$$

$$528 = \text{work rate}$$

$$\text{Work rate} = 528 \text{ kg} \cdot \text{m}^{-1} \cdot \text{min}^{-1} = 86.6 \text{ W}$$

Using metabolic calculations (see Table D.1) to determine % grade during walking on a treadmill

Available data:

A 54 yr old male who is moderately physically active

Weight: 190 lb (86.4 kg)

Height: 70 inches (177.8 cm)

Desired walking speed: $2.5 \text{ mi} \cdot \text{hr}^{-1}$ ($4 \text{ km} \cdot \text{hr}^{-1}$; $67 \text{ m} \cdot \text{min}^{-1}$)

Desired MET: 5 METs; determined from the exercise prescription

Formula: $\dot{V}\text{O}_2 = 3.5 + (0.1 \times \text{speed}) + (1.8 \times \text{speed} \times \% \text{ grade})$

1. Determine target $\dot{V}\text{O}_2$:

$$\text{Target } \dot{V}\text{O}_2 = \text{MET} \times 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

$$\text{Target } \dot{V}\text{O}_2 = 5 \times 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 17.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

2. Determine treadmill grade:

$$\dot{V}\text{O}_2 = 3.5 + (0.1 \times \text{speed}) + (1.8 \times \text{speed} \times \text{grade})$$

$$17.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 3.5 + (0.1 \times 67 \text{ m} \cdot \text{s}^{-1}) + (1.8 \times 67 \text{ m} \cdot \text{s}^{-1} \times \text{grade})$$

$$14 = (0.1 \times 67 \text{ m} \cdot \text{s}^{-1}) + (1.8 \times 67 \text{ m} \cdot \text{s}^{-1} \times \text{grade})$$

$$14 = 6.7 + (120.6 \times \text{grade})$$

$$7.3 = 120.6 \times \text{grade}$$

$$0.06 = \text{grade}$$

$$\% \text{ grade} = 6\%$$

Determination of net energy expenditure (EE), in kcals, for 30 min of exercise

1. Determine net $\dot{V}\text{O}_2$:

$$\text{Net } \dot{V}\text{O}_2 = \text{gross } \dot{V}\text{O}_2 - \text{resting } \dot{V}\text{O}_2$$

$$\text{Net } \dot{V}\text{O}_2 = 17.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} - 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 14 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

2. Convert relative rate of $\dot{V}\text{O}_2$ to absolute rate of $\dot{V}\text{O}_2$:

$$\text{Absolute rate of } \dot{V}\text{O}_2 \text{ in } \text{L} \cdot \text{min}^{-1} = (\dot{V}\text{O}_2 \text{ in } \text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}) (\text{body mass}) / 1,000$$

$$\text{Absolute rate of } \dot{V}\text{O}_2 \text{ in } \text{L} \cdot \text{min}^{-1} = (14.0) (86.4) / 1,000 = 1.21 \text{ L} \cdot \text{min}^{-1}$$

3. Convert absolute rate of $\dot{V}\text{O}_2$ in $\text{L} \cdot \text{min}^{-1}$ to $\text{kcal} \cdot \text{min}^{-1}$:

$$\text{Net caloric expenditure in } \text{kcal} \cdot \text{min}^{-1} = 1.21 \text{ L} \cdot \text{min}^{-1} \times 4.9 \text{ kcal} \cdot \text{L}^{-1} = 5.9 \text{ kcal} \cdot \text{min}^{-1}$$

$$\text{Net caloric expenditure for 30 min} = 5.9 \text{ kcal} \cdot \text{min}^{-1} \times 30 \text{ min} = 177 \text{ kcal}$$

Figure C.1 Examples of the application of selected metabolic equations. MET, metabolic equivalent; $\dot{V}O_2$, volume of oxygen consumed per unit time.

ESTIMATING $\dot{V}O_{2peak}$: THE FRIEND REGISTRY PREDICTION EQUATIONS

Participants in the FRIEND registry completed a CPET that directly measured peak oxygen uptake ($\dot{V}O_{2peak}$) (peak respiratory exchange ratio >1.0) by open-circuit spirometry in one of eight participating laboratories that used valid and calibrated equipment and testing procedures administered by experienced personnel ([2–5](#)). The prediction equations developed from the FRIEND registry to estimate $\dot{V}O_{2peak}$ ([Table C.2](#)) are clinically relevant as they have demonstrated accuracy and a lower average error between directly measured $\dot{V}O_{2peak}$ and predicted $\dot{V}O_{2peak}$ when compared to commonly used equations ([3–7](#)).

Table C.2 Equations for Predicting Maximal Oxygen Uptake ($\dot{V}O_2$ [$\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$])

Activity	Equation	Standard Error of Estimate
Cycle ergometry (5)	Nonsex specific: $\dot{V}O_{2\text{max}} = 1.74 \times [\text{Watts} \times 6.12 / \text{body weight (kg)}] + 3.5$ Males: $\dot{V}O_{2\text{max}} = 1.76 \times [\text{Watts} \times 6.12 / \text{body weight (kg)}] + 3.5$ Females: $\dot{V}O_{2\text{max}} = 1.65 \times [\text{Watts} \times 6.12 / \text{body weight (kg)}] + 3.5$ Formula for correcting the overestimation of METs by previously using the ACSM cycle formula. Nonsex specific: CORRECTED METs = ACSM-derived METs – $0.06 \times [\text{Watts} \times 6.12 / \text{body weight (kg)}] - 3.5$	None provided
Treadmill (4) walking or running	Nonsex specific: $\dot{V}O_{2\text{max}} = \text{speed (m} \cdot \text{min}^{-1}) \times (0.17 + \text{fractional grade} \times 0.79) + 3.5$	None provided
Cycle ergometry and treadmill (3)	Sex specific: $\dot{V}O_{2\text{max}} = 45.2 - (0.35 \times \text{age}) - [10.9 \times \text{sex (male = 1; female = 2)}] - [0.15 \times \text{weight (lb)}] + [0.68 \times \text{height (inches)}] - [0.46 \times \text{exercise mode (treadmill = 1; cycle ergometer = 2)}]$	$6.6 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$
Nonexercise (2)	Sex specific: gross $\dot{V}O_{2\text{max}} = 79.9 - (0.39 \times \text{age}) - [13.7 \times \text{sex (0 = male; 1 = female)}] - [0.127 \times \text{weight (lb)}]$	$7.2 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$
Patients with cardiovascular disease (6)	Sex specific: $\dot{V}O_{2\text{max}} = 16.18 - (0.22 \times \text{age}) - [3.63 \times \text{sex (male = 1; female = 0)}] + [0.14 \times \text{height (cm)}] - [0.12 \times \text{weight (kg)}] + [3.62 \times \text{mode (treadmill = 1; cycle = 0)}] - [2.70 \times \text{CABG (yes = 1, no = 0)}] - [0.31 \times \text{MI (yes = 1,$	$4.75 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$

Activity	Equation	Standard Error of Estimate
	no = 0)] + [0.37 × PCI (yes = 1, no = 0)] – [4.47 × HF (yes = 1, no = 0)]	
Treadmill for patients with heart failure (7)	Nonsex specific: $\dot{V}O_{2max} = \text{speed (m/min)} \times (0.17 + \text{fractional grade} \times 0.32) + 3.5$	None provided

ACSM, American College of Sports Medicine; METs, metabolic equivalents; $\dot{V}O_{2max}$, maximal volume of oxygen consumed per unit time.

APPLICATION OF VARIOUS METHODS FOR PRESCRIBING AEROBIC EXERCISE INTENSITY

Intensity is a fundamental element of the physical activity or exercise prescription. There is much scientific debate on the most accurate method of quantifying exercise intensity and to do so with a high level of precision would require real-time collection of biomarkers. Generally, intensity is measured as a percent of maximal aerobic capacity (e.g., $\dot{V}O_{2\max}$ or $\dot{V}O_2$ reserve) but maintaining the accuracy of the prescription would require regular maximal exercise tests to adjust for improvements in cardiorespiratory fitness as a result of exercise training (8). Accordingly, surrogate measures that align well with percent aerobic capacity, such as heart rate or perceived exertion are used (8). With regard to the use of maximal heart rate ($HR_{\max}$), when it is not determined from a maximal exercise test, it has been common practice to estimate $HR_{\max}$ using equations found in [Table 5.3](#). It is important to note that any prediction equation has large variance and should not be assumed to be a completely accurate estimate of one's $HR_{\max}$. The heart rate reserve method in many cases is a more accurate way of establishing a target heart rate rather than using a percent of $HR_{\max}$ (8). [Figure C.2](#) provides examples of methods used to monitor intensity.

Heart rate reserve (HRR) method

Available test data:

$$HR_{\text{rest}}: 70 \text{ beats} \cdot \text{min}^{-1}$$

$$HR_{\text{max}}: 180 \text{ beats} \cdot \text{min}^{-1}$$

Desired exercise intensity range: 50%–60%

$$\text{Formula: target heart rate (THR)} = [(HR_{\text{max}} - HR_{\text{rest}}) \times \% \text{ intensity}] + HR_{\text{rest}}$$

1. Calculation of HRR:

$$HRR = (HR_{\text{max}} - HR_{\text{rest}})$$

$$HRR = (180 \text{ beats} \cdot \text{min}^{-1} - 70 \text{ beats} \cdot \text{min}^{-1}) = 110 \text{ beats} \cdot \text{min}^{-1}$$

2. Determination of exercise intensity as %HRR:

Convert desired %HRR into a decimal by dividing by 100

$$\%HRR = \text{desired intensity} \times HRR$$

$$\%HRR = 0.5 \times 110 \text{ beats} \cdot \text{min}^{-1} = 55 \text{ beats} \cdot \text{min}^{-1}$$

$$\%HRR = 0.6 \times 110 \text{ beats} \cdot \text{min}^{-1} = 66 \text{ beats} \cdot \text{min}^{-1}$$

3. Determine THR range:

$$THR = \%HRR + HR_{\text{rest}}$$

To determine lower limit of THR range:

$$THR = 55 \text{ beats} \cdot \text{min}^{-1} + 70 \text{ beats} \cdot \text{min}^{-1} = 125 \text{ beats} \cdot \text{min}^{-1}$$

To determine upper limit of THR range:

$$THR = 66 \text{ beats} \cdot \text{min}^{-1} + 70 \text{ beats} \cdot \text{min}^{-1} = 136 \text{ beats} \cdot \text{min}^{-1}$$

$$\text{THR range: } 125 \text{ beats} \cdot \text{min}^{-1} \text{ to } 136 \text{ beats} \cdot \text{min}^{-1}$$

$\dot{V}O_2$ reserve ($\dot{V}O_{2R}$) method

Available test data:

$$\dot{V}O_{2\text{max}}: 30 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

$$\dot{V}O_{2\text{rest}}: 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

Desired exercise intensity range: 50%–60%

$$\text{Formula: Target } \dot{V}O_2 = [(\dot{V}O_{2\text{max}} - \dot{V}O_{2\text{rest}}) \times \% \text{ intensity}] + \dot{V}O_{2\text{rest}}$$

1. Calculation of $\dot{V}O_{2R}$:

$$\dot{V}O_{2R} = \dot{V}O_{2\text{max}} - \dot{V}O_{2\text{rest}}$$

$$\dot{V}O_{2R} = 30 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} - 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

$$\dot{V}O_{2R} = 26.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

2. Determination of exercise intensity as % $\dot{V}O_{2R}$:

Convert desired intensity (% $\dot{V}O_{2R}$) into a decimal by dividing by 100

$$\%\dot{V}O_{2R} = \text{desired intensity} \times \dot{V}O_{2R}$$

Calculate % $\dot{V}O_{2R}$

$$\%\dot{V}O_{2R} = 0.5 \times 26.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 13.3 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

$$\%\dot{V}O_{2R} = 0.6 \times 26.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 15.9 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

3. Determine target $\dot{V}O_{2R}$ range:

$$(\%\dot{V}O_{2R}) + \dot{V}O_{2\text{rest}}$$

To determine the lower target $\dot{V}O_2$ range:

$$\text{Target } \dot{V}O_2 = 13.3 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} + 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 16.8 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

To determine upper target $\dot{V}O_2$ range:

$$\text{Target } \dot{V}O_2 = 15.9 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} + 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 19.4 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

$$\text{Target } \dot{V}O_2 \text{ range: } 16.8 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \text{ to } 19.4 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

4. Determine MET target range (optional):

$$1 \text{ MET} = 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

Calculate lower MET target:

$$1 \text{ MET} / 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = X \text{ MET} / 16.8 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

$$X \text{ MET} = 16.8 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} / 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 4.8 \text{ METs}$$

Calculate upper MET target:

$$1 \text{ MET} / 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = X \text{ MET} / 19.4 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

$$X \text{ MET} = 19.4 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} / 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 5.5 \text{ METs}$$

5. Identify physical activities requiring energy expenditure within the target range from compendium of physical activities (7,9).

%HR_{max} (measured or estimated) method:

Available data:

A 45 yr old male

Desired exercise intensity: 70%–80%

Formula: $\text{THR} = \text{HR}_{\text{max}} \times \text{desired \%}$

Calculate estimated HR_{max} (if measured HR_{max} not available):

$$\text{HR}_{\text{max}} = 207 - (0.7 \times \text{age})$$

$$\text{HR}_{\text{max}} = 207 - (0.7 \times 45)$$

$$\text{HR}_{\text{max}} = 207 - 32 = 175 \text{ beats} \cdot \text{min}^{-1}$$

1. Determine THR range:

$$\text{THR} = \text{Desired \%} \times \text{HR}_{\text{max}}$$

Convert desired % HR_{max} into a decimal by dividing by 100

Determine lower limit of THR range:

$$\text{THR} = 175 \text{ beats} \cdot \text{min}^{-1} \times 0.70 = 123 \text{ beats} \cdot \text{min}^{-1}$$

Determine upper limit of THR range:

$$\text{THR} = 175 \text{ beats} \cdot \text{min}^{-1} \times 0.80 = 140 \text{ beats} \cdot \text{min}^{-1}$$

THR range: 123 to 140 beats · min⁻¹

% $\dot{V}\text{O}_2$ max (measured or estimated) method

Available data:

A 45 yr old female

Estimated $\dot{V}\text{O}_{2\text{max}}$: 30 mL · kg⁻¹ · min⁻¹

Desired $\dot{V}\text{O}_2$ range: 50%–60%

Formula: $\dot{V}\text{O}_{2\text{max}} \times \text{desired \%}$

Determine target $\dot{V}\text{O}_2$ range:

$$\text{Target } \dot{V}\text{O}_2 = \text{Desired \%} \times \dot{V}\text{O}_{2\text{max}}$$

Convert desired intensity (% $\dot{V}\text{O}_2$) into a decimal by dividing by 100

Determine lower limit of target $\dot{V}\text{O}_{2\text{max}}$ range

$$\text{Target } \dot{V}\text{O}_2 = 0.50 \times 30 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 15 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

Determine upper limit of target $\dot{V}\text{O}_{2\text{max}}$ range

$$\text{Target } \dot{V}\text{O}_2 = 0.60 \times 30 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} = 18 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

Target $\dot{V}\text{O}_2$ range: 15 mL · kg⁻¹ · min⁻¹ to 18 mL · kg⁻¹ · min⁻¹

Figure C.2 Examples of the application of various methods for prescribing exercise intensity. HR_{max} , maximal heart rate; HR_{rest} , resting heart rate; MET, metabolic equivalent; $\dot{V}O_2$, volume of oxygen consumed per unit of time; $\dot{V}O_{2max}$, maximal volume of oxygen consumed per unit of time; $\dot{V}O_{2R}$, percentage of oxygen uptake reserve; $\dot{V}O_{2rest}$, resting volume of oxygen consumed per unit of time. Adapted from ([1](#), [8](#)).

ONLINE RESOURCES

Compendium of Physical Activities: <https://pacompendium.com/>

Youth Compendium of Physical Activities: <https://www.nccor.org/nccor-tools/youthcompendium/>

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APPENDIX D

Reference List of Commonly Used Equations

The purpose of [*Appendix D*](#) is to serve as a section that can be used to reference common equations used when performing fitness assessments and exercise prescriptions. Metabolic and prediction equations are found in [*Appendix C*](#).

BODY COMPOSITION ([BOX D.1](#))

Box D.1 Generalized Skinfold Equations

Males

- **Seven-site formula** (chest, midaxillary, triceps, subscapular, abdomen, suprailiac, thigh)

Body density = $1.112 - 0.00043499 (\text{sum of seven skinfolds}) + 0.00000055 (\text{sum of seven skinfolds})^2 - 0.00028826 (\text{age})$ [SEE 0.008 or ~ 3.5 % fat]

StartLayout 1st Row Body density equals 1.112 minus 0.00043499 left parenthesis s u m of seven skinfolds right parenthesis 2nd Row plus 0.00000055 left parenthesis s u m of seven skinfolds right parenthesis squared 3rd Row minus 0.00028826 left parenthesis a g e right parenthesis left bracket upper S upper E upper E 0.008 or tilde 3.5 percent sign f a t right bracket

EndLayout

- **Three-site formula** (chest, abdomen, thigh)

Body density = $1.10938 - 0.0008267 (\text{sum of three skinfolds}) + 0.0000016 (\text{sum of three skinfolds})^2 - 0.0002574 (\text{age})$ [SEE 0.008 or ~ 3.4 % fat]

StartLayout 1st Row Body density equals 1.10938 minus 0.0008267 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000016 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.0002574 left parenthesis a g e right parenthesis left bracket upper S upper E upper E 0.008 or tilde 3.4 percent sign f a t right bracket

EndLayout

- **Three-site formula** (chest, triceps, subscapular)

Body density = $1.1125025 - 0.0013125 (\text{sum of three skinfolds}) + 0.0000055 (\text{sum of three skinfolds})^2 - 0.000244 (\text{age})$ [SEE 0.008 or ~ 3.6 % fat]

StartLayout 1st Row Body density equals 1.1125025 minus 0.0013125 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000055 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.000244 left parenthesis a g e right parenthesis left bracket upper S upper E upper E 0.008 or tilde 3.6 percent sign f a t right bracket

EndLayout

Females

- **Seven-site formula** (chest, midaxillary, triceps, subscapular, abdomen, suprailiac, thigh)

Body density = $1.097 - 0.00046971 (\text{sum of seven skinfolds}) + 0.00000056 (\text{sum of seven skinfolds})^2 - 0.00012828 (\text{age})$ [SEE 0.008 or ~ 3.8 % fat]

StartLayout 1st Row Body density equals 1.097 minus 0.00046971 left parenthesis s u m of seven skinfolds right parenthesis 2nd Row plus 0.00000056 left parenthesis s u m of seven skinfolds right parenthesis squared 3rd Row minus 0.00012828 left parenthesis a g e right parenthesis left bracket upper S upper E upper E 0.008 or tilde 3.8 percent sign f a t right bracket

EndLayout

- **Three-site formula** (triceps, suprailiac, thigh)

Body density = $1.0994921 - 0.0009929 (\text{sum of three skinfolds}) + 0.0000023 (\text{sum of three skinfolds})^2 - 0.0001392 (\text{age})$ [SEE 0.009 or ~ 3.9 % fat]

StartLayout 1st Row Body density equals 1.0994921 minus 0.0009929 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000023 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.0001392 left parenthesis italic a g e right parenthesis left bracket upper S upper E upper E 0.009 or tilde 3.9 percent sign f a t right bracket

EndLayout

- **Three-site formula** (triceps, suprailiac, abdominal)

Body density = $1.089733 - 0.0009245 (\text{sum of three skinfolds}) + 0.0000025 (\text{sum of three skinfolds})^2 - 0.0000979 (\text{age})$ [SEE 0.009 or ~ 3.9 % fat]

StartLayout 1st Row Body density equals 1.089733 minus 0.0009245 left parenthesis s u m of three skinfolds right parenthesis 2nd Row plus 0.0000025 left parenthesis s u m of three skinfolds right parenthesis squared 3rd Row minus 0.0000979 left parenthesis a g e right parenthesis left bracket upper S upper E upper E 0.009 or tilde 3.9 percent sign f a t right bracket

EndLayout

SEE, standard error of estimate.

Adapted from ([1](#), [2](#)).

CARDIORESPIRATORY FITNESS ASSESSMENT

Field Tests

Run/Walk Tests

1.5-mi run/walk test

$\dot{V}O_{2\max}(\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}) = 3.5 + 483 / 1.5 \text{mi time (min)}$ Modifying Above normal upper V With normal upper O Subscript 2 max Baseline left parenthesis m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts right parenthesis equals 3.5 plus 483 divided by 1.5 m i time left parenthesis min right parenthesis

12-min walk/run test

$\dot{V}O_{2\max}(\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}) = (\text{distance in meters} - 504.9) / 44.73$ Modifying Above normal upper V With normal upper O Subscript 2 max Baseline left parenthesis m upper L dot k g Superscript negative 1 Baseline dot min Overscript negative 1 Endscripts right parenthesis equals left parenthesis distance in meters negative 504.9 right parenthesis divided by 44.73

Rockport One-Mile Fitness Walking Test

$\dot{V}O_{2\max}(\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}) = 132.853 - (0.1692 \times \text{body mass in kg}) - (0.3877 \times \text{age in years}) + (6.315 \times \text{sex}) - (3.2649 \times \text{time in minutes}) - (0.1565 \times \text{HR})$ (SEE = 5.0 mL · kg⁻¹ · min⁻¹ ; sex = 0 for female , 1 for male) StartLayout 1st Row ModifyingAbove normal

upper V With' normal upper O Subscript 2 max Baseline left
 parenthesis m upper L dot k g Superscript negative 1 Baseline dot min
 Overscript negative 1 Endscripts right parenthesis equals 132.853
 minus left parenthesis 0.1692 times body mass in k g right parenthesis
 minus 2nd Row left parenthesis 0.3877 times a g e in years right
 parenthesis plus left parenthesis 6.315 times s e x right parenthesis
 minus left parenthesis 3.2649 times time in minutes right parenthesis
 minus 3rd Row left parenthesis 0.1565 times upper H upper R right
 parenthesis 4th Row left parenthesis upper S upper E upper E equals
 5.0 m upper L dot k g Superscript negative 1 Baseline dot min
 Overscript negative 1 Endscripts semicolon s e x equals 0 for female
 comma 1 for male right parenthesis EndLayout

6-min Walk Test

$V_{O_2 \text{ peak}} = V_{O_2 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}} = (0.02 \times \text{distance [m]}) - (0.191 \times \text{age [yr]}) - (0.07 \times \text{weight [kg]}) + (0.09 \times \text{height [cm]}) + (0.26 \times \text{RPP [} \times 10 - 3 \text{]}) + 2.45$

StartLayout 1st Row ModifyingAbove
 normal upper V With' normal upper O Subscript 2 peak Baseline equals
 ModifyingAbove normal upper V With' normal upper O 2 m upper L dot
 k g Superscript negative 1 Baseline dot min Overscript negative 1
 Endscripts equals left parenthesis 0.02 times distance left bracket
 normal m right bracket right parenthesis minus 2nd Row left
 parenthesis 0.191 times a g e left bracket y r right bracket right
 parenthesis minus left parenthesis 0.07 times weight left bracket k g
 right bracket right parenthesis plus left parenthesis 0.09 times height
 left bracket c m right bracket right parenthesis plus 3rd Row left
 parenthesis 0.26 times upper R upper P upper P left bracket times 10
 Superscript negative 3 Baseline right bracket right parenthesis plus
 2.45 EndLayout

where m = distance in meters; yr = year; kg = kilogram; cm =
centimeter; RPP = rate-pressure product (HR $\times$ SBP in mm Hg); SEE =
 $2.68 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$

SKELETAL MUSCLE FITNESS ASSESSMENT

Estimating 1 Repetition Maximum ([Table D.1](#))

TABLE D.1 RM percentage of 1 RM

Number of Completed Repetitions	Percentage of 1 RM
1	100
2	95
3	93
4	90
5	87
6	85
7	85
8	80
9	77
10	75
11	70
12	67
15	65

From Sheppard and Triplett ([14](#)).

For repetitions that are ≤ 10 repetitions, the following equations can be used. In the equations, W represents the weight lifted in pounds or kilograms and RTF for repetitions to fatigue.

Brzycki (3)

$$1RM = W / [1.0278 - (0.0278 \times RTF)]$$

Baechle et al. (4)

$$W \times [1 + (0.033 \times RTF)]$$

Epley (5)

$$(0.033 \times RTF \times W) + W$$

Landers (6)

$$(100 \times W) \div [101.3 - (2.67123 \times RTF)]$$

Adams (7)

$$W / (1 - 0.02 \times RTF)$$

Mayhew et al. (8)

$$W / (0.522 + 0.419 e^{-0.055 \times RTF})$$

O'Connor et al. (9)

$$0.025 (W \times RTF) + W$$

Reynolds et al. (10)

$$W / (0.5551 e^{-0.0723 \times RTF} + 0.4847)$$

EXERCISE PRESCRIPTION (BOXES D.2 AND D.3)

Box D.2 Summary of Methods for Prescribing Exercise Intensity Using Heart Rate (HR), Oxygen Uptake ($\dot{V}O_2$), and Metabolic Equivalents (METs)

- HRR method: target HR (THR) = $[(HR_{\text{max/peak}}^a - HR_{\text{rest}}) \times \% \text{ intensity desired}] + HR_{\text{rest}}$
- $\dot{V}O_{2R}$ method: target $\dot{V}O_{2R}^b = [(\dot{V}O_{2\text{max/peak}}^c - \dot{V}O_{2\text{rest}}) \times \% \text{ intensity desired} + \dot{V}O_{2\text{rest}}$
- HR method: target HR = $HR_{\text{max/peak}}^a \times \% \text{ intensity desired}$
- $\dot{V}O_2$ method: target $\dot{V}O_2^b = \dot{V}O_{2\text{max/peak}}^c \times \% \text{ intensity desired}$
- MET method: target MET^b = $[(\dot{V}O_{2\text{max/peak}}^c) / 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}] \times \% \text{ intensity desired}$

^a $HR_{\text{max/peak}}$ is the highest value obtained during maximal/peak exercise or it can be estimated via a prediction equation (see [Table 5.3](#)).

^b Activities at the target $\dot{V}O_2$ and MET can be determined using a compendium of physical activity ([11](#), [12](#)) or metabolic calculations ([13](#)) (see [Table D.1](#)).

^c $\dot{V}O_{2\text{max/peak}}$ is the highest value obtained during maximal/peak exercise or it can be estimated from a submaximal exercise test. See “The Concept of Maximal Oxygen Uptake” section in [Chapter 3](#) for the distinction between $\dot{V}O_{2\text{max}}$ and $\dot{V}O_{2\text{peak}}$. $HR_{\text{max/peak}}$, maximal or peak heart rate; HRR, heart rate reserve; HR_{rest} , resting heart rate; $\dot{V}O_{2\text{max/peak}}$, maximal or peak volume of oxygen consumed per unit of time; $\dot{V}O_{2R}$, oxygen uptake reserve; $\dot{V}O_{2\text{rest}}$, resting volume of oxygen consumed per unit of time.

Box D.3 Calculation of Metabolic Equivalents (METs), $\text{MET} \cdot \text{min}^{-1}$, and $\text{kcal} \cdot \text{min}^{-1}$

METs: an index of energy expenditure (EE). “A MET is the ratio of the rate of energy expended during an activity to the rate of energy expended at rest ... [One] MET is the rate of EE while sitting at rest ... by convention ... [1 MET is equal to] an oxygen uptake of 3.5 $[\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}]$ ”.

MET-min: an index of EE that quantifies the total amount of physical activity performed in a standardized manner across individuals and types of activities. Calculated as the product of the number of METs associated with one or more physical activities and the number of minutes the activities were performed (*i.e.*, $\text{METs} \times \text{min}$), usually standardized per week or per day as a measure of exercise volume.

Kilocalorie (kcal): the energy needed to increase the temperature of 1 kg of water by 1 °C. To convert METs to $\text{kcal} \cdot \text{min}^{-1}$, it is necessary to know an individual’s body weight, $\text{kcal} \cdot \text{min}^{-1} = [(\text{METs} \times 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1} \times \text{body weight in kg})/1,000] \times 5$. Usually standardized as kilocalorie per week or per day as a measure of exercise volume.

ADDITIONAL EQUATIONS ([TABLE D.2](#))

TABLE D.2 Commonly Used Equations for Estimating Maximal Heart Rate (HR_{max})

Author	Equation	Population
Astrand (15)	$HR_{\max} = 216.6 - (0.84 \times \text{age})$	Males and females age 4–34 yr
Tanaka et al. (16)	$HR_{\max} = 208 - (0.7 \times \text{age})$	Healthy males and females
Gellish et al. (17)	$HR_{\max} = 207 - (0.7 \times \text{age})$	Males and females in an adult fitness program with broad range of age and fitness levels
Gulati et al. (18)	$HR_{\max} = 206 - (0.88 \times \text{age})$	Asymptomatic middle-aged females referred for stress testing

Equations for Mechanical Work and Power

Work or $W = F \times D$, where F represents mass in pounds or kilograms and D represents displacement in feet or meters.

Power of $P = W/T$, where W is work (see above equation) and T is time lapsed in seconds or minutes.

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